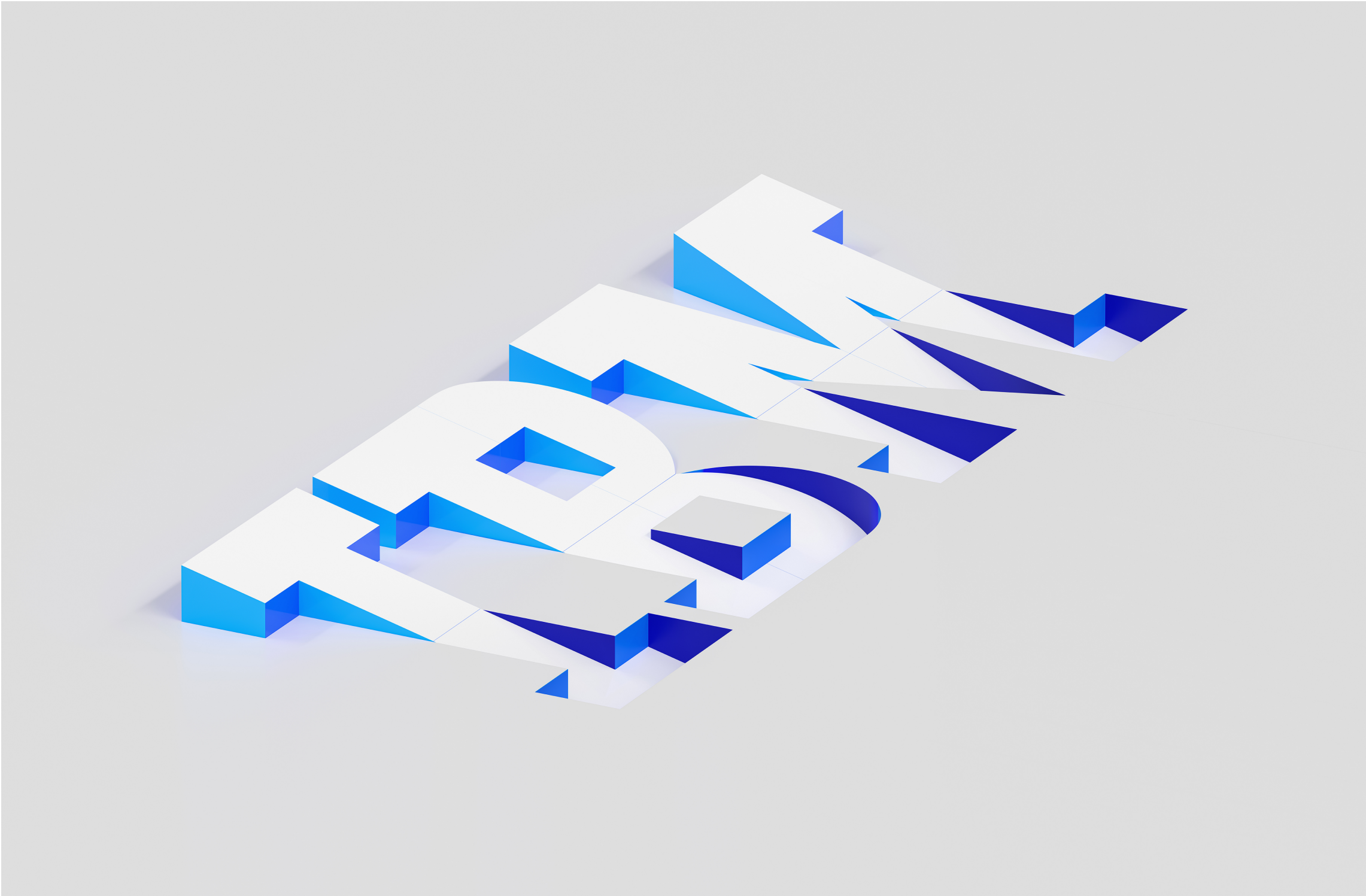


Power Virtual Server

Product guide



Edition notices

This PDF was created on 2026-09-11 as a supplement to *Power Virtual Server* in the IBM Cloud docs. It might not be a complete set of information or the latest version. For the latest information, see the IBM Cloud documentation at <https://cloud.ibm.com/docs/power-iaas>.

Getting started with IBM Power Virtual Server

IBM® Power® Virtual Server is an IBM Power server offering. You can use the Power Virtual Server to deploy a virtual server, also known as a logical partition (LPAR), in a matter of minutes. You can provision flexible, secure, and scalable compute capacity for Power enterprise workloads both on IBM Power Virtual Server in (IBM data center and in Client location) in your data center.

You get fast and flexible management that can be connected to access a stack of enterprise services from IBM – all with pay-as-you-use billing. Using these services you can easily adjust workloads with flexible compute capacity. You can quickly deploy a Power Virtual Server to meet your specific business needs and easily control workload demands. Avoid the large capital expenses or added risk when migrating your essential workloads and get started with Power Virtual Server today! For frequently asked questions about the Power Virtual Server, see [FAQ](#).

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server resources reside in IBM data centers with dedicated networking and Storage area network (SAN)-attached Fibre Channel storage. You can choose one of the regions that is listed in the specifications that are nearest to your data center. IBM Power clients who rely on client-owned infrastructure can now quickly and economically extend their Power IT resources off-premises.

In the data centers, the Power Virtual Servers are separated from the rest of the IBM Cloud servers with separate networks and direct-attached storage. The internal networks are fenced but offer connectivity options to IBM Cloud infrastructure or on-premises environments. This infrastructure design enables Power Virtual Server to maintain key enterprise software certification and support as the Power Virtual Server architecture is identical to certified on-premises infrastructure.

If you are creating or configuring a Power Virtual Server instance to support an SAP NetWeaver or SAP HANA workload, see [Planning your deployment](#) and [Deploying your infrastructure](#).

If you are creating or configuring a Red Hat OpenShift Cluster on Power Virtual Server, see [Cloud native development and application modernization by using Red Hat OpenShift on IBM Power Virtual Server](#).

IBM Power Virtual Server Private Cloud in Client location

IBM Power Virtual Server Private Cloud is an as-a-service offering that includes a prescriptive set of physical infrastructure (compute, network, and storage). The physical infrastructure or the pod is deployed in your data center. The pod is maintained and operated by IBM site reliability engineering (SREs) and managed through the IBM Cloud platform. Each pod is associated with an IBM Cloud Satellite location that is owned by your IBM Cloud account. This architecture helps to scale horizontally to add more pods according to your requirements. For more information about Satellite location, see [IBM Satellite location](#). To verify that the pre-installation requirements are met, see [Prerequisites for installing the pod](#).

IBM Power Virtual Server Private Cloud officially supports Red Hat Enterprise Linux (RHEL), IBM i, and IBM AIX®. With the support of these operating systems you can deploy several applications, including SAP HANA, SAP S/4HANA, and SAP NetWeaver. You can also run Red Hat OpenShift Container Platform to accelerate your application modernization efforts by using new cloud-native applications.



Note: Power Virtual Server does not provide Linux stock images. Bring your own Linux image (OVA format) and subscription. SUSE Linux Enterprise Server (SLES) and RHEL OVA images are supported. [Learn more](#)

Terminology

- **IBM Cloud account** - You can log on to IBM Cloud dashboard by using IBM Cloud account to access various IBM Cloud solutions, services, and offerings. To create an IBM Cloud account, see [Signing up for the IBM Cloud](#).
- **Location and region** - The global network of locations of IBM Cloud provides three tiers of regions: multizone regions (MZR), single-campus multizone regions, and data centers. To achieve low application latency, choose your nearest location and region. For details about the available IBM Cloud regions and data centers, review [Region and data center locations for resource deployment](#).
- **Satellite location** - A Satellite location is a representation of an environment in your infrastructure provider, such as a private cloud data center. Each IBM Power Virtual Server Private Cloud pod in your data center must be associated with a Satellite location to manage the lifecycle operations of the pod. You must create multiple and unique Satellite locations if you are planning to deploy multiple pods in your data center. For detailed information, see [Understanding Satellite location and hosts](#).
- **workspace** - A workspace is a container of all the virtual machines, storage volumes, network configurations, and so on, at a specific geographic region. Power Virtual Server workspaces are available from the **Resource list** in the Power Virtual Server user interface. Each workspace can contain multiple Power Virtual Server instances. Within each Satellite location, you can create multiple workspaces. For example, you can create two workspaces in the Satellite location of the North Data Center: one workspace for production workloads and another workspace for development tests.
- **pod** - a pod consists of IBM Power compute, storage, and network infrastructure that is built, installed, and maintained by IBM in your data center.

You can have multiple pods and each pod will be uniquely represented by an IBM Cloud Satellite location.

- **Control plane network and client data plane network** - Control plane network is the connection between IBM Cloud and client data center. The client data plane network is the connection between the client data center and the IBM Power Virtual Server Private Cloud pods. For more information, see the [Network overview for private cloud](#) topic.

Before you begin

Before you create your first Power Virtual Server instance, review the following prerequisites:

1. Create an IBM Cloud account if you don’t have an existing account. To create an IBM Cloud account, see [Signing up for the IBM Cloud](#).
2. Plan and specify your infrastructure requirements and review the estimated cost by using the [Power Virtual Server Estimate pricing](#) tool. Identify whether you want Power Virtual Server in IBM data center or in IBM Power Virtual Server Private Cloud in client location, or both.
3. Review the entire lifecycles of Power Virtual Server instances on [IBM Power Virtual Server in IBM data center](#) and [IBM Power Virtual Server in Client location](#).

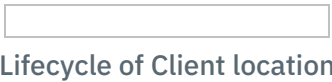


Lifecycle of a Power Virtual Server

IBM data center



Client location



Release notes for Power Virtual Server

Use these release notes to learn about the latest changes to IBM® Power® Virtual Server.

September 2026

11 September 2026

IBM Power System E1150 available in **CHE02** data center

The IBM Power System E1150 (9043-MRU) is available in the **CHE02** data center. For more information, see [Hardware specifications](#).

4 September 2026

Stock images support

- The following AIX stock images are available in the Power Virtual Server data centers:
 - AIX 7.3 TL4 SP2
 - AIX 7.3 TL3 SP3
 - AIX 7.2 TL5 SP13
- The following AIX stock images are removed from the Power Virtual Server data centers because the OS level is no longer supported:
 - AIX 7.3 TL4 SP1
 - AIX 7.3 TL3 SP2
 - AIX 7.2 TL5 SP12

August 2026

19 August 2026

Stock images support

- The following AIX stock images are available in the Power Virtual Server data centers:
 - AIX 7.3 TL4 SP1
 - AIX 7.2 TL5 SP12
- The following AIX stock images are removed from the Power Virtual Server data centers because the OS level is no longer supported:
 - AIX 7.3 TL4 SP0
 - AIX 7.2 TL5 SP11

6 August 2026

Power9 servers reach end of life on December 31, 2027

The Power9 servers (E980 and S922) in Power Virtual Server reach end of life on December 31, 2027. IBM is limiting new VSI deployments on Power9 servers. The following conditions apply:

- **If you had VSIs on Power9 servers before August 2026:** You can continue to deploy new VSIs on Power9 servers until November 2026, subject to available capacity. Your existing VSIs can continue to run until December 31, 2027.
- **If you did not have VSIs on Power9 servers before August 2026:** As of August 2026, you cannot deploy new VSIs on Power9 servers.

To migrate your VSIs to Power10 or Power11, open a [support ticket](#).

July 2026

31 July 2026

GRS support in data center pairs

Global Replication Services (GRS) is available in the `che03` and `in-mum-1` data center pair. For more information, see [Power Virtual Server regions that support GRS](#).

23 July 2026

Medium pod configuration in Client location

Power Virtual Server supports medium pods with an expandable modular architecture based on the standard 42U rack specification. A medium pod supports up to 6 compute racks for S1122 or E1150 system types, and up to 18 compute racks for E1180 (2CEC) system types. For more information, see [Medium pod configurations](#).

16 July 2026

Power Edge Router available in `MON01` data center

Power Edge Router (PER) is available in the `MON01` data center. For more information, see [Getting started with Power Edge Router](#).

June 2026

29 June 2026

GRS support in data center pairs

Global Replication Services (GRS) is available in the `che02` and `in-mum-2` data center pair. For more information, see [Power Virtual Server regions that support GRS](#).

24 June 2026

Disabling automated remote restart for a virtual server instance

Automated remote restart is enabled by default for all VSIs in the Power Virtual Server environment. You can disable automated remote restart by modifying the settings for a VSI. For more information, see [Disabling automated remote restart for a VSI](#).

Monitoring boot image import, boot image export, and virtual server instance (VSI) capture and export status in the UI

You can view the status of boot image import, boot image export, and VSI capture and export operations in the UI. This enhancement provides improved visibility into long-running image jobs before you start another image job. For more information, see [Viewing import boot image results](#), [Viewing export boot image results](#), and [Viewing the capture and export results](#).

Metadata service and trusted profiles support

The metadata service enables you to access instance metadata and use trusted profiles to call IAM-enabled services without managing API keys. For more information, see [Configuring and managing the metadata service for Power Virtual Server](#).

S1122 servers as dedicated hosts

You can provision S1122 servers as dedicated hosts. For more information see [Getting started with dedicated hosts on IBM Power Virtual Server](#).

Networking enhancements

- **DHCP service for Power Virtual Server** : You can enable or disable Dynamic Host Configuration Protocol (DHCP) services on subnets in your Power Virtual Server workspace. The DHCP service automatically provides IP addresses and network configuration information to VSIs that are configured to use DHCP. The DHCP service is available only on the IBM Power Virtual Server in Client location page.
- **Network routes**: The network routes feature is available for IBM Power Virtual Server in both IBM data center and in Client location. A network route defines the path that data packets follow to travel from a source to a destination across a network. You can use the network routes feature in Power Virtual Server to view implicit network routes and create or manage static routes in your Power Virtual Server workspaces. For more information, see [Creating and managing network routes in IBM Power Virtual Server workspaces](#).
- **Subnet ARP broadcast**: The Address Resolution Protocol (ARP) is available for IBM Power Virtual Server in both IBM data center and in Client location. ARP maps IP addresses to physical Media Access Control (MAC) addresses in a network. For more information, see [Configuring the ARP broadcast in Power Virtual Server subnets](#).

19 June 2026

in-mum-1 and in-mum-2 zones in IBM data center

Power Virtual Server offers the `in-mum-1` and `in-mum-2` zones as data centers in Mumbai, India. For more information about the data center capabilities, see the **Data center capabilities** section on the [Power with the benefits of distributed hybrid infrastructure](#) page.

9 June 2026

Private Cloud offering expansion

The IBM Power Virtual Server Private Cloud offering is available in Sweden and New Zealand.

May 2026

29th May 2026

Changes in public network connectivity

You can configure public network connectivity for your Power Virtual Server instances through IBM Cloud VPC, which provides both outbound and inbound network access. This approach offers enhanced security and integration with other IBM Cloud services. For detailed instructions about setting up public network connectivity, see [Public network connectivity for Power Virtual Server](#).

Stock images support

The RHEL 8.6 SAP stock image is removed from the Power Virtual Server data centers because the OS level is no longer supported by Red Hat.

April 2026

30 April 2026

ISO/IEC 20000 compliance

Power Virtual Server is ISO 20000 certified with other IBM Cloud services. For more information, see [Compliance certification](#).

Stock images support

- The RHEL 9.4 general-purpose stock image is removed from the Power Virtual Server data centers because the OS level is no longer supported by Red Hat.
- The IBM i 7.2 stock image is removed from the Power Virtual Server data centers because the OS level is no longer supported.

March 2026

25 March 2026

aSAPS metric available for SAP profiles

Accelerated SAPS (aSAPS) is available as a performance measurement attribute for SAP profiles. aSAPS is an advanced SAP performance unit that represents system capacity based on the SAP quote-to-cash (Q2C) benchmark. For more information, see [SAP profile attributes and performance metrics](#).

Deploying VSIs for SAP NetWeaver without SAP-certified profiles

You can deploy VSIs for SAP NetWeaver workloads without using an SAP-certified profile. For more information, see [Configuring a Power Virtual Server instance](#).

SAP Application Server profiles available on IBM Power S1122

SAP Application Server profiles are supported on the IBM Power S1122 system. For more information, see [SAP Application Server profiles](#).

IBM Cloud support for IBM Power Virtual Server Private Cloud in Client location

IBM provides advanced or premium IBM cloud support for IBM Power Virtual Server Private Cloud in Client location. For more information, see [Creating an estimate for Power Virtual Server resources in a client location](#).

AIX, IBM i, and SLES stock images added to OS Image Catalog

The following AIX, IBM i, and SLES stock images are available in the IBM OS Image Catalog:

- SLES 16 general purpose (SLES16)
- AIX 7.3 TL3 SP2
- IBM i 7.5 TR7
- IBM i 7.4 TR12 (refreshed)
- IBM i 7.3 TR13 (refreshed)
- IBM i COR (refreshed)

End-of-Life reminder for IBM i 7.2, RHEL 9.4 general purpose, and RHEL 8.6 SAP OS images



Important: The end of support date for the IBM i 7.2 operating system is 30 April 2026. Stock images for IBM i 7.2 are scheduled to be removed from the Power Virtual Server image library starting from 1 May 2026. Existing virtual server instances (VSIs) that are deployed using these stock images can continue to operate without interruption. To receive continued support and maintenance from IBM, update your VSIs to a supported operating system version. For more information about the IBM i end of support guidelines, see [IBM i Release Life Cycle](#). For more information about the available operating system stock images, see [Operating systems supported in IBM Power Virtual Server](#).



Important: The Red Hat Enterprise Linux (RHEL) end of support date for the RHEL 9.4 general purpose is 30 April 2026 and RHEL 8.6 SAP is 31 May 2026. Stock images for RHEL 9.4 general purpose are scheduled to be removed starting from 1 May 2026, and RHEL 8.6 SAP starting from 1 June 2026, from the Power Virtual Server image library. Existing virtual server instances (VSIs) that are deployed using these stock images can continue to operate without interruption. To receive continued support and maintenance from IBM, update your VSIs to a supported operating system version. For more information about the RHEL end of support guidelines, see [Red Hat Enterprise Linux Life Cycle](#). For more information about the available operating system stock images, see [Operating systems supported in IBM Power Virtual Server](#).

Private Cloud offering expansion

The IBM Power Virtual Server Private Cloud offering is available in Japan.

February 2026

23 Feb 2026

CHE02 and CHE03 data centers are available

CHE02 and **CHE03** are available as Power Virtual Server data centers in Chennai, India. For more information about the data center capabilities, see [Data center capabilities](#).

January 2026

14 Jan 2026

ENS High and Protected B certifications for Power Virtual Server

Power Virtual Server is Esquema Nacional de Seguridad (ENS) High and Protected B certified. For more information about ENS High and Protected B compliance certifications, see [ENS High](#) and [Protected B](#).

December 2025

- Virtual Persistent Memory (vPMEM) volumes are supported by IBM Power Virtual Server. You can attach a vPMEM volume to a Power Virtual Server instance (VSI) with AIX®, Linux®, or Linux (SAP HANA) operating systems on a Power10 or later systems. vPMEM volumes reduce the time it takes to restart SAP HANA virtual server instances. For more information see, [Attaching Virtual Persistent Memory in a Power Virtual Server instance](#).
- The virtual server instance (VSI) console supports copying and pasting text between your workstation and the VSI console. You can copy commands or scripts from Windows®, macOS®, or Linux®-based workstations and paste the copied text to an AIX®, IBM i, or Linux-based VSI console. For more information, see [Copying and pasting text in the VSI console](#).
- IBM Cloud Monitoring service can be used with IBM Power Virtual Server Private Cloud in Client location. For more information, see [Monitoring metrics for Power Virtual Server](#).
- Added the **SAO05** data center to the existing list of Power Virtual Server data centers. For more information, see [IBM Cloud® regions](#).
- The following AIX, RHEL, and SLES stock images are available in the IBM OS Image Catalog:
 - AIX 7.3 TL4 SP0
 - AIX 7.2 TL5 SP11
 - RHEL 9.6 for SAP HANA (RHEL9-SP6-SAP-HANA)
 - RHEL 9.6 for SAP NetWeaver (RHEL9-SP6-SAP-NETWEAVER)
 - SLES 15 SP7 for SAP HANA (SLES15-SP7-SAP)
 - SLES 15 SP7 for SAP NetWeaver (SLES15-SP7-SAP-NETWEAVER)
- Following the SUSE lifecycle policy, support for SLES 15 SP3 for SAP ends on 31 December 2025. The stock image for SLES 15 SP3 for SAP is scheduled for removal from the Power Virtual Server Image Catalog starting 31 December 2025.

November 2025

- You can set the preferred processor compatibility mode for a VSI when you create the VSI in the Power Virtual Server user interface. The option to set the preferred processor compatibility mode continues to be available by using the CLI, API, or Terraform. For more information about the processor compatibility mode for a VSI, see [How does the processor compatibility mode work in a VSI?](#).

October 2025



Note: End of Marketing: As of 31 October 2025, new deployments of VMware Solutions offerings are no longer available for new customers. Existing customers can still use and expand their active VMware® workloads on IBM Cloud®. For more information, see [End of Marketing for VMware on IBM Cloud](#).

September 2025

- You can use the Power Virtual Server user interface to edit the preferred processor compatibility mode for VSIs that are already deployed. The

processor compatibility mode determines the Instruction Set Architecture (ISA) version that is used by the processor and the platform-level features that are available to the VSI. For more information, see [Changing the preferred processor compatibility mode](#).

- You can change the tier of the replication-enabled primary volume. For more information, see [Changing the tier of a primary volume](#).
- You can use the IBM Cloud® carbon calculator to track, analyze, and manage your greenhouse gas (GHG) emissions for Power Virtual Server data from workspaces on Power10 or later systems. For more information, see [Using carbon calculator with Power Virtual Server](#).
- You can use the Power Virtual Server user interface to create and manage the instance snapshots for your VSIs. For more information, see [Creating a snapshot by using the Power Virtual Server user interface](#).
- You can deploy SR3 SAP HANA profiles on IBM Power11 processor-based servers. For more information, see [SAP certified profiles](#).
- The SLES 17 general purpose (SLES15-SP7) and RHEL 9.6 general purpose (RHEL9-SP6) OS stock images are available in the IBM OS Image Catalog. The images are supported on IBM Power9, Power10, and Power11 systems.

Client location

- You can connect your Power Virtual Server workspace and your data center network as a dynamic routing session to create a network peering connection. For more information, see [Creating a network peering connection](#).

August 2025

- The IBM i 7.4, 7.5, 7.6, and COR OS images in the Image Catalog are supported on Power10 and later systems.

July 2025

- Starting on 25 July 2025, you can deploy Power Virtual Server instances running the AIX, IBM i, RHEL, and SLES operating systems on IBM Power11 (subject to capacity availability). Refer to the [Operating systems supported in IBM Power Virtual Server](#) for more details.
- With IBM Power10 systems and later, you can assign an IBM i software tier to a virtual server instance (VSI). The IBM i software tier limits the size of the VSI, but not the physical system on which it runs. It controls the pricing tier for the IBM i operating system and licensed program products (LPPs). The IBM i software tier also defines the boundaries for the virtual processors and memory size. For more information, see [Assigning an IBM i software tier to an IBM i Power Virtual Server instance](#).
- You can set the visibility of an SSH key to improve the security and privacy of the SSH key. When you set the visibility of an SSH key, the SSH key is available only within a workspace or an account. For more information, see [Setting the access level of an SSH key in a Power Virtual Server workspace](#).

IBM data center

- You can use the routes feature in your PER-enabled Power Virtual Server workspaces to view implicit network routes and to create or manage static routes. For more information, see [Creating and managing network routes in IBM Power Virtual Server workspaces](#).
- The network security group feature is enhanced to provide the following additional capabilities:
 - You can clone an existing network security group to create a network security group. For more information, see [Cloning an NSG](#).
 - You can transfer members from one network security group to another. For more information, see [Moving members from one NSG to another NSG](#).
- You can enable the Address Resolution Protocol (ARP) to map IP addresses to physical Media Access Control (MAC) addresses in the Power Virtual Server network. For more information, see [Configuring the Subnet ARP Broadcast in Power Virtual Server subnets](#).
- You can integrate your Power Virtual Server workspace with IBM Cloud® Monitoring service to get the visibility to the performance and health of your applications, services, and platforms. For more information, see [Monitoring metrics for IBM Power Virtual Server](#).
- New server placement groups and shared processor pools are created with cloud resource names (CRNs), to provide benefits such as tagging and visibility in the IBM Cloud Resource list. The existing server placement groups and shared processor pools will be assigned a CRN by the end of August 2025.



Important: On 14 July 2025, the Power Virtual Server VPNaaS product reached its end of life and is no longer available for use. However, you can use the IBM Cloud VPC VPN to avoid VPN service interruptions. For assistance to upgrade or migrate to IBM Cloud VPC VPN, open a [support ticket](#) or engage with your Customer Support Manager (CSM).

- IBM Power Virtual Server integrates with IBM watsonx SaaS services to deliver a powerful and flexible infrastructure. The integrated infrastructure supports demanding workloads, runs advanced AI applications, and scales rapidly in a hybrid platform environment with automated AI

infrastructure setup and management. For more information, see [March 2025](#) for details), and you will incur monthly charges for any Direct Link connections that you continue to use. The pricing is based on the port speed of your connections. To avoid these charges, migrate your workspace to PER-enabled workspaces. The **CHE01** and **MON01** data centers continue to use Cloud Connections without any charges. For more information, see [Migrating to PER](#).

June 2025

IBM data center

- You can deploy SAP certified profiles on IBM Power10 or later servers by using a single custom OS boot image. For more information, see [SAP certified profiles](#).
- The single byte IBM i 7.6 stock image for IBM Power10 systems is available in the OS image catalog. The stock image is not supported on IBM Power9 systems.
- The migration of existing workspaces to cloud resource names (CRNs) in all the data centers except DAL14 is completed. For more information about CRNs, see [IBM cloud resource names](#).
- The following RHEL OS stock images are removed from the Power Virtual Server image library as the OS versions are no longer supported by Red Hat:
 - RHEL8.8 general purpose
 - RHEL9.2 general purpose
 - RHEL8.4 for SAP (HANA)
 - RHEL8.4 for SAP NetWeaver

May 2025

- If you have existing IBM Cloud Connections that are managed from non-PER enabled workspaces, you can view and delete the connections using the [IBM Cloud CLI](#) or [SAP NetWeaver profiles](#).

April 2025

IBM data center

 **Important:** The Red Hat Enterprise Linux (RHEL) end of support date for RHEL 9.2 general purpose, RHEL 8.8 general purpose, and RHEL 8.4 for SAP is 30 May 2025. Stock images for these OS versions are scheduled to be removed from the Power Virtual Server image library starting 31 May 2025. The existing virtual server instances (VSIs) that use the stock images can continue to operate without interruption. To get continued support and maintenance from IBM, update your VSIs to a supported OS version. For more information about the RHEL end of support guidelines, see [Red Hat Enterprise Linux Life Cycle](#). For more information about available OS stock images, see [FAQs](#).

- IBM i 7.3, 7.2 and Cloud Optical Repository (COR) stock images have been refreshed.
- IBM Power Virtual Server started the support of cloud resource names (CRNs) for new workspaces in [February 2025](#). Starting from April to June 2025, CRNs are being assigned to the existing Power Virtual Server workspaces. CRN identifiers are assigned to uniquely identify resources in the IBM Cloud, such as virtual server instances (VSIs), shared processor pools (SPPs), volumes, snapshots, and dedicated hosts.

The billing and metering plans are updated with the following changes:

- For billing, an instance of a resource is identified by using the associated CRN and not by the **Consumer ID** value.
- Each resource type has its own billing plan instead of billing all the resources under the **Workspace for Power Virtual Server** plan.
- In the billing plan, the value for the **Pricing Region** field of the instance is based on metro regions. However, the value for the **Location** field of the instance is based on the data center and zone regions.
- The costs of the snapshots are billed by using independent metrics for snapshots and are not included with the volume metrics.

The existing price of the resources does not change. The organization and visualization of the costs of the resources are updated to match the billing and metering plans for IBM Power Virtual Server Private Cloud in Client location that supports CRNs.

- The idle timeout for the virtual server instance (VSI) console (previously called VNC console) is extended to up to 2 hours from 30 minutes. For more information, see [Working with the VSI console](#).
- The new data center pair **WDC06** - **DAL14** supports global replication services. For more information, see [Locations that support global replication service](#).

- You can enable the Cloud Security Posture Management (CSPM) feature in your existing and new Power Virtual Server workspaces. For more information about enabling CSPM, see [Integrating Power Virtual Server with IBM Cloud Security and Compliance Center Workload Protection](#).

March 2025

- New AIX 7.3 TL3 SP0 and 7.2 TL5 SP9, IBM i 7.4 TR11 and 7.5 TR5, SLES 15 SP6 for SAP HANA (SLES15-SP6-SAP) and SLES 15 SP6 for SAP NetWeaver (SLES15-SP6-SAP-NETWEAVER) operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).

IBM data center

 **Important:** The End of Life date for the no-charge Cloud Connections, which IBM Power Virtual Server provides to clients to access IBM Cloud resources in their network, is now extended indefinitely beyond the original date of 18 June 2025. However, starting with 1 July 2025 metering charges are applied to any remaining Direct Link connections. After this date, you will incur monthly charges for any Direct Link connections, with pricing based on the port speed of your connections. To avoid these charges, migrate your workspace before 1 July 2025. For more information about the migration, see [Migrating to Power Edge Router](#).

- You can deploy SAP HANA CH2 and BH2 profiles on specific IBM Power servers. For more information, see [SAP HANA certified profiles](#).
- Shared Processor Pool (SPP) metering is optimized to improve the Total Cost of Ownership (TCO) for AIX and IBM i software licensing and DR scenarios. Virtual servers that are configured within an SPP will not incur additional cost for per VM core or the cost for high-use memory parts. For more information, see [Pricing for shared processor pool for IBM Power Virtual Server in IBM data center](#) and [Pricing for shared processor pool for IBM Power Virtual Server Private Cloud in Client location](#)

Client location

- The pricing for IBM Power Subscription plan to provision VMs with SAP bring your own license or Linux bring your own license images is now based on a flat subscription rate instead of metered usage charges. For more information, see [Using SLES within IBM Power Virtual Server](#) or [Using RHEL within the Power Virtual Server](#).

February 2025

 **Important:** The End of Support date for Cloud Connections has been extended from 18 April, 2025 to 18 June, 2025 to align with its end of life date. For more information about the End of Service and End of Life Notice, see [December 2024](#).

 **Note:** When you create a workspace by using the Terraform resource [ibm_resource_instance](#) or the CLI command [ibmcloud resource service-instance-create](#), for the **service name** field use the value `power-iaas` for IBM Power Virtual Server in IBM data center and for the **plan name** field use the value `power-virtual-server-group` or `power-virtual-server-private-group` for IBM Power Virtual Server Private Cloud in Client location.

- Power Virtual Server is now HITRUST CSF® compliant. For more information, see [Compliance certifications](#).

IBM data center

- Power Virtual Server is enhanced to provide complete billing transparency. With the enablement of IBM cloud resource names (CRNs), you can view the contribution of each resource to your monthly cloud expenditure. For more information, see [IBM cloud resource names](#). This enhancement does not introduce any changes in Power Virtual Server pricing. The IBM CRNs are displayed when you create new workspaces. Note that the existing workspaces are planned to be enhanced with CRNs by `June 2025` without disrupting your work.
- You can now use Network security groups (NSG) in your PER-enabled Power Virtual Server workspaces to control internal and external inbound traffic to your virtual network. You can define security rules for NSGs based on source IP addresses, ports, and protocols (TCP, UDP, ICMP, and Any). This new capability is available when creating new PER-enabled workspaces. Existing workspaces are planned to be migrated to support NSGs by `June 2025`. For more information, see [Network security groups](#).

January 2025

 **Important: End of Service reminder** - Effective `January 18, 2025`, IBM will no longer deliver standard support for the Power Virtual Server VPNaaS product. On `July 14, 2025`, the Power Virtual Server VPNaaS product will reach its end of life. If you are using Power Virtual Server VPNaaS product, you are encouraged to move to the [IBM Cloud VPC VPN](#) to avoid VPN service interruptions.

- Starting `January 22, 2025`, [virtual serial number \(VSN\)](#) and [Virtual host identifier](#) are supported in all the data centers.
- Global replication service supports new data center pair `LON04` - `LON06`. For more information, see [Locations that support global replication](#)

[service](#).

December 2024

IBM data center

- You can now assign a virtual serial number (VSN) to an IBM i VSI. With a VSN associated with your VSI, you need not pin the VSI to a host for licensing or entitlement purposes. For more information, see [virtual serial number \(VSN\)](#).
- You can now use automation to migrate an existing network to Power Edge Router (PER) through CLI. For more information, see [Migrating to PER](#).
- As the Power Virtual Server offering transitions to the PER integrated network solution, the ability to create new Power Virtual Server Cloud Connections across most data centers is disabled. You are encouraged to migrate to PER now if you have not done so already. For more information about PER, see [Getting Started with Power Edge Router](#). You can create Cloud Connections only in the **CHE01** and **MON01** data centers. For more information, see [IBM Power Virtual Server Cloud Connections](#).



Note: Support for Cloud Connections ends on **April 18, 2025** and the Cloud Connections solution will reach the end of life on **June 18, 2025**. For more information about End of Service Notice, see [July 2024](#).

- All IBM Power Virtual Server data centers are now PER-enabled, except for the **CHE01** and **MON01** data centers. For more information, see [Getting Started with Power Edge Router](#).
- A new data center **DAL14** is available. Following are some of the capabilities of the **DAL14** data center:
 - New Power Virtual Server node, [IBM Power System E1050 \(9043-MRX\)](#) which is a 4-socket server.
 - Power Virtual Server capabilities such as Power Edge Router network connectivity, IBM Cloud Monitoring, and Flexible IOPS.

For more information about your data center capabilities, see Power Virtual Server [overview](#) page in the IBM Cloud console.

- New RHEL 8.10 general purpose (RHEL8-SP10), RHEL 8.10 for SAP HANA (RHEL8-SP10-SAP-HANA), RHEL 8.10 for SAP NetWeaver (RHEL8-SP10-SAP-NETWEAVER), RHEL 9.4 for SAP HANA (RHEL9-SP4-SAP-HANA), RHEL 9.4 for SAP NetWeaver (RHEL9-SP4-SAP-NETWEAVER), SLES 15 SP6 general purpose (SLES15), and AIX 7.3 TL2 SP2 operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- Starting **February 24, 2025**, the Power Virtual Server dedicated host identifier will use the virtual host ID format. The virtual host ID has the standard format that is used to uniquely identify other Power Virtual Server resources. If you are using dedicated hosts and depend on specific resource identifiers such as API and CLI calls, you must update the logic to use the virtual host ID format; otherwise, the logic might fail. For more information, see [Understanding a Virtual host identifier](#).

Client location

- You can now enable GRS for asynchronous replication of data between the primary location infrastructure and the secondary location infrastructure. For more information, see [Global Replication Services in Client location](#).
- You can now access external connectivity to and from the virtual machines in a subnet. For more information, see [Use case 2: Multiple external connectivities](#).
- You can now discover infrastructure capabilities and important information that helps you to develop solutions by using Power Virtual Server on the Power Virtual Server [overview](#) page in the IBM Cloud console. For more information, see [Data center capabilities](#).
- The new AIX 7.3 TL2 SP2 operating system image is available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).

October 2024

- **IBM data center** You can now configure a virtual machine that contains a boot volume and deploy an IBM i virtual machine to support the attachment of large quantity of data volumes. You can configure a virtual machine available in all the data centers except for the virtual machines in the **CHE01** data center. For more information, see [Configuring large quantity of data volumes on IBM data center](#).
- **IBM data center** IBM Cloud Monitoring service is now available in the **CHE01** data center. For more information, see [Monitoring metrics for Power Virtual Server](#).
- **IBM data center** The refreshed RHEL 9.2 for SAP, RHEL 8.8 for general purpose and SAP, and RHEL 8.6 for general purpose and SAP OS images are available in the image catalog that supports SAP t-shirt profiles with more than 64 cores. For more information, see the [SAP documentation on OS](#) for IBM Power Virtual Server.



Note: The latest IBM i and AIX stock images were added to the OS image catalog in September. Hence, the previous versions of the service

pack will be removed from the catalog in the next thirty days. For more information about the OS versions, see [Operating systems supported in IBM Power Virtual Server](#). For new deployments, you must use the latest stock image in the catalog. Existing VSIs that use the previous stock images continue to operate without problems. However, it is recommended that you update your VSI to the latest service pack level for the most recent maintenance.

September 2024

- You can now discover data center capabilities and important information that helps you to develop solutions by using Power Virtual Server on the Power Virtual Server [overview](#) page in the IBM Cloud console. For more information, see [Data center capabilities](#).
- IBM data center** You can now enable the replication services for a volume from the user interface. For more information, see [Global Replication Services \(GRS\)](#).
- IBM data center** You can now configure a virtual machine that contains a boot volume and deploy an IBM i virtual machine to support the attachment of large quantity of data volumes available in **DAL10** and **WDC07** data centers. For more information, see [Configuring large quantity of data volumes on IBM data center](#).
- Client location** You can now select the **DAL10** and **WDC07** location pair to resize a replication-enabled primary volume from the primary site. The system then resizes the associated auxiliary volume on its corresponding remote site within 24 hours. For more information, see [Updating a primary volume](#).
- Client location** You can now select the SUSE Linux Enterprise Server (SLES) stock image when you register for full Linux subscription, for Power Virtual Server. For more information, see [Full Linux® subscription for IBM Power Virtual Server in Client location](#).
- Client location** You can now validate the import file against the SHA-256 checksum file from the user interface. For more information, see [Using the Power Virtual Server user interface to import a boot image](#) and [Using the Power Virtual Server user interface to capture and export a VM](#).
- You can now download, Secure Automated Backup with Compass by Cobalt Iron, from [IBM Cloud Catalog](#) for immediate deployment. The solution provides the environment for Cloud backup and recovery for IBM Power Virtual Server workloads. The offering is the only automated BaaS and recovery solution for IBM Power Virtual Server workloads that are powered by IBM Storage Protect, such as SAP HANA, Oracle, Db2 on AIX and Linux. For more information, see [Cobalt Iron Expands Access to Secure Automated Backup With Compass for IBM Power Virtual Server](#).
- IBM data center** new AIX 7.2 TL5 SP8 and RHEL 9.4 general purpose (RHEL9-SP4) operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- Client location** new AIX 7.3 TL2 SP1 and AIX 7.2 TL5 SP8 operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- New IBM i 7.5 TR4 and IBM i 7.4 TR10 operating system images are available for both IBM data center and Client location. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- New **IBM i COR** stock image is available for both IBM data center and Client location. While you are installing the image, if you encounter any issues, you can refer to the [Extended Base Support in Error](#) procedure. For more information, see [FAQs](#).



Important: End of Service Notice - End of service effective **October 31, 2024** will prevent the use of E880 hosts in the IBM Power Virtual Server offering. E880 hosts will no longer be accessible to establish new or existing workspaces. You must ensure that all existing workspaces are moved to another host. Power Virtual Server team is available to assist you. If you need assistance, [open a support ticket](#). If you choose not to take the recommended action and encounter issues, IBM will not be able to support you. The solution will reach its end of life on **October 31, 2024**.

August 2024

- IBM Cloud Monitoring service is now available in **MON01**. For more information, see [Monitoring metrics for Power Virtual Server](#).
- You can now migrate Db2 on AIX to Power Virtual Server. For more information, see [Migrating Db2 on AIX to Power Virtual Server](#).

July 2024

- IBM Cloud Monitoring service is now available in **TOR01**. For more information, see [Monitoring metrics for Power Virtual Server](#).
- Global replication service supports new data center pairs **SAO01** - **SAO04** and **MON01** - **TOR01**. For more information, see [Locations that support global replication service](#).
- You can now deploy Linux for SAP (HANA or Netweaver) OS images within IBM Power Virtual Server by using the SAP provided central image repository or by bringing your own customized image. For more information, see [SAP provided central image repository](#) and [Bring your own](#)

[customized Linux for SAP \(HANA or NetWeaver\) image.](#)

- You can now deploy SAP full system profiles on S1022 systems, if no virtual machines are deployed on the system. For more information, see the [SAP full system profiles](#).
- You can now deploy SAP HANA sr2 and sh2 profiles on S1022 and E1080 systems. For more information, see [SAP HANA sr2 and sh2 profiles](#).
- You can now provision Power Virtual Server workloads on Power10 E1080 (9080-HEX) hosts available in **DAL10** and **WDC07** data centers.



Important: End of Service Notice - End of marketing effective **October 19, 2024** will prevent new Cloud Connections from being established in existing workspaces. In turn, IBM will stop delivering standard support for Power Virtual Server Direct Link Connect in PER-enabled data centers by **April 18, 2025**. If you choose not to take the recommended action and encounter issues, IBM will not be able to support you. The solution will reach its end of life on **June 18, 2025**.

- Due to inherent configuration complexity of the current Power Virtual Server Cloud Connections which leverage Direct Link Connect service for connectivity to IBM Cloud, the direction going forward is to leverage the newly enabled Power Edge Router capability along with Transit Gateway service to connect Power Virtual Server workspaces with IBM Cloud resources for a better user experience, improved reliability, and significantly higher bandwidth. You can follow a detailed [migration plan](#) to move your existing workspace to a PER workspace if you have set up your network manually using the Cloud Connections solution. Power Virtual Server team is available to assist you. If you need assistance, open a [open a support ticket](#).

June 2024

- Power Virtual Server Private Cloud, an infrastructure as a service (IaaS) offering, is now available. It extends the capabilities and benefits of Power Virtual Server within your data center, with a prescriptive set of hardware and delivered in a consumption model with no upfront payment. For more information about Power Virtual Server Private Cloud and how to get started, see [Getting started with IBM Power Virtual Server](#).



Note: The content specific to IBM Power Virtual Server Private Cloud is marked with **Client location** tag and the content specific to IBM Power Virtual Server is marked with **IBM data center** tag.

- The RHEL 8.6 general purpose (RHEL8-SP6) stock image is removed from the Power Virtual Server data centers because the OS level is no longer supported by Red Hat.
- Global replication service supports the new data center pairs **OSA21** and **TOK04**; **SYD04** and **SYD05**. For more information, see [Locations that support global replication service](#).
- Access the dedicated host capability from the Power Virtual Server user interface. IBM Power S922 and S1022 servers can be provisioned for your dedicated use. For more information, see [Dedicated host](#).
- Snapshot usage information is now available using the API. For more information, see [Managing the shared processor pool](#).

May 2024

- Power Edge Router (PER) is now available in the **DAL13**, **TOR01**, and **WDC04** data centers. For more information, see [Getting Started with Power Edge Router](#).
- Power Virtual Server now offers refreshed images for IBM i 7.2 TR 9.
- The SLES 15 SP4 general purpose (SLES15-SP4) stock image is in the process of being removed from the Power Virtual Server data centers since the OS level is no longer supported by Red Hat.
- You can now see a **Consumer ID** in your [Billing and Usage](#) page. Using the consumer ID you can identify and get a detailed view at the resource level broken down by part metric. For more information, see [Consumer ID](#).
- Power Virtual Server is now Financial Services® Validated and has received a SOC 2 Type II report. For more information, see the [compliance](#) page.
- Flexible I/O operation per second (IOPS) is now available in the **TOR01** data center. For more information, see [Flexible IOPS](#).

April 2024

- Power Edge Router (PER) is now available in the **LON06** and **SYD04** data centers. For more information, see [Getting Started with Power Edge Router](#).
- Power Virtual Server now offers refreshed images for RHEL 9.2 general purpose, RHEL 9.2 for SAP, RHEL8.8 general purpose, and RHEL 8.8 for SAP. For more information, see the [SAP documentation on OS](#) for IBM Power Virtual Server and [Operating systems supported in IBM Power Virtual Server](#).
- AIX 7.1 will continue to be charged at standard usage rates until further notice [\[1\]](#). For additional information on AIX lifecycle and service

extensions, refer to [AIX support lifecycle information](#).

- Power Virtual Server on IBM Cloud is now HIPAA-ready. See the [Compliance certifications](#) for more details.

March 2024

- Reserve an IP address from the Power Virtual Server user interface. The IP address that you reserve are not assigned to a virtual server instance. For more information, see [Reserving IP addresses](#).
- New AIX 7.3 TL2, and 7.2 TL5 SP7 operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- New IBM i 7.5 TR3, IBM i 7.4 TR9, and IBM i COR [\[2\]](#) operating system images are available. For more information, see [Operating systems supported in IBM Power Virtual Server](#).
- A new data center **CHE01** is available. The following are some capabilities that differ for this data center:
 - **CHE01** supports up to 5 GB Direct Link connections compared to other data centers that supports up to 10 GB.
 - To use Transit Gateway, you need to connect to **CHE01** using a different data center that supports Transit Gateway. For more information, see [Managing IBM Cloud connections](#).
 - IBM Cloud Monitoring and Power Edge Router network connectivity are currently unavailable for **CHE01**.
- SUSE part numbers for each tier are available. For more information, see the part numbers table in [Pricing for Power Virtual Servers](#).
- Flexible I/O operation per second (IOPS) is now available in the **DAL12**, **DAL13**, **SAO01**, **SAO04**, **WDC04**, and **WDC06** data centers. For more information, see [Flexible IOPS](#).
- IBM Cloud Monitoring service is now available in **DAL13** and **FRA04**. For more information, see [Monitoring metrics for Power Virtual Server](#).
- PER is now available in the **OSA21**, **SYD05**, and **TOK04** data center. For more information, see [Getting Started with Power Edge Router](#).

February 2024

- PER is now available in the **SAO01** data center. For more information, see [Getting Started with Power Edge Router](#).
- You can now provision the SLES 15 SP5 images. You are required to install an additional [software package](#) as a prerequisite for SAP and Netweaver workloads.
- You can now choose to use IBM provided Linux images with your own license. While provisioning the boot image, select the OS image that is listed under the **Client supplied subscription** section. The stock non-Full Linux Subscription (FLS) images are available across all the data centers. For more information see, [Full Linux® subscription for Power Virtual Servers](#).
- Flexible IOPS is now available in the **FRA04**, **LON06**, and **MAD02** data centers. For more information, see [Flexible IOPS](#).
- GRS is now available in **WDC07** and **DAL10** data center pair.
- You can now migrate your workspace that uses a Direct Link connection to a PER workspace through a support ticket. For more information, see [Migrating to Power Edge Router](#).
- IBM Cloud Monitoring service is now available in **OSA21**, **DAL10**, and **WDC07**.

January 2024

- Certain stock images for SAP will be updated with new Ansible packages. Until then it is recommended that you do not use the following images to create a new Power Virtual Server instance for use with SAP workloads:
 - RHEL8-SP8-SAP
 - RHEL8-SP8-SAP-NETWEAVER
 - RHEL9-SP2-SAP
 - RHEL9-SP2-SAP-NETWEAVER
- The Power Virtual Server workspaces provisioned in London, São Paulo, Osaka, Washington D.C., Montreal, and Toronto will send events to activity tracker instances in their respective regions effective from 29 January 2024. For more information, see [Activity tracker regions](#).
- PER is now available in **DAL12** and **FRA04**. For more information, see [Getting Started with Power Edge Router](#).
- The direction going forward for VPN connectivity capability of the Power Virtual Server will be to use the existing [IBM Cloud VPC VPN](#) for a one cloud experience, improved reliability, and high availability connections.

The VPNAAS_CONNECT_APPLICATION_INSTANCE_HOURS part ID of the Power Virtual Server is designed only for Power Virtual Server and will be withdrawn by the end of March 2024.

Existing users can choose to continue to use Power Virtual Server VPN until the end of service or upgrade to their preferred IBM Cloud VPN for VPC service.

By January 18, 2025, IBM will no longer deliver standard support for the Power Virtual Server VPNaaS product, thus if you choose not to take

the recommended action and run into issues, IBM will not support with end of life for parts occurring six months later on July 14, 2025. Additionally, after March 2024 new connections can no longer be provisioned with the Power Virtual Server VPN thus you are encouraged to upgrade to the IBM Cloud VPC VPN at your earliest convenience and ideally before March 2024.

- IBM Cloud Monitoring service is now available in `SYD04`, `SAO04`, and `DAL12`.

2023

December 2023

New data center availability

- `MAD04` is available. It is a PER-enabled Power10 data center that supports IBM Cloud Monitoring service.
- `SAO04` is available for PER.
- **Cost estimator tool** - A new cost estimator tool for Power Virtual Server is available. You can access it from the Power Virtual Server [home page](#). To learn more about the cost estimator tool, see [Getting started with the cost estimator tool](#).
- **Dedicated hosts** - A new dedicated host capability is available. You can provision IBM Power S922 and S1022 servers for your dedicated use. For more information, see [Dedicated host](#).
- **VPC VPN service** - The VPC VPN is a robust service that replaces the legacy Power Virtual Server VPN. To learn more on VPC VPN, see [Creating a Virtual Private Cloud VPN connection](#).
- **Flexible IOPS** - Power Virtual Server now offers a tier-less storage service with the name Flexible IOPS. With Flexible IOPS, you can now change the IOPS level for your existing volumes and clone volumes to your choice of IOPS level, and much more. See: [Flexible IOPS](#).
- New RHEL versions are available. See [Operating systems supported in IBM Power Virtual Server](#) page for details on the latest version.
- **Update on the new SAP HANA large t-shirt profiles feature** - The latest RHEL 9.2 for SAP, RHEL 8.8 for general purpose and SAP, and current RHEL 8.6 for general purpose and SAP OS images are being updated to support the larger t-shirt profiles. Until further notice, use t-shirt profiles with less than 64 cores for RHEL 9.2, RHEL 8.8, and RHEL 8.6 OS images. For more information, see the [SAP documentation on OS](#) for IBM Power Virtual Server.
- **New GRS pairs support** - The respective data center pairs `MAD02` and `FRA04` along with `MAD04` and `FRA05` now supports GRS.

November 2023

- `WDC06` and `MAD02` data center are now available for PER. `MAD02` is a PER-enabled Power10 data center.
- Availability of IBM Cloud Monitoring service is now extended in `FRA04`, `FRA05`, `LON04`, `LON06`, `MAD02`, `SAO01`, and `TOK04`.

September 2023

- **Power Virtual Server and SysDig Integration** - You can now seamlessly monitor your platform metrics within the IBM Cloud environment by using the IBM Cloud Monitoring service. For detailed instructions on how to access and use this integration, refer to the documentation: [Monitoring metrics for Power Virtual Server](#).



Note: The data center where you can monitor your platform metrics currently are `WDC06`, `SYD05`, and `WDC04` with other data center coming soon.

- **Introduction of Power Virtual Server S1022 processor** - The Power10 S1022 (9105-22A) is now available in `WDC07` data center. As we progress, additional data centers are made available.



Note: SAP Netweaver is not certified for use with S1022 systems, making them suitable exclusively for nonproduction workloads.

- **IBM i 7.3 service extension pricing** - Effective 1 October 2023, IBM i 7.3 on Power Virtual Server is at the end of normal support and will be in service extension. Service extension pricing is automatically added to the IBM i charge for IBM i 7.3 usage.
- **Addition of New Global Replication Service data center pair** - Power Virtual Server expanded the GRS capabilities by adding a new data center pair - `DAL13` and `WDC04`. This enhancement is available on the Tier 1 storage support within the GRS framework. See: [Locations that support global replication service](#).

June 2023

- The Power Virtual Server enhances the management of stock images in the backend. This enhancement eliminates identical versions (copies)

IBM Power Virtual Server enhances the management of stock images in the dashboard. This enhancement eliminates redundant references (copies) of stock images and improves the performance of copying a stock image in your account by using an image reference. Find more information in the [FAQ doc page](#).

We are currently rolling out the feature to eliminate stock image copies in all data centers in phases. If your data center has already received the update, you notice that the export option for stock images is no longer available because you are using the reference of the stock image.

- IBM Power Virtual Server is sequentially rolling out a Power Edge Router (PER) solution that is available in **DAL10** data center.

PER improves network communication across different parts of the IBM network. See, [Getting Started with Power Edge Router](#) for more information.

May 2023

- IBM Power Virtual Server now supports two new data centers **WDC07** (Washington DC) and **SAO04** (São Paulo).
- IBM Key Protect is now supported on AIX and Linux workloads. For more information, see [Integrating Power Virtual Server with IBM Cloud Key Management Services](#).
- Effective 1 May 2023, AIX 7.1 on Power Virtual Server is covered for software support. In Power Virtual Server, starting March 2024 [\[2:1\]](#), Service Extension pricing is automatically added to the AIX charge for AIX 7.1 usage. AIX 7.1 Service Extension was announced on January 24, 2023. For more information see, [AIX support lifecycle information](#).

1. This update is regarding the service extension update in the [May 2023](#) release notes section. [↩](#)

2. See the update in the [April 2024](#) release note section. [↩](#) [↩](#)

What is IBM Power Virtual Server?

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

IBM® Power® Virtual Server is an IBM Power server offering. You can use the Power Virtual Server to deploy a virtual server, also known as a logical partition (LPAR), in minutes. You can provision flexible, secure, and scalable compute capacity for Power servers enterprise workloads on IBM Power Virtual Server in both IBM data center and Client location".

Review this information to understand the IBM Power Virtual Server in (IBM data center and Client location) architectures and to identify the hardware and software specifications:

- [IBM Power Virtual Server in IBM data center architecture](#)
- [IBM Power Virtual Server in Client location architecture](#)

Operating systems supported in IBM Power Virtual Server

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

IBM® Power® Virtual Server supports AIX, IBM i, and Linux operating systems. You can run these operating systems on your virtual server instances (VSIs), also known as logical partitions (LPARs). Each operating system (OS) runs natively on the IBM Power architecture and follows a specific version, update, and support requirements. This topic lists the supported OS versions, key technical details, and lifecycle information. The availability of specific operating systems depends on whether the Power system is deployed in an IBM data center or at a client location.

The choice between AIX, IBM i, and Linux depends on the type of the workload that you want to run on a VSI.

AIX

You can use AIX to run enterprise UNIX applications that require high availability, scalability, and system-level performance. For more information about AIX, see [IBM AIX](#).

For information about supported AIX versions for Power Virtual Server, see [Supported AIX versions for Power Virtual Server](#).

IBM i

You can use IBM i to run integrated business applications that rely on IBM® Db2®, Report Program Generator (RPG), or Control Language (CL) in a secure and stable environment. For more information about IBM i, see [IBM i](#).

For information about supported IBM i versions for Power Virtual Server, see [Supported IBM i versions for Power Virtual Server](#).

When you use IBM i for your VSI, review the following requirements:

- If you are using an older version of IBM i, you must upgrade the OS to a current support level and then migrate to Power Virtual Server. For more information about upgrading the IBM i OS, see [Fast path for software installation, upgrade, and deletion tasks](#).
- Install the appropriate program temporary fixes (PTFs) depending on the version of IBM i that you use. For more information about the minimum PTF levels, see [Minimum PTF levels for IBM i](#).
- The IBM i 7.4, 7.5, 7.6, and Cloud Optical Repository (COR) OS images support Power10 and later systems. However, you can create a custom image or upgrade your IBM i VSI by downloading the media from [Entitled Systems Support \(ESS\)](#) and navigating to the **My entitled software > IBM i Evaluation and NLV Download** section.
- When you upgrade an IBM i VSI that has a virtual serial number (VSN) assigned, you must contact IBM Support to update the IBM i licenses to match the new IBM i version. For more information about VSN, see [About VSN in IBM data center](#).

Linux

You can use Linux to run open source, cloud-native, or containerized workloads with flexibility by using the supported Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) distributions on Power Virtual Server. For more information about the supported Linux distributions, see [Enterprise Linux on Power](#).

For information about supported Linux versions for Power Virtual Server, see [Supported Linux versions for Power Virtual Server](#).

When you use Linux for your VSI, review the following Linux distribution options and requirements:

- Power Virtual Server in the IBM data center supports Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) distributions. The Linux stock images are available when you select the Full Linux Subscription (FLS) option or the bring-your-own-license (BYOL) option. An FLS image refers to an IBM-provided RHEL or SLES stock OS image that includes the full Linux subscription support that is delivered through IBM.

For more information about FLS, see [Full Linux® subscription for Power Virtual Server](#).



Important: SAP FLS stock images are available only in IBM data centers and are not available in the Power Virtual Server Private Cloud environments that are deployed at the client locations.

- To use your own license, select the OS image with the **-BYOL** suffix. On the *Create virtual server instance* page, these images are listed under the *Client supplied subscription* section. Alternatively, you can create your own customized Linux image in Open Virtualization Appliance (OVA)

format by using the Linux stock images that are available when you select Full Linux Subscription (FLS). For more information, see [Creating a custom Linux image in OVA format](#).



Note: Red Hat Enterprise Linux CoreOS (RHCOS) images are intended for use only with Red Hat OpenShift clusters, in accordance with the Red Hat licensing terms. Power Virtual Server does not support RHCOS as a general-purpose operating system.

The following IBM Power Systems are supported by Red Hat and SUSE:

- [IBM Power System E1180 \(9080-HEU\)](#)
- [IBM Power System E1150 \(9043-MRU\)](#)
- [IBM Power System S1122 \(9824-22A\)](#)
- [IBM Power System E1080 \(9080-HEX\)](#)
- [IBM Power System E1050 \(9040-MRX\)](#)
- [IBM Power System S1022 \(9105-22A\)](#)
- [IBM Power System E980 \(9080-M9S\)](#)
- [IBM Power System S922 \(9009-22A\)](#)
- [IBM Power System S922 \(9009-22G\)](#)

For additional Linux support, refer to your distribution’s documentation. For cloud-init configuration instructions, see [Installing and configuring cloud-init on Linux](#).

Stock images

A stock image is an IBM-supplied OS image that is ready for deployment on Power Virtual Server. The stock images are available for use when you deploy the VSIs. The latest version of each supported OS is provided for provisioning new Power Virtual Server instances. As an alternative, you can use your own backup copy of an OS image. This option is called bring-your-own-image (BYOI). For more information, see [Deploying a custom image within IBM Power Virtual Server](#).

For each major version of the OS that is enabled through the offering, Power Virtual Server provides a single stock image. Power Virtual Server typically provides stock images for the last three major versions of the supported OS. Any update to the OS stock image is planned only when the image level is validated for Power Virtual Server environment. IBM Cloud sends advance notifications before it removes the unsupported stock OS images.



Note: The VSIs can continue to run without any issues after the stock OS images are removed from the image catalog. However, you are advised to update the VSIs to the supported OS versions.

Supported AIX, IBM i, and Linux versions for Power Virtual Server

The following table lists the supported AIX, IBM i, and Linux OS versions that can be installed on Power Virtual Server instances. OS versions that are not listed in the following table are not supported and cannot be installed on a VSI.

Processor family	Supported AIX versions	Latest stock image versions
Power11	AIX 7.2 TL5 or later	AIX 7.3 TL4 SP2 AIX 7.3 TL3 SP3 AIX 7.2 TL5 SP13
Power10 [1]	AIX 7.2 TL5 or later	AIX 7.3 TL4 SP2 AIX 7.3 TL3 SP3 AIX 7.2 TL5 SP13
Power9 [2]	AIX 7.2 TL5 or later	AIX 7.3 TL4 SP2 AIX 7.3 TL3 SP3 AIX 7.2 TL5 SP13
Supported AIX versions for Power Virtual Server		
Processor family	Supported IBM i versions	Latest stock image versions

Power11	IBM i 7.4 or later	IBM i COR [3] [4] IBM i 7.5 TR7 IBM i 7.4 TR12
Power10	IBM i 7.3 or later	IBM i COR IBM i 7.5 TR7 IBM i 7.4 TR12 IBM i 7.3 TR13 [5]
Power9 [6]	IBM i 7.3 or later [7]	IBM i COR IBM i 7.5 TR7 IBM i 7.4 TR12 IBM i 7.3 TR13

Supported IBM i versions for Power Virtual Server		
Processor family	Supported RHEL versions	Latest stock image versions
Power11	- RHEL 9.6 general purpose and SAP - RHEL 8.10 general purpose	- RHEL 9.6 general purpose (RHEL9-SP6) - RHEL 9.6 for SAP HANA (RHEL9-SP6-SAP-HANA) - RHEL 9.6 for SAP NetWeaver (RHEL9-SP6-SAP-NETWEAVER) - RHEL 8.10 general purpose (RHEL8-SP10)
Power10	- RHEL 9.6 general purpose and SAP - RHEL 9.4 SAP - RHEL 9.2 SAP - RHEL 8.10 general purpose and SAP - RHEL 8.8 SAP	- RHEL 9.6 general purpose (RHEL9-SP6) - RHEL 9.6 for SAP HANA (RHEL9-SP6-SAP-HANA) - RHEL 9.6 for SAP NetWeaver (RHEL9-SP6-SAP-NETWEAVER) - RHEL 9.4 for SAP HANA (RHEL9-SP4-SAP-HANA) - RHEL 9.4 for SAP NetWeaver (RHEL9-SP4-SAP-NETWEAVER) - RHEL 9.2 for SAP HANA (RHEL9-SP2-SAP) - RHEL 9.2 for SAP NetWeaver (RHEL9-SP2-SAP-NETWEAVER) - RHEL 8.10 general purpose (RHEL8-SP10) - RHEL 8.10 for SAP HANA (RHEL8-SP10-SAP-HANA) - RHEL 8.10 for SAP NetWeaver (RHEL8-SP10-SAP-NETWEAVER) - RHEL 8.8 for SAP HANA (RHEL8-SP8-SAP) - RHEL 8.8 for SAP NetWeaver (RHEL8-SP8-SAP-NETWEAVER)
Power9 [8]	- RHEL 9.6 general purpose and SAP - RHEL 9.4 SAP - RHEL 9.2 SAP - RHEL 8.10 general purpose and SAP - RHEL 8.8 SAP	- RHEL 9.6 general purpose (RHEL9-SP6) - RHEL 9.6 for SAP HANA (RHEL9-SP6-SAP-HANA) - RHEL 9.6 for SAP NetWeaver (RHEL9-SP6-SAP-NETWEAVER) - RHEL 9.4 for SAP HANA (RHEL9-SP4-SAP-HANA) - RHEL 9.4 for SAP NetWeaver (RHEL9-SP4-SAP-NETWEAVER) - RHEL 9.2 for SAP HANA (RHEL9-SP2-SAP) - RHEL 9.2 for SAP NetWeaver (RHEL9-SP2-SAP-NETWEAVER) - RHEL 8.10 general purpose (RHEL8-SP10) - RHEL 8.10 for SAP HANA (RHEL8-SP10-SAP-HANA) - RHEL 8.10 for SAP NetWeaver (RHEL8-SP10-SAP-NETWEAVER) - RHEL 8.8 for SAP HANA (RHEL8-SP8-SAP) - RHEL 8.8 for SAP NetWeaver (RHEL8-SP8-SAP-NETWEAVER)

Supported Linux versions for Power Virtual Server		
Processor family	Supported SLES versions	Latest stock image versions

Power11	◦ SLES 16 general purpose ◦ SLES 15.7 general purpose and SAP ◦ SLES 15.6 general purpose and SAP	◦ SLES 16 general purpose (SLES16) ◦ SLES 15 SP7 general purpose (SLES15) ◦ SLES 15 SP7 for SAP HANA (SLES15-SP7-SAP) ◦ SLES 15 SP7 for SAP NetWeaver (SLES15-SP7-SAP-NETWEAVER) ◦ SLES 15 SP6 general purpose (SLES15) ◦ SLES 15 SP6 for SAP HANA (SLES15-SP6-SAP) ◦ SLES 15 SP6 for SAP NetWeaver (SLES15-SP6-SAP-NETWEAVER)
Power10	◦ SLES 16 general purpose ◦ SLES 15.7 general purpose and SAP ◦ SLES 15.6 general purpose and SAP ◦ SLES 15.5 SAP ◦ SLES 15.4 SAP	◦ SLES 16 general purpose (SLES16) ◦ SLES 15 SP7 general purpose (SLES15) ◦ SLES 15 SP7 for SAP HANA (SLES15-SP7-SAP) ◦ SLES 15 SP7 for SAP NetWeaver (SLES15-SP7-SAP-NETWEAVER) ◦ SLES 15 SP6 general purpose (SLES15) ◦ SLES 15 SP6 for SAP HANA (SLES15-SP6-SAP) ◦ SLES 15 SP6 for SAP NetWeaver (SLES15-SP6-SAP-NETWEAVER) ◦ SLES 15 SP5 for SAP HANA (SLES15-SP5-SAP) ◦ SLES 15 SP5 for SAP NetWeaver (SLES15-SP5-SAP-NETWEAVER) ◦ SLES 15 SP4 for SAP HANA (SLES15-SP4-SAP) ◦ SLES 15 SP4 for SAP NetWeaver (SLES15-SP4-SAP-NETWEAVER)
Power9 [9]	◦ SLES 16 general purpose ◦ SLES 15.7 general purpose and SAP ◦ SLES 15.6 general purpose and SAP ◦ SLES 15.5 SAP ◦ SLES 15.4 SAP	◦ SLES 16 general purpose (SLES16) ◦ SLES 15 SP7 general purpose (SLES15) ◦ SLES 15 SP7 for SAP HANA (SLES15-SP7-SAP) ◦ SLES 15 SP7 for SAP NetWeaver (SLES15-SP7-SAP-NETWEAVER) ◦ SLES 15 SP6 general purpose (SLES15) ◦ SLES 15 SP6 for SAP HANA (SLES15-SP6-SAP) ◦ SLES 15 SP6 for SAP NetWeaver (SLES15-SP6-SAP-NETWEAVER) ◦ SLES 15 SP5 for SAP HANA (SLES15-SP5-SAP) ◦ SLES 15 SP5 for SAP NetWeaver (SLES15-SP5-SAP-NETWEAVER) ◦ SLES 15 SP4 for SAP HANA (SLES15-SP4-SAP) ◦ SLES 15 SP4 for SAP NetWeaver (SLES15-SP4-SAP-NETWEAVER)

Supported Linux versions for Power Virtual Server

System software maps

System software maps are reference documents that outline the compatibility between different OS versions and IBM Power Systems hardware. For more information about AIX, IBM i, and Linux (SUSE and RHEL) versions that are compatible with various Power Systems, refer to the following software maps:

- [System software maps for AIX](#)
- [System software maps for IBM i](#)
- [System software maps for SUSE Linux Enterprise Server](#)
- [System software maps for Red Hat Enterprise Linux](#)

Licensing and software components

The license for the AIX and IBM i operating systems is a part of the overall cost for the workspace. You cannot use an existing license that you have already purchased.

You can use movable IBM i OS entitlements (IBM i Moveable Operating License (MOL)) to move your existing on-premises entitlements to the Power Virtual Server. Contact IBM support to know more about IBM i MOL. For the available support resources, see [Getting help and support](#).

Power Virtual Server supports multiple levels of RHEL and SLES. You can either use the stock Linux images that IBM provides with Full Linux

Subscription or bring your own custom Linux image with vendor-provided subscription.

For more information about IBM software licensing, see [IBM License Information documents](#).



Important: You must ensure compliance with all third-party software licenses that you use in your environment, including middleware, application software, monitoring tools, and any other non-IBM licensed components.

Operating system lifecycle information

Review the lifecycle information for the OSs that you deploy in the Power Virtual Server environment. You can use the following resources to plan upgrades, support, and long-term maintenance for your AIX, IBM i, and Linux-based VSIs:

- [AIX support lifecycle information](#)
- [IBM i release lifecycle](#)
- [Red Hat Enterprise Linux lifecycle](#)
- [SUSE Linux Enterprise Server product support lifecycle](#)

Unsupported operating system versions in Power Virtual Server

You can deploy various IBM i, AIX, and Linux operating systems in Power Virtual Server environment. However, any OS version that IBM no longer supports is considered unsupported, even if the OS version still runs on the Power Virtual Server infrastructure.

You must upgrade your VSIs to a supported OS to ensure compatibility with the platform features, maintain system reliability, and receive security updates and technical support.

Risks of using unsupported operating system versions

Unsupported OS versions might continue to work in the Power Virtual Server infrastructure, but they carry significant risks and do not qualify for support, including:

- **Limited feature compatibility:** Power Virtual Server automation features such as Live Partition Mobility (LPM) and remote restart might not work, which can reduce application uptime during host maintenance.
- **Server restart failures:** The standard Power Virtual Server restart function might not work, which can cause the VSIs to remain pinned to the host and disrupt availability.
- **Security vulnerabilities:** Unsupported OS versions might not receive critical security updates, which can increase exposure to security threats.

Customer responsibilities

If you run an unsupported operating system version, you are responsible for managing all the associated risks and issues that might occur. You are also responsible for maintaining, updating, and supporting any custom image that is based on an unsupported OS. In addition, you acknowledge that if IBM identifies a security vulnerability or a critical issue, IBM can suspend any VSI that is running an unsupported OS in the Power Virtual Server environment.

-
1. Power10 systems running firmware level FW1110 support AIX 7.2 TL5 and later. [↪](#)
 2. Not available for IBM Power Virtual Server Private Cloud in client locations. [↪](#)
 3. IBM i Cloud Optical Repository (COR) is a virtual image. You can deploy the image and use it as a Network File Server (NFS) to perform various IBM i tasks that require media. For more information on COR images, see [Cloud Optical Repository](#). [↪](#)
 4. For more information about performing an upgrade, see [57xxSS1 Option 1 or Option 3 in *ERROR - Tips Before Reinstallation](#). [↪](#)
 5. IBM i 7.3 on Power Virtual Server ended normal support on 1 October 2023 and is supported with paid service extension only. Refer to the [IBM i lifecycle](#) page for upcoming EoL dates and prepare to upgrade to a later version. [↪](#)
 6. Not available for IBM Power Virtual Server Private Cloud in client locations. [↪](#)
 7. IBM i 7.1 is supported on the IBM Power9 S924 and E980 models. [↪](#)
 8. Not available for IBM Power Virtual Server Private Cloud in client locations. [↪](#)
 9. Not available for IBM Power Virtual Server Private Cloud in client locations. [↪](#)

Integrating IBM Power Virtual Server with watsonx

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

IBM watsonx is a powerful AI platform that helps you unlock the full potential of your data by providing advanced capabilities in machine learning, data analytics, and natural language processing. You can use IBM watsonx to automate complex tasks, gain actionable insights from large datasets, and deliver personalized experiences.

Integrating IBM Power Virtual Server workloads with IBM watsonx SaaS services provides a powerful and flexible infrastructure. The integrated infrastructure supports demanding workloads, runs advanced AI applications, and scales rapidly in a hybrid platform environment with automated AI infrastructure setup and management. Some of the benefits of integrating Power Virtual Server with IBM watsonx include:

- Seamless hybrid platform integration
- AI-driven insights for business transformation
- Cost efficiency with cloud flexibility
- Simplified management and integration
- Security and compliance

For more information about Power Virtual Server and watsonx integration, see [IBM Power Virtual Server and watsonx integration](#).

Architecture

Architecture for IBM Power Virtual Server in IBM data center

IBM Power Virtual Server in IBM data center

IBM® Power® Virtual Server is located in the IBM data centers, distinct from the IBM Cloud servers with separate networks and direct-attached storage.

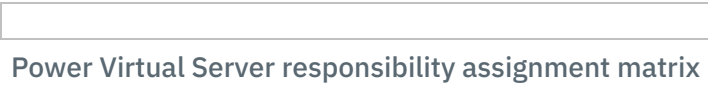
Review the following topics to understand the Power Virtual Server architecture, key features, and hardware and software requirements:

- [High-level architecture](#)
- [Key features](#)
- [Hardware specifications](#)
- [Data center capabilities](#)
- [Storage tiers](#)
- [Public and private networks](#)

High-level architecture

Power Virtual Server is located in the IBM data centers, distinct from the IBM Cloud servers with separate networks and direct-attached storage.

With the Power Virtual Server, you can quickly create and deploy one or more virtual servers (that are running either the AIX, IBM i, or Linux operating systems). After you provision the Power Virtual Server, you get access to infrastructure and physical computing resources without the need to manage or operate them. However, you must manage the operating system and the software applications and data. The following graphic represents a responsibility assignment (RACI) matrix for Power Virtual Server:



Key features

The key features for the Power Virtual Server are as follows.

Straightforward billing

Power Virtual Server uses a monthly billing rate that includes the licenses for the AIX, IBM i, or Linux operating systems. The monthly billing rate is pro-rated by the hour based on the resources that are deployed to the Power Virtual Server instance for the month. When you create the Power Virtual Server instance, you can see the total cost for your configuration based on the options that you specify. You can quickly identify what configuration options provide you with the best value for your business needs. For more information, see [Pricing](#).



Note: Bring your own Linux image (OVA format) and subscription. SLES and RHEL OVA images are supported. For more information, see [Using SLES within the IBM Power Virtual Server](#).

Infrastructure customization

You can configure and customize the following options when you create a Power Virtual Server:

- Number of virtual server instances
- Number of cores
- Amount of memory
- Data volume size and type
- Network interfaces

Bring your own image

IBM provides you with stock AIX and IBM i images when you create a Power Virtual Server. However, you can always [bring your own](#) custom AIX, IBM i, or Linux image that is tested and deployed.

Support for SAP NetWeaver or SAP HANA applications

When you provision a Power Virtual Server instance to support SAP NetWeaver applications, select a version of the IBM-provided AIX or Linux stock operating system image. When you provision a Power Virtual Server instance to support the SAP HANA applications, select a version of the IBM provided Linux® stock image. IBM i operating system and custom AIX and Linux images are not supported for SAP workloads. For more information about supported operating system versions, see [FAQ](#). For more information, see [Deploying your infrastructure](#).



Note: When you provision a Power Virtual Server to support SAP NetWeaver applications, select a version of the IBM-provided AIX operating system stock image. IBM i and operating system custom images are not supported for SAP workloads.

Support for deploying a Red Hat OpenShift Cluster

When you provision a Red Hat OpenShift Cluster on Power Virtual Server, it is easier to use the IBM™ provided automation to create the entire cluster of servers. You can install Red Hat OpenShift rather than individually provisioning Power Virtual Server instances. For instructions to deploy a Red Hat OpenShift Cluster on Power Virtual Server, including preparation of the required Operation System images for the cluster, see [Deploying Red Hat OpenShift Container Platform 4.x on IBM Power Virtual Server](#).

Hardware specifications

The following IBM Power server can host a Power Virtual Server:

- [IBM Power System S922 \(9009-22A\)](#)
- [IBM Power System E980 \(9080-M9S\)](#)
- [IBM Power System S1022 \(9105-22A\) \(Power10\)](#)
- [IBM Power System E1080 \(9080-HEX\)](#)
- [IBM Power System E1050 \(9043-MRX\)](#) ^[1]
- [IBM Power System S1122 \(9824-22A\)](#)
- [IBM Power System E1150 \(9043-MRU\)](#)



Tip: You can compare the performance of your current environment with the environment available through the Power Virtual Server. For more information, see the [IBM Power Performance Report](#).

Data center capabilities

You can check and compare your data center capabilities among three different data centers on the overview page of the [IBM Power Virtual Server](#) in the IBM Cloud console. You can also use the external interfaces such as API, CLI, and Terraform to check your data center capabilities.

For example, you can determine the support for the following capabilities in each data center:

- Machine types (Power9, Power10, or Power11)
- Global replication site pair(s)
- Dedicated host types

The IBM data centers have the following hardware components:

- [Computes](#)
- [Storage](#)
- [Networking](#)

Compute

The IBM data centers are configured with one or more of the following IBM Power Systems:

- [IBM Power System S922 \(9009-22A\)](#)
- [IBM Power System E980 \(9080-M9S\)](#)
- [IBM Power System S1022 \(9105-22A\) \(Power10\)](#)
- [IBM Power System E1080 \(9080-HEX\)](#)
- [IBM Power System E1050 \(9043-MRX\)](#) ^[1:1]
- [IBM Power System S1122 \(9824-22A\)](#)

- [IBM Power System E1150 \(9043-MRU\)](#)

For more information about your data center capabilities, see the overview page of the [IBM Power Virtual Server](#) in the IBM Cloud console.

Storage

IBM Power Virtual Server data centers are enabled with IBM FlashSystem®. For more information about IBM FlashSystem®, see [IBM FlashSystem®](#).

Networking

The networking in the IBM data centers are configured using the following Cisco switches:

- Cisco Nexus 9000 93180YC-EX (10G)
- Cisco Nexus 9000 93180YC-FX (Leaf 25G)
- Cisco Nexus 9000 9348GC-FXP (Leaf 1G)
- Cisco Nexus 9000 C9348GC-FXP (1G)
- Cisco Nexus 9000 C9336PQ (Spine 10G)
- Cisco Nexus 9000 N9K-C9364C (Spine 10G)
- Cisco Nexus 9000 C93180YC (10G)
- Cisco Nexus 9000 C93108TC-EX (1G)
- Cisco UCS - APIC controller
- Cisco ASR1001-HX Router
- Avocent ACS8048
- Avocent ACS8016
- Avocent ACS8032DAC-400

 **Note:** For a complete list of supported data centers, see [Creating a Power Virtual Server workspace](#).

Storage tiers

IBM Power Virtual Server offers you the option to select an I/O operation per second (IOPS) based storage according to your requirement. Flexible IOPS is a tier-less storage offering that removes the notion of disk type and replace it with a storage pool. Each of the storage pools supports multiple storage tiers. The storage tiers are based on different IOPS levels.

The following table shows the supported storage tiers with corresponding IOPS.

Tier level	IOPS	Performance
Tier 0	25 IOPS/GB	A 100-GB volume receives 2500 IOPS. This is 2.5x faster than tier 1 and 8.3x faster than tier 3.
Tier 1	10 IOPS/GB	A 100-GB volume receives 1000 IOPS. This is 3.3x faster than tier 3.
Tier 3	3 IOPS/GB	A 100-GB volume receives 300 IOPS.
Fixed IOPS	5000 IOPS regardless of size	A 100-GB volume receives 5000 IOPS.

Tier and IOPS mapping

 **Important:** The use of fixed IOPS is limited to volumes with a size of 200 GB or less, which is the break even size with Tier 0 (200 GB @ 25 IOPS/GB = 5000 IOPS).

Working with the APIs

Use the [Configure a private network subnet](#).

1. Available only at DAL14 data center. [↩](#) [↩](#)

Architecture for IBM Power Virtual Server Private Cloud in Client location

IBM Power Virtual Server Private Cloud in **Client location**

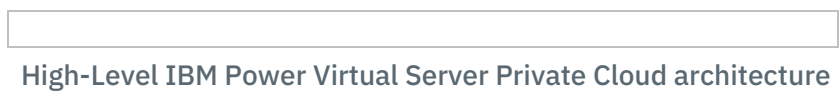
IBM® Power® Virtual Server is an as-a-service offering that includes a prescriptive set of physical infrastructure (compute, network, and storage). The physical infrastructure is deployed in your data center. The site reliability engineers (SREs) from IBM maintain and operates this infrastructure and manage it through the IBM Cloud platform.

To understand the Client location architecture, key features, and hardware and software requirements, review the following topics:

- [High-level architecture](#)
- [Key features](#)
- [Hardware and software specifications](#)
 - [Pods](#)
 - [Small pod configurations](#)
 - [Medium pod configurations](#)
 - [Supported Power11 servers](#)
 - [Operating systems](#)
 - [Storage](#)
 - [Storage tiers](#)
- [Network](#)
- [Data center capabilities](#)

High-level architecture

The following diagram provides a high-level architectural view of the IBM Power Virtual Server Private Cloud:



Key features

The key features for the Client location version of IBM Power Virtual Server are as follows:

- **Easy management and automation interfaces:** You can easily manage your Power Virtual Server resources by using GUI, CLI, API, or Terraform interfaces.
- **Bring your own image:** You can bring your own custom IBM AIX, Linux®, or IBM i image that is tested and deployed. Currently, the supported images include the following operating system images:
 - IBM AIX 7.2, or later
 - IBM i 7.4, or later and IBM i COR [\[1\]](#)
 - Red Hat Enterprise Linux (RHEL)
 - SUSE Linux Enterprise Server (SLES)
 - Red Hat Enterprise Linux CoreOS (RHCOS) for Red Hat OpenShift Container Platform
- **Dynamic resource adjustment:** You can configure and customize the following resources on the virtual server when you work with IBM Power Virtual Server in Client location:
 - Number of cores
 - Amount of memory
 - Storage volume size
- **Shared (capped and uncapped) and dedicated virtual machine :** When you deploy a virtual machine, you can choose one of the following options for the type of core:
 - **Shared capped:** The processor is shared among other virtual machines but the partition cannot use higher number of cores than the assigned numbers unlike shared uncapped processor partitions. This option is used mostly for licensing purposes.
 - **Shared uncapped:** The processor is shared among other virtual machines.
 - **Dedicated:** The processor is allocated for the specific virtual machine.

- **Colocation policies for virtual machines and volumes** : You can apply an affinity or anti-affinity policy to each virtual machine instance to control the server on which a new virtual machine is placed. You can build high availability infrastructure within a data center by using this feature.
- **Volume snapshot and clone operations** : You can capture full, point-in-time copies of the virtual machines or datasets. You can create delta snapshots, volume clones, and restore your disks by using IBM FlashCopy feature on Power Virtual Server.
- **Entitled processor-to-virtual-processor ratio** : The core-to-virtual core ratio can be in the range of 1:1 to 1:20. For shared processors, fractional cores are estimated up to the nearest whole number. For example, 1.25 cores is equal to 2 virtual cores.



Note: The create or update operations of a VSI can fail even when the core-to-virtual core ratio of 1:20 is satisfied. This issue occurs if the configured count of vCPU exceeds the maximum supported per-instance limit that is defined by the system policy.

Hardware and software specifications

For more information about how IBM Cloud region hosts connections from the pods for IBM Power Virtual Server in Client location, see [IBM Satellite location](#).

Pods

The following pod sizes are available:

- Small: 1 rack of IBM Power11 (S1122) processor.
- Medium: 2–4 racks of IBM Power11 (S1122, E1150, or E1180) processors.

You can expand the pod by adding more compute nodes up to a specific maximum number. This limit is related to the configuration size of the pod. For example, if you start the pod with 5 nodes, you can later add 3 more nodes. The pods are equipped with a spare compute node per compute type. For example, 1 compute node for each group of IBM Power E1180 processors. The spare nodes are used for maintenance or automatic high availability purposes.

You can use 100% of the core, memory, and storage of the nodes by excluding the spare nodes.



Note: The spare node is used by the IBM site reliability engineering (SRE) team for maintenance and is not available for your use.

Small pod configurations

A small pod has a 1x42U rack and S1122 system type is supported in the rack. You can expand the capacity of a small pod by adding servers of the same type that are already installed in the pod. You can add a minimum of one server, up to the maximum capacity of the rack.

[Table 1](#) illustrates the available configurations for server types and memory types on a small pod that has one rack. [Table 2](#) illustrates the available configurations for storage types on small pods with flash system storage options.

Server types	Min (2 TB option)	Min (4 TB option)	Max
Server quantity in a pod	5	5	9
Number of cores per server	60	60	60
Total number of cores	300	300	540
Usable cores	255	255	459

Small pod configuration.

For the S1122 server, the following memory configurations are available:

- **2 TB memory option per server** : A minimum 10 TB of memory and a maximum 18 TB of memory.
- **4 TB memory option per server** : A minimum 20 TB of memory and a maximum 36 TB of memory.

The small pod with one rack is available with FS 5300 TB flash system storage.

Storage types	FS 5300 TB
---------------	------------

Number of racks	1	1
Drives for each flash system	12	12
Capacity for each drive in TB	19.2	19.2
Number of flash systems in a pod	1	2
Total drives in a pod	12	24
Total capacity in TB	230	460
Usable capacity in TB	219	438
Usable capacity in TB at 2x compression	438	876

Small pod with flash system storage configuration.

Medium pod configurations

An expandable modular architecture is used to design a medium pod that supports the following pod configurations, based on the standard 42U rack specification:

- 1 management rack
- 1 storage rack
- 1 to 18 compute racks

A medium pod supports up to 6 compute racks for S1122 or E1150 system types and up to 18 compute racks for a E1180 (2CEC) system type.


Note: You can expand the capacity of a medium pod by adding compute racks of the same type that are already installed in the pod.

S1122 or E1150 pods are configured with a minimum of three base racks that include one compute rack, one storage rack, and one management rack. You can add up to five compute racks to make it a total of six compute racks.

An E1180 pod is configured with a minimum of four base racks that include two compute racks, one storage rack, and one management rack. You can add up to 16 compute racks to make it a total of 18 compute racks.

The following configurations and memory types are available for S1122, E1150, and E1180 (2CEC) pods:

Compute racks	Base configuration	1 additional rack		5 additional racks		Maximum configuration (6 racks)	
	Base rack	Min	Max	Min	Max	Min	Max
Servers in a pod, excluding LPM server	11	1	12	1	60	11	71
Cores per server	60	60	60	60	60	60	60
Usable cores per server	51	51	51	51	51	51	51
Total cores	660	60	720	60	3600	660	4260
Total Usable cores	561	51	612	51	3060	561	3621
Memory capacities							
2 TB	24	2	24	2	120	22	142

4 TB	44	4	48	4	240	44	284
8 TB	–	–	–	–	–	–	–
16 TB	–	–	–	–	–	–	–
32 TB	–	–	–	–	–	–	–

Medium pod configuration.							
Compute racks	Base configuration	1 additional rack		5 additional racks		Maximum configuration (6 racks)	
	Base rack	Min	Max	Min	Max	Min	Max
Servers in a pod, excluding LPM server	5	1	6	1	30	5	35
Cores per server	64	64	64	64	64	64	64
Usable cores per server	55	55	55	55	55	55	55
Total cores	320	64	384	64	1920	320	2240
Total usable cores	275	55	330	55	1650	275	1925

Memory capacities							
2 TB	–	–	–	–	–	–	–
4 TB	20	4	24	4	120	20	140
8 TB	40	8	48	8	240	40	280
16 TB	–	–	–	–	–	–	–
32 TB	–	–	–	–	–	–	–

Medium pod configuration.									
Compute racks	Base configuration	1 additional rack		Set of 3 additional racks		16 additional racks		Maximum configuration (18 racks) total	
	Base rack	Min	Max	Min	Max	Min	Max	Min	Max
Servers in a pod, excluding LPM server	2	1	2	1	6	1	30	2	35
Cores per server	128	128	128	128	128	128	128	128	128
Usable cores per server	107	107	107	107	107	107	107	107	107
Total cores	256	128	256	128	768	128	3840	256	4480
Total usable cores	214	107	214	107	642	107	3210	214	3745

Memory capacities									

2 TB	–	–	–	–	–	–	–	–	–
4 TB	–	–	–	–	–	–	–	–	–
8 TB	16	8	16	8	48	8	264	16	280
16 TB	32	16	32	16	96	16	528	32	560
32 TB	64	32	64	32	192	32	1056	64	1120

Medium pod configuration.

The medium pod is available with storage capacity of 460 TB or 920 TB FlashSystem (FS) storage for S1122, E1150, and E1180 server types. The following storage configurations are available for medium pods with FS storage options:

460 TB FS configurations	Minimum configuration	Maximum configuration
Number of FS in pod	1	5
Drives per FS	24	24
Capacity per drive (TB)	19.2	19.2
Total drives in pod	24	120
Total capacity (TB)	460	2300
Usable capacity (TB)	438	2190
Usable capacity at 2x compression (TB)	876	4380

Medium pod with FS storage configuration.		
920 TB FS configurations	Minimum configuration	Maximum configuration
Number of FS in pod	1	5
Drives per FS	48	48
Capacity per drive (TB)	19.2	19.2
Total drives in pod	48	240
Total capacity (TB)	920	4600
Usable capacity (TB)	876	4380
Usable capacity at 2x compression (TB)	1752	8760

Medium pod with FS storage configuration.

Supported Power11 servers

The following Power11 servers are supported:

- [IBM Power S1122](#)
- [IBM Power E1150](#)
- [IBM Power E1180](#)

Operating systems

The Power11 supports Linux, AIX, and IBM i operating system.

IBM Power Virtual Server in Client location provides a complete Red Hat Enterprise Linux (RHEL) offering experience with RHEL stock images. The offering includes support from IBM and access to RHEL bug fixes from Satellite servers that are hosted in IBM Cloud. Currently, you must bring your own licenses for all the other operating system images. For more flexibility, you can always bring your own custom Linux image that is tested and deployed. The AIX stock images are supported on the Power11 with AIX operating system.

Storage

For small pods, only the IBM Flash System FS5300 is supported. For more information, see [IBM Flash System FS5300](#).

 **Note:** You can extend the storage capacity of the pods, but you cannot add more storage controllers.

Storage tiers

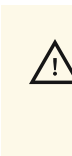
The storage tiers are based on I/O operations per second (IOPS). The performance of your storage volumes is limited to the maximum number of IOPS based on the storage volume capacity and storage tier.

Flexible IOPS is a tier-less storage offering that removes the notion of a disk type and replace it with a storage pool. Each of the storage pools supports multiple storage tiers. The storage tiers are based on different IOPS levels.

Table 5 shows the supported storage tiers with corresponding IOPS.

Tier level	IOPS	Performance
Tier 0	25 IOPS/GB	A 100-GB volume receives 2500 IOPS. This is 2.5x faster than tier 1 and 8.3x faster than tier 3.
Tier 1	10 IOPS/GB	A 100-GB volume receives 1000 IOPS. This is 3.3x faster than tier 3.
Tier 3	3 IOPS/GB	A 100-GB volume receives 300 IOPS.
Fixed IOPS	5000 IOPS regardless of size	A 100-GB volume receives 5000 IOPS.

Tier and IOPS mapping

 **Important:** The use of fixed IOPS is limited to volumes with a size of 200 GB or less, which is the break even size with Tier 0 (200 GB @ 25 IOPS/GB = 5000 IOPS).

For example, a 100 GB Tier 3 storage can receive up to 300 IOPs, and a 100 GB Tier 1 storage volume can receive up to 1000 IOPs. After the IOPs limit is reached for the storage volume, the I/O latency increases. Tier 3 storage is not suitable for production workloads. When you are choosing a storage tier, ensure that you consider not just the average I/O load, but more importantly the peak IOPs of your storage workload.

Network

The entire network subsystem can be divided into the following parts:

- **Control plane network traffic:** The control plane network is the connection between IBM Cloud and client data center. The IBM SREs establish this connection manually by using the Direct Link Connect (secure) or Virtual private network (VPN) between IBM Cloud and your Client location data center during the initial deployment.
- **Client data plane network traffic:** The client data plane is the connection between the client data center and the Power Virtual Server pods. Communication between each virtual machine within a pod is established by using private IP addresses. The IBM SREs configure the communication between virtual machines and the data center corporate network manually by using different network use cases. For more information, see [Network use cases](#). Use your own model for firewall and load balancer services. For any communication between your Client location data center and other IBM Cloud services, you must manually set up the network configuration.

For more information, see [Network overview](#).

Data center capabilities

You can check and compare the data center capabilities among three different infrastructure locations on the overview page of the [IBM Power Virtual Server](#) in the IBM Cloud console. You can also use the external interfaces such as API, CLI, and Terraform to check your data center capabilities.

For example, you can determine the support for the following capabilities in your infrastructure:

- Machine types (Power11)
- Global Replication Service (GRS)

-
1. IBM i Cloud Optical Repository (COR) is a virtual image that can be deployed and used as a Network File Server (NFS) to perform various IBM i tasks that require media. For more information on COR images, see [Cloud Optical Repository](#). [↵](#)

Locations

IBM Cloud® regions for IBM Power Virtual Server

IBM Power Virtual Server in IBM data center

The following table displays the appropriate region for your workspace:

The network round-trip latency, also known as round-trip time (RTT), is approximately 0.4 milliseconds when your workloads are running in Power Virtual Server and IBM Cloud Virtual Private Cloud (VPC), which are colocated in the same zone and connected through IBM Cloud Transit Gateway. RTT statistics might vary and are intended to help you plan the optimal placement of your cloud deployment.

Location	VPC region [1]	Power Virtual Server region	VPC zone	Power Virtual Server zone	Classic infrastructure [2]
Dallas	us-south	us-south	x	x	dal08 [4]
			x	x	dal09
			us-south-1	dal10	dal10
			us-south-2	dal12	dal12
			us-south-3	us-south	dal13
			us-south-3 [3]	dal14	dal14
Montreal	ca-mon	mon	x	mon01	mon01
San Jose	x	x	x	x	sjc03
	x	x	x	x	sjc04
Sao Paulo	br-sao	sao	br-sao-1	sao01	sao01
			br-sao-2	sao04	sao04
			br-sao-3	sao05	sao05
Toronto	ca-tor	tor	ca-tor-1	tor01 [5]	tor01
			ca-tor-2	x	tor04
			ca-tor-3	x	tor05
Washington DC	us-east	us-east	x	x	wdc03 [6]
			us-east-1	us-east	wdc04
			us-east-2	wdc06	wdc06
			us-east-3	wdc07	wdc07

Power Virtual Server regions and zones in North and South America

Location	VPC region [7]	Power Virtual Server region	VPC zone	Power Virtual Server zone	Classic infrastructure [8]
Amsterdam	x	x	x	x	ams03
Frankfurt	eu-de	eu-de	eu-de-1	x	fra02
			eu-de-2	eu-de-1	fra04
			eu-de-3	eu-de-2	fra05
London	eu-gb	lon	eu-gb-1	lon04	lon02
			eu-gb-2	x	lon04
			eu-gb-3	lon06	lon05
Madrid	eu-es	mad	eu-es-1	mad02	mad02
			eu-es-2	mad04	mad04
			eu-es-3	x	mad05

Milan	x	x	x	x	mil01
Paris	x	x	x	x	par01

Power Virtual Server regions and zones in Europe

Location	VPC region [9]	Power Virtual Server region	VPC zone	Power Virtual Server zone	Classic infrastructure [10]
Mumbai	in-mum	in-mum	in-mum-2 in-mum-1 in-mum-3	in-mum-1 in-mum-2 x	x x x
Chennai	x	in-che	x	che01 [11]	che01
Chennai	in-che	in-che	in-che-1 in-che-3 in-che-2	x che02 che03	x x x
Osaka	jp-osa	osa	jp-osa-1 jp-osa-2 jp-osa-3	osa21 x x	osa21 osa22 osa23
Singapore	x	x	x	x	sng01
Sydney	au-syd	syd	au-syd-1 au-syd-2 au-syd-3	x syd04 syd05	syd01 syd04 syd05
Tokyo	jp-tok	tok	jp-tok-1 jp-tok-2 jp-tok-3	x tok04 x	tok02 tok04 tok05

Power Virtual Server regions and zones in Asia Pacific



Tip: If you plan to create an application that uses both VPC and Power Virtual Server, review the zone mapping of your account. For example, create your VPC resources first, and then use the zone mapping to identify the universal zone for the VPC resources. This approach allows you to create the classic infrastructure resources in the same physical location. Classic infrastructure and IBM® Power® Virtual Server identify service locations by data center, while VPC identifies physical locations by universal zone name.



Note: When you create a workspace by using the Terraform resource [ibm_resource_instance](#) or the CLI command [ibmcloud resource service-instance-create](#), for the **service name** field use the value `power-iaas` for IBM Power Virtual Server in IBM data center and for the **plan name** field use the value `power-virtual-server-group` or `power-virtual-server-private-group` for IBM Power Virtual Server Private Cloud in Client location.

- For more information about the IBM Cloud regions that support VPC, see [Creating a VPC in a different region.](#) [↵](#)
- For more information about the list of classic data centers in IBM Cloud, see [Classic data centers.](#) [↵](#)
- IBM Cloud maps the VPC zone names by account and uses the universal zone names. For more information about the zone mapping by account, see [Zone mapping per account](#) and the tip after the table. [↵](#)
- IBM Cloud for government. [Learn more.](#) [↵](#)
- This is a single data center for the location. [↵](#)
- IBM Cloud for government. [Learn more.](#) [↵](#)
- For more information about the IBM Cloud regions that support VPC, see [Creating a VPC in a different region.](#) [↵](#)
- For more information about the list of classic data centers in IBM Cloud, see [Classic data centers.](#) [↵](#)
- For more information about the IBM Cloud regions that support VPC, see [Creating a VPC in a different region.](#) [↵](#)
- For more information about the list of classic data centers in IBM Cloud, see [Classic data centers.](#) [↵](#)

11. This is a single data center for the location. [↩](#)

Supported regions for IBM Power Virtual Server Private Cloud

Specific IBM Cloud® regions are available so that you can host connections from the pods for IBM Power Virtual Server Private Cloud in your data center.



Note: Selecting different IBM Cloud regions can affect computing pricing.

Client location The following IBM Cloud regions can be used for IBM Power Virtual Server Private Cloud. Select the IBM Cloud region that is closest to the physical location of your data center.

- Dallas (satcon_dal)
- Frankfurt (satcon_fra)
- London (satcon_lon)
- Madrid (satcon_mad)
- Osaka (satcon_osa)
- Sao Paulo (satcon_sao)
- Sydney (satcon_syd)
- Tokyo (satcon_tok)
- Toronto (satcon_tor)
- Washington, DC (satcon_wdc)

Create an IBM Cloud Satellite® location associated with a supported IBM Cloud region for use with IBM Power Virtual Server Private Cloud. The network latency between your data center and the selected IBM Cloud region must maintain a network round-trip time (RTT) of less than or equal to 200 milliseconds. For more information, see [Create a Satellite location overview](#).

Pricing

Pricing for IBM Power Virtual Server in IBM data centers

IBM Power Virtual Server in IBM data center

IBM® Power® Virtual Server in IBM data center provides a secure and unified billing system for the used hardware and software resources. The resources are metered that is the usage of the resources are tracked and measured based on the actual consumption. Metering is done based on resource allocation for billing purpose. The following list of resources are metered:

- Virtual server instances: CPU (in cores) with processor modes (capped, shared, or dedicated), and memory (in GB)
- Volumes: Storage volumes (in GB)
- Snapshots: Storage snapshots (in GB)
- Dedicated hosts: Dedicated hosts in host units

In addition to hardware resources, the licensed operating systems and the associated workloads are metered along with VSI resources. For more information about the billing of VSI resources, see [Operating systems](#) and [Linux for SAP workloads](#).

You can generate an estimate of the resources for IBM Power Virtual Server. The estimated cost might differ from the actual cost when you purchase the infrastructure or instances due to discounts and promotion codes. For more information, see [Generating an estimate for IBM Power Virtual Server](#).

IBM Power Virtual Servers are offered in selected regions with scale-out or scale-up logical partitions (LPARs). The IBM Power Systems that can host Power Virtual Servers have the following theoretical maximums:

**Important:** All prices that are mentioned in the topic are illustrative and do not represent the actual amounts that are used for billing. To calculate the exact pricing, use the [cost estimator](#).

Power Systems	Usable cores	Memory
E980 (9080-M9S)	143	Up to 15,307 GB [1]
S922 (9009-22A) [2]	15	Up to 942 GB
S1022 (9105-22A) [3]	33	Up to 1984 GB
E1080 (9080-HEX)	165	Up to 32 TB
E1050 (9043-MRX)	87	Up to 8,192 GB
S1122 (9824-22A)	51	Up to 1,904 GB

Theoretical maximum processors and memory

For more information about the systems that are available for your data center, see the overview page of the [Power Virtual Server](#) in the IBM Cloud console.

A theoretical maximum of a system depends on the data center. Also, the Power Virtual Server development team enforces the current available resources within each data center. With these processing maximums, the Power Virtual Server can meet any business workload requirement.

Billing details

In your invoice, use one of the following identifiers to identify the charges that apply to the deployed resource instances:

- [Consumer ID](#) The Consumer ID groups the billing usages that are under a single resource.
- [IBM cloud resource names \(CRNs\)](#) The IBM CRNs enable the identification and tracking of Power Virtual Server resources, such as volumes and snapshots, to manage and bill the usage of the resources.

Consumer ID

For metrics that are charged on a workspace instance, the **Consumer ID** groups billing usage under a VSI resource in that workspace. You can use the **Consumer ID** to identify the resource type and instance for each charge.

The **Consumer ID** has the following benefits:

- You can see a more detailed view of your bill by using the **Usage** page in the [Billing and Usage](#) portal.
- Charges are itemized by a resource type and an instance UUID in the Consumer ID field with the format `resource-type:resource-uuid`. Currently, only one metric is charged at the workspace level, which uses `pvm-instance` (VSI) as the resource type.

To view the usage details at the resource level, do the following steps:

- Open the [Billing and Usage](#) page in the IBM Cloud console.
- Click **Usage**, on the left navigation menu.
- Click **View plans** for the entry **Power Virtual Server Workspace**. A page that lists all your workspaces is opened.
- Click **View details** for a workspace. A page that lists the usage details of a selected workspace is opened.
- Scroll to the end of the page and click **View instance details**. A page that lists the usage details of the selected VSI is opened.

For more information on the billing and usage page, see [Billing and Usage documentation](#).

IBM Cloud Resource Names

IBM cloud resource names (CRNs) are identifiers that are assigned to uniquely identify resources within IBM Cloud. As CRNs are assigned to individual resources, a comprehensive bill is generated with individual Power Virtual Server resource billing.

When you create a resource, IBM Cloud assigns a CRN to the resource.

When you delete a workspace, the resources that are associated with the workspace are deleted before the workspace is deleted. If you use a dedicated host that is shared with a secondary workspace, remove the secondary workspace resources (VSIs) from the dedicated host, then delete the dedicated host workspace.

For more information about IBM Cloud CRNs, see [Cloud Resource Names](#).

**Note:** To view or search for resources that are provisioned in IBM Power Virtual Server by using the assigned `user tags`, see [Searching for Resources](#). The user tags are not included in the response for GET API and CLI requests.

The following table lists the Power Virtual Server resources that are enabled with CRN.

Logical Resource	IBM Power Virtual Server in IBM data center	IBM Power Virtual Server Private Cloud in Client location	Billable elements
General			
Workspace	Enabled	Enabled	VPN
Compute			
Virtual server instance	Enabled	Enabled	- Cores - Memory - SAP workload licenses - OS licenses
Shared Processor Pool	Enabled	Enabled	- SPP cores - SAP workloads
Server placement groups	Enabled	X	SAP workloads
Dedicated Host	Enabled	X	Dedicated host capacity

Storage			
Volume	Enabled	Enabled	- Standard volume storage - Image volume storage (onboarded by users) - Replicated volume storage - Service charges for Global replication services
Snapshot	Enabled	Enabled	Snapshot storage
Virtual Persistent Memory	Enabled	X	Virtual Persistent Memory storage

Power Virtual Server resources that are CRN enabled.

Monthly usage

IBM Power Virtual Server instances are charged at a monthly rate that is metered every hour. If you change resources in an LPAR during the month, the monthly bill for the LPAR reflects the resource change and the LPAR price aggregating all hourly charges before and after the change.

To reduce costs, capture a VSI as a volume-backed image and delete it when the VSI is not used. For more information about capturing a VSI, see [Capturing and exporting a VSI](#). For more information about deleting a VSI, see [Deleting a virtual server instance](#).

**Important:** All prices that are mentioned in the topic are illustrative and do not represent the actual amounts that are used for billing. To generate an estimated price, use the [Power Virtual Server Estimate pricing](#) tool. For more information, see [Generating an estimate](#).

In the following monthly usage example, the customer purchases a Power Virtual Server instance with the following configurations at a base price of \$250.57 per month (\$0.343 per hour):

- One core with 8 GB of memory
- A 150 GB disk
- AIX 7200-03-02 OS.

As the month progresses, the customer adds more memory. The new price for the LPAR is \$339.45 per month (\$0.465 per hour). The monthly bill is metered by the hour and reflects actual usage. The total amount combines the lesser rate that is charged for hours at the original configuration and the higher rate that is charged for hours at the updated configuration.

Hours elapsed in a month	Amount charged	LPAR description
300 hours	(300 hours x \$0.343)/month = \$103	1 core, 8 GB memory, 150 GB disk, AIX
430 hours	(430 hours x \$0.465)/month = \$200	1 core, 16 GB memory, 150 GB disk, AIX
730 hours (Monthly Total)	\$103 + \$200 = \$303 (Monthly Total)	1 core, 16 GB memory, 150 GB disk, AIX

Monthly LPAR charges

In this example, when the LPAR usage reaches 300 hours in a month, the LPAR memory is increased from 8 GB to 16 GB. The LPAR is metered by the hour. Out of the total 730 hours used in the month, the first 300 hours are metered at the lesser rate, while the remaining 430 hours are metered at the higher rate. This way of metering results in a total monthly cost of \$303.

Part number descriptions

A part number is associated with the license for the software product to be used in IBM Cloud. For detailed usage and billing information, refer to the part number descriptions. The part number descriptions are listed in your invoice or in the [IBM® Cloud billing](#) portal.

Refer to the following table to view the part number descriptions and the associated metric IDs.

Part description for virtual server instance group (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
---	--

Shared capped processor cores	
IBM Power S1122 shared capped core-hour	power-iaas-metric-S1122-core-capped
IBM Power E1080 shared capped core-hour	power-iaas-metric-E1080-core-capped
IBM Power S1022 shared capped core-hour	power-iaas-metric-S1022-core-capped
IBM Power E980 shared capped core-hour	power-iaas-metric-E980-core-capped
IBM Power S922 shared capped core-hour	power-iaas-metric-S922-core-capped
SAP NetWeaver workload - IBM Power S1022 shared capped core-hour	power-iaas-metric-netweaver-S1022-capped
Shared uncapped processor cores	
IBM Power S1122 shared uncapped core-hour	power-iaas-metric-S1122-core-shared
IBM Power E1080 shared uncapped core-hour	power-iaas-metric-E1080-core-shared
IBM Power S1022 shared uncapped core-hour	power-iaas-metric-S1022-core-shared
IBM Power E980 shared uncapped core-hour	power-iaas-metric-E980-core-shared
IBM Power S922 shared uncapped core-hour	power-iaas-metric-S922-core-shared
SAP NetWeaver workload - IBM Power S1022 shared uncapped core-hour	power-iaas-metric-netweaver-S1022-shared
Dedicated processor cores	
IBM Power S1122 dedicated core-hour	power-iaas-metric-S1122-core-dedicated
IBM Power E1080 dedicated core-hour	power-iaas-metric-E1080-core-dedicated
IBM Power S1022 dedicated core-hour	power-iaas-metric-S1022-core-dedicated
IBM Power E980 dedicated core-hour	power-iaas-metric-E980-core-dedicated
IBM Power S922 dedicated core-hour	power-iaas-metric-S922-core-dedicated
SAP NetWeaver workload - IBM Power S1022 dedicated core-hour	power-iaas-metric-netweaver-S1022-dedicated
AIX scale-out license core-hour	power-iaas-metric-aix-scale-out
AIX scale-up license core-hour	power-iaas-metric-aix-scale-up
SAP HANA workload balanced profile (bh1) - IBM Power E980 core-hour	power-iaas-metric-bh1-hana-core
SAP HANA workload compute optimized profile (ch1) - IBM Power E980 core-hour	power-iaas-metric-ch1-hana-core
SAP HANA workload custom profile (cnp) - IBM Power E980 core-hour	power-iaas-metric-hana-core
SAP HANA workload custom profile (cnp) - IBM Power E980 memory gigabyte-hour	power-iaas-metric-hana-memory
IBM i Cloud Storage Solutions instance core-hour	power-iaas-metric-ibmi-cos

IBM i PowerHA small instance core-hour	power-iaas-metric-ibmi-pha-small
IBM i PowerHA medium instance core-hour	power-iaas-metric-ibmi-pha-medium
IBM i P10 license core-hour	power-iaas-metric-ibmi-os-p10
IBM i P10 license - mobile core-hour - SWMA paid	power-iaas-metric-ibmi-os-p10-mol
IBM i P10 service extension core-hour	power-iaas-metric-ibmi-os-p10-sve
IBM i P30 license core-hour	power-iaas-metric-ibmi-os-p30
IBM i P30 license - mobile core-hour - SWMA paid	power-iaas-metric-ibmi-os-p30-mol
IBM i LPP core-hour	power-iaas-metric-ibmi-lpp
IBM i LPP core-hour - mobile core-hour - SWMA paid	power-iaas-metric-ibmi-lpp-mol
IBM i P10 LPP core-hour	power-iaas-metric-ibmi-lpp-p10
IBM i LPP P10 - mobile core-hour - SWMA paid	power-iaas-metric-ibmi-lpp-p10-mol
IBM i P30 LPP core-hour	power-iaas-metric-ibmi-lpp-p30
IBM i LPP P30 - mobile core-hour - SWMA paid	power-iaas-metric-ibmi-lpp-p30-mol
IBM i P05 software tier license core-hour	power-iaas-metric-ibmi-os-p05-tier
IBM i P10 software tier license core-hour	power-iaas-metric-ibmi-os-p10-tier
IBM i P20 software tier license core-hour	power-iaas-metric-ibmi-os-p20-tier
IBM i P30 software tier license core-hour	power-iaas-metric-ibmi-os-p30-tier
IBM i P05 software tier service extension core-hour	power-iaas-metric-ibmi-os-p05-sve-tier
IBM i P10 software tier service extension core-hour	power-iaas-metric-ibmi-os-p10-sve-tier
IBM i P20 software tier service extension core-hour	power-iaas-metric-ibmi-os-p20-sve-tier
IBM i P30 software tier service extension core-hour	power-iaas-metric-ibmi-os-p30-sve-tier
IBM i software tier LPP core-hour	power-iaas-metric-ibmi-lpp-tier
IBM i software tier PowerHA small instance core-hour	power-iaas-metric-ibmi-pha-small-tier
IBM i software tier PowerHA medium instance core-hour	power-iaas-metric-ibmi-pha-medium-tier
IBM i P10 software tier license - mobile-core-hour - SWMA paid	power-iaas-metric-ibmi-os-p10-mol-tier
IBM i P30 software tier license - mobile-core-hour - SWMA paid	power-iaas-metric-ibmi-os-p30-mol-tier
IBM i software tier LPP - mobile-core-hour - SWMA paid	power-iaas-metric-ibmi-lpp-mol-tier
IBM i P30 service extension core-hour	power-iaas-metric-ibmi-os-p30-sve

IBM i P30 - PowerHA instance core-hour	power-iaas-metric-ibmi-pha-p30
IBM i Rational Developer Studio instance-hour	power-iaas-metric-ibmi-rds
SAP HANA workload memory optimized profile (mh1) - IBM Power E980 core-hour	power-iaas-metric-mh1-hana-core
SAP NetWeaver workload - IBM Power S1022 memory gigabyte-hour	power-iaas-metric-netweaver-S1022-memory
IBM Power11 high-use scale-out memory gigabyte-hour	power-iaas-metric-p11-so-memory-highuse
IBM Power11 standard scale-out memory gigabyte-hour	power-iaas-metric-p11-so-memory-standard
IBM Power10 high-use scale-out memory gigabyte-hour	power-iaas-metric-p10-so-memory-highuse
IBM Power10 standard scale-out memory gigabyte-hour	power-iaas-metric-p10-so-memory-standard
IBM Power10 high-use scale-up memory gigabyte-hour	power-iaas-metric-p10-su-memory-highuse
IBM Power10 standard scale-up memory gigabyte-hour	power-iaas-metric-p10-su-memory-standard
IBM Power9 high-use memory gigabyte-hour	power-iaas-metric-p9-memory-highuse
IBM Power9 standard memory gigabyte-hour	power-iaas-metric-p9-memory-standard
Red Hat Enterprise Linux SAP RISE scale-out license core-hour	power-iaas-metric-rhel-sap-rise-scale-out
Red Hat Enterprise Linux SAP RISE scale-up license core-hour	power-iaas-metric-rhel-sap-rise-scale-up
Red Hat Enterprise Linux SAP scale-out license core-hour	power-iaas-metric-rhel-sap-scale-out
Red Hat Enterprise Linux SAP scale-up license core-hour	power-iaas-metric-rhel-sap-scale-up
Red Hat Enterprise Linux scale-out license core-hour	power-iaas-metric-rhel-scale-out
Red Hat Enterprise Linux scale-up license core-hour	power-iaas-metric-rhel-scale-up
SAP HANA workload small profile (sh2) - IBM Power E1050 core-hour	power-iaas-metric-sh2-hana-E1050-core
SAP HANA workload small profile (sh2) - IBM Power E1050 memory gigabyte-hour	power-iaas-metric-sh2-hana-E1050-memory
SAP HANA workload small profile (sh2) - IBM Power E1080 core-hour	power-iaas-metric-sh2-hana-E1080-core
SAP HANA workload small profile (sh2) - IBM Power E1080 memory gigabyte-hour	power-iaas-metric-sh2-hana-E1080-memory
SAP HANA workload small profile (sh2) - IBM Power S1022 core-hour	power-iaas-metric-sh2-hana-S1022-core
SAP HANA workload small profile (sh2) - IBM Power S1022 memory gigabyte-hour	power-iaas-metric-sh2-hana-S1022-memory
SAP HANA workload balanced profile (bh2) - IBM Power E1050 core-hour	power-iaas-metric-bh2-hana-E1050-core
SAP HANA workload balanced profile (bh2) - IBM Power E1050 memory gigabyte-hour	power-iaas-metric-bh2-hana-E1050-memory
SAP HANA workload balanced profile (bh2) - IBM Power E1080 core-hour	power-iaas-metric-bh2-hana-E1080-core
SAP HANA workload balanced profile (bh2) - IBM Power E1080 memory gigabyte-hour	power-iaas-metric-bh2-hana-E1080-memory

SAP HANA workload compute profile (ch2) - IBM Power E1050 core-hour	power-iaas-metric-ch2-hana-E1050-core
SAP HANA workload compute profile (ch2) - IBM Power E1050 memory gigabyte-hour	power-iaas-metric-ch2-hana-E1050-memory
SAP HANA workload compute profile (ch2) - IBM Power E1080 core-hour	power-iaas-metric-ch2-hana-E1080-core
SAP HANA workload compute profile (ch2) - IBM Power E1080 memory gigabyte-hour	power-iaas-metric-ch2-hana-E1080-memory
SAP HANA workload medium profile (mh2) - IBM Power E1080 core-hour	power-iaas-metric-mh2-hana-E1080-core
SAP HANA workload medium profile (mh2) - IBM Power E1080 memory gigabyte-hour	power-iaas-metric-mh2-hana-E1080-memory
SUSE Linux Enterprise Server SAP tier 1 instance-hour	power-iaas-metric-sles-sap-tier1
SUSE Linux Enterprise Server SAP tier 2 instance-hour	power-iaas-metric-sles-sap-tier2
SUSE Linux Enterprise Server SAP tier 3 instance-hour	power-iaas-metric-sles-sap-tier3
SUSE Linux Enterprise Server tier 1 instance-hour	power-iaas-metric-sles-tier1
SUSE Linux Enterprise Server tier 2 instance-hour	power-iaas-metric-sles-tier2
SUSE Linux Enterprise Server tier 3 instance-hour	power-iaas-metric-sles-tier3
SAP RISE - SAP HANA workload optimized profile (sr3) - IBM Power S1122 core-hour	power-iaas-metric-sr3-hana-S1122-core
SAP RISE - SAP HANA workload optimized profile (sr3) - IBM Power S1122 memory gigabyte-hour	power-iaas-metric-sr3-hana-S1122-memory
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power E1050 core-hour	power-iaas-metric-sr2-hana-E1050-core
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power E1050 memory gigabyte-hour	power-iaas-metric-sr2-hana-E1050-memory
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power E1080 core-hour	power-iaas-metric-sr2-hana-E1080-core
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power E1080 memory gigabyte-hour	power-iaas-metric-sr2-hana-E1080-memory
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power S1022 core-hour	power-iaas-metric-sr2-hana-S1022-core
SAP RISE - SAP HANA workload optimized profile (sr2) - IBM Power S1022 memory gigabyte-hour	power-iaas-metric-sr2-hana-S1022-memory
SAP HANA workload ultra-memory optimized profile (umh) - IBM Power E980 core-hour	power-iaas-metric-umh-hana-core
Virtual Tape Library terabyte-hour	power-iaas-metric-vtl

Part number descriptions for IBM Power Virtual Server.

Part description for shared processor pool group (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
Shared Processor Pool - IBM Power S1122 core-hour	power-iaas -metric-S1122-spp-cores
Shared Processor Pool - IBM Power S1022 core-hour	power-iaas-metric-S1022-cores-spp
Shared Processor Pool - IBM Power E1050 core-hour	power-iaas-metric-E1050-cores-spp
Shared Processor Pool - IBM Power E1080 core-hour	power-iaas-metric-E1080-cores-spp
Shared Processor Pool - IBM Power S922 core-hour	power-iaas-metric-S922-cores-spp

Shared Processor Pool - IBM Power E980 core-hour	power-iaas-metric-E980-cores-spp
Shared Processor Pool - IBM Power E880 core-hour	power-iaas-metric-E980-cores-spp
Part number descriptions for IBM Power Virtual Server.	
Part description for dedicated host group (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
IBM Power S922 (15 usable cores; 1 TB memory) dedicated host-hour	power-iaas-metric-S922-dedicated-host
IBM Power S1022 (33 usable cores; 2 TB memory) dedicated host-hour	power-iaas-metric-S1022-dedicated-host
IBM Power S1122 (51 usable cores; 2 TB memory) dedicated host-hour	power-iaas-metric-S1122-dedicated-host
Part number descriptions for IBM Power Virtual Server.	
Part description for volume group (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
Volume storage (tier 0: 25 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier0
Volume storage (tier 1: 10 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier1
Volume storage (tier 3: 3 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier3
Volume storage (tier 5k: 5,000 IOPS) gigabyte-hour	power-iaas-metric-volume-tier5k
Volume storage (low IOP) gigabyte-hour	power-iaas-metric-volume-lowiop
Asynchronous global replication storage gigabyte-hour	power-iaas-metric-volume-async-replication-service
Asynchronous replicated volume storage (tier 0: 25 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier0-async-replicated
Asynchronous replicated volume storage (tier 1: 10 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier1-async-replicated
Asynchronous replicated volume storage (tier 3: 3 IOPS per GB) gigabyte-hour	power-iaas-metric-volume-tier3-async-replicated
Asynchronous replicated volume storage (tier 5k: 5,000 IOPS) gigabyte-hour	power-iaas-metric-volume-tier5k-async-replicated
Part number descriptions for IBM Power Virtual Server.	
Part description for snapshot group (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
Snapshot storage (tier 0: 25 IOPS per GB) gigabyte-hour	power-iaas-metric-snapshot-tier0
Snapshot storage (tier 1: 10 IOPS per GB) gigabyte-hour	power-iaas-metric-snapshot-tier1
Snapshot storage (tier 3: 3 IOPS per GB) gigabyte-hour	power-iaas-metric-snapshot-tier3
Snapshot storage (tier 5k: 5,000 IOPS) gigabyte-hour	power-iaas-metric-snapshot-tier5k
Part number descriptions for IBM Power Virtual Server.	
Part description for vPMEM composite service (available on the IBM invoice)	Metric ID (available in the IBM Cloud catalog)
SAP workload - IBM Power E1050 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-E1050-vpmem
SAP workload - IBM Power E1080 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-E1080-vpmem
SAP workload - IBM Power S1022 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-S1022-vpmem
SAP application server workload - IBM Power S1022 virtual persistent memory volume gigabyte	power-iaas-metric-netweaver-S1022-vpmem

IBM Power10 high-use scale-out virtual persistent memory volume gigabyte-hour	power-iaas-metric-p10-so-vpmem-highuse
IBM Power10 standard scale-out virtual persistent memory volume gigabyte-hour	power-iaas-metric-p10-so-vpmem-standard
IBM Power10 high-use scale-up virtual persistent memory volume gigabyte-hour	power-iaas-metric-p10-su-vpmem-highuse
IBM Power10 standard scale-up virtual persistent memory volume gigabyte-hour	power-iaas-metric-p10-su-vpmem-standard
SAP workload - IBM Power E1150 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-E1150-vpmem
SAP workload - IBM Power E1180 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-E1180-vpmem
SAP workload - IBM Power S1122 virtual persistent memory volume gigabyte-hour	power-iaas-metric-sap-S1122-vpmem
SAP application server workload - IBM Power S1122 virtual persistent memory volume gigabyte-hour	power-iaas-metric-netweaver-S1122-vpmem
IBM Power11 high-use scale-out virtual persistent memory volume gigabyte-hour	power-iaas-metric-p11-so-vpmem-highuse
IBM Power11 standard scale-out virtual persistent memory volume gigabyte-hour	power-iaas-metric-p11-so-vpmem-standard
IBM Power11 high-use scale-up virtual persistent memory volume gigabyte-hour	power-iaas-metric-p11-su-vpmem-highuse
IBM Power11 standard scale-up virtual persistent memory volume gigabyte-hour	power-iaas-metric-p11-su-vpmem-standard

Part number descriptions for IBM Power Virtual Server.

Base instances

The base instance billing depends on your virtual instance options when you create a Power Virtual Server. The machine type, number of cores, and amount of memory all affect the base instance billing. When you create your VSI, the associated monthly rate is displayed. For more information, see [Creating a Power Virtual Server](#).

The pricing for memory and for vPMEM memory is calculated based on a ratio of 64 GB per core. For example, if you use more than 16 GB for 0.25 cores, you must pay a premium high-use RAM price for the excess memory. However, if you use up to 128 GB for 2 cores, you do not have to pay any premium memory price.

Operating systems

The Power Virtual Server pricing for AIX and IBM i includes license and IBM software maintenance.

Power Virtual Server provides AIX and IBM i stock images. The operating system version levels of the stock images are subject to change.

You can bring your own custom AIX or IBM i image to use on a Power Virtual Server instance, but you must purchase an operating system license for virtual server resources. The pricing for AIX and IBM i operating system license is not based on whether you use a custom image or a stock image.

You can select an IBM Power Virtual Server provided Linux® stock image or bring your own Red Hat Linux Enterprise (RHEL) and SUSE Linux Enterprise Server (SLES) image in OVA format. For a Linux subscription, you can opt to use a [full Linux® subscription](#) for Power Virtual Server or obtain the subscription for the Linux operating system directly from the vendor. For more information about how to create an OVA format Linux image, see [Creating a custom Linux image in OVA format](#).

If you bring your own image, you are charged for the image size and the storage tier that you use for the image. The cost of the storage unit (per GB) for stored boot images is the same as the cost of the storage tier (Tier 0 or Tier 3) in which your boot disks are deployed. To generate an estimated price, use the [Power Virtual Server Estimate pricing](#) tool. For more information, see [Deploying a custom image within IBM Power Virtual Server](#).

Linux for SAP workloads

You can deploy the following types of SAP workloads as VSIs:

- **SAP NetWeaver:** For SAP NetWeaver for Linux, the charges for hardware, processors, and memory are like other types of Linux deployments.

However, the SAP NetWeaver for Linux requires a different type of operating system and license to be deployed. So, the charges for these types of deployments are not the same as that of Red Hat Enterprise Linux for SAP (non-SAP OS). SAP NetWeaver can be deployed on S1022, E1080, S922, and E980 systems.

- **SAP HANA:** For SAP HANA, the billing parts for processor and memory are distinct. The charges on these distinct parts appear on the monthly invoices. The following types of systems, processors, and memory are supported for SAP HANA:
 - System types: S1022 (for workloads less than 2 TB of RAM), E1080 (for workloads more than 2 TB of RAM), or E980.
 - Virtual processor core type: dedicated
 - Memory types: scale-out or scale-up

For SAP HANA workloads, the charges for processor and memory parts are the same as compared to non-SAP HANA workloads.

 **Note:** The pricing is subject to change depending on the SAP HANA operational costs.

You can also bring your own SAP (HANA or NetWeaver) image with your own subscription. For more information, see [Deploying a Linux for SAP \(HANA or NetWeaver\) custom image](#).

Processor types

You are charged different rates based on the processor type that you choose for your VSI. **Dedicated processors** are priced the highest as they provide the best overall performance. **Shared capped processors** cost slightly more than **shared uncapped processors** because of their flexibility in addressing licensing restrictions. The processors are all charged on an hourly basis according to the machine type, processor type, and the number of cores that are used in a month.

Processor cores are charged at different hourly rates based on the core type (**Dedicated**, **Shared uncapped**, or **Shared capped**) and the machine type (S922, E1080, S1022, E1080, and S1122). For information on different processor type functions, see [What's the difference between shared capped and shared uncapped processor performance? How are they compared with dedicated processor performance?](#)

 **Important:** All prices that are mentioned in the topic are illustrative and do not represent the actual amounts that are used for billing. To generate an estimated price, use the [Power Virtual Server cost estimator](#) tool. For more information, see [Generating an estimate](#).

The following examples show the cost per system for general purpose VSIs in North and South America regions:

Number of cores (S922)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.51 (dedicated)	\$368.91
1	\$0.13 (shared uncapped)	\$92.33
1	\$0.19 (shared capped)	\$138.38
S922 processor type pricing example		
Number of cores (E980)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$1.77 (dedicated)	\$1291.28
1	\$0.44 (shared uncapped)	\$322.84
1	\$0.66 (shared capped)	\$484.26
E980 processor type pricing example		
Number of cores (S1022)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.59 (dedicated)	\$424.25
1	\$0.15 (shared uncapped)	\$106.06
1	\$0.22 (shared capped)	\$159.14
S1022 processor type pricing example		

Number of cores (S1122)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.62 (dedicated)	\$454.37
1	\$0.16 (shared uncapped)	\$113.59
1	\$0.23 (shared capped)	\$170.44

S1122 processor type pricing example

Pricing for Global Replication Services

The cost for the Global Replication Services (GRS) depends on the location where the replication-enabled volumes are active. When you create a replication-enabled volume, it is active in the location where it was created. When a failover event occurs, the auxiliary volume becomes the active volume.

The following costs are calculated for replication-enabled volumes:

- The cost of the basic services for the replication capability is based on the size of the replication-enabled volume under the part number `GLOBAL_REPLICATION_STORAGE_GIGABYTE_HOURS`.
- The cost of the storage is equal to two times the size of the volumes against two different storage part numbers. This cost is based on the tier of the volume against the primary volume.

Part numbers are used for calculating the cost of GRS based on the storage tier that is associated with the primary volume.


Note: The cost of the auxiliary volumes is not calculated separately and are part of the cost of the primary volume.

The cost of GRS is calculated for only one location at any point in time. The location where the volumes are not active does not incur any cost. If replication is broken among the primary or auxiliary volumes, the cost of the volumes is the same as the normal volumes on both primary and secondary locations. The cost is calculated based on the size of the volume according to the tier and no cost is considered for replication capability.

Pricing for shared processor pool

Shared processor pool (SPP) provides the capability to manage CPU cores efficiently. The number of cores that are assigned to an SPP depends on the entitled cores (EC) and virtual cores (VC) ratio. The EC:VP ratio varies by system type. For example, POWER10 systems allow different ratios as compared to POWER9 systems. For more information, see [Managing the shared processor pool](#).

SPP metering is optimized to improve the Total Cost of Ownership (TCO) for AIX and IBM i software licensing and disaster recovery (DR) scenarios. Virtual servers that are configured within an SPP do not incur additional cost per VSI core or the cost for high-use memory parts.

When you use SPPs, you pay for the maximum capacity of the SPP reserved cores. If a VM is assigned to the SPP, you are not billed for the cores that the VM uses. The cost of the cores that the VM uses are covered in the cost of the main core for the SPP.


Note: VMs might not use the reserved cores but they are considered for cost calculation because the reserved cores in SPP are exclusively reserved for your use.

If a VM is assigned to the SPP, additional resources that are used by a VM are billed with the cost of VM. Additional resources can include memory and license for operating systems or other workloads that are deployed in the VM.

Pricing for dedicated hosts

Dedicated hosts are priced based on the host type. The available dedicated hosts are IBM Power S922, IBM Power S1022, and IBM Power S1122. Each server type is metered by the hour and the price includes the entire capacity of the host.

Consider the following points for dedicated host pricing:

- You are not charged separately for SPPs that you deploy to the dedicated host.
- Software charges for the supported operating systems are metered and charged by the core.

To learn more about the dedicated host, see: [dedicated host](#).

Storage types

The Power Virtual Server charges based on three different storage types:

- **Data volumes** are the simplest form of volume that you create. You are billed based on the current volume capacity at the metering time. The following table shows an example of how you are billed based on your volume creation:

Volume size that you create	Volume size that you are billed
10 GB	10 GB
10+5 GB	15 GB

Calculation of data volume

- **Image backing volumes** are part of a boot image in your cloud-instance boot image catalog. You are billed based on the total volume sizes that are contained in the image. When the image has a single backing volume, you are billed based on the GB size of the single volume. When the image has multiple backing volumes, you are billed based on tallying up the sizes of all the image backing volumes. The following table shows an example of how you are billed based on your boot volume:

Image volume size	Single or multiple backing	You are billed
20 GB	Single backing volume	20 GB
Volume 1 (20 GB), volume 2 (10 GB)	Multiple backing volumes	30 GB

Calculation of image backing volume

- **Deployed VSI volumes** are created when you deploy a VSI with an image. The deployed VSIs get a copy of all the volumes in the image. Any additional data volumes attached to the deployed VSI are already accounted for under Data Volumes. The following table shows an example of how you are billed based on the VSIs that you deploy:

Image backing volume	You are billed
20 GB	20 GB
20 GB + 30 GB	50 GB

Calculation of deployed VSIs volume

- **Deployed VSI snapshots** are the snapshots of the volumes that are taken after the VSI is deployed. The size of the volume snapshot is related to the number of updates that are made to the VSI. When you take a snapshot for the first time, the size of the snapshot is a fraction of the size of one or more volumes of the VSI. The size of subsequent snapshots might increase based on the changes that are made to the original volume.

For an example, consider a VSI with a volume of 100 GB. The size of the first snapshot is 100 GB. The size of the second snapshot might be 1 GB.

**Note:** Snapshot sizes are not predictable as they are related to the updates made to the volume between two snapshots.

The following tables show examples of how different storage types affect the cost per system:

Volume storage	Hourly rate (Processor type)	Monthly cost (730 hours)
Tier 0	\$0.00029589	\$0.22
Tier 1	\$0.00024658	\$0.18
Tier 3	\$0.00012884	\$0.09
Tier 5k	\$0.00035507	\$0.26

Volume storage pricing example

Snapshot storage	Hourly rate (Processor type)	Monthly cost (730 hours)
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Tier 0	\$0.00008877	\$0.06
Tier 1	\$0.00007397	\$0.05
Tier 3	\$0.00003865	\$0.03
Tier 5k	\$0.00010652	\$0.08

Snapshot storage pricing example

- **IBM i software tier**: Starting with IBM Power10 systems and later, you can assign IBM i software tier to a VSI. The IBM i software tier limits the size of the VSI, but not the physical system it runs on. For example, a virtual server that is assigned to a P10 tier can run on either an S1022 or an E1080 server. The tier restricts the resource allocation for the VSI, not its hardware compatibility.

Use case of account billable storage

The following table shows the use case on how you are billed based on the storage that you use (assuming tier 1):

Name	Size	State
data-volume-1	20 GB	Available
data-volume-2	25 GB	In-use (attached to vm-1)
data-volume-3	100 GB	In-use (attached to vm-1)
data-volume-4	30 GB	Available
data-volume-5	60 GB	In-use (attached to vm-2)

Account billable for storage use case

Name	Size	Description
AIX-71-01	30 GB	1 backing volume
IBMi-74-001	100 GB + 30 GB	2 backing volumes
SLES-15-1	40 GB	1 backing volume

Account billable for storage use case

Name	Size	Description
vm-1 deployed AIX-71	30 GB	Volume of AIX-71 + data-volume-2 + data-volume-3 (volumes that are created from copying the deployed AIX-71 image, Data volumes are already accounted.)
vm-2 deployed IBMi-74-001	130 GB	Volume of IBMi-74-001 + data-volume-5 (volumes that are created from copying the deployed IBMi-74-001 image, Data volumes are already accounted.)

Account billable for storage use case

Total billable storage = 595 GB

- Data volumes: 235 GB
- Image volumes: 200 GB
- Deployed VSIs: 160 GB

You can generate an estimate of the resources for IBM Power Virtual Server. The estimated cost might differ from the actual cost when you purchase the infrastructure or instances due to discounts and promotion codes. For more information, see [Generating an estimate for IBM Power Virtual Server](#).

For more information about IBM Cloud regions that can host connections from the pods for IBM Power Virtual Server Private Cloud, see [IBM Satellite location](#).

Monthly usage

IBM Power Virtual Server Private Cloud instances are charged at a monthly rate that is prorated per hour. If you add resources to a virtual machine in the middle of the month, the bill for the virtual machine reflects the resource change at a per-hour prorated price.

**Important:** All prices that are mentioned in the topic are illustrative and do not represent the actual amounts that are used for billing. To generate an estimated price, use the [Power Virtual Server Estimate pricing](#) tool. For more information, see [Generating an estimate](#).

In the following example, the customer provisions an IBM Power Virtual Server Private Cloud instance. The instance has 1 core with 8 GB of memory, a 150 GB disk, and is running Red Hat Enterprise Linux (RHEL) operating system (OS). Assume the following hypothetical monthly prices:

- Cores: \$143.23 per month x 1 = \$143.23
- Memory: \$9 per month each GB x 8 = \$ 72.00
- Storage: \$0.216 per month each GB x 150 GB = \$ 32.40
- OS license: \$43.80 per month each core = \$ 43.80

Total cost per month = \$291.43

Factoring the example quantities and prices, the base price for the virtual machine is \$291.43 per month or an average of \$0.399 per hour. The base price is calculated for all the resources that are used. Assume that during the month, the customer allocates more memory to the virtual machine. At the end of the month, the new price for the virtual machine is higher to account for the extra memory. The new price amounts to \$0.498 per hour for all the resources that are used. The monthly bill is prorated by an hour for the resources that are deployed as shown in the following table.

In the table, the virtual machine resources are increased (after the resources reach 300 hours in the month) from 8 GB to 16 GB of memory. The price of the virtual machine is prorated by an hour for the final monthly price of \$303.

Hours elapsed in a month	Amount charged	Virtual machine description
300 hours	(300 hours x \$0.399)/month = \$119.70	1 core, 8 GB memory, 150 GB disk, RHEL
430 hours	(430 hours x \$0.498)/month = \$214.14	1 core, 16 GB memory, 150 GB disk, RHEL
730 hours (Monthly Total)	\$119.70 + \$214.14 = \$333.84 (Monthly Total)	1 core, 16 GB memory, 150 GB disk, RHEL

An example of monthly charges for a virtual machine

For the example, in this table you can see four distinct part numbers for the following resources in the invoice:

- The virtual cores (one for each core)
- The memory
- The volume storage (disk)
- The OS license (RHEL)

Part number descriptions

A part number is associated with the license for the software product to be used in IBM Cloud. For detailed usage and billing information, refer to the part number descriptions. The part number descriptions are listed in your invoice or in the [IBM® Cloud billing](#) portal.

Refer to the following table to view the part number descriptions and the associated metric IDs.

Part description (visible on the invoice from IBM)	Metric ID (visible in the IBM Cloud catalog)
IBM Power Systems Virtual Server virtual machine group	
IBM Power Systems E1180 virtual processor core hour - capped	ppcaas-metric-E1180-cores-capped

IBM Power Systems E1180 virtual processor core hour - dedicated	ppcaas-metric-E1180-cores-dedicated
IBM Power Systems E1180 virtual processor core hour - shared	ppcaas-metric-E1180-cores-shared
IBM Power Systems E1150 virtual processor core hour - capped	ppcaas-metric-E1150-cores-capped
IBM Power Systems E1150 virtual processor core hour - dedicated	ppcaas-metric-E1150-cores-dedicated
IBM Power Systems E1150 virtual processor core hour - shared	ppcaas-metric-E1150-cores-shared
IBM Power Systems S1122 virtual processor core hour - capped	ppcaas-metric-S1122-cores-capped
IBM Power Systems S1122 virtual processor core hour - dedicated	ppcaas-metric-S1122-cores-dedicated
IBM Power Systems S1122 virtual processor core hour - shared	ppcaas-metric-S1122-cores-shared
IBM Power Systems P11 scale-out memory gigabyte hours	ppcaas-metric-p11-2u-memory-standard
IBM Power Systems P11 scale-up memory gigabyte hours	ppcaas-metric-p11-4u-memory-standard
AIX operating system scale-up license core-hour	ppcaas-metric-aix-scale-up
AIX operating system scale-out license core-hour	ppcaas-metric-aix-scale-out
Red Hat Enterprise Linux operating system scale-up license core-hour	ppcaas-metric-rhel-scale-up
Red Hat Enterprise Linux operating system scale-out license core-hour	ppcaas-metric-rhel-scale-out
Red Hat Enterprise Linux operating system scale-up license core-hour - SAP workload	ppcaas-metric-rhel-sap-scale-up
Red Hat Enterprise Linux operating system scale-out license core-hour - SAP workload	ppcaas-metric-rhel-sap-scale-out
SUSE Linux Enterprise Server operating system tier 1 instance-hour	ppcaas-metric-sles-tier1
SUSE Linux Enterprise Server operating system tier 2 instance-hour	ppcaas-metric-sles-tier2
SUSE Linux Enterprise Server operating system tier 3 instance-hour	ppcaas-metric-sles-tier3
SUSE Linux Enterprise Server operating system tier 1 instance-hour - SAP workload	ppcaas-metric-sles-sap-tier1
SUSE Linux Enterprise Server operating system tier 2 instance-hour - SAP workload	ppcaas-metric-sles-sap-tier2
SUSE Linux Enterprise Server operating system tier 3 instance-hour - SAP workload	ppcaas-metric-sles-sap-tier3
IBM i P10 license core-hour	ppcaas-metric-ibmi-os-p10
IBM i P30 license core-hour	ppcaas-metric-ibmi-os-p30
IBM i P10 service extension core-hour	ppcaas-metric-ibmi-os-p10-sve
IBM i P30 service extension core-hour	ppcaas-metric-ibmi-os-p30-sve
IBM i LPP core-hour	ppcaas-metric-ibmi-lpp
IBM i P10 LPP core-hour	ppcaas-metric-ibmi-lpp-p10

IBM i P30 LPP core-hour	ppcaas-metric-ibmi-lpp-p30
IBM i P10 PowerHA instance core-hour	ppcaas-metric-ibmi-pha-p10
IBM i P30 PowerHA instance core-hour	ppcaas-metric-ibmi-pha-p30
IBM i PowerHA small instance core-hour	ppcaas-metric-ibmi-pha-small
IBM i PowerHA medium instance core-hour	ppcaas-metric-ibmi-pha-medium
IBM i P10 license - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-os-p10-mol
IBM i P30 license - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-os-p30-mol
IBM i LPP P10 - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-lpp-p10-mol
IBM i LPP P30 - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-lpp-p30-mol
IBM i LPP core-hour - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-lpp-mol
IBM i Db2 Web Query instance core-hour	ppcaas-metric-ibmi-db2wq
IBM i Cloud Storage Solutions instance core-hour	ppcaas-metric-ibmi-cos
IBM i Rational Development Studio instance-hour	ppcaas-metric-ibmi-rds
IBM i P05 software tier license core-hour	ppcaas-metric-ibmi-os-p05-tier
IBM i P10 software tier license core-hour	ppcaas-metric-ibmi-os-p10-tier
IBM i P20 software tier license core-hour	ppcaas-metric-ibmi-os-p20-tier
IBM i P30 software tier license core-hour	ppcaas-metric-ibmi-os-p30-tier
IBM i P05 software tier service extension core-hour	ppcaas-metric-ibmi-os-p05-sve-tier
IBM i P10 software tier service extension core-hour	ppcaas-metric-ibmi-os-p10-sve-tier
IBM i P20 software tier service extension core-hour	ppcaas-metric-ibmi-os-p20-sve-tier
IBM i P30 software tier service extension core-hour	ppcaas-metric-ibmi-os-p30-sve-tier
IBM i software tier LPP core-hour	ppcaas-metric-ibmi-lpp-tier
IBM i software tier PowerHA small instance core-hour	ppcaas-metric-ibmi-pha-small-tier
IBM i software tier PowerHA medium instance core-hour	ppcaas-metric-ibmi-pha-medium-tier
IBM i P10 software tier license - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-os-p10-mol-tier
IBM i P30 software tier license - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-os-p30-mol-tier
IBM i software tier LPP - mobile core-hour - SWMA paid	ppcaas-metric-ibmi-lpp-mol-tier
Virtual Tape Library terabyte-hour	ppcaas-metric-vtl

Part number descriptions for IBM Power Virtual Server Private Cloud in Client location.

Part description (visible on the invoice from IBM)	Metric ID (visible in the IBM Cloud catalog)
IBM Power Systems Virtual Server Private Cloud volume group	
Volume Storage Tier 0 gigabyte-hour	ppcaas-metric-volume-tier0
Volume Storage Tier 1 gigabyte-hour	ppcaas-metric-volume-tier1
Volume Storage Tier 3 gigabyte-hour	ppcaas-metric-volume-tier3
Volume Storage Tier 5k gigabyte-hour	ppcaas-metric-volume-tier5k
Asynchronous replicated volume storage (tier 0: 25 IOPS per GB) gigabyte-hour	ppcaas-metric-volume-tier0-async-replicated
Asynchronous replicated volume storage (tier 1: 10 IOPS per GB) gigabyte-hour	ppcaas-metric-volume-tier1-async-replicated
Asynchronous replicated volume storage (tier 3: 3 IOPS per GB) gigabyte-hour	ppcaas-metric-volume-tier3-async-replicated
Asynchronous replicated volume storage (tier 5k: 5,000 IOPS) gigabyte-hour	ppcaas-metric-volume-tier5k-async-replicated
Asynchronous global replication storage gigabyte-hour	ppcaas-metric-volume-async-replication-service

Part number descriptions for IBM Power Virtual Server Private Cloud in Client location.

Part description (visible on the invoice from IBM)	Metric ID (visible in the IBM Cloud catalog)
IBM Power Systems Virtual Server Private Cloud snapshot group	
Snapshot Storage Tier 0 gigabyte-hour	ppcaas-metric-snapshot-tier0
Snapshot Storage Tier 1 gigabyte-hour	ppcaas-metric-snapshot-tier1
Snapshot Storage Tier 3 gigabyte-hour	ppcaas-metric-snapshot-tier3
Snapshot Storage Tier 5k gigabyte-hour	ppcaas-metric-snapshot-tier5k

Part number descriptions for IBM Power Virtual Server Private Cloud in Client location.

Part description (visible on the invoice from IBM)	Metric ID (visible in the IBM Cloud catalog)
IBM Power Systems Virtual Server Private Cloud shared processor pool group	
IBM Power Systems S1122 virtual processor core-hour - Shared Processor Pool	ppcaas-metric-S1122-spp-cores
IBM Power Systems E1150 virtual processor core-hour - Shared Processor Pool	ppcaas-metric-E1150-spp-cores
IBM Power Systems E1180 virtual processor core-hour - Shared Processor Pool	ppcaas-metric-E1180-spp-cores

Part number descriptions for IBM Power Virtual Server Private Cloud in Client location.

For more information about unit prices for each metric ID, see [Where can I find the unit prices for the billing metrics?](#).

Base instances

The billing of a base instance depends on the options that you select when you create a virtual machine. The machine type, number of cores, and amount of memory affect the base instance billing. When you create your virtual machine, the associated monthly rate is displayed on the billing of the base instances.

The pricing for memory is calculated based on a ratio of 64 GB per core. For example, if you use more than 16 GB for 0.25 cores, you must pay a premium high-use RAM price for the excess memory. However, if you use up to 128 GB for 2 cores, you do not have to pay any premium memory price.

Operating systems

Linux and AIX operating systems are supported and only RHEL stock images are available. For more information about supported versions and distributions, see [Full Linux subscription for Power Virtual Server in Client location](#). When you select a RHEL Linux stock image, the pricing includes the full Linux subscription charges. By using the Full Linux subscription, you can subscribe to an updated or upgraded package from a Red Hat Satellite server. To subscribe, configure the provisioned virtual machine. The usage meter for RHEL full Linux subscription starts when you create the virtual machine. If you bring your own license, it is not metered or billed.

If you bring your own custom image, you are charged for the image size and the storage tier that you use for the custom image. After you deploy a stock image (and only after deployment), you are charged for the space that is used to store the stock image. The storage unit price (per GB) for the stored boot images is the same as the selected storage tier (Tier 0, Tier 1, or Tier 3) where your boot disks are deployed. To estimate the storage rates, use the [Power Virtual Server Estimate pricing](#) tool. To reduce costs, select the virtual machine that is not needed and delete it.

Processor types

Depending on the type of system that you choose for your virtual machine, you are charged different rates. Enterprise systems, such as E1180 or E1150, cost more because they provide more processing capacity per core. For example, the E1180 costs more than the E1150.

You can choose one of the following core types for your workload:

- **Dedicated virtual processor cores**
- **Shared capped virtual processor cores**
- **Shared uncapped virtual processor cores**

Different scenarios provide different benefits when you use each type of virtual processor core.

**Important:** All prices that are mentioned in the topic are illustrative and do not represent the actual amounts that are used for billing. To generate an estimated price, use the [Power Virtual Server Estimate pricing](#) tool. For more information, see [Generating an estimate](#).

In the following example, assume that the cost of different types of virtual processor cores is the same within each system type they belong to.

The following tables show how different processor types affect the cost of each system:

Number of cores (S1122)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.1962 (dedicated)	\$143.23
1	\$0.1962 (shared uncapped)	\$143.23
1	\$0.1962 (shared capped)	\$143.23
S1122 processor type pricing		
Number of cores (E1180)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.6866 (dedicated)	\$501.20
1	\$0.6866 (shared uncapped)	\$501.20
1	\$0.6866 (shared capped)	\$501.20
E1180 processor type pricing		
Number of cores (E1150)	Hourly rate (Processor type)	Monthly cost (730 hours)
1	\$0.2945 (dedicated)	\$215.00
1	\$0.2945 (shared uncapped)	\$215.00
1	\$0.2945 (shared capped)	\$215.00
E1150 processor type pricing		

Memory types

The system determines the cost of IBM Power Virtual Server Private Cloud based on the memory usage (in gigabytes) and the type of the system that is chosen for the virtual machine. Enterprise systems, such as E1180 or E1150, use scale-up memory with higher density, whereas S1122 systems use scale-up memory with lesser density. Currently, both scale-up and scale-out memory might cost you the same price. However, this pricing is subject to change in the future, depending on the cost of the memory chips. The following table shows how memory types affect the cost of each system:

Number of GBs	Memory type	Hourly rate (Processor type)	Monthly cost (730 hours)
1	IBM Power Systems scale-out memory	\$0.01232877	\$9.00
1	IBM Power Systems scale-up memory	\$0.01232877	\$9.00

Memory type pricing

Pricing for shared processor pool

Shared Processor Pool (SPP) metering is optimized to improve the Total Cost of Ownership (TCO) for AIX and IBM i software licensing and DR scenarios. Virtual servers that are configured within an SPP do not incur additional costs for each VM core.

In the IBM Power Virtual Server Private Cloud environment, a minimum core-to-virtual core ratio of 1:20 exists. This ratio stipulates the relationship between physical cores and virtual cores within the cloud infrastructure. Therefore, the calculation of the operating system license charges follows a distinct method within the IBM Power Virtual Server Private Cloud setup to accommodate the core-to-virtual core ratio efficiently.

**Note:** The create or update operations of a VSI fails even if the core-to-virtual core ratio of 1:20 is satisfied but the configured count of vCPU exceeds the maximum supported per-instance limit as defined by the system policy.

When you use SPPs, you pay for the maximum capacity of the SPP reserved cores. If a VM is assigned to the SPP, you are not billed for the cores that the VM uses. The cost of the cores that the VM uses are covered in the cost of the main core for the SPP.

**Note:** VMs might not use the reserved cores but they are considered for cost calculation because the reserved cores in SPP are exclusively reserved for your use.

If a VM is assigned to the SPP, additional resources that are used by a VM are billed with the cost of VM. Additional resources can include memory and license for operating systems or other workloads that are deployed in the VM.

When you use SPP in IBM Power Virtual Server Private Cloud, you pay for the following items:

- The maximum capacity of the SPP reserved cores that use the shared capped part number.
- The entitled capacity of the shared capped or uncapped part numbers when virtual server instance cores are deployed into the SPP. This amount is variable and depends on the entitled capacity.
- The operating system license, which is based on the following types of virtual server instances:
 - Capped SPP: pricing is calculated based on the entitled capacity.
 - Uncapped SPP: pricing is calculated based on the minimum value between the total number of virtual processors and the maximum capacity of the SPP. To get the total number of virtual processors, count the number of virtual processors that are associated with the partitions in an SPP. The partitions must be considered for each type of operating system such as AIX or IBM i.

Following table shows the details core pricing and OS license pricing of SPP in IBM Power Virtual Server Private Cloud.

Solution	Core-to-virtual core ratio	Core Pricing	OS license pricing
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IBM Power Virtual Server Private Cloud User-defined Pools	1:20	1. SPP capacity at shared capped processor rate 2. VM cores charged at EC (capped or uncapped) 3. Existing behavior – no change	Per OS type: Minimum of (sum of VP of VMs in pool or max capacity of pool); only for uncapped mode For IBM Power Virtual Server Private Cloud, these OS license charges are associated with the respective VM proportionate to VPs of that VM
Default Pool	1:20	1. No charge at SPP level 2. VM cores charged at EC (capped or uncapped) 3. Existing behavior – no change	

Shared processor pool for IBM Power Virtual Server Private Cloud

For more information, see [How to calculate the pricing for OS licensing in SPP](#).

Storage types

IBM Power Virtual Server Private Cloud charges are based on the following storage types:

- **Data volumes:** These volumes are basic forms that you can create. The billing depends on the volume capacity when metered. The following table shows an example billing based on your volume creation:

Volume size that you create	Volume size that you are billed
10 GB	10 GB
10+5 GB	15 GB

Calculation of data volume

- **Image backing volumes:** These volumes are part of a boot image in your cloud instance that is listed in the boot image catalog. Billing for image backing volumes is determined as follows:
 - Single volume billing - When the image has a single backing volume, you are billed based on the size (GB) of the single volume.
 - Multiple volume billing - When the image has multiple backing volumes, billing is based on tallying up the sizes of all the image backing volumes.

The following table shows an example of how you are billed based on your boot volume:

Image volume size	Single or multiple backing	Billed volume size
20 GB	Single backing volume	20 GB
Volume 1 (20 GB), volume 2 (10 GB)	Multiple backing volumes	30 GB

Calculation of image backing volume

- **Deployed virtual machine volumes:** When you deploy a virtual machine with an image, it gets a copy of all the volumes in the image. Extra data volumes added to the deployed virtual machine are billed separately as Data Volumes. The following table shows an example of the billing based on the virtual machines that you deploy:

Image backing volume	Billed volume size
20 GB	20 GB

20 GB + 30 GB	50 GB
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Calculation of deployed virtual machines volume

- **Deployed virtual machine snapshots** : The snapshots of the volumes are taken after the virtual machine is deployed. The size of the volume snapshot is related to the number of updates that are made to the virtual machine. When you take a snapshot for the first time, the size of the snapshot is a fraction of the size of the volumes of the virtual machine. The size of subsequent snapshots might increase based on the changes that are made to the original volume.

For an example, consider a virtual machine with a volume of 100 GB. The size of the first snapshot is 100 GB. The size of the second snapshot might be 1 GB.

 **Note:** Snapshot sizes are not predictable as they are related to the updates made to the volume between two snapshots.

The following tables show how different storage types affect the cost per system:

Volume storage	Hourly rate (Processor type)	Monthly cost (730 hours)
Tier 0	\$0.00029589	\$0.22
Tier 1	\$0.00024658	\$0.18
Tier 3	\$0.00012884	\$0.09
Tier 5k	\$0.00035507	\$0.26

Volume storage pricing

Snapshot storage	Hourly rate (Processor type)	Monthly cost (730 hours)
Tier 0	\$0.00008877	\$0.06
Tier 1	\$0.00007397	\$0.05
Tier 3	\$0.00003865	\$0.03
Tier 5k	\$0.00010652	\$0.08

Snapshot storage pricing

- **IBM i software tier** : Starting with IBM Power10 systems and later, you can assign IBM i software tier to a virtual server instance (VSI). The IBM i software tier limits the size of the VSI, but not the physical system it runs on. For example, a virtual server that is assigned to a P10 tier can run on either an S1022 or an E1080 server. The tier restricts the resource allocation for the VSI, not its hardware compatibility.

Use case of account billable storage

The following table shows the use case on how you are billed based on the storage that you use (assuming tier 1):

Name	Size	State or Description
data-volume-1	20 GB	Available
data-volume-2	25 GB	In-use (attached to vm-1)
data-volume-3	100 GB	In-use (attached to vm-1)
data-volume-4	30 GB	Available
data-volume-5	60 GB	In-use (attached to vm-2)
image-volume-1	100 GB	In-use (attached to vm-1)

image-volume-2	100 GB	In-use (attached to vm-2)
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Account billable for storage use case

Total billable storage = 595 GB

- Data volumes: 235 GB
- Image volumes: 200 GB
- Deployed virtual machine boot volumes: 200 GB

To generate an estimated price, use the [Power Virtual Server Estimate pricing](#) tool. For more information, see the following links:

- [Creating an estimate for Power Virtual Server resources in an IBM data center](#).
- [Creating an estimate for Power Virtual Server resources in a client location](#).

End of billing

The monthly billing cycle ends when you delete the virtual machine. If you scale your infrastructure up and down in response to workload requirements, your billing follows the timing of the virtual machine provision change. If you stop the virtual machine, the billing process is not stopped. Delete the virtual machine to stop the billing cycle.

**Important:** When your virtual machine is active but idle, you can use Dynamic Logical Partitioning (DLPAR) to resize it to a minimal state. You can drastically decrease the price per hour by reducing the core count and memory of the virtual machine.

Frequently asked questions

Review the following frequently asked questions about pricing:

Where do I find cost estimates for Power Virtual Server offerings?

You can generate an estimated price by using the [Power Virtual Server Estimate pricing](#) tool. For more information, see the following links:

- [Creating an estimate for Power Virtual Server resources in an IBM data center](#).
- [Creating an estimate for Power Virtual Server resources in a client location](#).

By default, the cost estimator shows hourly prices for the selected resources and their units of measure. For more information, see [Estimating your costs](#).

Where can I find the unit prices for the billing metrics?

You can view the unit prices for the billing metrics for IBM Power Virtual Server Private Cloud on the [IBM global catalog](#).

On the global catalog, complete the following steps to view the pricing for each metric ID:

- Log on to [IBM global catalog](#).
- Search the catalog for the resources for which you need pricing. The following options for search strings are available:
 - **Power Virtual Server Virtual Machine** for virtual machines that includes processor, memory, operating systems, and workloads.
 - **Power Virtual Server Volume** for Volumes.
 - **Power Virtual Server Snapshot** for Snapshots.
 - **Power Virtual Server Shared Processor Pool** for Shared Processor Pools. No pricing is associated with other types of resources.
- Click the selected resource type entry from the search results. The corresponding page opens.
- Click the Power Virtual Server Private Cloud billing plan from the left page:
 - **Power Virtual Server Private Cloud Virtual Machine Group** for virtual machines.
 - **Power Virtual Server Private Cloud Volume Group** for Volumes.
 - **Power Virtual Server Private Cloud Snapshot Group** for Snapshots.
 - **Power Virtual Server Private Cloud Shared Processor Pool** for Shared Processor Pools (SPP). Deployments with different satellite regions are displayed.
- Select the deployment with the name that ends with the satellite region for which you want the pricing. For example, the **Power Virtual Server Private Cloud PVM Instance Groupsatcon_dal** deployment displays the pricing for virtual machines in the satellite locations that are hosted in

the Dallas region.

- Click the **Pricing** tab on the right navigation page to view the pricings for different **Metric IDs**. Refer to the table with [Part definitions and metric IDs](#) to find the metric ID corresponding to the part description. Prices are listed for all IBM Cloud supported currencies.

How to calculate the pricing for OS licensing in the uncapped SPP?

To calculate the price for OS licensing in the uncapped SPP, consider the following conditions:

- If the total number of VPs associated with the partitions in an SPP is greater than the maximum capacity of the pool, then the cost is distributed proportionately for all VPs.
- If the total number of VPs associated with the partitions in an SPP is less than or equal to the maximum capacity of the pool, then the pricing is considered for each VP.

The following table shows how the pricing for OS licensing is calculated considering that the maximum pool capacity is 8 cores :

Example	VM	VP on VM	Total number of VPs versus Maximum pool capacity	Cost calculation
1	AIX VM1	4	6 < 8 (6 is considered)	4 x cost of AIX OS license
	AIX VM2	2		2 x cost of AIX OS license
2	AIX VM1	4	10 > 8 (8 is considered)	3.2 x cost of AIX OS license
	AIX VM2	2		1.6 x cost of AIX OS license
	AIX VM3	4		3.2 x cost of AIX OS license

Pricing for OS licensing in SPP in IBM Power Virtual Server Private Cloud

Is there any initial one-time payment before the pod infrastructure is installed in the Private Cloud data center?

No

In terms of costs, what do I pay for?

The monthly consumption charges are factored based on the **minimum committed spend** and metered consumption charges. If the consumption charges are less than or equal to the **minimum committed spend** , you pay the **minimum committed spend** amount. If the consumption charges exceed the **minimum committed spend** , you pay the metered consumption charges.

Can I change infrastructure configuration based on my requirements after my virtual servers are provisioned? How will it impact billing?

Yes, you can add extra nodes but cannot reduce the number of nodes. The billing gets updated from the date that the new nodes are added.

When does the billing begin?

Billing begins when an IBM representative sets up and registers the pod in the IBM Power Virtual Server Private Cloud data center.

Creating an estimate for Power Virtual Server resources in an IBM data center

Before you deploy a Power Virtual Server with cloud resources, such as virtual server instances, storage volumes, shared processor pools, virtual tape libraries, or dedicated hosts, it is important that you create an estimate to understand the associated costs. You can use the [cost estimator](#) to estimate the cost of the IBM® Power® Virtual Server resources before you deploy them in an IBM data center or in a client location.

You can create an estimate by specifying the parameters for your Power Virtual Server resources. You can customize the resource quantities based on your business requirement and view the estimated cost. Based on your selection, you can view the estimated cost of your Power Virtual Server resources. You can connect with your IBM Business Partner or contact IBM to place your order.

**Note:** You are not charged for creating an estimate. The resources that you estimate can be deployed only after a workspace is created.

Estimating Power Virtual Server resources in an IBM data center

You can create an estimate for Power Virtual Server resources in an IBM data center where the infrastructure is hosted and managed by IBM. To create an estimate, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. Search for **Power Virtual Server** in the search box for IBM Cloud catalog.
3. Click the **Power Virtual Server** tile.
4. Click **Estimate Cost**. You are redirected to the **Estimate cost** page for Power Virtual Server.
5. Select **IBM data center** from the **Location type** list.
6. Select an IBM Cloud region from the **Location** list.
7. From the **Estimate resource** section, select one of the following resources to learn about the available options and fields for generating an estimate. Each section provides detailed guidance to help you configure the resources based on your business needs. After you review the relevant section, return here to continue with the estimation process.
 - [Estimating a virtual server instance](#)
 - [Estimating a Power Virtual Server instance with an IBM i software tier](#)
 - [Estimating SAP workloads](#)
 - [Estimating a storage volume](#)
 - [Estimating a shared processor pool](#)
 - [Estimating a virtual tape library](#)
 - [Estimating a dedicated host](#)
8. In the Summary panel, review the estimated cost and click **Add to estimate**. The **Estimate** panel is displayed.
9. In the **Save Estimate** panel, select **Create New**.
10. Enter a name for the estimate in the **Name** field.
11. Optional: Enter a description for the estimate in the **Description (optional)** field.
12. Click **Save estimate**. You are redirected to the Estimate panel.
13. Click **View Estimate** to open the estimate in a new tab. You can use the **Action** menu to export the estimate in XLSX, CSV, or PDF format.

Estimating a virtual server instance

The following table explains the fields that you can use to create an estimate for a virtual server instance in a Power Virtual Server workspace:

Field	Action
Number of virtual servers	Specify the number of virtual server instances that you want to estimate in a Power Virtual Server workspace.
Operating system	Select an operating system from the list that meets your workload requirements.
Configure for Epic workloads (AIX)	Select this checkbox if you want to deploy epic workloads on E980, E1080, or E1180 Power Systems with Tier 1 storage and dedicated cores, at a shared-capped price. When you select this indicator, the other dependent fields are automatically filled. To learn more about epic workloads, see Configuring a VM for Epic workloads .
Add to a shared processor pool	Select this checkbox if you want to use shared processor pool. A shared processor pool is a pool of processor capacity that is shared between a group of virtual server instances (VM).
Machine type	Select the required Power system from the list.

Core type	Select the core type. The available options are: ◦ Shared uncapped: Shared among other clients. ◦ Shared capped: Shared, but resources do not expand beyond those that are requested (used mostly for licensing). ◦ Dedicated: Resources are allocated for a specific client (used for specific third-party considerations).
Cores	Specify the number of cores that you need. The minimum cores that you can specify is 0.25 for shared capped and shared uncapped core types. For dedicated core type, the minimum cores that you can specify is 1.
Memory (GiB)	Specify the memory size. The minimum memory size that you can specify is 2 GiB.
Storage tiers	Each estimated virtual server instance must have an associated boot volume. Total boot volume storage is multiplied by the number of estimated instances. You can choose from Tier 0, Tier 1, Tier 3, or Fixed IOPs. You cannot add a separate boot volume estimation. Hence, you must define the storage volume value by considering the boot volumes and data volume that you might need.

Fields and corresponding actions for estimating a VSI in Power Virtual Server

You can perform the following actions based on the type of OS:

- **IBM i**: Assign an IBM i software tier to your instance to restrict the number of CPUs and the amount of memory that can be assigned to the instance. For more information, see [Estimating an IBM i workloads with IBM i software tier](#).
- **Linux for SAP (HANA)**: Select an SAP certified profile. For more information, see [Estimating SAP workloads](#).

Estimating a Power Virtual Server instance with an IBM i software tier

A virtual software tier limits the capacity of the virtual machines (VMs) for a particular pricing tier, but the tiers can run on any system. The IBM i software tier is used for the software pricing of operating system (OS) and IBM i Licensed Program Products (LPP). When you assign an IBM i software tier to an instance, the number of CPUs and the amount of memory that can be assigned to the instance is restricted by the IBM i software tier. For more information, see [Assigning an IBM i software tier to an IBM i Power Virtual Server instance](#).

You can create an estimate of the Power Virtual Server instance with an IBM i software tier before you deploy the instance. To assign an IBM i software tier to an IBM i Power Virtual Server instance, meet the following prerequisites:

- IBM i image version is 7.3 or later
- VSN is assigned to the instance
- Machine type is IBM Power10 or later

The following table explains the fields that you can use to create a Power Virtual Server instance with an IBM i software tier:

Field	Action
Operating system	Select IBM i from the Operating system list.
Version	Select the required IBM i version 7.3 or later from the Version list.
Virtual serial number (VSN)	Set Virtual serial number (VSN) to Assigned to allocate a VSN to the VSI. VSN allocation enables IBM i software tiering options and does not impact the cost of the VSI.
IBM i licenses	Select the appropriate IBM i licenses. The available options are: – IBM i Cloud Storage Solution – IBM i PowerHA – Rational Dev Studio for IBM i
Machine type	Select an IBM Power10 or later Power system from the list.
Cores	Specify the number of cores that you need. The minimum cores that you can specify is 0.25 for shared capped and shared uncapped core types. For dedicated core type, the minimum cores that you can select is 1.

Memory (GiB)	Specify the memory size. The minimum memory size that you can specify is 2 GiB.
IBM i software tier	Based on the machine type, a list of supported IBM i software tiers is displayed. A text message with the recommended IBM i software tier is displayed after the IBM i software tier field based on the number of cores and memory size you select. You can select the recommended IBM i software tier from the IBM i software tier list or other options from the list.

Fields and corresponding actions for estimating a Power Virtual Server instance with IBM i software tier

Estimating SAP workloads

The following table explains the fields that you can use to create an estimate for a Power Virtual Server instance with SAP workloads:

Field	Action
Operating system	- Select Linux for SAP (HANA) in the IBM provided subscription section to use the IBM provided Linux subscription. - Select Linux for SAP (HANA) in the Client supplied subscription section to use your own license.
Linux image type	Select the required Linux image type. Available options are SUSE and Red Hat Enterprise.
Machine type	- Select the required Power system from the list to deploy an SAP HANA profile. - Select an IBM Power10 or later machine type from the list to deploy an SAP certified profile.
Advance Configuration	Set SAP RISE deployment to on to estimate the cost of a Power Virtual Server instance with an SAP certified profile - Standard RISE or Application Server.
Profile	Select an SAP HANA, a Standard RISE, or an Application Server profile.

Fields and corresponding actions for estimating an SAP workload in Power Virtual Server

**Note:** The SAP certified profiles are available with IBM Power10 or later servers.

To learn more on how to create an instance, see [Configuring a Power Virtual Server instance](#).

Estimating a storage volume

The following table explains the fields that you can use to create an estimate for a storage volume in a Power Virtual Server workspace:

Field	Action
Number of volumes	Specify the number of volumes that you need. The minimum number of volumes that you can specify is 1.
Tier	Select Tier 0, Tier 1, Tier 3, or Fixed IOPs from the Tier list.
Total storage (GB)	Specify the total storage. The minimum total storage that you can specify is 1 GB.

Fields and corresponding actions for estimating a storage volume in Power Virtual Server

Estimating a shared processor pool

The following table explains the fields that you can use to create an estimate for a shared processor pool in a Power Virtual Server workspace:

Field	Action
Number of pools	Specify the number of pools that you need. The minimum number of pools that you can specify is 1.
Machine type	Select the required Power system from the list.

Reserved cores	Specify the number of cores that you want to reserve.
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Fields and corresponding actions for estimating a shared processor pool in Power Virtual Server

To learn more about a shared processor pool, see [Managing a shared processor pool](#).

Estimating a virtual tape library

The following table explains the fields that you can use to create an estimate for a virtual tape library (VTL) in a Power Virtual Server workspace:

Field	Action
Number of VTLs	Specify the number of virtual tape library instances that you need. The minimum number of VTL that you can specify is 1.
Licensed repository capacity (TB)	Specify the size of your repository capacity that determines which software license is needed. The minimum licensed repository capacity that you can specify is 1 TB.
Machine type	Select the required Power system from the list.
Core type	Select the core type. The available options are: Shared uncapped : Shared among other clients. Shared capped : Shared, but resources do not expand beyond those that are requested (used mostly for licensing). Dedicated : Resources are allocated for a specific client (used for specific third-party considerations).
Cores	Specify the number of cores that you need. The minimum cores that you can specify is 0.25 for shared capped and shared uncapped core types. For dedicated core type, the minimum cores that you can select is 1.
Memory (GB)	Specify the memory size. The minimum memory size that you can specify is 2 GiB.
Storage tiers	Optional. You can click Add volume + to attach storage volumes to your estimate. You can choose from Tier 0, Tier 1, Tier 3, or Fixed IOPs. You cannot add a separate boot volume estimation. Hence, enter the storage volume by considering the boot volumes and data volume that you might need.

Fields and corresponding actions for estimating a virtual tape library (VTL) in Power Virtual Server

To learn more about virtual tape libraries, see [Managing a virtual tape library](#).

Estimating a dedicated host

The following table explains the fields that you can use to create an estimate for a dedicated host in a Power Virtual Server workspace:

Field	Action
Number of dedicated hosts	Specify the number of dedicated hosts that you want to reserve.
Machine type	From the list of available systems, select the Power system that you want to use as a dedicated host.

Fields and corresponding actions for estimating a dedicated host in Power Virtual Server

To learn more about dedicated host, see [Getting started with dedicated host](#).

Estimating a vPMEM volume

To create an estimate of a Virtual Persistent Memory (vPMEM) volume on a Power10 or later system, you must update the following fields:

Field	Action
Operating system	Select AIX or Linux for SAP (HANA) .
Image type	Select the required AIX or Linux image type.

Machine type	Select a Power10 or later system from the list.
Advance Configuration	Set Virtual persistent memory volumes to on. Select the existing vPMEM or you can create a vPMEM volume to add to the VSI.

Fields and corresponding actions for estimating a vPMEM volume

When you add a vPMEM volume to a VSI with AIX OS, the vPMEM volumes use memory in addition to the base memory. The total cost is calculated based on the total memory usage.

When you add a vPMEM volume to a VSI with SAP (HANA) profile, select the SAP certified profile that supports vPMEM volume. The total memory of the VSI that is the base memory with vPMEM memory, the overall memory consumption, and the cost of the memory remains the same.

Contacting IBM to place an order

To begin the ordering process for IBM Power Virtual Server through an IBM Business Partner, complete the following steps:

1. Share the downloaded estimate with your IBM Business Partner.
2. Initiate a discussion with the partner to:
 - Plan your infrastructure configuration
 - Define data-center requirements
 - Explore possible savings plans
 - Finalize the order

If you do not work with an IBM Business Partner, contact IBM through email or [book a meeting](#) with an IBM expert.

For more information about creating an estimate for Power Virtual Server Private Cloud resources, see [Creating an estimate for Power Virtual Server resources in a client location](#).

Creating an estimate for Power Virtual Server resources in a client location

You can use the [cost estimator](#) to estimate the cost of the IBM® Power® Virtual Server resources before you deploy them in a client location where the infrastructure is hosted and managed by IBM.

You can create an estimate by selecting the required compute and storage configurations based on your business requirement. Based on your selection and configuration, you can view the estimated cost of your Power Virtual Server private cloud infrastructure. You can connect with your IBM Business Partner or contact IBM to place your order.


Note: You are not charged for creating an estimate. The resources that you estimate can be deployed only after a workspace is created.

Estimating Power Virtual Server infrastructure in a client location

To create an estimate for Power Virtual Server infrastructure in a client location, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. Search for **Power Virtual Server** in the search box for IBM Cloud catalog.
3. Click the **Power Virtual Server** tile.
4. Click **Estimate Cost**. You are redirected to the **Estimate cost** page for Power Virtual Server.
5. Select **Client location** from the **Location type** list.
6. Select an IBM Cloud region that is near to your physical location from the **Location** list.
7. In the **Pricing plan** section, select the pricing plan that best suits your workloads. The available options are:
 - **Metered:** The metered pricing plan meters costs hourly for the allocated resources on a monthly basis and the workloads can scale to the full capacity of the infrastructure. However, the metered plan does not support SAP workloads.
 - **Subscription:** The subscription pricing plan provides a monthly subscription with access to 100% of the infrastructure capacity. The capacity is not metered. The subscription plan is best suited for SAP HANA and SAP NetWeaver workloads. However, the subscription plan does not support AIX or IBM i workloads. To use the subscription pricing plan, contact [IBM sales](#) or your IBM Business Partner representative.

8. In the **Infrastructure configuration** section, specify the infrastructure requirements, such as the machine type, number of systems, memory per system, and the number and type of storage controllers. The **Infrastructure overview** section displays the summary of the selected configuration. The *Deployment size* depends on the machine type, memory per system, and number of systems that you select.

Currently, IBM Power S1122 (2U), IBM Power E1150 (4U), and IBM Power E1180 (12U) servers are supported by specific memory. Each server type is shipped with a fixed number of cores.

 **Note:** Machine types cannot be mixed within a rack, and all systems must have the same amount of memory.

9. In the **Committed monthly spend** section, select the number of years from the **Contract commitment term** list. Choose **Advanced** or **Premium** IBM Cloud support.

The **Summary** panel displays the estimated costs for the contract commitment years based on your selection of IBM Cloud support.

 **Note:** A longer contract commitment term reduces the minimum committed spend value. You can contact IBM to learn more about the possible savings with the commitment plans.

10. In the Summary panel, review the estimated cost and click **Add to estimate**. The **Estimate** panel is displayed.
11. In the **Save Estimate** panel, select **Create New**.
12. Enter a name for the estimate in the **Name** field.
13. Optional: Enter a description for the estimate in the **Description (optional)** field.
14. Click **Save estimate**. You are redirected to the Estimate panel.
15. Click **View estimate** to open the estimate in a new tab. You can use the **Action** menu to export the estimate in XLSX, CSV, or PDF format.

Contacting IBM to place an order for Power Virtual Server Private Cloud in a client location

To begin the ordering process for IBM Power Virtual Server through an IBM Business Partner, complete the following steps:

1. Share the downloaded estimate with your IBM Business Partner.
2. Initiate a discussion with the partner to:
 - Plan your infrastructure configuration
 - Define data-center requirements
 - Explore possible savings plans
 - Finalize the order

If you do not work with an IBM Business Partner, contact IBM through email or [book a meeting](#) with an IBM expert.

 **Important:** Before you place the order for IBM Power Virtual Server Private Cloud, verify that the basic criteria to own a pod is met. For more information, see [Prerequisites for installing the pod](#).

High availability and disaster recovery

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

Both High Availability (HA) and Disaster Recovery (DR) are essential for business continuity. HA addresses local planned and unplanned outages for hardware or software. Disaster recovery addresses the catastrophic events during which the IBM Cloud region might become unavailable.

Review the following topics to understand the HA and DR for IBM data center and Client location offerings of IBM Power Virtual Server:

- [High availability and disaster recovery options in IBM data center](#)
- [High availability and disaster recovery options in Client location](#)

High availability and disaster recovery

IBM Power Virtual Server in **IBM data center**

Both High Availability (HA) and Disaster Recovery (DR) are essential for business continuity. HA addresses local planned and unplanned outages for hardware or software. Disaster recovery addresses the catastrophic events during which the IBM Cloud region might become unavailable.

Review the following topics to understand the HA and DR for IBM data center and Client location offerings of IBM Power Virtual Server:

- [High availability and disaster recovery options in IBM data center](#)
- [High availability and disaster recovery options in Client location](#)

High Availability and Disaster Recovery options in IBM data center

IBM® Power® Virtual Server supports various high availability and disaster recovery solutions that you can deploy in your environment. Host failure recovery is the default high availability solution in Power Virtual Server. You can also deploy advanced solutions, such as PowerHA SystemMirror for cluster management and PowerHA geographic mirroring for replication-based disaster recovery.

Host failure recovery

Power Virtual Server is built on the IBM Power enterprise infrastructure with redundant networking and storage area network (SAN) fabric capabilities. IBM Power Virtual Server monitors your infrastructure to ensure that hosts are responsive and operating correctly.

When a host fails unexpectedly, the virtual server instances (VSIs) on the failed host are automatically restarted on another available host. In some cases, you must manually recover the failed host.

The host failure recovery process restarts VSIs on a different available host. This process completely reboots the operating system. After the operating system restarts, you must restart your applications according to your standard boot procedures.

The automated remote restart feature enables host failure recovery by default for all VSIs in the Power Virtual Server environment. To disable this feature, you can set **Automated remote restart** to off during VSI creation. Alternatively, you can modify the settings on the Virtual server instance details page. For more information, see [Disabling automated remote restart for a VSI](#).

Host failure recovery:

- Does not restart a pinned VSI. Pinning VSIs to specific hosts results in extended downtime because recovery depends on the time required to repair the failed host. To minimize downtime, ensure that VSIs are not pinned to a host. For more information, see [Virtual server pinning and its impacts on VSI availability](#).
- Does not restart a VSI in a server placement group that uses an affinity policy and includes other VSIs that are hard-pinned to the host. Affinity policies require all VSIs in the group to remain on the same host. A hard-pinned VSI prevents VSIs in the group from moving to another host.
- Restarts the VSI on another host with a different physical serial number. If your software depends on serial numbers, consider using virtual serial numbers (VSNs) for IBM i. Check your independent software vendor (ISV) licensing policy to determine whether VSNs are appropriate.

PowerHA SystemMirror for AIX Standard Edition

PowerHA SystemMirror for AIX Standard Edition is available with a monthly subscription model. For more information, see [Standard Edition monthly pricing options](#).

After you purchase the software, you can download it from [Entitled Systems Support \(ESS\)](#). You can install PowerHA SystemMirror for AIX on the virtual server that is running in your Power Virtual Server environment. For installation instructions, see [Installing PowerHA SystemMirror](#).

Review the following information for implementing PowerHA SystemMirror for AIX in your Power Virtual Server environment:

- Select **Different Server** from the **Colocation Rules** field when you create the virtual servers that are part of the PowerHA SystemMirror cluster. Selecting **Different Server** ensures that the different logical partitions (LPARs) in the PowerHA SystemMirror cluster are not deployed on the same host.
- Select **On** from the **Shareable** field when you create storage volumes for the virtual servers that are part of the PowerHA SystemMirror cluster.
- You do not have access to the HMC, VIOS, and host system on Power Virtual Server. Therefore, PowerHA SystemMirror functions that require access to these capabilities, such as Resource Optimized High Availability (ROHA) and Active Node Halt Policy (ANHP), are not available. However, PowerHA SystemMirror 7.2.6 SP1 or later versions support ROHA functions. For more information about configuring and using ROHA

with Power Virtual Server, see [Resource Optimized High Availability in Cloud](#).



Note: Licenses that are purchased outside a subscription model are not eligible for use with Power Virtual Server.

Disaster recovery mechanisms

You can use Geographic Logical Volume Manager (GLVM) replication to implement disaster recovery between two AIX VSIs that are deployed in separate IBM Cloud data centers. For a complete tutorial, see [AIX Disaster Recovery with Power Virtual Server](#).

You can tune the Transmission Control Protocol (TCP) to improve wide area network (WAN) connection performance between AIX virtual machines. For more information, see [TCP tuning for AIX WAN connections](#).

You can implement disaster recovery mechanisms between two IBM i VSIs by using PowerHA geographic mirroring. For a complete tutorial, see [IBM i Disaster Recovery with IBM Power Virtual Server](#).

Business continuity through backup and restore

Client location Your application configuration and data are not backed up automatically. To recover from a disaster, IBM backs up the configuration data that is needed to rebuild a pod. The configuration data includes the virtual machine configurations and private cloud image repositories. However, you are responsible for backing up and restoring client data and client OS images.

IBM data center Your Power Virtual Server configuration and data are not backed up automatically. You can back up your virtual server to [Cloud Object Storage](#) as explained in [Backup strategies for Power Virtual Server](#). You can also restore your virtual server in case a critical failure occurs.

Importing and exporting images require significant processing power and network bandwidth. As a result, you can submit only one import or export request at a time. Any additional import or export requests are queued. Typically, you import or export system disks (AIX rootvg disks) that are less than 1 TB to facilitate the transfer to and from Cloud Object Storage. If your image is greater than 1 TB, the transfer might take longer and is more likely to fail or time out. The maximum uncompressed image that you can import or export is 10 TB.

High availability and disaster recovery options in Client location

IBM Power Virtual Server Private Cloud in **Client location**

For the pod software, IBM® Power® Virtual Server removes all single points of failure by implementing redundancy as follows:

- Redundancy in power supplies to the servers that are connected to different power sources.
- Redundancy in network adapters that are connected to redundant network switches.
- Redundancy in Fibre Channel adapters that are connected to redundant storage area network (SAN) fabrics.
- A dual-VIOS setup with redundant subcomponents for each Virtual I/O Server (VIOS).
- A dual-HMC setup is provided with medium size pod. However, a small pod configuration has only a single HMC. For small pods, the standby system is shared between the managing and managed nodes.
- Dual storage systems that allow host-based mirroring of customer data.

Host failure recovery

Power Virtual Server is built on the IBM Power enterprise infrastructure with redundant networking and storage area network (SAN) fabrics capabilities. IBM Power Virtual Server continuously monitors your infrastructure to ensure that hosts are responsive and operating correctly.

When a host fails unexpectedly, the virtual server instances (VSIs) on the failed host are automatically restarted on another available host. In some cases, manual recovery of the failed host is required.



Note: The host failure recovery process involves restarting the VSIs on alternate hosts and results in a complete reboot of the operating system. After the operating system is rebooted, the applications must be restarted to recover and resume according to your standard boot procedures.

Host failure recovery is enabled by default for all VSIs in the Power Virtual Server environment through the automated remote restart feature. To disable this feature, you can set **Automated remote restart** to off during VSI creation. Alternatively, you can modify the settings on the Virtual server instance details page. For more information, see [Disabling automated remote restart for a VSI](#).

Host failure recovery:

- Does not restart a pinned VSI. When you pin virtual server instances to specific hosts, the recovery depends on the time taken to repair the failed host, which results in extended downtime. To minimize downtime, ensure that VSIs are not pinned to a host. For more information, see [Virtual server pinning and its impacts on VSI availability](#).
- Does not restart a VSI in a server placement group that uses an affinity policy and includes other VSIs that are hard-pinned to the host. Affinity policies require all VSIs in the group to remain on the same host. A hard-pinned VSI prevents VSIs in the group from moving to another host.
- Restarts the VSI on another host with a different physical serial number. If your software depends on serial numbers, consider using virtual serial numbers (VSN) for IBM i, based on your independent software vendor (ISV) licensing policies.

For advanced high-availability functions, you can implement HA clusters and HA services such as PowerHA SystemMirror, on the virtual server instances. For HA clusters, you must use server placement groups with anti-affinity policy to enable the functions of the following items:

- All virtual server instances run on different systems.
- You have disks from different controllers.
- You can create mirror volume groups.

For client-managed applications, you must implement a high-availability strategy. For example, you can use solutions such as Red Hat Enterprise Linux High Availability or SUSE Linux Enterprise High Availability.

Disaster recovery mechanisms in IBM Power Virtual Server Private Cloud

Disaster recovery addresses the catastrophic events during which the IBM Cloud region might become unavailable.

For the IBM-managed software, the disaster recovery strategy remains the same as the high-availability strategy. For client-managed applications and software, you are responsible for the disaster recovery strategy. For example, you can use solutions such as Red Hat Enterprise Linux High Availability and SUSE Linux Enterprise Clustered Disaster Recovery.

Backup strategies

Backup for IBM data center Workloads in IBM data center

IBM Power Virtual Server in [IBM data center](#)

Learn more about different AIX, Linux, IBM i, and other workload backup strategies for IBM® Power Systems™ Virtual Server.

Secure automated backup with Compass for Power Virtual Server

IBM Cloud® Partner Cobalt Iron® provides an automated backup offering for AIX, Linux, IBM i, and other Power workloads on IBM IBM® Power® Virtual Server. The backup offering is called as **Secure Automated Backup with Compass® for PowerVS**. The **Secure Automated Backup with Compass® for PowerVS** is referred as “Backup Offering” throughout the topic.

The Backup Offering is powered by Cobalt Iron Compass and is accessible from the [IBM cloud catalog](#).

Deploying the Compass backup offering

To deploy the Backup Offering in an IBM data center, complete the following steps:

1. Set up the Backup Offering for your account for a region from the [IBM cloud catalog](#).

For Compass to create or manage VPCs, Transit Gateways, and other cloud networking resources, provide an IBM Cloud API key with the following cloud resource permissions for your IBM Cloud account:

- Transit Gateway (Resource Group specific): `transit.gateway.edit`
- VPC Infrastructure (All Resources): `is.vpc.vpc.list` and `is.vpc.vpc.read`
- VPC Infrastructure (Resource Group specific): `is.vpc.vpc.create`

2. Attach the Power Virtual Server workspace that you want to enable for the backup zone to the IBM Cloud Transit Gateway.

3. Complete the following steps to protect AIX and Linux systems:

- a. Access the Cobalt Iron Commander graphical interface to register new systems that need protection to Compass.
- b. Complete the following steps to download the software and to install the Compass agent inside the virtual server instance (VSI):
 - a. Log in to [Cobalt Iron Commander graphical interface](#)
 - b. In the Commander interface, go to the **Protected system**.
 - c. Download the Agent installer and install the agent on your IBM® Power® Virtual Server instance.

4. Complete the following steps to protect IBM i instance:

5. Access the Cobalt Iron Commander graphical interface to configure a Compass Virtual Tape Library (VTL).

6. In the **Admin** tab, select the **Virtual Tape Libraries** tile to open the **Create Virtual Tape Library** wizard.

When Compass creates the VTL, it also creates Challenge Handshake Authentication Protocol (CHAP) authentication for the Compass VTL.

7. Configure the Compass VTL on your IBM i instance as an iSCSI target device. Use IBM Navigator for IBM i, as it is the preferred way to configure iSCSI targets.

8. Use the Compass iSCSI VTL as the target for your Backup, Recovery, and Media Services (BRMS) or Save Object (SAV) backups and restores.

For more information about protecting IBM i with Compass, see [Compass Quick Start Guide for IBM i Data Protection](#).

Network architecture for deploying the Backup Offering

To deploy the Backup Offering, select one of the following architectures:

- [Single copy Backup Offering](#)
- [Dual copy Backup Offering](#)

Single copy Backup Offering

You can take a backup of your workload in a single data center by using a single copy Backup Offering,

By studying the network architecture diagram of single copy Backup Offering, you can understand the following concepts:

- The architecture of single copy Backup Offering deployed in IBM data center
- The requirements for AIX, Linux, and IBM i VSIs on Power to access the Compass backup servers through the IBM Cloud network

Compass Vaults are backup server instances that are preconfigured in IBM Cloud data centers.



Important: Do not deploy any additional cloud resources to the Backup Offering VPC.



Single copy Backup Offering network architecture

The Compass Backup as a Service (BaaS) VPC is created when the Backup Offering is provisioned. The BaaS VPC enables Virtual Private Endpoints (VPEs) for private IP connectivity to the managed backup server instances. When you set up the Backup Offering service, an automation process creates the following network segments:

- Local Transit Gateway, if it does not exist
- BaaS VPC for the dedicated use of the backup activity
- VPE for secure connectivity to each of the backup servers
- Security group with inbound rule, address prefix, and subnet

The Backup Offering VPC and the Power Virtual Server workspaces must exist in the same region and be connected by using the local Transit Gateway. You can connect your workloads in IBM Power Virtual Server Private Cloud in Client location to the Transit Gateway through the Direct Link connection. You can use VPN connection in place of the Direct Link connection.

Dual copy Backup Offering

You can protect your workload backups across two different data center regions by using the dual copy Backup Offering.

By studying the network architecture diagram of dual copy Backup Offering, you can understand the following concepts:

- The architecture of dual copy Backup Offering that is deployed in IBM data centers
- The requirements for AIX, Linux, and IBM i VSIs on Power to access the Compass backup servers through the IBM Cloud network

Compass Vaults are preconfigured backup servers in IBM Cloud data centers and they replicate data to preconfigured Compass Vaults in other regions.



Important: Do not deploy any additional resources to the Backup Offering VPC.



Dual copy Backup Offering network architecture

The Backup Offering VPC is used by Compass for backup operations and is deployed when the Backup Offering is provisioned. When you set up the Backup Offering service, an automation process creates the following network segments:

- Local Transit Gateway if it does not exist
- VPC for backup activity only
- VPE for each of the backup servers
- Security group with inbound rule, address prefix, and subnet

The Backup Offering VPC and the Power Virtual Server workspaces must exist in the same region and be connected by using the local Transit Gateway. You can connect your on-premises workloads to the Transit Gateway through the Direct Link connection. You can use VPN connection in place of the Direct Link connection.

Provisioning the Backup Offering service in IBM data center

Complete the following steps to set up the Backup Offering from the IBM Cloud catalog:

1. For Compass to create or manage VPCs, Transit Gateways, and other cloud networking resources, provide an IBM Cloud API key with the following cloud resource permissions for your IBM Cloud account:

- Transit Gateway (Resource Group specific) : `transit.gateway.edit`
- VPC Infrastructure (All Resources): `is.vpc.vpc.list` and `is.vpc.vpc.read`
- VPC Infrastructure (Resource Group specific): `is.vpc.vpc.create`

- 2. In the search box, type "Compass Backup" and click **Secure Automated Backup with Compass for PowerVS** tile.
- 3. Select a deployment location for your backup target.

 **Important:** Do not to deploy any additional resources to the Backup Offering VPC.

- 4. Specify the **Pricing plan**, **Service name**, **Resource group**, your **IBM Cloud API key** , and Compass organization name according to your business needs. Also, specify the VPC subnet IP address range that you want to use to access the Compass Vaults.
- 5. Click **Create**.
- 6. Compass creates and connects the Backup VPC to the Power Virtual Server workspace that you want to back up by using the local Transit Gateway. A local Transit Gateway is created if it does not exist.

For more information, see [Ordering IBM Cloud Transit Gateway](#) and [Using virtual private endpoints for VPC to privately connect to IBM Cloud Transit Gateway](#).

- 7. Click **Launch Compass UI** that will redirect you to the Cobalt Iron Compass Commander page where you need to complete the setup. For more information, see [Cobalt Iron documentation](#).

Pricing in IBM data center

When you use the Backup Offering, you are billed monthly through IBM Cloud for the stored amount of data that is backed up for the region for either single or dual copy. The backup data is the stored data after duplication and compression. For more information about pricing plans, see [Secure Automated Backup with Compass for Power VS](#). The page is accessible from the [IBM Cloud catalog](#). You can generate an estimate of the cost based on your expected usage from the **Summary** panel.

In your account, the following deployments are made:

- Connectivity is established between Power Virtual Server instances and the backup servers through a local Transit Gateway connection to the backup VPC. The automation when the Backup Offering is provisioned establishes this connection.
- Achieve the name resolution for the backup server connection by using the file on the agent system available at `/etc/hosts` path or by adding `CNAME` entries to the DNS server associated with the agent system.

Supported data centers

The single copy Backup Offering is available in the following data centers:

- DAL10
- DAL12
- DAL13
- FRA04
- FRA05
- LON04
- LON06
- MAD02
- MAD04
- OSA21
- SAO01
- SAO04
- SYD04
- SYD05
- TOK04
- WDC04
- WDC06
- WDC07

The dual copy Backup Offering is available in the following data center pairs:

Data Center 1	Data Center 2
---------------	---------------

DAL10	WDC07
DAL12	WDC06
MAD02	FRA04
MAD04	FRA05
SAO01	SAO04
OSA21	TOK04
SYD04	SYD05
DAL13	WDC04
LON04	LON06
Data center pairs that are availibile for Backup Offering	

Support for Compass Backup Offering

To get the support for the Backup Offering, contact Cobalt Iron. You must have the login credentials to Cobalt Iron to access the following pages:

- For more information, see the [Cobalt Iron documentation](#).
- For issues related to backup and restore, contact Cobalt Iron by opening a service ticket through `support.cobaltiron.com`.

For more information about the issues that are related to Power Virtual Server or IBM Cloud, see [Getting help and support](#).

Backup strategies for IBM i

IBM Power Virtual Server in `IBM data center`

IBM Power Virtual Server Private Cloud in `Client location`

Secure automated backup with Compass for IBM i

Cobalt Iron, which is an IBM Cloud Partner, provides an automated Backup Offering for IBM i instances of IBM Power Virtual Server. The Backup Offering is called **Secure Automated Backup with Compass for PowerVS** . It is powered by the Cobalt Iron Compass Tape Gateway (CTG) Virtual Tape Library (VTL) and is accessible from the IBM Cloud catalog.

Cobalt Iron CTG delivers the following capabilities that IBM i users need:

- Airtight security (including the unique Compass Zero Access® architecture)
- Intelligent automation
- Seamless integration with IBM i
- Flexible deployment
- Modern SaaS-native approach to IBM i data protection.

Compass is designed to simplify operations, strengthen security, and support hybrid deployments. Compass helps organizations to modernize their IBM i backup strategies by reducing complexity and risk.

The Compass VTL feature can be used with IBM® Backup, Recovery, and Media Services (BRMS) or with IBM i Save Object (SAV) commands to protect IBM i data.

Deploying the Compass backup offering for IBM i

Complete the following steps to deploy Compass for IBM i in an IBM data center:

- Set up secure automated backup with Compass for your account for a region from the IBM Cloud catalog.

For Compass to create or manage VPCs, Transit Gateways, and other cloud networking resources, provide an IBM Cloud API key with the following cloud resource permissions for your IBM Cloud account:

- Transit Gateway (Resource Group specific) : `transit.gateway.edit`
- VPC Infrastructure (All Resources): `is.vpc.vpc.list` and `is.vpc.vpc.read`
- VPC Infrastructure (Resource Group specific): `is.vpc.vpc.create`

2. Ensure that your Power Virtual Server workspace is connected to the Compass Vault IP addresses that are provided as a part of the Compass service provisioning. Complete the following steps to view the Compass Vault IP addresses:

Complete the following steps to find the specific IP address that is assigned for your Compass Vault in your IBM Cloud Data Center:

- a. Go to the Cloud Resource List in your IBM Cloud account and go to the Storage category.
- b. Double-click the Secure Automated Backup Compass cloud service.
- c. Click **Get Started**.
- d. Click **Step 2 – Setup Compass Name Resolution**.

The connectivity between your Power Virtual Server workspace and the Compass Vault IP addresses is provided through your IBM Cloud Transit Gateway.

3. Complete the following steps to protect IBM i instance:
4. Access the Cobalt Iron Commander graphical interface to configure a Compass Virtual Tape Library (VTL).
5. In the **Admin** tab, select the **Virtual Tape Libraries** tile to open the **Create Virtual Tape Library** wizard.
6. Create a new Compass VTL as an iSCSI target. The **Create Virtual Tape Library** wizard creates the library, drives, and virtual tape volumes for use by IBM i backup utilities. When Compass creates the VTL, it also creates Challenge Handshake Authentication Protocol (CHAP) authentication for the Compass VTL.
7. Configure the Compass VTL on your IBM i instance as an iSCSI target device. Use IBM Navigator for IBM i, as it is the preferred way to configure iSCSI targets.
8. Use the Compass iSCSI VTL as the target for your Backup, Recovery, and Media Services (BRMS) or Save Object (SAV) backups and restores.

For more information about protecting IBM i with Compass, see [Compass Quick Start Guide for IBM i Data Protection](#).

BRMS and ICC

Another IBM i backup strategy is to use IBM® Backup, Recovery, and Media Services (BRMS) and IBM Cloud Storage Solutions (ICC). Together, these products automatically back up your LPARs to IBM Cloud Object Storage. The ICC product can be integrated with BRMS to move and retrieve objects from remote locations, including Cloud Object Storage. In most cases, this process involves backing up to virtual tapes and image catalogs.



Note: You need extra storage for the LPAR to host the image catalogs until they are moved to Cloud Object Storage.

The typical IBM i customer uses the following flow to back up LPARs and objects:

1. Use the 5733-ICC product to connect to Cloud Object Storage (~2 times the disk capacity to hold the backup images).
2. Connect to IBM Cloud Object Storage by following the steps that are mentioned in [Using Cloud Object Storage](#).
3. Complete the back up to Cloud Object Storage by choosing the speed and resiliency that is required.
 - [Working with ICC](#)
 - [BRMS with Cloud Storage Solutions for i considerations and requirements](#)
 - [Data backup and recovery by using BRMS and IBM Cloud Storage Solutions for i](#)

For a complete tutorial on backing up and restoring IBM i VM data, see [Backing up and restoring data in an IBM i VM](#).

FalconStor StorSafe VTL

For more information, see [FalconStor StorSafe VTL](#).

Additional backup strategies

The additional backup strategies that you can use are as follows:

IBM teams and managed services

You can engage IBM teams and services to assist you throughout the migration lifecycle.

- [Expert Labs](#)
- [IBM Services for Cloud Migration](#)

Image capture

Image capture produces a storage FlashCopy of the logical partition (LPAR) and works on both AIX, Linux, and IBM i LPARs. You can use image capture to store VM images within your account (locally) as a part of your image catalog, directly to [IBM Cloud Object Storage](#), or both.

Importing and exporting images requires a considerable amount of processing power and network bandwidth. As a result, you can submit only one import or export request before it is queued. Typically, users import or export system disks (AIX rootvg disks) that are less than 1 TB to facilitate the transfer to and from Cloud Object Storage. If your image is greater than 1 TB, your transfer might take a long time and be prone to failure. The maximum image that you can import or export is **10 TB**.

Cloud Object Storage

The preferred ways to connect to Cloud Object Storage from a VM in Power Virtual Server are as follows:

1. In a PER workspace, attach the Power Virtual Server workspace to a Transit Gateway and directly access the Cloud Object Storage direct endpoint. See, [Attaching Transit Gateway to a PER workspace](#).
2. In a non-PER workspace that are in a multi-zone region (MZR) the best way to connect to Cloud Object Storage is as follows:
 - a. Create a [Virtual Private Cloud \(VPC\) with subnets](#) in the same region as your Power Virtual Server workspace.
 - b. Create a [Virtual Private Endpoint gateway](#) (VPE).
 - c. Connect the VPC to a [Transit Gateway](#).
 - d. [Create a cloud connection](#) to connect the non-PER Power Virtual Server workspace to the same Transit Gateway.

The Power Virtual Server would then use the VPE's IP address to connect to Cloud Object Storage. A Custom Resolver (CR) is needed by the Power Virtual Server to reach the Cloud Object Storage. If the VPE has multiple IP addresses, you can set up custom DNS and a custom hostname to connect to Cloud Object Storage.

3. Deploy a Nginx reverse proxy server in either the classic or VPC infrastructure.

Nginx is a mature, compact, and fast open source web server that excels at specialized tasks, including the reverse proxy server role. For information on setting up a Nginx reverse proxy server, see [Installing your Nginx reverse proxy](#).

Cloud Object Storage on AIX

IBM Power Systems that are running AIX 7.2 TL3, or later, have a script that is located in the path `/usr/samples/nim/cloud_setup`. The `cloud_setup` command installs the command-line environment for cloud storage services.

```
$ cloud_setup [-I | G | C] [-v]
```

-I: Install the necessary RPMs for universal CLI (supports Cloud Object Storage).
-G: Install the necessary RPMs for gsutil CLI (Google Cloud Storage).
-C: Install the necessary RPMs for cloud-init.
-v: Enable debug output.

1. To begin, copy the file to the system that requires AWS and give it run permission.
2. Enter the `cloud_setup -I` command to install the AWS CLI and all of the dependent RPMs.
3. After the installation is complete, you must configure `awscli` for access to Cloud Object Storage and provide the correct region (where your bucket Cloud Object Storage is defined) in the `aws --endpoint-url s3` command. In the following example, the **us-east** region is used:


```
$ # export PATH=$PATH:/opt/freeware/bin

# aws configure
AWS Access Key ID [None]: d197xxxxxxxxxxxxxxxxxxxxxxxxxxa4
AWS Secret Access Key [None]: f52a5xxxxxxxxxxxxxxxxxxxx001a33b74d8
Default region name [None]: us-east
Default output format [None]: json

# aws --endpoint-url https://s3.us-east.cloud-object-storage.appdomain.cloud s3 ls
2019-01-28 13:32:40 poweriaastest

# aws --endpoint-url https://s3.us-east.cloud-object-storage.appdomain.cloud s3 ls s3://poweriaastest
2019-01-28 13:33:42 6832 nimstat.sh
2019-01-28 15:05:25 1380725 yum-3.4.3-5.aix6.1.noarch.rpm
```

Veeam for AIX

Veeam for AIX provides simple physical server backup solutions for systems that are running in respective UNIX® operating systems. With them, IT organizations can provide industry-leading file-based backup and disaster recovery for their environments. For more information, see [Veeam Agents for IBM AIX](#).

FalconStor StorSafe VTL

For more information, see [FalconStor StorSafe VTL](#).

Migration strategies

Planning a workload migration to IBM® Power® Virtual Server

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

When workloads are deployed on a new system, you must pay attention to its configuration and tuning to achieve the expected performance. Power Virtual Server uses different IBM Power Systems depending on the location type.

IBM data center

- E980 (9080-M9S)
- S922 (9009-22A)
- S1022 (9105-22A)
- E1080 (9080-HEX)
- S1122 (9824-22A)

Client location

- S1122 (9824-22A)
- E1150 (9043-MRU)
- E1180 (9080-HEU)

For more information on hardware specifications that you might need, see [Hardware specifications for Power Virtual Server in IBM data center](#) and [Hardware and software specifications for IBM Power Virtual Server Private Cloud in Client location](#).

Supported AIX versions for Power Virtual Server

For AIX, Power Virtual Server supports only AIX 7.1, or later. If you use an unsupported version, it is subject to outages during planned maintenance windows with no advanced notification given. Your current AIX level and POWER processor family can help determine which migration path to follow.

Supported IBM i versions for Power Virtual Server

You must use the latest version of IBM i OS. If you are using an older version of IBM i, you must upgrade the OS to a current support level and then migrate to Power Virtual Server. For more information about upgrading the IBM i OS, see [Fast path for software installation, upgrade, and deletion tasks](#). For more information about the supported IBM i OS levels, see [IBM i Release Support](#).

Migration checklist for Power Virtual Server

Before you migrate to a newer IBM Power, review the following checklist:

- Plan the system migration.
- Install the latest required software and apply the available fixes.
- Set the appropriate processor compatibility mode for logical partitions (LPARs) before and after migration.
- Plan the virtual processor (VP) and entitlement for each LPAR to best fit your operation and performance requirements.
- Perform the steps provided in the I/O consideration guide.
- Consider contacting [IBM Technology Expert Labs](#) to ease the migration process.

Migrating to an IBM Power11

Learn more about migrating workloads from an existing IBM Power system to a Power11 system. Before you begin your migration, review the information in this section.

AIX and IBM i - You must update your LPAR operating system to the recommended levels before you migrate to a newer system. Installing only the minimum levels can make your partitions or workloads vulnerable to issues that are resolved in some of the latest updates. Hence it is recommended to install the required and latest updates from Fix Central before you start your migration.

Linux - You must migrate your Linux operating system level to a Power11-supported level.

For more information about the supported operating system levels and available stock images for Power11 systems, [Operating systems supported in IBM Power Virtual Server](#).

For more information about IBM Power11 performance and migration strategies, see the following articles:

- [System to IBM i mapping](#)
- [IBM i on Power - Performance FAQ](#)
- [IBM Power Systems Performance Report](#)
- [System software maps for SUSE Linux Enterprise Server](#)
- [System software maps for Red Hat Enterprise Linux](#)

Migrating to an IBM Power10 in an IBM data center

IBM data center

Learn more about migrating workloads from an existing IBM Power system to a Power10 system. Before you begin your migration, review the information in this section.

AIX and IBM i - You must update your LPAR operating system to the recommended levels before you migrate to a newer system. Installing only the minimum levels can make your partitions or workloads vulnerable to issues that are resolved in some of the latest updates. Hence it is recommended to install the required and latest updates from Fix Central before you start your migration. For IBM i the supported levels are IBM i 7.3 TR 11 and IBM i 7.4 TR 5, or later.

Linux - You must migrate your Linux operating system level to a Power10-supported level.

For more information about the supported operating system levels and available stock images for Power11 systems, see [Operating systems that are supported in IBM Power Virtual Server](#).

To learn more about IBM Power10 performance and migration strategies, see the following articles:

- [System to IBM i mapping](#)
- [IBM i on Power - Performance FAQ](#)
- [IBM Power performance resource center](#)
- [Supported Linux distributions and virtualization options for Power10 Linux on Power servers](#)

Migrating to an IBM Power9 in an IBM data center

IBM data center

Learn more about migrating workloads from your older IBM POWER to a Power9 system. Before you begin your migration, review the information in this section.



Tip: Installing only the minimum levels can make your partitions or workloads vulnerable to issues that are resolved in some of the latest updates.

- Power9 makes more efficient use of the 8 hardware SMT threads that are available per CPU (when it runs in *SMT8* mode). When you migrate from an older system, consider the use of SMT8. Also, consider reducing the allocation of CPUs (in dedicated CPU LPARs), or reducing VPs and CPU entitlement on shared CPU LPARs.
- Capacity planning is important when you are considering processor migration. When you are setting performance improvement goals and expectations, take note of the application behavior (for example, highly multi-threaded workloads vs single-threaded workloads).

To learn more about IBM Power9 performance and migration strategies, see the following articles:

- [Hints and tips for migrating workloads to IBM Power9](#)
- [IBM i on Power - Performance FAQ](#)
- [IBM Power Performance Report](#)

Lab services for Power Virtual Server

[IBM Technology Expert Labs](#) has service offerings available to assist you with resolving system, application, and database performance problems. Formal and informal training opportunities are also available where you can learn how to use performance tools and resolve issues on your own.

Migration strategies for AIX

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

Learn about the various AIX migration strategies for IBM® Power® Virtual Server.

Migrating an AIX-based Oracle database

To migrate an AIX-based Oracle database from IBM Power to Power Virtual Server, see [Oracle technical migration procedure](#).

Migrating Db2 on AIX to Power Virtual Server

See the use case on [Migrating Db2 on AIX to Power Virtual Server](#) for complete information.

Migrating client-managed environment to Power Virtual Server

See the [Move data from client-managed environment to Power Virtual Server](#) topic for complete information.

Migrating by using NIM on Power Virtual Server

You can configure your Network Installation Management (NIM) server on a Power Virtual Server and migrate an AIX virtual server instance from AIX 7.1 to AIX 7.2 or 7.3, and other operations.

See [Migrating by using NIM on Power Virtual Server](#) for more information.

Migration by using MKSYSB

You can migrate your data by using the `mksysb` command:

1. Back up the client-managed environment OS image with the `alt_disk_mksysb` command and transfer it to Cloud Object Storage.
2. Create a Power Virtual Server and import the image.
3. Restore the client-managed environment OS image.

For more information, see [Restoring an AIX mksysb image onto a Power Virtual Server instance](#).

Alternative AIX migration strategies

Some alternative AIX migration strategies include:

- `rsync` for nondatabase files
- Log shipping for databases

Migration strategies for IBM i

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

Learn about the various strategies to migrate your data and workloads to an IBM® Power® Virtual Server.

IBM i migration strategies

Learn about migration strategies that are specific to IBM i systems.

Recent IBM i migration enhancements:

1. BRMS enhanced to [support software data compression](#) for tape and virtual tape
2. [90-day software license for BRMS and ICC](#).

Backup Recovery and Media Services (BRMS) and Cloud Storage (ICC)

BRMS is an IBM i product that can be used to automate activities that help define and process your backup, recovery, and media management operations. The ICC product can be integrated with BRMS to move and retrieve objects from remote locations, including Cloud Object Storage.

The following high-level steps detail how to migrate your OS and data from an Client location system to the Power Virtual Server environment. Keep in mind that most of these steps can be automated by using BRMS and ICC.

1. Create your IBM i VM in the Power Virtual Server.
2. Create a virtual media and *IMGCLG* on the deployed VM.
3. Create a virtual optical and *IMGCLG* on the Client location system and perform an operating system save.
4. FTP the images to the Power Virtual Server.
5. Restore or slip the installation of the OS to get the base OS to the same level as it was on your Client location system.
6. Migrate your remaining data.

Refer to this documentation on migrating [IBM i to the Cloud by using BRMS and Cloud Storage Solutions with FTP](#).

For more information, see [Data backup and recovery by using BRMS and IBM Cloud Storage Solutions for i](#) and [BRMS with Cloud Storage Solutions for i considerations and requirements](#).

Logical Replication with Geographic Mirroring and PowerHA SystemMirror for i

GeoMirroring enables IBM i disk mirroring technology on multiple system environments and supports host-based and logical replication across geographically distant sites. Geo mirroring supports synchronous and asynchronous modes. You can integrate PowerHA SystemMirror (Enterprise Edition) for network monitoring and automated failover support.

- [Geographic mirroring](#)
- [IBM PowerHA SystemMirror for i: Using Geographic Mirroring](#)

Additional migration strategies

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IBM teams and managed services

You can engage IBM teams and services to assist you throughout the migration lifecycle.

- [Expert Labs](#)
- [IBM Services for Cloud Migration](#)

Aspera Technologies

IBM Aspera® on Cloud is a hosted service that quickly and reliably moves and shares your data sets across a hybrid platform environment. IBM Aspera can help transfer data to the IBM Cloud for later retrieval from the Power Virtual Server environment. For more information, see [IBM Aspera on Cloud](#).

Use the [Accelerated network transfer migration guide](#) powered by Aspera to accelerate the transfer of your migration content. It helps you to directly transfer the migration content to the Power Virtual Server workspace and you achieve faster time to value from IBM cloud.

IBM Cloud Object Storage

Cloud Object Storage can be used as an intermediary location to store files from your Client location environment. You can retrieve and send your files to the Power Virtual Server environment from this location. You must create Cloud Object Storage buckets to transfer data over the public internet or privately secured links. For more information, see [IBM Cloud Object Storage: FAQ](#).

To copy data from Cloud Object Storage to your AIX virtual machine (VM), you must install the [Amazon Web Services \(AWS\) CLI](#) by using either the **Yellowdog Updater, Modified (yum)** or **Pip installs Python pip (pip)** package managers. If you use the yum package manager, you can install the AWS CLI with the `yum install aws-cli` command. For the pip package manager, you can use `pip install awscli` to install the AWS CLI. After the installation, you can use the universal S3 commands that are supported by AWS to copy objects.

You can also find a script that ships (as a sample command) with AIX 7.2 TL3, or later, that simplifies the installation of yum, pip, and the AWS CLI. You can find the script at: `/usr/samples/nim/cloud_setup`.

- [Uses of AIX toolbox for open source software](#)
- [Python for AIX](#)

For an IBM i VM, you must use the [Cloud Storage Solution for IBM i product \(5733-ICC\)](#) to communicate with Cloud Object Storage and transfer data.

PowerVC images and Cloud Object Storage

If you have an environment with access to PowerVC, you can capture OVA images to easily migrate your data. Using the Power Virtual Server offering, you can provision a new virtual server based on an OVA image. To accomplish this, regardless of the operating system (OS), you must complete the following steps:

1. Create an OVA image on your local system.
2. Copy the OVA image to your **Cloud Object Storage** account.
3. [Deploy the OVA image and provision a new Power Virtual Server](#).

Using third-party tools for migration

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

You can use the third-party tools to complete your data migration. The following third-party tools are commonly used for data migration:

- Secure Automated Backup with Compass for PowerVS
- Double-Take® Move™ for AIX
- [FalconStor StorSafe VTL](#)
- [Rocket iCluster HA/DR](#)
- Maxava HA
- [Migrate 23](#)
- MIMIX

FalconStor StorSafe VTL

StorSafe VTL is an IBM-certified solution for migration, backup optimization, archive, and data recovery (DR).

StorSafe VTL is software that emulates physical tape drives and libraries to optimize backup and recovery. It works for both in the cloud and Client location. It enables simple migration from Client location to IBM® Power® Virtual Server. StorSafe VTL can backup IBM i or AIX workloads Client location, replicate to a cloud-resident StorSafe VTL, and then restore to a Power Virtual Server. Similarly, legacy tapes can be handled.

Following are some of the features of StorSafe VTL:

- Nondisruptive and secure process that can scale to PB-sized workloads.
- Deduplication and replication improve performance and reduce storage capacity.
- The same tool for Client location and cloud that enables both workload and tape migrations.
- System and application data are deduplicated and compressed before the data is transmitted over the network. This process accelerates migration even on the initial baseline transfer.
- Replicated tapes land in Cloud Object Storage and can be used by the cloud VTL as a backup baseline without copying the data.
- Migration uses existing backup processes and tools by using existing environment and knowledge.
- Data replication can occur over time as part of the backups with final sync replication completed fast during regular backup windows.
- FalconStor StorSafe VTL for Power Virtual Server can take advantage of IBM low-cost Cloud Object Storage storage for its repository.

Migrating an IBM i system

See the video that shows migration of an IBM i system from Client location to Power Virtual Server - [Migration of an IBM i system](#).

You can access StorSafe VTL from the [IBM Cloud catalog](#). In the search field, enter `FalconStor StorSafe VTL for PowerVS Cloud` or `FalconStor StorSafe`

FalconStor StorSafe VTL for IBM data center

FalconStor StorSafe VTL for Power Virtual Server comes bundled with OS and can be deployed from the IBM Cloud catalog with one click.

FalconStor StorSafe VTL for Client location

FalconStor StorSafe VTL for Client location provides flexible deployment options for the user:

- It can be installed on any X86 server that is on [Falconstor certification matrix](#).
- It can be installed as a VM in Hypervisors such as VMware, HyperV, Xen, and KVM.
- It can be installed inside Power Virtual Server as an LPAR.

Migration Services

Services to help migration are available from IBM, FalconStor, and Authorized Migration Business Partners. Reach out if you would like more information about using migration services or becoming a FalconStor Migration Partner at PowerVSMigration@falconstor.com.

If you need any help with deploying FalconStor StorSafe VTL, see [FalconStor VTL for IBM Deployment Guide](#).

For more information on FalconStor StorSafe VTL, see www.falconstor.com.

Back up and restore

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

You can back up your Client location environment and restore it to Power Virtual Server. In most cases, Cloud Object Storage and NFS servers serve as an intermediary to back up and restore data. The [AIX migration strategies](#) and [IBM i migration strategies](#) sections provide information on OS-specific migration strategies.

Replication

Host and OS-based logical replication

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IBM recommends that you use PowerHA SystemMirror (Enterprise Edition) with Geographic Mirroring (Geo mirroring) and Geographic Logical Volume Manager (GLVM). The following host and OS-based logical replication options exist depending on your OS:

- **AIX** - GLVM
- **IBM i** - Geo mirroring

Application-specific replication

Applications might have replication mechanisms that can sync multiple environments. These options are commonly used for application-specific replication:

- Db2 HADR
- [Rocket iCluster HA/DR](#)
- Maxava HA
- [Migrate 23](#)
- MIMIX
- Oracle Data Guard
- Oracle Goldengate

IBM® Power® Virtual Server Global Replication Services

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IBM® Power® Virtual Server supports Global Replication Services (GRS). You can use GRS that provides volume replication services based on Storage Area Network (SAN) as the foundation for building disaster recovery (DR) solutions. GRS is based on an asynchronous replication technology called IBM FlashSystem Global Mirror Change Volume (GMCV).

You can create a replication-enabled volume from the GRS-enabled location. The replication-enabled volume is considered as the primary volume. The replicated volume on the secondary location is known as the auxiliary volume.

GRS on Power Virtual Server has the following benefits:

- Maintains a consistent and recoverable copy of the data at a secondary location. The copy of the data is created with minimal impact to applications at your primary location.
- Synchronizes the data between the primary and secondary locations. The failover and fail back modes in the primary and secondary locations reduce the time that is required to return to the primary location after a planned or unplanned outage.
- Maintains redundant data centers in distant locations for fast recovery from disasters.
- Removes the dedicated networks that are overpriced for replication and avoids bandwidth upgrades.

Enabling GRS in the IBM Power Virtual Server allows asynchronous replication of data between two IBM Cloud regional data centers in which storage replication is enabled. The data center pairs are fixed and mapped in a one-to-one relationship mode in both the directions.

Terms and definitions

In the following table, the terms that are used throughout the document are defined.

Term	Definition
------	------------

Primary location	Location in which the volume is created.
Secondary location	Location in which the auxiliary volume for replication is created.
Primary volume	Initial instance of the replication volume in the primary location. This volume is visible to the user and managed by IBM Power Virtual Server.
Auxiliary volume	Instance of the replicated volume in the secondary location. When the auxiliary volume is onboarded, it is visible and managed by IBM Power Virtual Server.
IBM cloud resource names (CRNs)	Identifiers that are assigned to uniquely identify resources in IBM Cloud.

GRS terms and definitions

GRS in the Client location

Enabling GRS in the IBM Power Virtual Server Private Cloud enables asynchronous replication of data between the primary location infrastructure and the secondary location infrastructure. The two infrastructure locations have the identical set of capabilities that IBM Power Virtual Server in IBM data center provides.

The network configuration between the primary location and the secondary location for replication includes the following prerequisites:

- The network bandwidth must be greater than or equal to 10 Gbps.
- The network latency must be less than or equal to 200 ms.

During the first data synchronization of primary volumes and auxiliary volumes, the entire data from the primary volumes is copied to the auxiliary volumes. For subsequent synchronizations, only the changes between the two synchronization operations are copied. The effective Recovery Point Objective (RPO) depends on the capability of the underlying network throughput and the application characteristics. If the network throughput is insufficient to reach the defined RPO, the time between the synchronization increases.

 **Note:** When the defined RPO exceeds more than 8 hours, the IBM operations team alerts you.

Replication is not supported between the VMs in the IBM Power Virtual Server in IBM data center environment and the IBM Power Virtual Server Private Cloud in Client location environment.

Pricing for GRS

Part numbers are used for calculating the cost of GRS based on the storage tier that is associated with the primary volume. For more information, see [Pricing for Global Replication Services \(GRS\)](#).

Performance and I/O considerations for GRS

To maintain the required performance and predictable I/O response times when you use GRS, it is best to limit each volume group to 30 volumes or fewer. As the number of replicated volumes grows, the storage system must coordinate a consistent set of changes across all volumes. The more volumes involved in this coordination, the greater the impact on response times. This additional synchronization can introduce latency and affect application responsiveness.

The ideal number of volumes depends on your workload characteristics, including the operating system, I/O intensity, and the tolerance of your environment for increased response times.

Power Virtual Server regions that support GRS

The following table shows the location pairs that support replication.

Location 1	Location 2
che03	in-mum-1
che02	in-mum-2

mad02	eu-de-1 (fra04)
mad04	eu-de-2 (fra05)
us-east (wdc04)	us-south (dal13)
wdc06	dal12
wdc06	dal14
wdc07	dal10
osa21	tok04
syd04	syd05
sao01	sao04
mon01	tor01
lon04	lon06

The Power Virtual Server region pairs that are replication-enabled

Listing the Power Virtual Server regions that support GRS by using API or CLI commands

To get a list of Power Virtual Server regions that are replication-enabled, use the following API or CLI commands:

- **API:** [Get the disaster recovery site details for the current location](#).
- **CLI:** [ibmcloud pi disaster-recovery](#). Set the `--all-regions` flag as `true`.

**Note:** Verify whether the primary and secondary locations to be used for replication are enabled for replication services and are location pairs that support replication.

To verify whether two data centers are a location pair that support replication, you can use the code similar to the following example. In the following example, the `wdc07` and `dal10` data centers are a location pair that support replication:

```
$
{
  "location": "dal10",
  "replicationSites": [
    {
      "ReplicationPoolMap": [
        {
          "remotePool": "General-Flash-92",
          "volumePool": "General-Flash-83"
        }
      ],
      "isActive": true,
      "location": "wdc07"
    }
  ]
}
```

Power Virtual Server storage pools that support GRS

You can allocate replication-enabled volumes only from storage pools that support replication. To identify the storage pools that support replication, use the following API or CLI commands:

- **API:** [Storage capacity for all available storage pools in a region](#).

- CLI: [ibmcloud pi storage-pools](#)

If a storage pool supports replication, the `replication-enabled` property is set to `true`.

When you create a volume that is replication-enabled, the volume is created in one of the storage pools that supports replication. To update a volume for replication services at a later point, you must create the volume in a storage pool that supports replication. Otherwise, the update operation fails.

Creating a volume group for GRS

A volume group is a Power Virtual Server managed resource. By using the volume group, you can enable, disable, and manage a storage replication consistency group. The volume group contains the volumes that must be recovered during a disaster. You can add the replication-enabled volumes to a volume group. A replication-enabled volume can be a part of only one volume group at any time. In addition, all volumes in a volume group must be a part of the same storage pool.

When you create a volume group on the primary location, the storage replication consistency group is created in the storage backend in the primary and secondary locations. A storage backend is the storage subsystem that contains the storage pool and the storage controller. The storage replication consistency group stores the consistent copy for the volumes in the volume group. You cannot perform any operation on a consistency group. Instead, you must perform the operation on the volume group that represents the consistency group. Power Virtual Server does not directly manage the consistency group in the storage backend.

When you onboard a volume on the secondary location, if the volume group is not yet created, the volume group is created and then the volume that is onboarded is added to it. This volume group is visible and managed by Power Virtual Server on the secondary location. For more information, see [Onboarding an auxiliary volume](#).

A volume group is used to enable, disable, and manage a replication-enabled consistency group in storage volumes. By using a consistency group, you can perform operations on multiple volumes instead of managing each volume individually. The consistency group has a `state` property that indicates the state of the `copy` operation of the volumes in the group. Refer to the following table that lists different states of the replication-enabled consistency group in storage volumes.

State	Description
<code>inconsistent_stopped</code>	The primary volumes are accessible for <code>read</code> and <code>write</code> I/O operations but the auxiliary volumes are not accessible. This state indicates that copying the data from primary to auxiliary volume has stopped. Start the <code>copy</code> operation on the auxiliary volume to make it consistent with primary volume.
<code>inconsistent_copying</code>	The primary volumes are accessible for <code>read</code> and <code>write</code> I/O but the auxiliary volumes are not accessible and the <code>copy</code> operation is started. This state indicates that the <code>copy</code> operation has started on the consistency group that was previously in the <code>inconsistent_stopped</code> state.
<code>consistent_copying</code>	The primary volumes are accessible for <code>read</code> and <code>write</code> I/O operation. The auxiliary volumes contain a consistent copy of the data on the primary volumes. The data on the auxiliary volume can become outdated and so the data must be updated with the data on the primary volume. This state indicates that copying is in progress and auxiliary volumes are updated with the current copy of the primary volumes. The <code>consistent_copying</code> progress bar is an indicator to show the progress of copy operation. One cycle period of copy operation is expected to be between the range of 1-100%. In some cases, the copy operation can complete the activity before the progress bar reaches the maximum value of 100%.
<code>consistent_stopped</code>	The auxiliary volumes contain a consistent copy of the primary volumes, but can be outdated with the data on the primary volumes. The state indicates that the consistency group that was in a <code>consistent_copying</code> state was stopped.
<code>idling</code>	The primary and auxiliary volumes are operating in the primary role and are accessible for <code>read</code> and <code>write</code> I/O operations. The <code>idling</code> state indicates that the data from the primary or auxiliary volumes is not copied to the primary or auxiliary volumes in the replication pair because the replication process is disabled.
<code>idling_disconnected</code>	This state indicates that the volumes in the consistency group are operating in the primary role and can accept <code>read</code> and <code>write</code> I/O operations.
<code>consistent_disconnected</code>	This state indicates that the volumes in the consistency groups are operating in the non-primary role and you cannot perform <code>read</code> or <code>write</code> I/O operations.
<code>empty</code>	This state indicates that the volumes in the consistency group do not have any relationship with each other.

Different states of storage replication consistency group and their descriptions.

If you perform any operation on a volume group at one location, it affects the associated volume group on the other location. For example, consider that a volume group is stopped on a primary location. After sometime, the associated volume group on the secondary location is updated to reflect the replication status of the pair in the volume group. In this case, the `replicationStatus` field displays the state of the volume group as `disabled` on both primary and secondary locations.

Before you perform any operation on a volume group at one location, confirm the replication status of both the volume groups at the primary and secondary locations.

You can group the replication-enabled volumes that are similar in function. For example, volumes that are related to a specific workload can be grouped under one volume group. When additional replication-enabled volumes are created, you can add a volume to an existing volume group with similar volumes based on the function. You can create a new volume group for the volumes if their function is not same as the existing volumes.

Preparation for disaster recovery

Consider that you have virtual server instances with data volumes that are running workloads. If a failure occurs and if the data volumes are replicated, you can recover the data volumes from the secondary location. To enable the volume replication service, complete the following actions:

- [Actions on the primary site](#)
- [Actions on the secondary site](#)



Note: You must complete the actions on the primary site before you proceed with the actions on the secondary site.

Prerequisites

Complete the following prerequisites before you prepare a replication-enabled volume for disaster recovery (DR):

- Use the same IBM Cloud account ID to create two workspaces, each in the primary and secondary locations that supports GRS
- Ensure that two workspaces have different CRNs
- Do not define additional properties for the workspaces to denote that they contain replication-enabled volumes

Actions on the primary site

To enable volume replication on the primary site, complete the following steps:

1. [Create a volume for replication](#)
2. [Verify the replication status of the volume](#)
3. [Update an existing volume to support replication](#)
4. [Create a volume group](#)
5. [Verify the status of the volume group](#)

Creating a volume for replication

Use Power Virtual Server interface to create a virtual server instance with volumes that support replication.



Note: The boot volumes of virtual server instances that you create are always set to non-replication-enabled. You can provide a combination of replication- and non-replication-enabled volumes for a virtual server instance if they belong to the same storage pool. All the affinity policies that are related to a storage pool are valid for replication-enabled volumes. For more information, see [Configuring affinity policies](#).

You can also create a replication-enabled volume by using the following API or CLI commands:

- **API:** [ibmcloud pi volume create](#).

Set the values for the following parameters:

- `VOLUME_ID`: Set the value to the primary volume ID
- `--size`: Set the unit of measurement to gigabytes (GB)
- `replicationEnabled`: Set the flag to `True`

For more information about the replication properties of a volume, see [FAQ](#).

Verifying the replication status of a volume

After you create a replication-enabled volume, retrieve the details of the volume to verify the replication status of the volume. The `replicationEnabled`

property must be set to `true` . The replication-enabled status might not be enabled immediately as the creation of the volume is asynchronous. Continue to monitor the state of the volume. If the volume is in the `available` state and the replication-enabled status is not yet set to `true` , update the volume to set the replication-enabled status to `true` . For more information, see [Updating a volume to be replication-enabled](#) .

Use the following API and CLI commands to get the replication-status of a volume:

- **API:** [ibmcloud pi volume get](#). Set the value of the `VOLUME_ID` parameter to the primary volume ID.

Refer to the following table for the properties of a replication-enabled volume.

Property	Description
<code>consistencyGroupName</code>	Indicates the name of the consistency group when a volume is part of a volume group.
<code>masterVolumeName</code>	Indicates the name of the <code>master</code> volume in the storage. The storage controller auto-generates this name.
<code>mirroringState</code>	Indicates the mirrored state of the replication-enabled volume. This state is related to the current state of replication between the primary and the auxiliary volumes. For more information, see Status of volume groups .
<code>outOfBandDeleted</code>	Indicates the status of the replication-enabled volume when deleted. If the replication status is <code>disabled</code> on the primary volume and the auxiliary volume on the secondary location is not deleted within 24 hours, the status of the <code>outOfBandDeleted</code> property is set to <code>true</code> . In this state, you cannot perform any actions on the primary volume. When primary volumes are in this state, they are not billed.
<code>primaryRole</code>	Indicates the active volume in the primary and auxiliary volume. If this property value is set to <code>master</code> , the primary volume is the active volume in which you can perform I/O operations. If this property value is set to <code>aux</code> , the auxiliary volume is the active volume in which you can perform I/O operations. An inactive volume does not allow I/O operations to be performed on it. For a replication-enabled volume pair, the value of this property is the same.
<code>replicationEnabled</code>	Indicates the replication status of a volume. Set to <code>True</code> if the volume is replication-enabled.
<code>replicationStatus</code>	Returns the value of the replication status for a volume. If the returned value is <code>enabled</code> , the replication is active for the volume. If the returned value is <code>disabled</code> , the replication is inactive for the volume. If the returned value is <code>not-capable</code> , the volume is not replication-enabled and not associated with another volume on a different location.

Properties of replication-enabled volume and their descriptions.

Creating a volume group

Create a volume group (consistency group) to add replication-enabled volumes to it. You can create a volume group by using the following API or CLI commands:

- **API:** [ibmcloud pi volume-group create](#)

Set the value of the `VOLUME_ID` parameter to the primary volume ID. Provide a name for the primary volume group and the primary volume IDs to create the volume group.

A replication-enabled volume can be attached to a volume group when you create it. To attach the volume, provide the volume ID of the replication-enabled volume that is created on the primary location. Replication-enabled volumes can be attached to an existing volume group. For more information, see [Adding a replication-enabled volume to an existing volume group](#) .

Verifying the status of the volume group

Verify that the volume group is created successfully and it is in a `consistent_copying` state by using the following API or CLI commands:

- **API:** [ibmcloud pi volume-group storage-details](#)

Set the value of the `VOLUME_GROUP_ID` parameter to the primary volume group ID. Provide the primary volume group IDs to get the details.

Refer to the following table for the properties that you can use with the API and CLI commands for verifying the status of the volume group and its definitions.

Property	Description
----------	-------------

consistencyGroupName	Indicates the name of the replication consistency group that is created at the storage level. The name is same as the replication consistency group on the secondary location. The storage controller creates the name and not defined by the user.
auxiliary	Indicates whether the volume group is for the auxiliary volumes or for the primary volumes. If the volume group is for the auxiliary volumes on the secondary location, the returned value is true . If the volume group is for the primary volumes on the primary location, the returned value is false .
name	Indicates the name of the volume group that you provide when you created it in the primary location. In the secondary location, the name property is same as the consistencyGroupName property.
volumeIDs	Lists the volume IDs that are part of the volume group.
statusDescription	Assigned with a status, if any failure occurs when you add volumes to a volume group.
status	Indicates one of the following volume group states: available - ready to be managed error - encountered an error. Volume group is not manageable in this state and you can only perform a delete operation updating - update operation on the volume group is in progress creating - create operation on the volume group is in progress
replicationStatus	Indicates that replication status is active for the volume group. If this property value is set to disabled , replication is not active for the volume group.
state	Indicates the status of the consistency group.

Properties of a volume group and their descriptions.

The status of the volume group changes to **consistent_copying** state when the primary volume data is replicated to the auxiliary volume on the secondary location and is ready to continue the replication process. The Power Virtual Server manages the auxiliary volume after it is onboarded to the secondary location.

You must have the following information before you start onboarding the auxiliary volumes on the secondary location:

- CRN of the workspace in which the primary volumes are added
- Auxiliary volume names associated with the primary volumes that are replication-enabled. You can get the auxiliary volume names by querying the details of the primary volumes.

After you collect the CRN and auxiliary volume names, you can switch to the secondary location and the workspace in which the auxiliary volumes are located.

**Note:** The workspace must be part of the same IBM Cloud account ID as the primary location workspace.

Actions on the secondary location

Complete the following actions to onboard auxiliary volumes on the secondary location:

- [Onboarding an auxiliary volume](#)
- [Verifying the auxiliary volume for replication](#)

Onboarding an auxiliary volume

To manage the replicated volume on a remote location and perform volume recovery, onboard an auxiliary volume. For Power Virtual Server to manage the auxiliary volume, onboard the auxiliary volume to the secondary location.

**Note:** You must have the **editor** role access on the source and target Power Virtual Server workspaces to onboard the auxiliary volume. The source and target workspace must be created by using the same account ID.

Obtain the following information from the primary location to request for onboarding the auxiliary volumes on the secondary location:

- Have editor role access on both the primary and the secondary location workspaces
- Maintain the same IBM Cloud account ID on both primary and secondary location workspaces
- Fetch the cloud resource name (CRN) of the Power Virtual Server workspace instance in which the primary volumes are located (primary location)
- Fetch the auxiliary volume names from the **auxVolumeName** field of primary volumes in the primary location for onboarding

When onboarding an auxiliary volume, the following conditions apply:

- If a volume group of the auxiliary volume exists in the secondary location, the auxiliary volume is added to it.
- If a volume group of the auxiliary volume does not exist in the secondary location, the onboarding operation automatically creates a volume group. The volume group is associated with the auxiliary volume on the secondary location.

The onboarded auxiliary volume is added to this volume group. The volume group that is created on the secondary location is associated with the volume group on the primary location. The volume group on the primary location contains the primary volume that is associated with the auxiliary volume. To verify the replication status between the two primary and the auxiliary volumes, compare the consistency group name of the volume group on both the locations.



Note: Use the [ibmcloud pi workspace](#) CLI command to set the value of the workspace to the service workspace in which the auxiliary volumes are located. For example, `ibmcloud pi ws target AUXILIARY_WS_CRN` .

To onboard the auxiliary volume on the secondary site, use the following API or CLI commands:

- **API:** [ibmcloud pi volume onboarding create](#)

Specify the following parameters:

- source CRN: Specify the CRN parameter of the workspace in which the primary volume is located
- auxiliary volume: Specify the auxiliary volume name

Verifying the replication status of the auxiliary volume

An onboarding task ID is returned when you complete onboarding the auxiliary volumes on the secondary location. Use the task ID to check the status of the onboarding operation by using the following API and CLI commands:

- **API:** [ibmcloud pi volume onboarding get](#)

Refer to the following table for the properties that you can use with the API or CLI commands for verifying the replication status of the auxiliary volume and its definitions.

Property	Description
progress	Indicates the progress of the onboarding operation of the auxiliary volume in percentage
results	Contains the list of onboarded volume names or details of any failures that occurred during the onboarding operation of the auxiliary volume
status	Indicates the status of the volume onboarding operation. If the operation is successful, the return value is Success . If an error occurred during the onboarding operation, the return value is Failure .
Properties of an auxiliary volume and their descriptions.	

If the onboarding process of the auxiliary volume on the secondary location is successful, both the onboarded volume and volume group are present in the Power Virtual Server workspace on the secondary location. The resources in the secondary location have their own IDs. The IDs are different from the IDs and CRNs of the associated volumes on the primary location.

Obtain the status of the auxiliary volumes by using the auxiliary volume names. If the onboarding operation is successful, the auxiliary volume names are available. Verify the status of the auxiliary volumes by using the following API and CLI commands:

- **API:** [ibmcloud pi volume get](#)

Refer to the following table to verify whether the replication properties of the volume are as expected.

Property	Verifying the replication status
auxVolumeName	Matches the <code>auxVolumeName</code> value that is used to onboard the volume

auxiliary	Set to <code>true</code> because the volume is the auxiliary volume
consistencyGroupName	Matches the consistency group name of the volume group on the primary location
groupID	Returns the ID of the volume group
masterVolumeName	Matches the <code>masterVolumeName</code> value of the primary volume on the primary location
mirroringState	Set to <code>consistent_copying</code> state
primaryRole	Set to <code>master</code> as the primary volume is acting as the active volume
replicationEnabled	Set to <code>true</code>
replicationStatus	Set to <code>enabled</code>

Verifying the replication status of an auxiliary volume.

Using the `groupID` value of the auxiliary volume, query the details of the volume group. Verify whether the auxiliary volume is replication-enabled by using the following API and CLI commands:

- **CLI:** [ibmcloud pi volume-group get](#)
- **API:** [ibmcloud pi volume-group action](#). Set the `VOLUME_GROUP_ID` parameter to the auxiliary volume group ID, `--operation` flag to `stop`, and `--allow-read-access` flag to `True`.



Note: If you are using the CLI command to onboard an auxiliary volume, the service workspace must be set to the workspace in which the auxiliary volumes are located. For example, `ibmcloud pi ws target AUXILIARY_WS_CRN`.

Verify that the action that you perform on a volume group meets the current state of the volume group. Otherwise, an error message appears indicating that the volume group is not in the expected state.

When you perform an action on a volume group, an error is returned if the following state and actions on the volume group do not match:

- Stop a volume group when the `replicationStatus` is in the `disabled` state
- Start a volume group when the `replicationsStatus` is in the `enabled` or `available` state
- Reset a volume group that is not in the `error` state

Verifying the volume group for idling state

When you stop the volume group, the status of the consistency group is changed to `idling`, replication is disabled, and the auxiliary volumes allow I/O operations.

Get the volume group details to verify the status of the volume group by using the following API and CLI commands:

- **API:** [ibmcloud pi volume-group storage-details](#)

Fallback to the primary location

When the primary location is recovered, you can re-enable the primary volume group for replication. When the primary volume is ready for replication, you can fallback to the primary location. Complete the following steps to synchronize the data between the auxiliary volumes and primary volumes:

1. Turn off the virtual server instance on the secondary location
2. [Synchronize the I/O updates from auxiliary volume to primary volume](#)
3. [Stop the primary volume group to disable replication](#)
4. [Re-enable replication on the primary volume group](#)

After replication is re-enabled, you can start the virtual server instance and its workload in the primary location. When the primary server instance is the active instance, you can turn off the virtual server instance in the secondary location.

Synchronizing I/O updates from auxiliary volume to primary volume

To fallback to the primary volume, any I/O updates made to the auxiliary volume must be replicated to the primary volume. Start the primary volume group in the auxiliary mode to synchronize the data from auxiliary volume to primary volume.



Note: Use the [ibmcloud pi workspace](#) CLI command to set the service workspace to the workspace in which the auxiliary volumes are located. For example, `ibmcloud pi ws target AUXILIARY_WS_CRN`.

Use the following API or CLI commands to start the primary volume in auxiliary mode:

- **API:** [ibmcloud pi volume-group action](#). Set the `VOLUME_GROUP_ID` parameter to the auxiliary volume group ID, `--operation` flag to `start`, and set `--source` flag to `aux`.

Verify whether the `primaryRole` value of the volume group is set to `aux`. Monitor the state of the volume group until the replication of data from the auxiliary volume to the primary volume is completed. The change in the state of the primary volume group to `consistent_copying` value indicates that the data is synchronized from the secondary location to the primary location.

Stopping the primary volume group to disable replication

When you complete data replication from the auxiliary volume to the primary volume, it is time to fallback to the primary volume. To enable the primary volume, use the following API and CLI commands to stop the primary volume group and disable the replication:

- **API:** [ibmcloud pi volume-group action](#). Set the `VOLUME_GROUP_ID` parameter to primary volume group ID, `--operation` flag to `stop`, and `--allow-read-access` flag to `True`.

Wait for the `replicationStatus` parameter of the primary volume group to change to the `disabled` state. In this state, the primary and secondary locations do not replicate with each other. Therefore, I/O operations on the auxiliary volume do not replicate to the primary volume.

Reenabling replication on the primary volume group

To restart the primary volume group that is in the disabled state, use the `start` command. When you start the volume group to re-enable replication, the primary volumes of the primary volume group are the `master` volumes again.

- **API:** [ibmcloud pi volume-group action](#). Set the `VOLUME_GROUP_ID` parameter to primary volume group ID, `--operation` flag to `start`, and set `--source` flag to `master`.

Wait for the volume group to be replication-enabled and in a `consistent_copying` state. When replication is active, the I/O operations on the primary volume are replicated to the auxiliary volume on the secondary location.

Updating an existing volume as a replication-enabled volume

You can modify an existing volume to be replication-enabled if it is created in a storage pool that supports the replication process. To verify, query the details of the volume, get the details of the storage pool that contains the volume, and verify it with the list of storage pools that support replication. For more information about storage pools that support replication, see [Power Virtual Server storage pool that supports GRS](#).

Use the following API and CLI commands to query the volume details:

- **API:** [Perform an action on a volume group](#). Set the `VOLUME_ID` parameter to the primary volume ID. Using the API, query the volume details as defined in the following code:

```
$
Request Body:
{
  "replicationEnabled": true
}
```

- **CLI:** [ibmcloud pi volume-group action](#). Set the `VOLUME_ID` parameter to the primary volume ID, and query if the `--replication-enabled` flag is set to `True`.

Adding a replication-enabled volume to an existing volume group

You can add replication-enabled volumes to an existing volume group by using the following API and CLI commands:

- **API:** [ibmcloud pi volume-group update](#). Set the `VOLUME_GROUP_ID` parameter to the primary volume group ID, and set the `--add-member-ids` flag to the primary volume IDs that must be added to the volume group.

Disabling replication of a volume

When you disable replication of a primary volume, the associated auxiliary volume on the secondary location is deleted.

Failure to complete this procedure in the specified order might result in the loss of source volume data. First, you must complete the actions on the primary location and then complete the actions on the secondary location.

To disable replication of a primary volume, perform the following steps:

1. Actions on the primary location
 - a. [Remove the primary volume from the volume group](#)
 - b. [Delete the primary volume group, if empty](#)
 - c. [Disable replication for the primary volume](#)
 - d. [Verify whether the replication is disabled for the primary volume](#)
2. Actions on the secondary location



Note: You can perform these steps only if you complete the onboarding operation of the auxiliary volume.

1. [Remove the auxiliary volume from the volume group](#)
2. [Delete the auxiliary volume group, if empty](#)
3. [Delete the auxiliary volume](#)

Removing the primary volume from the volume group

Use the following API and CLI commands to remove the primary volume from the volume group:

- **API:** [ibmcloud pi volume-group update](#). Set the `VOLUME_GROUP_ID` parameter to the primary volume group ID, and set the `--remove-member-volume-ids` flag to the primary volume IDs that must be removed from the volume group.



Note: Use the [ibmcloud pi workspace](#) CLI command to set the service workspace to the workspace in which the primary volumes are located. For example, `ibmcloud pi ws target PRIMARY_WS_CRN`.

You can remove multiple replication-enabled volumes from a volume group.

Deleting an empty a primary volume group

Use the following API and CLI commands to delete the primary volume group if it is empty:

- **API:** [ibmcloud pi volume-group delete](#). Set the `VOLUME_GROUP_ID` parameter to a primary volume group ID that must be deleted.

Disabling replication for the primary volume

Use the following API and CLI commands to disable replication for the primary volume, if it is removed from the volume group:

- **API:** [ibmcloud pi volume action](#). Set the `VOLUME_ID` parameter to the primary volume ID that must be disabled and set the `--replication-enabled` flag to `False`.

Verifying whether replication is disabled for the primary volume

Use the following API and CLI commands to verify whether replication is disabled for the primary volume:

- **API:** [ibmcloud pi volume get](#). Set the `VOLUME_ID` parameter to the primary volume ID to query the replication status.

Removing auxiliary volume from the volume group

Use the following API and CLI commands to remove the auxiliary volume from the volume group:

- **API:** [ibmcloud pi volume-group update](#). Set the `VOLUME_GROUP_ID` parameter to an auxiliary volume group ID, and set the `--remove-member-volume-ids` flag to the auxiliary volume IDs that must be removed from the volume group.

Deleting an empty auxiliary volume group

Use the following API and CLI commands to delete the auxiliary volume group if it is empty:

- **API:** [ibmcloud pi volume-group delete](#). Set the `VOLUME_GROUP_ID` parameter to an auxiliary volume group ID that must be deleted.

Deleting an auxiliary volume

Use the following API and CLI commands to delete the auxiliary volume:

- **API:** [ibmcloud pi volume delete](#). Set the `VOLUME_ID` parameter to an auxiliary volume ID that must be deleted.

Changing the tier of a replication-enabled volume

You can change the tier of a replication-enabled volume if one of the following conditions is met:

- The volume is a primary volume and the `primaryRole` property of the volume is set to `master`.
- The volume is an auxiliary volume and the `primaryRole` property of the volume is set to `aux`.

Any attempt to change the tier of a replication-enabled volume fails, if one of the conditions are not met. The change in the tier of the replication-enabled volume does not synchronize with the paired volume.

You can change the tier of the replication-enabled volume by using the following API and CLI commands:

- API: [Perform an action on a Volume](#)
- CLI: [ibmcloud pi volume-action](#)

Modifying a replication-enabled primary volume

You can modify the attributes of a replication-enabled primary volume. To modify some of the properties of the primary and auxiliary volumes, you must perform related actions on the primary and secondary locations.

Changing the capacity of a primary volume

To change the capacity of a replication-enabled primary volume, complete the following steps:

1. Remove the primary volume from the volume group
2. Change the capacity of the primary volume
3. Add the primary volume back to the volume group

In the next 24 hours, the capacity of the auxiliary volume on the secondary location is changed.

For example, on the `dal10` and `wdc07` location pair, when you change the capacity of a replication-enabled primary volume on the primary location, the system changes the size of the auxiliary volume on the secondary location in the next 24 hours.

Use the following API and CLI commands to change the capacity of the replication-enabled primary volume:

-API: [ibmcloud pi volume update](#). Set the value in the `VOLUME_ID` parameter to the primary volume ID and set the `--size` value to be in GB.



Important: Do not resize a primary volume by disabling replication of the volume as it results in errors on the auxiliary volume on the secondary location.

Changing the bootable and shareable properties of a primary volume

To change the values of the `bootable` or `shareable` properties on the primary volume, use the following API or CLI commands:

-API: [ibmcloud pi volume update](#). Set the value of the `VOLUME_ID` parameter to the primary volume ID and set the value of the `--bootable` or `--shareable` flags to `True` or `False` value.



Note: A volume can be either shareable or bootable, but cannot be both.

After you change the `bootable` or `shareable` property values for a primary volume on the primary location, you must update the same property for the auxiliary volume on the secondary location.

Deleting a replication-enabled primary volume

To delete a volume or a replication-enabled primary volume, the status of the volume must indicate one of the following states: `available`, `error`, `error_restoring`, `error_extending`, or `error_managing`. In addition, you cannot delete the volume if it is in the `migrating` or `attached` state, belongs to a volume group, has snapshots, or if it is disassociated from its snapshots after a transfer.

To delete a primary volume, you must complete the actions on the primary and secondary locations:

- Complete the following actions on the primary location:
 1. [Remove the primary volume from its volume group, if the primary volume is associated with a volume group](#)
 2. [Delete the primary volume](#)
- Complete the following actions on the secondary location, if the auxiliary volume is onboarded on the secondary location:
 1. [Remove an auxiliary volume from its volume group, if the auxiliary volume is associated with a volume group](#)
 2. [Delete the auxiliary volume](#)



Note: If you do not delete the auxiliary volume from the secondary location, an out-of-band periodic check that occurs every 24 hours sets the auxiliary volume to the `ERROR` state. You cannot use the auxiliary volume when it is in the `ERROR` state. You can only delete the volume.

Removing the primary volume from its volume group

Use the following API or CLI commands to remove the primary volume from its volume group:



Note: Use the [ibmcloud pi workspace](#) CLI command to set the target workspace to the workspace in which the primary volumes are located. For example, `ibmcloud pi ws target PRIMARY_WS_CRN`.

- **API:** [ibmcloud pi volume-group update](#). Set the `VOLUME_GROUP_ID` parameter to the primary volume group ID, and set the `--remove-member-volume-ids` flag to the primary volume ID that must be removed from the volume group.

Deleting the primary volume

Use the following API and CLI commands to delete the primary volume:

- **API:** [ibmcloud pi volume delete](#). Set the `VOLUME_ID` parameter to a primary volume ID that must be deleted.

Deleting a replication-enabled auxiliary volume

If you delete an auxiliary volume, the associated primary volume is also deleted.

When you [disable the replication service on the primary volume](#) or [delete the primary volume](#), the replication relationship between the primary volume and the secondary volume is deleted in the storage backend. If the auxiliary volume on the secondary location is associated with a volume group, [remove the auxiliary volume from the volume group](#). Delete the auxiliary volume manually. If you do not delete the auxiliary volume from the secondary site, an out-of-band periodic check that occurs every 24 hours sets the auxiliary volume to an `ERROR` state. Confirm the status of the auxiliary volume by checking the `outOfBandDeleted` property of the auxiliary volume.

GRS impacts on other Power Virtual Server operations

The impacts of GRS on other Power Virtual Server operations are as follows:

- Image interface is not changed and replication is not supported for images
- Operation interface and the affinity policies for storage pool for virtual server instances are not changed. The replication and non-replication-enabled volumes support volume attachment and detachment.
- Snapshot and capture operations are allowed for a virtual server instance with replication and non-replication-enabled volumes when the volumes are from the same storage pool. The snapshot and capture operations are performed by using the non-replication-enabled volumes internally even though the volumes of a virtual server instance are replication-enabled.
- Cloned volume is replication-enabled when you clone a replication-enabled volume by default. During cloning, you can specify whether the cloned volume must be replication-enabled or not. If you want the cloned volume to be replication-enabled, specify the policy for the clone volume. You can use the following methods to clone a volume:
 - [Clone a volume by using CLI](#).
 - [Clone a volume by using Terraform](#).

Best practices for GRS

- Set the **Shareable** and **Bootable** flag explicitly on onboarded volumes, if required.
- Start the onboarding of auxiliary volumes only when the primary volumes and volume group are in a consistent copying state. You can get the volume details by using the [Power Virtual Server CLI](#)
- [Terraform](#).
- When you add or remove a primary or an auxiliary volume from a volume group from one location, perform the same operation from the other location to keep the data in sync.
- Delete the volumes from the primary and secondary location. Volumes are charged when you delete an auxiliary volume but fail to delete the primary volume.
- Use the primary location for all the volume operations and perform operations on the auxiliary volume on the secondary location only during failover.

Security and compliance

General Data Protection Regulation (GDPR)

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

The [General Data Protection Regulation](#) seeks to create a harmonized data protection law framework across the European Union (EU) and aims to give citizens back the control of their personal data. The GDPR imposes strict rules on those hosting and processing personal data anywhere in the world. Additionally, this regulation introduces rules that relate to the free movement of personal data within and outside of the EU.

With the GDPR, IBM® Power® Virtual Server clients can rely on the Power Virtual Server team's understanding and compliance with emerging data privacy standards and legislation. Power Virtual Server clients can also rely on IBM's wider ability to provide a comprehensive suite of solutions to assist businesses of all sizes with their own internal data governance requirements.

Online retention of client data

Client data is retained during the client's use of the workspace (that is, until the client's contract termination or expiration) by default. Additionally, clients can define shorter retention periods for some of their data within the workspace to align with their own organizational retention policies. All client data is removed from the workspace **1 business day** after the client's contract termination or expiration.

Backup retention of client data

Power Virtual Server does not provide backups of client data.

Destruction of client data

Power Virtual Server gives the client the ability to delete all online client data at their request or upon termination of the workspace. The secure destruction of all client data happens at the *end of life* of the associated infrastructure or cloud service.

Export of client data

IBM contractually commits, as required by various industry and government regulations to provide clients the capability to export their data, or that IBM provides the client with export copies of their data upon request. Power Virtual Server gives clients the ability to export data on their own.

Compliance certifications

IBM Power Virtual Server in [IBM data center](#)

The IBM® Power® Virtual Server provides programs and certifications that help you establish and strengthen compliance for a wide range of internationally recognized standards.

MeitY

The Ministry of Electronics and Information Technology (MeitY), Government of India, oversees cloud adoption for government and public sector organizations through its MeghRaj (GI Cloud) initiative. This program enables secure cloud services for public sector workloads across India.

Cloud service providers seeking to serve government customers must undergo MeitY's comprehensive empanelment process, which includes an independent assessment by the Standardization Testing and Quality Certification (STQC) Directorate. After successful empanelment, providers can participate in government and public sector cloud procurement programs.

IBM Power Virtual Server completed Cloud Service Provider (CSP) empanelment from MeitY for cloud services offered from the IBM Cloud Chennai, India Data Center in 2026. The empanelment confirms that Power Virtual Server complies with the stringent security and compliance standards set by the Government of India and is qualified for use by Indian public sector and government agencies.

For more information about MeitY, see [What is MeitY?](#).

BSI C5

The Cloud Computing Compliance Controls Catalogue (C5) is a cloud security framework developed by the German Federal Office for Information Security (BSI). C5 defines requirements that cloud service providers must meet to demonstrate a baseline level of security and transparency for cloud operations

and data processing.

BSI C5 reports are independent, third-party assessments performed by BSI-certified assessors. These reports provide assurance regarding data protection, information security, and cybersecurity controls. Power Virtual Server has successfully completed a BSI C5 assessment and maintains a current BSI C5 report.

For more information about C5, see [BSI Federal Office for Information Security](#) and [IBM Cloud® compliance: C5](#).

BSI C5 reports are available only upon request. To submit a request, use the [IBM Cloud Compliance Report Request Form](#).

ENS High

Power Virtual Server is Esquema Nacional de Seguridad (ENS) High certified. Spain's National Security Framework is a set of basic principles and minimum requirements for information security and data protection established by the Spanish government. The ENS was established by Royal Decree 311/2022, of 3 May. That royal decree was intended to modernize Spain's information security policy for both public sector agencies and private sector entities that provide services to the government agencies.

For more information about ENS High and the list of validated IBM Cloud services, see [Esquema Nacional de Seguridad \(ENS\)](#).

Protected B

The Canadian Centre for Cyber Security (CCCS) established the Government of Canada Cloud Service Provider (CSP) Information Technology Security (ITS) Assessment Program, a framework that is designed to evaluate the security of cloud services. The primary objective of this program is to assure the Government of Canada (GC) departments and agencies that cloud offerings meet the GC's public cloud security requirements for handling information and services that are classified up to the Protected B level.

Federal government contracts contain clauses with security requirements. These requirements specify the levels of security needed to safeguard sensitive information, assets, and work sites. Protected B is one of the classified security levels that applies to information or assets that, if compromised, could cause serious injury to an individual, organization, or government. Systems that handle Protected B data must meet enhanced security controls, covering access management, encryption, incident response, and physical protection to ensure the confidentiality, integrity, and availability of the information. For more information about Protected B, see [IBM Cloud® compliance: Protected B](#).

For more information about IBM Cloud Protected B Compliant services, see [GC Cloud Framework Agreements catalogue](#).

ISMAP

The Information System Security Management and Assessment Program (ISMAP) is a Japanese government program to assess the security of public cloud service providers (CSP). ISMAP approves independent third-party auditors who evaluate and register CSPs and their services to ensure they meet specific security requirements set by the Japanese government, enabling agency usage without the added burden and cost of agency-led assessments. Power Virtual Server is ISMAP compliant and is registered on the [ISMAP Website](#) (listing is in Japanese). The listing contains all Power Virtual Server ISMAP documents available for download in the Japanese language. For more information about IBM Cloud® ISMAP compliance certification, see [IBM Cloud® compliance: ISMAP](#).

HITRUST

The HITRUST CSF® is a framework for assessing and managing cybersecurity threats and safeguarding sensitive data such as protected health information (PHI). Power Virtual Server is HITRUST CSF® compliant. [Contact an IBM representative](#) to request HITRUST certification letters for Power Virtual Server with more detailed scope descriptions for IBM Cloud infrastructure services.

For more information, see [What is HITRUST?](#).

Financial Services® Validated

IBM designates IBM Cloud services as *IBM Cloud for Financial Services Validated* when the services are determined to materially implement the IBM Cloud framework for financial services control requirements.

For more information and the list of validated IBM Cloud services, see [IBM Cloud® for Financial Services®](#).

SOC

The System and Organization Controls (SOC) framework, developed by the American Institute of Certified Public Accountants (AICPA), is a standard for controls that protect information that is stored in the cloud. Certified public accountants (CPAs) audit cloud service providers (CSPs), resulting in internal control reports on the services provided by a service organization. SOC reports can help users assess and address the risks that are associated with an outsourced service.

SOC 1 is an audit of the internal controls at a service organization that is implemented to protect client-owned data involved in client financial reporting.

SOC 1 audits and reports are based on the Statement on Standards for Attestation Engagements (SSAE 18) and the International Standards for Assurance Engagements No. 3402 (ISAE 3402).

The following SOC reports are available for Power Virtual Server:

- SOC 1 type II
- SOC 2 type II
- [SOC 3 report](#)

SOC 1 and SOC 2 reports are available only upon request. To submit a request, use the [IBM Cloud Compliance Report Request Form](#).

ISO 27017:2015

The International Organization for Standardization (ISO) is an independent, non-governmental organization with a membership of 164 national standards bodies. ISO develops international standards that are voluntary, consensus-based, and market relevant. The goal is to ensure that products and services are safe, reliable, and of good quality.

Power Virtual Server provides services that are delivered from global data centers that are within the scope of the IBM Cloud™ IaaS ISO certification. The ISO certification covers a family of 4 standards as follows:

- ISO/IEC 27001:2013
- ISO/IEC 27017:2015
- ISO/IEC 27018:2019
- ISO/IEC 27701:2019

For more information, see [IBM Cloud® compliance: ISO 27017](#) and [Products in the scope of the IBM services information security management system \(ISMS\)](#).

PCI-DSS

To ensure consistent standards for merchants, the Payment Card Industry Security Standards Council established the Payment Card Industry (PCI) data security standards. These standards incorporate best practices to protect cardholder data, and they often require validation from a third-party Qualified Service Assessor (QSA). IBM is a Level 1 Service Provider for PCI DSS.

You are responsible for storing, processing, and transmitting cardholder data and might create cardholder data environments (CDEs) that can store, transmit, or process cardholder data by using IBM Cloud Platform services. You can use the IBM Cloud Attestation of Compliance (AOC) when you seek your own PCI DSS certifications. It is your responsibility to document and operate CDEs and applications that are built by using IBM Cloud Platform services in a PCI DSS-compliant manner.

Contact an [IBM representative](#) to request a PCI DSS Attestation of Compliance (AOC) or a Service Responsibility Matrix (SRM) guide for Power Virtual Server.

You can build PCI-DSS compliant environments and applications by using IBM Cloud. For more information, see [IBM Cloud PCI DSS Guidance](#).

PCI-DSS reports are available only upon request. To submit a request, use the [IBM Cloud Compliance Report Request Form](#).

HIPAA

The US Health Insurance Portability and Accountability Act (HIPAA) and the Health Information Technology for Economic and Clinical Health (HITECH) Act define standards for handling electronic healthcare transactions and information. Power Virtual Server on IBM Cloud is HIPAA-ready. You can build HIPAA-ready environments and applications by using Power Virtual Server. For more information, see [IBM Cloud® compliance: HIPAA](#).

If your company is a covered entity as defined by HIPAA, you must enable the HIPAA Supported setting if you run sensitive workloads that are regulated under HIPAA and the HITECH Act. By using this setting, you can filter on HIPAA Enabled services in the catalog, indicate to IBM that your account stores protected health information (PHI), and digitally accept the IBM Business Associate Addendum for covered entities. For more information, see [Enabling HIPAA support for your account](#).

ISO/IEC 20000

ISO 20000 is the international standard for [IT service management \(ITSM\)](#), developed jointly by the ISO and the International Electrotechnical Commission (IEC).

ITSM is the practice of delivering IT services that enable an organization to meet the needs of its employees and customers, and achieve its business goals. ISO 20000 outlines requirements, best practices, benchmarks, and guidance for planning, designing, implementing, maintaining, and continually improving a service management system (SMS).

Power Virtual Server is ISO 20000 certified alongside other IBM Cloud services. For more information, see the certificate at [ISO/IEC 20000 Scope of certification](#).

For more information about IBM Cloud ISO 20000 certification, see [ISO/IEC 20000](#).

European Union support

IBM® Power® Virtual Server has additional controls in place in the European Union (EU). The controls ensure that the access to your Power Virtual Server resources is restricted and controlled by the IBM Cloud® support team in the EU region. For more information about how to enable the setting for your IBM Cloud account, see [Enabling the EU Supported setting](#). For more information about the support process, see [Requesting support](#) for resources in the European Union.

Integrating Power Virtual Server with IBM Cloud Key Management Services

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**



Note: **Client location** To integrate IBM® Power® Virtual Server with IBM Cloud key management services, establish a connection from your virtual machine to IBM Cloud.

IBM provides two Cloud key management services that integrate with Power Virtual Server workloads:

1. [IBM Cloud® Hyper Protect Crypto Services \(HPCS\)](#) is a dedicated key management service and hardware security module (HSM) based on IBM Cloud. You can integrate HPCS with Power Virtual Server to securely store and protect encryption key information for AIX and Linux.
2. [IBM Key Protect](#) is a full-service multi-tenant encryption solution that allows data to be secured and stored in IBM Cloud™ with the envelope encryption techniques. You can integrate Key Protect with Power Virtual Server to securely store and protect encryption key information for AIX and Linux.

Using Hyper Protect Crypto Services (HPCS) and Key Protect for AIX



Note: HPCS and Key Protect are supported by AIX 7.3 TL1 for AIX logical volume encryption.



Important: Power-AIX integration for PKCS11/TDE integration for Oracle or Db2 workloads is not available currently. No impact to the volume-level encryption for AIX, Power-Linux with HPCS.

You can use Power Virtual Server to integrate with HPCS and Key Protect to use encryption on AIX file systems with `keysvrmgr` and `hdcryptmgr` command.

The `keysvrmgr` command manages the Object Data Manager (ODM) database entries that are associated with the encryption key server when the logical or physical volume uses the key server key-protection method for encryption. For more information, see [keysvrmgr Command](#).

The `hdcryptmgr` command helps to manage the cryptographic management of logical volumes (LV) and physical volumes (PV). For more information, see [hdcryptmgr Command](#).

Using Hyper Protect Crypto Services (HPCS) or Key Protect for Linux

You can use Power Virtual Server to integrate with HPCS or Key Protect to protect Linux Unified Key set up (LUKS) encryption keys from being compromised. Either key management service can act as the single point of control to enable or disable access to data across the enterprise. It is done by successively wrapping encryption keys, with the ultimate control being a master key that resides in a hardware security module (HSM).

For more information, see [Protect LUKS encryption keys with IBM Cloud Hyper Protect Crypto Services and Key Protect](#).

Additional support for configuring Hyper Protect Crypto Services or Key Protect

For any additional information and assistance on HPCS or Key Protect for AIX or Linux, [contact IBM](#).

Integrating Power Virtual Server with IBM Cloud Security and Compliance Center Workload Protection

IBM Cloud® Security and Compliance Center Workload Protection offers a comprehensive suite of security solutions to help your organization secure its cloud environments. Compliance Manager Workload Protection centrally manages your organization's security, risk, and compliance with regulatory standards and industry benchmarks. For more information about Compliance Manager Workload Protection, see [Getting started with IBM Cloud Security and Compliance Center Workload Protection](#).

In sectors such as financial services, continuous compliance in the cloud environment is crucial to protect customer and application data. Cloud Security Posture Management (CSPM) is one of the key features of the Compliance Manager Workload Protection service. When this feature is enabled in your workspace, the CSPM ensures that automatic compliance checks are integrated in your development workflow to mitigate such risks on a daily basis.

IBM Cloud® Security Posture Management (CSPM)

CSPM accelerates hybrid platform adoption by verifying security and regulatory compliance with automated compliance checks for IBM Cloud Framework. The automated compliance checks for Financial Services, Digital Operational Resilience Act (DORA), CIS IBM Cloud Foundations Benchmark, PCI, and many other industry-related or best practice standards. For more information, see [About IBM Cloud Security Posture Management \(CSPM\)](#). CSPM helps with:

- Continuous validation of compliance
- Vulnerability prioritization
- Detection and response to runtime threats
- Forensics and incident response

You can enable the CSPM feature in your new and existing Power Virtual Server workspaces. CSPM provides a unified platform to manage security and compliance across hybrid and multicloud environments and services.

Integrating CSPM with Power Virtual Server workspaces

To integrate Power Virtual Server with Cloud Security and Compliance Center, enable Cloud Security Posture Management (CSPM) feature in your existing or new workspaces:

- [Enabling CSPM in a new workspace](#)
- [Enabling CSPM in an existing workspace](#)

Enabling CSPM in a new workspace

To enable CSPM in a new Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Create a workspace**.
4. Select **IBM data center** as the location type.
5. Select an IBM data center from the **Location** list and click **Continue**.
6. In the Details section, provide a name for the workspace and select the resource group from the **Resource group** drop-down list. You can optionally provide User tags and Access management tags for the workspace.
7. Click **Continue**. The selected workspace details are displayed on the Summary page.
8. In the Integrations (Optional) section, the **Cloud security posture management** toggle switch is enabled by default. To disable CSPM, set **Cloud security posture management** to off.
9. Click **Finish**. The selected workspace details are displayed on the Summary page.
10. Select the **I agree to the Terms and conditions** checkbox and click **Create**.



Important: Integration costs are based on usage and vary based on the hourly consumption for nodes and virtual machines. You can review the estimated cost on the Summary page. To review the cost associated with CSPM, see [Security and Compliance Center Workload Protection](#) in **IBM Cloud catalog**.

Enabling CSPM in an existing workspace

To enable Monitoring or CSPM in an existing Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud](#) Power Virtual Server user interface.
2. Click **Workspaces** in the left navigation menu.

- 3. Select the workspace for which you want to enable Monitoring or CSPM. The Workspace details pane is displayed.
- 4. To enable CSPM, click **Add CSPM** in the Integrations section. The Add cloud security posture management pane is displayed with the predefined SCC Workload Protection instance, Trusted profile, and App Configuration instance.
- 5. Click **Edit** to change the name, location, and plan for the CSPM instance, and click **Save**.
- 6. Click **Create**.

Workload Protection agent in Linux® and AIX® on Power Virtual Server

After you provision an instance of the Compliance Manager Workload Protection service in IBM Cloud, you can deploy the Workload Protection agent on your Linux or AIX hosts on IBM® Power® Virtual Server. To provision an instance of Workload Protection, see [Managing the Workload Protection agent in Linux on Power Virtual Server](#) or [Managing the Workload Protection agent on AIX on Power Virtual Server](#) .

Workload Protection provides the following features to protect your stand-alone Linux or AIX hosts on Power Virtual Server.

Feature	On Linux hosts	On AIX hosts
Posture management	Scans host configuration files for compliance and benchmarks such as CIS Linux Benchmark	Scans host configuration files for compliance and benchmarks such as CIS AIX Benchmark
Host scanning	Scans host packages, detects associated vulnerabilities, and identifies the resolution priority based on available fixed versions and severity	
Threat detection and response	Identifies threats and suspicious activity based on application, network, and host activity by processing syscall events and investigates with detailed system captures	
Workload Protection agent features		

For more information about managing and deploying the Workload Protection agent on your Linux hosts, see [Managing the Workload Protection agent in Linux on PowerVS](#).

For more information about managing and deploying the Workload Protection agent on your AIX hosts, see [Managing the Workload Protection agent in AIX on PowerVS](#).

Use cases

Connecting IBM Power Virtual Server to Microsoft Azure with Megaport

This topic provides architectural solution guidelines for running IBM® Power® Virtual Servers and IBM Cloud with private connectivity between IBM Cloud and Microsoft Azure by using Megaport.

The solution is based on publicly available documentation from:

- [IBM Cloud](#)
- [Microsoft Azure](#)
- [Megaport](#)

Solution architecture overview

This section provides an overview of the solution architecture, and includes an overview of the solution components and cloud services that are used.

This solution provides private network connectivity between IBM Power Virtual Server and Microsoft Azure for high-performance peer-to-peer connection of cross-cloud workloads. The architecture uses private connection services through IBM Cloud Direct Link, which is coupled with Microsoft Azure ExpressRoute, and joined by the Megaport Cloud Router.



Power Virtual Server workspace

A container of all the virtual machines, storage volumes, network configurations, and so on, at a specific geographic region.

IBM Cloud Transit Gateway

Provides interconnectivity across various IBM Cloud platforms. You can create single or multiple transit gateways to connect VPCs, Power Virtual Server workspaces, or IBM Cloud classic infrastructure.

IBM Cloud Direct Link

A routed, OSI L3 service that offers a direct connection to the IBM Cloud Private network backbone, with low latency and uses BGP as the routing protocol.

Megaport Cloud Router (MCR)

A managed virtual router service that enables direct networking between endpoints on the Megaport network, which can be purchased without the user having a physical presence in that data center.

Virtual Cross Connect (VXC)

A connection between any two endpoints on the Megaport network. Most connections support capacity ranging from 1 Mbps to 10 Gbps.

Microsoft Azure ExpressRoute (ER)

Extends your Client location networks into the Microsoft cloud over a private connection with the help of a connectivity provider.

Microsoft Azure virtual network gateway

A service that enables you to establish connectivity between your **virtual network (vNet)** in Microsoft Azure and your Client location infrastructure.

Building the interconnectivity

To provide private connectivity to IBM Cloud, you can use IBM Cloud Direct Link (DL) and Transit Gateway (TGW). In Microsoft Azure, the equivalent offerings are called ExpressRoute and Virtual network gateway. In addition to these core cloud services, you need a connectivity provider. This example uses Megaport offerings, Megaport Cloud Router (MCR), and Virtual Cross Connects (VXC). This solution focuses on providing connectivity with IBM Power Virtual Server, but you might have other workload needs. IBM Cloud VPC, IBM Cloud classic infrastructure, and various VMware Solutions integrate in a similar way by using IBM Cloud Transit Gateway as the core routing service between the platforms.

The following diagram shows a high-level overview of the solution and explains how the solution components are used.



1. **IBM Cloud Transit Gateway** provides interconnectivity across various IBM Cloud platforms. You can create single or multiple transit gateways to connect VPCs, Power Virtual Server workspaces, or IBM Cloud classic infrastructure. New networks connected to a TGW are automatically made

available to every other network connected to it. You can also provision Unbound GRE tunnels to your TGW that can be used to connect various network appliances or either self-managed or IBM-managed VMware Cloud Foundation (VCF) instances to the TGW.

- 2. **IBM Cloud Direct Link** service is a routed, OSI L3 service. It offers a direct connection to the IBM Cloud Private network backbone, with low latency and uses BGP as the routing protocol. IBM Cloud Direct Link can be attached to most IBM Cloud IaaS platforms, or you can attach it to a Transit Gateway where it can provide private interconnectivity to all connections attached to a TGW. IBM Cloud does not provide built-in connection redundancy as part of the product. To establish redundant connectivity, you must acquire two connections on diverse cross-connect routers (XCRs) and configure BGP on each IBM Cloud Direct Link.
- 3. **Megaport Cloud Router (MCR)** is a managed virtual router service that enables direct networking between endpoints on the Megaport network. MCR can be purchased as a stand-alone service to route traffic between different cloud environments without the user having a physical presence in that data center. It can also be connected to user Ports to route pertinent traffic back to a physical location. BGP is used as the routing protocol to learn and advertise routes to its connections.
- 4. **A Port** is the physical point of connection between your organization’s network and the Megaport network. After you have configured a Port or an MCR, you can create **Virtual Cross Connects (VXCs)** to connect to services on the Megaport network. A VXC is a private point-to-point Ethernet connection between your A-End Port (MCR) and a B-End destination (CSPs, Ports, Megaport Marketplace services, or IX). In this scenario, the B-End destinations are IBM Cloud and Microsoft Azure.
- 5. **Microsoft Azure ExpressRoute (ER)** extends your Client location networks into the Microsoft cloud over a private connection with the help of a connectivity provider. With ExpressRoute, you can establish OSI L3 connections to Microsoft cloud services, such as Microsoft Azure and Microsoft 365. Redundancy is built in ExpressRoute in every peering location for high availability. ExpressRoute uses BGP as the routing protocol to advertise and learn routes from Microsoft Azure to the connected router, and to/from MCR in this example.
- 6. **Microsoft Azure virtual network gateway** is a service that enables you to establish connectivity between your **virtual network (vNet)** in Microsoft Azure and your Client location infrastructure. It acts as a router between your vNets and your Client location network (in this case, IBM Cloud).

Configuring the solution architecture

The following sections provide links to external resources with the detailed steps and examples on what is needed to configure and validate end-to-end connectivity:

- 1. [Configure IBM Power Virtual Server environment.](#)
- 2. [Configure a Megaport Cloud Router \(MCR\).](#)
- 3. [Configure IBM Cloud Direct Links and Virtual Cross Connections \(VXC\).](#)
- 4. [Configure Microsoft Azure ExpressRoute circuits.](#)

See the architecture diagram for [private connectivity between IBM Power Virtual Server and Microsoft Azure](#) to get a high-level information.

This solution uses the following networks as shown in the table:

Platform or service	Network details
Power Virtual Server	10.100.10/24
Microsoft Azure	10.200.10/24
Client location networks	10.0.10/24

Network details

Configuring the Power Virtual Server environment and Transit Gateway in IBM Cloud

IBM Cloud Account configuration

Obtaining IBM Cloud Account ID

To establish private connectivity by using cloud connectivity providers, you need to obtain the IBM Cloud Account ID. This 32-character unique identifier identifies your IBM Cloud account that owns the IBM Cloud Direct Link service.

To obtain your IBM Cloud Account ID, follow the instructions that are provided in the [IBM Cloud Docs pages](#).

Configuring the Power Virtual Server

Provisioning a workspace and instance

Power Virtual Server resources reside in IBM data centers with dedicated networking and Storage area network (SAN)-attached Fibre Channel storage. You

can choose one of the regions that is listed in the specifications that are nearest to your data center. In the data centers, the Power Virtual Servers are separated from the rest of the IBM Cloud servers with separate networks and direct-attached storage, and you can use IBM Cloud interconnectivity offerings to provide connectivity to IBM Cloud infrastructure or private cloud environments.

To provision and configure your IBM Power Virtual Server environment in IBM Cloud, you need to create the following IBM Power Virtual Server items:

- A workspace
- A private network subnet
- An SSH Key
- An instance

For more information, see IBM Cloud docs pages [how to configure networking](#) and [how to create Power Virtual Server](#).

Configuring IBM Cloud Transit Gateway

Provisioning a Transit Gateway

Transit gateway provides routed L3 private interconnectivity to your Power Virtual Server instances to other IBM Cloud IaaS platforms or your private networks via Transit Gateway that is attached to IBM Cloud Direct Link. To provision a Transit Gateway, you need to:

- Create an IBM Cloud Transit Gateway Instance

For more information on how to create a Transit Gateway, see the [IBM Cloud Docs pages](#).

Adding Power Virtual Server environment to Transit gateway

Before your Power Virtual Server workspace can use Transit Gateway, it must be added as a Transit Gateway connection. If required, you might optionally configure prefix filtering, if you do not want to advertise all Power PV subnets to the selected Transit Gateway.

To configure a Transit Gateway, you need to:

- Create a connection between TGW and your IBM Power Virtual Server workspace instance
- Configure prefix filtering on the Power Virtual Server Transit Gateway connection

For more information on how to add a Power Virtual Server connection to your Transit Gateway, see [IBM Cloud Docs pages](#).

Configuring hybrid platform connectivity with Megaport

Configuring Megaport portal

In this section, you configure the connectivity between Megaport and IBM Cloud. You need to:

- Create a Megaport Cloud Account
- Provision a Megaport Cloud Router
- Create a Megaport Virtual Cross Connect
- Create IBM Cloud Direct Link Authorization of Megaport VXC connection

This section provides an overview of the configuration steps. For a more detailed explanation, refer to [Megaport documentation](#).

Creating a Megaport account

The first step is to log in to the Megaport portal: <https://portal.megaport.com/login> and configure your account, or you can skip to the next step “ **Creating a Megaport Cloud Router**”.

Setting up a new Megaport Account (<https://docs.megaport.com/setting-up/>)

1. Create a company profile
2. Add technical support contact details
3. Set up a billing account and assign a user role to a finance contact for invoice purposes.
4. After you complete these steps, you can provision new services, test connectivity, and verify service pricing.

Creating a Megaport Cloud Router

The Megaport Cloud Router (MCR) is a managed virtual router service that establishes Layer 3 connectivity on the worldwide Megaport software-defined network (SDN). MCR instances are preconfigured in data centers in key global routing zones. An MCR enables data transfer between multi-cloud or hybrid

platform networks, network service providers, and cloud service providers.

An MCR joins two or more independent Virtual Cross Connect (VXC) services into a single routing domain, providing connectivity between all the VXCs attached to the MCR. In this example, two VXCs are needed; one for IBM Cloud Direct Link and one for Microsoft Azure ExpressRoute. VCXs are configured after the MCR is provisioned.

To create an MCR, you need to make a few decisions for the connectivity. The following table is a worksheet of some of the configuration options that you customize based on your solution.

Configurable Field	Description of Option	Example Choice
MCR Location	Geo Location to be used	Equinix EQ1
Rate Limit	What network speed you would like to order	1 Gbps to 10 Gbps
MCR Name	Specify a name for the MCR that is easily identifiable	IBM-Azure-MCR
Minimum Term	Select your term length	- No Minimum Term 12 Months 24 Months 36 Months
BGP State	Select whether BGP connections are enabled or shut down by default	Enabled

Megaport Cloud Router Worksheet

For a detailed explanation of the steps required, refer to [Megaport documentation](#).

To find details for IBM Cloud Direct Link connectivity locations, see the [IBM Cloud Docs pages](#).

Creating a connection from MCR to IBM Cloud Direct Link

IBM Direct Link lets you seamlessly create private connections between your Client location and cloud resources. Direct Link Connect with Megaport extends your organizations data center network to public clouds to offer better performance, higher throughput, and security to IBM Cloud environments.

Creating a Megaport connection to IBM Direct Link

To set up IBM Direct Link with Megaport, you need to create a Virtual Cross Connect (VXC) from your MCR in the Megaport Portal to the IBM Direct Link cloud location.

Design considerations when setting up your Direct Link to IBM Cloud:

Redundancy

The megaport supports redundant connections. In each of the Direct Link peering locations, Megaport established two diverse Direct Links with IBM Cloud. In the [Megaport Portal](#), the Ports are noted as Port 1 and Port 2. A redundant IBM Cloud Direct Link configuration requires two or more VXCs.

Local and Global routing options

The Local Routing option provides access to data centers with the same three-letter prefix as the Point of Presence (such as DAL, AMS, and MEL). The Global Routing option is required as an add-on to reach data centers outside of those locations. For more information on local and global routing options, see the [IBM cloud documentation](#).

Port speed

Port speed options for IBM Direct Link Connect are 50 Mbps, 100 Mbps, 200 Mbps, 500 Mbps, 1 Gbps, 2 Gbps, or 5 Gbps.

Creating a VXC to the IBM Direct Link Connect

A VXC is a point-to-point Layer 2 circuit between two endpoints that is mapped with a VLAN ID on each end. You can order VXCs to reach any destination in the Megaport network just like the VXCs used with the physical Ports.

This section describes how to connect to IBM Direct Link from an MCR. For more information on connecting from a Port, see [Connecting to IBM Cloud Direct Link](#).

You can also watch this [Private Connectivity to IBM Cloud](#) video, which shows how quick and easy to connect to IBM Cloud using Megaport.

See the documentation outlined in [Megaport documentation](#), by using the worksheet fields that you entered previously.

After your connection is added, the next steps are to authorize on the IBM Cloud Console.

Configurable Field	Description of Option	Example Choice
Cloud Connection	Choose your cloud provider	IBM Cloud
VXC Connection name	Unique name	IBM-Cloud-MCR-1
Rate Limit	The speed of your connection in Mbps	
Minimum Term	Select your term length	- No Minimum Term 12 Months 24 Months 36 Months
IBM Account ID	Your IBM Cloud account ID. Learn more on the Managing your account, resources, and access documentation	The account ID is a 32-character, unique account identifier.

Megaport Virtual Cloud Connection: IBM Cloud Worksheet

IBM Cloud Direct Link Configuration

Before following to this step, make sure that all services are being properly provisioned in Megaport.

Approving Megaport Direct Connection in IBM Cloud Portal

After the Megaport VXC has been successfully ordered in the previous step, you will be able to see the IBM Cloud Direct Link connection ready for approval. Take the following steps.

1. Log in to your [IBM cloud](#) account.
2. Click menu on the upper left of the page, then click **Interconnectivity**.
3. Click **Direct Link** in the navigation pane.
4. Click the name of the direct link that you want to change the connection.
5. Click the **Review** button.
6. Specify the resource group.
7. Choose your routing (local).
8. Click **Next**, then **Next** again on AS prepends and route filtering.
9. Click **Create** and wait for the status to change to **“Provisioned”**

Adding Direct Link Connection to IBM Cloud Transit Gateway

After the Megaport and IBM Cloud Direct Link have been provisioned successfully in the previous steps, the next step is to add the IBM Cloud Direct Link to the IBM Cloud Transit Gateway.

1. Log in to your [IBM cloud](#) account.
2. Click menu on the upper left of the page, then click **Interconnectivity**.
3. Click **Transit Gateway** in the navigation pane.
4. Click the name of the **Transit Gateway** that you want to change the connection type for.
5. Click Create Connection.

- a. In the drop-down list menu, select **Direct Link**.
6. Select the Direct Link connection and Connection name.
7. Click **Create**, and then wait for the **Available Status**.

Repeat the add connection steps to add Power Virtual Server

1. Click Create Connection.
2. In the drop-down list menu, select **Power Systems Virtual Server**.
3. Select the Power Virtual Server Location and Connection name.
4. Click Add, and then wait for the **Available Status**.

Verifying IBM Cloud and Megaport are sharing routing information

Verification of the network routes being shared can be done both in IBM Cloud and Megaport.

For IBM Cloud from the Transit Gateway interface, select the Routes tab. This interface displays the Power Virtual Server network subnet. Check for the network of Power Virtual Server “azure-power-test-1” and the learned routes.

1. Log in to your [IBM cloud](#) account.
2. Click menu on the upper left of the page, then click **Interconnectivity**.
3. Click **Transit Gateway** in the navigation pane.
4. Click the name of the **Transit Gateway** whose routes you want to check.
5. Click the **Routes** tab and generate a route report.
6. On the **Routes** tab, filter the Power Virtual Server allocated subnet ranges to see the learned routes.
7. On the **BGP** tab, filter the Power Virtual Server allocated subnet ranges to see the BGP AS-path.

For more information about Route Reports on Transit Gateway, see [IBM Cloud Docs pages](#).

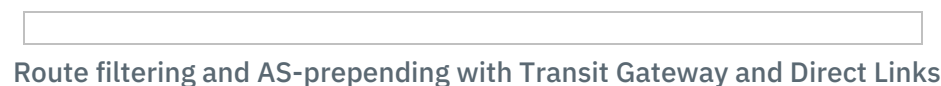
You can also check the routing and BGP tables by using Megaport Cloud Router Looking Glass.

1. Log in to your [Megaport](#) account.
2. Click Services on the upper tabs.
3. You can filter based on your MCR name.
4. Click the **Megaport Cloud Router Looking Glass** icon on the MCR.
5. Select your **BGP session towards IBM Cloud Direct Link** and click **Neighbor routes**. Select **Received** to see the routes the MCR has learned from IBM Cloud Direct Link. You might optionally filter based on the specific prefix, if you have several advertised routes.

For more information about MCR Looking Glass, see [Megaport documentation](#).

Performing route filtering and AS-path prepends

In some cases, you might need to filter some routes to be advertised to Megaport from IBM Cloud Transit Gateway connections, or you might want to prioritize your preferred routing paths with AS-prepends, if you have multiple Direct Links attached. The following diagram depicts an overview of how route filtering and AS-prepends work in the IBM Cloud interconnectivity solutions.



For route filtering, you can use either use Transit Gateway’s route filtering capabilities or apply route filtering on Direct Link, depending on where you want to filter the routes. With Transit Gateway, you can only filter learned prefixes (import) from the connections. With Direct link, you can apply route filters for both learned (import) and advertisements (export). You can also attach GRE tunnels to a Transit Gateway. With GRE tunnels, the route filtering is done with BGP from the connecting router/gateway.

For more information on route-filtering, see the IBM Cloud Docs pages for [Transit Gateway](#) and [Direct Link](#).

For AS-prepends, you can use Direct Link’s AS-prepend capabilities to alter the length of the learned (import) or advertised (export) AS-paths. When using GRE tunnels, it is recommended to connect to all Transit Gateway routers in the multizone region, and you might want to control your preferred GRE tunnels with BGP metrics/attributes. With GRE tunnels, the AS-prepends are done with BGP from the connecting router/gateway.

For more information AS-prepending, see the following IBM Cloud Docs pages for [Direct Link](#). For GRE tunnels, see the [Transit Gateway pages](#) in IBM Cloud Docs.

Creating ExpressRoute through Microsoft Azure portal

Complete the following steps to create and provision a Microsoft Azure ExpressRoute circuit.

1. Sign in to the [Microsoft Azure portal](#).
2. Create a new ExpressRoute circuit. On the Microsoft Azure portal menu, select **+ Create a resource**. Search for **ExpressRoute** and then select Create.
3. On the Create ExpressRoute page, provide the **Resource Group**, **Region**, and **Name** for the circuit. Then, select **Next: Configuration >**.
4. Enter the required values for the ExpressCircuit. For more information, see [Microsoft Azure documentation](#).
5. Select **Review + create** and then select **Create** to deploy the ExpressRoute circuit.

After provisioning is complete, view the circuits and properties.

1. Search for **ExpressRoute circuits** in the search box at the top of the portal.
2. Select your circuit and view the properties of the circuit by selecting it.
3. On the Overview page for your circuit, find the **Service Key**. Provide the service key to the service provider to complete the provisioning process. The service key is unique to your circuit.

For more information about Microsoft Azure ExpressRoute, see [Microsoft Azure documentation](#).

Creating a connection from the Megaport Cloud Router to Microsoft Azure ExpressRoute

Next, you can create a VXC from an MCR to Microsoft Azure ExpressRoute through the [Megaport Portal](#). You can create the VXC to Microsoft Azure from the MCR and establish either private or Microsoft peering.

To connect to ExpressRoute using MCR, you need an ExpressRoute service key that is obtained from the Azure Resource Manager (ARM) portal. Follow the steps in the Microsoft topic [Tutorial: Create and modify an ExpressRoute circuit](#) to get this key.

To create a VXC from a Megaport Cloud Router (MCR) to Microsoft Azure:

1. In the [Megaport Portal](#), go to the **Services** page and select the MCR you want to use.
2. Add a new **VXC connection** for the MCR. Click **+Connection**, click **Cloud**, and click **Azure ExpressRoute**.
3. Provide the Azure Service Key (from Microsoft Azure Portal, see [Microsoft Azure documentation](#)).

For more information, see [Megaport documentation](#).

Configuring peering and virtual network gateway through the Microsoft Azure portal

Creating and modify peering for an ExpressRoute circuit

Create a routing configuration for an Azure Resource Manager ExpressRoute circuit by using the Microsoft Azure portal. Follow the steps for Microsoft Azure private peering. You can also check the status, update, or delete and deprovision peerings for an ExpressRoute circuit. Follow the documentation steps in [Microsoft Azure documentation](#).

Configuring a virtual network gateway for ExpressRoute

Add a virtual network gateway for a virtual network (VNet). Follow the documentation steps in [Microsoft Azure documentation](#). For more information about virtual network gateways and gateway configuration settings for ExpressRoute, see [About virtual network gateways for ExpressRoute](#).

Connecting a virtual network to ExpressRoute circuits

Create a connection to link a virtual network to Microsoft Azure ExpressRoute circuits by using the Microsoft Azure portal. Follow the documentation steps in [Microsoft Azure documentation](#).

Verifying Microsoft Azure and Megaport are sharing routing information

Check the routing and BGP tables by using MCR Looking Glass.

1. Log in to your [Megaport](#) account.
2. Click **Services** on the upper tabs.
3. You can **filter** based on your MCR name.
4. Click the **MCR Looking Glass** icon on the MCR.
5. Select your **BGP session toward IBM Cloud Direct Link** and click **Neighbor routes**. Select **Received** to see the routes MCR has learned from IBM Cloud Direct Link. You might optionally filter based on the specific prefix, if you have several advertised routes.

For more information about MCR Looking Glass, see [Megaport documentation](#).

Verification of end-to-end connectivity

Checking routing tables and BGP

Check the routing and BGP tables by using MCR Looking Glass.

1. Log in to your [Megaport](#) account.
2. Click the **Services** tab.
3. You can filter based on your MCR name.
4. Click the **MCR Looking Glass** icon on the MCR.
5. Click the **Routes Table** tab and filter the routes that are assigned to IBM Power Virtual Server.
6. Click the **Routes Table** tab and filter the routes that are assigned to Microsoft Azure.
7. Click the **BGP Table** tab and filter the routes that are assigned to IBM Power Virtual Server.
8. Click the **BGP Table** tab and filter the routes that are assigned to Microsoft Azure.

For more information about MCR Looking Glass, see [Megaport documentation](#).

Connectivity tests

When the routes are correctly advertised between IBM Cloud, Megaport, and Microsoft Azure, you should have an end-to-end connectivity path between the platforms. You can verify connectivity with common tools such as ping or netcat.

From a Microsoft Azure virtual machine, use the following ping command to test the connectivity to a default gateway:

```
$ # ping 10.100.10.1
```

From a Microsoft Azure virtual machine, use the following netcat command to test the connectivity to a Power Virtual Server instance with IP address 10.100.10.10 to TCP port 22:

```
$ # nc -vz 10.100.10.10 22
```

Migrating Db2 on AIX to Power Virtual Server

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

IBM® Power® Virtual Server provides an excellent infrastructure and environment to relocate your Power on AIX assets to high performance virtual computing infrastructure.

As a DB2 administrator or infrastructure architect you get the option to leverage Db2 backup/restore procedures and Db2 HADR database synchronization commands.

You can create an off-prem backup of your Db2 database for disaster recovery, or fully migrate your Db2 workload from client-managed environment to a Power Virtual Server environment in a safe and secure IBM datacenter, quickly and easily using common Db2 tools.

To perform migration, see [Migrating Db2 on AIX to Power Virtual Server](#).

AIX and IBM i development and test

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

IBM® Power® Virtual Server is a great solution if you want to create a temporary sandbox environment to do testing and as a step before you deploy production applications.

For example, - You need a remote environment to test the software or hardware updates. - You need some system resources temporarily. - You need to evaluate, plan, and test next-generation hardware or operating system (OS) versions. - You want a temporary and isolated infrastructure for application testing. - You want to test hardware before a refresh.

Client needs

- Requires a remote environment to test updates
- Temporary need for system resources (consumption model)

- Evaluate, plan, and test next-generation hardware or operating system (OS) versions

Examples

- An initial step in the *Hybrid platform Journey*
- Testing complicated or large architecture changes without the need to procure the capital (for example, evaluating a scale-up E980 system deployment if the client's needs grow beyond scale-out image limits)
- Looking for a temporary and isolated infrastructure for application testing
- Wants to test hardware before a refresh

Technical solution

- IBM® Power® Virtual Server
- Direct Link Connect for optional connectivity into full IBM Cloud or from a private cloud environment
- VPN and Direct Link Dedicated for private cloud connectivity
- Cloud Object Storage for optional backup and custom image hosting or migration

AIX and IBM i production application hosting

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

The client wants to run an enterprise-level workload in an infrastructure as a service (IaaS) environment with the ability to use the IBM Cloud services.

Client needs

- Traditional IaaS value
- Modernize core business by using the IBM Cloud portfolio
- Flexible option to bursting and scaling their private cloud environment or to host temporary capacity
- Interested in or have an evolving strategy for a consumption-based option
- Looking to move away from operating a data center

Examples

- Oracle® and Db2 Workloads
- Custom Enterprise applications
- Modernization and extending the value of their enterprise systems with IBM Cloud native applications
- Customer acquiring a business and needs the temporary capacity while moving or merging multiple data centers

Solution components

- IBM® Power® Virtual Server
- PowerHA *(optional)*
- Mass data migration (MDM)
- Direct Link and Direct Link Dedicated for private cloud connectivity
- Cloud Object Storage for optional backup and custom image hosting

Hybrid Power Disaster Recovery (DR)

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

The client wants to use the IBM® Power® Virtual Server as a DR destination for their private cloud environment.

Client needs

- A first step into using infrastructure as a service (IaaS) environment before you commit to end-to-end, mission-critical application hosting.
- A mission-critical IBM Power environment with a DR solution to raise the level of security and reliability of their private cloud data center.
- A DR solution in a different geography.

Examples

- User is in need of reducing risk by adding a DR destination without acquiring or growing their data center and capital investment
- User has a long-term strategy for cloud migration and can use their DR environment to gain infrastructure as a service experience, and prepare for a production rollout
- Remote to remote DR site

Solution components

- Power Virtual Server
- PowerHA (*Optional*)
- Direct Link Connect and Direct Link Dedicated (*Optional*)
- Cloud Object Storage for custom image hosting

Deploying Oracle Workloads on AIX in IBM Power Virtual Servers

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

Learn how to deploy an Oracle database in an IBM® Power® Virtual Server instance.

Migrate OracleDB on AIX in IBM® Power® Virtual Server

To migrate an AIX-based Oracle database from IBM Power to Power Virtual Server, complete the procedure that is provided in the following resources:

- For more information about the procedure, see [Oracle technical migration procedure](#).
- For more information about instructions in a demo video, see [Migrating an Oracle Database from on-premises IBM Power to IBM Power Systems Virtual Server](#).
- For more information about deploying Oracle Database on IBM Power Virtual Server by using deployable architecture, see [Oracle on IBM Power Virtual Server](#).

Implementing Oracle RAC on AIX in Power Virtual Server

To deploy Oracle Real Application Clusters (RAC), use the information that is provided in the following resources:

- For more information, see [Oracle RAC implementation on IBM Power Systems Virtual Server](#).
- For more information about instructions in a demo video, see [Oracle RAC in IBM's PowerVS Hybrid Cloud](#).

Automating Oracle deployments in Power Virtual Server

To learn about automating Oracle RAC and Oracle database by using Ansible, use the information that is provided in the following resources:

- For more information, see [Using Ansible for Automation in IBM Power Environments Redbooks](#).
- For more information about instructions in a demo video, see [Automating Oracle and AIX Installation with Ansible Tools](#).

Implementing disaster recovery solutions for Oracle in Power Virtual Server

You can set up Oracle Data Guard on Power Virtual Server instances that are hosted in different data centers. For more information, see [Oracle Disaster Recovery on IBM Power Virtual Server](#).

Cloud native development and application modernization by using Red Hat OpenShift on IBM Power Virtual Server

IBM Power Virtual Server in **IBM data center**

Cloud-native applications are the applications that are **born in the cloud**, which means that the applications use microservice-based architecture, containers, and a corresponding container orchestration platform. Red Hat® OpenShift® is an enterprise Kubernetes platform enabling IT organizations to develop applications faster, deploy them reliably, and manage them efficiently.

In addition to developing new cloud-native applications, IT organizations are modernizing older, monolithic core business applications to support the faster delivery of new functions to the business.

For more information about application modernization on IBM Power, see [Field Guide to Application Modernization on IBM Power](#).

You can deploy Red Hat OpenShift on Power Virtual Server by using one of the following possible deployment methods:

- User-Provisioned Infrastructure (UPI): Uses a manual process specific to your environment that gives the administrator the ability to create and manage OpenShift nodes themselves. For more information, see [Installing OpenShift Container Platform on Power Virtual Server with UPI](#).

IBM data center

- Installer-Provisioned Installation (IPI): Creates all the necessary infrastructure services within the Red Hat OpenShift cluster and provides a turn-key solution. Choose this deployment method for an automated and faster installation. Additional benefits, including support for autoscaling, are available when you deploy Red Hat OpenShift by using IPI on PowerVS. For more information, see [Installing OpenShift Container Platform on Power Virtual Server with IPI](#).



Note: For IPI-related issues, you must first contact [Red Hat support](#). The Red Hat support team assists you with the issues and routes any underlying infrastructure problems to the appropriate team within IBM if necessary.

- Additionally, IBM developers created a set of playbooks to help you get started with a UPI installation. You can use these playbooks as-is, or take as examples. For more information, see the [GitHub Playbooks](#).

Client location

- IBM Power Virtual Server Private Cloud relies on the UPI method of installation. For more information about how to set up a OpenShift cluster in the Power Virtual Server Cloud infrastructure, see [Install Red Hat OpenShift on IBM Power Virtual Server Private Cloud infrastructure](#).

For application storage, you can deploy the IBM® Block Storage CSI driver for Power Virtual Server into Red Hat OpenShift clusters running on both IBM data center and Client location offerings of IBM Power Virtual Server. For more information, see [readme file](#).

SAP HANA deployment

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

SAP HANA on the IBM Cloud platform provides faster provisioning, affordable scaling, and reduced downtime. IBM® Power® Virtual Server provides Linux stock images for SAP HANA and SAP NetWeaver applications.



Note: You cannot transfer an OS license from a private cloud system to IBM Cloud.

Implementing network security groups for securing three-tier web architecture workloads in Power Virtual Server

IBM Power Virtual Server in **IBM data center**

This topic provides the architectural overview of network security groups (NSGs) with the help of illustrations and examples, and explains how NSGs work in your Power Edge Router (PER) and enhanced CRN-enabled IBM® Power® Virtual Server workspaces. A network security group (NSG) is used to define security rules to allow or deny specific network traffic that is related to resources provisioned in a Power Virtual Server workspace. For more information about NSG and its core components, see [Network security groups](#).

Running a three-tier web application in a Power Virtual Server workspace generally requires a network with the following components:

- Application Load Balancer (ALB) in a virtual private cloud (VPC)
- Multiple virtual server instances, such as web servers, application servers, and database (DB) servers

- Jump hosts for management access
- Database replication across Power Virtual Server workspaces
- Private endpoints like Domain Name System (DNS) and Secrets Manager

High-level architecture of a network

The following diagram provides a high-level architectural view of a sample network in your Power Virtual Server environment and illustrates how various subnets and components are connected. The diagram includes different virtual machines (VMs) such as web servers, database servers, and an application server. The network also includes components such as subnet ranges, jump hosts, transit gateway, and private endpoints (DNS and Secrets Manager).



The network components that are depicted in the high-level network architecture diagram are explained in the following table:

Component	Description
Transit Gateway (TGW)	Connects the VPC and your Power Virtual Server workspaces. The transit gateway enables secure and efficient communication between different subnets and zones. For more information about the transit gateway, see About IBM Cloud Transit Gateway .
Z1	Power Virtual Server workspace that contains the web servers (VMs), an application Server (VM2), and a database server (VM3) for production workload.
Z2	Power Virtual Server workspace that contains another database server (VM4) for database replication and disaster recovery (DR).
Application Load Balancer (ALB)	Distributes the incoming network traffic among multiple web server instances to balance the load and provides high availability and performance. The ALB is located in the virtual private cloud (VPC) 10.10.10.0/24 subnet. For more information, see About application load balancers .
Jump Host (VM1)	Provides secure management and administration access to Power Virtual Server resources. A jump host is also known as a bastion host. The jump host is located in the 10.10.11.0/24 subnet within the VPC.
Web Servers (VMs in Z1 Workspace)	Manages the incoming HTTP or HTTPS web requests. The web servers (VMs) are hosted in the 10.10.20.0/24 subnet in Z1 Workspace.
App Server (VM2 in Z1 Workspace)	Runs the application workload and communicates with the web servers (VMs) and the database server (VM3). The application server (VM2) is located in the same subnet as the web servers (10.10.20.0/24).
DB Server (VM3 in Z1 Workspace)	Acts as a database server and stores application data. The DB Sever (VM3) is located in the 10.10.21.0/24 subnet in Z1 Workspace . This instance of the database server is isolated from the web servers (VMs) and the application server (VM2).
DB Server (VM4 in Z2 Workspace)	Acts as a replica of the database server (VM3) in Z1 Workspace. The DB Server (VM4) is located in the 10.10.30.0/24 subnet in Z2 Workspace. Database replication between workspaces helps ensure redundancy and high availability.
DNS (Private Endpoint)	Used as a private endpoint in the network. The DNS provides internal domain name resolution securely within the cloud environment and is hosted in the 161.26.0.0/16 subnet.
Secrets Manager (IaaS Private Endpoint)	Used as an IaaS Private Endpoint to securely store and manage sensitive data such as API keys and login credentials. The Secrets Manager is hosted in the 166.8.0.0/14 subnet. For more information, see IBM Cloud Secrets Manager .

Network components

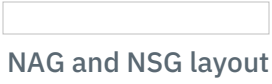
Network address groups and CIDR members

The following table lists the network address groups (NAGs) and the associated Classless Inter-Domain Routing (CIDR) members that are used as examples in this topic. A NAG simplifies the NSG rule implementation by using logical groups instead of individual IPs. For more information about NAGs, see [Network address groups](#).

NAGs	CIDR members
ALB (Load balancer)	10.10.10.0/24
MGMT (Management)	10.10.11.0/24
REPL (Replication)	10.10.30.0/24
IBM (Cloud services)	161.26.0.0/16 and 166.8.0.0/14
NAGs and CIDR members	

Network address group and network security group layout

The following diagram illustrates the NAGs and NSGs layout and how the traffic flows in the network.



The following table lists the NAGs and NSGs used in the NAG and NSG layout diagram, and explains the purpose of each component in the layout:

NAG/NSG	Associated component	Purpose
ALB NAG	ALB (10.10.10.0/24)	Contains the ALB and ensures that the ALB can communicate with the web servers (VMs) in the WEB NSG.
MGMT NAG	Jump host (10.10.11.0/24)	Contains the jump host (VM1) and controls SSH access to components in the production (Z1) and disaster recovery (Z2) workspaces.
REPL NAG	DR DB Server (10.10.30.0/24)	Contains the disaster recovery database server (VM4) and allows MySQL replication from the database server (VM3) in the DB NSG.
IBM NAG	DNS (161.26.0.0/16) and Secrets Manager (166.8.0.0/14) private endpoints	Contains the DNS and Secrets Manager private endpoints and allows DNS and HTTPS traffic.
WEB NSG	Web Servers (10.10.20.0/24)	Groups the web servers (VMs) to apply rules that allow HTTP traffic from the ALB and enable outbound HTTPS traffic to the application server (VM2) in the APP NSG.
APP NSG	App Server (10.10.20.0/24)	Groups the application server (VM2) to apply rules that allow HTTP traffic from the WEB NSG and MySQL traffic to the database server (VM3) in the DB NSG.
DB NSG	DB Server (10.10.21.0/24)	Groups the database server (VM3) to apply rules that allow MySQL traffic from the application server (VM2) in APP NSG and enable replication traffic to the disaster recovery database server (VM4) in the REPL NAG.
NAG and NSG layout		

Traffic flow across network components

The following table outlines the expected traffic flows among various network components by using different protocols and its purpose, as depicted in the NAG and NSG layout diagram.

Source	Destination	Protocol	Purpose
ALB	Web Servers (VMs)	HTTP	Distributes and routes incoming HTTP traffic to the web servers (VMs).
Jump Host (VM1)	Web Servers (VMs)	SSH	Controls SSH access to components in the production (Z1) and disaster recovery (Z2) workspaces.

Web Servers (VMs)	App Server (VM2)	HTTP	Sends HTTP requests to the app server (VM2) for further processing.
App Server (VM2)	DB Server (VM3)	MySQL	Allows traffic that is required for database operations. The app server (VM2) queries the database Server (VM3) to store or retrieve data.
DB Server (VM3)	DB Server (DR) (VM4)	MySQL	Allows data replication traffic between the database server (VM3) and the other database server (VM4) for disaster recovery and redundancy.
Web Servers (VMs) App Server (VM2) DB Server (VM3)	DNS Private Endpoint	DNS	Resolves domain names for internal and external communication.
Web Servers (VMs) App Server (VM2) DB Server (VM3)	Secrets Manager Private Endpoint	HTTPS	Stores and manages sensitive data, such as API keys and login credentials.

Traffic flow across network components

For more information about how network traffic and security rules are processed and prioritized when NSG is implemented in a Power Virtual Server workspace, see [Network traffic rules and precedence in NSG](#).

Defining NSG rules for web servers

This section outlines the rules that can be defined in an NSG to control what traffic is allowed or denied to the Web Servers. Consider the following requirements:

- Deny SSH traffic between the other web servers.
- Allow HTTP from the ALB.
- Allow SSH access from the jump host.
- Allow all traffic from the web servers.
- Allow return DNS packets from IBM resolvers.
- Allow return HTTPS packets from the Secrets Manager.
- Allow return HTTP packets from the application server.
- Define an implicit deny rule to restrict all the other unspecified traffic.

Based on these requirements, you can create an NSG and define the security rules with the following configurations. Each row in the following table represents an individual security rule.

Rule	Action	Protocol	Remote	Source	Destination	Flags
Deny SSH between the other web servers	Deny	TCP	WEB NSG	1-65535	22-22	–
Allow HTTP from the ALB	Allow	TCP	ALB NAG	1-65535	80	–
Allow SSH from the jump host	Allow	TCP	MGMT NAG	1-65535	22-22	–
Allow all traffic from the other web servers	Allow	Any	WEB NSG	–	–	–
Allow return DNS packets from IBM resolvers	Allow	UDP	IBM NAG	53	1-65535	–
Allow return HTTPS packets from the Secrets Manager	Allow	TCP	IBM NAG	443-443	1-65535	ACK,FIN

Allow return HTTP packets from the application server	Allow	TCP	APP NSG	80-80	1-65535	ACK,FIN
Implicit deny	Deny	Any	Any	–	–	–

NSG rules for web servers

Defining NSG rules for the application server

This section outlines the rules that can be defined in an NSG to control the traffic that is allowed or denied to the application server. Consider the following requirements:

- Allow HTTP traffic from the web servers.
- Allow SSH from the jump host.
- Allow all traffic from the other application servers.
- Allow return DNS packets from IBM resolvers.
- Allow return HTTPS packets from the Secrets Manager.
- Allow return packets from the database servers.
- Define an implicit deny rule to restrict all other unspecified traffic.

Based on these requirements, you can create an NSG and define the security rules with the following configurations. Each row in the following table represents an individual security rule.

Rule	Action	Protocol	Remote	Source	Destination	Flags
Allow HTTP from the web servers	Allow	TCP	WEB NSB	1-65535	80	–
Allow SSH from the jump host	Allow	TCP	MGMT NAG	1-65535	22-22	–
Allow all traffic from the other application servers	Allow	Any	APP NSG	–	–	–
Allow return DNS packets from IBM resolvers	Allow	UDP	IBM NAG	53	1-65535	–
Allow return HTTPS packets from the Secrets Manager	Allow	TCP	IBM NAG	443-443	1-65535	ACK,FIN
Allow return packets from the database servers	Allow	TCP	DB NSG	80-80	1-65535	ACK,FIN
Implicit deny	Deny	Any	Any	–	–	–

NSG rules for the application server

Defining NSG rules for database servers

This section outlines the rules that can be defined in an NSG to control the traffic that is allowed or denied to the database servers. Consider the following requirements:

- Allow MySQL traffic from the application server.
- Allow SSH traffic from the jump host.
- Allow all traffic from the other database servers.
- Allow return DNS packets from IBM resolvers.
- Allow return HTTPS packets from the Secrets Manager.
- Allow return database packets from the replication server.
- Define an implicit deny rule to restrict all other unspecified traffic.

Based on these requirements, you can create an NSG and define the security rules with the following configurations. Each row in the following table represents an individual security rule.

Rule	Action	Protocol	Remote	Source	Destination	Flags
Allow MySQL from the application server	Allow	TCP	APP NSG	1-65535	3306-3306	–

Allow SSH from the jump host	Allow	TCP	MGMT NAG	1-65535	22-22	–
Allow all traffic from the other database servers	Allow	Any	DB NSG	–	–	–
Allow return DNS packets from IBM resolvers	Allow	UDP	IBM NAG	53	1-65535	–
Allow return HTTPS packets from the Secrets Manager	Allow	TCP	IBM NAG	443-443	1-65535	ACK, FIN
Allow return database packets from replication servers	Allow	TCP	REPL NAG	3306-3306	1-65535	ACK, FIN
Implicit deny	Deny	Any	Any	–	–	–

NSG rules for database servers

For more information about setting up, configuring, and managing NSGs, see [Setting up network security groups in a workspace](#).

Networking

Configuring a private network subnet

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can configure a private network subnet when you create an IBM® Power® Virtual Server by providing your subnet a name and specifying a Classless Inter-Domain Routing (CIDR).

In a IBM data center, the private network subnet configuration depends on the networking configuration of the Power Virtual Server workspace. The networking configuration of the workspace can use one of the following approaches:

- Power Virtual Server workspace that is enabled with the [Power Edge Router \(PER\)](#). This configuration is the default for most locations if the workspace is created after mid-2023 and the workspace can use VPN Connections.
- Power Virtual Server workspace that is enabled with [Power Cloud Connections](#). In this configuration, PER is the default connection, and the workspace can use VPN Connections.

For more information about VPN connections, see [VPN connections](#).

When you specify a CIDR, the following values are automatically populated:

- A gateway
- An IP range
- DNS server

You must use a CIDR notation when you choose the IP ranges for your private network subnet. CIDR notation is defined in [RFC 1518](#) and [RFC 1519](#). A CIDR has the following format:

<IPv4 address>/<number>

For example, `192.168.100.14/24` represents the IPv4 address, `192.168.100.14`, and its associated routing prefix `192.168.100.0`, or equivalently, its subnet mask `255.255.255.0` (which has 24 leading 1 bit).

Configure a private network subnet by using UI

To create a new subnet, complete the following steps:



Note: You cannot assign the subnet that is already assigned to another virtual machine. However, when you edit the configuration of a virtual machine, you can assign the same subnet multiple times.

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. In the search box, type Power Virtual Server and click the Power Virtual Server tile.
3. In the navigation panel, click **Workspaces**. The workspaces page with a list of existing workspaces is displayed.
4. Click your workspace from the list to open the virtual server instance page.
5. Click **Subnets** in the left navigation panel, then click **Create subnet**. The new subnet panel is displayed.
6. Enter a name, CIDR value (for example, `192.168.100.14/24`), gateway number (for example, `192.168.100.15`), and the IP range values for the subnet.
7. Enter the **DNS server** value.

A **DNS server** value of `9.9.9.9` might not be reachable if you don't have a public IP. This issue can cause the LPAR to hang during the startup operation. You can use the default DNS server value of `127.0.0.1` to avoid this issue. You can add up to 20 DNS server IP addresses separated by commas.

8. Set **Advertise** to one of the following options:
 - **Enabled:** The static route is advertised to external connections. **Advertise** is set to **Enabled** by default.
 - **Disabled:** The static route is not advertised to external connections.

9. Set **Subnet ARP Broadcast** to one of the following options:

- **Enabled:** Distributes the Address Resolution Protocol (ARP) traffic in the Power Virtual Server network fabric.
- **Disabled:** Does not distribute the ARP traffic in the Power Virtual Server network fabric. **Subnet ARP Broadcast** is set to **Disabled** by default.

For more information about subnet ARP broadcast, see [Configuring the ARP broadcast in Power Virtual Server subnets](#).

10. Set **DHCP** to one of the following options:

- **Enabled:** Network interfaces that are configured to use Dynamic Host Configuration Protocol (DHCP) receive their IPv4 configurations automatically. **DHCP** is set to **Enabled** by default.
- **Disabled:** Network interfaces that are configured to use DHCP do not receive their IPv4 configurations automatically.

The DHCP service is available only in IBM Power Virtual Server in Client location. For more information about DHCP, see [DHCP network inside the pod](#).

11. You can also attach a primary and redundant cloud connection to the subnet to set up high availability in an IBM data center. For more information about high availability, see [Setting up high availability over cloud connections](#).

12. Click **Create subnet**.

Updating an existing subnet

Complete the following steps to update an existing subnet:

1. Click **Subnets** in the left navigation panel.
2. Click the subnet that you want to update from the displayed list.
3. Click **Edit details** to update the subnet.

You can attach or detach cloud connections to each of the subnets in the **Attached cloud connections** section in IBM data center.

Configure a private network subnet by using CLI

You can also create and configure a private network subnet by using the IBM command-line interface (CLI). Use the following command to create a private network subnet:

```
ibmcloud pi network-create-private NETWORK_NAME --cidr-block CIDR --ip-range "startIP-endIP[,startIP-endIP]" [--dns-servers "DNS1 DNS2"] [--gateway GATEWAY] [--json]
```

Reserving IP addresses

Use the reserved IP addresses to block Power Virtual Server from assigning a specific IP address to a virtual server instance.

You can complete the following tasks:

- Add IP address
- See the list of reserved IP addresses
- Delete the reserved IP address

An IP address that is present in the reserved list is not auto assigned to a virtual server instance.

Adding an IP address in the reserved list

To add an IP address into the reserved IP address list, complete the following steps:

- Open the [Subnets](#) page in the IBM Cloud.
- From the list of subnets that are created, click the subnet for which you want to reserve the IP address.
- Click **Reserve IP**.
- Enter your IP address in the **IP address** field.



Note: Make sure the IP address that you want to reserve falls in the IP range that is defined when the subnet is created.

- Provide a description of your reserved IP in the **Reserved IP description (optional)** field.

Networking considerations for IBM data center

You can establish a private network communication between the two Power Virtual Server instances with any one of the following approaches:

- 1. Use a PER-enabled workspace. See, [Getting started with PER](#).
- 2. Create and attach the subnet to a cloud connection and Transit Gateway.

Using CIDR notation

You can use CIDR notations if public subnets are enabled and are in use. You can specify CIDR notation for a public subnet with either an [RFC 1918](#) IP address space (`10.0.0.0/8` , `172.16.0.0/12` , or `192.168.0.0/16`) or an IP address space registered with the Internet Assigned Numbers Authority (IANA). You must plan carefully when you use a non-RFC 1918 IP address space. If a subnet contains non-RFC 1918 IP addresses, a virtual server instance (VSI) in that subnet might not reach certain IP addresses when public networks are also attached to the VSI.

The Power Virtual Server assigns internal IP addresses from the `192.168.0.0/16` CIDR block to your account to enable connectivity to public networks. The IP addresses from the `192.168.0.0/16` block that are assigned to your account for public network use cannot be used for private networks.

The number after the slash represents the bit length of the subnet mask. As a result, a lower number after the slash means more IP addresses are allocated. The following table lists the number of available addresses in a subnet based on its specified CIDR block capacity:

CIDR block capacity	Available IP addresses (WDC04, WDC06)	Available IP addresses (non-WDC)
/22	1019	1021
/23	507	509
/25	123	125
/26	59	61
/27	27	29
/28	11	13

Understanding CIDR notation caption

Creating and managing network routes in IBM Power Virtual Server workspaces

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

A network route is a defined path that data packets follow to travel from a source to a destination across a network. You can use the routes feature in the IBM® Power® Virtual Server to view implicit network routes and to create or manage static routes within your Power Virtual Server workspaces. Static routes are custom routes that control where the network packets are sent, depending on the destination Classless Inter-Domain Routing (CIDR) and the specified next hop address. The next hop is the IP address to which you want to route the network packet.

With static routes, you can build an overlay network by using CIDRs that are not associated with subnets in your Power Virtual Server workspace to implement high availability (HA) and disaster recovery (DR) strategies. You can also view external network routes to identify which external destinations are accessible from your Power Virtual Server workspace.



Note: The routes feature is available on PER-enabled Power Virtual Server workspaces and on Power Virtual Server in Client location. For more information about PER, see [Getting started with the Power Edge Router](#).

Creating static routes

To create a static route in your Power Virtual Server workspace, complete the following steps:

- 1. Log in to the [IBM Cloud catalog](#) with your credentials.
- 2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
- 3. Click **Workspaces** in the left navigation menu.
- 4. Select a workspace in which you want to create the static route. The Workspace details panel is displayed.

5. Click **View virtual servers**.
6. In the navigation pane, click **Networking > Routes**. The Routes page is displayed with a list of any existing static routes on the **Static routes** tab.
7. Click **Create static route** to create a static route. The Create static route panel is displayed.
8. Enter a name in the **Name** field.
9. Optional: Enter user tags in the **User tags (optional)** field.
10. Enter the destination IP address in the **Destination** field and the next hop IP address in the **Next hop** field.
11. Set **Advertise** to one of the following options:
 - **Enabled**: Default. If **Advertise** is set to **Enabled**, the static route is advertised to the external connections.
 - **Disabled**: If **Advertise** is set to **Disabled**, the static route is not advertised to the external connections.
12. Set **State** to one of the following options:
 - **Enabled**: Default. If **State** is set to **Enabled**, the static route is enabled within the network fabric.
 - **Disabled**: If **State** is set to **Disabled**, the static route is disabled within the network fabric.



Important: If you set **State** to **Disabled**, the static route is not advertised to the external connections, even if **Advertise** is set to **Enabled**.

13. Click **Create route**.



Note: The static route destination is currently limited to a single IP address (/32 CIDR) and must not overlap with the local subnet or any subnets that are derived from it.

The static route that you created is displayed on the **Static routes** tab. The State column indicates one of the following operational statuses of each static route:

- **Deployed**: Indicates that the static route is active and the next hop IP address exists within a subnet that is defined in the Power Virtual Server workspace. Routes in the deployed state are advertised externally when **Advertise** is set to **Enabled**.
- **Defined**: Indicates that the static route is configured but not yet deployed or active. A static route might be in the defined state when the specified next hop IP address does not currently exist in any subnet within the Power Virtual Server workspace.

Routes in the defined state are not used for forwarding network traffic and are not advertised externally even if **Advertise** is set to **Enabled**. However, after a subnet that includes the next hop IP address is created, the static route automatically becomes active and the status changes to **Deployed**.
- **Disabled**: Indicates that the static route is manually disabled by setting **Status** to **Disabled**. Routes in the disabled state are not used for forwarding traffic and are not advertised externally.

Modifying static routes

You can edit an existing static route to modify its name, destination IP address, next hop IP address, and the advertise and status options. To modify a static route in your Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the left navigation menu.
4. Select the workspace that contains the static route that you want to modify. The Workspace details panel is displayed.
5. Click **View virtual servers**.
6. In the navigation pane, click **Networking > Routes**. The Routes page is displayed with a list of any existing static routes on the **Static routes** tab.
7. Click the overflow menu (icon with 3 vertical dots) on the static route entry that you want to modify and select **Edit**. The Edit static route panel is displayed.
8. Modify the route details and click **Edit route**.



Note: You can modify the static routes by using the GUI, API, CLI, and Terraform interfaces.

Deleting static routes

To delete a static route in your Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the left navigation menu.
4. Select the workspace that contains the static route that you want to delete. The Workspace details panel is displayed.
5. Click **View virtual servers**.
6. In the navigation pane, click **Networking > Routes**. The Routes page is displayed with a list of any existing static routes on the **Static routes** tab.
7. Click the overflow menu (icon with 3 vertical dots) on the static route entry that you want to delete and select **Delete**. The Delete static route dialog is displayed.
8. Click **Delete** to delete the static route.



Important: You cannot recover a static route after it is deleted.



Important: When you delete a workspace, all the static routes that were created within that workspace are also deleted.

Viewing external routes

The external routes tab displays all external networks and subnets with which your Power Virtual Server workspace can communicate through the connected infrastructure, such as IBM Cloud Transit Gateway. The external routes are learned from external sources, such as other Power Virtual Server workspaces, VPCs, or other on-premises networks. You can use the external routes information to identify the external destinations that are accessible from your Power Virtual Server workspace.

To view the external routes, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the left navigation menu.
4. Select a workspace in which you want to create the static route. The Workspace details panel is displayed.
5. Click **View virtual servers**.
6. In the navigation pane, click **Networking > Routes**. The Routes page is displayed with a list of any existing static routes on the **Static routes** tab.
7. Click the **External routes** tab. The available external routes are displayed.

Configuring the ARP broadcast in Power Virtual Server subnets

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

The Address Resolution Protocol (ARP) is a network protocol that is used to map IP addresses to physical Media Access Control (MAC) addresses in a network. The IP address mapping must be established to enable the communication between endpoints in subnets or in a local network.

The **Subnet ARP broadcast** is a switch that is provided on a subnet page of a Power Virtual Server workspace. You can set **Subnet ARP broadcast** to **Enabled** or **Disabled**. When you set **Subnet ARP broadcast** to **Enabled**, the ARP request is broadcast across the subnet. You can use the broadcast request for ARP to distribute the ARP traffic in the Power Virtual Server network fabric. A Gratuitous ARP (GARP) request can update the host ARP caches when an IP address has a different MAC address. To accomplish GARP requests, you must broadcast an ARP request across the subnet. For example, when **Subnet ARP broadcast** is used with a cluster software such as Kubernetes, the network traffic is routed to the active cluster node.



Note: The **Subnet ARP broadcast** option is available on PER-enabled Power Virtual Server workspaces and on Power Virtual Server in Client location. For more information about PER, see [Getting started with the Power Edge Router](#).

Benefits of enabling the ARP broadcast in Power Virtual Server subnets

When you enable the **Subnet ARP broadcast** in Power Virtual Server subnets, dynamic communication between IP addresses is enabled and scalability and compatibility with various devices and systems improves. Hence, network administration is simplified because the automatic conversion of the IP address to MAC address is enabled. Thus, manual configuration of the network is reduced and network performance improves.

To get the benefits of the **Subnet ARP broadcast**, you must set the option to **Enabled** for a subnet of a Power Virtual Server workspace on the Subnet page. The ARP request is broadcast across the subnet. You can determine the following actions in the subnet:

- Identify the change in the MAC address of a device that is mapped to an IP address
- Identify a MAC address of an IP address that frequently moves from one MAC addresses to another
- Identify the current IP address of a silent IP address in the subnet

In the Power Virtual Server network fabric, when an IP address is reassigned to a different host without notifying the network fabric is called as a silent IP address. When you set the **Subnet ARP broadcast** to **Enabled**, the host with the silent IP address responds to the ARP request. The Power Virtual Server network fabric updates the MAC address of the silent IP address with the MAC address of the host that responded to the ARP request.

If the **Subnet ARP broadcast** option is disabled, the Power Virtual Server network fabric continues to forward the ARP request to the MAC address that is saved in the network fabric. The network fabric continues to forward the ARP request until the silent IP address endpoint exists. Also, the network traffic flow is optimized as the ARP request is sent to the MAC address associated with the silent IP address. The IP address moves to the new host through GARP or other requests. The endpoint then notifies the network fabric about the change in the current host.

Enabling or disabling ARP broadcast in Power Virtual Server subnets

To enable the Subnet ARP broadcast option when you create a new subnet, set **Subnet ARP broadcast** to **Enabled** on the Create subnet page. By default, the **Subnet ARP broadcast** option is set to **Disabled**. For more information, see [Configuring a private network subnet](#). You can enable or disable the subnet ARP option by using the GUI, CLI, or API.

To change the status of the **Subnet ARP broadcast** option for an existing subnet on the **Power Virtual Server** page, complete the following steps:

1. Sign in to the [IBM Cloud Portal](#).
2. Search for **Power Virtual Server** in the search box for IBM Cloud catalog.
3. Select **Workspace** from the left menu.
4. Select the workspace for which you want to edit the subnet details.
5. Select **Subnets** under Networking from the left menu.
6. Select the subnet for which you want to enable the **Subnet ARP broadcast** option. The **Subnet ARP Broadcast** field is set as **Enabled** or **Disabled**.
7. Click **Edit Subnet** in the Subnet details section.
8. Set **Subnet ARP Broadcast** to **Enabled** or **Disabled** on the Edit Subnets pane.

To change the status of the **Subnet ARP broadcast** option for an existing subnet, use the following API or CLI commands:

- **API:** Use the following APIs to enable or disable the ARP broadcast option:
 - When you create a subnet, use the [Create a new Network](#) API.
 - When you update a subnet, use the [Update a Network](#) API.

Set the `arpBroadcast` parameter to `enable` or `disable` value. By default, the `arpBroadcast` parameter is set to `disable` value.

- **CLI:** Use the following commands to enable or disable the ARP broadcast option:
 - When you create a subnet, use the command [ibmcloud pi subnet create](#).
 - When you update a subnet, use the command [ibmcloud pi subnet update](#).

Set the `arpBroadcast` parameter to `enable` or `disable` value. By default, the `arpBroadcast` parameter is set to `disable` value.

IBM® Power® Virtual Server integration with x86-based workloads

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

For a complete tutorial on the following scenarios, see [IBM Power Virtual Server integration with x86-based workloads](#):

- x86 VSI communication with a Power Virtual Server instance

- x86 VSI within a VMware cluster in IBM Cloud communication with a Power Virtual Server instance
- x86 VSI in Virtual Private Cloud communication with a Power Virtual Server instance

Networking for IBM data center

Network architecture diagrams

IBM Power Virtual Server in [IBM data center](#)

This topic describes typical network architectures that are used in the IBM® Power® Virtual Server network architecture and is not an exhaustive list of Power Virtual Server connection methods.

Power Virtual Server networking environment

When you create a Power Virtual Server, you can select a private or public network interface. For more information, see [Public and Private networks](#).

Power Virtual Server network architectures consist of one or more of the following networks:

- **IBM Cloud infrastructure networks:** While the following infrastructure network environments offer different features and are managed separately, they can be connected to each other to provide layer-3 IPv4 traffic flow:
 - Classic: Classic network resources include VLANs, subnets, and SSL Virtual Private Network (VPN) access. See [Network security architecture](#) for a description of the classic network components. Bring Your Own IP (BYOIP) is not supported.
 - Virtual Private Cloud (VPC): VPC network resources include subnets, floating IP addresses, security groups, and VPN gateways. For more information, see [About networking](#). See [Advanced networking for IBM Cloud VPC](#) for a one-hour course on VPC networking. BYOIP is supported.
 - IBM Power: Network resources include subnets. BYOIP is supported.
- **Overlay networks:** These networks exist in the IBM Cloud VMware Shared and VMware Dedicated offerings. While technically hosted in the IBM Cloud classic infrastructure environment, these networks are implemented in VMware NSX and are under your direct control, including the IP addressing schema. BYOIP is supported. Therefore, the IBM Cloud infrastructure networks cannot route the overlay networks; access is through tunnels.
- **External networks:**
 - Internet: Access the internet through resources that are hosted in any of the three IBM Cloud infrastructure environments.
 - Remote: Connect remote networks to your IBM Cloud networks. You can use the following services to connect to a remote network:
 - Internet VPN: Uses the public internet to connect remote networks and their IBM Cloud networks through a VPN. The VPN is terminated on gateway devices or a service within IBM Cloud.
 - Direct Link: Direct Link is a suite of offerings that enable the creation of direct and private connections between your remote, private cloud network and IBM Cloud, without traversing the public internet. For more information, see [Getting started with IBM Cloud Direct Link \(2.0\)](#).



Note: You can connect Direct Links to either a local or remote IBM Cloud Transit Gateway, which allows the private cloud network to access all networks that are connected to the IBM Cloud Transit Gateway.

Power Edge Routers

A Power Edge Router (PER) is a high-performance router that provides advanced routing capabilities for IBM® Power® Virtual Server users. For more information, see [Getting started with the Power Edge Router](#).

Power Edge Router use cases

Using a Power Edge Router (PER)-enabled workspace provides the following benefits:

- Improved performance with aggregated bandwidth of 400 Gbps.
- Direct access to the IBM Cloud services from the Power Virtual Server workspace.
- Direct access to the Power Virtual Server from a client-managed environment by using a Direct Link Connect or Direct Link Dedicated.

The following are some of the use cases of a PER-enabled Power Virtual Server workspace:

1. [Connecting a client-managed data center](#)
2. [Connecting to classic infrastructure](#)
3. [Connecting to Virtual Private Cloud](#)
4. [Connecting to IBM Cloud services](#)



Important: The use cases are enabled with base capabilities and can be customized to meet any specific requirements.

Connecting a client-managed data center

In this depiction, a client-managed data center uses a Direct Link connection to attach to the Transit Gateway.

1. You can attach a PER-enabled Power Virtual Server workspace to a Transit Gateway. The Transit Gateway enables the connectivity to the client-managed data center through the incoming Direct Link connectivity by using Border Gateway Protocol (BGP) routing protocol. The Direct Link connectivity is interconnected through the Transit Gateway.



Connecting an client-managed workspace with a PER-enabled Power Virtual Server by using the Transit Gateway.

2. You pay for the Direct Link connection that you use to connect your client-managed environment with the Transit Gateway. You can use up to 4 connections on a local Transit Gateway without any charges. For more information on Transit Gateway pricing, see: [Pricing for Power Edge Router](#).
3. You can attach a PER-enabled Power Virtual Server workspace that is located in different IBM Cloud Accounts through a Transit Gateway. The Transit Gateway enables the connectivity to the client-managed data center through the incoming Direct Link connectivity.



Connecting two IBM Cloud Accounts with a PER and Global Transit Gateway.

Connecting to classic infrastructure

Establish a connection between PER-enabled Power Virtual Server workspace and the classic infrastructure by local routing of Transit Gateway.

You do not pay extra charges for Transit Gateway. To establish the connection, the following conditions must be met:

- The workspace and the classic infrastructure must be in the same region.
- The workspace must be connected to a Transit Gateway.

For more information on Transit Gateway pricing, see: [Pricing for Power Edge Router](#).



Connecting PER workspace with classic.

The following two scenarios for connecting the classic infrastructure to a PER-enabled workspace are differentiated based on the Generic Routing Encapsulation (GRE) tunnel requirement:

- **When is a GRE tunnel not required?**

A GRE tunnel is not required to establish a connection from your PER-enabled workspace with classic infrastructure and to establish a connection when you do not have a custom IP address on your workspace.



Connecting PER and Transit Gateway without a GRE tunnel.

- **When is a GRE tunnel required?**

A GRE tunnel is required to establish a connection from your PER-enabled Power Virtual Server workspace to the classic infrastructure environment that is configured to use custom IP address ranges.

The GRE tunnel is used to wrap the custom IP address with another header so that it is routable through the IBM Cloud network underlay. For example, consider a Backend Customer Router (BCR) within the network underlay. The BCR allows only the IP address range 10.0.0.0/8 to pass through, while it drops the traffic that is originating from IBM Power Virtual Server VLAN subnets that use the IP range 172.16.0.0/12 or other IP ranges.

The GRE tunnel must be attached to a Transit Gateway for the routing between the IBM Power VS workspace and classic infrastructure.

For detailed steps, see [Configuring Generic Routing Encapsulation \(GRE\) tunnel](#).



Connecting PER and GRE through the Transit Gateway.

Connecting to Virtual Private Cloud

Establish a connection between a PER-enabled Power Virtual Server workspace and a Virtual Private Cloud (VPC) by using the Transit Gateway. To create a multi-connection network, connect the classic infrastructure with the Transit Gateway. Thus, a three-way connection is established between the PER-enabled workspace, VPC, and the classic infrastructure is established.

The pricing for this connection depends on the usage of a local or global Transit Gateway. For more information, see: [Pricing for Power Edge Router](#).



Connecting to IBM Cloud (Classic) services through Secure Endpoint (SE)

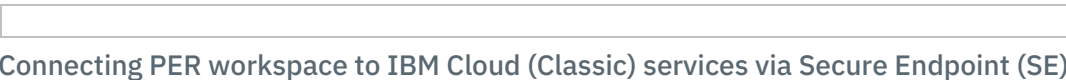
By default, a PER-enabled Power Virtual Server workspace can connect to the IBM Cloud Service Endpoints (CSE).

You can also connect a PER-enabled workspace seamlessly with the Secure Endpoints (SE) in the classic infrastructure environment. To establish the connection through a Transit Gateway, complete the following steps:

1. Connect a PER-enabled Power Virtual Server workspace to the classic infrastructure environment by using a Transit Gateway.
2. Connect Power Virtual Server to the Secure Endpoint (SE) of the IBM Cloud (Classic) services.

On a PER-enabled workspace that uses a custom IP address, the network is routed through a Network Address Translator (NAT) device. The NAT device converts the custom IP address into an IBM Cloud-supported IP address.

For information about Transit Gateway pricing, see [Pricing for Power Edge Router](#).



Connecting to IBM Cloud services using VPE with PER

You can connect a private network with IBM Cloud Services APIs by using one of the following options:

- **Public endpoints:** You can connect to resources in all regions in your account through IBM Cloud public network.
- **Virtual Private Endpoints (VPE):** You can connect by using a private IP address that is accessible only from your VPC.
- **Service endpoints (SE):** You can establish the connection by completing the following steps:
 1. Enable virtual routing and forwarding (VRF) option on your network routers.
 2. Enable service endpoints for your account.
 3. Establish the connection by using a private IP address that is accessible only through the IBM Cloud private network.

However, you must establish secure connectivity to Cloud Object Storage (COS) for backup and restoration, encryption services, key management, and monitoring or database-as-a-service solutions.

To use VPE or SE options, you do not need a public network and you can apply a network access control list (ACL) to control network access. In addition, you can also use DNS services.

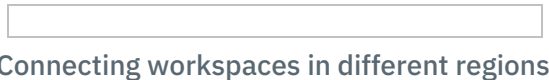


Connecting multiple workspaces across a data center

Use a local Transit Gateway to connect multiple Power Virtual Server workspaces that are in the same region. A PER-enabled workspace must be attached with the Transit Gateway that can exchange routes between two workspaces.

Use a global Transit Gateway to connect the workspaces that are present in different regions.

For more information about Transit Gateway charges for local and global routing, see [Pricing for Power Edge Router](#).



Getting started with the Power Edge Router

IBM Power Virtual Server in [IBM data center](#)

A Power Edge Router (PER) is a high-performance router that provides advanced routing capabilities for IBM® Power® Virtual Server users.

PER improves network communication across different parts of the IBM network. The PER solution creates a direct connection to the IBM Cloud Multi Protocol Label Switching (MPLS) backbone, making it easy for different parts of the IBM network to communicate with each other. The PER solution consists of two routers that enable an aggregate connectivity of 400 Gbps to each Power Virtual Server data center.

In a PER and enhanced CRN-enabled workspace, you can also define security rules for network traffic control using the network security group feature. For more information, see [Network security groups](#).



Note: The PER capability will be deployed in all the data centers over time. See [Data centers that support PER](#) for more information.

PER associates specific Power Virtual Server networks with unique MPLS route distinguishers (RDs). The association different networks to communicate with each other across the IBM Cloud MPLS backbone.

To facilitate communication between Power Virtual Server instances and other parts of the network, such as [Classic infrastructure](#), [Virtual Private Cloud \(VPC\)](#), and remote Power Virtual Server instances, the [Transit Gateway](#) is used.

By using PER solution, as a Power Virtual Server user you can access other IBM Cloud services, such as IBM Cloud domain name server (DNS), Network Time Protocol (NTP), and Cloud Object Storage. You can connect to these services without having to use proxies or virtual routers, as the PER solution includes a Network Address Translation (NAT) device that simplifies the access process.

The following network architecture diagram explains how the PER is integrated into the IBM Cloud environment:



The network traffic in a PER environment can flow in the following two ways:

- Accessing classic infrastructure through the Transit Gateway.
 - **1** - Traffic from ACI tenants is forwarded to the PER.
 - **2** - PER forwards the traffic to classic infrastructure services that use Transit Gateway.
- Accessing cloud services that can access the resources that are attached to each other.
 - **1** - Traffic from ACI tenants is forwarded to the PER.
 - **3** - Traffic from PER is forwarded to the NAT services with Service Gateway routers. The Service Gateway converts the destination addresses to ADN and CSE networks.
 - **4** - The converted traffic from NAT is forwarded to PER.
 - **2** - Traffic from PER is now forwarded to IBM Cloud PPRs for final delivery.

The automation of ACI, PER, and NAT Services provisioning in IBM data centers is designed to simplify network integration and accelerate connection time for IBM Power Virtual Server users in the IBM Cloud.

For detailed networking PER use cases and architecture diagrams, see [Power Edge Router use cases](#).

Considerations for using PER

- You cannot create a Cloud Connection or a VPN connection in a PER workspace.
- You can establish a connection between collocated workspaces if one colocation facility (colo) is PER-enabled (such as **DAL10**) and the second colo (**DAL12** / **DAL13**) uses [Direct Link](#). Both collocated workspaces must be connected to the same Transit Gateway.
- When a PER workspace is connected to a Transit Gateway, you can connect a Direct Link to the same Transit Gateway. You can use the connection to achieve an end-to-end connectivity from a network in your client-managed environment to the PER workspace.
- You can establish a connection between VPC and classic infrastructure with PER by adding them to the Transit Gateway.
- When you create private networks in a PER workspace, a maximum of one DNS server can be specified.
- A GRE (Generic Routing Encapsulation) tunnel is not supported in a PER workspace.
- You cannot create a non-PER workspace in a PER-enabled data center. However, you can still use your old non-PER workspaces that are existing in a PER-enabled data center that are created before PER rollout.
- In certain situations, local connection charges can apply when you connect from a client-managed location to Power Virtual Server. To ensure accurate pricing, it is important to use the cost estimator tool. See the [Pricing of Power Edge Router](#) to learn more about PER pricing.

Migrating to PER

The migration of an existing workspace to PER is supported through CLI by using the [ibmcloud pi workspace action](#) command with the following values:

- **per-migrate-start**: Initiates the migration of the workspace to PER.
- **per-migrate-validate**: Removes the Cloud Connections for the workspace from the network database. Use this command after you remove the Cloud Connections for the workspace from the Transit Gateway.



Note: Before you migrate the last workspace in the data center to PER, delete any Cloud Connections that are not attached to any networks. After the migration, Cloud Connections can only be deleted using the [IBM Cloud CLI](#) or [ibmcloud pi workspace action](#) command.

Use the CLI command `ibmcloud pi workspace get <WORKSPACE_ID> --json` to get the migration status of your workspace. In the CLI response, once the `migrationStatus` property changes to `migrating` and `state` property changes to `user-validation` you can continue to complete the next steps.

- Attach the workspace to the Transit Gateway to validate the connectivity of the workspace with other workspaces.
- Remove the Cloud Connections from the Transit Gateway.
- Remove the Cloud Connections from the network database by using the `per-migrate-validate` value with `ibmcloud pi workspace action` command.

Use the CLI command `ibmcloud pi workspace get <WORKSPACE_ID> --json` to view the Cloud Connection clean up progress of the migration process. In the CLI response, once the `migrationStatus` property is not included and `state` property is set to `Active` state, the migration process is complete and the workspace and network operations can be resumed.



Note: You cannot delete the workspace and add or delete the subnets that are attached to the workspace until you complete the migration process by using the `per-migrate-validate` command.

If an error occurs during the migration process, open a support ticket.

The PER status is displayed in the **PER status** column on the [Workspaces](#) page. Also, you can select a specific workspace to view its PER status on the **Workspace details** page.

If you are migrating the existing workloads to a new PER-enabled workspace, back up the data from the existing workspace, and restore the data to the new PER-enabled workspace.

Only `CHE01` data center is not PER-enabled. Therefore, use Cloud Connections for interconnectivity between other parts of the IBM network.

Complete the following steps to connect an existing non-PER workspace to an existing Transit Gateway by using the IBM Cloud command-line interface (CLI):

1. Use the `ibmcloud pi workspaces` command to list the Power Virtual Server workspaces in your account. Make note of the cloud resource name (CRN) for the workspace that you want to connect to the Transit Gateway.
2. Use the `ibmcloud tg gateways` command to list the Transit Gateways within your account. Make a note of the gateway ID that you want to connect to the Power Virtual Server workspace.
3. Use the `ibmcloud tg connection-create` command to create a new connection between the Transit Gateway and the PER-enabled workspace.

For example, `ibmcloud tg connection-create aaaa-bbbb-cccc-dddd-eeee --name powervs_per_fra02 --network-id crn:v1:bluemix:public:power-iaas:fra02:a/aaaa:bbbb:: --network-type power_virtual_server` is an executable command, where:

- Transit Gateway ID is `aaaa-bbbb-cccc-dddd-eeee`
- The Power Virtual Server workspace CRN is `crn:v1:bluemix:public:power-iaas:fra02:a/aaaa:bbbb::`

Creating a PER workspace

All Power Virtual Server data centers, except `CHE01` are PER-enabled. When you create a workspace in the PER-enabled data center, the workspace is PER-enabled by default.

To create a PER workspace, follow the steps that are mentioned in [Creating a Power Virtual Server workspace](#) and choose a PER-enabled data center.



Note: You can check whether a workspace is PER-enabled by selecting the workspace and viewing the details of the workspace. The PER-enabled workspace shows an information message on Transit Gateway.

You can create, delete, attach, detach, and update private networks by using the **Subnets** and **Virtual server instances** pages on a PER workspace, the same as with a non-PER workspace. However, private networks on PER workspaces in a PER-enabled data center, such as `DAL10`, use upgraded networking technology for higher performance, and seamless connectivity. See, [Configuring and adding a private network subnet](#) to complete a necessary operation.

Use Transit Gateway only to configure the virtual connections, as opposed to using Cloud connection.



Note: On a PER workspace, if you do not find the **cloud Connections** and **VPN connections** options in the left navigation of the user interface, then PER do not require or support these options.

On a PER workspace, you can do the following actions:

- Attach a network without creating a separate Cloud Connection such as Direct Link.

- Attach a connection to the IBM cloud network after you attach the Transit Gateway with your PER workspace.
- Connect to your network in the client-managed environment by creating a Direct Link. Attach the Direct Link with the Transit Gateway that is present on the PER workspace.

 **Important:** To delete the PER workspaces that are connected to the Transit Gateway, you must first delete the Transit Gateway connections.

Using IBM cloud services in a PER workspace

From your PER workspace, you can create a virtual server instance and attach subnets to it. These virtual server instances can then access the IBM Cloud resources such as Cloud Object Storage, Domain Name System (DNS), and other services that use the allocated IP addresses in the range `161.26.0.0/16`. See [IaaS endpoints](#) for more information.

 **Important:** You must use Direct endpoints to access Cloud Object Storage (COS) from a PER-enabled workspace. For more information about available Direct endpoints, see [Regional Endpoints](#).

 **Important:** Attach the workspace to the Transit Gateway if you want to connect your workspace with the VPC and classic infrastructure.

Attaching Transit Gateway to a PER workspace

Transit Gateway is required to connect with VPC and classic infrastructure. To attach a virtual server instance from a PER workspace with the Transit Gateway, follow the instructions in the [Ordering IBM Cloud Transit Gateway](#).

Select **Power Virtual Server** under connection to attach a virtual server instance that was created on a PER-enabled workspace. You can also add VPC and classic infrastructures as connection.

The connections attached to the Transit Gateway can communicate with each other. For example, a Power Virtual Server workspace and VPC added under the Transit Gateway connection can access the resources that are associated with each other.

 **Note:** Ensure that the classic infrastructure is Virtual Routing and Forwarding (VRF) enabled before you attach it to the Transit Gateway.

Managing PER workspace fallback scenarios

When you create a PER-enabled workspace, the underlying network device configuration might fail. In such cases, the system automatically falls back to a non-PER-enabled workspace.

A workspace that is created in this fallback scenario might support the Cloud Connection capability instead of the PER capability. In addition, the Transit Gateway service does not detect the fallback workspace as an available connection in the data center.

Verifying PER configuration for a workspace

To check whether your workspace is PER-enabled by using the Power Virtual Server user interface, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Click **Workspaces** in the left navigation menu.
3. Select the workspace for which you want to review the PER configuration. The Workspace details panel is displayed.
4. In the Workspace details panel, locate the **Power Edge Router (PER)** section. If PER is enabled for the selected workspace, the section displays Status as **Active**.

To verify PER status of a workspace by using the command-line interface (CLI), run the following command:

```
$ ibmcloud pi workspace get <WORKSPACE_ID>
```

If the workspace is PER-enabled, the Power Edge Router details are displayed in the command output. For example,

```
ID:                12345678-abcd-1234-abcd-1234567890ab
Name:              SAMPLE-WORKSPACE
Region:           dal10
Status:           active
Type:             off-premises
Capabilities:
  cloud-connections    false
  custom-virtual-cores true
```

```
dedicated-hosts      false
network-security-groups  true
open-systems-enabled  false
power-edge-router     true
power-vpn-connections  false
routes                true
shared-images         false
transit-gateway-connection  false
```

Power Edge Router Details:

```
State      active
Type       automated
```

Network Security Group Details:

```
State      active
```

 **Note:** The absence of the Power Edge Router (PER) section in the Workspace details panel and in the `$ ibmcloud pi workspace get <WORKSPACE_ID>` output indicates that the workspace is not PER-enabled and is likely configured to use Cloud Connection instead.

Resolving a fallback Cloud Connection workspace

To resolve a fallback Cloud Connection workspace, you can delete the workspace and create a new PER-enabled workspace. To create a new PER-enabled workspace, complete the steps provided in the [Creating a PER workspace](#) section.

 **Important:** Deleting a workspace does not automatically remove the failed underlying network configuration. To remove the failed network configurations, [open a support ticket](#).

Best practices for creating PER-enabled workspaces

To prevent scenarios where workspaces are created without PER capabilities, allow enough time between workspace creation requests when creating multiple PER-enabled workspaces in succession.

 **Tip:** A time gap between workspace creation requests reduces the likelihood of backend network configuration conflicts that can lead to PER-enabled workspace provisioning failures.

OS support in a PER workspace

AIX, IBM i, and Linux operating systems are supported in a PER workspace.

AIX and IBM i support on PER

AIX and IBM i operating systems operate in PER workspaces in the same way that they do in non-PER workspaces.

Full Linux Subscription with PER

 **Note:** See [Full Linux® subscription for Power Virtual Servers](#) for Power Virtual Server to register `RHEL84`, `SLES SP2`, `SLES SP3` images on a non-PER workspace.

Full Linux subscription `RHEL86` and `SLES15 SP4` images can be used in a PER workspace. Follow these instructions for a PER-enabled workspace to let the virtual server instance automatically register a full Linux subscription:

1. Create a private network.
 - a. Open the Power Virtual Server user interface from the IBM Cloud console.
 - b. Click **Subnets** under **Networking** in the left navigation menu.
 - c. Click **Create subnet**.
 - d. Enter a unique name and CIDR. Make sure the CIDR being used is not the same as another CIDR already in use or a subset of that CIDR. The host server for the satellite server will be unable to resolve a network conflict as a result.
 - e. Enter `161.26.0.10` in the **DNS server** field.
2. Create a virtual server instance. See, [Configuring a Power Virtual Server instance](#) for detailed instructions.
3. Attach the private network created in step 1.
4. Verify whether the registration is successful with the following commands:

For SUSE:

```
$ SUSEConnect -s
```

For RHEL:

```
$ subscription-manager status
```

CLI and API support with PER

PER uses the same existing Power Virtual Server network APIs and CLIs.

For more information, refer to the Power Virtual Server documentation on:

- API - [Create a cloud connection](#)

Data centers that support PER

The following table shows the data centers for Power Virtual Server that supports PER:

Location	Data centers	PER availability
Mumbai	in-mum-1	Yes
	in-mum-2	Yes
Chennai	CHE01	No
	CHE02	Yes
	CHE03	Yes
Dallas	DAL10	Yes
	DAL12	Yes
	DAL13	Yes
	DAL14	Yes
Frankfurt	FRA04	Yes
	FRA05	Yes
London	LON04	Yes
	LON06	Yes
Madrid	MAD02	Yes
	MAD04	Yes
Montreal	MON01	No
Osaka	OSA21	Yes
Sao Paulo	SAO01	Yes
	SAO04	Yes
Sydney	SYD04	Yes
	SYD05	Yes
Tokyo	TOK04	Yes
Toronto	TOR01	Yes
Washington DC	WDC04	Yes
	WDC06	Yes
	WDC07	Yes

Power Virtual Server supported data centers and its status for PER

Network security groups in IBM Power Virtual Server

IBM Power Virtual Server in [IBM data center](#)

A network security group (NSG) defines security rules to allow or deny specific network traffic for resources that are provisioned in an IBM® Power® Virtual Server workspace. You can create NSGs to inspect and filter network traffic between resources in Power Virtual Server workspaces.

Security rules in an NSG define which inbound network traffic is allowed or denied from reaching members. Members are one or more network interfaces (NIC) at the subnet and virtual server instance (VSI) level. All outbound or egress traffic is automatically allowed.

Using NSGs in your Power Virtual Server environment provides the following benefits:

- Enhances security and traffic control to limit unauthorized access to VSIs and network resources
- Supports creating specific security rules that are based on source, destination, port, and protocol (TCP, UDP, ICMP, and Any)
- Incurs no additional cost; NSG is included as part of Power Virtual Server workspaces
- Does not negatively affect overall network throughput or network latency for any NSG members

Existing workspaces can support NSG only after the data center where the workspaces are deployed is updated to use the new metering code. The metering code must be based on cloud resource name (CRN). For more information about the rollout schedule, see [Release notes](#).

NSG components

Review the following components of the NSG implementation in Power Virtual Server:

- [Network address groups](#)
- [Members](#)
- [Rules](#)

Network address groups

A network address group (NAG) is a collection of one or more external network address ranges, such as Classless Inter-Domain Routing (CIDRs), that are not directly related to provisioned network resources in your Power Virtual Server workspace. NAGs are used to categorize external network address ranges, such as those accessible over Transit Gateway or natively by PER, such as [IBM Cloud Private endpoints](#).

NAGs serve as reference points in NSG rules and simplify NSG configuration by grouping multiple external IP addresses, subnets, or virtual network segments under a single entity. Instead of listing multiple IP addresses or subnets in an NSG rule, you can refer to a NAG to apply the same rule to all addresses in that group.

Using NAGs provides the following benefits:

- Simplifies management of NSG rules for external network addresses
- Reduces complexity when you apply security policies across multiple addresses
- Improves scalability, as you can add new IP addresses, subnets, or virtual network segments to an NAG without modifying the NSG rules

Members

Members are one or more network interfaces that are provisioned in your Power Virtual Server workspace. A network interface can be identified by its Ethernet MAC address or its IPv4 address. Do not add the same network interface as a member to two different NSGs, one identified by the MAC address and the other by the IPv4 address. Doing so can lead to unexpected behavior.

An NSG is a collection of members and security rules. For more information, see [rules](#).



Note: An NSG member can only be associated with a single NSG. A provisioned network resource cannot belong to multiple NSGs concurrently.

Rules

Security rules define the inbound network traffic that is allowed or denied from reaching members of an NSG. By default, all inbound network traffic is denied unless security rules are explicitly defined. To learn more about defining rules to allow inbound traffic, see [Creating a rule on an NSG](#).



Note: All outbound traffic is automatically allowed.

Network traffic rules and precedence in NSG

Review the following topics to understand how network traffic and security rules are processed and prioritized in an NSG-enabled Power Virtual Server workspace:

- [Rules evaluation order](#)
- [NAG precedence in traffic matching](#)

Rules evaluation order

The order in which a rule identifies inbound network traffic to be allowed or denied depends on the following key factors:

- Deny rules are always evaluated first and take the highest precedence. If a deny rule matches the traffic, the traffic is immediately blocked and the evaluation stops, regardless of any existing allow rules.
- The system processes allow rules only after all deny rules are evaluated. If no deny rules are matched, the system proceeds to evaluate the allow rules.
- If multiple allow rules overlap, the first matching rule in the list is applied. Overlapping rules do not cause conflicts. For example,
 - **Rule A:** Allow TCP `All`.
 - **Rule B:** Allow TCP `Port 22`.

Result: Either rule might be applied, depending on which rule is matched first.

- Traffic that does not match any allow rules is denied by default, following the implicit deny rule.

NAG precedence in traffic matching

Review the following information to understand how inbound network traffic is matched to NAGs:

- Inbound network traffic is matched to the most specific CIDR in any custom NAGs. When allow or deny rules for an NAG are evaluated, the source IP address is matched against the most specific CIDR in a custom NAG and not the default NAG (`0.0.0.0/0`).
- If a source IP falls in a more specific NAG, it must have an explicit NSG rule allowing network traffic from that NAG.
- If a more specific CIDR exists in a custom NAG, relying on the default NAG (`0.0.0.0/0`) does not allow inbound traffic.

For example, consider a remote system and a custom NAG with the following configuration:

- Remote system with the IP address `10.55.55.2`.
- Existing custom NAG with the CIDR `10.55.55.0/24`.

To allow traffic from `10.55.55.2`, an NSG rule must explicitly allow traffic from the custom NAG (`10.55.55.0/24`). This traffic does not match against the default NAG (`0.0.0.0/0`).

For more information about working with NAGs, see [Creating and managing NAGs in a workspace](#).

Setting up network security groups in a workspace

When you provision a workspace in your Power Virtual Server environment, a default NSG and NAG are automatically created to allow bidirectional communication between the existing network interfaces in that workspace.

Review the following topics to set up, configure, and manage NSGs:

- [Enabling or disabling NSG on a workspace](#)
- [Creating and managing NSGs in a workspace](#)
- [Creating and managing NAGs in a workspace](#)
- [Managing rules in an NSG](#)
- [Adding members to an NSG and managing them](#)

Enabling or disabling NSG on a workspace

You can enable or disable the NSG feature on PER and enhanced CRN-enabled workspaces. However, you cannot enable NSG on non-PER, manual, VPN, or Cloud Connection workspaces.

To determine whether your Power Virtual Server workspace has the prerequisites to support NSGs, run the following IBM Cloud CLI command:

```
$ ibmcloud resource service-instance <WORKSPACE_CRN> -o json
```


Verify that the output contains the `resourceCRNs` property with a value of `true`:

```
$ "extensions": {  
  "resourceCRNs": true  
},
```

To enable or disable the NSG feature on an existing workspace, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Locate the workspace on which you want to enable or disable the NSG feature, click the overflow menu icon (three vertical dots), and select **View details**. The "Workspace details" panel is displayed.
5. In the "Workspace details" panel, set the **Network security groups** toggle to **Enabled** or **Disabled**.

When the NSG feature is enabled on a workspace, a default NSG with two rules is automatically created. The default NSG includes all existing network interfaces (members) in the workspace. The first rule allows all inbound traffic from other members in the **Default local addresses** NSG. The second rule allows all inbound traffic from the **Default external addresses** NAG.



Important: Before you disable the network security groups feature on a workspace, you must delete all the existing NSGs and NAGs, except the NSG and NAG that are tagged as **Default**.



Tip: You can delete the default rules to achieve complete isolation between members in the same NSG and from external addresses. When the default rules are deleted, all inbound traffic is blocked. To delete these rules, complete the steps that are listed in the [Deleting rules from an existing NSG](#) section.

Creating and managing NSGs in a workspace

You can create an NSG with the default configuration (no rules or members), or you can define inbound rules and members when you create the NSG.

Creating an NSG with the default configuration

When you create an NSG in your workspace, you are not required to define security rules or add members. However, after an NSG is created with default settings, you can add rules and members to it later.

By default, all inbound network traffic is denied and does not reach any member of the network security group. To create an NSG with the default configuration, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace in which you want to create the NSG. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.

Before you can create an NSG, you must enable the NSG feature on the workspace. If NSG is not enabled, the **Create network security group** button is unavailable. To enable NSG on a workspace, see [Enabling or disabling NSG on a workspace](#).

6. Click **Create network security group**. The "Create network security group" page is displayed.
7. In the **General** section, enter a name for the network security group in the **Name** field. Optionally, enter one or more tags in the **User tags** field to help organize and identify the NSG. Click **Continue**.
8. In the **Inbound rules (optional)** section, click **Continue** to skip adding rules.

If you skip this step, a warning is displayed: *"All network traffic will be denied to members."* This is expected behavior. By default, all inbound traffic is denied until rules are defined. You can add rules later.

9. In the **Members (optional)** section, click **Finish**, and then click **Create**.

Creating an NSG with inbound rules and members

You must explicitly define inbound rules to allow or deny traffic. To create and define an NSG with inbound rules and members, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace in which you want to create the NSG. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Click **Create network security group**. The "Create network security group" panel is displayed.
7. In the General section, enter a name in the **Name** field and click **Continue**.
8. In the Inbound rules (optional) section, click **Create rule**. The "Create rule" panel is displayed.
9. Select any of the following actions:
 - **Allow**: Use this option to allow network traffic from the specified Remote.
 - **Deny**: Use this option to create traffic exceptions by overriding the allow rules.
10. Select one of the following protocols based on the type of traffic that you want to control:
 - **Transmission Control Protocol (TCP)**: Ideal for reliable, connection-oriented traffic such as web (HTTP/HTTPS), SSH, RDP, FTP, and databases. This protocol guarantees that all packets reach the destination in the correct order.
 - **User Datagram Protocol (UDP)**: Ideal for fast, real-time, and connectionless traffic such as DNS, VoIP, network monitoring, video streaming, and gaming. This protocol does not guarantee delivery, order, or error correction.
 - **Internet Control Message Protocol (ICMP)**: Ideal for network diagnostics such as `ping` and `traceroute`. Unlike TCP or UDP, ICMP does not support data transmission between applications.
 - **Any**: Ideal for creating rules that apply to any protocol over a specific port or a set of ports. However, this option does not provide control over specific protocol-based security.
11. Depending on the protocol that you select from the list in the previous step, complete the following steps:

TCP

1. **Remote**: Select a network security group or network address group to allow or deny traffic for the rule. If you do not see the network address group that you want, click **Create network address group** to create one.
2. **Source port range**: Specify the range of ports from which the inbound traffic originates. The valid range is `1–65535`.
3. **Destination port range**: Specify the range of ports on which the inbound traffic is received. The valid range is `1–65535`.
4. **Allowed/Denied flags**: You can use the **Advanced** section to select one or more TCP flags from the **Allowed/Denied flags** list to evaluate the incoming packets based on the selected flags. If you do not select any flags, the network fabric does not apply flag-based filtering. When you select multiple flags, the network fabric applies the rule only to the packets in which all the selected flags are present simultaneously. Therefore, the incoming packet must contain every flag that you have selected for the rule to take effect. You can select the following TCP flags:

- `ack`
- `fin`
- `rst`
- `syn`

UDP

1. **Remote**: Select a network security group or network address group to allow or deny traffic for the rule. If you do not see the network address group that you want, click **Create network address group** to create one.
2. **Source port range**: Specify the range of ports from which the inbound traffic originates. The valid range is `1–65535`.

3. **Destination port range**: Specify the range of ports on which the inbound traffic is received. The valid range is **1–65535**.

ICMP

1. **Remote**: Select a network security group or network address group to allow or deny traffic for the rule. If you do not see the network address group that you want, click **Create network address group** to create one.
2. **ICMP Message type**: Select the required message type from the list. Supported types are:

- **all**
- **echo**
- **echo-reply**
- **source-quench**
- **time-exceeded**
- **destination-unreach**

Any

1. **Remote**: Select a network security group or network address group to allow or deny traffic for the rule. If you do not see the network address group that you want, click **Create network address group** to create one.

12. Click **Create rule**.

13. Click **Continue** to add members to the NSG.

14. In the Members (optional) section, click **Add member**. All existing virtual server instances in the workspace are listed. If no virtual servers are listed, create one by following the steps in [Creating an IBM Power Virtual Server](#).

Adding members to a network security group allows you to control inbound network traffic to the associated network interfaces. A member can be associated with only one network security group at a time.

15. Select the virtual server instance and click **Next**. A list of network interfaces that belong to the virtual server instance is displayed.

16. Select one or more network interfaces from the list and click **Add member**.

17. Click **Finish**.

18. Click **Create**.

The NSG is created and listed on the **Network security groups** tab.

After the NSGs are created, you can manage them by performing the following actions:

- [Renaming an NSG](#)
- [Cloning an NSG](#)
- [Deleting an NSG](#)

Renaming an NSG

To rename an NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG that you want to rename. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking** > **Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Click the overflow menu icon (three vertical dots) on the NSG that you want to rename, and select **Edit**. The "Edit network security group details" panel is displayed.
7. Enter a new name for the NSG in the **Name** field and click **Save**.

Cloning an NSG

When you clone an NSG, a new network security group is created with the same rules and member configurations as the original. To clone an NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG that you want to clone. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Click the overflow menu icon (three vertical dots) on the NSG that you want to clone, and select **Clone**. The "Clone network security group" dialog is displayed.
7. Enter a name for the NSG in the **Name** field and click **Clone**.

Deleting an NSG

You can delete any NSG that you created. However, you cannot delete the default NSG that is created when the NSG feature is enabled on a Power Virtual Server workspace.



Important: You cannot delete an NSG or NAG until all the associated rules and members are removed. For more information, see [Deleting rules from an existing NSG](#) and [Removing members from an NSG](#).

To delete an NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG that you want to delete. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Click the overflow menu icon (three vertical dots) on the NSG that you want to delete, and select **Delete**. The "Delete network security group" dialog is displayed.
7. In the confirmation field, enter the name of the NSG, and click **Delete**. The NSG is permanently deleted.



Important: When you delete a Power Virtual Server workspace, all NSGs in that workspace are also deleted.

Creating and managing NAGs in a workspace

NAGs are part of the inbound rules that define inbound traffic from network addresses external to your Power Virtual Server workspace to members of an NSG. You can create NAGs in your workspace and add CIDR addresses to them. You can create up to 10 NAGs in a single workspace.



Note: In addition to customer-defined NAGs, a default NAG consisting of the `0.0.0.0/0` CIDR is available.

To create an NAG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace in which you want to create the NAG. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed.
6. Select the **Network address groups** tab. A list of existing NAGs is displayed.
7. Click **Create network address group**. The "Create network address group" panel is displayed.

8. Enter a name in the **Name** field. Optionally, enter user tags in the **User tags (optional)** field.
9. Optionally, enter one or more CIDR addresses in the **CIDR** field to add external IP addresses as members. To add more CIDR addresses, click **Add another**.

Use the following guidelines when entering CIDR addresses:

- CIDR notation must follow the standards defined in [RFC 1518](#) and [RFC 1519](#).
- The valid CIDR format is `<IPv4 address>/<number>`. For example, `192.168.1.0/24` represents the IP range of `192.168.1.0 — 192.168.1.255` and `10.0.0.0/16` represents the IP range of `10.0.0.0—10.0.255.255`.



Note: The same CIDR cannot be used in more than one NAG. You can add a maximum of three members when creating an NAG, but you can add more members after the NAG is created.



Tip: You can use CIDR `161.26.0.0/16` to identify IBM Cloud IaaS private endpoints such as DNS, Linux® software repositories, and NTP. You can use CIDR `166.8.0.0/14` to identify IBM Cloud PaaS private endpoints such as IBM Cloud Databases.

10. Click **Create group**.

After you create an NAG, you can manage it by performing the following actions:

- [Renaming an NAG](#)
- [Deleting an NAG](#)

Renaming an NAG

To rename an NAG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NAG that you want to rename. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**.
6. Select the **Network address groups** tab. A list of existing NAGs is displayed.
7. Click the overflow menu icon (three vertical dots) on the NAG that you want to rename, and select **Edit**. The "Edit network address group details" panel is displayed.
8. In the **Name** field, enter a new name and click **Save**.

Deleting an NAG

To delete an NAG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NAG that you want to delete. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**.
6. Select the **Network address groups** tab. A list of existing NAGs is displayed.
7. Click the overflow menu icon (three vertical dots) on the NAG that you want to delete, and select **Delete**. The "Delete network address group" dialog is displayed.
8. Click **Delete** to initiate the deletion request. This action cannot be undone.

Creating and managing inbound rules in an NSG

You can manage inbound security rules in an NSG by performing the following operations:

- [Creating a rule on an NSG](#)
- [Cloning rules on an existing NSG](#)
- [Deleting rules from an existing NSG](#)

Creating a rule on an NSG

You can create rules on an NSG in two ways: when you initially create the NSG, or later by revisiting the "Network security group details" page. To create rules when you are creating an NSG, see [Creating an NSG with inbound rules and members](#).



Note: NSGs are stateless. Because all outbound traffic is automatically allowed, you must define rules for all return packets that flow into NSG members from other NSGs or NAGs.

To create rules on an existing NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG in which you want to create rules. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Select the NSG in which you want to create the rule. The "Network security group details" page is displayed.
7. In the Inbound rules section, click **Create rule**. The "Create rule" panel is displayed.
8. Complete the steps provided in the [Creating an NSG with inbound rules and members](#) section.

Cloning rules on an existing NSG



Note: To create an inbound rule with the same configuration as an existing inbound rule, you can clone the rules of an existing NSG. Only the rules in an NSG can be cloned, not the members.

When you clone an existing rule, all properties such as action (allow or deny), protocol, remote, and source and destination port ranges, are copied to the new rule. The cloning process ensures that all rules in an NSG are transferred when a member is moved from one NSG to another.

You can also customize the cloned properties of a rule to create a different rule efficiently and quickly. To clone an existing rule, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG with the rules that you want to clone. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Select the NSG that contains the rule that you want to clone. The "Network security group details" page is displayed.
7. In the **Inbound rules** section, select the tab (**TCP**, **UDP**, **ICMP**, or **Any**) that contains the rule that you want to clone.
8. Click the overflow menu icon (three vertical dots) on the rule that you want to clone, and select **Duplicate**. The "Create rule" panel is displayed.
9. Click **Create rule**. The new rule is created with the same configuration. You can also create a rule with different configurations by making the necessary changes before you click **Create rule**.

Deleting rules from an existing NSG

To delete a rule from an existing NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.

3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG with the rule that you want to delete. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Select the NSG with the rule that you want to delete. The "Network security group details" page is displayed.
7. In the **Inbound rules** section, select the tab (**TCP**, **UDP**, **ICMP**, or **Any**) that contains the rule that you want to delete.
8. Click the overflow menu icon (three vertical dots) on the rule that you want to delete, and select **Delete**. The "Delete network security group rule" dialog is displayed.
9. Click **Delete** to initiate the deletion request. This action cannot be undone.

Adding members to an NSG and managing them

You can manage members that are associated with an NSG by performing the following operations:

- [Adding members to a network security group](#)
- [Moving members from one NSG to another NSG](#)
- [Removing members from a network security group](#)

Adding members to a network security group

You can add members to an NSG to control the inbound network traffic to the associated network interfaces.

You can add members in two ways: when you initially create the NSG, or later by revisiting the **Network security group details** page of an existing NSG. To add members when you are creating an NSG, see [Creating an NSG with inbound rules and members](#).

To add members to an existing NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG for adding members. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Select the NSG to which you want to add members. The "Network security group details" page is displayed.
7. In the Members section, click **Add member**. Existing virtual server instances that are part of the workspace are listed. If virtual servers are not listed, you can create a virtual server by completing the steps provided at [Creating an IBM Power Virtual Server](#).

A member can only be associated with one network security group at a time.

8. Select the virtual server instance and click **Next**. All existing network interfaces for the virtual server instance are displayed.
9. Select one or more network interfaces from the list and click **Add member**.
10. Click **Create**.

Moving members from one NSG to another NSG

You can use the **Move** option to transfer members from one NSG to another NSG in some of the following situations:

- **Security policy changes**: When security policies or requirements evolve, you can use the Move option to apply updated rules to specific members without affecting other members in the original NSG.
- **Environment segmentation**: During network reconfiguration or restructuring, such as separating development, staging, and production environments, you can move members to different NSGs to achieve member isolation and meet compliance or operational requirements.
- **Access control adjustments**: When a member requires access to a different set of services or external networks, you can move the member to a new NSG with appropriate rules to ensure the required connectivity and permissions.
- **Incident response and troubleshooting**: When you respond to cybersecurity incidents or troubleshoot connectivity issues, you can move the affected member to a more restrictive or isolated NSG to contain the security threat and safely test network behavior.



Important: When you move a network interface (member) from one network security group to another, the change is applied instantly and does not require approval. After the member is successfully moved and associated with the new NSG, the rules of the new NSG take effect immediately.

To move a member from one NSG to another NSG, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, enter **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
4. Select the workspace that contains the NSG from which you want to move the members. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
6. Select the NSG from which you want to move the members. The "Network security group details" page is displayed.
7. In the Members section, click the overflow menu icon (three vertical dots) on the member that you want to move, and select **Move**. The "Move network interface" dialog is displayed.
8. From the **Target network security group** list, select the NSG to which you want to move the member.
9. Click **Move network interface**.

Removing members from an NSG

To remove members that are associated with an NSG, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
3. Select the workspace that contains the NSG from which you want to remove the members. The "Virtual server instances" page is displayed.
4. In the navigation panel, click **Networking > Network security groups**. The "Network security groups" page is displayed with a list of existing NSGs on the **Network security groups** tab.
5. Select the NSG from which you want to remove the members. The "Network security group details" page is displayed.
6. In the **Members** section, click the delete icon next to the member entry that you want to remove. The Delete network security group member confirmation message box appears.
7. Click **Delete** to initiate the deletion request. This action cannot be undone.



Note: When you remove a network interface or a subnet that is attached to a VSI, the NIC is also detached from all associated NSGs.

Quotas and limitations

NSGs have the following limitations in the Power Virtual Server environment:

- You can create a maximum of 10 NSGs and 10 NAGs in a single workspace. These quotas are defined at the data center level. To request an increase, open a [support ticket](#), subject to standard limits.



Important: The support tickets for quota increases are evaluated by Customer Support and the Power Virtual Server operations team. IBM makes the final decision at its discretion and might decline the request depending on resource availability and the business justification.

- A network interface can be identified by its Ethernet MAC address or IPv4 address. Do not add the same network interface to two different NSGs using different identifiers. Doing so can result in unexpected network traffic behavior and is not a supported configuration.
- A member can only be associated with one NSG at a time.
- You cannot add members to an NSG if their attached network interface (NIC) is assigned a public IP address. NSGs can only be applied to private networks.

Security considerations

When the NSG feature is disabled on a Power Virtual Server workspace, the system allows unrestricted communication between VSIs and subnets in that

workspace. Enabling the NSG feature on a workspace maintains this behavior but also allows you to control the network traffic with security rules to meet specific requirements.

Public network connectivity for Power Virtual Server

IBM Power Virtual Server in IBM data center

In some Power Virtual Server data centers, internet access is provided through alternative networking configurations because you cannot directly attach public subnets to the logical partition (LPAR) or virtual server instance (VSI) in these data centers. In these cases, you can use the virtual private cloud (VPC) infrastructure components and a transit gateway connection to provide both outbound and inbound network connectivity. A transit gateway connects VPCs and Power Virtual Server workspaces.

Configuring the public network for a Power Virtual Server instance

Complete the following steps to configure public network access for your LPAR or VSI:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. In the navigation panel, click **Workspaces**. The Workspaces page with a list of existing workspaces is displayed.
4. Click your workspace in the list to open the "Virtual server instances" page.
5. Create a VSI. For instructions, see [Configuring a Power Virtual Server instance](#).
6. Configure the outbound networking for the VSI. For instructions, see [Outbound network overview](#).
7. Configure the inbound networking for the VSI. For instructions, see [Inbound network overview](#). You can configure inbound networking only after you configure outbound networking.



Note: You must select the same region and zone as your VSI for all the configurations.

Outbound network overview

Power Virtual Server instances can access the internet through outbound network connections. The following architecture describes this setup:

1. Connect the Power Virtual Server workspace to a transit gateway. The transit gateway provides the connection between the Power Virtual Server instance and the VPC infrastructure.
2. Connect the transit gateway to a VPC that contains a network load balancer (NLB) that is configured in routing mode.
3. Route internet traffic from Power Virtual Server instances through a VPC subnet to the internet by using the NLB as a routing gateway. The VPC security groups that are associated with the NLB filter the traffic between the Power Virtual Server instance and the internet.
4. Attach the VPC subnet to a public gateway that provides the actual internet connectivity.

The following diagram illustrates the outbound network architecture to show how traffic flows from Power Virtual Server instances through the transit gateway, VPC, and NLB to reach the internet.



When an LPAR or VSI in a Power Virtual Server workspace sends traffic to the internet, it sends the data packets to the default gateway. The default gateway routes the traffic through the transit gateway to the VPC. The VPC routing table directs traffic to the NLB, which forwards the traffic through the public gateway to the internet. The return traffic follows the same path back to the Power Virtual Server instance.

Considerations for outbound connectivity

Planning the outbound connectivity for your Power Virtual Server instance includes the following considerations:

- A workspace with no external connections restricts LPAR and VSI communication to resources within the same Power Virtual Server workspace only.
- A workspace that is connected to a transit gateway links to a VPC with internet access and enables outbound connectivity for the workspace resources.
- Your Power Virtual Server instances use the network connectivity defined in the workspace configuration. As a result, you need not configure network access for individual LPARs or VSIs.

Configuring outbound network access for Power Virtual Server

Complete the following steps to configure outbound network access for Power Virtual Server:



Note: Ensure that a VSI is created before you configure outbound network access. For more information, see [Configuring public network for Power Virtual Server instance](#).

Step 1: Create the VPC infrastructure.

1. Create a VPC and subnet. For instructions, see [Creating a VPC and subnet](#).

When you create a VPC, set the following values:

- Select the same region as your Power Virtual Server workspace.
- Ensure that you select the **Create a default prefix for each zone** checkbox.

If the **Create a default prefix for each zone** checkbox is selected, IBM Cloud automatically creates default subnets in each available zone. You can use these default subnets for the NLB or create a custom subnet.

2. Record the subnet ID and Classless Inter-Domain Routing (CIDR) range of the subnet. Use the subnet ID and CIDR range when you create an NLB. For more information, see [Create a private NLB in routing mode](#).
3. Create a public gateway and attach the public gateway to the VPC subnet. For instructions, see [Creating public gateways](#).

Step 2: Create a private NLB in routing mode.

1. Create a private NLB in routing mode. For instructions, see [Creating a network load balancer with routing mode by using the UI](#).

When you create the NLB, set the following values:

- Select the same VPC and VPC subnet that you created in [Step 1: Create the VPC infrastructure](#).
- In the details section, select the type as **Private**.
- Set **Routing mode for VNFs** to **Enabled** to enable the routing mode for virtual network functions (VNFs).
- Leave the **Back-end pools** and **Front-end listeners** sections empty.
- In the **Security groups** section, select the VPC default security group.

2. Create a back-end pool for your NLB. For instructions, see [Working with network load balancer pools](#).

When you create the back-end pool for your NLB, set the following values:

- Leave the **Instance group** field empty.
- Select **Bypass** as the failsafe policy.

3. Configure a listener for your NLB. For instructions, see [Working with listeners](#).

When you create the listener for your NLB, select the back-end pool for your NLB as the default back-end pool.

Step 3: Create a transit gateway and connect to Power Virtual Server.

1. Create a transit gateway. For instructions, see [Creating a transit gateway](#).

When you create the transit gateway, set the following values:

- Select the default resource group.
- Ensure that **GRE enhanced route propagation** is set to **Disabled**. For more information, see [Generic Routing Encapsulation \(GRE\) connection considerations](#).
- Select **Local routing** for a single region.
- Choose a location that matches your Power Virtual Server workspace location.
- Create the transit gateway without adding network connections.

2. Connect the VPC to the transit gateway. For instructions, see [Adding a connection](#).

When you connect the VPC to the transit gateway, set the following values:

- Select **VPC** as the network connection.
- In the **Connection reach** area, click **Add new connection in this account**.
- Use the same region, which is the region of the transit gateway.

- Select your VPC in the **Available connection** section.

3. Connect the Power Virtual Server workspace to the transit gateway. For instructions, see [Adding a connection](#).

When you connect the Power Virtual Server workspace to the transit gateway, set the following values:

- Select **Power Systems Virtual Server** as the network connection.
- Select **Add new connection in this account** as the connection reach.
- Use the same region, which is the region of the transit gateway.
- Select your Power Virtual Server workspace in the **Available connection** section.

Step 4: Configure VPC routing.

1. Create a custom routing table. For instructions, see [Creating a routing table](#).

When you create the routing table, set the following values:

- Select your VPC.
- In the **Traffic** section, set the **Transit gateway** to on.
- Set **Advertise to** to **On**.

2. Create a route. For instructions, see [Creating a route](#).

When you create the route, set the following values:

- Select the same zone as your NLB.
- Set **Destination CIDR** to the following value:

0.0.0.0/0
- Set **Priority** to **2**, **Action** to **Deliver**, and **Next hop (IP address)** to the IP address of your NLB.
- Set **Advertise to** to **On**.

Step 5: Test outbound connectivity.

From the Power Virtual Server VSI console, run the following command to test the outbound connectivity:

```
ping -c4 <URL>
```

For example:

```
ping -c4 google.com
```



Note: If your outbound connectivity is not working, check whether your VPC security group is configured correctly. You must review the rules that allow only the required TCP protocols. For more information about security groups, see [About security groups](#).

Inbound network overview

You can enable inbound network connectivity to allow external clients on the internet to initiate connections to your Power Virtual Server instances. This inbound network connectivity requires the [outbound network configuration](#). The following architecture describes this setup:

1. Allocate and bind a public IP address range to your VPC in the same zone as your NLB.
2. Configure a secondary network interface within the guest operating system on your Power Virtual Server instance by using the public IP address.
3. Configure VPC route tables to direct incoming internet traffic for the public IP address to the NLB.
4. Forward internet traffic from the NLB through the transit gateway to your Power Virtual Server workspace.
5. Create a custom route in your Power Virtual Server workspace to send internet traffic for the public IP address directly to the private IP address of your Power Virtual Server instance.

The following diagram illustrates the inbound network architecture to show how internet traffic is routed through the VPC infrastructure to reach your Power Virtual Server instances.


Inbound network architecture

When a client on the internet sends traffic to your public IP address, the VPC subnet routing table delegates the traffic to the VPC routing table. The VPC routing table forwards the traffic to the private IP address of the NLB. The NLB forwards the traffic through the transit gateway to Power Virtual Server. The workspace routes the traffic directly to your Power Virtual Server instance by using its private IP address. The Power Virtual Server instance receives the traffic on the secondary interface that is configured with the public IP address.

The Power Virtual Server instance sends the response from the public IP address on the secondary interface. Power Virtual Server routes the response through the transit gateway to the VPC, which routes the response to the NLB. The NLB then forwards the response to the internet.

Configuring inbound network access for Power Virtual Server

Configure the inbound network by using the same VPC and subnet that you created in [Step 1: Create VPC infrastructure](#).



Note: Ensure that a VSI is created and the outbound network access is configured before you configure the inbound network access. For more information, see [Configuring public network for Power Virtual Server instance](#).

Step 1: Attach a public IP address range to the VPC.



Note: If you have a public IP address that is attached to your VPC, do not perform this step and go to [Step 2: Locate the NLB IP address](#).

1. Reserve a public address range and attach it to the VPC. For instructions, see [Creating public address ranges](#).

When you create the public address range, set the following values:

- Set the **Size** to the following value:

/32 (1 address)
- Retain the default value for **Geography**, **Region**, and **Resource group**.
- Set **Bind** to on. Select your VPC to attach the public address range.
- Set **Zone** to the same zone as the NLB that you created in [Step 2: Create a private NLB in the route mode](#).

2. Record the allocated public IP address. Use this IP address to access your Power Virtual Server VSI from the internet.

Step 2: Locate the NLB IP address.

Complete the following steps to locate the private IP address of the NLB that you created in [Step 2: Create a private NLB in the route mode](#).



Note: If you have not created an NLB, create an NLB in routing mode. For instructions, see [Step 2: Create a private NLB in the route mode](#).

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. From the **Navigation** menu, click **Infrastructure** > **Network** > **Load balancers**.
3. Select your NLB and record the private IP address.

Step 3: Configure a VPC routing table for the internet traffic.

1. Create a routing table to direct incoming internet traffic for the public IP address to the NLB. For instructions, see [Creating a routing table](#).

When you create the routing table, set the following values:

- Select your VPC.
- Set **Traffic source** to **Internet**.
- Set **Advertise to** to **Off**.

2. Create a route. For instructions, see [Creating a route](#).

When you create the route, set the following values:

- Set the **Zone** to the same zone as your NLB.
- Set **Destination CIDR** to the following value:

1.2.3.4/32
- Set **Priority** to **2**, **Action** to **Deliver**, and **Next hop (IP address)** to the IP address of your NLB.

- Set **Advertise** to **Off**.

Step 4: Create the transit gateway.

The transit gateway that you created for outbound connectivity already connects the VPC and Power Virtual Server. For more information, see [Step 3: Create transit gateway and connect to Power Virtual Server](#).

Step 5: Configure VPC routing for the transit gateway.

The transit gateway is already configured for outbound connectivity. For more information, see [Step 4: Configure VPC routing](#).

Step 6: Configure the VPC routing table that is associated with the NLB subnet.

Route the incoming traffic for the public IP address to the VPC subnet. Configure the default routing table and create a route for your VPC. For instructions, see [Creating a route](#).

When you create a route in the default routing table of your VPC, set the following values:

- Set the **Zone** to the same zone as your NLB.
- Set **Destination CIDR** to the following value:

1.2.3.4/32
- Set **Action** to **Delegate-VPC** and **Next hop (IP address)** to the IP address of your NLB.
- Set **Advertise** to **Off**.



Important: Do not modify the routing table that you created in [Step 4: Configure VPC routing](#).

Step 7: Configure a Power Virtual Server route.

Add a static route to your Power Virtual Server workspace that points to your VSI. For instructions, see [Creating and managing network routes in IBM Power Virtual Server workspaces](#).

When you create the static route, set the following values:

- Set **Destination** to the following value:

1.2.3.4/32
- Set **Next hop** to your VSI IP address.
- Set **Advertise** and **State** to **Enabled**.

This static route configuration routes traffic for the public IP address directly to the VSI.

Step 8: Configure the VSI network.

Because Power Virtual Server does not support network address translation (NAT), you must configure the public IP address as a secondary interface in the guest operating system of your VSI. Retain the existing private interface for normal Power Virtual Server subnet connectivity and add the public IP address to a secondary interface.

Configure the secondary interface in the guest operating system on your VSI, and not in the Power Virtual Server console. Complete this task in the operating system of your VSI that receives the routed internet traffic.

Step 9: Configure the security groups for the NLB to permit inbound internet traffic.

If the outbound connectivity is configured, the required security groups are already mapped to your NLB. Verify that the security groups allow the required inbound ports for your public IP address.

If the security groups that are mapped to your NLB do not permit the inbound internet traffic, configure the security groups to add the rules for inbound internet traffic. For instructions, see [Defining security group rules](#).

For new outbound or inbound connections, map the required security groups to the NLB. You can also adjust the rules to match your security requirements. For more information about security groups, see [About security groups](#).

Step 10: Test the public IP address

Test the public IP address from an external network by using a protocol that is allowed by your network security group.

For example:

```
ping -c4 <IP_ADDRESS>
```

or

```
ssh <IP_ADDRESS>
```

With outbound and inbound network connectivity configured, your Power Virtual Server instance can communicate with external networks by using the VPC infrastructure and a transit gateway.

VPN connections

IBM Power Virtual Server in [IBM data center](#)

The Virtual Private Cloud (VPC) provided Virtual Private Network (VPN) service provides private and low-cost connectivity to the IBM Cloud services for improved reliability and high availability. You can use the Secure Shell (SSH) and other client-managed applications that are running on your host to access your virtual server instances through private IP addresses.

IBM Cloud offers the following two VPN options:

- *VPN for VPC* for site-to-site gateways to safely and securely connect from client-managed environment to resources in VPC, Power, and classic infrastructure.
- *Client VPN for VPC* for client-to-site servers that allow remote devices to secretly connect to the VPC network in a secure manner.



Important: On 14 July 2025, the Power Virtual Server VPNaaS product reached its end of life and is no longer available for use. If you are using Power Virtual Server VPNaaS product, you are encouraged to move to the IBM Cloud VPC VPN to avoid VPN service interruptions. For assistance to upgrade or migrate to IBM Cloud VPC VPN, open a [support ticket](#) or engage with your Customer Support Manager (CSM).

For more information about the available VPN options, see the VPC documentation on [VPNs for VPC overview](#).

Creating a Virtual Private Cloud VPN connection

Complete the following steps for creating a VPC VPN connection:

1. Create a VPC resource.
2. Create a Site-to-Site VPN gateway in VPC.
3. Attach the VPN connection to the Power Virtual Server workspace by using one of the following methods:
 - Use a Transit Gateway in a Power Edge Router (PER) workspace.
 - Use a Cloud connection in a non-PER workspace.



Note: It is recommended that you create a direct cloud connection between the VPC and the Power Virtual Server. Adding a Transit Gateway is feasible, but it incurs extra charges. The cloud connection setup is not required in a PER-enabled workspace.

Architecture diagram

Configuring VPC VPN in a PER workspace



Configuring VPC VPN in a PER workspace

1. Define the client-managed subnet in the address prefix for the VPC.
2. Define a routing table with Transit Gateway and VPN gateway.

Configuring VPC VPN in a non-PER workspace



Configuring VPC VPN in a non-PER workspace

1. Define the client-managed subnet in the address prefix for the VPC.
2. Define a routing table with Direct Link and VPN gateway.

3. Attach Direct Link to workspace subnet.
4. You can choose to attach a Transit Gateway along with the Direct Link, but it incurs extra charges.

Procedure

1. Create a VPC resource. Complete the steps that are documented in [Using the IBM Cloud console to create VPC resources](#).
2. Create a Site-to-Site VPN gateway in VPC. Complete the steps documented in [About site-to-site VPN gateways](#).



Note: To create a VPN connection, use a policy-based VPN.

3. Attach the VPN connection to the Power Virtual Server workspace. Use one of the following procedures that suit your needs:
 - For a PER-enabled workspace, see: [Attaching Transit Gateway to a PER workspace](#).
 - For a non-PER enabled workspace, see: [Creating IBM Cloud connections](#).

Considerations for configuring VPC VPN in a non-PER workspace

You must note the following considerations when you configure VPC VPN in a non-PER workspace:

- Use a policy-based VPN when you configure the VPN connections.
- Add the subnets that are created in the Power Virtual Server to the following lists:
 - Local Classless Inter-Domain Routing (CIDR) list of IBM Cloud VPC
 - Peer CIDR list of IBM Cloud VPC in your client-managed environment
- Enable VPN gateway and traffic source for Direct Link and transit gateway in the Edit Traffic window under the routing table of IBM Cloud VPC.
- Complete one of the following configurations:
 - Select Direct Link that is enabled with transit gateway in the Edit Traffic window under the routing table of IBM Cloud VPC.
 - Disable the configuration for different Power Virtual Server workspaces that are in the same region.

Changing from VPNaaS to the VPC VPN service

To use the VPC VPN service, you must switch from Power Virtual Server VPN as a service (VPNaaS) to the VPC VPN service. The following use case illustrates the steps to set up the VPC VPN connection and confirms that the VPC VPN service is enabled:

1. Create a VPC connection in the same data center as the Power Virtual Server by using the same account as Power Virtual Server. Complete the following configurations:
 - a. Select the **VPN server** and **VPN gateway** values for **Accepts routes from** to configure the default routing table. This configuration allows the traffic between the virtual servers that are members of the VPC subnet and the devices on the remote VPN connection.
 - b. Create a second routing table to complete the following configurations:
 - a. Select the values for VPN server, VPN gateway, and transit gateway.
 - b. Select **Advertise to** under Transit Gateway.

For more information, see [Getting started with Virtual Private Cloud \(VPC\)](#).

2. Create a VPN connection between the VPC and the remote connection of the existing VPNaaS. Complete the following steps:
 - a. Add the workspace CIDRs to the list of local CIDRs in the VPN connection.
 - b. Add the workspace CIDRs to the peer CIDRs list in the remote VPNaaS connection.

You must consider the following conditions when you create the VPN connection:

- Use a policy-based VPN in all the configurations of the VPN connection.
- Subnets for the remote VPN, VPC, and workspace must be distinct.
- Subnets cannot be shared or overlapped.

For more information about VPN options, see [About site-to-site VPN gateways](#).

3. Create a transit gateway by completing the following steps:
 - a. Add VPC to Transit Gateway.
 - b. Click **Create** under **Routing tables** to create a routing table. You can see the list of VPC, CIDR, and the CIDR for the remote VPN.

For more information about transit gateways, see [Getting started with IBM Cloud Transit Gateway](#).

Consider the following assumptions and start the switch from VPNaaS to VPC VPN:

- The connectivity between a virtual server on the VPC and a system on the remote VPN is working.
- The CIDRs are advertised through the transit gateway.



Note: The workspace and the remote VPN are not connected until the switch to VPC VPN is completed.

4. Delete the VPNaaS gateway. You must select the VPNaaS connections that are attached to the workspace.
5. Migrate the workspace to PER. You must remove the active cloud connections attached to the workspace on the other subnets. For more information, see [Migrating to PER](#).
6. Connect to the transit gateway after the workspace is PER-enabled.
7. Verify that CIDRs appear in the routing table. Complete the following steps to verify CIDRs:
 - a. Generate the routing table from the **IBM Cloud** console > **Infrastructure** > **Network** > **Routing tables**.
 - b. Select the VPC associated with your VPN gateway.
 - c. Click the name of the routing table that you want to use.
 - d. Verify that the CIDRs for the workspace appear with the existing CIDRs in the routing table.

Additional information

- [Connecting IBM VPC to IBM Power Virtual Servers and IBM Cloud Object Storage](#).

Network security for Power Virtual Server

IBM Power Virtual Server in **IBM data center**

The infrastructure provides virtual LAN (VLAN) isolation between different tenants, which are enforced at Virtual I/O Server (VIOS) and physical switches and routers.

Default firewall ports

The IBM® Power® Virtual Server network security architecture relies on a set of fixed firewall ports open on the *Juniper vSRX* firewalls:

- 22 (SSH)
- 443 (HTTPS)
- 992 (IBM i5250 emulation SSL)
- ICMP traffic



Note: There are plans to add the ability to dynamically configure the firewall rules in the future.

The following firewall ports are also open, typically used for IBM i logical partitions (LPARs):

- 2005
- 2007
- 2010
- 2012
- 9470
- 9475
- 9476

The port 6443 is also open for miscellaneous purposes. However, the port 6443 is not open for the WDC04 and DAL13 data centers.

If you need extra ports to be opened, use a customer-specific firewall option. The option is available by using an IBM Cloud firewall, such as Vyatta, Juniper vSRX, or FortiGate, and by connecting to Power Virtual Server by using [Direct Link Connect](#). To understand the Power Virtual Server connection methods, see [Network architecture diagrams](#).

Former IBM data center networking configurations

Migrating the existing network configurations to Power Edge Router

IBM Power Virtual Server in [IBM data center](#)

Power Edge Router (PER) is enabled for all Power Virtual Server data centers except `CHE01`. Two migration paths are available depending on how your workspace network was originally configured: self-service automation or by opening a support ticket.

Before you begin

You must have access to Direct Link to complete your PER migration. For more information about required access roles, see [Access role requirements for Power Virtual Server](#).

- If you configured your workspace network without using a support ticket, use the [ibmcloud pi workspace action](#) CLI command to automate the migration. For more information, see [Migrating to PER](#).
- If you manually configured your subnets and Direct Link through a support ticket, migrate your workspace to PER by opening a support ticket. See [Migrating to PER through a support ticket](#).

Determining your Direct Link configuration method

If you are unsure whether you manually configured subnets and Direct Link for your workspace, complete the following steps to verify your configuration method.

Identifying automatically configured Cloud connections

To confirm that you have an automatically configured Cloud connection, check the following two conditions:

1. The connection appears on the **Cloud connections** page in the Power Virtual Server workspace. To verify, complete the following steps:

- a. Log in to the [IBM Cloud catalog](#).
- b. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
- c. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
- d. From the Workspaces page, select your workspace.
- e. Click **Cloud connections** in the left navigation panel. The Cloud connections page is displayed.

If the **Cloud connections** option does not appear, your workspace is already PER-enabled, and this procedure does not apply to you. If your connections are listed here, continue to the next condition.

2. The Direct Link details page displays the message "This Direct Link is read-only". To verify, complete the following steps:

- a. Log in to the [IBM Cloud console](#).
- b. From the navigation menu, select **Infrastructure > Network > Direct link**. The Direct Link page is displayed.
- c. Click the Direct Link connection that has the same name as your Cloud connection.
- d. In the **Details** section, check for the read-only message.

If both conditions are met, your Direct Link connection was automatically configured through Cloud connections. Use the self-service automation to migrate to PER.

Identifying manually configured Direct Link 2.0

To confirm that you have a manually configured Direct Link 2.0, check the following conditions:

1. The connection does not appear on the **Cloud connections** page in the Power Virtual Server workspace. To verify, complete the following steps:

- a. Log in to the [IBM Cloud catalog](#).
- b. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
- c. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
- d. From the Workspaces page, select your workspace.

- e. Click **Cloud connections** in the left navigation panel. The Cloud connections page is displayed.

If the **Cloud connections** option does not appear, your workspace is already PER-enabled. If no connections are listed or your Direct Link is not shown here, your Direct Link might be manually configured.

2. The Direct Link details page does not display the message "This Direct Link is read-only". To verify, complete the following steps:

- a. Log in to the [IBM Cloud console](#).
- b. From the navigation menu, select **Infrastructure > Network > Direct link**.
- c. Click the Direct Link connection.
- d. In the **Details** section, verify that the read-only message is not displayed.

If both conditions are met, your Direct Link connection was manually configured through a support ticket. In this case, you must also migrate to PER by opening a support ticket.

Migrating to PER through a support ticket

To migrate your workspace to PER through a support ticket, complete the following steps:

1. Create a [support case](#) in the IBM Cloud Support Center.
2. Select **Power Virtual Server** under **Topic**, and select **Power VS Network Related** under **Subtopic**.
3. In the **Subject** field, provide a brief description of your migration requirements.
4. In the **Description** box under additional information, provide the following information:
 - The CRN (cloud resource name) of the workspace that you want to migrate
 - The subnet configuration details for the workspace
 - The Direct Link connection details associated with the workspace subnets

The Power Virtual Server operations team processes the support ticket by configuring PER and other network devices in parallel with the Direct Link configuration. When the PER configuration is complete, the team notifies you through a ticket update that the PER configuration is ready for your validation and testing.

5. Click **Next**.
6. Schedule a maintenance window during which you must provision the Transit Gateway to complete the PER network configuration. For more information, see [PER use cases](#). To route the network traffic through PER, complete the following steps:
 - a. In the [IBM Cloud console](#), from the navigation menu, select **Infrastructure > Network > Direct link**.
 - b. Click the Direct Link connection to which your workspace is connected. The connection details page for the selected connection is displayed.
 - c. Under **Shortcuts**, click **Go to Route filters**.
 - d. On the **Route filters** page, set the following filters:
 - a. On the **Import route filters** tab, set the **Default** value to **Deny all import routes**.
 - b. On the **Export route filters** tab, set the **Default** value to **Deny all export routes**.

Setting both filters to deny blocks the Direct Link routes, which redirects the network traffic through the PER network.

Repeat the preceding steps for each Direct Link connection.

7. Verify that the network connection is working correctly. For example, from each VSI, you can ping the IBM Cloud DNS resolvers that are accessible over PER-enabled workspaces by running the `ping 161.26.0.10` or `ping 161.26.0.11` command.

If you encounter any PER connectivity issues, revert to the Direct Link path by setting the **Import route filters** default back to **Permit all import routes** and the **Export route filters** default back to **Permit all export routes**. This unblocks the Direct Link routes. Then disconnect the workspace from the Transit Gateway.

8. After successful verification, follow the steps in [Deleting a Direct Link](#) to remove the Direct Link connections, and then notify IBM by updating the ticket.
9. The Power Virtual Server team marks the workspace as **Migrated** and closes the ticket.
10. Open a support ticket to remove the backend device configuration, and then delete the pre-migration workspace.



Important: Before you delete the pre-migration workspace, you must open a support ticket to remove the backend device configuration first.

After migration, if you need to make additional network configuration changes that require backend device configuration, complete the following actions:

- **Create a subnet:** In the Power Virtual Server user interface, create the subnet, and then open a support ticket so that the Power Virtual Server operations team can configure the subnet on the backend devices.
- **Update or delete a subnet:** First, open a support ticket so that the Power Virtual Server operations team can update or remove the configuration on the backend devices. Then, update or delete the subnet in the Power Virtual Server user interface.

Configuring FLS after migration

If you run Full Linux Subscription (FLS) on virtual server instances (VSIs) in the migrated workspace, you must update the DNS configuration after migration. FLS requires IBM DNS IP addresses to function correctly.

You can update the DNS configuration in two ways:

1. FLS works if you update the migrated subnets with DNS IP addresses.

Update the migrated subnet by adding DNS IP addresses, for example, `161.26.0.10` or `161.26.0.11`.

Run the following command:

```
ibmcloud pi netu <network_id> --dns-servers "127.0.0.1 161.26.0.10 161.26.0.11"
```

Where `127.0.0.1` refers to the local stub resolver on the VSI.

2. For existing VSIs that are deployed by using migrated subnets, add the following DNS IP addresses, or the specific DNS IP addresses for your environment, to the `/etc/resolv.conf` file:

- `nameserver 161.26.0.10`
- `nameserver 161.26.0.11`

Network architecture diagrams for non-PER

IBM Power Virtual Server in [IBM data center](#)

This topic describes typical network architectures that are used in the Power® Virtual Server network architecture and is not an exhaustive list of Power Virtual Server connection methods.



Important: IBM Cloud Direct Link (2.0) Connect is available in all current locations. IBM clients can always contact Megaport connectivity services directly to procure their services.

Power Virtual Server networking environment

When you create a Power Virtual Server, you can configure a private network subnet. For more information, see [Configuring a private network subnet](#).

Power Virtual Server network architectures consist of one or more of the following networks:

- IBM Cloud infrastructure networks - While the following infrastructure network environments offer different features and are managed separately, they can be connected to each other to provide layer-3 IPv4 traffic flow:
 - Classic - Classic network resources include VLANs, subnets, and SSL Virtual Private Network (VPN) access. See [Network security architecture](#) for a description of the classic network components. Bring Your Own IP (BYOIP) is not supported.
 - Virtual Private Cloud (VPC) - VPC network resources include subnets, floating IP addresses, security groups, and VPN gateways. For more information, see [About networking](#). For more information about a course on VPC networking, see [Advanced networking for IBM Cloud VPC](#). BYOIP is supported.
 - Power Systems - Network resources include subnets. BYOIP is supported.
- Overlay networks - These networks exist in the IBM Cloud VMware Shared and VMware Dedicated offerings. While technically hosted in the IBM Cloud classic infrastructure environment, these networks are implemented in VMware NSX and under your direct control, including the IP addressing schema. BYOIP is supported. Therefore, overlay networks cannot be routed by the IBM Cloud infrastructure networks; access is through tunnels.
- External:
 - Internet - Access the internet through resources that are hosted in any of these three infrastructure environments.

- Remote - Connect remote networks to your IBM Cloud networks. You can use the following services to connect to a remote network:
 - Internet VPN - Uses the public internet to connect remote networks and their IBM Cloud networks through a VPN. The VPN is terminated on gateway devices or a service within IBM Cloud.
 - Direct Link - Direct Link is a suite of offerings that enable the creation of direct, private connections between your remote network in the client-managed environment and IBM Cloud, without traversing the public internet. For more information, see [Getting started with IBM Cloud Direct Link \(2.0\)](#).



Note: You can connect Direct Links to either a local or remote IBM Cloud Transit Gateway, which allows the network in the client-managed environment to access all networks that are connected to the IBM Cloud Transit Gateway.

Non Power Edge Router use cases

These use cases describe the following deployment topologies:

- Connecting Power Virtual Server to IBM Cloud classic infrastructure by using IBM Cloud Direct Link (2.0). Typical use cases for this topology are as follows:
 - Use IBM Cloud classic x86 resources to create tiered applications across different hardware platforms, that is, x86 application servers and Power database servers.
 - Build a backup and restore environment based on [IBM Spectrum Protect Cloud Blueprints](#) for both IBM Spectrum Protect and IBM Spectrum Protect Plus topologies. Also, see [AIX Backups with Power Virtual Server](#).
- Connecting Power Virtual Server to the IBM Cloud VPC infrastructure environment by using Direct Link (2.0) Connect. A typical use case for this topology is to use IBM Cloud VPC x86 resources to create tiered applications across different hardware platforms, that is, x86 application servers and Power database servers.
- Connecting Power Virtual Server to the network in the client-managed environment by using MegaPort or Direct Link (2.0) Connect. A typical use case for this topology is that you require access to your Power virtual servers from your external networks, such as networks in your client-managed environment. This topology uses MegaPort services or Direct Link (2.0) Connect.
- Connecting two Power Virtual Server environments by using MegaPort or Direct Link Connect. This topology connects two or more Power virtual server environments together by using MegaPort services or Direct Link (2.0) Connect. Connecting two or more environments together enables use cases, such as disaster recovery.
- Connecting Power Virtual Server to a network in the client-managed environment through the IBM classic infrastructure by using private SSL connection and a jump server. This is a specific use case for connecting to the classic environment so that the SSL VPN connection can be used to access your Power Virtual Servers for operations and administration tasks.
- Connecting Power Virtual Server to a network in the client-managed environment through the IBM Cloud classic infrastructure by using an internet IPsec VPN connection. This use case describes how to connect to the classic environment so that an IPsec VPN connection can be used to access your classic and Power Virtual Servers. Typically, this network architecture is used for small production environments or proof-of-concept, development, and test purposes.
- Connecting Power Virtual Server to a network in the client-managed environment through an IBM Cloud classic infrastructure by using a private Direct Link. A Direct Link enables your remote networks to connect to IBM Cloud over a private connection that does not use public networks.

Multiple topologies, described in this document, can be layered to create a topology that suits your deployment needs.

Connecting to classic infrastructure by using Direct Link (2.0)

In this deployment topology, a Direct Link is used to connect your Power Virtual Server networks to IBM Cloud resources hosted in a classic infrastructure environment. You can either [order Direct Link \(2.0\) Connect](#) connections, or use IBM Cloud connections.



Complete the following steps to implement this scenario:

1. Define the IP address schema of your Power Virtual Server subnets in your Power Virtual Server environment. Your Power Virtual Server instances are hosted on these subnets. Ensure that you do not overlap the subnets with your IBM Cloud classic private subnets, or the IP addressing schema that is used for the services network. For instructions, see [Configuring and adding a private network subnet](#).
2. Order [Direct Link \(2.0\) Connect](#), or provision an [IBM Cloud connection](#) instance from the Power Virtual Server interface to connect the Power Virtual Server router to the IBM Cloud classic infrastructure. This connection is established and operated by the IBM Cloud team and you have no direct access or control.

Consider the following information before you proceed:

- Ensure that you select the virtual routing and forwarding (VRF) option when you provision a Direct Link. The VRF option allows multiple instances of a routing table to exist in a router and to work simultaneously. With VRF, each IBM Cloud account classic environment network is

segmented within its own routing table. For more information, see [Virtual routing and forwarding on IBM Cloud](#).

- When this connection is established, the Border Gateway Protocol (BGP) is configured automatically by the IBM Cloud team in the following way:
 - The Power Virtual Server router advertises your Power Virtual Server subnets to the classic infrastructure environment.
 - The classic infrastructure advertises only your IBM Cloud classic private subnets to the Power Virtual Server router.
 - The classic infrastructure filters the following IP addresses from the BGP advertisements that are coming from the Power Virtual Server router because these IP addresses are used by the services networks: `10.0.0.0/14`, `10.198.0.0/15`, `10.200.0.0/14`, `169.254.0.0/16`, `224.0.0.0/4` and any IP ranges that are assigned to your IBM Cloud classic private subnets.

 **Important:** You can use the `10.x.x.x` range if there is no conflict with an IBM Cloud back-end `10.x.x.x` service. You must contact IBM Support if you want to resolve the IP address conflict by using a Network Address Translator (NAT) device or IP aliasing method. However, IBM does not recommend by using the `10.x.x.x` range when you create a network.

3. Identify the IBM Cloud classic private subnets and the IP address schema that is assigned to you when you ordered bare metal or virtual server instances that are hosted on your IBM Cloud classic private subnets. Connect to the required resources and services.

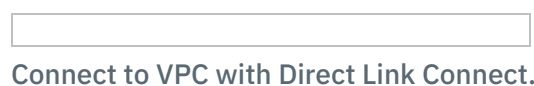
All the routers in all classic infrastructure data centers and point-of-presence (POP) points across the global IBM Cloud backbone are connected to the Power Virtual Server router by using the Direct Link Connect VRF-enabled connection.



Note: Each IBM Cloud Direct Link workspace is not redundant. However, diversity can be engineered by using multiple Direct Links and BGP. For more information, see [Models for diversity and redundancy in Direct Link \(2.0\)](#).

Connecting to VPC by using Direct Link Connect

In this deployment topology, a Direct Link (2.0) is used to connect your Power Virtual Server networks to resources that are hosted in the IBM Cloud VPC infrastructure.



Complete the following steps to implement this scenario:

1. Define the IP address schema of your Power Virtual Server subnets in your Power Virtual Server environment. Your Power Virtual Server instances are hosted on these subnets. Ensure that you do not overlap these subnets with your IBM Cloud classic private subnets or the IP addressing schema that is used for the services network. For instructions, see [Configuring and adding a private network subnet](#).
2. Order [Direct Link \(2.0\) Connect](#), or provision an [IBM Cloud connection](#) instance from the Power Virtual Server interface to connect the Power Virtual Server router to the IBM Cloud classic infrastructure. This connection is established and operated by the IBM Cloud team and you have no direct access or control.

When this connection is established, the BGP is configured by the IBM Cloud team in the following way:

- The Power Virtual Server router advertises your Power Virtual Server subnets to the router in the VPC infrastructure.
 - The router in the VPC infrastructure advertises only your IBM Cloud VPC subnets to the Power Virtual Server router.
 - The router in the VPC infrastructure filters the following networks from the BGP advertisements coming from the Power Virtual Server router because these IP addresses are used by the endpoint networks: `169.254.0.0/16`, `224.0.0.0/4`, `166.9.0.0/16`, and any IP ranges that are assigned to your IBM Cloud VPC subnets.
- 3. Identify the IBM Cloud VPC subnets and the IP address schema that is assigned to you when you ordered VPC services. The cross-connect routers (XCRs) connects to the endpoint networks by using the VPC implicit router that provides routing functions to each VPC, and allows each VPC to have access to its own copy of the IPv4 address space. The Multi-Protocol Label Switching (MPLS) VPN works with Direct Link and classic infrastructure environments. For more information, see [Network isolation, data packet flow, and the role of an implicit router in a VPC](#). You cannot access some endpoint networks (services and infrastructure services) from your Power Virtual Server subnets. The types of endpoint networks are as follows:
 - Service endpoints - Allows connection to IBM Cloud services available through Domain Name System (DNS) names in the cloud.ibm.com domain and resolves to `166.9.x.x` addresses.
 - Infrastructure services - Allows connection to IBM Cloud services from the adn.networklayer.com domain and resolves to `161.26.0.0/16` addresses. Services that you can reach include:
 - DNS resolvers - `161.26.0.10` and `161.26.0.11` (Windows VS is set up with `161.26.0.7`; Linux with `161.26.0.7` and `161.26.0.8`)
 - Ubuntu and Debian mirrors - mirrors.adn.networklayer.com or `161.26.0.6`
 - Network Time Protocol (NTP) - time.adn.networklayer.com or `161.26.0.6`. This is the same IP address as the Ubuntu and Debian mirrors.

- IBM Cloud Object Storage

4. Create Virtual Private Endpoints (VPEs) for services of interest such as DNS, NTP, and IBM Cloud Object Storage. For more information, see [VPE supported services](#).
5. If you want to access the endpoint network, use a VPC virtual server instance to proxy any required endpoint services that do not support VPEs.

However, each IBM Cloud Direct Link service is not redundant diversity can be engineered by using multiple Direct Links and BGP. For more information, see [Models for diversity and redundancy in Direct Link \(2.0\)](#).

Connecting to a client-managed environment by using Megaport

In this deployment topology, [Megaport](#) or Direct Link Connect is used to provide connectivity from your (remote) network in the client-managed environment to your Power Virtual Server subnets.



IBM Cloud Connect is a managed network service that uses Megaport services. This service is available only in the United States. You can also use Megaport to connect your network in the client-managed environment to Power Virtual Server directly.

Review the following characteristics about Megaport connectivity services:

- Megaport operates a global network infrastructure that enables on-demand connectivity to hundreds of global services in Asia Pacific, North America, Europe, and the Middle East.
- A port is the physical point of connection between your organization's network and the Megaport network. While a single data center connection is possible, the best practice is to select two different port locations to provide redundancy.
- Megaport has a number of cloud service providers including IBM Cloud.
- Virtual Cross Connects (VXCs) provide connections between any of the locations and services on the Megaport network. Ordering a VXC (by using the Megaport portal or API) allows you to connect into the Power virtual server environment and optionally, the classic/VPC infrastructure environments or other clouds.



Important: Megaport connectivity services are available in DAL12, DAL13, FRA05, LON06, MON01, SYD05, OSA21, WDC04, and WDC06 data centers.

Complete the following steps to implement this scenario:

1. Define the IP address schema of your Power Virtual Server subnets in your Power Virtual Server environment. Your Power Virtual Server instances are hosted on these subnets. Ensure that you do not overlap these subnets with your IBM Cloud classic private subnets or the IP addressing schema that is used for the services network. For instructions, see [Configuring and adding a private network subnet](#).
2. Procure the [Megaport](#) VXC connections to connect your (remote) network in the client-managed environment to the Megaport network.
 - Open an IBM Support case against Power Virtual Server to receive a service ID or a virtual cross-connect (VxC) identifier from IBM.
 - Engage with Megaport to procure the connection (VxC) to Power Virtual Server Port at Megaport. Although a single data center connection between the network in the client-managed environment and the Megaport network is possible, best practice is to select two different port locations to provide redundancy.
3. Open an IBM Support case against the Power Virtual Server team to configure the Megaport network to the Power Virtual Server router by using VXCs. Remember to include the following pieces of information in your case:

\$ Customer name and contact:
 Customer account ID
 Service ID (VxC Identifier):

Customer network subnet:
 Customer router IP Address:
 Power Virtual Server customer network IP address:
 Power Virtual Server network ASN: 64999 for WDC04 and 64997 for DAL13
 Customer Network ASN:

Customer subnets to be advertised:
 Power Virtual Server customer Private Network ID (1):
 Power Virtual Server customer Private Network ID (2):
 Power Virtual Server customer Private Network ID (3):

The Power Virtual Server router is the default gateway for your Power virtual server instances. This router is operated by the Power Virtual Server team and you have no direct access or control.

Connecting two Power virtual server environments

In this deployment topology, [Megaport](#) or Direct Link (2.0) Connect is used to provide connectivity between Power virtual server environments at two different data centers.



IBM Cloud Connect is a managed network service that uses Megaport services. This service is available only in the United States. You can also use Megaport to connect your network in the client-managed environment to Power Virtual Server directly.

The key features of this Connect-to-classic topology are as follows:

1. Define the IP address schema of your Power Virtual Server subnets in your Power Virtual Server environment. Your Power Virtual Server instances are hosted on these subnets. Ensure that you do not overlap these subnets with your IBM Cloud classic private subnets or the IP addressing schema that is used for the services network. For instructions, see [Configuring and adding a private network subnet](#).
2. Procure the [Megaport](#) VXC connections to connect your (remote) networks in the client-managed environment to the Megaport network.
 - Open an IBM Support case against Power Virtual Server to receive a service ID or a virtual cross-connect (VxC) identifier from IBM.
 - Engage with Megaport to procure the connection (VxC) to Power Virtual Server Port at Megaport. When connecting a Power IaaS Location-1 to Power IaaS Location-2 by using Megaport, you might need a [Megaport Cloud Router \(MCR\)](#) unless network connectivity is through a customer router. Consult a Direct Link Connect or Megaport representative for specific network requirements.
3. Open an IBM Support case against the Power Virtual Server team to perform the network configuration to connect the Megaport network to the Power Virtual Server router by using VXCs. Remember to include the following pieces of information in your case:

\$ Customer name and contact:
Customer account ID
Service ID (VxC Identifier):

Customer network subnet:
Customer router IP Address:
Power Virtual Server customer network IP address:
Power Virtual Server network ASN: 64999 for WDC04 and 64997 for DAL13
Customer Network ASN:

Customer subnets to be advertised:
Power Virtual Server customer Private Network ID (1):
Power Virtual Server customer Private Network ID (2):
Power Virtual Server customer Private Network ID (3):

The Power Virtual Server router is the default gateway for your Power virtual server instances. This router is operated by the IBM Cloud team and you have no direct access or control.

Connecting to a network in the client-managed environment by using a private SSL connection and a jump server

The IBM Cloud SSL VPN service is a feature of the IBM Cloud classic infrastructure, which enables you to manage your classic resources, remotely, over the IBM Cloud Private network. An SSL VPN connection from your location to the private network allows out-of-band management and server rescue through an encrypted VPN tunnel. For more information, see [About VPN](#).

The IBM Cloud SSL VPN service can access only your classic private IP subnets. Therefore, you cannot use the SSL VPN feature to access your Power Virtual Server instances directly from your workstation. Instead, a jump server or bastion host is used to access your Power Virtual Server instances from your network in the client-managed environment.



Note: This deployment topology builds on the [Connect-to-classic](#) architecture.

Complete the following steps to implement the connectivity using jump server or bastion host:

1. Ensure that you meet the following prerequisites:
 - Ensure that your IBM Cloud account has the required permissions, a VPN password is configured, and you have access to the subnets that are

hosting your resources. For instructions, see [Activating or deactivating SSL VPN access for a user](#).

- Ensure that your workstation or laptop at your client-managed location has the stand-alone VPN client installed. For instructions, see [standalone VPN clients \(Windows, Linux, and Mac OS X\)](#)
 - Ensure that you have internet access and the stand-alone client has a connection to the endpoint for the location that is hosting your classic resources. For a complete list, see [Available VPN endpoints](#).
2. Deploy a jump server or bastion host with your preferred operating system. You can connect from your workstation or laptop at your location to the private IP address of your jump server or bastion host by using Remote Desktop Protocol (RDP) or Secure Shell Protocol (SSH).
 3. Establish a connection from the jump server or bastion host to your Power Virtual Server instances by using SSH.



Note: This option is typically used to manage infrastructures and is not recommended for production workloads.

Connecting to a network in the client-managed environment by using an internet IPsec VPN connection

Although individual Power Virtual Server instances can have internet access, there is no site-to-site IPsec VPN service that connects your Power Virtual Server subnets to your remote networks currently available.

Connect to client-managed environment with IPsec VPN.



This deployment topology uses the IBM Cloud classic infrastructure gateway appliance to provide an internet-connected IPsec VPN gateway to enable a site-to-site VPN connection to your Power Virtual Server resources.

- The IBM Cloud gateway appliance allows you to selectively route private and public network traffic through a full-featured, enterprise-level firewall that is powered by the software features of VyOS, JunOS, or any other operating system (Bring Your Own Appliance) that you choose.
- All appliance features are customer-managed.
- By using the IBM Cloud GUI, CLI, or API, you can select your VLANs, and hence, the associated subnets, that you want to associate with your gateway appliance. Associating a VLAN with a gateway appliance reroutes (or trunks) that VLAN and all of its subnets to your appliance, giving you control over filtering, forwarding, and protection.
- A gateway appliance is attached to two nonremovable transit VLANs, one each for your public and private networks.

Complete the following steps to implement this scenario:

1. [Set up an IBM Cloud gateway appliance](#) to establish an IPsec VPN connection from your remote location to the IBM Cloud classic infrastructure.
2. Complete all steps that are mentioned in [Connect to classic infrastructure by using IBM Cloud \(2.0\) Direct Link](#). This includes configuring Power Virtual Server private subnets and provisioning a Direct Link (2.0) Connect or an IBM Cloud connection instance with the [VRF option](#).

The IBM Cloud classic infrastructure environment VRF contains the following routes:

- Subnets that are assigned to you by IBM Cloud for use in your classic environment.
- Your Power Virtual Server subnets advertised by the Power Virtual Server router.

However, it does not contain routes to your remote networks. Static routes or a routing protocol, such as BGP, shares routes between your remote network and the gateway appliance. The routes that are advertised by the gateway appliance include your Power Virtual Server subnets.

3. Configure a Generic Routing Encapsulation (GRE) tunnel between the gateway appliance and the Power Virtual Server router as this router doesn't have routes for your remote networks that are advertised to it through the IBM Cloud-side router. Within the GRE tunnel, static routes are configured between the Power Virtual Server router and the gateway appliance. For detailed steps, see [Configuring Generic Routing Encapsulation \(GRE\) tunnel](#).

For a tutorial on site-to-site VPN connectivity, see [IBM Power Virtual Server Virtual Private Network Connectivity](#).

Additionally, the x86 servers can be deployed on to the subnets contained within the associated and nonassociated VLANs to build complex topologies, such as three-tier architectures.

Connecting to a network in the client-managed environment by using a private Direct Link connection

To connect your external networks to your Power Virtual Server subnets, you can use the Direct Link service; however, you cannot use this connectivity for a direct connection as both your Power Virtual Server subnets and your remote networks are considered to be external networks to the VRF and direct external-to-external traffic is prohibited. Therefore, in this deployment topology, a gateway appliance and GRE tunnels are used to provide connectivity.

This topology focuses on Direct Link (2.0) service that has the following offerings:

- Direct Link Dedicated - A single-tenant, fiber-based, cross-connect into the IBM Cloud Private network.
- Direct Link Connect - Uses a service provider who is already connected to the IBM Cloud Private network to deliver multi-tenant, high capacity links, also known as network-to-network interfaces.

Currently, Direct Link (2.0) is not available in all locations. For more information, see [Locations and providers](#).

If Direct Link (2.0) is not available at a suitable location, then see [Getting started with IBM Cloud Direct Link on Classic](#).



This deployment topology uses the IBM Cloud classic infrastructure gateway appliance to provide a GRE gateway. For more information, see [Getting started with IBM Cloud Gateway Appliance](#).

- The IBM Cloud gateway appliance allows you to selectively route private and optionally public network traffic through a full-featured, enterprise-level firewall that is powered by the software features of; VyOS, JunOS, or any other operating system (Bring Your Own Appliance) you choose.
- All appliance features are customer-managed.
- By using the IBM Cloud GUI, CLI, or API, you can select your VLANs, and hence, the associated subnets that you want to associate with your gateway appliance. Associating a VLAN with a gateway appliance reroutes (or trunks) that VLAN and all of its subnets to your appliance, giving you control over filtering, forwarding, and protection.
- A gateway appliance is attached to two nonremovable transit VLANs, one each for your public and private networks.

Complete the following steps to implement this scenario:

1. [Set up an IBM Cloud Gateway Appliance](#).
2. Complete all steps that are mentioned in [Connect-to-classic infrastructure by using IBM Direct Link](#). It includes configuring Power Virtual Server private subnets and provisioning a Direct Link (2.0) Connect or an IBM Cloud connection instance with the VRF option.
3. The IBM Cloud classic infrastructure environment VRF contains the following routes:
 - Subnets that are assigned to you by IBM Cloud for use in your classic environment.
 - Your Power Virtual Server subnets advertised by the Power Virtual Server router.

However, it does not contain routes to your remote networks. Static routes or a routing protocol, such as BGP, shares routes between your remote network and the gateway appliance. The routes that are advertised by the gateway appliance include your Power Virtual Server subnets.

4. Configure a GRE tunnel between the gateway appliance and the Power Virtual Server router as this router doesn't have routes for your remote networks that are advertised to it through the IBM Cloud-side router. Within the GRE tunnel, static routes are configured between the Power Virtual Server router and the gateway appliance. For more information, see [Configuring Generic Routing Encapsulation \(GRE\) tunnel](#).
5. Set up a separate Direct Link (2.0) connection between your remote network (client-managed environment) and the IBM Cloud classic infrastructure. See [Getting started with IBM Cloud Direct Link \(2.0\)](#).

Key elements of IBM Cloud Direct Link (2.0) include the following:

- Requires BGP to establish the routes to a customer's remote network.
 - Each IBM Cloud Direct Link service is not redundant but, diversity can be enabled over multiple Direct Links along with BGP.
 - Ensure that IP subnet overlaps do not exist between the infrastructure environments and the remote networks.
6. Configure a GRE tunnel between the gateway appliance and a tunnel endpoint that is connected to your external network to enable the remote network connectivity. For more information, see [Configuring Generic Routing Encapsulation \(GRE\) tunnel](#).

For tutorials based on some of the topologies described, see [IBM Power Virtual Server Virtual Private Network Connectivity](#).

Connecting two Power Virtual Server environments by using IBM Cloud Transit Gateway

In this deployment topology, a connection through IBM Cloud Transit Gateway is used to provide connectivity between Power virtual server environments at two different data centers. With IBM Cloud Transit Gateway, you can also interconnect your Power Virtual Servers to the IBM Cloud classic and VPC infrastructures, keeping traffic within the IBM Cloud network. Transit Gateway enables you to connect your otherwise disconnected private networks, such as classic, VPC, and Direct Link. In addition, you can establish connection between multiple Power Virtual Server workspaces across different data centers.

The following network architecture allows connectivity between multiple Power Virtual Server locations with high availability (HA) and disaster recovery (DR) solutions.



Key features are as follows:

- Access and connectivity between two different Power Virtual Server locations in the same region (for example DAL12 to DAL13) to support HA through replication.
- Access and connectivity between two different Power Virtual Server locations in different regions (for example DAL12 to WDC06) to support DR through replication.

Complete the following steps to implement this scenario:

1. Create an [IBM Cloud Transit Gateway](#) to enable the virtual connections.
2. [Create an IBM Cloud connection](#) with Transit Gateway enabled.
3. Connect your Power Virtual Servers that are located in data center 1 and data center 2 through the IBM Cloud network by using a Transit Gateway.

After the Transit Gateway connection is established, different IBM Cloud networks are connected to each other.

IBM Power Virtual Server Cloud Connections

IBM Power Virtual Server in IBM data center

Important announcement

Effective 1 July 2025, metering charges are applied to any Direct Link connections that you continue to use. For more information, refer to the [Cloud announcement](#). Depending on the port speed of your connections, the charges vary from USD 100 to USD 5000. To avoid these charges, migrate your workspace to the PER-enabled workspaces before 1 July 2025. The **CHE01** data center continue to use Cloud Connections without any charges.

To connect the Power Virtual Server workspaces with the IBM Cloud, use the [Power Edge Router](#) (PER) capability that is integrated with the Transit Gateway service without any additional charge. PER-enabled workspaces provide improved user experience and increased bandwidth.

If you have manually configured the network in your workspace, see [Migrating the existing network configurations to Power Edge Router](#). Alternatively, you can use the `ibmcloud pi workspace action` CLI command to migrate the existing Power Virtual Server workspaces to PER-enabled workspaces. For more information, see [Migrating to PER](#).

If you need assistance with the migration, you can open a [support ticket](#).

You can use Power Virtual Server Cloud Connections to connect your Power Virtual Server instances to IBM Cloud resources on IBM Cloud classic network and Virtual Private Cloud (VPC) infrastructures. Power Virtual Server Cloud Connection creates a Direct Link (2.0) Connect instance to connect your Power Virtual Server instances to the IBM Cloud resources within your account. For cross-account connectivity, use IBM Transit Gateway to interconnect your Power Virtual Server to the IBM Cloud classic and Virtual Private Cloud (VPC) infrastructures. The speed and reliability of the Direct Link connection extends your Power Virtual Server network to the IBM Cloud network and offers more consistent and higher-throughput connectivity, while keeping the network traffic within IBM Cloud.



Important: You can have a maximum of two IBM Cloud (Power Virtual Server Direct Link Connect) per account per Power Virtual Server data center. To create a Power Virtual Server IBM Cloud, you must have the required access to create the connections. For more information, see [Access roles requirements for Power Virtual Server](#).

Support for Power Virtual Server workspaces with Power Virtual Server Cloud Connections

The Power Virtual Server supports multiple workspaces from the same account. However, a specific IBM Cloud connection can be used by only one workspace.



Note: When you perform multiple Power Virtual Server Cloud Connection tasks, actions within a task can time out. When the timeout occurs, the tasks are completed in the background and the status might not change immediately.

Creating Power Virtual Server Cloud Connections

The creation of new Power Virtual Server Cloud Connections across most of the data centers is disabled. You can create Cloud Connections only in the **CHE01** data center. If you do not have the authorization and attempt to create a Cloud Connection (Direct Link Connect), a link is generated. Direct Link is not operational until an authorized IBM Cloud account user with the required authority must authorize the connection by using [IBM Cloud Console Direct Link Portal](#).

To create Power Virtual Server Cloud Connection from the **CHE01** data center, complete the following steps:

1. Go to the Power Virtual Server user interface and click **Cloud connections**.
2. On the Cloud connections page, click **Create connection**.
3. In the **Resource details** section, complete the following steps:
 - a. Specify a connection name and select a connection speed. Follow these guidelines for setting the speed:
 - The maximum connection speed is 10 Gbps.
 - You can select 10 Gbps speed only when you are creating a new connection.
 - If you select 10 Gbps as the required speed, the GRE tunneling option is disabled.
 - You cannot modify a Cloud connection with 10 Gbps to be GRE capable by reducing the speed.
 - You cannot modify the speed of an IBM Cloud connection when the speed is set to 10 Gbps at the time of creation.
 - b. Select **Enable global routing** if you need access to other data centers outside your Power Virtual Server region. For example, you can use global routing to share workloads between dispersed IBM Cloud resources, such as Dallas to Tokyo, or Dallas to Frankfurt. If you want to enable IBM Transit Gateway for the Cloud connection, then global routing option is not required.
 - c. Select **Enable IBM Transit Gateway** to interconnect your Power Virtual Server to the IBM Cloud classic and Virtual Private Cloud (VPC) infrastructures and to keep traffic within IBM Cloud. IBM Cloud Transit Gateway connects the private networks, such as classic, VPC, and Direct Link. For more information, see [Getting started with IBM Cloud Transit Gateway](#).

IBM Cloud Transit Gateway is available in **WDC04**, **DAL12**, **DAL13**, **DAL14**, **LON04**, **LON06**, **FRA04**, **FRA05**, **SAO01**, **SYD04**, **TOK04**, **TOR01**, **MON01**, **SYD05**, and **OSA21** data centers.



Note: You need to configure IBM Cloud Transit Gateway in **WDC06** manually. Select **Transit Gateway** as the network connection type instead of **Direct resources** while you create Direct Link (2.0) Connect to use IBM Cloud Transit Gateway. Complete your connection by submitting an IBM Support case to the Power Virtual Server team. For more information, see [Ordering Direct Link Connect](#).

4. In the **Virtual connections** section, you can establish a connection between multiple Power Virtual Server workspaces across different data centers by using an IBM Cloud Transit Gateway. You can create virtual connections that are directly attached to the Direct Link gateway, or you can choose to connect an IBM Cloud Transit Gateway and then create a connection from it to your network (VPC, classic). You must create an [IBM Cloud Transit Gateway](#) to enable virtual connections. Select the virtual connections checkbox to continue. This setting is required if you selected the **Enable IBM Transit Gateway** checkbox in the previous step.
5. In the **Subnets** section, click **Attach existing** to attach an existing subnet to the connection. For A GRE tunnel, a connection must be attached to a subnet. You can create a new subnet. If you enable IBM Cloud Transit Gateway, you can configure the GRE tunnel by using the IBM Cloud Transit Gateway interface. For more information, see [Configuring subnets](#). The table **Understanding CIDR notation caption** in the [Configuring subnets](#) topic lists all the subnets that are attached to the IBM Cloud connection.



Note: Attaching a subnet to an IBM Cloud connection is required as the network traffic is routed over the connection. You must route Power Virtual Server private network subnets over IBM Cloud Direct Link to allow connectivity between Power Virtual Server instances and the IBM Cloud network. Attaching a subnet to Cloud connections allows Power Virtual Server VM to VM communication and for the VMs that are located in the same subnet or different subnet within the workspace to communicate.



Note: If the Cloud connection is attached to a Transit Gateway, prepare for [Direct Link](#) and [Transit Gateway](#) changes to advertise service network routes.

6. Review the summary and the terms and conditions. Then, click **Create** to create an IBM Cloud connection.

Modifying IBM Cloud connections

When you create or edit a subnet, you can attach an existing IBM Cloud connection to the subnet. For more information, see [Configuring and adding a private network subnet](#).

To view or edit IBM Cloud connections, complete the following steps:

1. From the Power Virtual Server dashboard, click **Cloud connections** in the left navigation page.
2. Click the Cloud connection that you want to configure. The corresponding **Connection details** page appears.
3. If a Transit Gateway is enabled for the connection, you can view the **Managed with IBM Transit Gateway** details.

4. Click the **Edit details** icon.



Note: If you have an existing IBM Cloud connection with a speed of 5 Gbps or less, you cannot increase the speed limit to 10 Gbps. Also, you cannot modify an IBM Cloud connection with 10 Gbps to be GRE-capable by reducing the speed.

5. Modify the details, review the pricing changes, and click **Save edits**.

Deleting an IBM Cloud connection

To delete an IBM Cloud connection, complete the following steps:

1. From the Power Virtual Server dashboard, click **Cloud connections** in the left navigation window. A list of IBM Cloud connections that are currently configured is displayed.
2. To delete a specific IBM Cloud connection, click the **Delete** icon in the last column of the table.

When you delete a virtual server instance, it deletes the private networks (subnets) and other resources. Consider the following before you delete a virtual server instance:

- You must delete the Cloud Connection (Direct Link) from the IBM Cloud Transit Gateway interface before you delete the Cloud Connection.
- When there are more than one virtual server instances that use the Cloud Connection, and you delete one, you can delete the Cloud Connection from the existing server instances.



Note: Effective 1 July 2025, metering charges are applied to any Direct Link connections that you continue to use. If you have existing IBM Cloud Connections that are managed from non-PER enabled workspaces, you can now view and delete them using the [IBM Cloud CLI](#) or [Creating IBM Cloud connections](#).

A GRE tunnel subnet supports addressing for GRE tunnels. It is used for the tunnel source IP, local IP, and remote IP. The first half of the subnet IP range (**s1**) is used for source IP addresses and the second half (**s2**) is used for local and remote IP addresses. GRE tunnels use the first IP from **s1** as the source IP. The local IP is the first IP of **s2** and the remote IP is the second IP of **s2**.

GRE configuration example

Suppose you choose **10.148.252.83** as your destination IP address, which is a private IP of your vRealize Automation (VRA -IBM Cloud vSRX, Vyatta, or VMWare™ NSX Edge) and **172.16.3.0/29** as the GRE subnet:

GRE Destination IP: 10.148.252.83 (VRA private IP)
GRE Subnet: : 172.16.3.0/29 (GRE subnet that you chose)
PowerVS source IP : 172.16.3.1 mask 255.255.255.255
PowerVS tunnel IP : 172.16.3.5

Then, you must configure the GRE tunnel in your VRA as follows:

GRE Destination IP: 172.16.3.1/32 (PowerVS Tunnel End-point Destination IP)
VRA source IP : 10.148.252.83
VRA tunnel IP : 172.16.3.6
VRA ASN : 64880
PowerVS ASN : 64999



Note: In general, the overlay and underlay network must not be on the same subnet, so to make sure your configuration that is stated earlier is valid on your gateway appliance. You can change the subnet mask of the GRE subnet from /29 to /30 so that it excludes 172.16.3.1 from the overlay network.



Note: By default, gateway appliances send the traffic for the 172 network to the default gateway, which is generally the public subnet gateway of the gateway appliance. Make sure to add a static route for the destination PowerVS source IP address to be sent to the private subnet gateway of the gateway appliance.

You must configure the VRA with BGP protocol for route advertising so that the subnets can reach through the GRE tunnel. The ASN numbers are pre-assigned in the Power Virtual Servers and you can't choose another number.

Migrating an existing network configuration



Note: You can continue to have your existing configuration that is managed by using the Power Virtual Server IBM Support case process. You are not required to migrate to a Power Virtual Server network.

If you want to use the new features that are offered by network automation, you can migrate your existing network configuration by creating a Power Virtual Server operations IBM Support case.

Considerations for network configuration migration

When you are planning for migration, review the following considerations:

- Be sure to factor a maintenance window into your schedule.
- Migration might require configuration changes to a private cloud configuration.

Direct Link Connect for IBM Power Virtual Server

IBM Power Virtual Server in [IBM data center](#)

Use [IBM Cloud® Direct Link \(2.0\) Connect](#) to configure your private network with IBM® Power® Virtual Server. The Direct Link (2.0) Connect service creates a seamless connection that allows access to IBM Cloud resources from your Power Virtual Server instance.



Note: New connectivity requests for Direct Link Connect for Power Virtual Server are no longer supported. If you use existing connections, migrate the connection of your Power Virtual Server workspace in Client location to IBM Cloud Direct Link and PER if PER is available in your environment.

Direct Link (2.0) Connect provides the following advantages:

- Support for connections to multiple IBM Cloud accounts from a single direct link.
- Support for multiple VPCs (without classic access) from a single direct link within the same account.

Before you begin

Before you order Direct Link Connect, make sure that you review the following considerations and satisfied the prerequisites:

- Verify that your IBM Cloud account has the correct authorization to order the Direct Link (2.0) Connect service.
- Review [Direct Link prerequisites](#). Also, review [Routing considerations](#), and [Setting up high availability](#), as needed for your particular deployment.
- IBM Cloud highly recommends that you establish a second, diverse direct link to prevent outages, whether unplanned, or planned due to maintenance.
- A 10 Gbps connection is available only using the Direct Link (2.0) Connect offering.
- Direct Link (2.0) Connect is available in all current locations.
- Currently, there isn't a migration path from Direct Link on Classic (1.0) to Direct Link (2.0). You must order a new Direct Link (2.0) Connect connection.

Pricing

The Power Virtual Server offering includes a highly available connection to IBM Cloud services at no cost for each customer per data center. You can also select the global routing option for these direct links at no cost.

Ordering Direct Link (2.0) Connect

To order Direct Link Connect, complete the following steps:

1. Follow Direct Link Connect [ordering instructions](#) with these exceptions:
 - In the **Gateway** section, select the same location as the Power Virtual Server instance. Each location requires its own direct link. For example, if you order a direct link for Frankfurt 04, you cannot establish a connection to the VMs in Frankfurt 05.



Important: For the list of Direct Link Connect PowerVS locations, see [Providers and locations](#).

- For **Routing**, select **Local** to access all of the data centers that are connected at the location that you specified. Select **Global** to access all of the IBM Cloud data centers in the world.
- For **Provider**, select **IBM POWER VS** from the list.
- For **Speed**, select the link speed to meet your workload requirements. The recommended speed is 1 Gbps.

- For the list of **Ports**, select a port that has the least number of connections. If you have multiple direct links, you must choose different ports for each connection.
- In the **Billing** section, select **Unmetered**.
- In the **BGP** section:
 - For **BGP peering subnet**, you can select **Auto-select IP** to auto-select an IP address from range `169.254.0.0/16`, or manually enter addresses in a specific range to avoid conflict with an existing connection.
 - For **BGP ASN**, enter `64999` as the BGP ASN number for the Direct Link Connect location unless a different ASN number is required as indicated on the provisioning page. For example, the BGP ASN number for the WDC04 location is `64995`. For 10 Gbps ports that are not GRE capable, you must use the BGP ASN number `64997`.
 - Do not try to change the BGP ASN number to **64995**. You must contact the IBM Power support team to handle your request to change the BGP ASN number.

Direct Link Connect site	BGP ASN
Dallas 10	4206000072
Dallas 12	64999
Dallas 13	64999
Washington 4	64995
Washington 6	64999
Washington 7	4206000068
Frankfurt 4	64999
Frankfurt 5	64999
London 6	64999
Toronto 1	64999
Montreal 1	64999
São Paulo 4	4206000076
Chennai 01	4206000092

BGP ASN number for specific Connect sites

2. Read and agree to the [Direct Link prerequisites](#), then click **Create**.

After your Direct Link connection order is submitted, go to **Interconnectivity > Direct Link** to view the status of your order. The **Direct Link** page lists all existing Direct Link connections.

3. To [complete your connection](#), submit an [IBM Support case](#) to the Power Virtual Server team. In the description field, enter the following details.

\$ Customer name:

Customer account ID:

Data center details:

Direct Link Connect subnet:

IBM Cloud IP address:

Power Virtual Server network IP address:

Direct Link Connect VLAN:

IBM Cloud ASN:

Power Virtual Server network ASN:

Power Virtual Server Private Network (subnet) Name (1):

Power Virtual Server Private Network (subnet) Name (2):

Power Virtual Server Private Network (subnet) Name (3):

Direct Link Connect 2.0 Virtual Connections:

Classic Network Virtual Connection: Yes/No

VPC Virtual Connection 1: VPC name and ID

VPC Virtual Connection 2: VPC name and ID

 **Important:** The **Power Virtual Server network** autonomous system number (ASN) is the same as your Border Gateway Protocol (BGP) ASN. The IBM Cloud network team generates the **IBM Cloud ASN** and adds it to the IBM Cloud support case. The IBM Cloud network team also generates the IP addresses. Your private network name is your Power Virtual Server private network subnet name.

The Power Virtual Server support case is closed when the Direct Link Connect connection is configured to communicate with your Power Virtual Server instance.

Setting up high availability over Direct Link Connect

Your Direct Link connections are location-specific. By default, IBM Cloud Direct Link is not a redundant service. You must order a separate Direct Link Connect instance for redundancy.

To set up a high availability through Direct Link Connect, complete the following steps:

- 1. Order two instances of Direct Link Connect. For instructions, see [Ordering Direct Link Connect](#).

In the **BGP** section, select a port from a separate port group for the redundant Direct Link Connect instance. Both Direct Link Connect instances must be on separate ports to connect to different Power Virtual Server routers, thus, ensuring redundancy.

The following example shows the Direct Link Connect ports for the DAL12 data center. The ports that end with **1-1** and **1-2** belong to one port group, and the ports ending with **2-1** and **2-2** belong to another port group. If you select a port from the first port group, you must select a port from the second port group for the redundant Direct Link Connect instance. That is, if you select **-1-1** for the first Direct Link Connect instance, you must select **-2-1** or **-2-2** for the second Direct Link Connect instance.

Provider

IBM POWER VS *only available choice

Speed

5 Gbps

BGP and connections

Choose the specific network connections that you want to make with your direct link. You can edit these connections after the direct link is provisioned.

Ports	# of direct links i
☐ IBM POWER VS-CLOUD EXCHANGE SL-DAL12-1-1	1
☐ IBM POWER VS-CLOUD EXCHANGE SL-DAL12-1-2	0
☐ IBM POWER VS-CLOUD EXCHANGE SL-DAL12-2-1	0
☐ IBM POWER VS-CLOUD EXCHANGE SL-DAL12-2-2	0

BGP peering subnet i

☒ Auto-select IP ☐ Manual-select IP

Range

169.254.0.0/16 *only available choice

BGP ASN

Specify private ASN

IBM ASN

13884

BGP and connections

Border Gateway Protocol (BGP) sessions are configured for the Direct Link Connect service in such a way that when a fault is detected on a Direct Link Connect instance, the BGP routes traffic to an alternative Direct Link Connect instance. For 10 Gbps connections, use the new ports that are not GRE capable. Ports that are GRE capable can use only up to a 5 Gbps speed.

Refer to the following table to identify the ports and port groups that you must select for the redundant Direct Link Connect instance:

Data center	Network provider	Port group 1	Port group 2 for redundancy
LON04	IBM Power VS	IBM POWER VS-CLOUD EXCHANGE SL-LON04-1-1 IBM POWER VS-CLOUD EXCHANGE SL-LON04-1-2	IBM POWER VS-CLOUD EXCHANGE SL-LON04-2-1 IBM POWER VS-CLOUD EXCHANGE SL-LON04-2-2
LON06	IBM Power VS	SL-LON06-IBMPowerLAASLITE-1-1 SL-LON06-IBMPowerLAASLITE-1-2 xPowerVS-LON04-10G-NOGRE-1-1 [1]	SL-LON06-IBMPowerLAASLITE-2-1 SL-LON06-IBMPowerLAASLITE-2-2 xPowerVS-LON04-10G-NOGRE-1-2 [2]
FRA05	IBM Power VS	SL-FRA05-IBMPowerLAASLITE-1-1 SL-FRA05-IBMPowerLAASLITE-1-2 xPowerVS-FRA05-10G-NOGRE-1-1 [3]	SL-FRA05-IBMPowerLAASLITE-2-1 SL-FRA05-IBMPowerLAASLITE-2-2 xPowerVS-FRA05-10G-NOGRE-1-2 [4]
FRA04	IBM Power VS	SL-FRA04-IBMPowerLAASLITE-1-1 SL-FRA04-IBMPowerLAASLITE-1-2 xPowerVS-FRA04-10G-NOGRE-1-1 [5]	SL-FRA04-IBMPowerLAASLITE-2-1 SL-FRA04-IBMPowerLAASLITE-2-2 xPowerVS-FRA04-10G-NOGRE-1-2 [6]
WDC04	IBM Power VS	NNI-LINK-SL-WDC04-IBMPowerLAASLITE-1-2 xPowerVS-WDC04-10G-NOGRE-1-1 [7]	NNI-LINK-SL-WDC04-IBMPowerLAASLITE-2-2 xPowerVS-WDC04-10G-NOGRE-1-2 [8]
WDC06	IBM Power VS	IBM POWER VS-CLOUD EXCHANGE SL-WDC06-1-1 IBM POWER VS-CLOUD EXCHANGE SL-WDC06-1-2	IBM POWER VS-CLOUD EXCHANGE SL-WDC06-2-1 IBM POWER VS-CLOUD EXCHANGE SL-WDC06-2-2
WDC07	IBM Power VS	SL-WDC07-IBMPowerLAASLITE-1-1 SL-WDC07-IBMPowerLAASLITE-1-2	SL-WDC07-IBMPowerLAASLITE-2-1 SL-WDC07-IBMPowerLAASLITE-2-2
DAL10	IBM Power VS	SL-DAL10-IBMPowerLAASLITE-1-1 SL-DAL10-IBMPowerLAASLITE-1-2	SL-DAL10-IBMPowerLAASLITE-2-1 SL-DAL10-IBMPowerLAASLITE-2-2
DAL12	IBM Power VS	IBM POWER VS-CLOUD EXCHANGE SL-DAL12-1-1 IBM POWER VS-CLOUD EXCHANGE SL-DAL12-1-2 xPowerVS-DAL12-10G-NOGRE-1-1 [9]	IBM POWER VS-CLOUD EXCHANGE SL-DAL12-2-1 IBM POWER VS-CLOUD EXCHANGE SL-DAL12-2-2 xPowerVS-DAL12-10G-NOGRE-1-2 [10]
DAL13	IBM Power VS	SOFTLAYER-IBMPowerIAASLITE-1-1 SOFTLAYER-IBMPowerIAASLITE-1-2 xPowerVS-DAL13-10G-NOGRE-1-1 [11]	SOFTLAYER-IBMPowerIAASLITE-2-1 SOFTLAYER-IBMPowerIAASLITE-2-2 xPowerVS-DAL13-10G-NOGRE-1-2 [12]
SYD04	IBM Power VS	SL-SYD04-IBMPowerLAASLITE-1-1 SL-SYD04-IBMPowerLAASLITE-1-2 xPowerVS-SYD04-10G-NOGRE-1-1 [13]	SL-SYD04-IBMPowerLAASLITE-2-1 SL-SYD04-IBMPowerLAASLITE-2-2 xPowerVS-SYD04-10G-NOGRE-1-2 [14]
TOK04	IBM Power VS	SL-TOK01-ZENLAYER-3-1 01675-TK4-X-I-0002 TKY/TKY/LE-275454	SL-TOK01-ZENLAYER-4-1
OSA21	IBM Power VS	SL-OSA01-ATTOKYO-1-1	SL-OSA01-ATTOKYO-2-1
SAO04	IBM Power VS	SL-SAO04-POWERVSCX-1-1 SL-SAO04-POWERVSCX-1-2	SL-SAO04-POWERVSCX-2-1 SL-SAO04-POWERVSCX-2-2
CHE01	IBM Power VS	IBM-CHE01-POWERCX-1-1 IBM-CHE01-POWERCX-1-2	IBM-CHE01-POWERCX-2-1 IBM-CHE01-POWERCX-2-2

Port and Port groups for redundant Direct Link instances

2. Select the remaining options and create the Direct Link Connect instance as described in [Ordering Direct Link \(2.0\) Connect](#).

Routing considerations for VPC

If traffic is sent from the Power Virtual Server to your private cloud public IP address, and if the virtual server instance has a public floating IP, you might need a special configuration in the VPC. Otherwise, the traffic goes through the virtual machine's public interface instead of a private interface.

For proper VPC configuration, the private cloud IP address must meet the following requirements:

- The IP address range for a private network must be within the following blocks, which are reserved by the Internet Assigned Numbers Authority (IANA):
 - Class A — 10.0.0.0 — 10.255.255.255 (16,777,216 total hosts)
 - Class B — 172.16.0.0 — 172.31.255.255 (1,048,576 total hosts)
 - Class C — 192.168.0.0 — 192.168.255.255 (65,536 total hosts)
- VM instances within the VPC must not have a floating IP.
- Use the **Delegate-VPC** action to create a route to the private cloud public subnet in the VPC default routing table.

For more information, see [Routing considerations for IANA-registered IP assignments](#).

-
1. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 2. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 3. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 4. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 5. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 6. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 7. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 8. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 9. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 10. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 11. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 12. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 13. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)
 14. This port is not GRE capable, but supports a speed of 10 Gbps. [↵](#)

Networking for Client location

Network overview

IBM Power Virtual Server Private Cloud in Client location

Establishing the connectivity between the pod in your data center and IBM® Power® Virtual Server IBM Cloud requires meticulous planning, site preparation, meeting the prerequisites, understanding the network architecture, and others.

Setting up a network involves the following two networks:

- [Control plane network](#) - for communication between the IBM Cloud and the pod that is located in your data center.
- [Data plane network](#) - for accessing virtual servers.

Control plane network

The IBM Power Virtual Server Private Cloud network architecture requires connectivity between the IBM Cloud and the pod that is located in your IBM Power Virtual Server Private Cloud in client location data center. This connectivity is the main communication channel between the Service Broker, IBM Cloud, and pod. This communication channel is known as a control plane network. Through this channel, the Service Broker can start the APIs on PowerVC for Power Virtual Server for the virtual machine lifecycle operations.

The control plane network is set up with redundancy, providing multiple network paths, to ensure that the pods are connected to the IBM Cloud regions reliably. The control plane network can be set up by using an IBM Cloud Direct Link 2.0 or Virtual private network (VPN). For IBM Cloud Direct Link 2.0 connection, IBM orders the device that uses an IPsec over VPN (last-mile connectivity) to connect to the IBM Power Virtual Server Private Cloud in client location pod in the IBM Cloud region.

Before the pod installation, provide the required network-specific information so that either the IBM Cloud Direct Link 2.0 connection or VPN connection can be established.

Impact of unplanned network outage

When an IBM Power Virtual Server Private Cloud infrastructure that is deployed in a client location encounters an unplanned network outage, the communication link is interrupted. This outage results in the loss of primary and secondary management connections (Direct Link or site-to-site VPN) to the IBM Cloud. The following table outlines the implications of an IBM Power Virtual Server Private Cloud infrastructure that is operating in the disconnected mode.

Capability	Behavior in disconnected mode
Client workload and data	Client workload is operational and you can access all the data.
GUI or API for read operations	GUI is operational and uses the most recent cached data.
GUI or API for write operations, such as creating virtual machine (VM) or volumes	You cannot perform write operations to the Power Virtual Sever resources until the connectivity to the control plane network is restored.
Command-line interface (CLI)	You can perform read operations. However, you cannot perform write operations until the connectivity to the control plane network is restored.
Billing and metering	Metering uses the most recent cached data. If the infrastructure is disconnected, you cannot perform write operations until the connectivity to the control plane network is restored.
IBM remote support	IBM remote support team cannot remotely connect to the IBM Power Virtual Server Private Cloud infrastructure until the connectivity to the control plane network is restored.
Dynamic Host Configuration Protocol (DHCP) services for client data networks	DHCP services are provided by the resident network in the IBM Power Virtual Server Private Cloud infrastructure and do not require a connection to the IBM Cloud.

Impact of unplanned network outage

IBM Cloud Direct Link 2.0 Connect

The IBM Cloud Direct Link 2.0 Connect is an IBM Cloud technology to connect to the remote data centers. This technology connects a Virtual Private Cloud

(VPC) on IBM Cloud and a router connected to the pod that is located in your data center. The IBM Cloud Direct Link 2.0 Connect uses an IPsec over VPN (also known as last-mile connectivity) to connect to the pod. Thus, the IBM Cloud Direct Link network goes through the data center environment at a minimum level and is connected to the pod's router. The control plane network is set up with redundancy, providing multiple network paths, to ensure that the pods are connected to the IBM Cloud regions reliably. IBM orders an IBM Cloud Direct Link 2.0 Connect for your account by using the third-party network provider details that you might have provided earlier (like Megaport or Equinix). The attributes, such as service keys, are gathered from the ordering process. After the ordering is complete, the corresponding third-party network providers can establish the remote connectivity between the IBM Cloud and the data center.

The IBM Cloud Direct Link 2.0 Connect can be viewed as an alternative to a traditional site-to-site VPN solution. This connectivity can provide secure, private, consistent, and higher-throughput connectivity between a remote network and IBM Cloud environments. For more information about IBM Cloud Direct Link Connect, see [Getting started with IBM Cloud Direct Link](#).

The connectivity between IBM Cloud and pod has two parts:

- Shared connectivity between the IBM Cloud network infrastructure and the third-party network provider infrastructure.
- An internet-based connectivity between the third-party network provider infrastructure and the data center. This connectivity is referred to as last-mile connectivity and might involve an IPsec VPN. Contract with a service provider to provide the last mile connection.

To establish this connectivity, you, IBM, and the third-party network provider must collaborate.

Virtual Private Network

The Virtual Private Network(VPN) between IBM Cloud and the pod can be established in the following ways:

- **Site-to-Site VPN connectivity** — A VPN gateway can be set up on the VPC that houses the service broker. This gateway provides an internet-facing IP address. You can establish a VPN client on your data center that can provide an internet-facing IP address. This connectivity forms an IPsec VPN tunnel between the two end points. Then, the VPN client on your data center can be extended to the IBM Power Virtual Server Private Cloud routers.
- **VPN connectivity by using IBM Cloud classic environment** — A Juniper vSRX Virtual Firewall can be deployed in the classic environment on the IBM Cloud. This setup can be connected to the VPC that houses the service broker through a transit-gateway. Alternatively, a similar VPN hardware or software environment can be set up on your data center environment. This setup can be connected through an IPsec VPN tunnel and extended to the IBM Power Virtual Server Private Cloud routers. Your data center infrastructure must be connected to the routers on the pod. This connectivity must be as minimal as possible. You can set up one or more virtual routers in your environment and set up a Border Gateway Protocol (BGP) deployment by using the IBM Power Virtual Server Private Cloud routers.

Data plane network

The data plane network is enabled when the pods in your data center are connected to your IBM Power Virtual Server Private Cloud network infrastructure. You must be able to access the virtual servers (logical partitions) in the pod through your network instead of IBM Cloud.

The pod contains all the software components that are required to communicate to the IBM Cloud. The components include service software such as HMC and PowerVC, and storage and network devices such as IBM Power Virtual Server network fabric and router.

When you create a network within a pod, ensure that the network does not overlap with other existing networks in the same workspace within the pod. If you create an overlapping network, an error message is displayed.

If you create an overlapping network through an API, an error message similar to the following example is displayed:

```
{
  "errors": "Subnet overlap between uni/tn-VPN502/BD-BD_VPN502_V504_PRIV_3302/subnet-[192.168.31.1/24] and uni/tn-VPN502/BD-BD_VPN502_V504_PRIV_3301/subnet-[192.168.31.1/24] in VRF uni/tn-VPN502/ctx-VPN502_V504_VRF. It is not possible to have overlapping subnets across different Bridge Domains within the same VRF."
}
```

As part of the network planning, you can review the following use cases and identify the use cases that are applicable to you. You can communicate about such requirements before the installation so that you do not have to open separate support tickets to implement the use-cases and configurations.

- **Connectivity between the LPARs within a pod** - You can create an internal private network and deploy LPARs on them with no external connectivity. These LPARs are accessible to each other due to the automated host communication, which configures the network fabric inside the pod. For more information, see [Connectivity between the LPARs within a pod](#).
- **Outbound-only external connectivity via DNAT** - You can create external networks in the pod. Connectivity for these networks must be configured through a manual ticket process. After the connectivity configuration, LPARs can be deployed on them and they can access external networks. You can always deploy LPARs with both internal and external networks as separate network interfaces. For more information, see [Outbound-only external connectivity via DNAT](#).
- **Bi-directional external connectivity via BGP** - You can deploy an external network within the pod to establish the communication between the endpoints that are in the IBM Power Virtual Server Private Cloud network environment by using BGP. For more information, see [Bi-directional external connectivity via BGP](#).

- **Bi-directional external connectivity via Static Route** - You can deploy an external network within the pod to establish the communication between the endpoints that are in the IBM Power Virtual Server Private Cloud network environment by using static route similar to BGP. For more information, see [Bi-directional external connectivity via static routes](#).
- **Bi-directional external connectivity - L2Out** - When you establish this type of external connectivity, it bypasses the router and directly connects to the network fabric. This type of connectivity is useful if the same IP address space is required on both internal and external networks. All the other types of external networks involve two distinct subnets. For more information, see [Bi-directional external connectivity via L2Out](#).

Network architecture diagrams

Figure 1 shows the overall view of the network architecture.



About network peering connection

You create a network peering connection when you connect your Power Virtual Server workspace and your data center network. When you create a network peering connection, you can enable file sharing, use distributed applications, and work collaboratively between the two networks. By using the network peering connection, you can make your Power Virtual Server workspace more scalable and efficient.

To create a network peering connection, you must inform the IBM Power Virtual Server Private Cloud support team before the pod setup. You can enable the network peering connection only if the data network in your infrastructure supports bidirectional external connectivity. For more information about bidirectional external network use cases, see [Network use cases](#).

A peer network interface is a physical network endpoint in the IBM Power Virtual Server Private Cloud network fabric that is used for external network connectivity. Based on the type of data network in your infrastructure, the IBM Power Virtual Server Private Cloud support team defines one or more peer network interfaces during the pod setup. You can use a peer network interface to logically configure the network peering connections.

You can connect your Power Virtual Server workspace and your data center network as a dynamic routing session through BGP over each of the physical peer interfaces. To create a network peering connection, collect the following network information from your network engineering team:

- Network interface name
- BGP Autonomous System Number (ASN) for your network
- IBM BGP ASN
- IBM IPv4 Classless Inter-Domain Routing (CIDR)
- IPv4 CIDR for your network
- VLAN ID

The BGP connection exports the workspace host virtual server instance (VSI) IP addresses as `/32` routes. If these routes are not required, you can add a deny rule in the route filter for each address on the network peer.



Note: If your enterprise network advertises the same prefixes from multiple connections with equal cost, Power Virtual Server Private Cloud honors all those links as equal-cost multi-path routes.

Creating a network peering connection

After you collect the required details from your network engineering team, complete the following steps to create a network peering connection:

1. Log in to [IBM Cloud catalog](#) with your credentials.
2. In the search box, type Power Virtual Server and click the Power Virtual Server tile.
3. In the navigation panel, click **Workspaces**. The Workspaces page with a list of existing workspaces is displayed.
4. Click your workspace that you want to update from the list to open the virtual server instance page.
5. Click **Network peering** under the **Networking** section in the left pane. The Network peering page appears.
6. Click **Create peering connection**. The Create peering connection page appears.
 - a. Specify a name in the **Name** field.
 - b. Select a network interface from the **Network interface** list.
 - c. Select the type of network from the **Peer network type** list.
 - d. Click **Continue**.
 - e. On the **BGP** tab, update the fields with the network information that is collected from your network engineering team.
7. Agree to the terms and conditions in the Summary pane.
8. Click **Create**.



Note: You can create the network peering connection only if you use the details that are provided by your network engineering team. Otherwise, the network peering connection is not created.

Import and export route filters

Complete the following steps to allow data exchange between the data center network and the network peering connection:

1. Click **Network peering** under the **Networking** section in the navigation pane of the **Power Virtual Server** window. The Network peering page appears.
2. Click a network peering connection.
3. Set the **Default filter behavior** on the **Import route filters** tab to one of the following values:
 - **Permit all import routes**: The data center network can access only the virtual server instances (VSIs) within the Classless Inter-Domain Routing (CIDR) that is marked as allowed and not in the other CIDRs. This option is the default value.
 - **Deny all import routes**: The data center network cannot access the VSIs within the CIDR that is marked as denied but can access the VSIs in all the other CIDRs.
4. Set the **Default filter behavior** on the **Export route filters** tab to one of the following values:
 - **Permit all export routes**: The VSIs within the CIDR that is marked as allowed can access the data center network but the VSIs in all the other CIDRs cannot access the data center network. This option is the default value.
 - **Deny all export routes**: The VSIs within the CIDR that is marked as denied cannot access the data center network but the VSIs in all the other CIDRs can access the data center network.



Note: If a CIDR is configured with both permit and deny actions in a route filter, the rule with the higher priority takes precedence, and the other action is ignored.

Preparing the data center

Prerequisites for installing the pod

IBM Power Virtual Server Private Cloud in **Client location**

Proper planning is essential for the successful setup and use of IBM® Power® Virtual Server pods. It ensures that you have everything you need and that you meet all the prerequisites for your pod.

IBM owns the responsibility to install, upgrade, and update the hardware and software for the pod infrastructure. IBM coordinates with you for any dependencies to ensure that your IBM Power Virtual Server Private Cloud in client location data center meets all the prerequisites before the pod installation.

IBM provides a pre-installation checklist after you place an order for IBM Power Virtual Server Private Cloud in client location and also during the pre-installation stage. You must complete the items in the pre-installation checklist to prepare your IBM Power Virtual Server Private Cloud environment for the pod installation. An IBM representative conducts an installation readiness review to ensure that you understand the prerequisites.

Preorder checklist

Before you place the order, you must confirm if the following criteria to install the pod in your data center are met:

- Identify an IBM Cloud region closest to the physical location of your data center where the pod resides. The network latency between your data center and the selected IBM Cloud region that will run the control plane must maintain a network round-trip time (RTT) of less than or equal to 200 milliseconds. For more information, see [Network latency](#).
- Confirm that the site and environment requirements to place the pod are met. For more information, see [Site-planning and environmental requirements](#).

Power Virtual Server requirements

Before you connect to the IBM Cloud, create a workspace, or perform other pre-installation activities, you must complete the following tasks or keep the following information in hand:

- You must have an active IBM Cloud account ID. To create an IBM Cloud account ID, see [Setting up your IBM Cloud account](#).
- Create a Satellite location associated with an IBM Cloud region. Select the IBM Cloud region that is closest to the physical location of your data

center. The network latency between your data center and the selected IBM Cloud region must maintain a network round-trip time (RTT) of less than or equal to 200 milliseconds. For more information, see [Create a Satellite location overview](#).

Site-planning and environmental requirements

Before you place the order, you must prepare your data center site to meet the following site and environmental requirements:

- The data center can accommodate all the 42U racks in a pod. For example, a small pod contains one 42U rack and a medium pod might contain two or four 42U racks.
- The floor of the data center is able to withstand the weight of the racks and other equipment. For more information about rack dimensions and weight, see [Site access requirements](#).
- Ensure heating, ventilation, and air conditioning (HVAC) setup for front-to-back airflow and humidity control is ready. For more information, see [Environmental requirements](#).
- The data center site meets the power source requirements for rack connector and load requirements. For more information, see [Power requirements](#).
- The data center is in a secured location with restricted access. For more information, see [Site readiness](#).

The following data center site requirements must be met before the pod is delivered to your data center:

- Ensure that the IBM site reliability engineering (SRE) operations team has the security and access approval to operationally manage the pod infrastructure.
- Ensure that the power cables and network cables are routed to the location where the pod will be installed. The power cables must have connectors that are compatible with rack PDU. For more information, see [Supported PDU power cord](#).
- Identify a network administrator and hire a service provider who can participate in establishing the connections to IBM Direct Link connection or VPN connection and to provide last mile connectivity.

Networking requirements

The following network requirements must be completed before the pod is delivered to your data center:

- After the order is placed, IBM provides a pre-installation checklist. Complete the pre-installation checklist to provide network-specific information, such as Autonomous system numbers (ASN), Service key, and others. For more information, see [Network requirements](#).
- Review and identify the available network use cases and communicate your requirements to IBM before you proceed with installation. For more information, see [Network use cases](#).

Network validation and account activation

You must complete the following steps to validate the network functions and to activate the IBM Power Virtual Server Private Cloud account:

- After the physical cabling is completed, provide the initial configuration to set up the data plane network to the IBM site reliability engineering (SREs) team.
- The SRE team configures the network for network validation testing. The network validation testing ensures the functionality of the control plane network.
- The SRE team conducts provisioning testing over the control plane by simulating a user request from the IBM Cloud. To complete the remaining network connectivity after the installation of the pod, you can contact IBM Support Center through the ticketing system. For more information, see [Getting Support](#).
- After testing is complete, the SRE team activates the account specifying the period from when the billing starts.
- The SRE team works with you to ensure the visibility of the pod Satellite location in your IBM Power Virtual Server Private Cloud account.
- Perform a provisioning test by using the service broker. The provisioning test is successful if the following statements are true:
 - A virtual machine is successfully provisioned.
 - The virtual machine has an IP address that is assigned to it.
 - You can log on to the virtual machine and can run basic commands to inspect the environment settings, such as check memory, CPU, and storage capacity.

Site readiness

IBM Power Virtual Server Private Cloud in Client location

Determine whether to construct a new site or to modify an existing site. The electric power and communication facilities must be available in adequate quantities for operation. If these facilities are inadequate, contact the utility company to determine whether additional services can be made available.

Ensure that the IBM® Power® Virtual Server pod is protected with restricted access that is consistent with your company data protection and physical

access control policies.

For more information, see [Site preparation and physical planning](#).

Site access requirements

IBM Power Virtual Server Private Cloud in Client location

Site planners can use this information to assess the physical site and operational requirements that are necessary to prepare your site for new IBM® Power® Virtual Server racks. Work with the IBM representative to ensure that these prerequisites are completed before the pods are delivered to your data center.

Review the following requirements that your IBM Power Virtual Server Private Cloud site must fulfill:

- Accommodate a fully configured 42U rack that is 7 ft tall.
- Provide secured and controlled access to the Power Virtual Server racks in your IBM Power Virtual Server Private Cloud site.
- Provide the flooring that supports the weight of the Power Virtual Server racks.

Refer Table 1 to understand rack dimensions and weight.

Rack	Width	Depth	Height	Weight(Empty)
Rack only	600 mm (23.6 in.)	1070 mm (42.1 in.)	2020 mm (79.5 in.)	166 kg (365 lb)
Rack with two standard doors	600 mm (23.6 in.)	1132 mm (44.6 in)	2020 mm (79.5 in.)	177 kg (391 lb)
Rack with rear door, heat exchanger (dry), and standard doors	600 mm (23.6 in.)	1231 mm (48.5 in.)	2020 mm (79.5 in.)	210 kg (463 lb)
Rack with high-end appearance front door and rear door	600 mm (23.6 in.)	1210 mm (47.3 in.)	2020 mm (79.5 in.)	181 kg (398 lb)

Rack dimensions and weight

Refer to Table 2 to understand rack characteristics and capacity.

Characteristics	Maximum weight	EIA unit capacity
Dynamic (rolling)	1134 kg (2500 lb)	18 kg (40 lb) or EIA average
Static	1678 kg (3700 lb)	32 kg (70 lb) or EIA average
Seismic certified	1170 mm (2580 lb)	20 kg (45 lb) or EIA maximum

EIA: Energy Information Administration

Rack characteristics and capacity

Floor load calculator: You can use [Floor load calculator](#) to determine the distributed load that a server, expansion unit, or migration tower places on a subfloor (for example, in a data center with a raised floor) or on an above-ground or nonraised floor structure.

For generic prerequisites and planning guidelines for Power10, see the following topics:

- [Site preparation and physical planning](#)
- [IBM Power as Rack Specification](#)

Environmental requirements

IBM Power Virtual Server Private Cloud in Client location

Your Client location site must have adequate cooling and humidity control for IBM® Power® Virtual Server racks. To know the recommended and allowed

values of the required parameters, see Table 1:

Environment (operating) (1)		
Properties	Recommended	Allowable (2,3,4)
ASHRAE class		A3 (Fourth edition)
Airflow direction	Front-to-back	
Temperature	18.0°C – 27.0°C (64.4°F – 80.6°F)	5.0°C – 40.0°C (41.0°F – 104.0°F)
Low-end moisture	-9.0°C (15.8°F) dew point	-12.0°C (10.4°F) dew point and 8% relative humidity
High-end moisture	60% relative humidity and 15°C (59°F) dew point	85% relative humidity and 24.0°C (75.2°F) dew point
Maximum altitude		3050 m (10,000 ft)
Allowable environment (nonoperating) (5)		
Temperature	5°C - 45°C (41°F - 113°F)	
Relative humidity	8% to 85%	
Maximum dew point	27.0°C (80.6°F)	

Recommended and allowed environmental parameter values

To set up the operating environment, IBM suggests using the recommended values from Table 1. Over prolonged use, this operating environment becomes reliable and energy-efficient.

To set up a short-term operating environment, you can use the allowable values from Table 1. In an operating system, if you are using the allowable values for the required parameters, use an equipment that is functionally tested.

When allowable values for the required parameters in the operating environment are used for continuous operation, it can overload the equipment.



Note: If you want to use any allowable configurations of ASHRAE A3 to operate at the maximum limits, contact IBM Support.

If altitude of the Client location site exceeds by 900 m (2953 ft), derate the maximum allowable temperature by 1°C (1.8°F) for every 175 m (574 ft). The maximum allowable elevation is 3050 m (10000 ft).

To set the minimum and maximum humidity levels, see the following guidelines:

- The minimum humidity level is the maximum absolute humidity of the -12°C (10.4°F) dew point and 8% relative humidity. These levels intersect at approximately 25°C (77°F). Based on the temperature, set the minimum humidity level as follows:
 - If the temperature is less than 25°C (77°F), the dew point (-12°C) represents the minimum humidity level.
 - If the temperature is greater than 25°C (77°F), the relative humidity (8%) represents the minimum humidity level.
- The maximum humidity level is the minimum absolute humidity of the -12°C (10.4°F) dew point and 8% relative humidity.

The following minimum requirements apply to data centers that are operated at a low relative humidity:

- Data centers that lack electrostatic discharge (ESD) floors and permit the use of non-ESD shoes might consider increasing humidity levels. This action is required because the risk of generating 8 kV increases slightly at 8% relative humidity, when compared to 25% relative humidity.

- All mobile furnishings and equipment must be made of conductive or static dissipative materials and must be bonded to the ground.
- The maintenance personnel must wear a wrist strap when coming into contact with information technology (IT) equipment during hardware maintenance. The wrist strap must be properly functioning and grounded.

Before installation, the equipment is in the powered-down state when it is removed from the container. The allowable nonoperating environment is provided to define the environmental range that an unpowered system can experience for a short time without sustaining damage.

For more information, see Table 1 - 5 in [Environmental design criteria](#).

Power requirements

IBM Power Virtual Server Private Cloud in **Client location**

Your Client location site for IBM® Power® Virtual Server must be provisioned with A-side and B-side power redundancy that meets the Power Virtual Server rack connector and load requirements. Table 1 shows the rack connector and load requirements.

Be sure to read all caution and danger statements provided in the Safety notices before you perform the procedures. For more information about safety notices containing the caution and danger information, see [Safety notices](#). Read any additional safety information that comes with your system or optional device before you install the device.

Corresponding PDU	Rack line cord feature code	Wall plug	Rated voltage (V AC) line to line	Phase	Rated amperage	Location
ECJJ (C19) and ECJN (C13)	#6492	IEC 60309, 2P+G, 60 A	200 - 240 V	1	48 A	US, Canada, Latin America(LA), and Japan
	#6491	IEC 60309, P+N+G, 63 A	230 V	1	63 A	EMEA
	#6489	IEC 60309, 3P+N+G, 32 A	380 – 415 V	3 Wye	32 A per phase	Europe, Middle East, and Asia (EMEA)
	#6667	PDL	380 – 415 V	3 Wye	32 A per phase	Australia and New Zealand
	#6653	IEC 60309, 3P+N+G	380 - 415 V	3 Wye	16 A per phase	Switzerland
	#ELC2	IEC 60309, 3P+N+G, 30 A	380 - 415 V	3 Wye	30 A (derated to 24 A) per phase	United States, Canada, Mexico, and Japan (European Style Power for US type countries)
ECJL (C19) and ECJQ (C13)	#ECJ6	CS8365C	200 - 240 V	3 Delta	50 A or line (40 A derated)	United States, Canada
	#ECJ7	IEC 60309, 3P+G	200 - 240 V	3 Delta	60 A or line (48 A derated)	United States, Canada, Latin America, and Japan

Rack connectors with load capacity requirements

Table 2 shows the number of PDUs and the corresponding power cords that are required for different configurations.



Note: If you have selected the configuration with 4 PDUs, you can include 6 power drops in the facility plan to increase the number of PDUs in future.

Rack type	PDU feature codes	Number of PDUs	Number of Power S1122 servers	Number of Power S1150 servers	Number of Power S1122 and S1150 servers	Number of Power S1180 servers (no mixing)
ESMP (Small)	ECNJ or ECJQ** (Default)	4	8 x S1122 (includes management server)	N/A	N/A	N/A
		6	3 extra S1122	(C19 PDUs are required)	N/A	N/A
	ECJJ or ECJL**	4	1 (management server)	3	N/A	N/A
		6	2 x S1122 that is equivalent to 1 MRX. (Optional)	5	2 x S1122 that is equivalent to 1 MRX.	N/A
EMPD + 4651 or 4652 (Medium rack 1 or 2)	ECJJ or ECJL**	4	6	3	2 x S1122 that is equivalent to 1 MRX.	1x2 enclosures
		6	8	4	2 x S1122 that is equivalent to 1 MRX.	N/A
EMPD + 4653 or 54 (Medium rack 1 or 2)		4	8	3	N/A	1x2 enclosures
		6	12	4	N/A	2x2 enclosures

PDUs and power drops requirements



Note: ** You can place an order for ECJJ or ECJL PDUs only if there is an order with ER3E identifier (specifies 9043-MRU) or EN31 identifier (specifies 4983-AH8).

For more information about IBM power cords, see [Supported PDU power cord](#).

Power calculators: The [IBM Systems energy estimator](#) is a web-based tool for estimating power requirements for IBM systems. This tool estimates typical power requirements (watts) for a specific system configuration under normal operating conditions.

Network requirements

IBM Power Virtual Server Private Cloud in Client location

To facilitate the IBM® Power® Virtual Server pod infrastructure connectivity, you must evaluate the following network requirements:

- The data center site must provide network cables to connect the IBM Power Virtual Server Private Cloud network infrastructure and the data network at the site.
- The site must provide two uplink cables to connect the IBM Power Virtual Server Private Cloud network infrastructure to the IBM Cloud region through IBM Direct Link connections or through VPN connections.
- Contract with a service provider to:
 - Provide redundant connections to the IBM Direct Link connection or VPN connection.
 - Provide the last mile connection from the point-of-presence (PoP) of your service provider to the customer data center.

For network architecture diagrams, see [Network architecture diagrams](#).

Setting up a network has two parts:

- [Control plane network](#) - for communication between IBM Cloud and the pod.
- [Data plane network](#) - for accessing virtual servers.

Safety notices

IBM Power Virtual Server Private Cloud in Client location

These statements provide the caution and danger information that are used in this document.

Important

Each caution and danger statement in this documentation is labeled with a number. You can use this number to cross-reference an English-language **CAUTION** or **DANGER** statement with converted versions of the caution or danger statement in the [Safety Notices Translation document \(PN G229-9054\)](#).

For example, if a caution statement is labeled **D005**, conversions for that caution statement are in the Safety Notices conversion document under **D005**. Be sure to read all caution and danger statements in this documentation before you complete the procedures. Read any additional safety information that comes with your system or optional device before you install the device.



Danger: Overloading a branch circuit is potentially a fire hazard and a shock hazard under certain conditions. To avoid these hazards, ensure your system electrical requirements do not exceed branch circuit protection requirements. Refer to the information that is provided with your device or the power rating label for electrical specification **(D002)**



Danger: If the receptacle has a metal shell, do not touch the shell until you have completed the voltage and grounding checks. Improper wiring or grounding might place dangerous voltage on the metal shell. If any of the conditions are not as described, STOP. Ensure that the improper voltage or impedance conditions are corrected before proceeding. **(D003)**



Danger: Hazardous voltage, current, or energy levels are present inside any component that has the label that is attached. Do not open any cover or barrier that contains this label. **(L001)**



Danger: When you are working on or around the system, observe the following precautions:

Electrical voltage and current from power, telephone, and communication cables are hazardous. To avoid a shock hazard:

- If IBM supplied the power cord or power cords, connect power to this unit only with the IBM provided power cord. Do not use the IBM provided power cord for any other product.
- Do not open or service any power supply assembly.
- Do not connect or disconnect any cables or do the installation, maintenance, or reconfiguration of this product during an electrical storm.
- The product might be equipped with multiple power cords. To remove all hazardous voltages, disconnect all power cords.
 - For AC power, disconnect all power cords from their AC power source.
- When connecting power to the product ensure all power cables are properly connected.
 - For racks with AC power, connect all power cords to a properly wired and grounded electrical outlet. Ensure that the outlet supplies proper voltage and phase rotation according to the system rating plate.
- Connect any equipment that is attached to this product to properly wired outlets.
- When possible, use one hand only to connect or disconnect signal cables.
- Never turn on any equipment when there is evidence of fire, water, or structural damage.
 - Do not attempt to turn on power to the machine until all possible unsafe conditions are corrected.
 - Assume that an electrical safety hazard is present.
 - Do all the continuity, grounding, and power checks that are specified during the subsystem installation procedures to ensure that the machine meets safety requirements.

- Do not continue with the inspection if any unsafe conditions are present.
- Before you open the device covers, unless instructed otherwise in the installation and configuration procedures: Disconnect the attached AC power cords, turn off the applicable circuit breakers that are located in the rack power distribution panel (PDP), and disconnect any telecommunications systems, networks, and modems.



Danger: Connect and disconnect cables as described in the following procedures when installing, moving, or opening covers on this product or attached devices.

To Disconnect:

1. Turn off everything (unless instructed otherwise).
2. For AC power, remove the power cords from the outlets.
3. Remove the signal cables from the connectors.
4. Remove all cables from the devices.

To connect:

1. Turn off everything (unless instructed otherwise).
2. Attach all cables to the devices.
3. Attach the signal cables to the connectors.
4. For AC power, attach the power cords to the outlets.
5. Turn on the devices.

Sharp edges, corners, and joints might be present in and around the system. Use care when handling equipment to avoid cuts, scrapes, and pinching. **(D005)**



CAUTION: The weight of this part or unit is between 18 and 32 kg (39.7 and 70.5 lb). It takes two persons to safely lift this part or unit. **(C009)**



CAUTION: Data processing environments can contain equipment transmitting on system links with laser modules that operate at greater than Class 1 power levels. For this reason, never look into the end of an optical fiber cable or open receptacle. Although shining light into one end and looking into the other end of a disconnected optical fiber to verify the continuity of optic fibers might not injure the eye, this procedure is potentially dangerous. Therefore, verifying the continuity of optical fibers by shining light into one end and looking at the other end is not recommended. To verify continuity of a fiber optic cable, use an optical light source and power meter. **(C027)**



Danger: High Leakage Current. Earth connection is essential before connecting supply. **(LT020)**



CAUTION: A hot surface nearby. **(L007)**



DANGER: Multiple power cords. The product might be equipped with multiple AC power cords or multiple DC power cables. To remove all hazardous voltages, disconnect all power cords and power cables. **(L003)**

(R001 part 1 of 2):



DANGER: Observe the following precautions when working on or around your IT rack system:

- Heavy equipment—personal injury or equipment damage might result if mishandled.
- Always lower the leveling pads on the rack cabinet.
- Always install stabilizer brackets on the rack cabinet unless the earthquake option is to be installed.
- To avoid hazardous conditions due to uneven mechanical loading, always install the heaviest devices in the bottom of the rack cabinet. Always install servers and optional devices starting from the bottom of the rack cabinet.



- Rack-mounted devices are not to be used as shelves or workspaces. Do not place objects on top of rack-mounted devices. In addition, do not lean on rack-mounted devices and do not use them to stabilize your body position (for example, when working from a ladder).

- Each rack cabinet might have more than one power cord.
 - For AC-powered racks, be sure to disconnect all power cords in the rack cabinet when directed to disconnect power during servicing.
- Connect all devices installed in a rack cabinet to power devices installed in the same rack cabinet. Do not plug a power cord from a device that is installed in one rack cabinet into a power device that is installed in a different rack cabinet.
- An electrical outlet that is not correctly wired might place hazardous voltage on the metal parts of the system or the devices that attach to the system. It is the responsibility of the customer to ensure that the outlet is correctly wired and grounded to prevent an electrical shock.

(R001 part 2 of 2):



CAUTION:

- Do not install a unit in a rack where the internal rack ambient temperatures exceed the manufacturer's recommended ambient temperature for all your rack-mounted devices.
- Do not install a unit in a rack where the air flow is compromised. Ensure that air flow is not blocked or reduced on any side, front, or back of a unit used for air flow through the unit.
- Consideration should be given to the connection of the equipment to the supply circuit so that overloading of the circuits does not compromise the supply wiring or overcurrent protection. To provide the correct power connection to a rack, refer to the rating labels on the equipment in the rack to determine the total power requirement of the supply circuit.
- (For sliding drawers.) Do not pull out or install any drawer or feature if the rack stabilizer brackets are not attached to the rack or if the rack is not bolted to the floor. Do not pull out more than one drawer at a time. The rack might become unstable if you pull out more than one drawer at a time.
- (For fixed drawers.) This drawer is a fixed drawer and must not be moved for servicing unless specified by the manufacturer. Attempting to move the drawer partially or completely out of the rack might cause the rack to become unstable or cause the drawer to fall out of the rack.



CAUTION: Removing components from the upper positions in the rack cabinet improves rack stability during relocation. Follow these general guidelines whenever you relocate a populated rack cabinet within a room or building.

- Reduce the weight of the rack cabinet by removing equipment starting at the top of the rack cabinet. When possible, restore the rack cabinet to the configuration of the rack cabinet as you received it. If this configuration is not known, you must observe the following precautions:
 - Remove all devices in the 32U position (compliance ID RACK-001) or 22U (compliance ID RR001) and above.
 - Ensure that the heaviest devices are installed in the bottom of the rack cabinet.
 - Ensure that there are little-to-no empty U-levels between devices that are installed in the rack cabinet below the 32U (compliance ID RACK-001) or 22U (compliance ID RR001) level, unless the received configuration specifically allowed it.
- If the rack cabinet you are relocating is part of a suite of rack cabinets, detach the rack cabinet from the suite.
- If the rack cabinet you are relocating was supplied with removable outriggers they must be reinstalled before the cabinet is relocated.
- Inspect the route that you plan to take to eliminate potential hazards.
- Verify that the route that you choose can support the weight of the loaded rack cabinet. Refer to the documentation that comes with your rack cabinet for the weight of a loaded rack cabinet.
- Verify that all door openings are at least 760 x 230 mm (30 x 80 in.).
- Ensure that all devices, shelves, drawers, doors, and cables are secure.
- Ensure that the four leveling pads are raised to their highest position.
- Ensure that there is no stabilizer bracket that is installed on the rack cabinet during movement.
- Do not use a ramp that is inclined at more than 10 degrees.
- When the rack cabinet is in the new location, complete the following steps:
 - Lower the four leveling pads.
 - Install stabilizer brackets on the rack cabinet or in an earthquake environment bolt the rack to the floor.
 - If you removed any devices from the rack cabinet, repopulate the rack cabinet from the lowest position to the highest position.
- If a long-distance relocation is required, restore the rack cabinet to the configuration of the rack cabinet as you received it. Pack the rack cabinet in the original packaging material, or equivalent. Also, lower the leveling pads to raise the casters of the pallet and bolt the rack cabinet to the pallet.

(R002)

Network use cases

Review the common network use cases within the network architecture of IBM® Power® Virtual Server.

Use case 1: Private network within a pod

With this use case, you can establish a private network within a pod that allows communication between the applications that are located in the pod. You can establish a private network within the pod by using the IP address allocation method, Classless Inter-Domain Routing (CIDR).

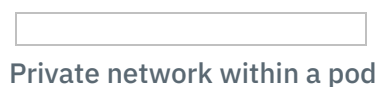
You can deploy virtual machines in a pod that has a default configuration by using one of the following patterns:

- **Affinity:** In this pattern, virtual machines are deployed on the same physical host. Therefore, the virtual machines can communicate with each other on the same host through the attached Ethernet switch.
- **Anti-affinity:** In this pattern, virtual machines are deployed on different physical hosts. A custom configuration is required on the externally connected Ethernet switch to enable communication between virtual machines that are deployed on different physical hosts.

Example

You have a database server and a web server that need to communicate exclusively with each other. You can connect both servers to the same private network to enable communication between them.

Figure 1 describes the private network within a pod type of network setup.



Use case 2: Multiple external connections

With this use case, you can access the external connectivity to and from the virtual machines in a subnet.

To access the external connectivity, you must create peer interfaces and establish an external subnet in a workspace. The VMs deployed on the workspace can access the external environment.

To access the external connectivity, complete the following steps:

Step 1: Create a peer interface

To create a peer interface, complete the following steps:

1. Create a workspace.
2. Open a [support ticket](#) to create a peer interface. The support team sets up a peer interface with one of the following combinations:
 - Layer 2
 - Layer 2 and Layer 3 with BGP route
 - Layer 2 and Layer 3 with static route
 - Layer 3 BGP route
 - Layer 3 static route

The peer interface combination is based on the type of data network cabling in your infrastructure. If you need a specific combination of peer interface, you must specify the combination in the ticket.

Step 2: Create an external subnet

To create an external subnet, use one of the peer interfaces configured in **Step 1**.

Step 3: Deploy VMs to access external connectivity

To configure your workspace, open a [support ticket](#).

You can deploy VMs on your workspace after you receive the confirmation from the support team about the completion of configuring your workspace.

Bidirectional external connectivity through BGP

With this use case, you can establish a network that allows communication between the applications within the pod and with the destination points on the external network. By setting up the Layer 3 Firewall rules, you can allow both inbound and outbound connections. Configure the BGP manually between the pod router and the corporate network. By using the BGP configuration, establish a connection between the private network and the corporate network. To configure BGP manually, contact the Support Center. For more information, see the [Getting support](#) section.

Example

You have a database server that is running inside the pod. You need to access the database server from another application that resides outside the pod but within your corporate network. With Layer 3 inbound access, you can route the traffic and apply your corporate firewall or routing rules to access the database server. The corporate network can access the pod subnets by using a BGP connection.

Figure 3 describes the bidirectional external connectivity through BGP type of network setup.



Bidirectional external connectivity through static routes

With this use case, you can establish a network that allows communication between applications within the pod and with destination points on the external network. Using Layer 3 firewall rules, you can allow both inbound and outbound connections. Establish a static route between edge routers that are within the pod and to the next hop router that is within the corporate network. The static route establishes a connection between the pod subnet and the corporate network. To establish the static route connectivity manually, contact the Support Center. For more information, see the [Getting support](#) section.

Example

You have a database server that is running inside the pod. You need to access the database server from another application that resides outside the pod but within your corporate network. With Layer 3 inbound access, you can route the traffic and apply your corporate firewall or routing rules to access the database server. Your corporate network can access the pod subnets through a static route connection.

Figure 4 describes the bidirectional external connectivity through static routes type of network setup.



Bidirectional external connectivity - Layer 2

With this use case, you can create a network that allows communication between applications within the pod and with destination points on the external network. Integrate a Layer 2 firewall with one of your existing corporate networks to allow the outbound connections. In this type of network, bidirectional external connectivity bypasses the router and connects to the IBM Power Virtual Server network fabric. You can establish this type of connectivity when you want the same IP address space on both internal and external networks. All other external networks involve two distinct subnets.

Figure 5 describes the bidirectional external connectivity by using Layer 2 firewall type of network setup.



Use case 3: Network connectivity for full Linux subscription

With this use case, you can establish a network between a virtual machine in the pod and the Red Hat Satellite server on IBM Cloud. The virtual machine has the stock image of Red Hat Enterprise Linux (RHEL) with full Linux subscription.

Connect the virtual machine in the pod to a proxy network in the corporate network environment. Connect the proxy network to the Red Hat Satellite server on IBM Cloud by using either Direct Link or VPN connection. The virtual machine in the pod can access the Linux® satellite server to retrieve software fixes and other artifacts.

The network connectivity for full Linux subscription can be established by providing the network requirements before the installation of the pod. For more information, see [Network requirement](#). After the pod is installed, you can configure the network connectivity for full Linux subscription by using the Support Center ticketing system. For more information about full Linux subscription, see [Full Linux subscription for Power Virtual Server Private Cloud](#). For more information about contacting the Support Center, see the [Getting support](#) section.

Figure 6 describes the network connectivity between a virtual machine and a Red Hat Satellite server on IBM Cloud setup.



Use case 4: DHCP network inside the pod

The Power Virtual Server Dynamic Host Configuration Protocol (DHCP) service provides automated IP address assignment and network configuration for virtual server instances (VSIs) in your workspace.

You can enable or disable the DHCP service on any subnet in your Power Virtual Server workspace either when you create a new subnet or after the subnet is created. When DHCP is enabled on a subnet, network interfaces that are configured to use DHCP automatically receive their IPv4 configurations, including the IP address, subnet mask, gateway, domain name system (DNS) servers, and maximum transmission unit (MTU) settings. When you create a new subnet, the DHCP service is enabled by default.



Note: The operating system (OS) of your logical partition (LPAR) must be configured to use the DHCP services.

Network latency

IBM Power Virtual Server Private Cloud in Client location

The network latency between your data center and the corresponding IBM Cloud region must maintain a network round-trip time (RTT) of less than or equal to 200 milliseconds.

You can note the IP addresses for the IBM Cloud region that you want to test from the following table:

IBM Cloud region	IP address
Dallas	52.117.39.146, 169.48.134.66, 169.63.36.210
Frankfurt	149.81.188.122, 158.177.88.18, 161.156.38.122
London	158.175.120.210, 141.125.97.106, 158.176.139.66
Madrid	13.120.67.114, 13.121.67.98, 13.122.67.106
Osaka	163.68.73.50, 163.69.65.242, 163.73.67.10
Sao Paulo	163.107.67.18, 163.109.71.82, 169.57.144.42
Sydney	130.198.65.82, 135.90.66.194, 168.1.58.90
Tokyo	161.202.104.226, 128.168.67.106, 165.192.108.10
Toronto	163.74.65.138, 163.75.70.50, 169.53.160.154
Washington, DC	169.63.123.154, 169.63.110.114, 169.62.13.2, 169.60.123.162, 169.59.152.58, 52.117.93.26

IBM Cloud Region and IP address

To check the latency requirements of a connection, complete the following steps:

1. From a computer in the data center where you place the pod, ping the IP address of the IBM Cloud region that is closest to the physical location of your data center. Run the ping test from a compute endpoint as close as possible to the potential region where the pod is placed.

For example,

```
$ ping <ip_address>
```

2. Close the connection when the transmission of a few packets is complete. For example, from the command-line, enter `ctrl+c`.
3. In the **ping statistics** output, note the average (avg) round-trip distance in milliseconds (ms) between the host and the IBM Cloud region. Compare whether the connection meets the latency requirement of less than or equal to 200 milliseconds (≤ 200 ms).



Note: If you are selecting the IBM Direct Link 2.0 Connect to connect the Virtual Private Cloud (VPC) on IBM Cloud and the router on pod, the minimum speed must be 1 Gbps.

The following example meets the latency requirements as the round-trip time is 77.716 ms:

```
$ --- 169.xx.xxx.xxx ping statistics ---
25 packets transmitted, 25 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 48.131/77.716/181.397/27.893 ms
```

The following example does not meet the latency requirements as the round-trip time is 217.37 ms:

\$ --- 158. xx.xxx.xxx ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 138.453/217.370/419.901/108.211 ms

Creating, managing, and deploying

Managing identity and access management (IAM) for IBM® Power® Virtual Servers

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IAM enables you to securely authenticate users, control access to Power® Virtual Server resources with resource groups, and allow access to specific resources for a set of users with access groups. IAM is your one-stop shop for all user and resource management in the IBM Cloud.



Important: Client location To display the **Infrastructure capacity** navigation menu for the IBM Power Virtual Server Private Cloud when you use a custom role with the `power-iaas.pod-capacity.view` IAM action, ensure that you have a `Viewer` role that is assigned in the IAM Access Management service.

For more information about IAM, review the following information:

- [Getting started with IAM](#)
- [Managing resource groups](#)
- [Streamlined access management with access groups](#)
- [IAM access concepts](#)

Platform access roles

You can use platform access roles to enable users to complete tasks on IBM Cloud resources, such as creating users or adding services.

The following table displays the IAM platform access roles and the corresponding type of control that is allowed by the Power Virtual Server:

Platform access role	Type of access allowed
Viewer	View instances and list instances.
Operator	View instances and manage aliases, bindings (IBM Power Virtual Server Private Cloud in client location only), and credentials.
Editor	View instances, list instances, create instances, and delete instances.
Administrator	View instances, list instances, create instances, delete instances, and assign policies to other users.

IAM platform access roles

Service access roles

You can use the service access roles to define the actions that the users can perform on Power Virtual Server resources. The following table displays the IAM service access roles and the corresponding actions that a user can complete by using the Power Virtual Server:

Service access role	Description of actions
Reader	View all resources (such as SSH keys, storage volumes, and network settings). You cannot modify the resources.
Manager	Configure all resources. You can perform the following actions: - Create instances - Increase storage volume sizes - Create SSH keys - Modify network settings - Create boot images - Delete storage volumes

To see the complete list of actions for each specific role, see the [IAM roles and actions](#) page in IBM Cloud documentation.

Resources supported for Power Virtual Server IAM access policies

When you assign access to the Power Virtual Server service, you can scope access to any of the following resources:

- All resources
- Specific resources, which support the following selections:
 - Resource group
 - Service instance

The access management tags are supported only on Power Virtual Server workspaces. The Power Virtual Server service ignores the access management tags that are attached to the individual resources in a workspace.



Note: Although you can select a **Resource type** from the **Attribute type** list, it is not supported. Any roles and actions that are assigned to the **Resource type** are ignored.

Access role requirements for Power Virtual Server

Power Virtual Server requires extra access for features such as Direct Link, Transit Gateway service, and Virtual Private Cloud. You might require these extra access based on your resource requirements. For example, to create a Cloud connection, you need **access** to the Direct Link service.

The following table displays the additional access roles that are required for the corresponding type of services that are allowed by Power Virtual Server:

Additional access role	Resources Attributes
Editor, Manager, Operator, Reader, Viewer	Power Virtual Server service
Editor, Manager, Operator, Reader, Viewer, VPN Client	VPC Infrastructure Services service
Editor, Operator, Viewer	Transit Gateway service
Reader, Viewer	All resources in account (Including future IAM enabled services)
Editor, Operator, Viewer	Direct Link service
Viewer	All resource group
Viewer	Satellite service Client location

Additional access roles

User access scenarios

For more information about managing and assigning access by using IAM policies, see [Managing access to resources](#).

Trusted profiles for Power Virtual Server

A trusted profile is an IAM identity for a compute resource. IBM Cloud IAM trusts your Power Virtual Server virtual server instances (VSI) as a compute resource, which allows the VSI to call the API endpoints of other IBM Cloud services without storing API keys or credentials on the instance.

Benefits of using trusted profiles

Trusted profiles provide the following security benefits for Power Virtual Server:

- **Enhanced security:** Eliminates the risk of credential exposure by removing the need to store API keys or passwords on your VSIs.
- **Simplified credential management:** Eliminates the need to rotate, update, or manage static credentials across your VSIs.
- **Dynamic authentication:** Applications in the guest OS can generate tokens on demand based on the VSI identity. Because tokens are short-lived, access credentials are always current and tied to the specific instance.
- **Reduced attack surface:** Prevents compromised VSIs from exposing long-lived credentials to access other resources.

- **Precise access control**: Controls which IBM Cloud services each VSI can access through IAM policies and access groups assigned at the individual VSI level.

Managing trusted profiles

You manage trusted profiles through IBM Identity and Access Management (IAM). You can create, configure, and manage trusted profiles from the [IBM Cloud Trusted Profiles UI](#). You can associate each profile with one or more Power Virtual Server VSIs and assign access policies to control which IBM Cloud services and resources the VSIs can access.

Creating a trusted profile for Power Virtual Server

To use trusted profiles with your Power Virtual Server VSIs, you must create a trusted profile in IAM. You can configure rule-based policies for the VSIs or specify an individual VSI that can use a trusted profile.

For more information about creating and managing a trusted profile, see [Establishing trust with Power Virtual Server compute resources by using the console](#).

Configuring and managing the metadata service for Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

The IBM® Power® Virtual Server metadata service provides information about your virtual server instances (VSIs) and enables identity operations for trusted profile authentication. When the metadata service is enabled on a VSI, you can retrieve information about that VSI by sending REST API requests from the VSI. Use the identity endpoints of the metadata service to authenticate with IAM-enabled services without storing credentials on the instance. Access to the metadata service API endpoint is available only within the Power Virtual Server environment.

Metadata service concepts

You can retrieve metadata about the VSI on which the metadata service is enabled, including the VSI name, CRN, and workspace information. For more information about all metadata that the metadata service returns, see [Configuring the metadata service interface on IBM i](#). Disabling access to the metadata service After disabling the metadata service, you must remove the TCP/IP interface and line description to complete cleanup. For more information, see [Troubleshooting on IBM i](#). Overwriting the boot disk with a captured image OS-level configuration of the metadata service network interface is required after you overwrite the boot disk. For more information, see [Reconfiguring the metadata service interface after a custom image overwrite on IBM i](#).

Scenario	AIX
Creating a VSI with metadata service access enabled	The metadata service interface is automatically configured with IP address 169.254.169.253 . No additional configuration is required.
Enabling access to the metadata service on an existing VSI	You can enable access only when the VSI is in the Shutoff state. After you enable the metadata service access and start the VSI, the metadata service interface is automatically configured. If you use AIX 7.1, you must configure the interface after the VSI boots. For more information, see Validating and recovering network connectivity on AIX .
Disabling access to the metadata service	You can disable access only when the VSI is in the Shutoff state. The <code>--metadata-service-force</code> CLI option and the <code>forceDisable</code> API parameter are not supported for AIX. When you disable the metadata service, the system removes the IP address, but the network adapter entries might remain in the Defined state. You can clean up the remaining network adapter entries. For more information, see Cleaning up network interfaces after disabling the metadata service on AIX .
Overwriting the boot disk with a captured image	After the boot disk is overwritten, you must reconfigure the metadata service interface. For more information, see Validating and recovering network connectivity on AIX .

Metadata service configuration requirements by operating system

Scenario	Linux
Creating a VSI with metadata service access enabled	The metadata service interface is automatically configured with IP address 169.254.169.253 . No additional configuration is required.
Enabling access to the metadata service on an existing VSI	You can enable access only when the VSI is in the Shutoff state. After you enable the metadata service access and start the VSI, the metadata service interface is automatically configured.
Disabling access to the metadata service	When you use the <code>--metadata-service-force</code> CLI option or the <code>forceDisable</code> API parameter, you must remove the network interface configuration. Specific commands depend on your Linux distribution's network management tools. For more information, see Cleaning up network interfaces after disabling the metadata service on Linux .
Overwriting the boot disk with a captured image	Before you copy the disk image, you must back up and clear the <code>udev</code> rules files (network interface configuration files) that contain MAC address bindings. This removes network persistence rules and prevents connectivity issues when the image is restored. For more information, see Reconfiguring the metadata service interface after image capture on Linux .

Metadata service configuration requirements by operating system

Enabling access to the metadata service for a new VSI by using the user interface

You can enable access to the metadata service when you create a VSI by using the Power Virtual Server user interface.

To enable access to the metadata service when you create a VSI, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. From the Workspaces page, select a workspace in which you want to create a VSI. The "Virtual server instances" page is displayed.
5. Click **Create instance**. The "Create virtual server instance" page is displayed.
6. In the General section, expand **Advanced configurations** and set **Metadata service** to on. The **Default trusted profile (optional)** field and the **Auto link** toggle are displayed.

You can optionally configure the following settings:

- **Default trusted profile:** To grant the VSI access to IAM-enabled services through a trusted profile, click **Select trusted profile**. The "Select default Trusted Profile" dialog is displayed. Select a trusted profile from the list and click **Select Trusted Profile**.
- **Auto link:** Set the **Auto link** toggle:
 - **Enabled** Default. The VSI cloud resource name (CRN) is automatically added to the selected trusted profile and the profile is available to the VSI when the VSI starts.
 - **Disabled:** If the selected trusted profile is not configured with rule-based filters, you must manually link the trusted profile to the VSI by using the IAM Trusted Profile UI, CLI, or API before the VSI can use the trusted profile.

 **Note:** If the selected trusted profile is created from an enterprise account template, by default the child account cannot edit the trusted profile. This causes the **Auto link** operation to fail. To avoid this error, disable **Auto link** and either create the trust relationship directly in IAM, or add a rule to the trusted profile template that grants access to VSIs from the child account. For more information, see [Creating trusted profile templates](#).

7. Click **Continue** and complete the remaining steps to create the VSI. For more information about creating a VSI, see [Creating a Power Virtual Server instance](#).

Enabling access to the metadata service for an existing VSI by using the user interface

 **Note:** When you enable access to the metadata service API endpoint for an existing VSI, the VSI must be in the **Shutoff** state.

To enable access to the metadata service for an existing VSI, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. Select the workspace that contains the VSI for which you want to enable the metadata service. The "Virtual server instances" page with a list of existing VSIs is displayed.
5. Select the VSI for which you want to enable the metadata service. The "Virtual server instance details" page for the selected VSI is displayed.

If you enable access to the metadata service for a VSI that is in the **Active** state, you might need to complete additional steps to configure the network. For more information about configuration steps, see [Configuration requirements by operating system](#).

6. On the **Overview** tab, go to the Metadata service section, and click the **Edit** icon. The "Edit Metadata service details" panel is displayed.
7. Set **Metadata service** to **Access enabled**. The **Default trusted profile (optional)** field and the **Auto link** toggle are displayed.

You can optionally configure the following settings:

- **Default trusted profile:** To grant the VSI access to IAM-enabled services through a trusted profile, click **Select trusted profile**. The "Select default Trusted Profile" dialog is displayed. Select a trusted profile from the list and click **Select Trusted Profile**.
- **Auto link:** Set the **Auto link** toggle:
 - **Enabled** (default): The VSI CRN is automatically added to the selected trusted profile and the profile is available to the VSI when the VSI starts.
 - **Disabled:** If the selected trusted profile is not configured with rule-based filters, you must manually link the trusted profile to the VSI by using the IAM Trusted Profile UI, CLI, or API before the VSI can use the trusted profile.



Note: If the selected trusted profile is created from an enterprise account template, by default the child account cannot edit the trusted profile. This causes the **Auto link** operation to fail. To avoid this error, disable **Auto link** and either create the trust relationship directly in IAM, or add a rule to the trusted profile template that grants access to VSIs from the child account. For more information, see [Creating trusted profile templates](#).

8. Click **Save**.

Disabling access to the metadata service by using the user interface

You can disable access to the metadata service for a VSI on which the service is currently enabled. By default, access to the metadata service is disabled when you create a VSI.

AIX VSIs must be in the **Shutoff** state before you can disable access to the metadata service.

To disable access to the metadata service for a VSI, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. Select the workspace that contains the VSI for which you want to disable the metadata service. The "Virtual server instances" page with a list of existing VSIs is displayed.
5. Select the VSI for which you want to disable the metadata service. The "Virtual server instance details" page for the selected VSI is displayed.
6. On the **Overview** tab, go to the Metadata service section and click the **Edit** icon. The "Edit Metadata service details" panel is displayed.
7. Set **Metadata service** to **Access disabled**.

If you disable access to the metadata service for a VSI that is in the **Active** state, you might need to configure the network interface. For more information about configuration steps, see [Configuring and troubleshooting metadata service connectivity](#).

8. Click **Save**.

Enabling or disabling access to the metadata service by using the CLI

You can use the Power Virtual Server CLI to enable or disable access to the metadata service when you create a VSI or update an existing VSI.

Before you begin, install the IBM Cloud CLI and the Power Virtual Server CLI plug-in. For more information, see [Installing the IBM Power Virtual Server CLI](#)

[plug-in](#).

Enabling or disabling access to the metadata service when you create a VSI by using the CLI

Run the [ibmcloud pi instance create](#) command and set the `--metadata-service` option to `True`. Access to the metadata service is disabled by default.

```
ibmcloud pi instance create test-instance-1 --image IMAGE_ID --subnets SUBNET_ID --processors 0.5 --processor-type shared --memory 4 --metadata-service true
```

You can also specify a default trusted profile for your VSI. When you specify a default trusted profile, the `--default-trusted-profile-autolink` option is set to `true` by default. When auto-link is enabled, the specified trusted profile is automatically linked to the VSI. The following example shows the VSI creation command with `--default-trusted-profile-autolink` option set to `false`.

```
ibmcloud pi instance create test-instance-2 --default-trusted-profile "Profile-9fd84246-7df4-4667-94e4-8ecde51d5ac5" --default-trusted-profile-autolink false
```

Enabling or disabling access to the metadata service for an existing VSI by using the CLI

To enable access to the metadata service, the VSI must be in the **Shutoff** state. To disable access to metadata service for an AIX VSI, the VSI must be in the **Shutoff** state. The `--metadata-service-force` option is not supported for AIX.

Run the [ibmcloud pi instance update](#) command and specify the instance ID. To enable access to the metadata service, set the `--metadata-service` option to `true`. To disable access, set the `--metadata-service` option to `false`. The following example shows a CLI command that enables access to the metadata service:

```
ibmcloud pi instance update INSTANCE_ID --metadata-service true
```

Enabling or disabling access to the metadata service by using the API

You can use the Power Virtual Server API to enable or disable access to the metadata service when you create a VSI or update an existing VSI.

Enabling or disabling access to the metadata service when you create a VSI by using the API

By default, the metadata service is disabled when you create a VSI by sending a `POST /pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances` request.

You can enable access to the metadata service by specifying the `metadataService` parameter and setting `enabled` to `true`.

This example shows how to enable access to the metadata service during VSI creation by using the API:

```
curl -X POST "$power_api_endpoint/pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances" \
-H "Authorization: Bearer $iam_token" \
-d '{
  "imageID": "7200-05-09",
  "keyPairName": "test1",
  "memory": 2,
  "networks": [
    {
      "networkID": "public-192_168_194_2-6-VLAN_2044"
    }
  ],
  "pinPolicy": "",
  "placementGroup": "",
  "procType": "shared",
  "processors": 0.25,
  "serverName": "vsnTest2",
  "storageType": "tier3",
  "sysType": "s922",
  "volumeIDs": [],
  "metadataService": {
    "enabled": true
  }
}'
```

The response shows that the `metadataService` parameter is set to `true` when you enable access to the metadata service. You can also verify the metadata service setting by sending a `GET /pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances/{pvm_instance_id}` request.

Enabling or disabling access to the metadata service for an existing VSI by using the API

To enable access to the metadata service, the VSI must be in the **Shutoff** state. To disable access to metadata service for an AIX VSI, the VSI must be in the **Shutoff** state. The `forceDisable` option is not supported for AIX.

To enable or disable access to the metadata service from an existing VSI, send a `PUT /pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances/{pvm_instance_id}` request and specify the `metadataService` parameter. By default, the `enabled` option is set to `false`. To enable access to the metadata service, set the `enabled` option to `true`.

The following example shows how to enable access to the metadata service for an existing VSI:

```
curl -X PUT "$power_api_endpoint/pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances/{pvm_instance_id}" \
-H "Authorization: Bearer $iam_token" \
-d '{
  "metadataService": {
    "enabled": true
  }
}'
```

The response shows that the `metadataService` parameter is set to `true` when you enable access to the service. You can also verify the metadata service setting by sending a `GET /pcloud/v1/cloud-instances/{cloud_instance_id}/pvm-instances/{pvm_instance_id}` request.

Disabling auto-link by using the API

To disable auto-link by using the API, set the `autolink` value for the `defaultTrustedProfile` parameter to `false`. The following example shows the `defaultTrustedProfile` parameter with the `autolink` option disabled.

```
$ "defaultTrustedProfile": {
  "autolink": false,
  "target": {
    "id": "Profile-9fd84246-7df4-4667-94e4-8ecde51d5ac5"
  }
},
```

Identity operations

You can use the metadata service to obtain an identity access token and then generate an IAM access token. You can use these tokens to access the metadata and IAM-enabled IBM Cloud APIs.

Identity access token

An identity access token is a security credential that you use to access the Power Virtual Server metadata and identity services. An identity access token is a signed token that contains claims based on information about the VSI.

**Important:** Communication between the VSI and the metadata service remains within the Power Virtual Server environment. You can access the APIs only from a Power Virtual Server VSIs. You cannot access the metadata and Identity APIs by using the Power Virtual Server user interface or CLI.

To obtain the identity access token, send a `PUT https://api.metadata.power-iaas.cloud.ibm.com/identity/v1/token` request to the [Updating trusted profiles](#).

Creating an IBM Power Virtual Server instance

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can create and configure Power Virtual Server instances by setting up a workspace, selecting boot images, and configuring machine profiles. You can also manage storage volumes and define network interfaces to deploy virtual server instances (VSIs) in your cloud environment.

Creating a Power Virtual Server workspace

To create a Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Create a workspace**.
4. Select **Location type** as Client location or IBM data center.

For Client location location types, select your satellite location from the **Satellite Location** list.

For IBM data center location types, select the IBM Cloud region that is closest to your physical location from the **Satellite Location** list. For the list of available IBM Cloud regions, see [IBM Cloud regions](#).

5. In the **Details** section, provide a name for the workspace and select a resource group.
6. Click **Continue**. Review the workspace details and estimated cost on the Summary page.
7. Select **I agree to the Terms and conditions** checkbox.
8. Click **Create**. You are redirected to the **Workspaces** page where you can select an existing workspace.

For more information about the appropriate region for your workspace, see [IBM Cloud regions](#).



Note: When you create a workspace by using the Terraform resource [ibm_resource_instance](#) or the CLI command [ibmcloud resource service-instance-create](#), for the **service name** field use the value `power-iaas` for IBM Power Virtual Server in IBM data center and for the **plan name** field use the value `power-virtual-server-group` or `power-virtual-server-private-group` for IBM Power Virtual Server Private Cloud in Client location.

Configuring a Power Virtual Server instance

To create a VSI, you must first create a [Power Virtual Server workspace](#) and then select it from the Workspaces list. The created workspaces are listed under **Workspaces** in the navigation panel of the Power Virtual Server user interface. Select the workspace for which you want to create an instance. Complete the following steps to create a virtual server instance:

1. Click **Virtual server instances** in the navigation panel. The VSIs that are associated with the selected workspace are displayed.



Important: Select a workspace to display the VSIs that were provisioned. If the displayed information appears outdated, refresh the page to view the latest data. For more information, see the FAQ page [What should I do when I do not see the latest information in the UI](#).



Note: **Client location** If an error occurs immediately after you create or open the VSI, you must delete it.

2. To create a new instance, click **Create instance**.
3. In the General section, enter a name for the VSI in the **Instance name** field.
4. Optional: Enter user tags in the **User tags (optional)** field.
5. Set the number of VSIs that you want to create in the **Number of instances** field. The default is 1, and you can create up to five VSIs at a time. If you set more than one instance, additional options are displayed.



Tip: The total due per month is dynamically updated in the **Order Summary** based on your selections. You can create a cost-effective Power Virtual Server instance that satisfies your business needs.

6. Choose an existing SSH key or create one to securely connect to your Power Virtual Server.
7. Optional: Expand **Advanced Configurations** to set more options for your VSI.
 - **Virtual server pinning:** By default, the **Virtual server pinning** option is set to off. When you enable this option, the VSI is kept on its current host; however, downtime can occur during planned and unplanned outages. To select a pinning type, set **Virtual server pinning** to on, and then select **Soft** or **Hard**.

For more information about VSI pinning, see [Virtual server pinning](#).

- **Metadata service:** By default, the **Metadata service** option is set to off. When you enable access to the metadata service on a VSI, you have access to the metadata and identity APIs. You can use the metadata service to access information about a VSI and access IAM-enabled services. For more information, see [Configuring and managing the metadata service for Power Virtual Server](#).
8. Complete the **Boot image** fields.

You can create or provision a VSI without an initial boot image volume. VSIs without a boot volume are helpful in [high availability and disaster recovery](#) use cases. You can create a VSI without a boot volume and then attach a cloned or replicated volume to restore the backed-up VSI.

Select the **Deploy empty virtual server instance** checkbox to provision a VSI without a boot volume. For more information, see [Provisioning a virtual machine without an initial boot volume](#).



Important: You can create a VSI without a boot volume for AIX, IBM i, and Linux operating systems. For Linux, an IBM-provided subscription is required. If you assign a virtual serial number (VSN) to a VSI without storage, you must assign the VSN to the VSI before you attach the boot volume and start the VSI.

You can set the affinity policies for storage pools. For more information, see [Configuring affinity policies](#).

When you select **Boot image**, the Power Virtual Server user interface allows you to select the boot images from a set of available stock images or from a custom image in your image catalog. Custom images are images that you can import from IBM Cloud Object Storage or create from a VSI capture.

When you select a stock image, you must also select the storage tier and the storage pool. When you select a custom image, the new VSIs are deployed into the same storage tier and pool where the image resides. You must select a storage type for stock images. Currently, you cannot mix **Tier 1** and **Tier 3** storage types. For more information, see [Storage tiers](#).



Note: **Client location** If you select a custom image from a local catalog, the VSIs are deployed on a single storage tier.

If you select AIX as the boot image, the Power Virtual Server user interface provides you with an option to configure the VSI for Epic workload. For more information about Epic workloads, see [Configuring a VSI for Epic workloads](#).

If you select IBM i as the boot image, the Power Virtual Server user interface provides you with the following options:

- Add the following licenses to your VSI:

- IBM i Cloud Storage Solution
- IBM i PowerHA
- Rational Dev Studio for IBM i

Adding a license increases the service cost. The selected licenses are injected to your VSI. You can install specific solutions on your VSI, and the licenses are automatically applied. If you want to use these licensed programs on your IBM i VSI, you must order these licenses through Power Virtual Server. You cannot use existing licenses in your VSI.

- Select the IBM i software tier from the **IBM i software tier** list. To select an IBM i software tier, you must select an image with OS version 7.3 or later from the **Boot image** field and set the **Virtual serial number (VSN)** as assigned.



Note: Assigning a VSN is not supported on a VSI with IBM i version 7.1 or earlier.

If you select Full Linux Subscription (FLS) images, the Power Virtual Server user interface provides you with an option to pass in user data or scripts during the first boot runtime. When you finish entering the user data for Linux images, the system does validation checks on the input. No validation checks are done for AIX and bring your own license images. For more information, see [Passing user-defined scripts](#).

Cloud Optical Repository (COR) is a virtual image that can be deployed and used as a Network File Server (NFS) to perform various IBM i tasks that require media. This virtual optical image includes a collection of the media necessary for various IBM i tasks, for all supported IBM i releases. With the COR image deployed, a second Power Virtual Server instance can be deployed on the same VLAN that is set up as the client and pointed to the COR (target) NFS Server Instance. For more information about COR images, see [Cloud Optical Repository](#).

To deploy a VSI for SAP HANA workload, select one of the following options from the **Operating system** list:

- Select **Linux for SAP (HANA)** in the IBM provided subscription section to use the IBM provided Linux subscription.
- Select **Linux for SAP (HANA)** in the Client supplied subscription section to use your own license.

To deploy an SAP certified profile from the **Standard RISE** or **Application Server** tabs, set **SAP RISE deployment** to on in the **Advanced Configurations* section. The **SAP RISE deployment** option is enabled only if you select the OS as **Linux for SAP (HANA)** and the machine type as IBM Power10 or later in the Profile section.

To deploy a VSI for SAP NetWeaver workloads without an SAP certified profile, complete the following steps:

- a. In the Boot image section, select **Linux for SAP (NetWeaver)** from the **Operating system** list.
- b. Select the appropriate SAP NetWeaver image from the **Image** list.
- c. Select a storage tier from the **Tier** list.
- d. Select a storage pool from the **Storage pool** field.
- e. Click **Done editing** to continue to the Profile section.

9. Complete the **Profile** fields by selecting the **Machine type**, the number of **Cores**, the amount of **Memory (GB)**, and **Core type**.



Important: The core-to-virtual core ratio is 1:1. For shared processors, fractional cores round up to the nearest whole number. For example, 1.25 cores are equal to 2 virtual cores. For more information about processor types, see [What's the difference between shared capped and shared uncapped processor performance? How do they compare to dedicated processor performance?](#). If the machine type is S922 and the operating system is IBM i, IBM i supports a maximum of 4 cores per VSI.

When you use an AIX stock image as the boot volume, a console session is required to set the initial root user password. Without completing this step, SSH login is disabled. For more information, see [How to create a new AIX VSI with SSH keys for root login](#).

You must complete the following prerequisites to assign an IBM i software tier to a Power Virtual Server instance:

- Select an IBM i image with version 7.3 or later from the **Image** list under the **Boot image** section.
- Select an IBM Power10 or later server type from the **Machine type** list.
- Complete the following steps to assign a VSN to the instance:
 - Edit the **Virtual serial number (VSN)** field. The VSN summary pane appears.
 - Select **Auto-assign** or **Select from retained VSNs** to assign a VSN.

The supported IBM i software tiers are displayed in the **IBM i software tier** list based on the machine type that you select. The recommended IBM i software tier is displayed in the **IBM i software tier** field based on the number of cores and the memory size. You can select the IBM i software tier that is displayed in the **IBM i software tier** field or other options from the list.

To deploy an SAP workload, complete the following steps:

- a. Select a **Machine type** from the list.
 - To deploy an SAP HANA profile, you can select a machine type from the list.
 - To deploy an SAP certified profile, you must select an IBM Power10 or later machine type from the list.
- b. Select a profile to deploy an SAP HANA profile.
- c. Select a profile from the **Standard RISE** or **Application Server** tab to deploy an SAP certified profile. The **Standard RISE** and **Application Server** tabs are enabled only when **SAP RISE deployment** is set to on in the **Advanced Configurations** section.

The **SAP RISE deployment** option is enabled when you select **Linux for SAP (HANA)** from the Operating system list and IBM Power10 or later from the Machine type list.

10. Optional: Expand **Advanced Configurations** to configure more settings for your VSI.

- **Specify preferred processor compatibility mode**: By default, this option is set to off. To use a specific processor compatibility mode, set **Specify preferred processor compatibility mode** to on, and then select the processor compatibility mode from the **Preferred processor compatibility mode** list.

The "Virtual server instance details" page of a deployed VSI displays the preferred and effective processor compatibility modes that are set for a VSI.

The preferred processor compatibility mode is the processor mode in which you want the VSI to operate. By default, Power Virtual Server sets the preferred processor compatibility mode to the highest mode that is supported by the targeted host type for the VSI.

The effective processor compatibility mode is the processor mode that is in use for the VSI. The physical host in which the VSI runs determines the effective processor compatibility mode.



Note: The effective processor compatibility mode for the VSI might not match the preferred mode that is selected. If the operating system installed in the VSI does not support the preferred processor compatibility mode, the hypervisor can set the effective mode to a lesser mode than the preferred mode. However, the hypervisor cannot set the effective mode to a higher mode than the preferred mode. For more information about processor compatibility modes, see [How does the processor compatibility mode work in a VSI?](#).

- **Automated remote restart**: Automated remote restart is enabled by default for all VSIs in the Power Virtual Server environment. This feature automatically restarts your VSI on another available host if the current host fails unexpectedly. To disable this feature, set **Automated remote restart** to off during VSI creation. Alternatively, you can modify the settings on the "Virtual server instance details" page. For more information, see [Disabling automated remote restart for a VSI](#).

Automated remote restart does not restart pinned VSIs. Pinning VSIs to specific hosts results in extended downtime because the recovery depends on the time that is taken to repair the failed host. To minimize downtime, ensure that VSIs are not pinned to a host and are enabled for automated remote restart. For more information about VSI pinning, see [Virtual server pinning and its impacts on VSI availability](#).

Automated remote restart also does not restart a VSI in a server placement group that uses an affinity policy and includes other VSIs that are hard-pinned to the host. Affinity policies require all VSIs in the group to remain on the same host. A hard-pinned VSI prevents the VSIs in the

group from moving to another host.

 **high availability** (HA) and **disaster recovery** (DR) policies are configured to protect the workload after a host failure. For more information, see [High availability and disaster recovery](#).

11. Click **Continue**.

12. Complete the **Storage volumes** fields to attach or create new volumes and associate them with the VSI.

Expand **Advanced Configurations**, and then set **Configure for large quantity volumes** to **Enabled** to support more than 127 (up to 500) volumes. This setting is at a VSI-level that remains unmodifiable upon provisioning.

 **Note:** You cannot create or attach volumes larger than 2047 GB on IBM i-based VSIs. However, machine types E890 and E1080 are optimized to support the attachment of a higher number of volumes on IBM i-based VSIs. For more information, see [Configuring for large quantity of volumes](#).

13. Define your **Network interfaces** by adding a public network, private network, or both. Public networks are not available at all IBM data center locations. When you add an existing private network, you can choose a specific IP address or have one auto-assigned.

 **Important:** When you choose to provide a specific IP address, ensure that the IP address is not listed under [reserved IP](#).

 **Note:** For an AIX VSI, network interface controllers (NICs) are assigned based on the order in which you specify them during creation. To display the information about all the network interfaces after provisioning, open the AIX console and type `ifconfig -a`.

14. Accept the **Terms of Use** and click **Create instance** to provision a new Power Virtual Server. To view your boot images, go to **Boot images** after you provision the instance.

About VSN in IBM data center

You can assign a VSN to a VSI.

 **Note:** Assigning a VSN is not supported on a VSI with IBM i version 7.1 or earlier.

VSN has the following characteristics:

- It is a unique seven-digit identifier.
- It is used for licensing and tracking the usage of the VSI.
- It can be associated with only one VSI.
- It can be assigned to a new or an existing VSI.
- It can be assigned to an empty VSI in **IBM data center**.

For more information, see [Assigning the virtual serial number to a logical partition](#).

A VSI moves across systems with its associated VSN. Therefore, when a VSN is associated with a VSI, you do not need to pin the VSI to a host for licensing or entitlement purposes.

VSN is a unique identifier and can be assigned only to one VSI at a time. If you deploy more than one VSI assigning the same VSN, only one VSI receives the VSN, and the remaining VSIs are deployed without a VSN.

You can view the details of a VSN that is associated with a VSI on the "Virtual server instance details" page. You can also view the details of the VSNs associated with the VSIs for the workspace on the "Virtual serial numbers" page. The VSNs are either in **assigned** or in **retained** state.

 **Important:** When you upgrade an IBM i VSI that has a VSN assigned, you must contact IBM Support to update the IBM i licenses to match the new version.

Mapping requirement for VSNs

Before you can use VSNs in Power Virtual Server, you must complete a one-time mapping of your IBM customer number to your IBM Cloud account ID. After you complete the mapping, you can use VSNs with any VSI that you deploy in that account. For more information, see [IBM Cloud Power Virtual Server - using virtual serial numbers and IBM customer numbers](#).

Assigning a VSN to a new VSI

You can assign a VSN only to a VSI with IBM i OS. To assign a VSN when you create a VSI, select **IBM i** as the operating system. The **Virtual serial number** field is enabled. The default VSN value is **None**.

Edit the default VSN value and select one of the following options:

- **Auto-assign**: Assigns a system-generated VSN to your VSI only if your `IBM Cloud account ID` is mapped with your `customer number` in the Entitled System Support (ESS).
- **Select from retained VSNs**: Displays a list of the retained VSNs. You can select a VSN in the `retained` state and assign it to your IBM i VSI.

For more information about creating a VSI, see [Configuring a Power Virtual Server instance](#).

Assigning a VSN to an existing VSI

To assign a VSN to an existing IBM i VSI, complete the following steps:

1. Shut down the IBM i VSI that you plan to assign the VSN.
2. Open the VSI to access the "Virtual server instance details" page.
3. Edit the **Virtual serial number** field.
4. From the **VSN assignment** list, select one of the following options:
 - **None**: Does not assign a VSN to the VSI.
 - **Auto-assign**: Assigns a system-generated VSN to the VSI.
 - **Select from retained VSNs**: Displays the list of retained VSNs that can be selected.
5. Click **Save**.
6. Power on the VSI.

Releasing or retaining a VSN

When you delete a VSI, you can either retain the VSN or release it. When you edit the VSI details to change the VSN, you can retain the existing VSN or release it. To release or retain a VSN, complete the following steps:

1. Click the delete icon from the "Virtual server instance details" page. The "Delete virtual server instance" window is displayed.
2. Edit the details of a VSI by clicking the overflow menu (three vertical dots) on the far right of the VSI entry from the "Virtual server instance details" page. The "Edit virtual server instance" window is displayed.
3. Set **Release the VSN attached to the VM** to **Enable** or **Disable** on the page to release or retain the VSN:
 - **Enable**: (Default) Deletes the VSI and attaches the VSN to the VSI that is released and no longer associated with your account. By default, the **Release the VSN attached to the VM** is enabled.
 - **Disable**: Deletes only the VSI and retains the VSN that continues to be associated with your account. The retained VSN is moved to the retained VSN pool.

The following table provides more information about each Power Virtual Server instance field.

Field	Description
-------	-------------

General	Instance name: Specify a name for your VSI. Number of Instances: Specify the number of instances that you want to create for the Power Virtual Server. You can apply placement groups only when you are creating a single VSI. If you choose the Machine type as E980, you can choose an anti-affinity policy with a maximum of 2 VSIs. Placement group: If you are creating only one instance, you can choose the placement group to control the selection of the host to host the instance. Select a placement group from the list. To create a new placement group, select one of the following options: Same server : Select this option to place the VSI on the same host. Different servers : Select this option to place the VSI on a different host. Colocation rules: If you specify more than one instance, you can select the following naming conventions and colocation rules: No preference: Select this option if you do not have a hosting preference. Same server: Select this option to host all instances on the same server. A placement group is automatically created. The instance name that is previously provided is used as the group name and cannot be edited. Different server: Select this option to host each instance on a different server. You can use this option if you are concerned about a single-server outage that might affect all Power Virtual Server instances. A placement group is automatically created. The instance name that is previously provided is used as the group name and cannot be edited. Numerical prefix: Select this option to add numbers before the name of the virtual server. If, for example, the first Power Virtual Server name is <i>Austin</i> the next name for the virtual instance is <i>1Austin</i>. Numerical postfix: Select this option to add numbers after the name of the virtual server. If, for example, the first Power Virtual Server name is <i>Austin</i> the next name for the virtual instance is <i>Austin 1</i>. Virtual server pinning: Select this option to pin your VSI. You can choose either a Hard or Soft pinning policy. Learn more. Note: When you create multiple instances of the virtual server, you must select On from the Shareable field for each data volume that you add. If you do not want the data volume to be shareable, you can add the data volume after you create the virtual server. For IBM i OS, you cannot have shareable data volumes.
Machine type	Specify the machine type. The machine type that you select determines the number of cores and memory that is available. For more information about hardware specifications, see S922 and E980 (Data centers other than Dallas and Washington) .
Cores	The core-to-virtual core ratio is 1:1. For shared processors, fractional cores round up to the nearest whole number. For example, 1.25 cores are equal to 2 virtual cores.
Memory	Select the amount of memory for the Power Virtual Server. If you choose to use more than 64 GBs of memory per core, a higher price is charged. For example, when you choose one core with 128 GBs of memory, the regular price for the first 64 GBs is charged. After the first 64 GBs (64 - 128 GBs), a higher price is charged.
Boot image	Select a version of the IBM-provided AIX or IBM i operating system stock image. You can also select Linux stock images for SAP HANA and SAP NetWeaver applications. For the SAP stock images, you must set an SSH key when you create the VSI. You can access the VSI only through SSH after provisioning. You can also set a password by using the <code>passwd</code> command during the first SSH access. By setting a password, you are able to access the instance in the UI console. You can also deploy your own custom image of AIX, IBM i, or Linux. IBM also provides a community-supported CentOS image under the Linux operating system. However, IBM support is not available for this image. For CentOS support, see the CentOS forum or FAQ page. Power Virtual Server now supports Linux (RHEL and SLES) stock images for non-SAP applications. To provision Power Virtual Server instance that supports SAP HANA and SAP NetWeaver applications, see Provisioning your IBM® Power® Virtual Server. Important: When you use an AIX stock image as the boot volume, a console session is required for the initial setting of the root user password. Without completing this step, SSH login as root appears as being <i>disabled</i>. For IBM i operating system licensing information, see IBM i License Program Products (LPP) and Operating System (OS) feature bundles.
Attached volumes	You can either create a new data volume or attach an existing one that you defined in your account. Create volume: Click Create volume to create a new data volume for your Power Virtual Server instance. If you want to allow multiple virtual instances to write data to the same data volume, you must click On under Shareable. Attached Volume: You can select an existing data volume from the Attached volumes list. If a previously used data volume does not appear, it might exist under a different account or resource instance.
Public Networks	Select this option to use an IBM-provided public network. Cost is associated when you select this option. Public networks are not available at all IBM data center locations. Learn more
Private Networks	Click Add to identify a new private network for the virtual server. If you already added a private network, you can select it from the list. For more information, see Configure a private network subnet.

Power Virtual Server instance fields

Virtual server pinning and its impacts on VSI availability

VSIs can be optionally pinned to a host. Pin VSIs only when necessary, because pinning negatively affects VSI availability during planned maintenance activities. VSI pinning restricts the movement of VSIs during the maintenance operations, host failures, and other restart events.

Virtual server pinning policies

The behavior of the VSI depends on the following **Virtual server pinning** policies that you select:

- **Soft:** If you select **Soft** pinning, Power Virtual Server automatically migrates the VSI back to the original host after the host returns to its operating state by using live partition migration (LPM).
- **Hard:** If you select **Hard** pinning, the movement of the VSI from the host does not occur during compute host failures and maintenance activities. When you use Hard pinning, consider protecting your workload by using a separate *High availability* (HA) and *Disaster recovery* (DR) solution.



Important: When you change the pin policy of a VSI from **Hard** to **Soft** or disable the pin policy, consider enabling **Automated remote restart** on the VSI. The automated remote restart setting restarts the VSI on another available host if an IBM Power host fails.

For more information about VSI pinning, see [What does VSI pinning do?](#).

Impacts of pinning a VSI

Enabling **Virtual server pinning** directly affects the VSI availability posture. The impacts of pinning a VSI are detailed in the following events:

- [Maintenance of Power Virtual Server](#)
- [Unplanned host failures](#)

Use the following table to determine when the VSI incurs downtime during planned maintenance and host failure events based on the type of pinning policy that is selected for the VSI.

Virtual server pinning policies	VSI downtime during planned maintenance	VSI downtime during host failure
Soft	Yes	No
Hard	Yes	Yes
None	No	No

VSI availability posture during planned maintenance or host failure events

Maintenance of Power Virtual Server

IBM Power Virtual Server operations team performs planned maintenance activities based on the requirements of the IBM Power Virtual Server infrastructure. During planned maintenance, the operations team cannot use the LPM feature to move VSIs that are set to a pinning policy.

To perform a planned maintenance activity, the IBM Power Virtual Server operations team coordinates with the VSI owner to temporarily remove pinning and perform an LPM migration without downtime. The owner of the VSI must approve the temporary removal of VSI pinning. If the owner of the VSI does not approve, the VSI is shut down until the maintenance operation is completed. After the maintenance operation is completed, the owner must restart the VSI.

Unplanned host failures

When a host fails, the selected pinning policy determines the recovery process of the VSIs that were previously operating on the failed host:

- If the VSI is set to **Soft**, it automatically restarts on the available compute resources.
- If the VSI is set to **Hard**, it cannot be automatically restarted on the available compute resources. Such VSIs remain unavailable until the IBM Power Virtual Server operations team resolves the issues that are related to host failure. Based on the type of the issue, the IBM Power Virtual Server operations team might require extra time to resolve the issue.



Note: Considering the impacts of pinning a VSI, pin a VSI only if it is necessary. You can use IBM i VSNs to retain the same serial number throughout the lifecycle of a VSI independent of the compute host on which the VSI runs. For more information, contact your independent software vendor (ISV).

Reusing volume names or VSI names in Power Virtual Server

To deploy a Power Virtual Server VSI, specify any name. If you delete a VSI and want to create a new VSI with the same name, wait at least an hour after

the original VSI is deleted before you use the same name.

For example, you create a VSI with the name TEST-VSI and you delete this VSI later. The name `TEST-VSI` is not immediately available for reuse. Before you attempt to reuse the name TEST-VSI, wait at least an hour after the VSI is deleted.

Implementing SAP NetWeaver and SAP HANA in the Power Virtual Server environment

You can deploy SAP NetWeaver on an AIX or Linux® operating system, and SAP HANA on a Linux operating system, in your Power Virtual Server environment. You must consider several SAP-specific infrastructure requirements to run SAP applications on Power Virtual Server instances. For more information, see [Planning your deployment](#) and [Deploying your infrastructure](#).

Consider an IBM Power server E980 that is running in a multiple VSI environment with at least one SAP HANA production system. You can deploy up to sixteen VSIs per physical server with dedicated or dedicated-donating processor cores. Each concurrently running VSI must be configured according to the workload and must fulfill the SAP HANA Hardware Configuration Check Tool (HWCCT) key performance indicators (KPIs). You must also consider the minimum number of CPU cores and memory of VSIs as described in SAP Note 2188482. For more information, see [SAP support Launchpad](#). You must have an SAP ID to access this web page.

Configuring a VSI for Epic workloads

You can configure your VSI to deploy Epic workloads when you select AIX as your operating system.

To configure a VSI for Epic workloads, select the **Configure for Epic workloads** checkbox on the **Boot image** tile. You can verify whether the deployed VSI supports Epic workloads by checking the corresponding "Virtual server instance details" page. On the "Virtual server instance details" page, the **Deployment type** field must be set to **Epic**.



Important: In the "Virtual server instance details" page, for the VSIs on which Epic workloads are supported, you must not create or attach volumes from Tier 3 to avoid performance issues. For the VSIs on which Epic workloads are supported and are in a shut-down state, you must not change the core type to any value other than `dedicated` to avoid performance issues.

The following table describes the VSI configuration differences between workloads that support Epic and those that do not:

VSI deployed for	Storage volume	Core type	Machine type
Non-Epic workloads	Tier 1 or Tier 3	Shared uncapped, shared capped, or dedicated	S922 or E980
Epic workloads	Always Tier 1	Always dedicated	E980 or E1080

VSI configuration difference that supports non-Epic and Epic workloads



Note: By default, Epic VSIs are not pinned. Unpinned Epic VSIs can be used for nonproduction workloads. For production Epic VSIs, enable pinning to avoid performance issues.

You can choose to configure a VSI for Epic workloads only when you select AIX as your operating system. The other conditions that apply are as follows:

1. Epic workloads are supported on AIX 7.2 and later. You cannot choose AIX 7.1.
2. The supported storage volume tier is Tier 1. You can change or attach a Tier 3 storage volume. However, changing the tier might lead to performance issues.
3. Supported machine types are E980 or E1080. You cannot select S922.
4. Supported core type is dedicated. You can switch to another core type, but doing so might lead to performance issues.

Configuring affinity policies

You can use the user interface to set the affinity policies for storage pools only when the total number of VSIs in your account is less than 100. If your account has more than 100 VSIs, you must use the CLI or API to set the volume affinity policies.

Select one of the following Storage pool options:

- **Auto-select pool:** Use this option to allow the system to automatically select a storage pool for the storage tier with sufficient capacity.
- **Affinity:** Use this option to identify the storage pool that must be used to place the boot volumes based on an existing VSI or storage volume from your account. The new storage volumes for the VSI are placed in the same storage pool where the affinity object resides. If you are using a PVM instance as the affinity object, the storage pool that is selected is based on the root (boot) volume of the PVM instance.

- **Anti-affinity:** Use this option to identify one or more storage pools that you want to exclude from getting selected to place the boot volumes. The boot volumes are placed based on one or more existing VSIs or storage volumes from your account. When you select a storage pool to create the custom image storage volumes, the storage pools in which the list of anti-affinity objects reside are not selected. If you are using VSIs as anti-affinity objects, the storage pools are excluded based on the root (boot) volume of each PVM instance that you specify.

To learn more about the flexible tier offering of Power Virtual Server, see [Storage tiers](#).

For more information about affinity and anti-affinity policy, see [What does it mean to set an affinity or anti-affinity rule?](#).



Note: If you add volumes to be created and attached to your new VSI during creation, all the volumes are provisioned in the same selected storage pool. Volumes can be created in different storage pools after the VSI is provisioned.

Provisioning a virtual machine without an initial boot volume

You can create and deploy a VSI without an initial boot volume.

The VSIs without a boot volume can be used for cloning operations. These VSIs are not bootable until a boot volume is attached after provisioning. The following table shows which images are deployed based on your OS selection:



Note: When you attach the boot volume after provisioning of the VSI, the boot image still shows the OS-specific image without the boot volume name.

OS selected	Image deployed
AIX	AIX (Empty image) image without boot volume
IBM i	IBM i (Empty image) image without boot volume
Linux	You must select one of the following images: • Linux - SUSE (Empty image) • Linux - RedHat Enterprise (Empty image) image without boot volume
Linux for SAP (HANA)	Provision of VSI without boot volume is not supported
Linux for SAP (NetWeaver)	Provision of VSI without boot volume is not supported
Client supplied subscriptions	Provision of VSI without boot volume is not supported for these OSs

Provision of VSIs without boot volume based on the OS selection.

When you select the **Deploy empty virtual server instance** checkbox, you can provision a VSI without a boot image and boot volume. Review the following table to understand how the selection of the **Deploy empty virtual server instance** checkbox works along with the provisioning of the large quantity of data volumes:

Features	Deploy empty virtual server instance checkbox is clear	Deploy empty virtual server instance checkbox is selected
Boot image and volume	Provision a VSI with a boot image and boot volume	Provision a VSI without a boot image and boot volume.
Creating new volume during VSI provisioning	Create up to 10 volumes from the VSI provisioning page. To create volumes in bulk, use the Storage volumes page in the Power Virtual Server user interface.	You cannot create a volume and attach it to the VSI during initial provisioning. You can create up to 10 volumes after provisioning.
Attaching existing volume during VSI provisioning	Attach up to 500 existing data volumes	You cannot attach any volumes to the VSI during initial provisioning. You can attach one boot volume and up to 500 data volumes after provisioning.

Attaching from multiple storage tiers	Supported. However, using multiple storage tier volumes increases the risk of a failed clone operation.	Supported. However, using multiple storage tier volumes increases the risk of a failed clone operation.
Boot volume	Boot volume is attached while provisioning. Click three dots on any data volume and set it as the boot volume. However, you cannot set shareable volumes as boot volumes.	Boot volume is attached after provisioning. Click three dots on any data volume and set it as the boot volume. However, you cannot set shareable volumes as boot volumes.

Provisioning a VSI with or without a boot volume.

Configuring large quantity of data volumes on IBM data center

IBM data center

While provisioning, you can configure your VSI to enable it to attach or detach more than 127 (up to 500) data volumes from the user interface.



Note: IBM i virtual machines in all data centers except **CHE01** support the configuration of a large quantity of data volumes. Configuring the large quantity of data volumes is supported only on IBM data center.

Limitations of large quantity volumes

Review the following limitations when you configure a large quantity of volumes:

- Complete the operations, such as deploy, attach, detach, or delete, in a sequential order to avoid any performance delays.
- Use the image catalog to capture the virtual machines with large quantity volumes. Using Cloud Object Storage (COS) or any other cloud option might result in delays. The delay depends on the volume size and network speed.
- When you attach a large quantity of volumes in a single request, displaying the status value from **available** to **attaching** might be delayed. Wait for the **attach** operation to complete before you select the attached volumes for other operations.
- Provisioning an IBM i VSI with small data volumes (even fewer than 10 volumes in some cases) can cause a 3-5 hour delay if bulk-volume operations are ongoing on the storage controller. During this delay, the VSI remains in the **building** state and cannot be modified.

Attaching virtual persistent memory in a Power Virtual Server instance

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

The virtual persistent memory (vPMEM) is nonvolatile memory that retains data even when the system is powered off. vPMEM is an enhancement to the IBM Power advanced virtualization platform (PowerVM) that enables the system to configure the persistent volumes by using the existing DRAM (dynamic RAM) technology. The vPMEM feature is available on IBM Power10 or later systems with IBM AIX®, Linux®, or Linux (SAP HANA) operating systems.



Note: vPMEM is not supported on IBM i-based virtual server instances (VSIs).

In the [IBM Cloud Resource List](#), you can identify your vPMEM volume under **Compute** resources. If you add a vPMEM volume when you create a VSI and add user tags to the VSI, the tag is also added to the vPMEM volume. If you add a vPMEM volume to an existing VSI, you can add user tags only to the vPMEM volume.

Before you begin

Before you create and add the vPMEM volumes to a VSI, review the following information:

- When you add vPMEM volumes to an AIX or Linux-based VSI, the vPMEM volumes use memory in addition to the base memory.
- Review the following considerations before adding a vPMEM volume to an SAP-certified VSI in an IBM data center:
 - The selected SAP-certified profile must support vPMEM volumes.
 - The VSI profile memory is the total available memory and includes the vPMEM volume.
 - You cannot resize the VSI after it is provisioned with an SAP-certified profile.



Note: You can deploy VSIs by using SAP-certified profiles only in IBM data centers. SAP-certified profiles are not supported for VSIs that are

deployed in IBM Power Virtual Server Private Cloud environments.



Important: To add a vPMEM volume to an existing VSI or to delete a vPMEM volume from the VSI, the VSI must be in the **Shutoff** state.

Adding a vPMEM volume during VSI provisioning

To create and add a vPMEM volume to a VSI when you provision a VSI in a Power10 or later system, update the following fields on the Create virtual server instance page.

Field	Action
Operating system	Select AIX®, Linux®, or Linux for SAP (HANA).
Image	Select an image from the list.
Machine type	Select Power10 or later system.
Advance Configuration	1. Set Virtual persistent memory volume to on. 2. Click Create volume to create a vPMEM volume. The Create virtual persistent memory volume pane appears. 3. Update the fields that are defined in the Creating a vPMEM volume topic.

Fields and the corresponding actions to create a VSI with vPMEM volume

Adding a vPMEM volume to a VSI

On the Virtual server instances page, complete the following steps:

1. Click the VSI to which you want to add the vPMEM volume.

2. Complete the following steps on the **Overview** tab of the VSI page:

a. Click **Create volume** under the Virtual persistent memory volumes section. The Create virtual persistent memory volume pane appears.

b. Update the fields to create a vPMEM volume. For more information, see [Creating a vPMEM volume](#).

Creating a vPMEM volume

Update the following fields to create a vPMEM volume in the Create virtual persistent memory volume pane.

Field	Action
Name	Specify a name for the vPMEM volume.
User tags	Optional. Enter user tags.
Size	Increment or decrement the size of the vPMEM volume. The values are calculated in gibibyte (GiB).
Total estimated cost	Review the total estimated cost of your vPMEM volume.
Create	Create the vPMEM volume. The Create field is enabled only if you agree to the estimated cost by selecting the I agree to the Terms and conditions checkbox.

Fields and the corresponding actions to create a vPMEM volume



Note: You cannot resize the vPMEM volume after you create the volume.

Creating and adding a vPMEM volume by using API and CLI commands

Use the following API and CLI commands to create and add a vPMEM volume to a VSI:

- API: Use the [Create a vPMEM volume to be attached to this PVM Instance](#) API to create a vPMEM volume for a VSI with AIX®, Linux®, or Linux (SAP HANA) OS.
- CLI: Use the [ibmcloud pi instance create](#) command to create a vPMEM volume. You can use [ibmcloud pi instance vpmem-volume](#) to perform the `attach`, `detach`, `get`, or `list` operations.

Deleting a vPMEM volume



Note: You can delete a vPMEM volume from a VSI only if the VSI is in the **Shutoff** state.

To delete a vPMEM volume, complete the following steps:

1. Click the VSI from which you want to delete the vPMEM volume on the Virtual server instances page.
2. Click the delete icon that is associated with the vPMEM volume to be deleted from the Virtual persistent memory volumes section.

If the vPMEM volume is associated with the VSI with AIX or Linux OS, the vPMEM volume memory is reduced from the total memory and the total cost is reduced.

If the vPMEM volume is associated with the VSI with Linux for SAP (HANA) OS, the vPMEM volume memory is returned to the VSI. The total cost does not change.

Deleting a vPMEM volume by using API and CLI commands

Use the following API and CLI commands to delete a vPMEM volume:

- API: Use the [Delete a vPMEM volume attached to this PVM Instance](#) API command to delete a vPMEM volume.
- CLI: Use the [ibmcloud pi instance delete](#) to delete a vPMEM volume.

Pricing for vPMEM volumes

The cost of a vPMEM volume is calculated based on the OS type of the VSI. Refer to the following examples to calculate the price for a vPMEM volume that is associated with a VSI with Linux for SAP (HANA), AIX®, or Linux® operating systems:

- For a VSI with the SAP (HANA) profile, the memory for vPMEM volume is allocated from the total memory of the SAP (HANA) profile. For example, consider a VSI with `sh2-25x1000` SAP profile that has 1000 GB memory. You add a vPMEM volume of 200 GB to the VSI. The 200 GB memory is allocated to the vPMEM volume from the `sh2-25x1000` profile size and 800 GB memory is retained with the VSI. So, the VSI with 800 GB memory has a vPMEM volume of 200 GB attached to it. No additional cost is incurred. When you delete the vPMEM volume, the memory that is used from the total profile size is returned and the cost remains the same.
- For a VSI with AIX or Linux OS, the total cost is calculated based on the total memory usage. For example, consider a VSI with 1000 GB memory. When you add a vPMEM volume of 200 GB to the VSI, the total memory changes to 1200 GB so the cost is calculated for 1200 GB memory. When you delete the vPMEM volume, the memory is reduced from the total memory and the total cost reduces.

For more information about the part numbers for vPMEM volumes, see [Pricing for IBM Power Virtual Server in IBM data centers](#).

Limitations of vPMEM

vPMEM has the following limitations:

- You can create and add vPMEM volume to a VSI only on IBM Power10 or later systems with AIX®, Linux®, or Linux (SAP HANA) operating systems.
- You can add or detach a vPMEM volume to a VSI only when the VSI is shut down.
- You can add one vPMEM volume to a VSI with SAP (HANA) profile and add four vPMEM volumes to a VSI with AIX or Linux operating system.
- You cannot add a vPMEM volume to a VSI with the IBM i operating system.
- You cannot resize a VSI with SAP (HANA) profile if a vPMEM volume is added to the VSI.
- You cannot edit the size of a vPMEM volume.
- You cannot enable global replication services (GRS) on vPMEM volumes.

Getting started with dedicated hosts on IBM Power Virtual Server

A dedicated host on IBM® Power® Virtual Server in IBM data center is a virtualized, single-tenant, dedicated server. It provides you with maximum control over workload placement and flexible post-provisioning options. A dedicated server that hosts from Power Virtual Server is designed to provide total isolation, control, and security for your heavy mission-critical workloads.

Using the dedicated host capability on Power Virtual Server in IBM data center you can provision an entire server for your sole use, significantly expanding your computing options. You can handle mission-critical workloads with complete isolation, control, and security.

Using a dedicated host provides extra flexibility in two ways:

- It allows you to create virtual server instances and control their placement, and
- It offers a unique shared processor pool capability.

A dedicated host on a Power Virtual Server enables you to optimize your cloud infrastructure by using single-tenant servers to manage software licensing costs. Most importantly, dedicated hosts enhance isolation from other users in a cloud environment to ensure that your operations run smoothly and securely.

Dedicated hosts on a Power Virtual Server provide the following features:

- Ability to reserve a compatible server for exclusive use.
- Flexibility to provision virtual server instances onto a dedicated host.
- Flexibility to manage resource utilization, including an increased Virtual Processor to Entitled Capacity ratio of up to 20:1.
- Detailed user interface view of host capacity information such as: core, memory capacity, and consumption.
- Ability to share dedicated hosts with all or a subset of workspaces on the same account, and the flexibility to control access to dedicated hosts across your organization.
- Ability to set custom names for dedicated hosts and dedicated host groups.

Pricing for dedicated hosts

The hourly billing for a dedicated host on a Power Virtual Server covers the entire server.

You can create a cost estimate of a dedicated host before you deploy it by using the [cost estimator](#) or see the [pricing](#) page to learn more.

Primary workspace for a dedicated host

A Power Virtual Server primary workspace is the owning tenant of a dedicated host with full control over the associated resources.

A primary workspace can delete or share the dedicated resources that it owns under the IBM Cloud account. Primary workspace users can control sharing permissions for dedicated resources with all or a subset of their workspaces within the same account. You can use a primary workspace to reserve a dedicated host.

Billing for dedicated hosts is charged to the primary workspace.

Secondary workspace for a dedicated host

A Power Virtual Server secondary workspace is the one with which the dedicated resources are shared.

A secondary workspace can deploy virtual server instances on the shared dedicated resources. You can use a secondary workspace to grant access to the resources in the dedicated host. However, a secondary workspace cannot further share a resource that it doesn't own with another workspace.

Access and authorization for dedicated hosts

You can set up authorization for usage of a Power Virtual Server dedicated host for other workspaces by using IBM Cloud IAM Service Authorization capabilities.

As an administrator of your organization, you can assign users the following supported roles:

Role	Description
Reader	View the workspaces that a dedicated host is shared with.
Manager	Perform create, read, update, or delete actions on dedicated hosts, including: • Reserve and delete dedicated hosts • Share and unshare with other workspaces • Manage any other aspects of dedicated hosts.

Roles and permissions in a dedicated host on a Power Virtual Server

Creating a dedicated host group

A dedicated host group in your IBM Cloud account is where all your reserved dedicated hosts are contained.

The workspace from where you initiate the creation of a new dedicated host group becomes the primary workspace. You can select active workspaces in your IBM Cloud account to share making them as secondary workspaces.

To create a dedicated host group, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Select a workspace.
3. Click **Dedicated hosts** on the left navigation menu. Here, you can see all the existing dedicated hosts.
4. Click **Reserve host**.
5. Click **Create new**.
6. Enter a custom name on the **Dedicated host group name** field.
7. (Optional) Grant secondary access to the workspaces to share the dedicated host group.



Note: Only workspaces in the same region as the primary workspace and in an active state can be selected.

8. Click **Save**.



Important: You can click a dedicated host group to access its details, create more dedicated hosts, share, or stop sharing with secondary workspaces.

Sharing or unsharing dedicated host groups among workspaces

You can choose to grant workspaces with secondary access when you [create a dedicated host group](#) and from the dedicated host group details page. You can stop sharing them from the dedicated host group details page.

To share workspaces, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Select a workspace.
3. Click **Dedicated hosts** on the left navigation menu. Here, you can see all existing dedicated hosts.
4. Click a dedicated host group to open the details page.
5. Click **Share** and select the workspaces that you want to share with.



Note: Only workspaces in the same region as the primary workspace and in an active state can be selected.

6. Click **Save**.

To stop sharing dedicated host access with a secondary workspace, ensure that the selected workspace does not have any provisioned resources on the dedicated host.

Reserving a dedicated host

You can allocate an entire dedicated host exclusively to a single tenant by reserving the host.

To reserve a dedicated host in Power Virtual Server, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Select a workspace.
3. Click **Dedicated hosts** on the left navigation menu. Here, you can see all existing dedicated hosts.
4. Click **Reserve host**.
5. Select a dedicated host group from the available list of host groups already created.
6. Click **Create new** if no dedicated host group is available.
7. Enter your **Host name** and select the **Machine type** from the drop-down list.



Note: Dedicated hosts cannot be reserved in host groups where the current workspace has secondary access.

8. Click **Finish** and view the summary of estimated cost.
9. Click **Create**.



Important: You can click into a dedicated host to access its details, create and manage virtual server instances, created shared processor pools, or release the host.

Deploying a virtual server in a dedicated host

You can create a virtual server instance in a single-tenant environment on a dedicated host. The virtual server instance that is provisioned on a dedicated host can be deployed with any value up to a 20:1 ratio of Virtual Processor (VP) to Entitled Capacity (EC).

To deploy a virtual server in a dedicated host, complete the following steps:

1. Click a dedicated host to create a new virtual server instance.
2. Set **Deploy to dedicated host** to ON.
3. Follow the instructions in [Configuring a Power Virtual Server instance](#) page for detailed instructions.



Note: Virtual server instances that are deployed on the dedicated host are not billed for core and memory charges.

Creating SPP in a dedicated host

A shared processor pool (SPP) is a pool of processor capacity that is shared between a group of virtual server instances (VSIs). You can create an SPP in a single-tenant environment on a Power Virtual Server dedicated host. The SPPs provisioned on a dedicated host can be deployed with any value up to a 20:1 ratio of Virtual Processor (VP) to Entitled Capacity (EC).

To create an SPP in a dedicated host, complete the following steps:

1. Click a dedicated host to create a new shared processor pool.
2. Set **Deploy to dedicated host** to ON.
3. Follow the instructions in the [Managing shared processor pool](#) page for detailed instructions.



Note: When you use a dedicated host to deploy SPPs, the SPPs are not billed for the reserved capacity.

Releasing a dedicated host

To release a dedicated host is to delete it from your account. You must delete all the associated resources that are provisioned on the host before you release it.

You can release a dedicated host when no resources are deployed on the host.

To release a dedicated host, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Select a workspace.
3. Click **Dedicated hosts** on the left navigation menu. Here, you can see all existing dedicated hosts.
4. Select the dedicated host that you want to release. You must remove all associated resources that are provisioned on the host before you release it.
5. Click **Release host**.



Note: When you release the last dedicated host in a dedicated host group, the group is also deleted. Empty dedicated host groups are not allowed on a workspace.

Maintenance in dedicated host

When IBM needs to perform maintenance on a dedicated host, its virtual server instances (VSIs) are live migrated to another dedicated host during the maintenance operation window.

If a dedicated host encounters a server failure, IBM restarts the affected VSIs on another dedicated host, subject to the available capacity. This VSI restart capability does not protect against data center, storage, or network failures. For such scenarios, you can implement appropriate business continuity measures, such as high availability (HA) and disaster recovery (DR) solutions.



Note: When VSIs are relocated to another dedicated host for maintenance, you are not charged for an extra dedicated host.

Using the virtual host identifier for dedicated hosts

You can use the virtual host identifier (ID) to get information about your Power Virtual Server dedicated hosts. The virtual host ID follows a standard format to uniquely identify other Power Virtual Server resources.



Important: The Power Virtual Server dedicated host identifier format is updated to use the virtual host ID format. If you are using dedicated hosts and depend on specific resource identifiers such as API and CLI calls, you must update the logic to use the virtual host ID format. Otherwise, the logic might fail.

To map a dedicated host with VSIs or SPP resources, you must use the values of the `host` attribute of the VSI or the `hostID` attribute of the SPP in the `hostReference` attribute of the dedicated host.

To fetch the virtual host ID of a dedicated host, use the following API or CLI commands:

- API command – [get the list of all hosts](#) with the `?host_reference=true` query parameter.
- CLI command – [get the host information](#) with the `--json` option.

The following code examples provide the old format and new format of the virtual host ID after the format is updated to the standard format:

Example 1: /v1/hosts/{host_id} format

Old format

```
$ [
  {
    "capacity": {
      "cores": {
        "available": 14.75,
        "reserved": 5.25,
        "total": 20,
        "used": 0
      },
    },
    "memory": {
      "available": 930.83,
      "reserved": 93.17,
      "total": 1024,
      "used": 0
    }
  },
  "displayName": "myHost3",
  "hostGroup": {
    "access": "primary",
    "href": "/v1/host-groups/b1d71c2c-54fd-49d7-b061-44b663106fec",
    "name": "Fred_HG"
  },
  "id": 123,
  "state": "up",
  "status": "enabled",
  "sysType": "s922"
}
```

New format

```
$ [
  {
    "capacity": {
      "cores": {
        "available": 14.75,
```



```
      "reserved": 5.25,
      "total": 20,
      "used": 0
    },
    "memory": {
      "available": 930.83,
      "reserved": 93.17,
      "total": 1024,
      "used": 0
    }
  },
  "displayName": "myHost3",
  "hostGroup": {
    "access": "primary",
    "href": "/v1/host-groups/b1d71c2c-54fd-49d7-b061-44b663106fec",
    "name": "Fred_HG"
  },
  "id": "d750956e-e22c-4e4d-b0d1-a126c16d1cd0",
  "state": "up",
  "status": "enabled",
  "sysType": "s922"
}
]
```

Example 2: /v1/host-groups/{host_group_id} format

Old format

```
$ [
  {
    "creationDate": "2024-11-12T16:29:35.000Z",
    "hosts": [
      "/v1/hosts/123"
    ],
    "id": "d50dd63a-dfda-46a9-854f-e05f461f24ad",
    "name": "nonCrnEnabledHG",
    "primary": "85e6515b-9137-487f-9155-b96aee04b089",
    "secondaries": []
  }
]
```

New format

```
$ [
  {
    "creationDate": "2024-11-12T16:29:35.000Z",
    "hosts": [
      "/v1/hosts/d750956e-e22c-4e4d-b0d1-a126c16d1cd0"
    ],
    "id": "d50dd63a-dfda-46a9-854f-e05f461f24ad",
    "name": "nonCrnEnabledHG",
    "primary": "85e6515b-9137-487f-9155-b96aee04b089",
    "secondaries": []
  }
]
```

Viewing IBM Power Virtual Server pod details in Client location

IBM Power Virtual Server Private Cloud in Client location

You can view the Client location pod details such as the total number of cores, memory, and number of system units. You can also view the storage

capacity that you have used from the total storage per controller.

To view the pod details, complete the following steps:

1. In the Power Virtual Server interface, click **Client locations**. The **Client locations** page displays the total number of cores, memory, storage, and system units that are available for each of the Power Virtual Server pod.
2. Click a Power Virtual Server pod to view extra details such as the number of racks, satellite location of the pod, the total number of physical and usable cores, and the total number of systems of each system type.
3. The **Storage capacity** section displays the storage capacity that you have used from the total available storage. The **Storage capacity** information is displayed for each controller.

Generating and using an SSH key

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can set up one or more Secure Shell (SSH) keys for root login when you create a new virtual machine (VM). The keys are loaded into the **authorized_keys** file in the root directory. By using SSH keys, you can securely log in to a VM. You must use the available operating system options to create SSH keys. To generate SSH keys on a Linux® or Mac OS system, for example, you can use the standard `ssh-keygen` tool.

In this example, the user created a public key on a Linux-based IBM Cloud compute instance by using the `ssh-keygen` tool:

```
$ cat .ssh/id_rsa.pub
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUcg
9 testkey
```

To use an SSH key with a VM-create operation, you must add the public key to the IBM® Power® Virtual Server instance by using the `ibmcloud pi ssh-key create KEY_NAME --key KEY` command.

To add the generated public key and to set its visibility by using the `visibility` attribute, enter the following command (by replacing the example value with your own public key) :

```
$ ibmcloud pi ssh-key create testkey --key "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUcgTI9Dee8zzaA
9 testkey" --visibility "account"
SSHKey created: testkey
```

To confirm that the key was successfully added, use the `ibmcloud pi ssh list` command:

```
$ ibmcloud pi ssh list
Name      Key      Visibility CreationDate
testkey   ssh-rsa AAAAB3NzaC1y...UlzYr3u+79n9 testkey  account  2019-07-26T18:21:56.030Z
```

Setting the visibility of an SSH key in a Power Virtual Server workspace

In a Power Virtual Server workspace, when you create the SSH key, you can set the visibility of an SSH key as `Workspace` or `Account` . When you set the visibility of an SSH key as `Workspace` , the SSH key is available only in the workspace where it is created and not available in the other workspaces. When you set the visibility of an SSH key as `Account` , the SSH key is available in all the workspaces under the account. However, you can delete the SSH key only from the workspace in which it is created. Therefore, the security and privacy of the SSH key is improved.



Note: You cannot access the SSH key between IBM Power Virtual Server Private Cloud locations, even if the `visibility` property of an SSH key is set to `Account` .

Complete the following steps to create and set the visibility of an SSH key:

1. Open a workspace to add an SSH key from the Power Virtual Server user interface.
2. Click **SSH keys** under **Compute**. The list of SSH keys that are associated with the workspace are displayed on the SSH key page.
3. Click **Create SSH Key** . The new SSH key page appears.
4. Enter a unique name in the **Key name** field.

5. Select one of the following options from the Access setting section to set the visibility of the SSH key:
- **Workspace:** The SSH key is visible only in the workspace in which it is created.
 - **Account:** The SSH key is visible to all the workspaces in the account. However, you can delete the SSH key only from the workspace in which it is created.
6. Enter a description about the SSH key in the **Description** field.
7. Enter the SSH key in the **Public key** field.
8. Click **Add SSH key**.

The SSH key is created and listed in the data table on the SSH key page. The **Access setting** column in the data table displays the visibility of the SSH key as **Workspace** or **Account**.

Creating an AIX VM instance

You can create an AIX VM instance with a configured SSH key by using the Power Virtual Server CLI or the console. When you use an AIX stock image as your boot volume, the root password is not set. You must connect to the AIX VM and set the root password for the system. Without completing this step, SSH login as **root** appears as disabled. If you have public network access to the AIX VM, you can use telnet from a private cloud system and set the root password. For more information, see [IBM AIX V7.2 documentation](#).

Creating an AIX VM with a configured SSH key using UI

You must [generate a public SSH key](#) before you can create an AIX VM with a configured SSH key.

1. Ensure that you have the necessary account permissions and device access. Only the account owner, or a user with the **Manage Users** classic infrastructure permission, can adjust the permissions. For more information, see [Service access roles](#).
2. Click **Virtual server instances** from the left navigation in the Power Virtual Server user interface.
3. Click **Create instance**.
4. Under the **Virtual servers** section, select **Add SSH Keys**.
5. Enter a **Key name** and your previously generated **Public key**.
6. Click **Add SSH key** to add the SSH key.
7. Complete the rest of the fields to successfully create a new instance with a configured SSH key.

Creating an AIX VM with a configured SSH key using CLI

You can create a new VM with the public key with the following command by replacing the options with your own:

```
$ ibmcloud pi instance-create keytest-vm --image AIX-7200-03-03 --memory 5 --networks "cloud.ibm.com" --processors 1 --processor-type shared --key-name testkey
```

In this example, the `ibmcloud pi instance-create` command created a new AIX VM with an IP address of `172.16.7.16`. You can now SSH to the AIX VM from a connected system, which is configured with the private key for `testkey`.

```
$ ssh root@172.16.7.16
Enter passphrase for key '/home/keytest/.ssh/id_rsa':
Last login: Fri Jul 26 16:53:22 CDT 2019 on ssh from 10.150.0.11
*****
*                               *
*                               *
* Welcome to AIX Version 7.2!   *
*                               *
*                               *
* Please see the README file in /usr/lpp/bos for information pertinent to *
* this release of the AIX Operating System.                             *
*                               *
*                               *
*****
# oslevel -s
7200-03-03-1914
```

You can find the `testkey` value in the **authorized_keys** file:

```
$ cat .ssh/authorized_keys
ssh-rsa
```

```
AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUc9 testkey
```

Debugging the connection

If the SSH connection to the VM instance is failing, use the `-vvv` option of **ssh** command to determine the reason of the connection failure. Each additional verbose flag in the `-vvv` option increases the verbosity of the ssh client session and generates a large amount of data. Therefore, use a script session to capture the ssh log data so that log data is also sent to the **Standard output (STDOUT)**.

Run the following commands to start an SSH client debug session:

```
$ # /usr/bin/script /tmp/ssh.{host}.debug
# /usr/bin/ssh -vvv {hostname/IP_of_ssh_server}
```

After you end the SSH client debug session, close the script session by pressing **Ctrl+D** or by entering the **exit** command.

Additional support information

Refer to the following support pages for additional information for different use cases:

- [Debugging sshd without impacting existing sshd sessions](#)
- [Using IPsec rules to filter network traffic](#)
- [Downloading and installing or upgrading OpenSSL and OpenSSH](#)

Modifying an IBM Power Virtual Server instance

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

After you create a virtual server instance (VSI) in your IBM® Power® Virtual Server workspace, you can modify some of its configurations to meet your business and workload needs. You can update the VSI name, change the preferred processor compatibility mode, adjust its pinning state and placement group, change its capacity, such as the core type, number of cores, virtual cores, and memory. You can also manage the storage volumes and reconfigure the network interfaces that are attached to the VSI.

Modifying a VSI by using the Power Virtual Server user interface

To modify a VSI after you create it, complete the following steps:

- Open the Power Virtual Server user interface in [IBM Cloud](#).
- Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
- Select the workspace that contains the virtual server instance that you want to modify. The "Workspace details" panel is displayed.
- Click **View virtual servers**. The "Virtual server instances" page is displayed.
- Select the VSI that you want to modify. The "Virtual server instance details" page is displayed for the selected VSI.

From the "Virtual server instance details" page, you can use the **Overview**, **Storage**, and **Networking** tabs to modify specific components of a VSI. Depending on your requirements, you can perform the following modifications:

- [Changing the VSI name](#)
- [Changing the preferred processor compatibility mode](#)
- [Changing the pinning state and server placement group](#)
- [Resizing the capacity of a VSI](#)
- [Changing licenses for an IBM i-based VSI](#)
- [Managing the storage volumes](#)
- [Adding or removing a public network](#)
- [Enabling or disabling access to the metadata service](#)
- [Disabling automated remote restart for a VSI](#)

Changing the VSI name

To change the name of the VSI, complete the following steps:

1. On the **Overview** tab, click the **Edit** icon in the Virtual server instance details section. The "Edit virtual server instance details" panel is displayed.
2. Enter the new name in the **Name** field.
3. Click **Save**.

Changing the preferred processor compatibility mode

In virtualization environments, a VSI can operate in different processor compatibility modes. These modes determine the Instruction Set Architecture (ISA) version that is used by the processor and the platform-level features that are available to the VSI. The "Virtual server instance details" page on the **Overview** tab displays the preferred and effective processor compatibility mode that is currently set for the VSI. The following processor compatibility modes are displayed:

- **Preferred:** The processor mode in which you want the VSI to operate. By default, Power Virtual Server sets the preferred processor compatibility mode to the highest mode that is supported by the targeted host type for the VSI.
- **Effective:** The processor mode that is currently in use for the VSI. The physical host on which the VSI runs determines the effective processor compatibility mode.

You cannot dynamically change the effective processor compatibility mode of a VSI. To change the effective processor compatibility mode, you must first change the preferred processor compatibility mode of the VSI, shut down the VSI, and then start the VSI. When the VSI starts, the hypervisor attempts to set the effective processor compatibility mode to match the preferred mode that you have specified for the VSI.



Important: You must select the processor compatibility mode that the operating system that you use supports. If you set a preferred processor compatibility mode that the operating system in the VSI does not support, the VSI does not boot correctly and might enter the Error state. To resolve this issue, [open a support case](#).



Note: The effective mode might differ from the preferred mode if the host does not support the preferred mode that you have specified for the VSI.

You can use the GUI, CLI, API, or Terraform to set the preferred processor compatibility mode when you create the VSI. For more information, see [Configuring a Power Virtual Server instance](#).

To change the preferred processor compatibility mode of a VSI by using the Power Virtual Server user interface, complete the following steps:

1. On the **Overview** tab, click the **Edit** icon in the Virtual server instance details section. The "Edit virtual server instance details" panel is displayed.
2. Select the preferred processor compatibility mode from the **Preferred processor compatibility mode** list.
3. Click **Save**.

After you change the preferred processor compatibility mode of a VSI, you must shut down and then start the VSI for the changes to take effect. Only restarting the VSI does not activate the selected preferred processor compatibility mode.

To shut down the VSI, complete the following steps:

1. From the "Virtual server instance details" page of the selected VSI, select **OS Shutdown** from the overflow menu (⋮). The "OS shutdown confirmation" dialog is displayed.
2. Click **Shutdown** to proceed. A notification is displayed to indicate that the shutdown process has started.

To start the VSI, complete the following steps:

1. From the "Virtual server instance details" page of the selected VSI, select **Start** from the overflow menu (⋮). A notification is displayed to indicate that the VSI has started.
2. Click the **Refresh** icon to see the change.

For more information about the processor compatibility mode, see [How does the processor compatibility mode work in a VSI?](#)

Changing the pinning state and server placement groups

You can use VSI pinning to control the movement of VSIs during disasters and other restart events. You can either choose soft pin or hard pin to pin a VSI to the host on which it is running. For more information about VSI pinning, see [What does VSI pinning do?](#).

Server placement groups provide control over the host or server on which a new VSI is placed. By using server placement groups, you can build high availability within a data center. For more information, see [Managing server placement groups](#).



Important: When you change the pinning policy of a VSI from **Hard** to **Soft** or disable the pinning policy, consider enabling automated remote restart setting on the VSI. The automated remote restart setting restarts the VSI on another available host if an IBM Power host fails.

To change the pinning state and server placement group of the VSI, complete the following steps:

1. In the **Overview** tab, click the **Edit** icon in the Placement section. The "Edit placement" panel is displayed.
2. Select the new pinning policy from the **Virtual server pinning** list and the placement group from the **Placement group** list.
3. Click **Save**.

Resizing the capacity of a VSI

You can edit a VSI to resize its capacity, including the memory, core type, and number of physical and virtual cores.

You can scale up and scale down the memory and core counts of the VSI according to your workload requirements. When the VSI is active, you can resize the memory and core counts to a maximum of eight times the specified values. When the VSI is provisioned, you can resize the memory and core counts to a minimum of 1/8 times the specified values. However, you cannot resize the memory to less than 2 GB and core count to less than 0.25 cores. You can resize the memory and core count beyond the 8x and 1/8x boundaries when the VSI is shut down. The following table shows an example of the recalculated values:

Specified value when the VSI instance was provisioned	Minimum resize value (must be greater than or equal to 0.25 cores, 2 GB memory)	Maximum resize value
4 core and 8 GB memory	0.5 cores and 2 GB memory	32 cores and 64 GB memory

Resizing VSI core count and memory when the VSI is active		
Specified value when the VSI instance was provisioned	Minimum resize value (must be greater than or equal to 0.25 cores, 2 GB memory)	Maximum resize value
4 core and 8 GB memory	You can specify any value that is greater than 0.25 cores, 2 GB memory	You can specify any value that is smaller than the available resources in the host

Resizing VSI core count and memory when the VSI is shut down		
--	--	--

To resize an existing VSI that was created before 15 December 2020 to an 8x ratio of memory and core counts, you must first shut down the VSI, resize it, and then activate it. Resize the VSI at least once when the VSI is shut down to enable the 8x ratio. If you shut down and activate the VSI, the 8x ratio of memory and core counts are not enabled.

To edit a VSI and resize its capacity, complete the following steps:

1. On the **Overview** tab, click the **Edit** icon in the Capacity section. The "Edit capacity" panel is displayed.
2. Select the new value and size for the VSI memory, cores, and virtual cores. The total estimated cost for the new selection is displayed for your review.
3. Click the terms and conditions link to read the IBM Cloud Terms of Use. To continue, select the **I agree to the Terms and conditions** checkbox and click **Save**.



Tip: If the VSI that you are editing is in an inactive state, you can change the **Core type** to Dedicated, Shared uncapped shared, or Shared capped.

Adding or removing additional software licenses for an IBM i-based VSI

After you provision an IBM i-based VSI with the base IBM i OS license, you can modify the VSI to add or remove additional software licenses. To add or remove additional software licenses for the VSI, complete the following steps:

1. From the "Virtual server instances" page, select the IBM i-based VSI that you want to modify. The "Virtual server instance details" page is displayed for the selected VSI.
2. On the **Overview** tab, click the **Edit** icon next to Software licenses in the IBM i licensing section. The "Edit licenses" panel is displayed.
3. Select the additional software licenses that you want to add. To remove a license, clear the corresponding checkbox. The total estimated cost for the selected licenses is displayed. The available options are:
 - **IBM Cloud Storage Solutions for IBM i**

- **IBM PowerHA SystemMirror for IBM i**
- **IBM Rational Development Studio for IBM i**

4. Click the terms and conditions link to read the IBM Cloud Terms of Use. To continue, select the **I agree to the Terms and conditions** checkbox and click **Save**.

 **Tip:** You can also use the IBM i licensing section to change or assign a virtual server number (VSN) for an IBM i-based VSI. The VSI must be in the **Shutoff** state before performing this operation.

 **Note:** To apply or remove an IBM i software key, the VSI must be active and in the running state. If you have already run a resize operation, you must wait until the resize operation completes and the VSI returns to the **Active** state.

Managing storage volumes on a VSI

You can attach storage volumes to a VSI from different storage tiers and pools. However, you cannot attach storage volumes to the same storage pool where the root (boot) volume of the VSI is deployed. To attach mixed storage volumes, you must modify the VSI and set the **storagePoolAffinity** property to **false**. For more information, see [How to set a VSI to allow attaching mixed storage?](#).

Adding additional storage volumes to a VSI

After you create a VSI, you can modify the VSI to include additional storage volumes, by adding new volumes or attaching existing ones.

To modify a VSI to add additional storage volumes, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
3. Select the workspace that contains the virtual server instance to which you want to add additional volumes. The "Workspace details" panel is displayed.
4. Click **View virtual servers**. The "Virtual server instances" page is displayed.
5. Select the VSI to which you want to add additional volumes. The "Virtual server instance details" page is displayed for the selected VSI.

Depending on whether you want to attach an existing storage volume or create a new one for the VSI, follow the steps in the relevant section below:

- [Attaching an existing volume to a VSI](#)
- [Creating a new storage volume for the VSI](#)

Attaching an existing volume to a VSI

To attach an existing volume to a VSI, complete the following steps:

1. On the **Storage** tab, click **Attach existing** in the Storage volumes section. The "Attach storage volumes" panel is displayed with a list of existing storage volumes.
2. From the list of the existing storage volumes, select the storage volumes that you want to attach to the VSI. If the **Mixed storage pools in virtual server instances** warning message is displayed, select the following checkboxes:
 - **I acknowledge snapshots and clones will fail when performed on a set of volumes that are on mixed pools**
 - **I acknowledge volume replication across mixed storage pools will require managing multiple volume replication groups when enabled**
3. Click **Attach volume**.

 **Note:** Attaching storage volumes to a VSI is an asynchronous operation. Before you use the storage volume, refresh the "Virtual server instance details" page and check the status of the selected storage volume to verify that the volume is attached to the VSI. If the volume is not attached, use the **Attach existing** option again from the **Storage** tab.

Creating a storage volume for a VSI

To create a storage volume and attach it to a VSI, complete the following steps:

1. On the **Storage** tab, click **Create volume** in the Storage volumes section. The "Create volume" panel is displayed.
2. Enter the **Name** and **User tags** (optional) for the storage volume.

3. Select the required storage tier from the **Tier** list.
4. Enter the **Number of volumes** and **Size** of the storage volume.
5. Set **Shareable** to **On** to allow multiple virtual instances to write data to the same data volume.
6. Select one of the following **Storage pool** options:
 - **Auto-select pool:** Use this option to allow the system to automatically select a storage pool, for the required storage tier, that has sufficient capacity.
 - **Affinity:** Use this option to select a VSI or an existing volume as the affinity object. The new volume is created in the same storage pool where the affinity object resides. If you are using the VSI as an affinity object, the storage pool that is selected is based on the VSI's root (boot) volume.
 - **Anti-affinity:** Use this option to specify one or more existing VSIs or one or more volumes as the anti-affinity objects. The new volume is created in a different storage pool than the storage pool where one or more anti-affinity objects reside.

For more information about affinity and anti-affinity policies, see [What does it mean to set an affinity or anti-affinity rule?](#).



Note: In the API, for create volume feature the properties `antiAffinityVMInstances` and `antiAffinityVolumes` are used to specify anti-affinity objects. You can specify only one object type for affinity or anti-affinity objects, either VSIs or Volumes. For more information about storage volumes APIs, see [Create a new data volume](#) and [Create multiple data volumes from a single definition](#).

7. Optional: Set **Volume replication with GRS** to on to enable the volume for asynchronous replication. The name of the primary location (data center) is displayed. From the **Secondary data center** list, select the required secondary location.

If the **Mixed storage pools in virtual server instances** warning message is displayed, select the following checkboxes:

- **I acknowledge snapshots and clones will fail when performed on a set of volumes that are on mixed pools**
 - **I acknowledge volume replication across mixed storage pools will require managing multiple volume replication groups when enabled**
8. Click the terms and conditions link to read the [IBM Cloud Terms of Use](#). To continue, select the **I agree to the Terms and conditions** checkbox and then click **Create and Attach**.



Note: You can create a storage volume by specifying any name of your choice. If you want to reuse the storage volume name, you must delete the existing storage volume with the same name. If the existing volume is a replication-enabled volume, follow the steps to [delete a primary volume](#). After you delete the original volume, you must allow a minimum of one hour to create a new volume with the same name.

You can create a replication-enabled volume from any site. The site from which the replication is initiated contains what is referred to as the `primary volume`; the remote site volume is referred to as the `auxiliary volume`. For more information, see [Global Replication Services \(GRS\)](#). The GRS details of the volume are displayed when you click the storage volume.



Note: The primary volume and the auxiliary volume are mapped into one-to-one relationship mode in both directions. These two sites are fixed and are in replication partnership in both directions. For more information, see [Setting up GRS](#).

For more information about updating the volume groups for GRS, see [Updating a volume group](#).

Verify that the action that you perform on a volume group meets the current state of the volume group. Otherwise, an error message is displayed indicating that the volume group is not in the expected state.

The error message is displayed when you perform the following actions:

- Stop a volume group when the `ReplicationStatus` is in the `disabled` state.
- Start a volume group when the `ReplicationStatus` is in the `enabled` state or `available` state.
- Reset a volume group that is not in the error state.

Detaching storage volumes from a VSI

To detach storage volumes from a VSI, complete the following steps:

1. On the **Storage** tab, select the storage volumes to detach from the Storage volumes section.
2. Click **Detach**. The "Confirm detach" window is displayed.



Note: Detached storage volumes are not deleted automatically and continue to incur charges until you delete them. For more information

about deleting a volume, see [Deleting a volume](#).

3. To confirm, click **Detach**.



Note: The user interface might display a failure message when you attempt to detach a volume from a virtual server instance. In such cases, you need to reload the page after a brief delay to see whether the volume is successfully detached. Make another detach request if the volume is still not detached.

Resizing a storage volume

You can resize a storage volume after you create it. However, resizing is not immediately available after you deploy a VSI.

For more information about resizing a replication-enabled volume, see [Updating a primary volume](#).

IBM data center For IBM i 7.3 and later versions, you can resize a volume to increase the volume capacity, but this action requires an initial program load (IPL) to recognize the new volume size.



Important: Before you perform the IPL operation, you must run the macro to ensure that the volume resize operation is complete, then proceed with the IPL operation. For more information, see [Dynamically increasing the size of a SAN LUN](#). If you do an IPL operation before the resize operation is complete, an extra IPL is required.

If you cannot accommodate downtime, you can add extra volumes. You can attach a maximum of 127 volumes to the VSI.



Important: **IBM data center** Any volume that is included in a snapshot cannot be resized. To resize a volume that is included in a snapshot, you must first delete all the snapshots the volume is a part of.

To resize a storage volume after you create it, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
3. Select the workspace which contains the virtual server instance that you want to modify to resize its storage volumes. The "Workspace details" panel is displayed.
4. Click **View virtual servers**. The "Virtual server instances" page is displayed.
5. Select the VSI with the attached storage volume that you want to resize. The "Virtual server instance details" page is displayed for the selected VSI.
6. Click the **Storage** tab.
7. From the Storage volumes section, click the overflow menu (:) on the volume entry that you want to resize and select **Edit**. The "Edit storage volume" panel is displayed.
8. In the **Size (GB)** field, specify the new size for the storage volume. You can also modify the **Name** and **Tier** fields to update the name and performance tier of the selected volume.



Note: The size of a volume cannot be decreased once it is created. The maximum size of a volume that can be created is 72551 GB.

9. Optional: Set the **Shareable** or **Bootable** property of the storage volume to **On**. You cannot set both options to **On** at the same time.
10. Click the terms and conditions link to read the [IBM Cloud Terms of Use](#). To continue, select the **I agree to the Terms and conditions** checkbox and then click **Save**.



Note: In an AIX VSI, if you resize your boot storage volume, run the `chvg -g rootvg` command.

Deleting a volume

When you modify a VSI to update its configuration, you can use the **Storage** tab to only detach a storage volume from the VSI. Detached storage volumes are not deleted automatically and continue to incur charges until you delete them. To delete a storage volume, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#).
2. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.

3. Select the workspace which contains the storage volume to delete. The "Workspace details" panel is displayed.
4. Click **View virtual servers**. The "Virtual server instances" page is displayed.
5. In the navigation panel, click **Storage > Storage volumes**. The "Storage volumes" page is displayed with a list of existing storage volumes on the **Volumes** tab.
6. Click the overflow menu (:) on the volume entry to delete and select **Delete**. The "Confirm delete" dialog is displayed.
7. Click **Delete**.



Note: You cannot recover a storage volume after it is deleted.



Note: To delete a volume or a replication-enabled primary volume, the status of the storage volume must be `available`, `error`, `error_restoring`, `error_extending`, or `error_managing`. Also, the storage volume cannot be deleted if it is in the state of migrating, attached, belongs to a group, has snapshots, or is disassociated from its snapshots after a transfer.

For more information about deleting a primary volume, see [Deleting a primary volume](#).

Deleting a virtual server instance

You must delete a VSI manually. To delete all VSIs, delete the workspace or delete a subset of the VSI.

To delete a VSI, complete the following steps:

1. On the "Virtual server instances" page, locate the VSI that you want to delete and use one of the following methods:
 - Click the overflow menu (:) for the VSI entry and select **Delete**.
 - Click the VSI name to open the "Virtual server instance details" page, then click the **Delete** icon.

The "Delete virtual server instance" dialog is displayed.

2. On the "Delete virtual server instance" dialog, set **Delete data volumes attached to this instance** to on. When you enable this option, the following actions occur:
 - Data volumes that are attached only to this VSI are deleted.
 - Data volumes that are attached to multiple VSIs are not deleted.
 - Any auxiliary volumes on the secondary site are also deleted.
3. To confirm, type the name of the VSI that you want to delete.
4. Click **Delete virtual server instance**.



Note: The deletion process begins immediately and cannot be undone. You cannot recover a VSI after it is deleted.



Important: You cannot delete a VSI if it has one or more associated snapshots. You must delete all the associated snapshots before you delete the VSI. For more information about how to delete a snapshot, see [Deleting a snapshot](#).

Adding or removing a public network

IBM data center

Public networks are not supported in all the IBM data center locations. If your VSI is in a location that supports public networks, you can set **Public networks** to **On** or **Off** in the "Virtual server instances" page. When you toggle a public network off and then on, the Power Virtual Server user interface regenerates new internal and external IP addresses. You need to check the Power Virtual Server user interface for the new internal IP address (that maps to the external IP address). You must add a network interface controller (NIC) and point it to the new internal IP address. For information about how to add or remove an interface, see [How to add or remove a network interface from an AIX virtual machine \(VM\)](#) or [How to add or remove a network interface from an IBM i virtual machine \(VM\)](#).



Note: You cannot toggle a public network off if there are no other defined networks.

Enabling or disabling access to the metadata service

Access to the metadata service for a VSI is disabled by default. For an existing VSI, you can enable or disable access to the metadata service by using the Power Virtual Server user interface, CLI, or API.

When you enable access to the metadata service, you can access information about a VSI and access IAM-enabled services. For more information about enabling or disabling access to the metadata service, see the following topics:

- [Enabling access to the metadata service for an existing VSI](#)
- [Disabling access to the metadata service](#)

Disabling automated remote restart for a VSI

Automated remote restart is enabled by default for all VSIs in the Power Virtual Server environment. To disable automated remote restart for a VSI, complete the following steps:

1. On the **Overview** tab, click the **Edit** icon in the Virtual server instance details section. The "Edit virtual server instance details" panel is displayed.
2. Set **Automated remote restart** to **Disabled**.
3. Click **Save**.

Automated remote restart is disabled for the VSI.



high availability (HA) and *disaster recovery* (DR) policies are configured to protect the workload after a host failure. For more information, see [High availability and disaster recovery](#).



Note: The hard pin policy for a VSI takes precedence over the automated remote restart setting for the VSI; therefore, a hard-pinned VSI is not restarted automatically.

Detecting problems by using the System Reference Code (SRC)



Note: SRC is only supported for AIX and IBM i virtual machines.

A system reference code (SRC) is a set of eight alphanumeric characters that identifies the name of the system component that detects the error codes and the reference code. The error codes and the reference code describe the error condition. When the Power Virtual Server instance detects a problem, an SRC number is displayed along with a timestamp in the "Server details" page. You can use the SRC to resolve the issue yourself. If you are contacting support to resolve a problem, the SRC number might help the hardware service provider better understand the problem and to provide the solution.

IBM data center For an IBM i VSI, the SRC number can be progress code, operation code, or software code. For more information, see the [System Reference Code list](#) in the IBM i documentation. For AIX VSIs, the SRC numbers are progress codes that provide information about the stages that are involved in powering on and performing initial program load (IPL). AIX SRCs refresh once in 2 minutes. For more information, see [AIX IPL progress codes](#).

Use cases

The following are some of the scenarios that you might face when you make a request for resizing the memory of a Power Virtual Server instance:

Your request for resizing both memory and CPU fails

When you attempt to resize the memory and the CPU of a deployed virtual server instance through a single request, it might fail due to the following reasons:

- No free memory is available on the host.
- No free memory is available on the logical partition as the resources on it are running.
- The free memory that is available on the logical partition is less than the required value indicated in the resizing request.
- Multiple attempts are made for resizing.



Note: Currently, there is no preference for memory or CPU on what can be resized first. If the first request processing fails, the second request also fails automatically.

Example : Consider that the allocated memory for the logical partition is 4 GB. You make a request to reduce the memory value to 2 GB. The logical partitioning is using up to 3.2 GB for running resources in it. So, logical partitioning does not have free 2 GB memory for resizing. In such a scenario, it is possible that both the CPU and memory resize might fail.

Your request for resizing the memory, but you get a partial resize

When you attempt to resize the memory of a deployed virtual server instance through a request, it might partially resize or might fail due to the following reasons:

- No free memory is available on the host.
- No free memory is available on the logical partition as the resources are running on it.
- The free memory that is available on the logical partition is less than the required value indicated in the resizing request.
- Multiple attempts are made for resizing.

Example: Consider that the allocated memory for the logical partition is 4 GB. You make a request to reduce the memory value to 2 GB. The logical partitioning is using up to 3 GB for running resources in it and so can free up only 1 GB memory. Then, the partial resize can reduce the memory to 3 GB.

 **Important:** In the current cloud environment, it might take up to 1.5 hours for the change in memory to be updated. All the places that are referring to the memory of the logical partition are to be updated. Hence, if you attempt to repeat the resize request, the consecutive retries fails until all referencing tables are updated.

Managing virtual appliances

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

Virtual appliance is a bring-your-own license model, where independent software vendors (ISV) can offer OVA (ISV software plus operating system of your choice) for quick deployment of IBM® Power® Virtual Server workloads. It is an appliance-as-a service that allows seamless management and metering of PowerVS. Software support is handled directly by the ISVs for virtual appliances.

With virtual appliances, you can use extra services within Power Virtual Server such as virtual tape libraries. For more information, see [Managing Virtual tape library](#).

Onboarding your virtual server images for Power Virtual Server

Independent software vendors can sell software on the IBM Cloud platform by enrolling their account in the Partner Center and by completing the onboarding process from the IBM Cloud catalog. If you are an ISV and want to sell your software, see [Selling on IBM Cloud](#).

As part of the onboarding process for selling Power Virtual Server software, sellers must provide a public (shareable) virtual server image asset. To provide a public virtual server image for Power Virtual Server, complete the following steps:

1. Create your [Power Virtual Server instance](#).
2. Create an instance of [IBM Cloud Object Storage](#) and upload your image to a bucket.
3. Create your [HMAC credentials](#).
4. [Open a support case](#) so that the Power Virtual Server product management team can convert your image into a stock image. Include your HMAC credentials and bucket details in the support case.

Importing a boot image

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

You can import a custom operating system (OS) image from an IBM Cloud Object Storage (COS) bucket by using the IBM® Power® Virtual Server user interface, CLI, or API, instead of using an IBM-provided stock image. After you import the boot image, you can use it to create a Power Virtual Server virtual server instance (VSI).

When you import a boot image, you select a storage tier and storage pool for the image. Power Virtual Server creates the boot volumes from the imported boot image in the storage pool that you specify during the import. You cannot change the storage tier or storage pool after you import the image. A VSI can have disks from multiple storage types. All Power Virtual Server data centers support Tier 0, Tier 1, Tier 3, and Fixed IOPs storage types.

 **Important:** Boot image import and export are long-running, asynchronous operations. Power Virtual Server monitors these operations across all workspaces in your account. You can run only one import or export operation at a time in a workspace. You cannot start another operation in that workspace until the current operation is complete.

Before you begin

Before you import a boot image, verify that you meet the following prerequisites:

- Upload your boot image file to an IBM Cloud Object Storage bucket. Supported file formats are `.ova`, `.ova.gz`, `.tar`, `.tar.gz`, and `.tgz`. For more information, see [Create some buckets to store your data](#).
- Generate Hash-based Message Authentication Code (HMAC) credentials for your COS instance. For more information, see [Using HMAC credentials](#).
- Create or use an existing Power Virtual Server workspace.

Importing a boot image by using the Power Virtual Server user interface

To import a boot image by using the Power Virtual Server user interface, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. Select the workspace to which you want to import the boot image. The "Virtual server instances" page of the selected workspace is displayed.
5. Click **Boot images** in the navigation panel and click **Import image**. The "Import boot image" panel is displayed.
6. In the "Import boot image" panel, complete the following steps in the **Source** section:

- a. Select the image OS from the **Image OS** drop-down list.

If you select a SAP HANA or SAP NetWeaver image OS, you must select the self-certification checkbox.

- b. From the **Region** drop-down list, select the region that contains your COS bucket.

- c. Enter the file name of the image in the **Image filename** field. The image file name must not contain spaces.

The supported file name extensions are `.ova`, `.ova.gz`, `.tar`, `.tar.gz`, and `.tgz`. You can compress image files by using `gzip`. For example:
`Aix_7200-03-02-1846_cldrdy_112018.ova.gz`.

- d. Enter the name of your bucket in the **Bucket name** field. If your image file is stored in a subfolder in the bucket, specify the full path by using the `bucketName/optional/folders` format.

To identify your bucket name, go to **Navigation menu** > **Resource list** > **Storage** and click your Cloud Object Storage instance name. Your buckets are listed on the **Buckets** tab.

- e. Copy the `access_key_id` value from your COS service credentials and paste it into the **HMAC access key** field.

To find your service credentials, go to **Navigation menu** > **Resource list** > **Storage**, click your Cloud Object Storage instance name, and then go to **Service credentials** > **View credentials**.

- f. Copy the `secret_access_key` value from your COS service credentials and paste it into the **HMAC secret access key** field.

- g. Set **Validate import with checksum file** to **On** to verify the imported file against the checksum file. You must store the checksum file and the boot image file in the same COS bucket.

You can generate the checksum file when you export the boot image file to the IBM Cloud Object Storage bucket. The checksum file name is based on the name of the boot image file and uses the `.sha256` file extension.

If you create your own boot image, you can create a checksum file and store it with your boot image in the same bucket. You can generate the checksum file by running the `shasum -a 256 <filename>` or `sha256sum <filename>` command.

For more information about generating a checksum file, see [Using the Power Virtual Server user interface to capture and export a VM](#).



Note: Validating the import file against the checksum file might increase the import time.

7. Click **Next**.
8. Enter the following information in the **Destination** section:
 - a. In the **Custom image name** field, enter a name for the imported boot image.

- b. Optional: In the **User tags** field, enter tags to organize or categorize this boot image.



Note: User tags are visible to all users in the account. Do not include sensitive data in tag names.

- c. In the **Tier** field, select one of the following storage type options:

- **Tier 0 (25 IOPs / GiB)**
- **Tier 1 (10 IOPs / GiB)**
- **Tier 3 (3 IOPs / GiB)**
- **Fixed IOPs (5000 IOPs)**



Note: A VSI cannot use Tier 1 and Tier 3 storage types simultaneously. For more information, see [Storage tiers](#).

- d. In the **Storage pool** field, select a storage pool option.

Power Virtual Server places one or more custom image storage volumes in the storage pool based on the option that you select: **Auto-select**, **Affinity**, or **Anti-affinity**. The boot volume of any VSI that you deploy by using this image is created in the same storage pool. For more information about storage volumes, see [Adding and managing storage volumes](#).

The following storage pool options are available:

- **Auto-select:** Creates the storage volume in a storage pool with sufficient capacity automatically.
- **Affinity:** Identifies the storage pool for the boot volumes, based on an existing VSI or storage volume from your account. Power Virtual Server stores the custom image storage volumes in the same storage pool in which the affinity object exists. If the affinity object is a VSI, Power Virtual Server determines the storage pool based on the boot volume of the VSI.
- **Anti-affinity:** Identifies one or more storage pools that you want to exclude. Power Virtual Server does not consider the excluded storage pools when placing boot volumes. The pools to exclude are identified based on existing VSIs or storage volumes in your account. When you select this option, Power Virtual Server excludes the storage pools where the anti-affinity objects exist when it creates the custom image storage volumes. If you use VSIs as the anti-affinity objects, Power Virtual Server excludes the storage pools based on the boot volume of each VSI.

9. Click **Import image**.



Note: Large boot images might take longer to import. You might experience a delay before you receive a confirmation message.

The new boot image is listed on the **Boot images** page.

Importing a boot image by using the Power Virtual Server CLI

To import a boot image, run the [ibmcloud pi image import](#) command. To verify that the import is complete, run the [ibmcloud pi image list](#) command.

If you are using a bring-your-own-license (BYOL) SAP HANA or SAP NetWeaver image, add the `--import-details` option to the [ibmcloud pi image import](#) command to identify the image as SAP-certified.

Importing a boot image by using the Power Virtual Server API

To import a boot image from IBM Cloud Object Storage by using the API, use the [Create a cos-image import job](#) method with the required properties in the request body: `imageName`, `imageFilename`, and `bucketName`. For private buckets, you must specify `accessKey` and `secretKey`, which are the HMAC access key and secret key for your COS instance. For more information, see [Using HMAC credentials](#).

```
$ curl -X POST \
https://us-east.power-iaas.cloud.ibm.com/pcloud/v1/cloud-instances/$CLOUD_INSTANCE_ID/cos-images \
-H "Authorization: Bearer $TOKEN" \
-H "CRN: $CRN" \
-H "Content-Type: application/json" \
-d '{
  "imageName": "my-image-name",
  "imageFilename": "my-os-image-file.ova.gz",
  "bucketName": "my-cos-bucket-name",
  "region": "us-east",
  "accessKey": "my-cos-access-key",
  "secretKey": "my-cos-secret-key",
  "storageType": "tier3"
}
```

```
}'
```

If you are using a bring-your-own-license (BYOL) SAP HANA or SAP NetWeaver image, include the `importDetails` object in the request body to identify the image as SAP-certified:

```
"importDetails": {
  "licenseType": "byol",
  "product": "Hana",
  "vendor": "SAP"
}
```

Viewing boot image import results

After you start a boot image import, the **Status** column on the **Boot images** page shows the import progress. To view details, click **View details** to open the "Ongoing job status" dialog. The dialog shows the following information:

- Job ID
- Operation type
- Input resource
- Creation time
- Steps completed

To view the boot image import job details by using the CLI, use the [ibmcloud pi image import-show](#) command.

To view the boot image import job details by using the API, use the [Get detail of last cos-image import job](#) method.

Downloading a boot image from Cloud Object Storage

To download the boot image after you import it, complete the following steps:

1. In the IBM Cloud dashboard, go to **Resource list** and click your **Cloud Object Storage** instance.
2. In the bucket where you stored your boot image, click the boot image file.
3. Click **Download objects**.

For more information about the Cloud Object Storage CLI command, see [Download an object](#).

Related information

For information about exporting custom boot images, see [Exporting a boot image](#).

For information about capturing and exporting virtual server instances, see [Capturing and exporting a virtual server instance](#).

Exporting a boot image

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

You can export a custom boot image from your image catalog to IBM Cloud Object Storage by using the IBM® Power® Virtual Server user interface, CLI, or API. Use the **Export boot image** feature to back up or archive a custom boot image, or to copy it to a different workspace or account.

 **Note:** You cannot export IBM-provided stock images. Only custom boot images are available for export. Custom boot images include images that you imported by using the **Import image** function and images that you captured by using the **Capture and export** function.

 **Important:** Boot image import and export are long-running, asynchronous operations. Power Virtual Server monitors these operations across all workspaces in your account. You can run only one import or export operation at a time in a workspace. You cannot start another operation in that workspace until the current operation is complete.

Before you begin

Before you export a boot image, verify that you have the following prerequisites:

- A Power Virtual Server workspace with at least one custom boot image in your image catalog.
- An IBM Cloud Object Storage bucket to export the custom boot image to. For more information, see [Create some buckets to store your data](#).
- Hash-based Message Authentication Code (HMAC) credentials that are generated for your COS instance. For more information, see [Using HMAC credentials](#).

Exporting a boot image by using the Power Virtual Server user interface

To export a boot image from your image catalog by using the Power Virtual Server user interface, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. In the search box, type **Power Virtual Server** and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. Select the workspace that contains the boot image that you want to export. The "Virtual server instances" page of the selected workspace is displayed.
5. Click **Boot images** in the navigation panel.
6. Click the overflow menu for the boot image that you want to export and select **Export**. The "Export boot image" panel is displayed.
7. In the "Export boot image" panel, complete the following steps:
 - a. From the **Region** drop-down list, select the region that contains your COS bucket.
 - b. In the **Bucket name** field, enter the name of the bucket to which you want to export the image. If you want to export the image to a subfolder in the bucket, specify the full path by using the `bucketName/optional/folders` format.

To identify your bucket name, go to **Navigation menu > Resource list > Storage** and click your Cloud Object Storage instance name. Your buckets are listed on the **Buckets** tab.
 - c. Copy the `access_key_id` value from your COS service credentials and paste it into the **HMAC access key** field.

To find your service credentials, go to **Navigation menu > Resource list > Storage**, click your Cloud Object Storage instance name, and then go to **Service credentials > View credentials**.
 - d. Copy the `secret_access_key` value from your COS service credentials and paste it into the **HMAC secret access key** field.
 - e. Set **Generate checksum file** to **On** to generate a checksum file.

The checksum file is created and placed in the IBM Cloud Object Storage bucket with the exported boot image. The checksum file name is based on the name of the boot image file and uses the `.sha256` file extension. After the export completes, run the `shasum -a 256` command to verify the integrity of the exported boot image file.



Note: The maximum image size that you can export is 10 TB.

8. Click **Export**. The "Export boot image" dialog is displayed with the export operation details.
9. Review the information and click **Export** to confirm.

The boot image export job is submitted. You can monitor the progress in the **Status** column on the **Boot images** page. For more details, see [Viewing boot image export results](#).

Exporting a boot image by using the Power Virtual Server CLI

To export a boot image, run the [ibmcloud pi image export](#) command. To verify that the boot image is exported, run the [ibmcloud pi image export-show](#) command.

Exporting a boot image by using the Power Virtual Server API

To export a boot image to IBM Cloud Object Storage by using the API, use the [Add image export job to the jobs queue](#) method with the required properties in the request body: `accessKey` and `bucketName`. You can also include the optional properties `region` and `secretKey`. For more information, see [Using HMAC credentials](#).

```
$ curl -X POST \
https://us-east.power-iaas.cloud.ibm.com/pcloud/v2/cloud-instances/$CLOUD_INSTANCE_ID/images/$IMAGE_ID/export \
-H "Authorization: Bearer $TOKEN" \
-H "CRN: $CRN" \
-H "Content-Type: application/json" \
```

```
-d '{
  "bucketName": "my-cos-bucket-name",
  "accessKey": "my-cos-access-key",
  "region": "us-east",
  "secretKey": "my-cos-secret-key"
}'
```

Viewing boot image export results

After you start a boot image export, the **Status** column on the **Boot images** page shows the export progress. To view details, click **View details** to open the "Ongoing job status" dialog. The dialog shows the following information:

- Job ID
- Operation type
- Input resource
- Creation time
- Steps completed

To view the boot image export job details by using the CLI, use the `ibmcloud pi image export-show` command.

To view the boot image export job details by using the API, use the [Get detail of last image export job](#) method.

Related information

For information about importing custom boot images, see [Importing a boot image](#).

For information about capturing and exporting virtual server instances, see [Capturing and exporting a virtual server instance](#).

Deploying a custom image within IBM Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can bring your own customized AIX or IBM i operating system (OS) image to deploy within an IBM® Power® Virtual Server.



Note: You cannot transfer an OS license from a private cloud system to a Power Virtual Server. The license cost is factored into the overall hourly billing rate.

The basic steps that are involved in deploying an instance by using a custom image are:

1. Create the custom image.
2. Store the image in your **Cloud Object Storage** account.
3. Point the Power Virtual Server console to the image in the **Cloud Object Storage** and deploy the Virtual Server instance.

Before you begin

Before you can use a custom image as the boot volume, review the following information:

- You must have a basic understanding of [IBM Cloud Object Storage](#) concepts.
- If you do not have an existing AIX or IBM i image, you can use IBM® PowerVC™ from your private cloud environment to capture and export an image for use with a Power Virtual Server. For more information, see [Capturing a virtual machine](#) and [Exporting images](#). To capture and export an image by using IBM PowerVC, the PowerVC private environment must contain N_Port ID Virtualization (NPIV) data volumes. The Power Virtual Server does not support captured images from environment with shared Storage Pools (SSP) vSCSI data volumes.
- Alternatively, if you have deployed a virtual server instance, you can capture it and redeploy a new virtual server instance. To accomplish this, you can use the [IBM Cloud® CLI](#) to capture a virtual server instance.
- You must verify that your AIX, IBM i, or Linux OS technology level is supported on the Power hardware that you selected in the **Machine Type** field.

For complete tutorials on migrating your AIX and IBM i workloads to Power Virtual Server, see [Migration strategies for AIX](#) and [Migration strategies for IBM i](#).

The supported AIX and IBM i operating system versions depend on the IBM Power hardware that you select for the Power Virtual Server.



Important: If you are running AIX 6.1 or IBM i 6.1, or earlier, you must first upgrade the OS to the current support level before you migrate to the Power Virtual Server.

To view a list of the supported AIX, IBM i, and Linux operating system technology levels, see the following information:

AIX

The Power Virtual Server offering supports the following versions of AIX operating system depending on the IBM Power server:

IBM data center

Machine type	Supported AIX versions
- S922 (9009-22A) - E980 (9080-M9S)	AIX 7.1 or later
- E1080 (9080-HEX) - S1022 (9105-22A)	AIX 7.1 TL5 or later
S1122 (9824-22A)	AIX 7.2 TL5 SP8 AIX 7.3 TL2 SP2 AIX 7.3 TL3 SP0

Supported AIX versions in IBM data center

Client location

Machine type	Supported AIX versions
- S1122 (9824-22A) - E1150 (9043-MRU) - E1180 (9080-HEU)	AIX 7.2 TL5 SP8 AIX 7.3 TL2 SP2 AIX 7.3 TL3 SP0

Supported AIX versions in client location

When you view the system software maps, refer to the AIX 7.1, AIX 7.2, and AIX 7.3 information. If you use an unsupported version, it is subject to outages during planned maintenance windows with no advanced notification given.

For more information about which versions of AIX are compatible with various Power Systems, see the [System to AIX maps](#).



Note: For more information about end of service pack support (EoSPS) dates, see [AIX support lifecycle](#).

IBM i

If you are using an older version of IBM i, you must upgrade the OS to a current support level and then migrate to Power Virtual Server. For more information about upgrading the IBM i OS, see [Fast path for software installation, upgrade, and deletion tasks](#). For more information about the supported IBM i OS levels, see [IBM i Release Support](#).

IBM data center

Machine type	Supported IBM i versions
- E1022 (9105-22A) - E1080 (9080-HEX)	IBM i 7.3 or later

Supported IBM i versions in IBM data center

Machine type	Supported IBM i versions
- S1122 (9824-22A) - E1180 (9080-HEU)	IBM i 7.4 or later

Supported IBM i versions in client location

 **Note:** The IBM Power Virtual Server in Client location supports IBM i 7.4, or later and IBM i COR.

For more information, see:

- [S1022 \(9105-22A\) IBM i software map](#)
- [E1080 \(9080-HEX\) IBM i software map](#)
- [S1122 \(9824-22A\) IBM i software map](#)
- [E1180 \(9080-HEU\) IBM i software map](#)

Learn more about the [IBM i PTF minimum levels](#) and [IBM i release lifecycle](#)

 **Note:** IBM i 7.3, and later, supports up to 127 storage volumes per VM. IBM i 7.3 VSIs are at the end of support and are in service extension. Therefore, additional Service Extension fees apply.

Linux

SUSE Linux Enterprise (SLES) and Red Hat Enterprise Linux (RHEL) are supported by the appropriate IBM Cloud environment cloud-init packages. Download the appropriate cloud-init package and perform the following steps to configure it:

- [Installing and configuring cloud-init on SLES.](#)
- [Installing and configuring cloud-init on RHEL.](#)

Power server E1080 (9080-HEX) and S1022 (9105-22A) supports RHEL 8.4 or later and SLES 15 SP3 or later versions.

 **Note:** For SAP applications, ensure that you use an IBM stock OS image for SAP. These images are certified for SAP application use. To learn more about SAP applications with PowerVS, see the [Must-Reads](#) before you start deployment.

Creating an IBM Cloud Object Storage bucket

- Type **object storage** into the catalog's search box and select **Cloud Object Storage**.
- Give the service a name, add tags (if wanted), select your pricing plan and click **Create**.
- After you click **Create**, you are taken to the **Cloud Object Storage** landing page. Select **Create Bucket**.
- From here, you are automatically redirected to the workspace where you can start creating buckets. Your Object Storage instances are listed under **Storage** in the **Resource List**. The terms *resource instance* and *workspace* refer to the same concept, and can be used interchangeably.
- Choose a unique name. All buckets in all regions across the globe share a single namespace. Ensure that you have the correct permissions to create a bucket.

 **Note:** When you create buckets or add objects, be sure to avoid the use of Personally Identifiable Information (PII). PII is information that can identify any user (natural person) by name, location, or any other means.

- Select **Regional resiliency** and then choose the location where you want your data to be physically stored. The **Regional resiliency** option distributes the data across multiple devices within a single metropolitan area. Currently, Power Virtual Server supports regional resiliency and one single data center, which is **che01**. The Cross Region resiliency option is not supported.
- Choose the *storage class* of the bucket, which is a reflection of how often you expect to read the stored data and determines billing details. Follow the **Create** link to create and access your new bucket.

 **Tip:** Buckets are a way to organize your data, but they are not the only way. Object names (often referred to as object keys) can use one or

more forward slashes for a directory-like organizational system. You then use the portion of the object name before a delimiter to form an object prefix, which is used to list related objects in a single bucket through the API.

Objects are limited to 200 MB when uploaded through the console unless you use the Aspera High-Speed Transfer plug-in. Larger objects (up to 10 TB of uncompressed image) can also be split into parts and uploaded in parallel using the API. Object keys can be up to 1024 characters in length, and avoid any characters that might be problematic in a web address. These special characters (? , = , < , and so on) might cause unwanted behavior if not URL-encoded. For more information, see the [Cloud Object Storage tutorial](#).

Generating secret and access keys with Hash-based Message Authentication Code (HMAC)

- 1. You can generate secret and access keys when you create the service credentials for the IBM Cloud Object Storage. To create the service credentials, you must have **Writer** access for the **Object Storage** bucket.
- 2. Select **New credential** under **Service credentials** in the **Cloud Object Storage** page.
- 3. Complete all of the wanted fields for adding a credential. Remember to check **Include HMAC Credential** for obtaining a **Hash-based Message Authentication Code (HMAC)** credential.
- 4. Find your new service credential in the service credentials table.

To view your credential information, such as your secret and access keys, click the arrow that is next to the **View credentials** field. For more information, see [Service credentials](#) and [Bucket permissions](#).

Using a custom boot image to provision a new instance

Complete the following steps to provision a new instance by using a custom boot image. For more information about importing a custom boot image by using the Power Virtual Server CLI, see [Importing a boot image](#). Large boot images take time to successfully import. You might experience a delay before you receive a confirmation message.

- 1. Before you create a new Power Virtual Server instance, go to **Boot images** and click **Import**.
- 2. After you click **Import**, refer to the following table to complete the **Import boot image** fields:

 **Important:** The **Image file name** field supports the following formats: *.ova*, *.ova.gz*, *.tar*, *.tar.gz* and *.tgz*.

Field	Description
Catalog image name	Enter the name that you want displayed in your catalog.
Storage type	Select whether you want Tier 1 or Tier 3 for the storage type. A VM cannot have disks from both Tier 1 and Tier 3 storage types. For more information, see Storage tiers .
Region	Select the region from the drop-down list.
Image file name	Enter the file name of the image. The image file name must not contain spaces. Supported file formats are <i>tar</i> and <i>ova</i> . You can compress image files by using <i>gzip</i> . The supported file name extensions are <i>.ova</i> , <i>.ova.gz</i> , <i>.tar</i> , <i>.tar.gz</i> and <i>.tgz</i> . You must use the private endpoint domain. For example, <i>Aix_7200-03-02-1846_cldrdy_112018.gz</i> .
Bucket name	Sub folders can be used and specified as <i>bucketName/optional/folders</i> . Optional folders are created automatically if they don't exist. Optional folders can be added during an export image operation to Cloud Object Storage. To identity your bucket name, select Menu icon ≡ > Resource list > Storage > Cloud Object Storage name > Buckets .
Cloud Object Storage access key	To identify your access key, select Menu icon ≡ > Resource list > Storage > Cloud Object Storage name > Service credentials > View credentials . Copy the <i>access_key_id</i> value and past it into this field.

Cloud Object Storage secret key

To identify your secret key, select **Menu icon**  > **Resource list** > **Storage** > **Cloud Object Storage name** > **Service credentials** > **View credentials**. Copy the `secret_access_key` value and paste it into this field.

Importing a boot image

1. Return to **Virtual server instances** and provision a new Power Virtual Server instance. Click the arrow in the appropriate boot image tile to see your custom boot image.

 **Tip:** To download your image at a later point, go to the **Resource List** in the Power Virtual Server user interface. Once there, access your **Cloud Object Storage** workspace. In the bucket where your image is stored, select the image file that you want to download and select **Download objects**. See [Download an object](#) for the Cloud Object Storage CLI command.

Managing server placement groups

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Server placement groups provide you with control over the host or server on which a new virtual server instance (VSI) is placed. By using server placement groups, you can build high availability within a data center.

You can apply an affinity or anti-affinity policy to each VSI within a server placement group. After you create a placement group, you can provision a new VSI in the placement group. When you set a placement group with an affinity policy, all VSIs in the placement group are provisioned on the same server. When you set a placement group with an anti-affinity policy, all VSIs in the placement group are provisioned on different servers.

 **Note:** When a host with VSIs fails unexpectedly, the VSIs do not automatically restart on another host if they belong to an affinity-based server placement group and one or more VSIs in the group are hard-pinned to the failed host. Affinity policies require all VSIs in the group to remain on the same host. A hard-pinned VSI prevents the VSIs in the group from moving to another host.

 **Important:** You can use the user interface for placement groups only when the total number of VSIs in your account is less than 100. If your account has more than 100 VSIs, you must use the CLI or API to create placement groups.

Creating server placement groups by using the Power Virtual Server user interface

You can create server placement groups to provision VSIs with an affinity or anti-affinity policy. Each placement group that you create must have a unique name. You can create a maximum of 25 placement groups. If you need to create more than 25 placement groups, raise a support ticket to increase the maximum limit. Use the following API to create a placement group: [Getting help and support](#).

To add a VSI to a placement group, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. From the Workspaces page, select a workspace. The Virtual server instances page is displayed.
5. Click the **Server placement groups** tab. The Server placement groups page is displayed with a list of existing placement groups.
6. From the placement group table, click the placement group name. The placement group details page is displayed.
7. In the **Virtual server instances in placement group** section, you can create a new VSI or add an existing VSI:
 - To create a new VSI, click **Create new** and follow the steps in [Creating a Power Virtual Server](#).
 - To add an existing VSI, click **Add existing**. The Add existing virtual server instance panel is displayed with a list of available VSIs. Select one or more VSIs from the table, and click **Add**.

 **Note:** VSIs that you select must be on the same server or different server based on the colocation policy of the placement group. A VSI can belong to only one placement group, and VSIs that are already placed in a different placement group are disabled in the table.

Managing the shared processor pool

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

A Shared Processor Pool (SPP) is a pool of processor capacity that is shared between a group of virtual server instances (VSI).

In SPP, reserved cores can be adjusted based on availability, unlike a VSI with a fixed processing capacity.

The following table shows how an SPP is used to reduce the licensing cost when you pay per core:

Use of SPP	VSI 1	VSI 2	Reserved cores in Pool (User defined)	License requirement per core
No	Maximum cores = 5 Mode = Dedicated	Maximum cores = 6 Mode = Dedicated	NA	5+6 = 11
Yes	Maximum cores = 5 Mode = Shared/Uncapped Entitled capacity = 4.25	Maximum cores = 6 Mode = Shared/Uncapped Entitled capacity = 5.25	Maximum cores = 10	10, Determined by the reserved cores in a pool

SPP helps to reduce the licensing cost

The benefits of using an SPP are as follows:

- Control licensing costs by limiting the number of processors an uncapped partition can use, reducing the number of software licenses.
- A better overall ability to manage processor resources.

Power Virtual Server always has at least one defined SPP as the default pool. You can add up to 63 more SPPs to a single Power Virtual Server host. A set of VSIs that belongs to the same machine type (host) can use and share an SPP.

Managing the core-to-virtual core ratio

Client location

For IBM Power Virtual Server Private Cloud, the minimum core-to-virtual core ratio is 1:20. The minimum Entitled Capacity (EC) must be 0.05 and can be incremented by 0.05.



Note: The create or update operations of a VSI fails even if the core-to-virtual core ratio of 1:20 is satisfied but the configured count of vCPU exceeds the maximum supported per-instance limit as defined by the system policy.

IBM data center

In a Power Virtual Server environment with Power10 or later systems, you can provision a VSI in an SPP with up to 3.0 virtual cores (VC). For any Entitled Capacity (EC) up to 3.0 cores, you can select the ceiling of cores in VC up to 3.0 cores. For EC more than 3.0, VC is rounded off to the next higher integer value. VC is always equal or greater to EC.

For example, the following table shows how your VC is determined based of EC selection:

User-defined EC value	Possible VC value
EC is set to 1.5	VPs can be 2 or 3
EC is set to 2.25	VP is 3
EC is set to 3.5	VPs is 4

Example showing core-to-virtual core ratio

Review the following points when you use core-to-virtual core ratio:

- You can set EC in increments of 0.25.
- You can set VC in increments of 1.
- The ratio can be set only on a user-defined SPP.
- For non-dedicated hosts on Power10, you can set up to 3.0 cores.
- For Power9 and for virtual machines with entitled capacities greater than 2, the core-to-virtual core ratio is 1:1.

You can specify the host affinity and anti-affinity between two or more SPPs with SPP placement groups. For more information, see [Configuring shared processor pool placement group](#).

Pricing for Shared Processor Pool

When you use SPPs, you can optimize the cost by using reserved cores that are shared by VSIs in the SPP. For more information about the pricing for VSIs within SPPs, see the following topics:

- [Pricing for shared processor pool](#) if you are using IBM Power Virtual Server in IBM data center
- [Pricing for shared processor pool](#) if you are using IBM Power Virtual Server Private Cloud in Client location

Configuring a shared processor pool

SPP is a separate resource within a Power Virtual Server workspace. You can manage the entire SPP lifecycle with the Power Virtual Server user interface.

The SPP and your Power Virtual Server workspace are linked and thus shared resource pools are not visible from different accounts. You can create one or more SPPs per workspace.

Create an SPP by specifying the following parameters:

- A unique name
- Host group or the machine type
- Number of processors to reserve.

When you define these parameters, a backend process determines the best host for the new SPP.



Note: If the SPP is not configured successfully on the host, the SPP is not allocated with any processing cores. These SPPs must be removed manually, as they are not automatically deleted.

Creating a shared processor pool

To create an SPP, complete the following steps:

1. Go to **Shared processor pools** in the Power Virtual Server user interface under **Compute**.
2. Click **Create pool**.
3. In the **Create new shared processor pool** window, define the following preferences based on your requirements:

Field	Description
Name	Enter a name that is unique within your cloud account. Use a name of minimum 2 characters and a maximum of 12 characters. Special characters are not allowed except underscore ('_').
Add to a pool placement group	Select the checkbox if you want to deploy the SPP directly into an existing pool placement group. If the pool has the required affinity relation with other pools, the best practice is to deploy the pool directly into the placement group. You must create the pool placement group first. It prevents the pool from being deployed on a host that does not satisfy the affinity requirements, and having to move it later.
Select machine type	Specify the machine type. For more information about hardware specifications, see Hardware specifications .
Reserved processing cores [1]	IBM data center For Power Virtual Server, the core-to-virtual core ratio is 1:1 by default. Client location For IBM Power Virtual Server Private Cloud, the minimum core-to-virtual core ratio is 1:20. The minimum entitled capacity must be 0.05 and can be incremented by 0.05.

Creating a new SPP fields and descriptions

4. Click **Create**.

Updating or deleting a shared processor pool

To update an SPP, navigate to **Compute > Shared processor pools**. Then, click the SPP that you want to edit.

You can update the following details of an existing SPP:

- Name of the SPP: Follow the same naming conventions that you used when you created an SPP. For more information, see [Creating a shared processor pool](#).
- Number of cores: You can update the number of reserved cores based on resource availability and allocation.

To delete an SPP, navigate to **Compute > Shared processor pools**. At the end of the SPP row that you want to delete, click the delete icon. Before you delete, ensure that VSIs do not exist in the SPP. If VSIs are present in the SPP, delete all VSIs in the SPP.

Managing a VSI of a shared processor pool

When you deploy a VSI, you can choose an SPP instead of a host. As you cannot move a VSI into or out of an SPP, you can deploy a VSI directly into an SPP as a best practice. You can also deploy a VSI into a server placement group; the selected SPP host must be compatible with the affinity policy of the placement group that you selected.



Note: When you deploy multiple VSIs simultaneously, a new server placement group is created automatically.

During any planned maintenance activity or when you want to perform a remote restart operation, ensure that the VSIs are linked to an SPP based on your requirements.

Deploying a VSI into a shared processor pool

To add VSIs to an existing SPP, complete the following steps:



Important: **Client location** Use the API and CLI for bulk deployment of VSIs in an SPP. However, you cannot provision more than 3 VSIs within an SPP in both small and medium pods.

1. Open the **Virtual server instances** page in the Power Virtual Server user interface.
2. Click **Create instance**.
3. Complete the input fields under the **General** tile based on your requirement.
4. Select the checkbox **Add to a Shared processor pool**.
5. Select an existing SPP.
6. Continue with the process of creating a VSI. For more information, see [Configuring a Power Virtual Server instance](#).

Assigning an existing VSI to a shared processor pool

You can assign a VSI to a shared processor pool (SPP) only during the deployment of the VSI. If you deployed a VSI without assigning the VSI to an SPP, you cannot assign the VSI to an SPP directly. Instead, you must export the VSI as an image and then deploy a new VSI from that image. During the deployment, you can assign the new VSI to a new SPP or an existing SPP.

To assign an existing VSI to an SPP, complete the following steps:

1. Identify an existing SPP that you want to use or create an SPP by following the steps in the [Creating a shared processor pool](#) section.
2. Identify the existing VSI that you want to assign to an SPP and then capture and export the VSI by following the steps in the [Using the Power Virtual Server user interface to capture and export a VSI](#) section.



Important: When you capture and export the VSI, ensure that you select **Image catalog** as the export destination because exporting to Image catalog allows faster VSI deployment. If you select **Cloud Object Storage** as the export destination, you must import the image again from the Cloud Object Storage, which can increase the VSI deployment time.

3. Deploy a new VSI to an SPP by following the steps in the [Deploying a VSI into a shared processor pool](#) section. When you deploy the VSI, ensure that you make the following selections:
 - Select the **Add to a Shared processor pool** checkbox and then select the SPP that you want to use from the list.
 - In the Boot image section, select the VSI image that you exported in step 2 from the **Image** list.

Complete all the remaining steps in the [Deploying a VSI into a shared processor pool](#) section to complete the VSI deployment.



Tip: After you verify that the new VSI is deployed successfully, you can delete the VSI that was used for the *Capture and export* operation and the exported image to prevent ongoing charges and optimize resource utilization. For more information, see [Deleting a virtual server instance](#).



Important: The VSI capture, image export, and image import operations are restricted to one operation at a time in each Power Virtual Server workspace. If one of these operations is submitted successfully, then another operation cannot be submitted until the previous operation is complete.

Configuring a shared processor pool placement group

You can configure an SPP with the SPP placement group (SPP PG) to control the host in which SPPs are deployed.



Note: SPP PGs are different from server placement groups. They serve the same purpose but cannot combine resource types.

You must define the following parameters to define an SPP PG:

- A unique name - The SPP PG name must be unique in your cloud instance.
- Policy - You can set the colocation policy of the SPP PG as the same server (affinity) or a different server (anti-affinity).

The following table explains how the host selection is determined based on the different options of SPP PG policy selection:

SPP PG Policy	Host selection	Result
Same server (Affinity)	The new SPP is configured on the same host as determined by the existing PG SPP members.	The new SPP is automatically added as a member of the PG policy.
Different server (Anti-affinity)	Configure the new SPP on a different host than all host identified by all the existing PG members.	The new SPP is automatically added as a member of the PG policy.

Host selection when SPP PG is empty and non-empty

SPP PG Policy	Host selection	Result
Same server (Affinity)	A new SPP is configured on a host that is selected based on the backend processing	The new SPP is automatically added as a member of the PG policy
Different server (Anti-affinity)	A new SPP is configured on a host that is selected based on the backend processing	The new SPP is automatically added as a member of the PG policy

Host selection when SPP PG is empty and non-empty

Creating a shared processor pool placement group

To create an SPP PG, complete the following steps:

1. Go to **Shared processor pools** in the Power Virtual Server user interface under **Compute**.
2. Click the **Pool placement groups** tab.
3. Click **Create group**.
4. In the **Create new pool placement group** window, enter the following details:

Field	Description
Name	Enter a name that is unique within your cloud account. Use a minimum of 2 characters and a maximum of 12 characters. Special characters are not allowed except underscore ('_').
Policy	Same Server (Affinity) Different server (Anti-affinity)

Create a new SPP PG field description

5. Click **Create**.

You receive a notification when the new SPP PG is created.

Adding a shared processor pool to a placement group

You can add an SPP as a member to an existing SPP PG if the host of the SPP follows the affinity policy of the group. If the PG policy is set to **Same server** (affinity), all SPPs that are added as a member of the placement group must remain on the same host. If the PG policy is set to **Different server** (anti-affinity), the SPP is deployed on a different host other than all its group members.

To add an SPP to an existing SPP PG, complete the following steps:

- 1. Go to **Shared processor pools** in the Power Virtual Server user interface under **Compute**.
- 2. Click the **Pool placement groups** tab.
- 3. Select an existing pool placement group to which you want to add an SPP.
- 4. Select one of the following options:

Field	Description
Add existing	You can choose from existing SPPs.
Create new	You can create an SPP from scratch by defining pool name, machine type, and number of cores. For more information, see Create a new SPP

Add SPP to an SPP PG: field description

You will receive a notification after the new SPP PG is created.

Deleting a shared processor pool placement group

To delete an SPP PG, complete the following steps:

- 1. Go to **Shared processor pools** in the Power Virtual Server user interface under **Compute**.
- 2. Click the **Pool placement groups** tab.
- 3. Click the SPP PG that you want to delete.
- 4. In the details page, click the delete icon.



Note: Deleting a placement group does not delete its members or change the host in which the placement groups are deployed.

- 1. The Reserved processing cores are the number of cores that are reserved for processing and must always be a whole number. [↩](#)

Managing a Falconstor Virtual Tape Library

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

FalconStor Virtual Tape Library (VTL) is an optimized backup and deduplication solution that provides tape library emulation, high-speed backup or restore, data archival to supported S3 clouds for long-term storage, global data deduplication, enterprise-wide replication, and long-term cloud-based container archive, without requiring changes to the existing environment.

Here are some of the StorSafe VTL Benefits

- Up to 95% data reduction. FalconStor reduces the storage capacity required by backup data in the client-managed environment and in the cloud by up to 95%.
- Ransomware Protection. Recover data from any point in time with air-gapped, immutable backups.
- Offsite Protection. For offsite protection, StorSafe VTL exports virtual tapes as physical tapes, and keeps track of their location with a management dashboard. Virtual tapes can remain in the library for fast restores or can be stubbed when exported to local object storage.
- High Performance. Improve both backup and recovery.
- Modernize. FalconStor is 100% compatible with existing backup software, hardware, and operational procedures.

VTL is offered through the IBM Cloud Catalog for both IBM data center and Client location with the following names:

- [FalconStor StorSafe VTL for Power On-Premises](#) and
- [FalconStor StorSafe VTL for PowerVS Cloud](#).

For more information on how to create and configure an IBM® Power® Virtual Server running the FalconStor VTL software in the IBM Cloud®, see [FalconStor VTL for IBM Deployment Guide](#).

Migrating by using Falconstor StorSafe VTL

See [FalconStor StorSafe VTL](#) for migration steps.

Capturing and exporting a VSI

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can capture and export a virtual server instance (VSI) by using the IBM® Power® Virtual Server user interface, CLI, or API. When you capture a VSI, the boot volume and any attached data volumes are saved as a volume-backed image. The image is stored in new volumes on the storage providers. You can export the captured image to the image catalog, an IBM Cloud Object Storage bucket, or both. When an image is exported, the volumes of the image are copied and packaged in an Open Virtualization Appliance (OVA) file. The OVA file is compressed by using *gzip* before it is uploaded to your selected destination.

When you capture and export a VSI, you can choose the image catalog, Cloud Object Storage, or both as destinations. The image catalog is stored on the IBM Power storage area network (SAN). IBM Cloud Object Storage is encrypted and dispersed across multiple geographic locations, and accessed over HTTP by using a REST API. IBM Cloud Object Storage uses the distributed storage technologies that are provided by the IBM Cloud Object Storage System (formerly Cleversafe). After capturing an image to the image catalog, you can export it to Cloud Object Storage as a separate operation. You can also deploy the captured image to create a clone of the VSI by using a different network configuration.



Note: The maximum image size that you can export to Cloud Object Storage is 10 TB.

The IBM® Power® Virtual Server Job feature tracks long-running asynchronous operations like VSI capture, image export, and image import across multiple workspaces in your cloud account.



Important: The VSI capture, image export, and image import features are restricted to one operation at a time per Power Virtual Server workspace. Wait until one operation is complete before you submit a new operation (VSI Capture, Image Export, and Image Import).



Note: You are charged different rates based on whether you export to the image catalog or Cloud Object Storage.

Before you capture an IBM i VSI, ensure that any buffer I/O memory is flushed (written) to the disk by using the following command:

```
CHGASPACT ASPDEV(*SYSBAS) OPTION(*FRCWRT)
```

Capturing and exporting a VSI by using the Power Virtual Server user interface

Complete the following steps to capture and export a VSI:

1. Log in to the [IBM Cloud catalog](#) with your IBM credentials.
2. In the search box, type Power Virtual Server and click the **Power Virtual Server** tile.
3. Click **Workspaces** in the navigation panel. The Workspaces page with a list of existing workspaces is displayed.
4. Select your workspace from the displayed list of workspaces.
5. Click **Virtual server instances** in the navigation panel.
6. Select the VSI that you want to capture. Ensure that the VSI is in an active state. The **Virtual server instance details** page is displayed.
7. On the **Virtual server instance details** page, click options menu (3 vertical dots) and select **Capture and export**. The Capture and export virtual server pane is displayed.
8. In the **Capture contents** section, the boot volume of the selected VSI is automatically selected. You can optionally select any additional data

volumes that are attached to the VSI.



Note: Any volumes that are not shown are not on the same storage pool as the boot volume and cannot be included in the capture.

9. Click **Next**.

10. In the **Destination** section, select the export destination for the captured image:

- **Image catalog:** Stores the image in your workspace image catalog for future deployments
- **IBM Cloud Object Storage:** Exports to IBM Cloud Object Storage

Image catalog

If you selected **Image catalog** as the export destination, complete the following steps:

1. Enter an image name in the **Image name** field.
2. Optional: enter tags in the **User tags** field.

IBM Cloud Object Storage

If you selected **IBM Cloud Object Storage** as the export destination, complete the following steps:

1. Select the **Region** where your IBM Cloud Object Storage bucket is located.
2. In the **Bucket name** field, enter the name of the bucket where you want to export the image. Optionally, you can specify folder paths within the bucket.
3. In the **HMAC access key** field, provide your HMAC access key. For more information, see [Generating HMAC credentials](#).
4. In the **HMAC secret access key** field, provide your HMAC secret access key.
5. Set **Generate checksum file** to **On** to generate a checksum file.

The checksum file is created and placed in the IBM Cloud Object Storage bucket along with the exported image. The checksum file name is based on the name of the image file and has the file extension `.sha256`. Use the command `shasum -a 256` to verify that the copied file is correct.

If you are creating your own image, you can create a checksum image and place it with your own image in the same bucket. One of the ways to generate the checksum image is by using the command `shasum -a 256 <filename>` or `sha256sum <filename>`.



Note: Generating a checksum file might increase the image capture and export time.

11. Click the terms and conditions link to read the IBM Cloud Terms of Use. To continue, select the **I agree to the Terms and conditions** checkbox.

12. Click **Capture and export**. The **Capture & export virtual server** dialog is displayed with information about the capture operation. Review the information and click **Capture and export**.

Capturing and exporting a VSI by using the Power Virtual Server CLI

To capture and export a VSI, use the [ibmcloud pi instance-capture](#) command. You can export it to the image catalog, Cloud Object Storage, or both.

```
ibmcloud pi instance-capture INSTANCE_ID --destination DEST --name NAME [--volumes "VOLUME1 VOLUME2"] [--access-key KEY] [--secret-key KEY] [--region REGION] [--image-path TYPE]
```

Capturing and exporting a VSI by using the Power Virtual Server API

To capture and export a VSI to the image catalog or IBM Cloud Object Storage by using the API, use the [Add a capture pvm-instance to the jobs queue](#) method.

Viewing the capture and export results

After you click **Capture and export** in the confirmation dialog, a notification is displayed confirming that the capture and export process has started. A **Capture and export in progress** progress bar is displayed on the **Overview** tab of the selected virtual server instance.

You can monitor the progress of the operation by clicking **View details** next to the progress bar.

To view details of a capture and export job by using the CLI, use the `ibmcloud pi instance capture show` command.

To view details of a capture and export job by using the API, use the `Get detail of last capture job` method.



Important: If you select large volumes, the export process can take a significantly long time.

When the capture and export operation is complete, another notification is displayed indicating the status of the operation.

To view your exported image:

- If you exported your volume-backed image to IBM Cloud Object Storage, navigate to your Cloud Object Storage bucket to view the image.
- If you exported your volume-backed image to the image catalog, navigate to the **Boot images** page to view the image.

To see your newly exported image in the image catalog by using the CLI, use the `ibmcloud pi image-list-catalog` command:

```
ibmcloud pi image list-catalog [--long] [--json]
```

To see your newly exported image in Cloud Object Storage by using the CLI, use the `ibmcloud cos list-objects` command:

```
ibmcloud cos list-objects --bucket BUCKET_NAME [--delimiter DELIMITER] [--encoding-type METHOD] [--prefix PREFIX] [--starting-token TOKEN] [--page-size SIZE] [--max-items NUMBER] [--region REGION] [--json]
```

Related information

For information about exporting boot images from your image catalog to IBM Cloud Object Storage, see [Exporting a boot image](#).

For information about importing boot images from IBM Cloud Object Storage, see [Importing a boot image](#).

Downloading fixes and updates

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

You must use the AIX [Service Update Management Assistant \(SUMA\)](#) or the IBM i `Send PTF Order (SNDPTFORD)` command to download fixes and updates from the IBM Fix Central website.

If you'd like to download fixes and updates, you must complete one of the following actions:

- Put your virtual machine (VM) on the public network. You can add a public network [during](#) or [after](#) the provisioning stage. Depending on your VM, see the [SUMA tasks and the command-line](#) section for information on the **suma** command or the [SNDPTFORD command](#).
- Set up another VM as an AIX [NIM server](#) or an IBM i [Network installation Server](#).
- Set up another public-facing VM with an [HTTP/HTTPS proxy](#).

Ordering fixes and updates for AIX VMs

Learn more about ordering fixes and updates for AIX VMs by using SUMA.

Understanding SUMA

SUMA sets up an automated interface to download fixes and updates from a fix distribution website to your system. You can configure SUMA to periodically check the availability of specific new fixes and technology levels. With SUMA, you don't have to manually retrieve maintenance updates.

When you configure SUMA in an AIX logical partition (LPAR) or as the NIM master, SUMA establishes a connection to the fix distribution website and downloads the available service update.

To verify that SUMA can get through to the IBM fix servers, run the following command (from the AIX system where you want to download the fixes):

```
$ /usr/esa/bin/verifyConnectivity -tw
```

If the tests fail, work with your network security team to determine why you are unable to access the servers.

You can access the SUMA configuration by running the [suma command](#) or by using the `SMIT suma` fast path. When you create a SUMA policy, you must specify one of the following request types that specifies the type of download:

- **PTF**: Specifies a request to download a program temporary fix (PTF), such as U813941. Only certain PTFs can be downloaded as an individual file set. This limitation applies to PTFs that contain either the *bos.rte.install* or *bos.alt_disk_install.rte* file sets as well as those that are released in between Service Packs (SP). Otherwise, you must download the technology level (TL) or SP.
- **TL**: Specifies a request to download a specific TL (such as 7200-02).
- **SP**: Specifies a request to download a specific SP (such as 7200-02-00).
- **Latest**: Specifies a request to download the latest fixes. This value returns the latest SP or TL as specified in the **FilterML** attribute.

Configuring SUMA to use the proxy settings

Before you run the `suma` command to download any updates, ensure that the AIX LPAR is authenticated to access the internet and that your LPAR knows in which data centers it is running.

To make the LPAR aware in which data centers it is running, run the following command:

```
#echo "COUNTRY_CODE = **" >> /var/suma/data/config.suma
```

```
#export SUMA_COUNTRY_CODE=**
```

Where `**` is the country code of your DC based on the following table:

POD Location	Country	Country code
Sao Paulo	Brazil	BR
Washington & Dallas	USA	US
Toronto & Montreal	Canada	CA
London	UK	GB
Frankfurt	Germany	DE
Madrid	Spain	ES
Sydney	Australia	AU
Tokyo & Osaka	Japan	JP
Chennai	India	IN

POD location with their respective country code

To verify that the LPAR is connected to the internet, enter the following command:

```
$ suma -x -a Action=Preview -a RqType=LatestCopy
```

The `suma` command allows you to preview only the download operation. When you run this command, files are not downloaded. If the LPAR is not authenticated to access the internet, the command returns an error message. For information on troubleshooting SUMA error messages, see [Troubleshooting SUMA error messages](#).

Complete the following steps to configure SUMA to use the proxy settings:

1. Ensure that the *bos.ecc_client.rte* file set is installed on the AIX LPAR by running the following command:

```
$ lspp -h bos.ecc_client.rte
```

2. Determine whether the `config_conn_path` command is available in the `bos.ecc_client.rte` file set by running the following command:

```
$ lspp -w /usr/ecc/bin/config_conn_path
```

3. Configure your proxy settings by completing the following steps:

- a. Run the `smit srv_conn` command.
- b. Select **Create/Change Service Configuration** and press **Enter**.
- c. Select **Create/Change Primary Service Configuration** and press **Enter**.
- d. Set the following fields in the SMIT interface:

Configuring proxy settings:

```
$ Type or select values in entry fields.
Press Enter AFTER making all changes.

                                     [Entry Fields]
Connection type                      [HTTP_Proxy]
Test service configuration           [Yes]

If type is DIRECT_INTERNET, no entry required.

If type is HTTP_PROXY,
  IP address                        [xx.xx.xx.xx]
  Port number                       [5026]
  Authentication user ID            []
  Authentication password requested interactively.
```

Where, `xx.xx.xx.xx` is the IP address of the proxy and `5026` is the port number that is used to connect to the proxy settings. When you press **Enter**, a test connection determines whether the AIX LPAR is authenticated to access the internet by using the proxy settings. The common values for proxy port number are `3138` or `8080`.

- 4. Run the `smit suma_config_base` command to access the SUMA base configuration SMIT interface. Verify the fields that are shown in the **Base Configuration** screen capture.



Important: For the **Fixserver protocol** field, *https* is the only option. For the **Download protocol** field, *http* is the default option. You can change the default option to *https* for a secure connection. If you set the **Download protocol** to *https*, the downloads are slower but more secure because *HTTP* provides multi-threaded performance and *HTTPS* provides single-threaded performance.

Base SMIT configuration:

```
$          Base Configuration

Type or select values in entry fields.
Press Enter AFTER making all changes.

                                     [Entry Fields]
Screen output verbosity              [Info/Warnings/Errors] +
Logfile output verbosity             [Verbose]                +
Notification email verbosity         [Info/Warnings/Errors] +
Remove superseded filesets on Clean? Yes                        +
Remove duplicate base levels on Clean? Yes                      +
Remove conflicting updates on Clean? Yes                        +
Fixserver protocol                   https                      +
Download protocol                     http                      +
Maximum log file size (MB)           [1]                      #
Download timeout (seconds)           [180]                     #
```

SUMA tasks and the command line

The `suma` command can be used to perform various operations on a SUMA task or policy. An **RqType** parameter specifies the type of download that is being requested, such as a TL, SP, or Latest. You can use several flag options with the `suma` command to complete the following tasks:

- Create
- Edit
- List
- Schedule
- Unschedule
- Delete

To create and save a SUMA task by using the command line, enter the following command:

```
$ suma -w -a DisplayName=' AIX72TL2SP2' -a FilterML='7200-00'
```


The command returns a task ID after the successful creation of a SUMA task:

```
$ Task ID 10 created.
```

To create and schedule a task that downloads the latest fixes and adds a policy label through the **DisplayName** field (useful when you are listing policies through SMIT), enter the following command:

```
$ suma -s "30 2 15 * *" -a RqType=Latest \
-a DisplayName="Latest fixes - 15th Monthly"
```

Using the Electronic Service Agent (ESA) to download fixes and updates for IBM i VMs

Learn how to configure a [Universal Connection to IBM services](#). If you don't want to assign a public IP to an IBM i VM, you can use a multi-hop connection (proxy server) to download fixes and updates. You can also use the `SNDPTFORD` command and a network installation server to do the same.

The SNDPTFORD command

You can use the `SNDPTFORD` command to order and receive IBM-supplied fixes (or PTFs) for the IBM i environment and IBM-supplied applications. You can use this command over the electronic customer support configuration that uses TCP/IP connectivity through *Universal Connection*. You can use the `SNDPTFORD` command to order the following types of fixes and related information:

- Separate or accompanying cover letters
- Individual fixes
- Multiple fixes
- Cumulative PTF packages
- PTF groups
- PTF summaries
- Cross-reference summary lists
- Preventive service planning tips

For example, the following command sends a request for PTF numbers SI12345 and SI12346:

```
$ SNDPTFORD PTFID((SI12345) (SI12346))
```

For more information, see [Send PTF Order \(SNDPTFORD\)](#) and [Ordering fixes by using the Send PTF Order command](#). If you can't put your IBM i VM on the network, set up a network install server as instructed in the next section.

Setting up an IBM i network install server

Before you can install or upgrade an IBM i system through the network, you must [set up a network installation server](#). The network installation server contains images of the IBM i operating system, its internal code as well as licensed programs and PTFs.

Creating a snapshot, restoring a snapshot, and cloning a volume in Power® Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can capture full, point-in-time copies of the logical volumes or data sets by using IBM® Power® Virtual Server. You can create delta snapshots, volumes-clones, and restore your disks by using *FlashCopy* feature by IBM, Power® Virtual Server APIs.

Creating a snapshot

With snapshots, you can create a relationship between your source disks and a target disk at a defined time, for example **T1**. Target disks are created as part of creating a snapshot operation. The snapshot operation tracks the changes that are made to the source disk after **T1** time. Thus, you can restore the source disks to their **T1** state at a later time.

You can create and manage snapshots in your Power Virtual Server workspace by using the GUI, API, or CLI. Snapshot names must be unique for your workspace.

Creating a snapshot: A use case

You can create a snapshot for different scenarios. Consider the following use case:

An administrator plans to upgrade the middleware. If the upgrade fails, the administrator must restore the source disk to its original state. The administrator completes the following steps:

1. Starts a snapshot of the source disk that contains the middleware by using the GUI, API, or CLI. A snapshot of the source disk is created.
2. Upgrade the middleware.

If the upgrade fails, the administrator restores the source disk by using the snapshot. If the upgrade succeeds, deletes the snapshot.

You can start multiple snapshot operations. These concurrent snapshot operations occur on different sets of disks.

Prerequisites for creating a snapshot

- Before you take a snapshot of a volume, it is recommended that you quiesce all the applications that access the volume. This action is required to ensure that all the data is moved to the disk to prevent loss of data.
- If you are creating a snapshot of a volume that is shared among multiple virtual server instances, you must quiesce all the applications on all these virtual server instances. If you are not sure how to quiesce your applications, it is recommended to shut down any virtual server instances that are attached to the volumes included in the snapshot.
- To reuse a snapshot name, delete the existing snapshot with that name first.
- You can include a volume in the snapshot if the volume belongs to the same storage pool.
- You can include a volume in the snapshot if the volume is not in the **error** state and the health state of the volume is in the **ok** state.

Creating a snapshot by using the Power Virtual Server user interface

To create a snapshot by using the Power Virtual Server user interface, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#) and log in with your credentials.
2. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
3. Select a workspace in which you want to create the snapshot. The Workspace details panel is displayed.
4. Click **View virtual servers**. The Virtual server instances page is displayed.
5. In the navigation panel, click **Storage > Instance snapshots**. The Instance snapshots page is displayed with a list of any existing snapshots.
6. Click **Create instance snapshot**. The Create instance snapshot page is displayed.
7. In the General section, enter a name for the snapshot in the **Name** field.
8. Optional: Enter user tags and description for the snapshot in the **User tags (optional)** and **Description** fields.
9. Click **Continue**.
10. In the Volumes section, select the required virtual server instance from the **Select virtual server instance** list. Any storage volumes that are attached to the selected virtual server instance are displayed.
11. Select the storage volumes to include in the snapshot and click **Finish**. The summary of the selected snapshot configuration and its total estimated cost is displayed in the Summary panel.



Note: You can select multiple storage volumes to include in the snapshot. However, all the selected volumes must belong to the same storage pool.

12. Optional: Click **Add to estimate** to add the cost information to an existing estimate or to create a new estimate.
13. Click the terms and conditions link to read the [IBM Cloud Terms of Use](#). To continue, select the **I agree to the Terms and conditions** checkbox and click **Create**.

After the snapshot is successfully created, a success message is displayed on the screen.

The snapshot that you created is displayed in the Instance snapshots page with the following details:

- **Name:** Snapshot name.
- **Volumes:** Number of storage volumes that are included in the snapshot.
- **Description:** Snapshot description.

- **Virtual server instance:** Name of the virtual server instance to which the selected storage volumes were attached at the time of the snapshot creation.
- **Created:** Timestamp that indicates when the snapshot was created.
- **Status** and **Status details:** These two columns display the current status of a snapshot. A snapshot can have one of the following statuses:
 - **Available:** The snapshot was successfully created and is available for restore operations.
 - **Error:** An error has occurred while creating the snapshot. The snapshot cannot be recovered and must be deleted.
 - **Restoring:** A restore operation is in progress.
 - **Restore error:** An error has occurred during the restore process. You can use the **Restore** option from the overflow menu to retry the restore process. To revert a snapshot with *Restore error* status to its last working state, you can use the **Rollback** option from the **Actions** menu on the snapshot's details page.
 - **Restore reverted:** The restore attempt was reverted. A storage volume that is included in the snapshot might be involved in another snapshot operation. You can retry the restore once the other snapshot operation is complete.
 - **Rolling back:** A rollback operation is in progress. The rollback operation reverts the virtual server instance to its last working state.
 - **Rollback error:** An error has occurred during the rollback process.
- **Last update:** Timestamp that indicates when the snapshot was last updated.

Creating a snapshot by using API or CLI

You can create a snapshot by using the following API and CLI:

- **API:** [ibmcloud pi snapshot create](#)

You must provide values for the following parameters in the API and CLI:

- **name**: Required. Specifies the name of the snapshot. The **name** must be a unique value for each snapshot in the workspace.
- **description**: (Optional) Describes the snapshot.
- **volumeIDs**: (Optional) Includes a list of one or more volumes that you must include in the snapshot. If the parameter is not specified, all the volumes that are attached to the virtual server instance (VSI) are included in the snapshot.

You can use the [Get a list of flashcopy mappings of a given volume](#) API to confirm that the operation to create a snapshot of a volume is completed before you retry the resize operation.

Considerations for creating a snapshot

You must consider the following considerations when you create a snapshot to avoid the operation failure:

- You must attach all the volumes that are included in the snapshot to the same virtual server instance.
- When you create a snapshot by using the API or CLI, the **volumeIDs** parameter is optional. If the **volumeIDs** option is not specified, all the volumes that are attached to the virtual server instance are included in the snapshot.
- If the volumes that are attached to the virtual server instance are not in the same storage pool, you must create the snapshot for each storage pool. You must add the volume IDs in a storage pool to the **volumeIDs** option when you create the snapshot for the storage pool.



Important: If the create a snapshot operation fails, the snapshot state is set to the **error** state on the Instance snapshots page. You can delete the snapshot so that you do not perform any operations on it.

Restrictions for creating a snapshot

You must consider the following restrictions when you create a snapshot to avoid the operation failure:

- You cannot create a snapshot of a virtual server instance if its attached volumes are currently undergoing cloning or snapshot operations.
- You cannot modify the shareable or bootable properties of any volume that is included in an existing snapshot.
- You cannot run a snapshot operation from different virtual server instances at the same time for the same shared volume.
- You cannot resize a volume if the volume is included in a virtual server instance snapshot. To resize the volume, you must delete all the snapshots that contains the volume.
- You cannot detach the volumes from the virtual server instance if the volumes are included in an instance snapshot. To detach the volumes, you must delete all the snapshots that contains the volumes.

Restoring a snapshot

The restore operation restores the volumes that are included in a virtual server instance snapshot to the source disks. When the restore operation is in progress, Power Virtual Server creates a backup snapshot, which can be used if the restore operation fails. If the restore operation succeeds, the backup snapshots are deleted. If the restore operation fails, the snapshot status is set to `restore-error`. You can use the `restore` or `rollback` operations to restore the snapshot to its original state. Use the `restore_fail_action` query parameter with the `retry` value to retry the restore operation. To roll back the snapshot to a previous disk state, use the `restore_fail_action` query parameter with the `rollback` value. When the restore operation fails, the virtual server instance enters the **Error** state. For more information about `restore_fail_action` query parameter, see [Restoring a snapshot by using API or CLI](#).

Prerequisites for restoring a snapshot

Ensure that you meet the following prerequisites before you start the `restore` operation:

- By default, you must shut down the virtual server instance before you start the `restore` operation.
- To restore the data on a live virtual server instance, use the `force restore` operations. The following conditions must be met:
 - Quiesce the applications.
 - Do not run I/O operations on the volume disks.



Important: If you do not meet the conditions for `force restore` operation, it results in data corruption.

- If you are restoring an instance snapshot of a volume that is shared among multiple virtual server instances, you must quiesce all the applications on all the virtual server instances. When the restore operation is in progress ensure that I/O operations are not running on the shared volumes.
- If you are not sure how to quiesce your applications, shut down any virtual server instances that are attached to the volumes that you want to include in the snapshot.

Restoring a snapshot by using the Power Virtual Server user interface

To restore a snapshot by using the Power Virtual Server user interface, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#) and log in with your credentials.
2. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
3. Select the workspace which contains the snapshot that you want to restore. The Workspace details panel is displayed.
4. Click **View virtual servers**. The Virtual server instances page is displayed.
5. In the navigation panel, click **Storage > Instance snapshots**. The Instance snapshots page is displayed with a list of existing snapshots.
6. From the list of available snapshots, select the snapshot that you want to restore. The snapshot details page is displayed.
7. Select **Restore**. The Confirm restore dialog is displayed.



Important: To restore an instance snapshot, you must first shut down all the related virtual server instances, including the source and any shared volumes that are attached to the instance snapshot. If you cannot shut down the virtual server instances, quiesce all the related applications and set **Force restore** to **Enabled**.

8. Click **Restore**.

When the restore begins, a notification is displayed to indicate that the restore process has started and a progress bar displays the completion percentage. After the snapshot is restored successfully, the **Status details** column displays the Restore successfully completed status.



Note: A failed restore operation changes the snapshot status to **Restore error**. You can use the **Restore** option from the overflow menu to retry the restore process. To revert a snapshot with *Restore error* status to its last working state, select **Rollback** from the **Actions** menu on the snapshot's details page.

Restoring a snapshot by using API or CLI

You can restore a snapshot by using the following API and CLI:

- **API:** [ibmcloud pi snapshot-restore](#)

You must provide values for the following parameters in the API and CLI:

- `SNAPSHOT_ID`: Required. Specify a unique identifier for the snapshot
- `force`: (Optional) Specify the value of the flag as `True` or `False`. By default, the flag is set to `False`. This flag is set to `true` only if the virtual server

instance is shut down. By default, the virtual server instance must be shut down before you start a snapshot restore operation. When the `force` flag is set to `True`, the prerequisite of the virtual server instance being shut down is relaxed.

- **restore**: (Optional) Use the `restore_fail_action` query parameter only if a previous restore operation results in setting the snapshot to the `restore-error` status. The `restore_fail_action` query parameter accepts the following `retry` or `rollback` values:
 - If the snapshot status is set to `retry`, then again the failed restore operation is attempted.
 - If the snapshot status is set to `rollback`, then the volumes that were failed to be restored are rolled back to their original state.

If you are using the `force` option, you must ensure that all the applications that access the volumes in the instance snapshot are quiesced and no I/O operations are running on the volume disk.

Use cases for restoring a snapshot

Consider the following scenarios to restore a snapshot:

- Restoring all of the volumes that are included in a virtual server instance snapshot
- Running multiple virtual server instance restore operations
- Retrying a virtual server instance restore operation if it originally failed
- Rolling back a virtual server instance to its original volume state

Considerations for restoring a snapshot

- Before the system begins the `restore` operation of the snapshot, a temporary backup snapshot of the volumes is created for internal use only. If the snapshot `restore` operation fails, the temporary backup is used by the system to restore the virtual server instance to the state before the `restore` operation failed.
- When `restore` operation on a virtual server instance snapshot fails, the virtual server instance status changes to `restore-Error` state. You can retry the `restore` operation or you can request to roll back the virtual server instance to its original state it was in before the `restore` operation failed.
- If you perform a `restore` operation on the shared volume, all the virtual machines that are sharing the volume are restored.
- After successful `restore retry` operation, the virtual server instance state might not reset from `Error` state. Use `Reset State` feature on the GUI to reset the virtual server instance state.

Restrictions for restoring a snapshot

- Volume restore operation is not allowed if any copy operations are running on the target volume. The volumes in the instance snapshot are validated to ensure that no copy operations are running against them. Use the API to [get a list of FlashCopy mappings for a volume](#).
- When you perform a `restore` operation, the snapshot task status changes to `restoring`, the volume status is changed to `reverting`, and no other snapshot operations are allowed.
- If you perform a `restore` operation on the shared volume on one virtual machine, you cannot perform the snapshot, restore, clone, or capture operations on other virtual machines that use the shared volume.
- Before you perform a `restore` operation on a boot disk, the virtual machine must be shut down and the status must change to `shutoff` state.
- You cannot restore a virtual server instance snapshot if the snapshot was recently created and the `FlashCopy` operations are still running in the background. The `FlashCopy` operations must first get completed. Use the API to [get a list of FlashCopy mappings for a volume](#).

Deleting a snapshot

You can delete a snapshot by using the GUI, API, or CLI.



Important: You cannot delete a virtual server instance if it has one or more associated snapshots. You must delete all the associated snapshots before you can delete the virtual server instance.

Deleting a snapshot by using the Power Virtual Server user interface

To delete a snapshot using the Power Virtual Server user interface, complete the following steps:

1. Open the Power Virtual Server user interface in [IBM Cloud](#) and log in with your credentials.
2. Click **Workspaces** in the navigation panel. The Workspaces page is displayed with a list of existing workspaces.
3. Select the workspace which contains the snapshot that you want to delete. The Workspace details panel is displayed.
4. Click **View virtual servers**. The Virtual server instances page is displayed.

- 5. In the navigation panel, click **Storage > Instance snapshots**. The Instance snapshots page is displayed with a list of existing snapshots.
- 6. From the list of available snapshots, select the snapshot that you want to delete. The snapshot details page is displayed.
- 7. Select **Delete** from the **Actions** menu. The Confirm delete dialog is displayed.
- 8. Type the snapshot name that is displayed in the Confirm delete dialog and click **Delete**.

 **Important:** You cannot recover a snapshot after it is successfully deleted.

Deleting a snapshot by using API or CLI

You can delete a snapshot by using the following API and CLI:

- **API:** [ibmcloud pi snapshot-delete](#).

You must provide values for the following parameters in the API and CLI:

- `SNAPSHOT_ID`: Required. Specify a unique identifier for the snapshot

Metering and pricing of snapshot

Each snapshot that you create is monitored during every hour and charged depending on the disk space that is requested for the snapshot. The space that is used by a snapshot is charged at 30% of the base rate.

For example, consider that you have **M** disks on a virtual server instance that add up to 600 GB of space. The **M** disks are used as source disks for snapshots. The following charges are applicable on **M** disks:

- If you create one snapshot, you are charged for the disk space that is used by the base 600 GB of **M** disks plus 30% of 600 GB of disk space. That is, 600 GB (space of M disks) + 180 GB (30% of 600 GB) = 780 GB of disk space.
- If you create another snapshot by using same disks, you are charged for the disk space that is used by **M** disks. That is, 600 GB (space of M disks) + (30% of 600 GB) + (30% of 600 GB) = 960 GB of disk space.
- The pricing of a snapshot is calculated based on the volume size and storage tier of the individual source volumes that are present in the snapshot. For example, consider the following table to calculate the cost of a snapshot that has volume A and volume B.

Volume name	Volume size	Tier type	Cost of a volume
Volume A	100 GB	Tier 1	30% of 100 GB
Volume B	30 GB	Tier 0	30% of 30 GB

An example to calculate the cost of a snapshot

The cost of the snapshot is the sum of the costs of volume A and volume B.

Creating a volumes-clone

Creating a volumes-clone request

The clone operation creates a full copy of the volume. You can select multiple volumes and start a group clone operation. When multiple volumes are selected, the clone operation ensures that a consistent data copy is created.

The clone operation continues to copy data from the source disks to target disks in the background. Depending on the capacity of the source disks and the amount of data to be copied, the clone operation can take a significant amount of time.

Prerequisites for creating a volume-clone request

Ensure that you meet the following prerequisites before you start to create a volumes-clone request:

- A volumes-clone request name must be unique for your workspace. If you want to reuse a volumes-clone request name, delete the existing volumes-clone request with that name, and then reuse the name.
- Volumes to be included in the volumes-clone request must be located in the same storage pool.
- Volumes to be included in the volumes-clone request must not be in the `error` state and the health state of the volume must be `ok`.
- Quiesce the applications that access the set of volumes that you want to include in the volumes-clone request.

Creating the volumes-clone request by using API and CLI

The **Prepare action** operation completes the preparatory work for creating the group snapshot of volumes that are getting cloned.

Prerequisites for creating a volume-clone request by using API and CLI

Ensure that you meet the following prerequisites before you start the **Prepare action** operation:

- A minimum of two volumes
- A minimum of one volume to be in the **in-use** state
- A unique volumes-clone name



Note: When a **Prepare action** operation is in progress, you cannot trigger another **Prepare action** operation for the same set of volumes. A **Prepare action** operation must be followed by **Start** and **Execute** operations. If you are not using the existing **Prepare action** operation, you must cancel the existing operation before you start the next **Prepare action** operation.

You can initiate the **Prepare action** operation by using the following API and CLI:

- **API:** [ibmcloud pi volume clone create](#)

You must provide values for the following parameters in the API and CLI:

- **name**: Required. Provide a name to be assigned to the volumes-clone request. The **name** must be a unique value for each of the existing volumes-clone requests.
- **volumeIDs**: Provide the list of volume identifiers or names to be cloned. A minimum of two volume IDs are required and at least one of the volumes must be in the **in-use** state. All the volumes to be included in the volumes-clone request must be in the same storage pool.

When the **Prepare action** operation progresses to **100%**, the **Prepare** status changes depending on the following conditions:

- If **Prepare action** completes successfully, the status changes to **prepared**.
- If **Prepare action** fails to complete successfully, the status changes to **failed**.

To check the status of the **Prepare action** operation, use the API or CLI with **volumesCloneID** option.

Considerations for creating the volumes-clone request by using API and CLI

When a clone operation is performed on a volume that is in the **in-use** state, Power Virtual Server creates a consistent group snapshot and re-creates the copy of the cloned volume by using the group snapshot.

Restrictions for creating the volumes-clone request by using API and CLI

- A minimum of 2 volume IDs are required and at least one of them must be in the **in-use** state.



Note: To clone a single volume, you can use the [ibmcloud pi volume clone-async](#) CLI. To check on the status of the single volumes-clone, you can use the [ibmcloud pi volume clone-async get](#) CLI.

- You cannot modify the source or target disk attributes such as disk size, while the clone operation is in progress.
- You cannot attempt to clone a volume if the volume belongs to a consistency group. A volume belongs to a consistency group after you run the volumes-clone start request, or the volume is included in a snapshot **create** operation where the flash copying in the background is completed.
- A volume cannot be cloned from one storage pool to a different storage pool.

Starting a volumes-clone request

This step allows the consistency group to start the **FlashCopy** operation.



Note: As a prerequisite, the volumes-clone request must be in the **Prepared** state.

Starting a volumes-clone request by using API and CLI

The initial status of the volume-clone request is set to **Starting**.

You can start the volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone start](#)

You must provide the unique identifier of a volumes-clone request in the `volume_clones_id` parameter. The identifier can be the volumes-clone ID or the volumes-clone name.

When you start the volumes-clone request, the initial status is set to `Starting`. When the group snapshot is created, the request status changes to `Available`. The `start` operation of the volumes-clone request is synchronous. When the API call returns to the client, the volumes-clone request status changes to `Available` unless an error occurs. If an error occurs during the `start` operation, the status of the volumes-clone request changes to `Failed`. The reason for failure is specified in the error that is returned with the `start` operation response. The prepared snapshot data is removed so that you can clone the same set of volumes again. If you cancel the `start` operation of a volumes-clone request, the status of the volumes-clone request changes to `Cancelling`. You can cancel a volumes-clone request when the volumes-clone request is in the `Available` state.

Considerations for starting a volume-clone request

When a clone operation is performed on a volume that is in the `in-use` state, Power Virtual Server creates a consistent group snapshot and re-creates the copy of the cloned volume by using the group snapshot.

Restrictions for starting a volume-clone request

You must consider the following restrictions before you start the volumes-clone request:

- A minimum of two volume IDs are required and at least one of them must be in the `in-use` state.
- When a clone operation is in progress, you cannot modify the source or target disk attributes, such as disk size.

Executing a volumes-clone request

This step performs the remaining execution operation to create the cloned volumes from the available group snapshot.

Prerequisites for executing a volumes-clone request

The volumes-clone request must be in the `Available` state.

Executing volumes-clone request by using API and CLI

After you start the `execute` operation, the initial `percentComplete` will be 0% and the status is set to `executing`.

You can execute the volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone execute](#)

You must provide values for the following parameters in the API and CLI:

- `volume-clones-id`: Required. The unique identifier of a volumes-clone request. The identifier can be the volumes-clone ID or the volumes-clone name.
- `name`: Required. The base name of the new cloned volumes. Cloned volume names are prefixed with `clone-` and suffixed with `-#####-N`. In the suffix, `#####` is a five-digit random number and `N` is an incremental number that starts with 1.
- `rollbackPrepare`: (Optional) Default value is `False`. When the flag is set to `False`, the system runs the failed rolls back clone activity but the prepared snapshot is left as-is. When the flag is set to `True`, the system runs the failure rollback clone activity and removes the prepared snapshot.
- `userTags`: (Optional) Provide the list of user tags to be attached to the cloned volumes.
- `targetReplicationEnabled`: (Optional) Use this flag to enable or disable replication for the cloned volumes. By default, the flag uses the replication status of the source volumes that are getting cloned. When set to `False` the cloned volumes are not enabled for replication. When set to `True` the cloned volumes are enabled for replication.



Note: The storage pool where the source volumes reside must be enabled for replication, if the source volumes are not enabled for replication and you are enabling replication for the cloned volumes.

- `targetStorageTier`: Provide the name of the storage tier for the cloned volumes. Use this field to clone a set of volumes from one storage tier to a different storage tier. Cloned volumes must remain in the same storage pool as the source volumes.

Considerations for executing a volumes-clone request

You must consider the following restrictions and considerations before you start the volumes-clone request:

- A volumes-clone request is considered complete when the status of the volumes-clone request is changed to one of the `completed`, `failed`, or `cancelled` state.
- When you start the `volumes-clone execute` request, the initial status is set to `executing`.
- When the `execute` operation is completed successfully, the volumes-clone request status changes to `completed`.
- If an error occurs during the volume cloning process, any artifacts that are created by the cloning process are removed. The group snapshot is

returned to the `available` state so that you can retry the `execute` operation.

- You can cancel a volumes-clone request when it is in the `executing` state.
- If you cancel the `execute` operation of a volumes-clone request, the status of the volumes-clone request is changed to `cancelling`.

Restrictions for executing a volumes-clone request

You can control the behavior of a failed `execute` operation by using the `rollbackPrepare` option. The default value of the `rollbackPrepare` option is `false`. The default value leads the failed `execute` operation to roll back the clone activity and does not remove the prepared snapshot. If the `rollbackPrepare` option is set to `true`, it leads the failed `execute` operation to roll back the clone activity and removes the prepared snapshot.

Cancelling a volumes-clone request

The `Cancel a volumes-clone request` operation starts the cleanup activity that performs the cleanup of the preparatory clones and snapshot volumes.

Prerequisites for cancelling a volume-clone request

The volumes-clone request must be in the `Available` state.

Cancelling volumes-clone request by using API and CLI

After you start the `cancel` operation, the initial `percentComplete` will be 0% and the status is set to `cancelling`.

You can cancel the volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone cancel](#)

You must provide values for the following parameters in the API and CLI:

- `volume-clones-id`: Required. The unique identifier of a volumes-clone request. The identifier can be the volumes-clone ID or the volumes-clone name
- `Force`: (Optional) Default status is `False`. The `cancel` operation is allowed only if the status is set to `prepared` or `available`. When the status is set to `True`, the `cancel` operation is allowed when the status is set to `NOT completed`, `cancelling`, `cancelled`, or `failed`.

Considerations for cancelling volumes-clone request

You must consider the following considerations before you cancel the volumes-clone request:

- If you cancel the volumes-clone request, the initial status of the volumes-clone request is set to `Cancelling`. The status changes to `Cancelled` state when all the clean-up work is completed successfully.
- You can cancel a volumes-clone request when the volumes-clone request is in the `executing` state. The `force` option must be set to `True` in the `cancel` volumes-clone request.
- If the volumes-clone request is in the `Executing` state, any artifacts that are created by the clone operation process are removed.

Deleting a volumes-clone request

When a volumes-clone request is no longer required, you can delete the request by initiating the volumes-clone delete operation.

Prerequisites for deleting a volumes-clone request

The volumes-clone request must be in one of these final statuses: `Completed`, `Failed`, or `Cancelled`.

Deleting a volumes-clone request by using API and CLI

You can delete the volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone delete](#)

You must provide values for the following parameters in the API and CLI:

- `volume-clones-id`: Required. The unique identifier of a volumes-clone request. The identifier can be the volumes-clone ID or the volumes-clone name.
- `Force`: (Optional) Force the cancellation of a volumes-clone request. The `cancel` operation is allowed only if the status is set to `prepared` or `available`. When the status is set to `True`, the `cancel` operation is allowed when the status is set to `NOT completed`, `cancelling`, `cancelled`, or `failed`.

Considerations for deleting a volumes-clone request

If the volumes-clone request is not in one of the final statuses that is required for deletion, you can cancel the request, do any remaining cleanup, and transition the status to `Cancelled`. If the volumes-clone request is in the `Cancelled` status, the request can be deleted.

Getting the status for a volumes-clone request

This API request returns detailed information about the status of the volumes-clone request. For example, the list of source volumes that are getting cloned and the list of cloned volumes when the `execute` operation is successfully completed.

Prerequisites for getting the status for a volumes-clone request

None.

Getting the status of volumes-clone request by using API and CLI

You can get the status of volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone get](#)

Provide a value for the `volume-clones-id` field. The value is the unique identifier of a volumes-clone request. The identifier can be the volumes-clone ID or the volumes-clone name.

Considerations for getting the status of volumes-clone request by using API and CLI

None.

Restrictions for getting the status of volumes-clone request by using API and CLI

None.

Getting the list of volumes-clone request

The API request provides a list of all the volumes-clone requests in a workspace and the status of each request.

Prerequisites for getting the list of volumes-clone request

None.

Getting a list of volumes-clone request by using API and CLI

You can get a list of volumes-clone request by using the following API and CLI:

- **API:** [ibmcloud pi volume clone list](#)

Provide one of the values to the `filter` parameter from the following list:

- `prepare` - includes status values (preparing, prepared)
- `start` - includes status values (starting, available)
- `execute` - includes status values (executing, available-rollback)
- `cancel` - includes status values (cancelling)
- `completed` - includes status values (completed)
- `failed` - includes status values (failed)
- `cancelled` - includes status values (cancelled)
- `finalized` - included status values (completed, failed, cancelled)

Restrictions for getting a list of volumes-clone request by using API and CLI

You must consider the following restrictions and considerations before you get the list of volumes-clone requests:

- The detailed information about the source volumes or cloned volumes is not provided.
- The list can be limited by using the `filter` parameter. The filter is based on the status of the ‘filter’ parameter.

Frequently asked questions about creating a snapshot, restoring a snapshot, and cloning a volume in Power® Virtual Server

The following are some of the frequently asked questions on snapshots and cloning that are documented on the FAQ page:

- [What are the key differences between a snapshot and a clone?](#)
- [Are there any initial snapshot requirements in terms of storage?](#)
- [Does the snapshot and volume clone support any safeguard policy?](#)
- [Can you tell me more about the backup process by using the PowerHA Toolkit for IBM i?](#)

Managing events logs and notifications

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

The IBM® Power® Virtual Server logs all the events and notifications from the IBM Cloud console. You can access these events and notifications in the **Event logs** page.

Consider the following points when you are working with the event logs and notifications:

- The **Event logs** page shows a maximum of 400 entries from the current and previous months.
- You can choose to enable or disable notifications for events on all workspaces. For more information, see [Customizing the log notification settings](#).
- If you disable the notifications, you can still access the logs from the **Event logs** page.
- You can customize notifications; for example, you can enable or disable notifications from concurrent users within the same workspace.
- Your changes to the notification settings are associated with your user account and apply to all workspaces.
- Your every action or operation does not create a log that you can view in the Event logs page.

Accessing the event logs

You can access a maximum of 400 events for the current and previous months in the IBM Cloud console. To see the event logs, complete the following steps:

1. Log in to the [IBM Cloud catalog](#) with your credentials.
2. In the catalog's search box, type **Power Virtual Server**, and click the **Power Virtual Server** tile.
3. Click **Select workspace** on the left navigation under workspace of the IBM® Power® Virtual Server user interface to select from a list of previously created workspaces. If you do not have a workspace, click **Create a workspace**.
4. Click **Event logs** to see the list of event logs and notifications.

Customizing the log notification settings

You can stop the notifications or stop getting notifications from other concurrent users in the same workspace. To customize the log notification settings, complete the following steps:

1. Access the **Event logs** page from the user interface. For more information, see [Accessing the event logs](#).
2. Click **Log notification settings**.
3. Toggle the **Show notifications based on the settings below** button to on or off based on your requirement.

Toggle notification button status	Description
On	Shows notification based on other subset parameters that you can choose.
Off	All notifications are disabled.

Notification toggle button options

4. When the **Show notifications based on the settings below** button toggle is set to on, you can further toggle the **Show for events requested by others** button to on or off based on your requirements.

Toggle workspace notification button status	Description
On	Shows notification based on events that are requested or performed by concurrent users in the same workspace.
Off	Shows notifications for events that are requested or performed by you for the current workspace. It also hides notifications based on events that are requested or performed by concurrent users in the same workspace.

Workspace notification toggle button options

5. Click **Save**

Setting up Terraform for IBM Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Terraform on IBM Cloud enables predictable and consistent provisioning of IBM Cloud services so that you can rapidly build complex, multitier cloud environments following Infrastructure as Code (IaC) principles. Similar to using the IBM Cloud CLI or API and SDKs, you can automate the provisioning, update, and deletion of your IBM® Power® Virtual Server instances by using HashiCorp Configuration Language (HCL).

**Tip:** Looking for a managed Terraform on IBM Cloud solution? Try out [IBM Cloud Schematics](#). With Schematics, you can use the Terraform scripting language that you are familiar with, but you do not have to worry about setting up and maintaining the Terraform command-line and the IBM Cloud Provider plug-in. Schematics also provides pre-defined Terraform templates that you can easily install from the IBM Cloud catalog.

Installing Terraform and configuring resources for Power Virtual Server

Before you can create an authorization by using Terraform, make sure that you have completed the following prerequisites:

- Make sure that you have the [required access](#) to create and work with the Power Virtual Server resources.
- Install the Terraform CLI and configure the IBM Cloud Provider plug-in for Terraform. For more information, see the tutorial for [Getting started with Terraform on IBM Cloud](#). The plug-in abstracts the IBM Cloud APIs that are used to complete this task.
- Create a Terraform configuration file that is named `main.tf`. In this file, you define resources by using HashiCorp Configuration Language. For more information, see the [Terraform documentation](#).

1. Create a Power Virtual Server instance by using the `ibm_resource_instance` resource argument in your `main.tf` file. The Power Virtual Server instance in the following example is named `test_instance` and the resource is `ibm_pi_instance`.

```
resource "ibm_pi_instance" "test-instance" {
  pi_memory      = "4"
  pi_processors  = "2"
  pi_instance_name = "test-vm"
  pi_proc_type   = "shared"
  pi_image_id    = "${data.ibm_pi_image.powerimages.id}"
  pi_key_pair_name = ibm_pi_key.key.key_id
  pi_sys_type     = "s922"
  pi_cloud_instance_id = "51e1879c-bcbe-4ee1-a008-49cdba0eaf60"
  pi_pin_policy   = "none"
  pi_health_status = "WARNING"
  pi_network {
    network_id = data.ibm_pi_public_network.dsnetwork.id
  }
}
```

2. After you finish building your configuration file, initialize the Terraform CLI. For more information, see [Initializing Working Directories](#).

```
# terraform init
```

3. Create a Terraform execution plan. The Terraform execution plan summarizes all the actions that need to be run to create the Power Virtual Server instances:

```
# terraform plan
```

4. Create the Power Virtual Server instances in IBM Cloud.

```
# terraform apply
```

5. From the [IBM Cloud resource list](#), select the Power Virtual Server instance that you created and note the instance ID.

What's next?

Now that you have successfully created your Power Virtual Server instance with Terraform on IBM Cloud, you can visit the [Power Terraform registry](#) to

perform more tasks by using Terraform.

Returning the pod to IBM

IBM Power Virtual Server Private Cloud in **Client location**

If you want to discontinue the Client location services at the end of subscription tenure or because of any other scenario that is permitted by the contract with IBM, you must return the IBM Power Virtual Server Private Cloud infrastructure as explained in the following steps:

1. Open a ticket with [IBM Support](#) and state your request to discontinue the IBM Power Virtual Server Private Cloud services.
2. Back up all the application data and settings from the virtual server instances.
3. Delete all your virtual server instances and workspaces.
4. Delete the corresponding Satellite location instances after IBM completes the steps for pod teardown.

After you open the support ticket, IBM works with you to complete all the necessary operations from the IBM Cloud for the pod teardown, such as:

- Archiving any non-client operating system data.
- Securely and properly disposing of client OS and application residual data according to appropriate rules and regulations.
- Unregistering the pod from your IBM Cloud Satellite location.
- Decommissioning the IBM Direct Link connection that is used for control plane traffic and regional IBM Cloud resources that are associated with the pod.

IBM onsite service engineers visit your data center to perform the physical uninstallation and packing the infrastructure for return along with logistical support.

Working with the VSI console

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

The Virtual Server Instance (VSI) console in the IBM® Power® Virtual Server environment is also known as VTERM. You can use the VSI console to interact with the AIX®, IBM i, and Linux®-based Power virtual servers. The VSI console performs several management and troubleshooting activities. The VSI console provides the following benefits:

- Provides access to the AIX, IBM i, and Linux-based VSIs (Linux, SAP HANA, and SAP NetWeaver) to perform various OS-related tasks such as installation, configuration, troubleshooting, and recovery
- Debugs VSI boot issues
- Resets the Dedicated Service Tools (DST) password for the IBM i standard user

Starting the VSI console for a Power Virtual Server instance

To start the VSI console of a Power Virtual Server instance from the IBM Cloud UI after you [create the instance](#), complete the following steps:

1. Open the [Power Virtual Server user interface](#) and log in with your credentials.
2. Click **Virtual server instances** from the left navigation pane in the Power Virtual Server user interface. The Virtual server instances page is displayed. a list of existing virtual server instances that are associated with the selected workspace is displayed on the **Servers** tab.
3. Select a virtual server instance to open the Virtual server instance details page.
4. Click **VM actions** in the Virtual server details page and select **Open console** from the drop-down list. The VSI console opens in a new browser window. Ensure that your browser settings allow pop-up windows.



Note: When you open the console for an IBM i-based Power Virtual Server instance through the IBM Cloud GUI, the Console Setting dialog is displayed. You can optionally select a language other than English and click **Open console** to open the VSI console. To change the language, see [Changing the VSI console language](#).

To open the VSI console of a Power Virtual Server instance by using the IBM Cloud CLI, complete the following steps:

1. To get the URL that you can use to open the VSI console, use the [ibmcloud pi instance console](#) IBM Cloud CLI command. In the following command, `INSTANCE_ID` is a unique identifier or the name of the instance and must be replaced with the actual ID of your VSI.

```
$ ibmcloud pi instance console get INSTANCE_ID
```

2. Copy the URL that is generated in the previous step and paste it into the address bar of a web browser. The console window for the VSI is displayed.

Closing a VSI console session

To close a VSI console session for a Power Virtual Server instance, close the browser window or the tab.

Changing the VSI console language

When you open the VSI console of a Power Virtual Server instance that is running IBM i as the boot image, you can change the console language from English to another language.

To change the VSI console language using the Power Virtual Server user interface, complete the following steps:

1. Open the [Power Virtual Server user interface](#) and log in with your credentials.
2. Click **Virtual server instances** from the left navigation pane in the Power Virtual Server user interface. The Virtual server instances page is displayed. a list of existing virtual server instances that are associated with the selected workspace is displayed on the **Servers** tab.
3. Select an IBM i-based virtual server instance to open the Virtual server instance details page.
4. Click **VM actions** in the Virtual server details page and select **Open console** from the drop-down list. The Console Setting dialog is displayed.
5. Select a language from the **Console language** drop-down list.
6. Click **Open console** to open the console using the language that you selected in the previous step.

To change the VSI console language from the IBM Cloud CLI, run the [ibmcloud pi instance console update](#) command. In the following command, `INSTANCE_ID` and `CODE` must be replaced with the ID of your VSI and the associated language code:

```
$ ibmcloud pi instance console update INSTANCE_ID --code CODE
```

To view the languages available for an IBM i-based Power Virtual Server instance, you can use the following IBM Cloud CLI command:

```
$ ibmcloud pi instance console list
```



Note: Changing the VSI console language on an IBM i-based Power Virtual Server instance terminates any open console sessions before a new console session is started.

Copying and pasting text in the VSI console

The copy-and-paste functionality of the VSI console enables bidirectional text transfer between Windows®, macOS®, and Linux-based workstations, and AIX, IBM i, and Linux-based VSI consoles. You can copy text, such as commands and large scripts, and paste the copied text into the VSI console. You can also copy logs or error messages from the console and paste them into your workstation for troubleshooting or creating support tickets.

Copying and pasting text from your workstation to the VSI console

To copy and paste text from your workstation to the VSI console, complete the following steps:

1. Select the text that you want to copy on your workstation.
2. Use the following key combination based on your operating system to copy the selected text:
 - Windows: Press `Ctrl+C`.
 - Mac: Press `Command+C`.
 - Linux: Press `Ctrl+C`.
3. From the Virtual server instance details page, click the overflow menu (icon with 3 vertical dots) and select **Open Console**. The VSI console window is displayed.
4. Click the **Show/hide the control bar** icon, which is located on the left side of the VSI console window.
5. Click the **Clipboard** icon. The Clipboard panel is displayed.
6. Paste the copied text into the VSI console by completing the following steps:

- a. Type or paste the copied text in the text field of the Clipboard panel. To paste, use the following key combination based on your operating system:

- Windows: Press **Ctrl+V**.
- Mac: Press **Control+V**.
- Linux: Press **Ctrl+V**.

- b. After the text appears in the Clipboard panel, click the **Clipboard** icon to return to the VSI console, or click directly inside the console window, which remains visible in the background. Then, press the same key combination to paste the text at the cursor in the console window.

Copying and pasting text from the VSI console to your workstation

To copy and paste text from the VSI console to your workstation, complete the following steps:

1. From the Virtual server instance details page, click the **VM actions** list and select **Open Console**. The VSI console window is displayed.
2. Click the **Show/hide the control bar** icon, which is located on the left side of the VSI console window.
3. Click the **Clipboard** icon. The Clipboard panel is displayed.
4. To copy text from the VSI, select the text that you want to copy, and the selected text is automatically copied to the clipboard. However, for the IBM i-based VSI console, press **Ctrl+C** or **Command+C** to copy the selected text.
5. To paste the copied text to your workstation, use the following key combination based on your operating system:
 - Windows: Press **Ctrl+V**.
 - Mac: Press **Command+V**.
 - Linux: Press **Ctrl+V**.



Tip: You can use the **Full screen** option on the control bar to open the VSI console in the full-screen mode.

Copy and paste limitations

The copy and paste functionality in the VSI console has the following limitations:

- The copy and paste functionality from the clipboard into the VSI console might not work if the Caps Lock key is turned on.
- The copy and paste functionality in the VSI console currently supports English only.
- Special characters such as **^V**, **[[200~**, **{}**, or **[[201~** might appear when you paste text from your workstation into the VSI console of an AIX- or Linux-based VSI that uses bash as the command shell and is configured to use bracketed-paste mode. For more information, see [enable-bracketed-paste](#).

Sharing the VSI console session

When multiple users open the VSI console for the same Power Virtual Server instance, the same VSI console (VTERM) session is shared among all users. When the last user closes the VSI console, the underlying console session is terminated. Reopening the VSI console creates a new VTERM and establishes a new console connection to the VSI on the host.

VSI console considerations

To use the VSI console, consider the following factors:

- Power Virtual Server attempts to keep the browser window for the VSI console active and open for up to 2 hours. After the VSI console is idle for 2 hours, the console connection is terminated.
- The connection to the VSI console can be interrupted or disconnected for various reasons, including a disruption in the network connectivity between your workstation and the Power Virtual Server host. When the VSI console connection is interrupted, Power Virtual Server attempts to reconnect and restore the connection automatically.
- If multiple users connect to the same IBM i-based Power Virtual Server instance and one of the users changes the console language and opens the VSI console, the console that is in use by the other users is terminated and requires the console to be reopened.

For more information about how to use the VSI console with IBM i, see [Tips for working with the IBM i console](#).

Custom OS Images

Establishing connectivity to IBM Cloud Object Storage buckets to import custom images

IBM Power Virtual Server Private Cloud in Client location

You can maintain a set of operating system images in your IBM Cloud Object Storage buckets. These images are referred to as custom images. You can import these custom images into the boot volumes of the logical partitions (LPARs).

To import stored custom images from the **Boot images** page, click **Import image** option. Enter the following information in the **Import boot image** page to complete the following steps:

- Custom image name to be used during virtual machine provisioning
- Storage tier number
- Storage pool preferences
- Region of your Cloud Object Storage instance
- File name of the custom image to be imported as specified in the Cloud Object Storage instance
- Your Cloud Object Storage bucket name
- Hash-based Message Authentication Code (HMAC) secret key and access key

IBM Power Virtual Server Private Cloud processes the custom image file as follows:

- Downloads the image file from the Cloud Object Storage instance.
- Imports the image file into the attached storage.
- Converts the image file into an image volume.
- Copies the image volume as a boot volume to deploy the LPAR.

The following diagram shows the architecture for importing custom images that are stored in the IBM Cloud Object Storage buckets:



AIX

Creating an AIX virtual machine (VM) with SSH keys for root login

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can set up one or more Secure Shell (SSH) keys for root login when you create new AIX virtual machines (VM). The keys are loaded into the root's **authorized_keys** file. SSH keys allow you to securely log in to a VM. You must use the available operating system options to create SSH keys. To generate SSH keys on a Linux® or Mac OS system, for example, you can use the standard `ssh-keygen` tool.

Generating an SSH key

In this example, the user created a public key on a Linux-based IBM Cloud compute instance by using the `ssh-keygen` tool:

```
$ cat .ssh/id_rsa.pub
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUcgTI9Dee8zzaA
9 testkey
```

To use an SSH key with a VM-create operation, you must first add the public key to the IBM® Power® Virtual Server instance by using the `ibmcloud pi key-create` command. To add the generated public key, enter the following command (replacing the example value with your own public key):

```
$ ibmcloud pi key-create testkey --key "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUcgTI9Dee8zzaA
9 testkey"
SSHKey created: testkey
```

To confirm that the key was successfully added, use the `ibmcloud pi keys` command:

```
$ ibmcloud pi key testkey
Name      Key                                     CreationDate
testkey   ssh-rsa AAAAB3NzaC1y...UIzYr3u+79n9 testkey 2019-07-26T18:21:56.030Z
```

Creating an AIX VM instance

You can create an AIX VM instance with a configured SSH key by using the Power Virtual Server CLI or the console. When you use an AIX stock image as your boot volume, the root password is not set. You must connect to the AIX VM and set the root password for the system. Without completing this step, SSH login as **root** appears as being *disabled*. If you have public network access to the AIX VM, you can use telnet from a private cloud system and set the root password. For more information, see [IBM AIX V7.2 documentation](#).

Creating an AIX VM with a configured SSH key using UI

You must [generate a public SSH key](#) before you can create an AIX VM with a configured SSH key.

1. Ensure that you have the necessary account permissions and device access. Only the account owner, or a user with the **Manage Users** classic infrastructure permission, can adjust the permissions. For more information, see [Classic infrastructure permissions](#) and [Managing device access](#).
2. Click **Virtual server instances** from the left navigation in the Power Virtual Server user interface.
3. Click **Create instance**.
4. In the **Virtual servers** section, select **Add SSH Keys**.
5. Enter a **Key name** and your previously generated **Public key**.
6. Click **Add SSH key** to add the SSH key.
7. Complete the rest of the fields to successfully create a new instance with a configured SSH key.

Creating an AIX VM with a configured SSH key using CLI

You can create a new VM with the public key with the following command (replacing the options with your own):


```
$ ibmcloud pi instance-create keytest-vm --image AIX-7200-03-03 --memory 5 --networks "cloud.ibm.com" --processors 1 --processor-type shared --key-name testkey
```

In this example, the `ibmcloud pi instance-create` command created a new AIX VM with an IP address of `172.16.7.16`. You can now SSH to the AIX VM from a connected system, which is configured with the private key for `testkey`.

```
$ ssh root@172.16.7.16
Enter passphrase for key '/home/keytest/.ssh/id_rsa':
Last login: Fri Jul 26 16:53:22 CDT 2019 on ssh from 10.150.0.11
*****
*                                     *
*                                     *
* Welcome to AIX Version 7.2!                *
*                                     *
*                                     *
* Please see the README file in /usr/lpp/bos for information pertinent to *
* this release of the AIX Operating System.      *
*                                     *
*                                     *
*****
# oslevel -s
7200-03-03-1914
```

You can find the `testkey` value in the `authorized_keys` file:

```
$ cat .ssh/authorized_keys
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQCTuQnQOc2k4zaGzE7b3xUMCjUy++s/9O9HE4fXSm7UNKoTY39zjQ8mhOwaA3HEo12tOdzdFDYHHWNOYufCcFFk61CAL6HyQGGClib1nFc1xUc9
9 testkey
```

Debugging the connection

If the SSH connection to the VM instance is failing, use the `-vvv` option of `ssh` command to determine the reason of the connection failure. Each additional verbose flag in the `-vvv` option increases the verbosity of the ssh client session and generates a large amount of data. Therefore, use a script session to capture the ssh log data so that log data is also sent to the **Standard output (STDOUT)**.

Run the following commands to start an SSH client debug session:

```
$ # /usr/bin/script /tmp/ssh.{host}.debug
# /usr/bin/ssh -vvv {hostname/IP_of_ssh_server}
```

After you end the SSH client debug session, close the script session by pressing **Ctrl+D** or by entering **exit** command.

Additional support information

Refer to the following support pages for additional information for different use cases:

- [Debugging sshd without impacting existing sshd sessions](#)
- [Using IPSec rules to filter network traffic](#)
- [Downloading and installing or upgrading OpenSSL and OpenSSH](#)

Replicating an AIX configuration in IBM® Power® Virtual Server using an AIX mksysb image

Learn how to create and restore an AIX `mksysb` image onto an IBM® Power® Virtual Server instance.

A simple way to migrate a local AIX environment (or workload) into Power Virtual Server is to restore an operating system `rootvg` backup over an existing image and then migrate the data. The `rootvg` backup is created with the AIX `mksysb` command. The restored mksysb image applies the AIX configuration details while preserving the Power Virtual Server deployed storage and networking resources.

**Note:** When the restored AIX configuration is active, various methods can be used to migrate applications and related data. Those methods are outside the scope of this page.

Ensure that you have experience in AIX administration and are familiar with deploying Power Virtual Server instances using the Power Virtual Server user interface or the IBM Cloud CLI.

Considerations before creating the mksysb image on the source AIX instance

- Ensure that the installed RSCT package is at version 3.2.6.2 or later. The `/opt/rsct/install/bin/ctversion` command can be used to display the installed version. For more information see, [Recommended Reliable Scalable Cluster Technology \(RSCT\) package levels for imported AIX images](#).
- The AIX environment should be running an AIX version and technology level that is in standard support. For AIX levels that are under an extended support model, arrangements should be made to obtain an extended support agreement to cover the use of the AIX level in Power Virtual Server.

Consult the [General AIX support lifecycle information](#) and [Operating systems supported in IBM Power Virtual Server](#) for more details.

The AIX instance that is deployed in the Power Virtual Server should be at the same AIX version as the source AIX instance. For example, if the source instance is at AIX 7.2, then the deployed Power Virtual Server instance should also be a 7.2 based image.

 **Important:** The AIX technology levels (TL) do not need to match, but you might consider by using the latest TL that is available from the Power Virtual Server stock images.

- If you plan to capture an AIX system that is restored from an `mksysb` backup as a custom image, you must install the cloud-init packages before starting the capture. An AIX system that is restored from an `mksysb` backup might not include the cloud-init packages, which can result in a captured image without cloud-init. The cloud-init packages are required to ensure that the Power Virtual Server instances that are created from the image initialize correctly.
- Ensure that that NPIV file sets are installed in the AIX environment as Power Virtual Server VMs use the NPIV storage virtualization model. This can be checked by using the `lspp` command as follows.

```
$ ``screen
#
# lspp -l devices.vdevice.IBM.vfc-client.rte
Fileset          Level   State   Description
-----
Path: /usr/lib/obj repos
devices.vdevice.IBM.vfc-client.rte
              7.1.5.38   COMMITTED Virtual Fibre Channel Client Support

Path: /etc/obj repos
devices.vdevice.IBM.vfc-client.rte
              7.1.5.38   COMMITTED Virtual Fibre Channel Client Support

#
``
```

- Verify that the AIX initab file does not contain entries with dependencies on unique aspects of the source environment that would not be present in Power Virtual Server. Otherwise, the converted Power Virtual Server AIX instance might not boot. Similarly, if other boot time actions exist such as NFS file system mounts, those might need to be disabled.
- Power Virtual Server uses IBM Storage with the built-in AIX MPIO driver. Ensure that the I/O configuration of the source LPAR does not conflict with the use of AIX MPIO.
- Delete any temporary files or files, especially large files that are not necessary on the target Power Virtual Server AIX instance. The `-e` and `-x` `mksysb` options can also be used to exclude unneeded directories and file systems in rootvg. This reduces the size of the `mksysb` image for transfer to Power Virtual Server.

Creating the mksysb image on the source AIX instance

Refer to the [mksysb](#) documentation for full details on the command usage. Ensure that there is sufficient file system space to hold the produced `mksysb` image. Generally, 10 to 15 GB is sufficient depending on additional non-AIX data added to the rootvg.

In the following example, `mksysb` creates the image in `/tmp`. The `-i` builds the image from the latest rootvg details and the `-b` option can potentially improve the performance when creating the `mksysb` image. The `-X` `mksysb` option expands `/tmp` if necessary for the boot image. This can be omitted if the available space is known to be sufficient.

```
$ # create the mksysb image
mksysb -i -X -C -b 512 /tmp/my-mksysb
```

Once the `mksysb` image is created, the storage volume capacity that is necessary to restore of the image can be extracted as follows. In this example, a 25 GB volume would be necessary.

```
$ #
# restore -qf ./my-mksysb ./bosinst.data
x ./bosinst.data
# fgrep -p target_disk_data bosinst.data|fgrep SIZE_MB
    SIZE MB = 25600
# rm bosinst.data
#
```

Also, capture the image checksum for verification once the image is transferred to a Power Virtual Server AIX instance for restoration.

```
#
# cksum ./my-mksysb
1999777825 1861746688 ./my-mksysb
#
```

Creating a Power Virtual Server AIX instance for conversion with an mksysb image

An AIX instance is required in Power Virtual Server that can be converted with the mksysb image.

The instance must be at the same AIX version as the AIX instance where the mksyb image was created. Power Virtual Server provides a set of stock images that can be used to facilitate the conversion. When deployed, the AIX instance needs some updates to host the mksysb image. Choose the stock image and deploy a new Power Virtual Server AIX instance via the user interface or CLI with the cpu, memory, and network details that are required for the final converted instance.

For example, if an AIX 7.2 based mksysb image is being used, choose the `7200-05-05` stock image. When deploying the instance, attach an additional storage volume with sufficient capacity for the mksysb image restore. Use the information from the bosinst.data file to get the minimum required capacity in Megabytes. The following is an example AIX 7.2 deployed instance with an additional 30 GB volume (hdisk1).

The following details should be observed with the deployed AIX instance.

```
#
# oslevel -s
7200-05-05-2246
# lspv
hdiskO      00fa00d66c59c9d7      rootvg      active
hdiskl      none                None
# bootinfo -s hdiskl
30720
#
```

To make room for the mksysb image, disk space must be freed in the instance to hold it. This can be done by removing the `/usr/sys/inst.images` file system and creating a new one called `/mksysb-staging`.

This should result in sufficient space for most mksysb image use cases. If more space is required, then a new larger storage volume needs to be attached to the instance with the Power Virtual Server user interface and a JFS2 file system will need to be created on it. The following example removes `/usr/sys/inst.images` and creates a new 12 GB `mksysb-staging` file system.

```
#
# umount /usr/sys/inst.images
# rmfs -r /usr/sys/inst.images
rmlv: Logical volume repo00 is removed.
#
```

```
#
# crfs -v jfs2 -g rootvg -m /mksysb-staging -a size=12G
File system created successfully.
12582324 kilobytes total disk space.
New File System size is 25165824
# mount /mksysb-staging
# mount
```

node	mounted	mounted over	vfs	date	options
/dev/hd4	/	jfs2	Oct 11 17:47	rw,log=/dev/hd8	
/dev/hd2	/usr	jfs2	Oct 11 17:47	rw,log=/dev/hd8	
/dev/hd9var	/var	jfs2	Oct 11 17:47	rw,log=/dev/hd8	
/dev/hd3	/tmp	jfs2	Oct 11 17:48	rw,log=/dev/hd8	
/dev/hd1	/home	jfs2	Oct 11 17:48	rw,log=/dev/hd8	

```

/dev/hd1admin /admin      jfs2  Oct 11 17:48 rw,log=/dev/hd8
/proc        /proc        procfs Oct 11 17:48 rw
/dev/hd10opt  /opt          jfs2  Oct 11 17:48 rw,log=/dev/hd8
/dev/LLivedump /var/adm/ras/livedump jfs2  Oct 11 17:48 rw,log=/dev/hd8
/ahafs       /aha          ahafs Oct 11 17:49 rw
/dev/fslv00   /mksysb-staging jfs2  Oct 11 18:35 rw,log=/dev/hd8
#

```

In the preceding example, hdisk1 is the free storage volume where the mksysb image is restored.

Once the Power Virtual Server instance is created, the mksysb image is placed in the /mksysb-staging directory. Transferring the mksysb image to the `/mksysb-staging` directory depends on your connectivity options to the IBM Cloud and Power Virtual Server workspace.

For example, if you are able to access your Power Virtual Server instance over a network connection from your local system where the mksyb image resides, you might use scp.

For example, `# scp ./my-mksysb root@xxx.xxx.xxx.xxx:/mksysb-staging`

Use the `cksum` command to confirm the `my-mksysb` image file was successfully transferred.

Now the mksysb image can be restored onto the attached free storage volume that will become the new rootvg boot disk with the configuration from the source AIX instance. This is done by using the [alt_disk_mksysb](#) command. In the following mksysb restore example, `alt_disk_mksysb` sets hdisk1 as the boot disk for subsequent boots.

```
$ alt_disk_mksysb -c /dev/vty0 -d hdisk1 -m /mksysb-staging/my-mksysb
```

Once this completes successfully, an `lspv` indicates that hdisk1 is now the altinst_rootvg.

```

#
# lspv
hdisk0      00fa00d66c59c9d7      rootvg      active
hdisk1      00c939202144313b      altinst_rootvg

```

The AIX `bootlist` command can be used to confirm that hdisk1 is now the active AIX boot disk.

```

#
# bootlist -m normal -o
hdiskl blv=hd5 pathid=0
hdiskl blv=hd5 pathid=1
hdiskl blv=hd5 pathid=2
hdiskl blv=hd5 pathid=3
hdiskl blv=hd5 pathid=4
#

```

The AIX instance can be rebooted to the new hdisk1 rootvg.

```
sync;sync;shutdown -Fr
```

Once the AIX instance restarts, it has been converted with the mksysb image. An `lspv` shows hdisk1 as the new rootvg.

```

#
# lspv
hdisk0      00fa00d66c59c9d7      old_rootvg
hdisk1      00c939202144313b      rootvg      active
#

```

Before proceeding, consider running some basic tests on your new converted Power Virtual Server AIX instance.

1. Run `oslevel -s` to confirm that the converted lpar matches the AIX level from the source system where the mksysb backup was created. More detailed verification can optionally be done by using the `lspp -h` command.
2. In the user interface, you can test by changing the CPU and RAM with your instance. You might also try adding and removing a small storage volume. These tests confirm that the instance is properly interacting with PowerVM dynamic LPAR operations for live resource updates to the instance.

Now the `old_rootvg` storage volume (hdisk0) can be detached from the instance and deleted by using the user interface. First, remove the volume from the AIX configuration with the `rmdev` command.

```
# rmdev -dl hdisk0
```

Before detaching the `old_rootvg` volume (hdisk0), you need to set the new rootvg volume (hdisk1) to bootable and the `old_rootvg` volume to not bootable in Power Virtual Server.

Complete the following steps in Power Virtual Server user interface under the details of the AIX instance in the `Attached volumes` section:

- Change the bootable indicator of the new rootvg volume from **off** to **on**.
- Set the bootable indicator of the old boot volume from **on** to **off**.

This allows the old boot volume to be detached and deleted in the Power Virtual Server user interface. This can also be done by using the IBM Cloud CLI.

When the old boot volume is detached and deleted by using the user interface, the conversion of the AIX instance by using the `mksysb` is complete with a single rootvg volume. Now the instance is ready for installation of applications and related data.

Additional Information Resources

- [Getting Started with IBM Power Systems Virtual Servers](#).
- [Power Virtual Server CLI Reference](#).

Moving data from client-managed environment to IBM Power Virtual Server

IBM Power Virtual Server in `IBM data center`

IBM Power Virtual Server Private Cloud in `Client location`

Depending on your network bandwidth and data size constraints, the process of moving the data volume group is as simple as creating an *Open Virtualization Appliance (OVA) file* or an *mksysb* image (root volume group), and creating a set of *savevg* images for volume group data. By using an *OVA* file or an *mksysb* image, you can build or provision a VM and then migrate the data volume groups of the virtual machine (VM) by using the **restvg** command.

You can use the following methods to back the cloud data that is managed by you and move the data to IBM® Power® Virtual Server.

Migrating volume group data by using the *savevg* command

A volume group is a collection of physical volumes of various capacities and types. When a physical volume is assigned to a volume group, the physical blocks of storage media are organized into physical partitions. You can specify the capacity of the physical partition when you create the volume group. You can use built-in AIX *savevg* and *restvg* commands to back up and restore non-root volume groups. The *savevg* and *restvg* commands simplify the creation of new volume groups and file systems on the new VM.

The *savevg* command finds and backs up all files that belong to a specified volume group. The volume group must be varied-on, and the file systems must be mounted. The *savevg* command uses the data file that the *mkvgdata* command creates.

Use the following command to find and back up all files in a specific volume group.

```
# savevg -f <destination path> -i <non root vg files to be backed up>
```

For example,

```
$ # savevg -f /home/admin01/datavg_bkup -i datavg
```

Backing up multiple volume groups by using the *mkvgdata* and *restvg* commands

Small systems might require only one data volume group to contain all the physical volumes. For a non-root user you might want to create separate volume groups, for security reasons, because each volume group can have its own security permissions. If a volume group stops working, other volume groups remain active. Hence, separate volume groups are easier to maintain.

Run the following commands to back up multiple volume groups:

```
# lsvg -o | xargs -i mkvgdata {}  
# tar -cvf /backup/vgdata.tar /tmp/vgdata
```

To generate an output file that contains information about the volume group, run the *mkvgdata* command for each online volume group to generate the output file. This output file is located in the `/tmp/vgdata` directory. You can compress and store this output file in the `/backup` file system directory as shown in the following example. This output file contains information about all volume groups, logical volumes, and file systems that can be used as a single image. This image can be transferred or stored within an *mksysb* backup image if the `/backup` directory is on the root volume group.

Run the following command to re-create the volume groups, logical volumes, and file systems:

```
$ # tar -xvf /backup/vgdata.tar
```

Now edit the `/tmp/vgdata/{volume group name}/{volume group name}.data file` and look for the line with `VG_SOURCE_DISK_LIST=`. Change the line to have *hdisks*, *vpaths*, or *hdiskpowers* based on your requirement.

For example,

```
$ # restvg -r -d /tmp/vgdata/{volume group name}/{volume group name}.data
```

Migrating raw partitions by using the dd command

The output file of the `savevg` command can be restored by using the `restvg` command. A `savevg` backup file is compact in comparison to the physical volumes in the volume group. If the environments have several TBs of data, the prescribed method of moving volume group data by using the `savevg` command might present a disadvantage while considering transference and restoration procedures.

You can use the `savevg` command to back up volume groups. All logical volume information, Journaled File System (JFS), and JFS2 mounted file systems are archived. However, you cannot use the `savevg` command to back up raw logical volumes.

Use the following methods to back up and restore the contents of a file system:

- 1. Unmount the file system.
- 2. Save the raw logical volume content into a file by running the following command:

```
$ # dd if=/dev/lvname of=/file/system/lvname.dd
```

This command creates a copy of the logical volume named **lvname** to a file named **lvname.dd** in the file system `/file/system`. Make sure that the specified directory where the output file is stored (`/file/system` in the example) has enough available disk space to hold a full copy of the logical volume. For example, if the logical volume capacity is 100 GB, you need 100 GB file system space for the logical volume copy.

On the destination server, re-create the logical volume and the file system. If you are using an unmounted file system, run the following command to restore the backup copy:

```
$ # dd if=/file/system/lvname.dd of=/dev/lvname
```

After you run the `dd` command to mount the file system, you can access the contents of the original file system.

Adding or removing a network interface from an AIX virtual machine (VM)

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Starting from IBM PowerVC Version 1.2.2, IBM PowerVC can dynamically add a network interface controller (NIC) to a VM or remove a NIC from a VM. IBM PowerVC does not set the IP address for new network interfaces that are created after the machine deployment. Any removal of a NIC results in freeing the IP address that was set on it. You must remove and add again the AIX VM network interface if you choose to disconnect the Power Virtual Server AIX VM from a public network.

 **Note:** When you toggle a public network off and then on, the Power Virtual Server user interface regenerates new internal and external IP addresses. You need to check the Power Virtual Server user interface for the new internal IP address to complete this procedure.

Removing a network interface from an AIX VM

- 1. Use the `ifconfig` command to remove the network interface from the AIX VM. In the following example, `en0` is the public interface.

```
ifconfig en0 down detach
```

- 2. Next, run the `rmdev` command to remove the device from the AIX system.

```
rmdev -dl en0
```


3. Use the `cfgmgr` command to reconfigure the device.

Adding a network interface to an AIX VM

When you toggle a public network off and then on, a new Virtual Ethernet Adapter (VEA) is created on your VM. You must identify the network interface that the new VLAN is associated with before reconfiguring the IP address.



Note: Every time you toggle a public network off and then on, the system creates multiple network interfaces. As a result, the `ifconfig -a` command will show several unused network interfaces after toggling the public network. To remove these stale network interfaces, use `ifconfig` and `rmdev` as described in the previous section.

1. After you toggle a public network off and then on, look up the list of networks and write down the public network VLAN ID. To find the public network VLAN ID, see [Get a network's current state or information](#).
2. Use the `lsdev -Cc adapter` command to generate the list of adapters.
3. Use either the `ifconfig -a` or `netstat -in` command to see all of the network interfaces.
4. To find the network interface that has the public network VLAN ID, enter `entstat -d ent X | grep VLAN` (where *X* is the adapter number).

To add a network interface (for example, *en0*) and point it to the new internal IP address (as shown in the Power Virtual Server user interface), you can use `smitty mktcpip`. You can also use the AIX command line to perform the same task by using the [mktcpip command](#) (replacing the values with your own):

```
/usr/sbin/mktcpip -h power-systems-virtual-instance -a 192.168.103.12 -m 255.255.255.240 -i en0 -t N/A -g 192.168.103.1 -D 0.0.0.0
```

To manipulate domain name server (DNS) entries for local resolver routines in the system configuration database, see the [namerslv command](#).

Recommended Reliable Scalable Cluster Technology (RSCT) package levels for imported AIX images

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

RSCT is a set of software components that together provides a comprehensive clustering environment for AIX®, Linux®, Solaris, and Windows® operating systems. RSCT is the infrastructure that is used by various IBM products to provide clusters with improved system availability, scalability, and ease of use. For the IBM® Power® Virtual Server offering, RSCT 3.2.1 is the minimum package level that is required for an imported AIX image (provides IPv6 support). However, the Power Virtual Server development team recommends that you use [RSCT 3.2.6.2](#) for optimal performance.



Important: The RSCT `nodeid` is not rebuilt if you are deploying an AIX VM from a network installation management (NIM) server without `cloud-init`, and RSCT is installed.

Resource Management Control (RMC)

The [RMC subsystem](#) is the scalable backbone of RSCT that provides a generalized framework for managing resources within a single system or a cluster. Its generalized framework is used by cluster management tools to monitor, query, modify, and control cluster resources. RMC provides a single monitoring and management infrastructure for both RSCT peer domains and management domains.

The RMC status of the AIX VM is presented in the Power Virtual Server dashboard as a health status. The RMC health status can be either `OK` or `Warning`. Warning statuses happen when the RMC subsystem between your VM and the management system is not connected.

The RMC connection between your VM and the system management service is configured when you create the AIX VM. When you deploy an AIX VM, the IPv6 management interface is injected into the VM. If you remove or overwrite this interface, an RMC connection is not possible. The following procedures can cause the injected IPv6 management interface to be lost after you deploy the AIX VM:

- Attach a different boot volume to the VM and boot from it
- Use the `mksysb restore` operation from a private cloud VM
- Use `smitty` to remove the IPv6 interface

Diagnosing and recovering from a missing IPv6 link local address

One of your AIX VM network interface controllers (NICs) must include an IPv6 link local address to connect to the Novalink host. If the desired NIC does

not have an associated IPv6 link local address, you must perform a recovery procedure.

Diagnosing a missing IPv6 link local address

Enter the `ifconfig -a` command on your AIX VM terminal to see whether one of your NICs shows an IPv6 link local address (`2001:1234:5723:ABCD:5678:D14E:DBCA:0764/64`). The following example is a NIC without an associated IPv6 link local address:

```
$ en0: flags=1e084863,480<UP,BROADCAST,NOTRAILERS,RUNNING,SIMPLEX,MULTICAST,GROUPRT,64BIT,CHECKSUM_OFFLOAD(ACTIVE),CHAIN>
inet 192.168.2.104 netmask 0xfffff0 broadcast 192.168.2.111
```

If one of your NICs does not contain an IPv6 link local address, continue on to the next section.

Using a Power Virtual Server boot image to recover from a missing IPv6 link local address

1. Enter the `lspp -L rsct.*` command to ensure that an operating system (OS) modification did not affect the RSCT file set level. For more information, see [Verifying RSCT installation on AIX nodes](#).



Important: Power Virtual Server supports 3.2.1 as the minimum release. If you redeployed an AIX image with a package that is older than 3.2.1, you must upgrade RSCT first.

2. If you still have the Power Virtual Server deployed boot image, complete the following steps:
 - a. Boot to the original Power Virtual Server boot image.
 - b. Rerun the `ifconfig -a` command. The output includes the configured IPv6 link local address.



Note: If you removed the IPv6 link local address from the original boot image configuration, you can read the AIX `cloud-init` logs to find the IPv6 address.

- c. Open the `/var/log/cloud-init-output.log` file.
 - d. Use the `grep` command to search for the IP injection. There is an IPv4 address and an IPv6 link local address.
3. Using the IPv6 address for the host, [refresh the RMC services](#):

```
$ /opt/rsct/bin/rmcctrl -p
/opt/rsct/bin/rmcrefreshMD -s ctrmc
```

4. *(Optional)* If you altered the `nodeid` (that is unique to RMC), it can impact RMC. For example, when you are using PowerHA and trying to copy your `nodeid` details from a private cloud deployment that is not supported. Begin by rebuilding the node:

```
$ odmdelete -o CuAt -q name=cluster0 to remove 'cluster0' entry from the CuAt ODM.
```

5. *(Optional)* On the AIX VM, enter the `cat /etc/ct_node_id` command and save the output.
6. *(Optional)* To create new `nodeid` and restart RMC services:

```
$ /usr/sbin/rsct/install/bin/recfgct
```

7. *(Optional)* To build a `nodeid`, run the `/opt/rsct/bin/rmcctrl -p` command if not already done in step 3.



Tip: If these recovery steps do not restore the RMC status to **active** and its health to **OK**, open a case with [support](#).

Recovering from a missing IPv6 link local address when using your own boot image

Complete the following steps to recover from a missing IPv6 link local address:

1. Before you shut down the AIX VM, grab the IPv6 details.
2. Gather the IP interface and adapter configuration details by using `smitty`.
3. Run the `/usr/sbin/rsct/bin/lsnodeid` command to grab the `nodeid`.
4. Confirm that the `nodeid` matches the output from `cat /etc/ct_node_id` and `cat /var/ct/cfg/ct_node_id`.

- 5. Power down the AIX VM and restart it by using your custom boot image.
- 6. Re-create the same network configuration as on the IBM boot image by using `smitty`.
- 7. Validate that the RSCT packages are at least `3.2.1` or later by running the `lspp -L rsct.*` command.
- 8. Run the `/usr/sbin/rsct/bin/lsnodeid` command. This will not match the data from the IBM boot image.
- 9. Generate a `nodeid` to match the original boot ID. Start RMC again and wait 15 minutes.

```
$ /usr/sbin/rsct/bin/rmcctrl -p
odmdelete -o CuAt -q name=cluster0 (optional before the recfgct command)
/usr/sbin/rsct/install/bin/recfgct -I value from /etc/ct_node_id (on the original IBM boot disk collected above)
```

- 10. Run the `/usr/sbin/rsct/bin/lsnodeid` command.
- 11. The output should be the same as the output from the IBM boot image.

Setting up a Network Installation Management (NIM) server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can provision an AIX virtual machine (VM) and use it as a NIM server for troubleshooting purposes or for downloading fixes and updates.

- 1. Determine the hostname and IP address of the system by using the following two commands, `hostname` and `ifconfig -a`.

Determining your hostname and IP address:

```
$ # hostname
mpb-test-aix
# ifconfig -a
en0: flags=1e084863,814c0<UP, BROADCAST, NOTRAILERS, RUNNING, SIMPLEX, MULTICAST, GROUPT, 64BIT, CHECKSUM_OFFLOAD (ACTIVE),
LARGESEND, CHAIN>
    inet 192.168.2.228 netmask 0xfffff00 broadcast 192.168.2.255
    tcp_sendspace 262144 tcp_recvspace 262144 rfc1323 1
enl: flags=1e084863, 814c0<UP, BROADCAST, NOTRAILERS, RUNNING, SIMPLEX, MULTICAST, GROUPT, 64BIT, CHECKSUM_OFFLOAD (ACTIVE),
LARGESEND, CHAIN>
    inet6 fe80::205f: dfff:fe3e:dc56/64
    tcp_sendspace 262144 tcp_recvspace 262144 rfc1323 1
sit0: flags=8100041<UP, RUNNING, LINKO>
    inet6 ::96
lo0: flags=e08084b, c0<UP, BROADCAST, LOOPBACK, RUNNING, SIMPLEX, MULTICAST, GROUPT, 64 BIT, LARGESEND, CHAIN>
    inet 127.0.0.1 netmask 0xff000000 broadcast 127.255.255.255
    inet6 ::1%1/128
    tcp_sendspace 131072 tcp_recvspace 131072 rfc1323 1
#
```

- 2. Using the information from the previous step, add an entry for your hostname and IP address into `/etc/hosts`. For example, `echo "192.168.0.15 aix-7100-05-04" >> /etc/hosts`.
- 3. Confirm that a valid `bos.vendor.profile` exists by copying the `/usr/lpp/bosinst/bos.vendor.profile` into the `installp/ppc` directory of the source repository. For example, `cp /usr/lpp/bosinst/bos.vendor.profile /usr/sys/inst.images/installp/ppc/`.
- 4. Run the following command:

```
$ `nim_master_setup -a device=/usr/sys/inst.images -a mk_resource=no`
```

Creating a NIM master:

```
$ # echo "192.168.2.228 mpb-test-aix" >> /etc/hosts
# nim_master_setup -a device=/usr/sys/inst.images -a mk_resource=no

##### NIM master setup #####
During script execution, lpp_source and spot resource creation times may vary.
To view the install log at any time during nim_master_setup, run the command: tail -f /var/adm/ras/nim.setup in a separate screen.
Creating image.data file...
```

5. Once completed, the NIM master file set are installed and the basic resource objects created. The administrator is now able to add more NIM clients and define resources.

NIM master installation summary:

\$ +-----+				
Summaries				
+-----+				
Installation Summary				

Name	Level	Part	Event	Result
bos.sysmgt.nim.master	7.2.3.17	USR	APPLY	SUCCESS

For more information, see [Setting up NIM to boot into maintenance mode](#). If you are unfamiliar with this process, create a [new support case](#).

Additional information

- [Migrating Clients by using NIM on PowerVS](#) (: external).

IBM i

Configuring your IBM i VSI

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

The following instructions describe the first boot and usage of your IBM i-based virtual server instance (VSI).



Note: VSIs that runs the IBM i 7.3 or later stock images, support up to 127 storage volumes per VSI by default. However, you can configure the VSIs to support up to 508 storage volumes per VSI.

First boot of IBM i

When you use IBM i 7.5 or later stock OS image on an IBM® Power® Virtual Server, you must first reset the Dedicated Service Tools (DST) password for the IBM i standard user `QSECOFR` through the console. This step needs to be completed within an hour of VSI deployment for the TCP configuration to finish.

The IBM i standard user `QSECOFR` is set to the password `QSECOFR`, and must be changed to a password with 15 characters (no blank spaces) through the VSI console. For a refresh of VSI console usage with IBM i, see [Tips for working with the IBM i console](#).



Note: For steps on changing the DST password, see [Changing the DST password](#).

Accessing the VSI console

To open the VSI console on the first boot, complete the following steps:

1. Go to the **IBM Power Virtual Server** instance in the Power Virtual Server user interface and select your IBM i-based VSI. The Virtual server instance details page is displayed.
2. Click **VM actions** and select **Open console** from the drop-down list. The Console Setting dialog is displayed.
3. Select **037 English** from the **Console language** drop-down list.
4. Click **Open console**. The IBM i console opens in a new window. Ensure that your browser settings allow pop-up windows.



Important: The idle timeout for the VSI console is 2 hours.

To disconnect from the VSI console session, close the web browser window that is titled 'noVNC'.

If you lose internet connectivity, your VSI console session (provided through noVNC) displays an error and does not auto-connect when internet connectivity is restored.

To restore the VSI console session, complete the following steps:

1. Return to the IBM Power Virtual instance.
2. Click **VM actions** in the Virtual server details page and select **Open console** from the drop-down list. The session is restored and the 'Connected (encrypted)' message is displayed

Alternatively, if you have multiple IBM i-based VSIs you can use the IBM Cloud CLI to return the VSI console session URL, such as using a shell command loop:

```
$ ibmcloud pi workspace list
ibmcloud pi workspace target <<IBM_POWER_VS_WORKSPACE_CRN>>
for i in $(ibmcloud pi instance list --json | jq -r '.pvmlInstances.[]' | select(.operatingSystem.type=="ibmi").id'); do echo "" && ibmcloud pi instance console get "$i"; done
```

Changing the DST password

When the VSI console is loading and the IBM i console screen is shown (IBM i 7.5 or later), complete the following steps to reset the password of the IBM i standard user:



Note: Releases before IBM i 7.5 is presented on the System Sign-on screen first, where the `QSECOFR` password needs to be reset.

1. In the VSI console window, the IBM i VSI waits on **Dedicated Service Tools (DST) Sign On** screen, type `QSECOFR` followed by clicking **PF5** at the end of the console window to open the change password screen for the IBM i standard user.
2. The **Change Service Tools User Password** screen loads, and the cursor is on `Current password . . . .`, enter `QSECOFR`.
3. Press the TAB key to move the cursor to the `New password . . . .` field and enter a 15 character password (no blanks).
4. Repeat the step on the `New password (to verify) . . . .` field and enter the new password again.
5. Press ENTER to continue.
6. The **IPL or Install the System** screen is shown.



Note: Multiple attempts are permitted, if the warning message is shown about locking the user then click **PF3** at the bottom of the console window and start again.

The IBM i VSI is now ready to be configured and prompts to select whether to perform an Initial Program Load (IPL), Install the OS, or Perform an Automated Install of the OS.

The default is to select Option 1 'Perform an IPL'.

Reviewing software agreements

When the installation is complete, the **System Sign-On** screen is displayed. Follow the [Changing the DST password](#) procedure to change `QSECOFR` password. After the system sign-on is complete, accept the Licensed Program Software Agreements.

1. After you login, the screen automatically changes to the work with **Software Agreements** screen.
2. To accept the license agreements from the console, press **5=display** each license agreement and press ENTER.
3. Press **F14** to accept each agreement.
4. Once you accept all agreements press ENTER, the IBM i main menu is displayed.
5. When the IBM i main menu is shown, the cloud-init configuration of the network and injection of the license keys begin. The cloud-init configuration process can take up to 5 minutes.
6. Confirm that the system has the [Minimum PTF levels for IBM i](#).

Tips for working with the IBM i console

- The standard IBM i user/password is `QSECOFR/QSECOFR`.
- IBM i uses function keys extensively. At the bottom of the console, you can see **PF1** through **PF12**. To get to **PF13** to **PF24**, click the **Next...** button.
- If you see a red **X** in the console during the configuration process, use your keyboard's **CONTROL** button to exit.
- You can use **CONTROL+W** to end a hung session. If this happens, you must perform a bypass by clicking **PF18** and logging on again.
- It is recommended to first shut down the system before you restart it.
- If you are using a Mac computer, the **Page Down** key is the same as **FN + Down Arrow**.
- `CRTUSRPRF` and `CHGUSRPRF` use the same new security rules as `QSECOFR`, requiring at least a 15 character password.

Verifying licenses and network configuration

To resolve an issue with operating system licenses that are expired when you deploy IBM i images, perform the following steps on the IBM Power Virtual Server console:

1. After you change the default login password for OS in an IBM i VSI, the Operating System License Information screen is displayed. If you see the error message, `No operating system licenses are available at this time.` wait for 15 min.
2. Press **F5** to retry license verification and to continue with the sign-on process.
3. Accept Software Agreements.
4. Run IBM i command `STRSBS QBASE` or `STRSBS QCTL` to exit restricted state.
5. Run IBM i command `CFGTCP` to verify whether the TCP/IP configuration is complete.

If the stock image is deployed without displaying the Operating System License Information screen, run the `STRSBS QBASE` command within one hour of the VSI deployment. For additional help, contact the support team. For more information, see [Getting Help and Support](#).

To verify that `cloud-init` configured your IP addresses correctly, on the **IBM i main menu** screen type the `cfgtcp` command in the console window (on line Selection or command ==>), press ENTER, choose **1**, and press ENTER again.

One or more IP addresses with "Line Description" as `CLOUDINIT<<0..n>>` must be shown and match the network interfaces for the attached IBM Power VS network subnets that are shown in the IBM Power Virtual Server workspace. If the IP addresses match, then `cloud-init` configuration ran successfully.

See the following example of the `cloud-init` configuration verification:


```
$ Work with TCP/IP Interfaces
                                     System: KCW73A
Type options, press Enter.
1=Add 2=Change 4=Remove 5=Display 9=Start 10=End

   Internet      Subnet      Line      Line
Opt  Address      Mask      Description  Type
___  _____  _____  _____  ____
___  127.0.0.1     255.0.0.0  *LOOPBACK   *NONE
___  10.1.1.5       255.255.255.192  CLOUDINIT1  *ELAN
___  192.168.128.138 255.255.255.248  CLOUDINIT2  *ELAN

                                     Bottom
F3=Exit   F5=Refresh  F6 Print list  F11-Display interface status
F12=Cancel F17=Top      F18=Bottom
```

For external IP addresses, if you do not see the external IP address in the **Work with TCP/IP Interfaces** screen, wait approximately 10 minutes, open another terminal, and ping the external IP address. The external address must match what is shown in the Power Virtual Server user interface within your instance's **Server details** page. Contact support or delete and reprovision your IBM i VSI if the ping doesn't return anything.

Lastly, on the **IBM i main menu** screen type the `WRKLCINF` command in the console window (on line Selection or command ==>), press ENTER, then click **PF11** at the bottom of the console window to display the usage information. The Usage Limit, Usage Count and Peak Usage values must be populated for each license.

When you have verified your network and license key configuration, you can initial program load (IPL) the LPAR.

If license keys for the operating system or any IBM i licensed program Products (LPP) are not applied, follow instructions in the [PowerVS license key issues](#) document.

If this is an upgraded system that contained license Keys before, allow a weekend (Saturday-Sunday) to process the updated keys before you collect the [PowerVS license key issues](#) information.

 **Note:** The IBM i VSI might result in a timeout error, if you update the IBM i VSI license and memory together.

Changing the System Service Tools (SST) and Dedicated Service Tools (DST) passwords

The SST and DST passwords are expired by default. Complete the following tasks to get into SST and change the passwords.

 **Note:** For more information on user ID types, see [Managing service tools user IDs](#).

To change the System Service Tools (SST) and Dedicated Service Tools (DST) passwords, complete the following steps:

1. Go to the **IBM Power Virtual Server** instance in the Power Virtual Server user interface and click your IBM i VSI. The Virtual server instance details page is displayed.
2. Click **VM actions** in the Virtual server details page and select **IBM i operations** from the drop-down list. The IBM i operations dialog is displayed.
3. Select **(21) Active dedicated service tools** option from the list displayed on the **Control Panel functions** tab.
4. Click **Run action**.

 **Note:** For more information on user ID types, see [Managing service tools user IDs](#).

Managing the IBM i VSI instance

You can use the **IBM i operations** option to manage advanced VSI operations and configuration. To run the control panel functions or boot operations, complete the following steps:

1. Go to the **IBM Power Virtual Server** instance in the Power Virtual Server user interface and click your IBM i VSI. The Virtual server instance details page is displayed.
2. Click **VM actions** in the Virtual server details page and select **IBM i operations** from the drop-down list. The IBM i operations dialog is displayed.

 **Note:** The **IBM i operations** option is available only for the IBM i VSIs.

3. In the IBM i operations dialog, select one of the following actions that are listed on the **Control panel functions** as per your requirements:

Job operations:

- **(21) Activate dedicated service tools** – Activates dedicated service tools (DST) and makes it available on the system console display. You can use DST from the system console display to work with Licensed Internal Code (LIC), disk units, configuration and resources, and to verify devices and communications.
- **(34) Retry Dump** – Retries a failed main storage dump (MSD) operation. You can run this function when the VSI is hung during the MSD operation or when you want to reattempt the initial program load (IPL) operation without losing the original dump information.
- **(66) Enable remote service** - Activates the communications port that is used by a remote service session or operations console.
- **(67) Disk unit IOP reset/reload** - Initiates an I/O processor (IOP) dump operation and a disk unit IOP reset or reload operation. The function is enabled only when specific system reference codes (SRCs) are displayed on the control page and the associated IOP supports a reset or reload operation. This function is not available for all system types.
- **(70) IOP Control storage dump** - Saves the contents of the service processor control storage into a nonvolatile storage for potential use from an error log file.
- **(22) Dump restart** - Copies main storage data and processor data to the disk. Run this function only when an MSD operation is necessary, for example, after a suspended (system hang) condition or after an operating system failure.



Note: You must not shut down the system before the MSD operation. This action can cause data loss.

4. In the IBM i operations dialog, configure one of the following options that are listed on the **Boot operations** tab as per your requirements:

Server boot mode

- **A-** Perform an IPL operation from a disk by using the copy A of the system LIC.
- **B-** Perform an IPL operation from a disk by using the copy B of the system LIC.
- **C-** Reserved for IBM lab use only.
- **D-** Perform an IPL operation from a media other than the load-source disk. Perform an alternative IPL operation for code installation support.

Server operating mode

- **Normal** - Allows you to access the operating system and perform an unattended IPL operation. After the system power-on, operating the system in **Normal** (unattended) mode requires no operator intervention during the IPL operation. When you turn on the system in **Normal** mode, the system performs the IPL operation and presents the **Sign On** display on all available display stations. The operator cannot change the internal system environment during the IPL operation. Dedicated service tools (DST) and the operating system do not present any menus or options during this IPL operation.
- **Manual** - Allows you to access DST and perform an attended IPL operation. After the system power-on, operating the system in Manual (attended) mode means that an operator uses the **Operation** page or the control page to direct the system for special needs. During the IPL operation in **Manual** mode, DST and the operating system present menus and prompts you to change the internal system environment. These modifications can include entering debug mode for service representatives to diagnose difficult problems. For more information, see [Operating mode of an IPL](#).

5. Click **Run Action**.

IBM i language support

The VSI console of an IBM i VSI supports the English language by default and provides the following language support:

- IBM i stock OS images support English (2924) or English DBCS (2984). You can install secondary languages and change the primary language on an IBM i VSI.
- The VSI console also supports Japanese if it is installed as a primary or secondary language.



Note: If you select Japanese as the language in the Console Setting dialog before you start the IBM i VSI console, the system does not install the Japanese language. Installing Japanese as a primary or secondary language is a separate task that you must perform on an IBM i VSI.

- To change the service tools language of your IBM i system, see [Changing the service tools language of your system or logical partition](#).
- The COR stock OS image can be used to deploy a second IBM i VSI. The second IBM i VSI that is deployed contains the language media that can be used to install primary and secondary languages on other IBM i VSI.

For more information about IBM i network and software installation, see:

- [Setting up an IBM i network install server](#)
- [Globalization and IBM i software installation](#)

Considerations for IBM i and Global Replication Services (GRS)

If you plan to use Global Replication Services to replicate your IBM i VSI volumes, review the recommended volume group limits and their impact on I/O response times. For more information, see [Performance and I/O considerations for GRS](#).

Assigning an IBM i software tier to an IBM i Power Virtual Server instance

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IBM i software tiers are a licensing model that classifies IBM Power Virtual Servers by processor characteristics to determine the price and feature set for the IBM i operating system (OS) and other software.

On IBM Power10 and later systems, you can assign an IBM i software tier to a virtual server instance (VSI). The IBM i software tier limits the capacity of the VSI, but you can select any physical system that supports the assigned tier. For example, a virtual server that is assigned to a P10 tier can run on either an S1022 or an E1080 server. The tier restricts the resource allocation for the VSI but not its hardware compatibility.

The IBM i software tier determines the pricing tier for the IBM i OS and Licensed Program Products (LPPs). The tier also defines the limits for the following resources:

- Maximum number of virtual processors
- Maximum memory

You can adjust the number of CPUs and the amount of memory without powering off the VSI if the values are in the IBM i software tier limits. To increase the capacity beyond the current IBM i software tier limits, you must complete the following steps:

1. Power off the VSI.
2. Change the IBM i software tier to a tier that supports your required number of virtual processors and memory.

For more information, see [Supported resource limits by the IBM i software tier](#).

When you change the software tier, the billing for the IBM i software tier automatically changes to reflect the selected tier.

You can generate an estimate of Power Virtual Server resources with an IBM i software tier before you deploy the resources. For more information, see [Estimating a virtual server instance](#).

To assign an IBM i software tier to an IBM i Power Virtual Server instance, complete the following steps:

1. Create a VSI by following the instructions at [Configuring a Power Virtual Server instance](#).
2. Select an IBM i image with version 7.3 or later from the **Image** list in the **Boot image** section.
3. Select an IBM Power10 or later server type from the **Machine type** list in the **Profile** section.
4. Click the **Edit** icon for the **Virtual serial number (VSN)** field in the **Profile** section. The virtual serial number summary pane appears.
5. Select **Auto-assign** or **Select from retained VSNs** to assign a VSN.



Note: You can assign a VSN only to a VSI that runs IBM i version 7.2 or later.

The **IBM i software tier** list displays only the tiers that are compatible with the machine type that you select. The **IBM i software tier** list also shows the recommended tier based on the number of cores and memory that you configure. You can select the IBM i software tier that is displayed in the **IBM i software tier** field or other options from the list.

Supported resource limits by the IBM i software tier

IBM i software tiers are available in different licensing models such as P05, P10, P20, and P30. Each IBM i software tier defines the supported resource limits. These limits help align resource usage with software licensing costs. You must understand the supported limits of resources for planning your VSI capacity and ensure compliance with IBM i licensing models.

The following table lists the maximum number of virtual processors and memory for each IBM i software tier:

IBM i software tier	Maximum virtual processors	Maximum memory
P05	1	64 GB
P10	4	1 TB
P20	12	1 TB
P30	Unlimited	Unlimited

Maximum virtual processors and memory for each IBM i software tier

Supported IBM i software tiers on Power Virtual Server

You can assign IBM i software tiers only to compatible Power Virtual Server system types. The following table lists the IBM i software tiers that are compatible with each system type.

System types	Supported IBM i software tiers				
	P05		P10	P20	P30
S1022	Yes		Yes	Yes	No
S1122	Yes		Yes	Yes	No
E1080	No		Yes	Yes	Yes
E1180	No		Yes	Yes	Yes

Supported IBM i software tiers on Power Virtual Servers



Note: IBM i is not supported on IBM E1050 and E1150 system types.

If you deploy multiple IBM i VSIs with the P20 software tier on IBM S1022 hosts, the deployment might fail during an IBM data center upgrade. However, deploying a single IBM i VSI with the P20 software tier on an S1022 host is supported.

Connecting to an IBM i virtual machine (VM)

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Learn how to connect to an IBM i VM after configuring your system. Make sure to review [Configuring your IBM i virtual machine \(VM\)](#) before connecting to an IBM i VM.



Important: For a complete list of firewall ports that are available for IBM i VMs, see [Network security](#). If you plan on ordering [Direct Link Connect on Classic](#) or already have it, port forwarding is not needed.

Installing and configuring IBM i Access Client Solutions (ACS)

Before you begin, see [Install IBM i Access Client Solutions](#).

Using SSH tunneling to allow ACS to connect over the public IP

The public IP address blocks most ports. As a result, you need to use SSH tunneling or configure your certificates and use SSL to allow ACS to connect over public IP.

Before you use an SSH tunnel, you must create a user profile with `USRCLS(*SECOFR)` specified or enable the `QSECOFR` user profile. To enable the `QSECOFR` user profile, edit the `/QOpenSys/QIBM/ProdData/SC1/OpenSSH/etc/sshd_config` configuration file, and uncomment `PermitRootLogin yes`.

After the `QSECOFR` user profile is enabled, start the SSHD server on the VM:

```
$ endtcpsvr server(*SSHD)instance(*all)
strtcpsvr server(*SSHD)
```

On a Linux® or Mac system, you would run a command similar to the following example:

```
$ ssh -L 50000:localhost:23 -L 2001:localhost:2001 -L 449:localhost:449 -L 8470:localhost:8470 -L 8471:localhost:8471 -L 8472:localhost:8472 -L 2007:localhost:2007 -L 8473:localhost:8473 -L 8474:localhost:8474 -L 8475:localhost:8475 -L 8476:localhost:8476 -L 2003:localhost:2003 -L 2002:localhost:2002 -L 2006:localhost:2006 -L 2300:localhost:2300 -L 2323:localhost:2323 -L 3001:localhost:3001 -L 3002:localhost:3002 -L 2005:localhost:2005 -o ExitOnForwardFailure=yes -o ServerAliveInterval=15 -o ServerAliveCountMax=3 <myuser>@<myIPaddress>
```

 **Note:** You might have to type `sudo` in front of the `ssh` command if the system denies you permission.

For further information on information and mapping, see [TCP/IP Ports Required for IBM i Access and Related Functions](#) .

If you are on a Windows® system, continue with [Setting up and configuring PuTTY on a Windows system](#) , otherwise see [Starting TCP servers](#) .

Setting up and configuring PuTTY on a Windows system

Install [PuTTY](#) onto your system. PuTTY is used for the SSH tunnel on a Windows system.

1. Open **Session** under **Category:**.
2. Enter your system's **IP address** and select **SSH** as the **Connection type**.
3. Enter **22** as the port number.
4. Under the **Connection** category, select **Connection>SSH>Tunnels**.
5. Add your **Source port** number and **Destination**. For example, you can chose 50000 as the source port number.

 **Tip:** Do not change the source port numbers. When telnetting, avoid making the source port the same as the destination.

6. Click **Add** to add your source port to the forwarded port list.

 **Important:** You need to repeat step **3** to step **6** to add all of the following ports: 23, 449, 8470, 8471, 8472, 8473, 8474, 8475, 8476, 2003, 2002, 2006, 2300, 2323, 3001, 3002, and 2005.

7. After you add all of the necessary port numbers, check your populated list.
8. Click back on the **Session** category and give your session a name under **Saved Sessions**. Click **Save**.
9. Your saved session appears under **Saved Sessions** after you click **Save**. Select your session and click **Open** to start a PuTTY session to your system.


Seeing your list of saved sessions

10. You are prompted to accept a key on first use, and then presented with a log-in prompt. Use your IBM i session profile and password.
11. Configure you ACS client or **IBM i Access for Windows Client** to use the SSH tunnel. In both clients, you must select **Configure** from the **Communications** menu.
12. Change the IP address to `127.0.0.1` on **port 23**.
13. Press **OK** to save the changes. The client restarts and connects.

Starting the TCP servers

Start the required TCP servers on your IBM i operating system by completing the following tasks:

1. To allow SSH connections, enter the following command:

```
$ strtcpsvr server(*SSHD)
```

2. To start the IBM Navigator for i (iNav) and Digital Certificate Manager (DCM) GUIs, enter the following command:

```
$ strtcpvr server(*HTTP) httpvr(*ADMIN)
```

3. To get a 5250 console from ACS, start Telnet:

```
$ strtcpvr server(*TELNET)
```

Starting a 5250 session on your IBM i VM from ACS

To get a 5250 session on your IBM i VM from ACS, either configure the virtual devices or enable autoconfig. To enable autoconfig, complete the following steps by using the IBM i VM:



Note: For IBM i 7.5, it is required that you first use `CHGSSTSECA` to set `SECSYSVAL *YES` (to allow security system value changes).

1. Enter the `cfgtcp` command.
2. Select option **20** (Configure TCP/IP applications).
3. Select option **11** (Configure TELNET).
4. Select option **10** (Autoconfigure virtual devices).
5. Select `QAUTOVRT` with option **2** (Change).
6. Change the value from **0** to the number of auto-configured consoles that you want to be able to connect concurrently.
7. Go to the IBM i VM and start the telnet server for the console:

```
$ strtcpvr server(*TELNET)
```

After you complete these steps, you can get to a console from ACS. Additionally, you can get to *iNav/DM* by pointing your browser to the following address:

```
$ https://127.0.0.1:2005/ibm/console/login.do?action=secure
```

To enable ICC to use an SSL connection to IBM Cloud Object Storage, which IBM Cloud Object Storage requires, see [Configuring Cloud Storage Solutions file transfer encryption](#).

Configuring ACS

After starting ACS, create a system configuration (*sysconfig*).

1. Configure a server for *localhost*. For example, **port 50000** is forwarding to **port 23**. Go into the 5250 session configuration and change the port from **23** to **50000**.
2. Return to the IBM i terminal and enter your credentials, including *System*, *User*, and *Password*.

Setting up an IBM i network install server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Before you can install or upgrade an IBM i system through the network, you must set up a network installation server. The network installation server contains images of the IBM i operating system, its internal code, licensed programs, and PTFs. To share virtual optical images through a network, the network installation server must meet the following requirements that are documented in the subsequent sections.

To learn more about the options available on upgrading the IBM i operating system, see [Performing IBM i Upgrades on Power Virtual Servers](#).

Configuration

To complete the configuration of an IBM i network installation server, see [Virtual optical storage by using the Network File System](#) and [IBM i Network Install](#).

Cloud Optical Repository (COR) is a virtual image that can be deployed and used as a Network File Server (NFS) to complete various IBM i tasks that require media. This virtual optical image includes a collection of the media necessary for various IBM i tasks, for all supported IBM i releases. With the COR image deployed, a second Power Virtual Server Instance can be deployed on the same VLAN that is set up as the client and pointed to the COR (target) NFS

Server Instance. For more information on COR images, see [Cloud Optical Repository](#).

How to add or remove a network interface from an IBM i virtual machine (VM)

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Since IBM PowerVC Version 1.2.2, IBM PowerVC can dynamically add a network interface controller (NIC) to a virtual machine (VM) or remove a NIC from a VM. IBM PowerVC does not set the IP address for new network interfaces that are created after the VM deployment. Any removal of a NIC results in freeing the IP address that was set on it. You must remove and re-add the IBM i VM network interface if you choose to disconnect the Power Virtual Server IBM i VM from a public network.



Note: When you toggle a public network off and on, the IBM® Power® Virtual Server user interface regenerates new internal and external IP addresses. You need to check the Power Virtual Server user interface for the new internal IP address to complete this procedure.

Removing a network interface from an IBM i VM

You can change the TCP/IP address of your IBM i VM while the TCP/IP is active. However, you must deactivate the TCP/IP interface. For more information about the complete set of instructions, see [Changing the TCP/IP Address of the IBM i System](#).

1. Before you change a TCP/IP address, determine whether it has any associated routes. Select **Option 8** from the **NETSTAT *IFC** screen to view the routes.



Tip: For all the **non-*DIRECT** routes that are listed on the screen, you can display the routes and print it for reference. Select **Option 5** to display the details of the route. Press **F6** to print the details for reference for when the route must be re-added.

2. Run the `CFGTCP` command and select **Option 2** to work with your TCP/IP routes. Select **Option 4** next to the routes that you need to remove. When you remove the routes, the communication that is happening over the route is terminated.
3. To make the changes, you must deactivate and remove the interface before you add it back. Select **Option 10** next to the interface that you need to deactivate on the **NETSTAT *IFC** screen.
4. To remove the interface after deactivation, run the `CFGTCP` command and select **Option 1** from the menu. Select **Option 4** next to the interface that you need to remove.



Removing a network interface

1. You must vary off the Line description (LIND) after you remove the interface.

Adding a network interface to an IBM i VM

When you toggle a public network off and on, the Power Virtual Server user interface regenerates new internal and external IP addresses. You must check the Power Virtual Server user interface for the new internal IP address that maps to the external IP address. Point the new interface at the new internal IP address.

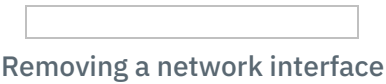
1. After routes and interfaces are removed, create the new configuration in the reverse order. To open the **ADDTCPIFC** screen, run the `CFGTCP` command and select **Option 1**.



Note: In most of the configurations, you might need to update only the first three fields.

2. Add the new internal IP address that you obtained from the Power Virtual Server user interface to the **Internet address** field. You can type the IP address over the quotation marks.
3. Add the new subnet mask to the **Subnet mask** field.
4. Complete the remaining fields by using your reference printout. The LIND value must be the same as the LIND value defined on the interface that was removed earlier.
5. Activate the new interface by selecting **Option 9** from the **NETSTAT *IFC** screen.
6. To verify that the new interface is active, ping the IP address of the new interface from the command line. If the ping responds, the interface is working correctly.

7. Add the new routes that use the new interface, if any. You can add the new routes by selecting **Option 2** from the `CFGTCP` menu, type **1** in the **Opt** column, and press the **Enter** key.



Verify that the local adapter address of the line description matches the MAC address of the newly added adapter address from the Power Virtual Server user interface.

For example, Find the MAC address in the line description by using `DSPLIND LIND(xxxxxxxx)` command. The line description is the name that matches the **Internet address** field value that was added earlier. Use the **NETSTAT *IFC** screen to view the line descriptor.



Minimum PTF levels for IBM i

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Install the following program temporary fixes (PTFs) depending on the version of IBM i that is being used:

- IBM i 7.3 - MF99213 (TR13) and SJ02851, SJ04128
- IBM i 7.4 - MJ99312 (TR12) and SJ08183, SJ04127
- IBM i 7.5 - MJ99407 (TR7) and SJ08081
- IBM i 7.6 - MJ99501 (TR1) and SJ08079

 **Important:** Ensure that your IBM i image includes the necessary code levels for the Power system that you are deploying. To verify the minimum supported code level, see the [IBM i Platform Support Summary](#).

For more information on installing PTF packages, see [Installing cumulative PTF packages](#). The `SNDPTFORD` command can also be used to send PTFs to the system, see [Send PTF Order \(SNDPTFORD\)](#).

When you use an IBM i custom image, these PTFs must be installed and the software that is required for `Cloud-Init`. For more information, see [Cloud-Init Support for IBM i](#). An overview of IBM i fix concepts and terms is found at [Fixes concepts and terms](#).

 **Important:** To learn more about the options available to upgrade the IBM i operating system, see [Performing IBM i Upgrades on Power Virtual Servers](#).

Required PTFs for iSCSI Virtual Tape Library (VTL)

You must install the program temporary fixes (PTFs) depending on the version of IBM i that you are using. To view PTF for each release along with the configuration procedures, see [IBM i Support for iSCSI VTL](#).

Best practices for using IBM i stock images

1. When you select an IBM i stock image, you must verify that the latest group of PTFs is installed, especially the Hiper and Security group. For more information, see [IBM i Group PTFs with level](#).
2. Refer to the [IBM PSIRT blog](#) for the latest Product Security Incident Response.
3. You can subscribe to valuable product notifications at [My Notifications](#).
4. Cloud Optical Repository (COR) is a virtual image that can be deployed and used as a Network File Server (NFS) to perform various IBM i tasks that require media. This virtual optical image includes a collection of the media necessary for various IBM i tasks, for all supported IBM i releases. With the COR image deployed, a second Power Virtual Server instance can be deployed on the same VLAN that is set up as the client and pointed to the COR (target) NFS Server instance. For more information on COR images, see [Cloud Optical Repository](#).
5. English (2924) and English DBSC (2984) versions are included with IBM i stock images. To install other languages, you can use the NFS stock image that is mentioned in the preceding best practice.

IBM i License Program Products (LPP) and Operating System feature bundles

When you select IBM i, in addition to the IBM i operating system, many other IBM i features are included in the IBM® Power® Virtual Server offering.

The following LPP, and IBM i operating system features are included in the IBM® Power® Virtual Server offering:

- 5770-SS1 IBM i processor and unlimited users
- 5770-SS1: IBM i Option 23, OptiConnect
- 5770-SS1: IBM i Option 44, Encrypted Backup Enablement
- 5770-SS1: IBM i Option 45, Encrypted ASP Enablement
- 5770-SS1 IBM i Option 18 Media & Storage Extensions
- 5770-SS1 IBM i Option 26 Db2 Symmetric Multiprocessing
- 5770-SS1 IBM i Option 27 Db2 Multisystem
- 5770-SS1 IBM i Option 38 PSF for IBM i Any Speed Printer Support
- 5770-SS1 IBM i Option 41 HA Switchable Resources
- 5770-SS1 IBM i Option 42 HA Journal Performance
- 5770-SS1 IBM i Options: all other IBM i Options that are included with IBM i at no additional charge. Examples:
 - Option 8, AFP Compatibility Fonts
 - Option 12, Host Servers
 - Option 13, System Openness Includes
- 5770-DG1: HTTP Server for i
- 5770-JV1: Developer Kit for Java
- 5770-NAE: Network Authentication Enablement for i
- 5733-SC1: Portable Utilities for i
- 5770-TC1: TCP/IP
- 5770-TS1: Transform Services for i
- 5770-UME: Universal Manageability Enablement for i
- 5770-XE1: IBM i Access for Windows operating system
- 5733-ACS: IBM i Access Client Solutions
- 5733-ARE: IBM Administration Runtime Expert
- 5798-FAX: IBM Facsimile Support for i
- 5770-SM1: IBM System Manager for i
- 5770-DFH: IBM CICS Transaction Server for i
- 5770-MG1: IBM Managed System Services for i
- 5770-AF1: Advanced Function Printing for i
- 5761-AMT: Rational Application Management Toolset
- 5761-AP1: Advanced DBCS Printer Support
- 5733-B45: AFP Font Collection for i
- 5770-BR1: Backup, Recovery, and Media Services
- 5770-BR2: Backup, Recovery, and Media Services
- 5761-DB1: System/38 Utilities
- 5761-CM1: Communications Utilities
- 5761-DS2: Business Graphics Utility
- 5648-E77: InfoPrint Fonts
- 5769-FN1: AFP DBCS Fonts
- 5769-FNT: AFP Fonts
- 5722-IP1: Infoprint Server for i
- 5770-JS1: Advanced Job Scheduler for i
- 5770-PT1: Performance Tools
- 5770-QU1: Query for i
- 5770-ST1: Db2 Query Manager and SQL Dev Kit for i

- 5733-XT2: XML Toolkit
- 5770-XH2: IBM i Access for web
- 5770-XW1: IBM i Access Family

Each LPP in the package contains all the features that's includes the optional features. For example, the **5770-BR1** solution includes the network feature and the advanced features in addition to the base product.

These additional LPPs for IBM i, which are available and can be included in your VM instance. To include one or more such LPPs, complete the following steps:

1. Go to **Virtual server instances** in the Power Virtual Server user interface and click your instance.
2. Click the **Edit details** option in the **Server details** page. A menu appears.
3. Select the wanted licenses that you want to include in your VM instance. Currently, you can purchase the following licenses through Power Virtual Server:
 - IBM i Cloud Storage Solutions (5773-ICC)
 - IBM i PowerHA (5770-HAS)
 - Rational Development Studio for IBM i (5770-WDS)

Each of these programs is individually priced. The selected licensed programs are injected to your VM instance. Or install specific licensed program solutions on your VM instance. If you want to use these programs on your IBM i VM instance, you must order these licensed programs through Power Virtual Server you cannot use existing licenses in your VM instance.

4. Check the service agreement box and click **Save edits and order** to complete the instance modification process and to accept the price.
5. View the **Server details** page to verify your instance modification.

Passport Advantage Software

In general, IBM programs that are acquired through Passport Advantage can be brought to the Power Virtual Server offering. One example of an IBM i program that is acquired through Passport Advantage is Rational Developer for i (RDi). For RDi, you can bring your current Rational Developer for i (RDi) license to the Power Virtual Server offering whether RDi was obtained through the Power Hardware channel or through Passport Advantage. If you do not currently have a RDi license and need the product, obtain a license by using Passport Advantage, and bring it to the Power Virtual Server offering.

Other examples of IBM software that is acquired through Passport Advantage are WebSphere MQ, Db2 Connect, and Lotus Notes. For more information, see [licensing and Passport Advantage](#).

Linux

Full Linux® subscription for IBM Power Virtual Server in IBM data center

IBM Power Virtual Server in **IBM data center**

The full Linux® subscription feature enables Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) support through IBM.

The full Linux subscription also provides, via activation keys, access to OS interim fixes and updates for Power servers hosted on an IBM satellite server within the IBM Cloud environment. Extra charges apply when setting up a Full Linux subscription account.

To register for the full Linux subscription, you must select one of the stock operating system (OS) images that are provided by IBM. IBM provides RHEL and SLES stock OS images for SAP and non-SAP applications. To know more about the SLES versions that are supported, see [Operating systems supported in IBM Power Virtual Server](#).

The full Linux subscription feature OS file name starts with the Red Hat or SUSE distribution name, **RHEL...** or **SLES...**.



Note: If you plan to use your own license, select the OS image that is suffixed with **-BYOL**. On the VM Provisioning page, these images are listed under the **Client supplied subscription** section.

Setting up full Linux subscription

To set up full Linux subscription for your account, complete the following steps:

1. [Configuring a Cloud connection](#)
2. [Creating a proxy](#)
3. [Creating and configuring a Power virtual machine \(PVM\)](#)
 - a. RHEL
 - b. SLES
4. [Configuring a proxy](#)
5. [Customizing the VMs by using the cloud-init script \(RHEL and SLES\)](#)

Step 1: Configuring a Cloud connection

Ensure that the following requirements are met when you create a Cloud connection: Create a Cloud connection with the following parameters:

- Ensure that you set up a Cloud connection between the IBM® Power® Virtual Server instance and the IBM Cloud classic infrastructure by using a private network.
- The Power Virtual Server in each region and zone must have its own cloud connection.
- You have the details of this private network for further configuration.

For more information about creating a Cloud connection, see [Managing Cloud connections](#).

Step 2: Creating a proxy

Create a proxy setup by completing the following steps. This proxy is set up in a virtual private cloud (VPC) environment.

1. Deploy a proxy instance based on your requirement. For more information, see [Enabling proxy protocol](#).
2. Open the Security groups for the VPC by navigation to the IBM Cloud dashboard > VPC Infrastructure > Networks > Security groups. For more information about security groups, see [security groups](#).
3. Select the security group that is attached to your proxy and add these port numbers: 443, 8443, 80, and 3128. Ensure that the IP address that you are asked to specify is the subnet CIDR of your private network for the Power Virtual Server instance. To view the subnet CIDR in the IBM Cloud dashboard, go to the zone, click **Subnets**, identify the subnet that you are using, and copy the CIDR.
4. After the proxy is created, log in to the proxy instance and locate the subnet CIDR by completing the following steps:
 - a. Open the command-line interface (CLI) and run the `ssh root@<public IP of proxy>` command.
 - b. Run the `route -n` command to identify the subnets. The subnet of your proxy instance is listed after the private gateway in the command output.

Example subnets are: 10.240.65.0, 10.209.155.192.

c. To get the CIDR, run the `ip route` command. Example CIDRs are: 10.240.65.0/24, 10.209.155.192/26.

d. Note the CIDR number as you need it, as described in [step 3](#) to establish connections from the VM to the proxy.

Step 3: Creating and configuring a Power virtual machine (PVM)

1. Create a Power Virtual Server instance (also known as PVM instance) by using Power Virtual instance GUI, CLI, or API with the following requirements.
 - a. SSH key
 - b. Public network
 - c. Private network that you created in [step 1](#)
2. In the proxy instance CLI, start an SSH connection to your RHEL or SLES PVM instance by using the PVM's external IP address in the following command:

```
ssh root@<external IP address>
```

Example: ssh [root@1.2.3.4](#)

Ensure to use a default gateway in the private subnet when you are provisioning a virtual server instance.

3. Add a network route from the PVM instance to the proxy instance that you configured in [step 2](#).



Note: Ensure the private network that is used during the VMs provisioning is set as the default gateway and not the public subnet.

- Run the `ip route` command and add the CIDR subnet of proxy through the private gateway of PVM.

- The CIDR subnet of the proxy is the value that you got after performing [step 2](#).
- You can view the private gateway of PVM in the server details of the PVM.

Example: Run the `ip route` command and add the CIDR subnet IP address 1.5.3.9/24 through the private gateway IP address 5.9.7.1.



Note: If your VM is restarted, these IP routes are removed and the IP routes must be added again.

Step 4: Configuring a proxy instance

Set up a proxy configuration by completing the following steps:

1. Log in to your proxy instance by running the `ssh root@<public IP of proxy>` command.
2. Ensure that another IP route is created within the proxy instance. This IP route allows the IP addresses that are within the range of the private IP gateway that you created for your VM. The IP addresses are used to connect your VM to the proxy gateway.
 - Run the `ip route` command and add the CIDR number of that private network that you derived in Step 1 via the gateway of proxy.
 - You can view the private VM gateway in the [IBM Cloud](#) website. Select the PowerVS VM that you created in [step 3](#), and go to the attached private network. The CIDR number is listed as the last variable.
 - Run the `route -n` command to view the gateway of the proxy.
3. A connection is now established with the proxy and the PVM. You can ping the connection in both the proxy instance and the PVM instance, as follows:
 - In the proxy instance, you can ping the private IP of the PVM by using the `ping <private IP of powervs vm>` command.
 - In the PVM instance, you can ping the private IP of the proxy by using the `ping <private IP of proxy>` command.
4. A squid (proxy caching server) must be installed on the VM.
5. Set up squid by completing the following steps:
 - a. Install the squid base by running the following commands:
 - `sudo yum update -y`
 - `sudo yum install epel-release`
 - `sudo yum install squid`

You must have root authority to run these commands. After the installation completes, the squid config file is stored in the `/etc/squid/squid.conf` location.

b. Use the configuration provided in the following example for your reference. Values that are represented in the example configuration denote the following:

- Acl localnet src 192.168.0.0/16: the IP ranges of the IBM private network that the proxy accepts.
- Acl ibmprivate dst 161.26.0.0/16 and Acl ibmprivate dst 166.8.0.0/14: the IP ranges of the IBM networks this proxy will be going to this location of the RHEL satellite servers and SLES RMT servers.
- Acl SSL_ports port 443 8443 : Ports that are used for communication with RHEL satellite servers and SLES RMT servers.
- http_port 3128: the port that is listening for squid. This is used as the communication port from your Power Virtual Server VM to the squid proxy.

```
$ # Recommended minimum configuration:
#

# Example rule allowing access from your local networks.
# Adapt to list your (internal) IP networks from where browsing
# should be allowed
acl localnet src 10.0.0.0/8    # RFC1918 possible internal network
acl localnet src 172.16.0.0/12 # RFC1918 possible internal network
acl localnet src 192.168.0.0/16 # RFC1918 possible internal network
acl localnet src fc00::/7     # RFC 4193 local private network range
acl localnet src fe80::/10    # RFC 4291 link-local (directly plugged) machines
acl localnet src all
acl ibmprivate dst 161.26.0.0/16
acl ibmprivate dst 166.8.0.0/14
acl FTP_ports port 21 20 1025-65535
acl SSL_ports port 443 8443
acl Safe_ports port 80      # http
acl Safe_ports port 21      # ftp
acl Safe_ports port 443     # https
acl Safe_ports port 22
acl Safe_ports port 8443
acl Safe_ports port 70      # gopher
acl Safe_ports port 210     # wais
acl Safe_ports port 1025-65535 # unregistered ports
acl Safe_ports port 280     # http-mgmt
acl Safe_ports port 488     # gss-http
acl Safe_ports port 591     # filemaker
acl Safe_ports port 777     # multiling http
acl CONNECT method CONNECT

#

# Recommended minimum Access Permission configuration:
#

# Deny requests to certain unsafe ports
http_access deny !Safe_ports

# Deny CONNECT to other than secure SSL ports
http_access deny CONNECT !SSL_ports

# Only allow cachemgr access from localhost
http_access allow localhost manager
http_access deny manager

# We strongly recommend the following be uncommented to protect innocent
# web applications running on the proxy server who think the only
# one who can access services on "localhost" is a local user
#http_access deny to_localhost

# INSERT YOUR OWN RULE(S) HERE TO ALLOW ACCESS FROM YOUR CLIENTS
## Example rule allowing access from your local networks.

# Adapt localnet in the ACL section to list your (internal) IP networks
# from where browsing should be allowed
http_access allow localnet
http_access allow localhost
```



```
# And finally deny all other access to this proxy
# http_access deny all

# Squid normally listens to port 3128
#http_port 10.220.54.36:3128
http_port 3128

# Uncomment and adjust the following to add a disk cache directory.
#cache_dir ufs /var/spool/squid 100 16 256

# Leave coredumps in the first cache dir
coredump_dir /var/spool/squid

#
# Add any of your own refresh_pattern entries above these.
#
refresh_pattern ^ftp:      1440  20%  10080
refresh_pattern ^gopher:  1440  0%   1440
refresh_pattern -i (/cgi-bin/|\?) 0    0%    0
refresh_pattern .          0    20%  4320
```

- c. Save the squid config file and start squid by using the `sudo systemctl start squid` command.
- d. You can stop squid by using the `sudo systemctl stop squid` command.
- e. You can restart squid by using the `sudo systemctl restart squid` command.

Step 5: Customizing the VMs by using the cloud-init script (RHEL and SLES)

You can customize your RHEL and SLES VMs by running the cloud-init script.

1. Log in to your RHEL or SLES VM, by using the `ssh root@<External IP">` command.
2. A readme file is generated in the `/usr/share/powervs-fls/powervs-fls-readme.md` location. For relevant information about variables that are required to run the cloud-init script, see the readme file.
3. Run the cloud-init script that is located in the `/usr/local/bin/` location. Run the script as `root`. You can switch the user to `root` by using the `sudo su root` command.

- a. For RHEL, run the following command:

```
. /usr/local/bin/rhel-cloud-init.sh -a Activation_Key -u Capsule_server_url -p Proxy_IP_and_port -o Org -t Deployment type
```

For information about the cloud-init script options and values, refer to the readme file that is generated in step 2 (`/usr/share/powervs-fls/powervs-fls-readme.md` location).

-a = activation key

-u = the URL of the RHEL capsule server you are registering against

-p = "private_ip_of_powervs_vm":3128

-o = organization associated with the activation key

-t = deployment type of this VM (RHEL, RHEL-SAP, RHEL-SAP-NETWEAVER)

- b. For SLES, run the following command:

```
. /usr/local/bin/sles-cloud-init.sh -s RMT_Server_address -p private_ip_of_proxy_vm:3128
```

-s = the URL of the SLES Repository Mirroring Tool (RMT) server that you are using for registration

-p = "private_ip_of_powervs_vm":3128

If the system is already registered, you need to unregister the system before registering again by running the following two commands: `SUSEConnect --de-register`

```
SUSEConnect --cleanup
```

If the registration fails and the following note appears in the logs:

```
Could not register with SLES RMT servers, Please verify connection to the proxy server. If proxy is verified, it may be an issue with SLES RMT servers undergoing maintenance. If so, please try executing the command again within a few minutes
```

Then, it might be a network issue and you should attempt to register again later.

Passing user-defined scripts

When you select a Full Linux Subscription (FLS) boot image while provisioning a Power Virtual Server instance, you get the option to pass in user data during first boot runtime.

In the user data you can pass the custom content that allows you to customize the startup configuration for the specific instance.

The user data that you pass should follow the following conditions:

- It should be uncompressed.
- It should start with `#cloud-config`.
- It should not exceed 63 Kb in size.

Full Linux® subscription for IBM Power Virtual Server Private Cloud

IBM Power Virtual Server Private Cloud in **Client location**

The full Linux® subscription feature enables Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) support through IBM.

The full Linux subscription uses activation keys to provide access to interim fixes and updates for the operating system (OS) for IBM® Power® Virtual Server. The Power Virtual Server is hosted on an IBM satellite server within the IBM Cloud environment. Extra charges apply for setting up a Full Linux subscription account.

To register for the full Linux subscription, you must select one of the stock OS images that are provided by IBM. IBM currently provides RHEL and SLES stock OS images for non-SAP applications.

To know more about the SLES versions that are supported, see [Operating systems supported in IBM Power Virtual Server](#).

To select the OS images, click **Boot image > Operating system** on the Create virtual server instance page. To use your own license and supply your own subscription, select the images under Client supplied subscription. Select the OS images based on the following subscription types:

- To use the full Linux subscription, select the OS image with the file name that starts with `RHEL` or `SLES`.
- To use your own license and supply your own subscription, select the OS image that is suffixed with `-BYOL`.

You can import your own custom image when you plan to supply your own subscription.

For SAP applications, you must use an IBM stock OS image for SAP that is certified for using SAP applications. The bring your own images feature is not supported. To learn more about SAP applications with Power Virtual Server before you start deployment, see [Must-Reads](#).



Important: The full Linux subscription feature for bring your own license or custom images is not available on the Create virtual server instance page under **IBM provided subscription** section. For Linux through IBM Cloud support, you must purchase a full Linux subscription from IBM Cloud. In addition, you must deploy one of the stock OS images for full Linux subscription that is available in the Power Virtual Server image catalog. If you bring your own license, you must use a bring your own license stock image or a custom image. To get support for a bring your own license stock image or a custom image, contact the OS image provider or your Linux subscription provider.

Setting up full Linux subscription

Complete the following steps to set up full Linux subscription for your account:

- [Configuring a data plane network](#)
- [Creating a proxy](#)
- [Install squid base](#)
- [Completing the setup](#)

Configuring a data plane network

The pod is configured with a control plane network and data plane network when onboarded. The full Linux subscription setup is done through the data plane network. The data plane network connects the pod with your data center infrastructure. The data plane network and the control plane network do not interact with each other. Hence, for completing the full Linux subscription, use the data plane network on the pod. Set up connectivity between your data center and IBM Cloud either by using a site-to-site VPN for virtual private cloud (VPC) or by using IBM Cloud Direct Link 2.0.

Creating a proxy

A proxy setup is set up in a virtual private cloud (VPC) on IBM Cloud with a Virtual Switch Interface (VSI). Connect this VPC on IBM Cloud to the control plane network through IBM Cloud Direct Link 2.0 Connect or VPN connection.



Note: The CentOS image is recommended for the proxy VSI.

To create a proxy setup, complete the following steps:

1. Open the Security groups for the VPC by navigation to the IBM Cloud dashboard > VPC Infrastructure > Networks > Security groups. For more information about security groups, see [About security groups](#).
2. In the default Security group that is attached to your proxy, add 443, 8443, 80 and 3128 ports.
3. On the VSI, enable a floating IP temporarily. In the proxy instance, start a Secure Shell (SSH) connection by using this temporary IP address in the `ssh` command in the following format:

```
ssh root@<external IP address>
```

For example, ssh [root@1.2.3.4](#)

Install squid base

In the VSI, install squid by using the following commands:

1. `sudo yum update -y`
2. `sudo yum install epel-release`
3. `sudo yum install squid`

You must have root authority to run these commands. After the installation completes, the `squid config` file is stored in the `/etc/squid/squid.conf` location.

Configure the squid by using the following commands:

1. `acl localnet "<CIDR of a subnet that you will deploy in the pod>"` for example, `10.140.129.217/29`
2. `acl ibmprivate dst 161.26.0.0/16`
3. `acl ibmprivate dst 166.8.0.0/14`
4. `acl SSL_ports port 8443`

Also, look at the other entries and make necessary changes according to your environment.

Save the `squid config` file and restart the squid service by using the following commands:

1. `sudo systemctl enable squid`
2. `sudo systemctl stop squid` (optional)
3. `sudo systemctl start squid`

Completing the setup

To complete the setup process, follow these steps:

1. Deploy a network in the pod.
2. Connect the network externally by using a ticketing process with Border Gateway Protocol (BGP).
3. Deploy the LPAR (RHEL or SLES) for completing the full Linux subscription.
4. Connect to the LPAR by using one of the following methods:
 - From the console on the browser.
 - By using the `SSH` command from your data network.
5. Test the internal private address of the VSI on the VPC by using the `ping` command. For example, `ping 10.240.0.4`
6. To register your LPAR with the RHEL or SLES subscription on the satellite server, open the `powervs-fls-readme.md` file that is stored in the path `/usr/share/powervs-fls` and use the following command in the file:

```
/usr/local/bin/rhel-cloud-init.sh
```

```
/usr/local/bin/sles-cloud-init.sh
```

7. One of the parameters for the command represents the proxy IP. Set this proxy IP to the internal private IP of your proxy VSI. For example,

`10.240.0.4` . Set the port to 3128.

To check whether the setup is complete and the subscription is successful, check the log files, `/var/log/powervs-fls.log` and `powervs-fls-dev.log` . When the setup is completed successfully, you can use the commands, such as `yum update -y` , `yum search <package>` , `yum install <package>` .

Passing user-defined scripts

When you select a Full Linux Subscription (FLS) boot image while provisioning a Power Virtual Server instance, you can pass in user data during the first boot runtime.

In the user data, you can include custom content to tailor the startup configuration for the specific instance.

The custom user data must meet the following conditions:

- It is uncompressed.
- It starts with `#cloud-config` .
- It must be within 63 Kb in size.

Using SLES within IBM Power Virtual Server

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

You can deploy a Linux® virtual machine (VM) by using one of the IBM stock operating system (OS) images, or you can bring your own Linux image (in OVA format).

You can choose from the following options:

- Register for a full Linux subscription.
- Use your own Linux subscription from a Linux vendor.

If you choose to register for full Linux subscription, an extra charge applies to your provisioned VM for Linux support through IBM. A full Linux subscription requires use of one of the stock OS images that are provided by IBM. In the image menu, select **IBM provided subscription** to choose one of the IBM stock images. For more information on how to provision and register by using a full Linux subscription, see [Full Linux subscription for Power Virtual Server](#).



Note: If you plan to use your own license, select the OS image that is suffixed with `-BYOL` . On the VM Provisioning page, these images are listed under the **Client supplied subscription** section.

The IBM® Power® Virtual Server provides Linux (RHEL and SLES) stock images for SAP and non-SAP applications. To know more about the SLES versions that are supported, see [Operating systems supported in IBM Power Virtual Server](#).



Important: To ensure that you are running the supported OS levels and to plan for upgrades to the latest levels, refer to [SLES](#) OS release lifecycle information.

If you do not choose to use the full Linux subscription for Power Virtual Server you must obtain the subscription directly from the vendor and bring your image. After you deploy your Linux VM, you must log in to the VM and register it with the Linux vendor’s satellite server. To reach the Linux vendor satellite servers (where you can register and obtain packages and fixes), you must attach a public network to your VM.

When you create an OVA image, you must include the appropriate Power Virtual Server environment `cloud-init` packages. Download the appropriate `cloud-init` package from [Installing and configuring cloud-init](#).

To run SUSE Linux Enterprise Server on Power Virtual Server, your hardware must meet the minimum memory requirements. A minimum of 1024 MB of memory is required for a minimal installation. On machines with more than two processors, add 512 MB per CPU. The memory values specified are valid only for the installation of the operating system. The actual memory requirement in production depends on the workload of the system. For more information about the SLES hardware recommendations, see [Installation on IBM POWER](#).

Client location The IBM Power Subscription plan can provision VMs with SAP bring your own license or Linux bring your own license images. The costs are based on a flat subscription rate and not on the metered instance usage. If you deploy VMs in your data center without ordering the IBM Power Subscription plan, the costs are based on the metered instance usage. When you register the infrastructure, if the metered plan is selected, you cannot provision VMs with SAP images and the costs are based on the instance usage.

Registering and purchasing a subscription to SLES



Note: You cannot contact the SUSE-based repository and download the appropriate software packages without first enabling your SLES subscription.

1. To buy a SUSE subscription, see [How to Buy](#).
2. To register your system, see [Registering an Installed System](#).

Capturing and importing a SLES image

To use SLES within the Power Virtual Server, you can use the [IBM Power Virtualization Center \(PowerVC\)](#) to capture your Linux image, then [import it](#) as an Open Virtualization Appliance (OVA) file. You must also bring your own license. If you cannot use PowerVC to capture an image, see the [Power OVA image capture](#) instructions.

Linux networking

To connect a Linux virtual machine (VM) to the public internet, you must add a public network when you provision a Power Virtual Server. You must set up a Linux-based NAT gateway on a public-facing Linux VM if you have Linux VMs that do not need an internet-facing external IP address. For more information, see [19.6 Basic Router Setup](#) and [Linux NAT\(Network Address Translation\) Router Explained](#).

When you are configuring a Source NAT (SNAT) gateway between your public and private networks, ensure that the TCP checksum offload option is disabled. You must also set the maximum transmission unit (MTU) value to 1450 on the network interface that is connected to the private network. To ensure that the interface checksum offloading and MTU settings of the network interface are persistent whenever the virtual machine is restarted, you need to modify the configuration files of your network interface.



Note: The TCP checksum offload option must be disabled on the private network interface of the SNAT Gateway and virtual Ethernet device must be of the type `ibmveth`. You do not need to change the TCP checksum offload option for public network interface. IBM® Power® Virtual Server VMs are deployed by using `ibmveth` devices only.

You can verify that the device interface type is `ibmveth` by using the following command:

```
$ ethtool -i <interface name> | grep driver
```

The following instructions are applicable to SLES 15. If you need additional help to configure network interfaces, refer to the SLES documentation.

1. Identify the name of the private network interface that you want to modify. Use the following command to identify the network interface names based on the IP address that is assigned to the network interface:

```
$ ip -4 a s (for IPv4 address)
ip -6 a s (for IPv6 address)
```

2. Edit the `ifcfg-<NIC>` file (where NIC is the network interface name that is identified in step 1).

```
$ SLES: /etc/sysconfig/network/ifcfg-<NIC>
```

- Add or modify the following lines:

```
$ For SLES:
MTU='1450'
ETHTOOL_OPTIONS='-K <NIC> rx off'
```

3. Restart the VM.
4. After the restart operation is complete, verify that the MTU value and the checksum offloading setting are correct.
 - Verify the checksum offloading setting by running the following command:

```
$ ethtool -k eth0
Features for eth0:
rx-checksumming: off
tx-checksumming: off
<cut>
```



Note: The `ethtool` command sets both the rx-checksumming and tx-checksumming options to off when one of these options is disabled.

Verify the MTU value by running the following command:

```
$ ip link show eth0
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc fq_codel state UNKNOWN mode DEFAULT <...>
```

Configuring SNAT in the Power Virtual Server environment

Most organizations are allotted a limited number of publicly routable IP addresses from their ISP. Due to this limited allowance, administrators must find a way to share access to internet services without giving limited public IP addresses to every node on the LAN. To learn more, see [Forward and NAT rules](#).

SNAT router configuration

Complete these steps to accurately configure your SNAT router.

1. Deploy a SLES LPAR on a public network.
2. Create subnets that require the SNAT function to get internet access.
3. Use the following commands to allow private network traffic to be accessible for SNAT-ing:

```
$ iptables -A FORWARD -i eth1 -j ACCEPT
iptables -A FORWARD -o eth1 -j ACCEPT
```

 **Important:** These commands assume that the network device for the public network is `eth0`, and `eth1` for the private network.

You can permanently set **IP forwarding** by editing the `/etc/sysctl.conf` file:

1. Find and edit the following line within the `/etc/sysctl.conf` file (replacing `0` with `1` if required): `net.ipv4.ip_forward = 1`.
2. Update the `sysctl.conf` file by entering the following command: `sysctl -p /etc/sysctl.conf`.
3. Finally, configure the source NAT by entering the following command: `iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE`.

Configuring Linux VMs to use a SNAT router

1. [Deploy the Linux VMs](#) that will be using the SNAT router to access the internet. Make sure that the SNAT router is routing the attached private networks.
2. Set the default router for your Linux VM to the SNAT router IP on the private network.

Using RHEL within the Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can deploy a Linux® virtual machine (VM) by using one of the IBM stock OS images, or you can bring your own Linux image (in OVA format).

You can choose from the following options:

- Register for a full Linux subscription.
- Use your own Linux subscription from a Linux vendor.

If you choose to register for full Linux subscription, an extra charge applies to your provisioned VM for Linux support through IBM. A full Linux subscription requires use of one of the stock operating system images that are provided by IBM. In the image menu, select **IBM provided subscription** to choose one of the IBM stock images. For more information on how to provision and register by using a full Linux subscription, see [Full Linux subscription for Power Virtual Server](#).

 **Note:** If you plan to use your own license, select the OS image that is suffixed with `-BYOL`. On the VM Provisioning page, these images are listed under the **Client supplied subscription** section.

The IBM® Power® Virtual Server provides Linux (RHEL and SLES) stock images for SAP and non-SAP applications. To know more about the SLES versions that are supported, see [Operating systems supported in IBM Power Virtual Server](#).

 **Important:** To ensure that you are running the supported OS levels and to plan for upgrades to the latest levels, refer to [RHEL](#) OS release lifecycle

information.

To view the certification details in the Red Hat catalog, see [IBM Power System E980 \(9080-M9S\)](#) and [IBM Power System S922 \(9009-22A\)](#).

If you do not choose to use the full Linux subscription for Power Virtual Server you must obtain the subscription directly from the vendor and bring your image. After you deploy your Linux VM, you must log in to the VM and register it with the Linux vendor's satellite server. To reach the Linux vendor satellite servers (where you can register and obtain packages and fixes), you must attach a public network to your VM.

When you create an OVA image, ensure that the image includes the correct version of RHEL image with cloud-init version from March 2021, or later. If you are using an earlier RHEL image, download the appropriate cloud-init and configure it as in the steps that are documented at [Installing and configuring cloud-init on Linux](#).



Note: You can use the [pvsadm tool](#) to convert the RHEL 8.3 Qcow2 images to OVA image. The pvsadm tool is an open source tool and not an IBM-supported product. If you experience any issues with this tool, you can [open an issue](#) within the pvsadm tool GitHub repository. If you use any tool for RHEL releases that are in extended support, ensure that only the Extended Update Support (EUS)-related packages are downloaded and packaged by the tool.

Client location

The IBM Power Subscription plan can provision VMs with SAP bring your own license or Linux bring your own license images. The costs are based on a flat subscription rate and not on the metered instance usage. If you deploy VMs in your data center without ordering the IBM Power Subscription plan, the costs are based on the metered instance usage. When you register the infrastructure, if the metered plan is selected, you cannot provision VMs with SAP images and the costs are based on the instance usage.

Registering and purchasing a subscription to RHEL



Note: You cannot contact the Red Hat-based repository and download the appropriate software packages without first enabling your RHEL subscription.

1. To buy an RHEL subscription, see [Red Hat Enterprise Linux Server](#).
2. To register your system, see [Quick Registration for RHEL](#).

Capturing and importing an RHEL image

To use RHEL within Power Virtual Server, you can use the [IBM Power Virtualization Center \(PowerVC\)](#) to capture your Linux image, then [import it](#) as an Open Virtualization Appliance (OVA) file. You must also bring your own license. If you cannot use PowerVC to capture an image, see the [Power OVA image capture](#) instructions.

Linux networking

To connect a Linux virtual machine (VM) to the public internet, you must add a public network when you provision a Power Virtual Server. You must set up a Linux-based Network Address Translation (NAT) gateway on a public-facing Linux VM if you have Linux VMs that do not need an internet-facing external IP address. For more information on NAT router, [Linux NAT Router Explained](#).

When you are configuring a Source NAT (SNAT) gateway between your public and private networks, ensure that the TCP checksum offload option is disabled. You must also set the maximum transmission unit (MTU) value to 1450 on the network interface that is connected to the private network. To ensure that the interface checksum offloading and MTU settings of the network interface are persistent whenever the virtual machine is restarted, you need to modify the configuration files of your network interface.



Note: The TCP checksum offload option must be disabled on the private network interface of the SNAT Gateway and virtual Ethernet device must be of the type `ibmveth`. You do not need to change the TCP checksum offload option for public network interface. IBM Power Virtual Server VMs are deployed by using ibmveth devices only.

You can verify that the device interface type is `ibmveth` by using the following command:

```
$ ethtool -i <interface name> | grep driver
```

The following instructions are applicable to RHEL version 8.1 and later. If you need more help to configure network interfaces, refer to the Red Hat documentation.

1. Identify the name of the private network interface that you want to modify. Use the following command to identify the network interface names based on the IP address that is assigned to the network interface:

```
$ ip -4 a s (for IPv4 address)
ip -6 a s (for IPv6 address)
```

2. Edit the `ifcfg-<NIC>` file (where NIC is the network interface name that is identified in step 1).

```
$ RHEL: /etc/sysconfig/network-scripts/ifcfg-<NIC>
```

- Add or modify the following lines:

```
$ For RHEL:
MTU=1450
ETHTOOL_OPTS="-K <NIC> rx off"
```

3. Restart the VM.
4. After the restart operation is complete, verify that the MTU value and the checksum offloading setting are correct.
 - Verify the checksum offloading setting by running the following command:

```
$ ethtool -k eth0
Features for eth0:
rx-checksumming: off
tx-checksumming: off
<cut>
```



Note: The `ethtool` command sets both the rx-checksumming and tx-checksumming options to off when one of these options is disabled.

Verify the MTU value by running the following command:

```
$ ip link show eth0
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc fq_codel state UNKNOWN mode DEFAULT <...>
```

Configuring Network Address Translation (NAT) in the Power Virtual Server environment

Most organizations are allotted a limited number of publicly routable IP addresses from their ISP. Due to this limited allowance, administrators must find a way to share access to internet services without giving limited public IP addresses to every node on the LAN. RHEL 8 uses the nftables utility, instead of iptables, to set up complex firewalls. For instructions on setting up NAT on RHEL, see [Configuring NAT using nftables](#).

Before running the `iptables` commands, complete the following steps to ensure that the configuration settings persist after the virtual machine is restarted.

1. Use the following command to list all the zones:

```
$ firewall-cmd --get-zones
block dmz drop external home internal nm-shared public trusted work
```

2. Use the following command to list the network interfaces for zones:

```
$ firewall-cmd --zone=public --list-all
```

3. To turn on the masquerade option that will set the configuration settings persist after the restart operations, run the following command:

```
$ firewall-cmd --zone=public --add-masquerade --permanent
success
```

4. Restart the system firewall:

```
$ systemctl restart firewalld
```

5. Verify whether the configuration is successful:

```
$ #sudo firewall-cmd --zone=public --query-masquerade
yes
```

Complete these steps to accurately configure your Source NAT (SNAT) router:

1. Deploy an RHEL LPAR on a public network.
2. Create subnets that require the SNAT function to get internet access.
3. Use the following commands to allow private network traffic to be accessible for SNAT-ing:

```
$ iptables -A FORWARD -i eth1 -j ACCEPT
iptables -A FORWARD -o eth1 -j ACCEPT
```

 **Important:** These commands assume that the network device for the public network is eth0, and eth1 for the private network.

You can permanently set **IP forwarding** by editing the `/etc/sysctl.conf` file:

1. Find and edit the following line within the `/etc/sysctl.conf` file (replacing 0 with 1 if required):

```
$ net.ipv4.ip_forward = 1
```

2. Update the `sysctl.conf` file by entering the following command:

```
$ sysctl -p /etc/sysctl.conf
```

3. Finally, configure the source NAT by entering the following command:

```
$ iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

Configuring Linux VMs to use a SNAT router

1. [Deploy the Linux VMs](#) that use the SNAT router to access the internet. Make sure that the SNAT router is routing the attached private networks.
2. Assign the SNAT router IP (eth1 as in the previous example) on the private network as the default router on your newly created Linux VM.

Creating a custom Linux image in OVA format

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can deploy a Linux® virtual machine (VM) in your workspace with a custom Linux image and receive support from a Linux vendor. When you are provisioning a VM, select a **Linux-Client supplied subscription** for your own Linux image (OVA format) and subscription.

If you bring your own image, you must obtain the subscription for Linux directly from the vendor. After you deploy your Linux VM, you must log in to the VM and register it with the Linux vendor's satellite server. To reach the Linux vendor satellite servers (where you can register and obtain packages and fixes), you must attach a public network to your VM. To learn more about the registration process, see [Registering and subscribing to SLES](#) or [Registering and subscribing to RHEL](#).

How to create an OVA format Linux image

Learn how to create an OVA image of a Linux operating system and import it into the IBM® Power® Virtual Server environment. You can use PowerVC or VIOS to capture an image.

Using PowerVC to capture and import an OVA image

If you've deployed PowerVC in your private cloud environment, you can use it to [Capture a virtual machine](#) and create an OVA image. After you create the OVA image, upload it to your Cloud Object Storage account and import it into the Power Virtual Server environment.

Capturing an image from VIOS

The [create_ova RPM](#) contains scripts that create a virtual disk image of a `mksysb` backup, raw disk file, or disk volume and packages the content into a consumable Open Virtual Appliance (OVA) package. To use this capture method, it is required that the root file system be present on a single disk. When you use the VIOS disk capture capability, you must obtain the appropriate disk volume name of the client VM that you are trying to capture. For more information about finding the disk configuration of a VIOS client, see [mkvdev command](#). **You must shut down your Linux LPAR for this method to work. Otherwise, you might encounter disk errors and the OVA image might not boot.**



Note: The `create_ova` RPM also contains the `create_ova` man page and license. You must install the RPM on VIOS releases, which are before VIOS 3.1.2.0. The `create_ova` command is provided as a system command on VIOS release 3.1.2.0, or later.

To see the contents of the RPM package, enter the `rpm` command as shown in the following example:

```
$ # rpm -qlp /tmp/create_ova-1.0-2.aix7.2.ppc.rpm
/opt/freeware/doc/create_ova-1.0
/opt/freeware/doc/create_ova-1.0/create_ova.pdf
/opt/freeware/licenses/create_ova-1.0
/opt/freeware/licenses/create_ova-1.0/LICENSE
/opt/ibm/sysmgmt/cloudrdy
/opt/ibm/sysmgmt/cloudrdy/EXTRAS/pv-1.6.0-1.aix6.1.ppc.rpm
/opt/ibm/sysmgmt/cloudrdy/LICENSE
/opt/ibm/sysmgmt/cloudrdy/bin/cloud_setup
/opt/ibm/sysmgmt/cloudrdy/bin/create_ova
/opt/ibm/sysmgmt/cloudrdy/bin/print_ovf
/opt/ibm/sysmgmt/cloudrdy/doc/create_ova.pdf
/usr/share/man/man1/create_ova.1
```

Once you obtain the correct disk name (through virtual adapter mapping), you can create a virtual disk image and package the contents into an OVA. After the RPM is installed, the man page and the executable (`create_ova`) are available in the normal paths. Note that a link is made to `/usr/bin/create_ova`, so there is no need to set the user path. If you decide to perform an uninstall, any links, files, or directories that are tracked by the RPM for this package are removed. The following example contains a list of sample commands and output:



Important: You can upload the `ova.gz` file into your Cloud Object Storage account. Once you upload it, go to the Power Virtual Server user interface and import the OVA image from your Cloud Object Storage account.

```
$ ssh (isotopes-vios2)

IBM Virtual I/O Server

login: padmin
padmin's Password:
Last login: Sun May 10 17:41:00 CDT 2020 on /dev/pts/0

$ lsmap -vadapter vhost18
SVSA          Physloc          Client Partition ID
-----
vhost18      U8233.E8B.100121P-V8-C21      0x0000000f

VTD          isotopes13_dsk1
Status       Available
LUN          0x8100000000000000
Backing device  isotopes13_lv1
Physloc
Mirrored     N/A

$ oem_setup_env
# create_ova -o /datafs -d risotopes13_lv1 -t sles -e -f /use rhel for RHEL
Initializing resources ...

Checking for resource group ROOTVG...
Checking for resource group PIPEVIEWER...already installed.
Checking /datafs space requirement...done

Checking for resource group linux_20200511101424.img...
20480+0 records in1.2MiB/s] [10.6MiB/s] [=====>] 99% ETA 0:00:00
20480+0 records out
 20GiB 0:32:15 [10.6MiB/s] [10.6MiB/s] [=====>] 100%
41943040+0 records in
41943040+0 records out
done

Checking for resource group linux_20200511101424.ova.gz...
Checking /datafs space requirement...done
a ./linux_20200511101424.ovf 4 blocks
a ./linux_20200511101424.img 41943040 blocks
```

20GiB 0:49:23 [6.91MiB/s] [6.91MiB/s] [======>] 100%

Done verifying resources.

```
# ls -l /datafs/linux_20200511101424.ova.gz
-rw-r--r--  1 root   staff   1890363097 May 11 2020 /datafs/linux_20200511101424.ova.gz
```


SAP

Deploying a Linux for SAP (HANA or NetWeaver) custom image

IBM Power Virtual Server in [IBM data center](#)

The Linux® for SAP (HANA or NetWeaver) operating system (OS) image can be deployed within an IBM® Power® Virtual Server by using IBM® provided central image repository for SAP offerings. You can bring your own SAP (HANA or NetWeaver) image with your own subscription, import the image to your workspace, and create a virtual machine by using the image.

You can import an image by using the Power Virtual Server interface, CLI, API, or Terraform.

When you import a Linux for SAP (HANA or NetWeaver) custom image through the interface, select the self-certification checkbox to confirm that the image is an SAP image. For more information about importing an image, see [Using the Power Virtual Server user interface to import a boot image](#).



Note: The IBM provided central image repository for SAP offerings cannot be used to refresh the stock images.

An imported SAP image is stored into a single storage pool and can be deployed only in that single storage pool. To deploy the SAP image into other storage pools, you must import the image into those storage pools. If you use APIs for image import, you can define the storage pool where the images can be stored.

SAP certified profiles

IBM Power Virtual Server in [IBM data center](#)

You can deploy SAP certified profiles on IBM POWER9 or later servers by using a single custom OS boot image. You can use the GUI, CLI, API, and Terraform to deploy SAP certified profiles.

SAP certified profiles for SAP HANA

The SAP certified profiles for SAP HANA are a set of profiles with defined attributes such as CPU cores and RAM.

You can deploy SAP HANA on the following IBM® Power® server types:

- **Power11:** S1122, E1150, and E1180
- **Power10:** S1022, E1050, and E1080
- **Power9:** E980

IBM Power9 and a few Power10 sr2 certified profiles are discouraged for new deployments. Users are advised to migrate to IBM Power11 certified profiles, selecting the appropriate profile based on memory and core requirements.

Power11 profiles

SR3 SAP HANA profiles are SAP RISE profiles, which are available for IBM Power11 processor-based servers. For more information about SAP certified profiles for SAP HANA, see [IBM Power Virtual Server certified profiles for SAP HANA](#).

Power10 profiles

The following SAP HANA profiles are available for IBM Power10 processor-based servers:

Profile	Profile type	Description
SR2	SAP RISE	SR2 profiles are SAP RISE profiles
SH2	Small	Suitable for balanced workloads that require less CPU and storage consumption
CH2	Compute	Suitable for CPU-intensive workloads, such as high web traffic, production batch processing, and front-end web servers
BH2	Balanced	Suitable for midsize databases and common cloud applications with moderate traffic

MH2	Very high memory	Suitable for server Online Analytical Processing (OLAP) databases, such as SAP NetWeaver
-----	------------------	--

Power10 SAP HANA certified profiles

SR2 profiles also support custom combination of physical CPU cores and memory. You must deploy custom SR2 profiles by using the CLI or API only. You cannot switch from an SR2 profile to a different SAP HANA profile. When an SR2 profile is deployed, you can edit the core value and memory capacity of the virtual machine by using the UI. Combinations that are certified by SAP for productive usage are documented in the [SAP Note 2947579 - SAP HANA on IBM Power Virtual Servers](#) and in the [Certified and Supported SAP HANA Hardware](#) directory. SR2 profiles are not available with the E980 machine type.

Power9 profiles

The following SAP HANA profiles are available for IBM Power9 processor-based servers:

Profile	Profile type	Description
CNP	Custom	Ideal for test or development use only. These profiles are not intended for production use and are not supported or certified for SAP production.
USH1	Small	Suitable for balanced workloads that require less CPU and storage consumption.
UMH	Ultra Memory	Provides the highest vCPU-to-memory ratios for serving in-memory OLTP databases, such as SAP HANA.
BH1	Balanced	Suitable for midsize databases and common cloud applications with moderate traffic.
CH1	Compute	Suitable for CPU-intensive workloads, such as high web traffic, production batch processing, and front-end web servers.

Power9 SAP HANA certified profiles

For more information about SAP certified profiles for SAP HANA, see [IBM Power Virtual Server certified profiles for SAP HANA](#). For more information about pricing, see the following topics:

- [Pricing for IBM Power Virtual Server in IBM data centers](#).
- [Pricing for IBM Power Virtual Server Private Cloud](#)

SAP profile attributes and performance metrics

The following table describes the fields displayed for an SAP profile in the Power Virtual Server user interface, along with their configuration details and associated performance indicators:

Field	Description
Profile	Profile identifier that represents the profile type, number of cores, and memory allocation
Cores	Number of virtual CPU cores that are allocated to the profile
RAM	Memory allocation for the profile in GiB (gibibytes)
SAPS	Hardware-independent unit of measurement that represents the performance of a system configuration in the SAP environment
aSAPS	Advanced SAP performance unit that represents the system capacity based on the SAP quote-to-cash (Q2C) benchmark
Price	Hourly cost that is associated with the profile

SAP profile attributes and performance metrics

For more information about SAP Application Performance Standard (SAPS) and advanced SAPS (aSAPS), see [SAPS/aSAPS](#).

SAP Application Server profiles

When you use SAP Application Server profiles, you can deploy a certified profile for SAP Application Servers or SAP NetWeaver on Power Virtual Server. The SAP Application Server profiles are available on IBM Power S1022 and S1122 systems.

To deploy a certified profile for SAP Application Servers or SAP NetWeaver by using CLI, API, or Terraform interfaces, specify an sr2 or sr3 profile ID that matches an SAP Application Server profile.

You can use the following commands to get the details of an SAP profile:

- **CLI** : [ibmcloud pi image](#)
- **API** : [Get list of SAP profiles](#)



Note: The `Type` property of an SAP Application Server profile has the value `sap-rise-app` .

For more information about SAP Application Server profiles, see [SAP Application Server certified instances on IBM Power Virtual Server](#).

Generating an estimate of a Power Virtual Server instance

When you are generating an estimate of a Power Virtual Server instance, you can select SAP certified profiles. For more information about generating an estimate of a Power Virtual Server instance with SAP certified profiles, see [Estimating SAP workloads](#).

Creating a Power Virtual Server instance

When you are creating a Power Virtual Server instance, you can select SAP certified profiles. For more information about creating a Power Virtual Server instance with SAP certified profiles, see [Configuring a Power Virtual Server instance](#).

SAP full system profiles

IBM Power Virtual Server in [IBM data center](#)

SAP HANA profiles that use all or most of the system resources are referred to as 'full system profiles'. You can deploy SAP HANA using full system profiles on a system if no virtual machines are deployed on the system.

No additional billing or metering is charged for full system profiles. The virtual machines with full system profiles are metered against the SAP HANA profile family in the same manner as any other profile from the family. For example, a profile such as sh2-24x1900 is metered by using the core and memory parts for the SH2 family that has 24 cores and 1900 GB of memory.



Note: SAP custom profiles are not supported on Power10 systems.

For more information about pricing, see [Pricing for Power Virtual Servers](#).

Observability

Using carbon calculator to view and export greenhouse gas emission data for Power Virtual Server

IBM Power Virtual Server in [IBM data center](#)

You can use the IBM Cloud® carbon calculator to view and export carbon emission data on Power Virtual Server. The carbon calculator is a dashboard to track, analyze, and manage greenhouse gas (GHG) emissions that is associated with your IBM Cloud services. Carbon emission data is available for IBM Power10 or later systems. To view and export the GHG emissions data for an account, you must have the account owner, administrator, or viewer access on the billing service.

The IBM Cloud carbon calculator performs the following functions:

- Displays the GHG emission data for an account. The GHG emission data is classified based on total energy consumption by an account, region, location, or service.
- Reports the overall energy consumption of an account and the efficiency of data centers.
- Tracks the GHG emissions that are associated with an account.
- Identifies hot-spots such as services, locations, or resource groups that reports the highest emissions to mitigate the carbon footprint.

For more information about carbon calculator, see [Working with IBM Cloud's carbon calculator](#).

IBM Cloud and sustainability

As organizations and users work together to achieve better outcomes for their business and for the environment, digital technologies are a growing source of the GHG emissions. The four areas that are the main source for the GHG emissions are: data centers, extensive data analytics, security, and internet usage. For more information about sustainability, see [IT sustainability beyond the data center](#).

For more information about environmental, social, and governance (ESG) initiatives of IBM, see [IBM Releases 2023 Impact Report](#). For more information about how different geographical areas address sustainability challenges, see [The State of Sustainability 2024 report](#).

FAQs for carbon calculator on Power Virtual Server

What is GHG emission?

GHG emission stands for greenhouse gas emission. For more information about carbon calculator, see [Carbon calculator terminology](#) and [What are greenhouse gas \(GHG\) emissions?](#).

How is GHG measured in IBM Cloud?

You can use the IBM Cloud carbon calculator for a detailed view of your cloud GHG emissions. For more information about GHG emissions, see [FAQ for IBM Cloud Carbon Calculator](#).

What methodology does IBM Cloud carbon calculator use?

IBM Cloud carbon calculator uses a calculation methodology that follows the GHG protocol. For more information, see [IBM Cloud carbon calculator methodology](#).

Who can view the carbon calculator data?

To view and export the GHG emissions data for an account, you must have the account owner, administrator, or viewer access on the billing service. You must be an account holder or have a billing administrator role to grant access to other users for the billing service. For more information, see [FAQ for IBM Cloud Carbon Calculator](#).

How to view the Power Virtual Server GHG emissions data on carbon calculator?

On the [IBM Cloud carbon calculator](#) GUI, select the **Power Virtual Server** option from the **Filter by server** list to view the Power Virtual Server GHG emissions data.

For Power Virtual Server, GHG emissions are not displayed individually for individual services. The data is a combined report of GHG emissions for workloads that run on Power Virtual Server instances (VSIs).

You can create a resource group and add the Power Virtual Server VSIs for a specific workload to the resource group. View the Emissions by resource group widget on the carbon calculator GUI to view the carbon emissions for the resource group that you created. For more information about creating a resource

group, see [Managing resource groups](#).

Why is data not displayed on the carbon calculator?

The data for your workspace is not displayed on the carbon calculator or you are not be able to see the data for your workspace because of the following reasons:

- Only the data from your workspaces on IBM Power10 or later systems is used to calculate the carbon emissions. If your workspace is on IBM Power9 or previous generations, the carbon emissions data is not available.
- You must have the required permissions or roles to view the data on the carbon calculator. For more information about assigning access, see [Assigning access to account management services](#).
- If you run workloads in a zero emission data center, emissions that are calculated by using the market-based methodology are set to zero and no data is displayed. To view emissions that are calculated by using the location-based methodology, select the **Display unadjusted emissions** checkbox on the [IBM Cloud carbon calculator](#) dashboard. For more information about the market-based and location-based methodology, see [Market-based and location-based methodology](#).

How to view and export the GHG emissions for an account in the carbon calculator?

To view and export the GHG emissions data for an account, you must have the account owner, administrator, or viewer access on the billing service. For more information, see [Tracking account emissions with carbon calculator](#).

How to extract the GHG emissions data by using APIs?

You can use APIs related to the carbon calculator to extract the GHG emissions data. For more information, see [Carbon Calculator API](#).

Monitoring platform metrics for IBM Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can monitor platform metrics from resources in your IBM® Power® Virtual Server workspace by enabling [IBM Cloud® Monitoring](#) for basic platform metrics or by installing the [Cloud Monitoring agent for Linux operating systems](#). The Linux based agent can fetch over 100 infrastructure metrics.

IBM Cloud Monitoring is an enterprise-grade monitoring service to view, alert, and troubleshoot an application. IBM Cloud Monitoring powered by Sysdig is used by enterprise development and IT teams that build, ship, and run business-critical applications at scale.

Platform metrics overview

Platform metrics are metrics that are exposed by enabled-monitoring services and the platform in IBM Cloud. You can monitor metrics from enabled-monitoring services on the IBM Cloud in the region where the service is available. For more information, see [Working with platform metrics](#).

You can view the platform metrics for Power Virtual Server after you create an IBM Cloud Monitoring instance in the same region that is enabled for monitoring the platform metrics. For more information, see [Enabling platform metrics](#).

Before you create an IBM Cloud Monitoring instance, consider the following points:

- You can configure only one instance of the IBM Cloud Monitoring service per region on your platform.
- To monitor platform metrics, verify whether the IBM Cloud Monitoring instance and the Power Virtual Server instance are provisioned in the same region.
- The IBM Cloud Monitoring-enabled instance collects the platform metrics automatically for monitoring.

Monitoring agent for Linux VMs

You must provision the IBM Cloud® Monitoring service instance in the IBM Cloud. Then, you can deploy the Monitoring agent on your Linux hosts in a Power Virtual Server workspace to collect the data and platform metrics from the active VMs.

The Monitoring service instance can collect over 100 metrics that includes additional CPU, memory, file, file system, and network data points. The metrics that are collected from the VMs are routed to the Sysdig backend and then displayed on the IBM Cloud Monitoring dashboards for the selected account. You can configure the metrics to be monitored in each environment. For more information about deploying, updating, and troubleshooting the Monitoring agent, see [Managing the IBM Cloud Monitoring Linux agent on a PowerVS workspace](#). For more information about configuring the metrics for your environment, see [Monitoring Linux on a PowerVS workspace](#).

By default, the Linux-based agent collects core infrastructure and network time series metrics. You can use the collected metrics to monitor the application

host. For more information about a list of collected metrics, see [Metrics Available for non-orchestrated environments](#).

For more information about basic platform metrics, see [Platform metrics overview](#).

Integrating IBM Cloud Monitoring service with Power Virtual Server workspaces

The IBM Cloud® Monitoring system is a cloud-native and container-intelligence management system. You can get the information about the performance and health of your applications, services, and platforms through the IBM Cloud Monitoring system. It offers administrators, DevOps teams, and developers a full-stack telemetry with advanced features to monitor and troubleshoot, define alerts, and design custom dashboards. For more information, see [IBM Cloud Monitoring](#).

Before you enable the IBM Cloud Monitoring service, you can review the estimated cost on the Summary page and generate an estimate. For more information about generating an estimate, see [Cloud Monitoring](#).

You can register a Power Virtual Server workspace as an observability instance on the IBM Cloud Monitoring system. Integration costs are based on the usage and vary due to the hourly consumption of nodes and virtual machines. Review the cost that is associated Monitoring services in the IBM Cloud catalog. For more information about the costs associated with a Power Virtual Server workspace monitoring, see [Pricing](#).

Enabling IBM Cloud Monitoring service in a Power Virtual Server workspace

To enable the IBM Cloud Monitoring service in an existing Power Virtual Server workspace, complete the following steps:

1. Log in to the [IBM Cloud](#) Power Virtual Server user interface.
2. Click **Workspaces** in the navigation panel.
3. Select a workspace on which you want to enable the IBM Cloud Monitoring service. The Workspace details pane is displayed.
4. Click **Monitoring** in the Integrations section to enable the IBM Cloud Monitoring service.
5. If the IBM Cloud Monitoring service is already enabled on the workspace, click **View Monitoring** to view the monitoring metrics.

The **Monitoring** option is enabled by default when you create an IBM Power Virtual Server workspace.

For more information about the regions that support IBM Cloud Monitoring, see [Regions for IBM Cloud Monitoring](#).

Creating an IBM Cloud Monitoring instance

Create an IBM Cloud Monitoring instance and enable the platform metrics to capture various performance metrics.

To create an IBM Cloud Monitoring instance, complete the following steps:



Important: To monitor platform metrics, select the region where your Power Virtual Server workspace is provisioned.

1. Log in to [IBM Cloud](#) console.
2. Search for **IBM Cloud Monitoring** and select it.
3. Select your location and enter your custom values for **Service name** field and other fields.
4. Select the **Enable** indicator for **IBM platform metrics**.
5. Select the license agreements indicator and click **Create**.

You can also create the IBM Cloud monitoring instance from the **Integration (Optional)** section when you create a workspace, if no IBM Cloud Monitoring instance is already created for that region.

Viewing metrics

To view the metrics dashboards, access the user interface of the IBM Cloud Monitoring in the following ways:

- [Access the IBM Cloud monitoring user interface from your Power Systems Virtual Server workspace](#) .
- [Access the IBM Cloud monitoring user interface from the **Observability** page](#).



Important: To view metrics in your dashboard, you must enable the platform metrics of the IBM Cloud Monitoring instance.

Accessing metrics from Power Virtual Server workspace

From the navigation panel of the Power Virtual Server user interface, complete the following steps:

1. Click **workspaces**.
2. Select the workspace for which a monitoring instance is available.

- From the **workspace details** page, click **Launch monitoring**. The IBM Cloud Monitoring dashboard opens.
- Click **Dashboards > Dashboard Library > IBM** and select your dashboard to view.

Accessing metrics from the Observability page.

To access the dashboard, complete the following steps:

- Log in to [IBM Cloud](#) console.
- Expand the navigation panel.
- Click **Resource list**.
- Click **Observability > Monitoring**.
- Click the instance.
- Click **Open dashboard**. The IBM Cloud Monitoring dashboard opens.
- Click **Dashboards > Dashboard Library > IBM** and select your dashboard to view.

Power Virtual Server metrics dictionary

CPU utilization

The CPU utilization of a Power Virtual Server instance in percentage.

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_cpu_util
Metric Type	gauge
Value Type	percent
Segment by	ibm_service_instance, ibm_resource

Table 1: CPU utilization metrics of a VM

Memory utilization

The memory utilization of a Power Virtual Server instance in percentage.

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_mem_util
Metric Type	gauge
Value Type	percent
Segment by	ibm_service_instance, ibm_resource

Table 2: Memory utilization metrics of a VM

Incoming network bytes

The incoming bytes of a Power Virtual Server instance per network interface (or per MAC address).

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_network_incoming_bytes
Metric Type	counter
Value Type	byte

Segment by	ibm_service_instance, ibm_resource, ibm_power_iaas_pvm_instance_network_mac_address
------------	---

Table 3: Incoming byte metrics of a VM per network interface

Outgoing network bytes

The outgoing bytes of a Power Virtual Server instance per network interface.

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_network_outgoing_bytes
Metric Type	counter
Value Type	byte
Segment by	ibm_service_instance, ibm_resource, ibm_power_iaas_pvm_instance_network_mac_address

Table 4: Outgoing byte metrics of a VM per network interface

Disk read bytes

The total disk read bytes at Power Virtual Server instance level.

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_disk_read_bytes
Metric Type	counter
Value Type	byte
Segment by	ibm_service_instance, ibm_resource

Table 5: Total disk read bytes metrics at a VM

Disk write bytes

The total disks write bytes at Power Virtual Server instance level.

Metadata	Description
Metric Name	ibm_power_iaas_pvm_instance_disk_write_bytes
Metric Type	counter
Value Type	byte
Segment by	ibm_service_instance, ibm_resource

Table 6: Total disk write bytes metrics at a VM

Attributes for segmentation

See the global and additional attributes that are available for segmentation.

Global attributes

The following global attributes are available for segmenting all the metrics that are listed in the metrics dictionary:

Metric label name	Metric description	Valid values
-------------------	--------------------	--------------

ibm_ctype	Type of Cloud	public.
ibm_location	Location of the monitored resource	All the supported data center such as WDC06
ibm_resource_group_name	Resource group that is associated to the service instance.	All the resource groups that are available in your account
ibm_scope	The extent of the data samples that are considered.	IBM Cloud account ID
ibm_service_name	Name of the service that generates this metric.	power-iaas

Table 7: Global segmentation attributes

Additional attributes

The following additional attributes are available for segmenting all the metrics that are listed in the metrics dictionary:

Metric label name	Metric description	Valid values
ibm_service_instance	The workspace ID	Power System Power Virtual Server workspace ID
ibm_service_instance_name	The workspace name	Defined name of the workspace
ibm_resource_type	The type of Power Virtual Server resource for which metric is available	"pvm-instance"
ibm_resource	ID of a resource	"pvm-instance" ID
ibm_resource_name	Name of the resource	The name of the Power Virtual Server instance
ibm_power_iaas_pvm_instance_network_mac_address	The MAC address of the network interface that is attached to the Power Virtual Server instance	Valid mac address

Table 8: Additional segmentation attributes

IBM Cloud monitoring limitations

IBM Cloud Monitoring has the following limitations:

- The Ipv6 interface usage metrics of a Power Virtual Server instance are for internal management network. Additionally, these metrics are available even though you did not configure it.
- When the memory utilization cannot be determined due to various reasons such as communication problem with the Power Virtual Server instance, the memory utilizations can show 100%.
- The memory utilization is zero if the Power Virtual Server instance is in a shut-off state.
- Metrics are available for Power Virtual Server instance that are deleted. Based on the IBM Cloud Monitoring retention policy, you can see the historical platform metrics of deleted instance. For more information, see [Sysdig documentation on retention limit](#).
- Metrics of Power Virtual Server instances are not available if the instance was never initialized before, error in the instance, or issue with the host.

Additional information

- Learn more about the Sysdig dashboard user interface, see [About the Dashboard UI](#).
- See the [IBM Cloud monitoring](#) documentation in IBM Cloud.
- Refer to [sample pricing](#) in IBM Cloud Monitoring documentation.

IBM Cloud Logs events for Power Virtual Server

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

IBM Cloud Logs is a scalable logging service that stores and manages logs. You can use IBM Cloud Logs to monitor and audit events and operations across your Power Virtual Server resources.

IBM® Power® Virtual Server automatically generates log events that report on activities that affect your resources. These log events can appear in different formats, including unstructured text, JSON, delimiter-separated values, and key-value pairs. You can use these log events to investigate abnormal activity, track critical actions, and meet regulatory audit requirements.

For more information, see [Getting started with IBM Cloud Logs](#).

Management events

The following events are auto-generated when you create, read, update, or delete Power Virtual Server resources, such as instances, networks, and volumes.

Instance events

The following events are generated when you list or read Power Virtual Server instances:

Event	Description
power-iaas.event.list	Lists all events from a cloud instance
power-iaas.event.read	Reads an event from a cloud instance

Instance events

Image events

The following events are generated when you import, export, update, or delete images in a workspace:

Event	Description
power-iaas.image.delete	Deletes an image
power-iaas.image.export	Exports an image
power-iaas.image.import	Imports an image
power-iaas.image.list	Lists all images
power-iaas.image.read	Reads an image
power-iaas.image.update	Updates an image

Image events

Network events

The following events are generated when you create, read, update, delete, or list network resources such as network address groups, network peers, security groups, and interfaces:

Event	Description
power-iaas.network-address-group.create	Creates a network address group
power-iaas.network-address-group.delete	Deletes a network address group from a workspace
power-iaas.network-address-group.list	Lists all network address groups for a workspace
power-iaas.network-address-group.read	Reads a network address group

power-iaas.network-address-group.update	Updates a network address group
power-iaas.network-interfaces.create	Creates a network interface
power-iaas.network-interfaces.delete	Deletes a network interface
power-iaas.network-interfaces.list	Lists all network interfaces
power-iaas.network-interfaces.read	Reads a network interface
power-iaas.network-interfaces.update	Updates a network interface
power-iaas.network-peer-route-filter.create	Creates a network peer route filter
power-iaas.network-peer-route-filter.delete	Deletes a network peer route filter
power-iaas.network-peer-route-filter.read	Reads a network peer route filter
power-iaas.network-peer.create	Creates a network peer
power-iaas.network-peer.delete	Deletes a network peer
power-iaas.network-peer.list	Lists all network peers
power-iaas.network-peer.read	Reads a network peer
power-iaas.network-peer.update	Updates a network peer
power-iaas.network-security-group.clone	Clones a network security group
power-iaas.network-security-group.create	Creates a network security group
power-iaas.network-security-group.delete	Deletes a network security group from a workspace
power-iaas.network-security-group.list	Lists all network security groups for a workspace
power-iaas.network-security-group.read	Reads a network security group
power-iaas.network-security-group.update	Updates a network security group
power-iaas.network.create	Creates a network (public or private)
power-iaas.network.delete	Deletes a network
power-iaas.network.list	Lists all networks
power-iaas.network.read	Reads a network
power-iaas.network.update	Updates a network
power-iaas.peer-interfaces.list	Lists all interfaces for a network peer

Network events

Power Virtual Server events

The following events are generated when you create, start, stop, capture, update, or delete Power Virtual Server instances and their associated resources:

Event	Description
power-iaas.pvm-instance-virtual-serial-number.create	Assigns a virtual serial number to a Power Virtual Server instance
power-iaas.pvm-instance-virtual-serial-number.delete	Unassigns a virtual serial number from a Power Virtual Server instance
power-iaas.pvm-instance-virtual-serial-number.read	Reads the virtual serial number information of a Power Virtual Server instance
power-iaas.pvm-instance-virtual-serial-number.update	Updates a virtual serial number of a Power Virtual Server instance
power-iaas.pvm-instance-vmem-volume.create	Creates a vmem volume to be attached to a Power Virtual Server instance
power-iaas.pvm-instance-vmem-volume.delete	Deletes a vmem volume that is attached to a Power Virtual Server instance
power-iaas.pvm-instance-vmem-volume.list	Lists all vmem volumes that are attached to a Power Virtual Server instance
power-iaas.pvm-instance-vmem-volume.read	Reads a vmem volume that is attached to a Power Virtual Server instance
power-iaas.pvm-instance-vmem-volume.update	Updates a vmem volume that is attached to a Power Virtual Server instance
power-iaas.pvm-instance.capture	Captures a Power Virtual Server instance as an image
power-iaas.pvm-instance.create	Creates a Power Virtual Server instance
power-iaas.pvm-instance.delete	Deletes a Power Virtual Server instance
power-iaas.pvm-instance.immediate-shutdown	Shuts down a Power Virtual Server instance immediately
power-iaas.pvm-instance.list	Lists all Power Virtual Server instances and instance snapshots
power-iaas.pvm-instance.monitor	Lists all supported console languages
power-iaas.pvm-instance.operation	Triggers an operation on a Power Virtual Server instance
power-iaas.pvm-instance.read	Reads a Power Virtual Server instance and restores an instance snapshot
power-iaas.pvm-instance.renew	Restarts a Power Virtual Server instance
power-iaas.pvm-instance.snapshot	Creates a Power Virtual Server instance snapshot
power-iaas.pvm-instance.start	Starts a Power Virtual Server instance
power-iaas.pvm-instance.stop	Stops a Power Virtual Server instance
power-iaas.pvm-instance.unknown	Triggers an unknown action on a Power Virtual Server instance
power-iaas.pvm-instance.update	Updates a Power Virtual Server instance
power-iaas.pvm-instance-network.create	Attaches a network to a Power Virtual Server instance
power-iaas.pvm-instance-network.delete	Detaches a network from a Power Virtual Server instance
power-iaas.pvm-instance-network.list	Lists all networks that are attached to a Power Virtual Server instance
power-iaas.pvm-instance-network.read	Reads the information of a network that is attached to a Power Virtual Server instance

power-iaas.cloud-instance.read	Reads the current state and configuration of a cloud instance
--------------------------------	---

Power Virtual Server events

Secure Shell (SSH) key events

The following events are generated when you create, update, or delete SSH keys in a workspace:

Event	Description
power-iaas.sshkey.create	Creates an SSH key
power-iaas.sshkey.delete	Deletes an SSH key
power-iaas.sshkey.list	Lists all SSH keys
power-iaas.sshkey.read	Reads an SSH key
power-iaas.sshkey.update	Updates an SSH key

Secure shell (SSH) key events

Data volume events

The following events are generated when you create, attach, clone, read, update, or delete data volumes and volume groups:

Event	Description
power-iaas.volume-group.delete	Deletes a cloud instance volume group
power-iaas.volume-group.list	Lists all volume groups
power-iaas.volume-group.post	Creates a volume group
power-iaas.volume-group.read	Lists the remote-copy volume relationships for a volume group
power-iaas.volume-group.reset	Resets an operation on a volume group
power-iaas.volume-group.start	Starts a volume group
power-iaas.volume-group.stop	Stops a volume group
power-iaas.volume-group.unknown	Triggers an unknown operation on a volume group
power-iaas.volume-group.update	Updates a volume group
power-iaas.volume-onboarding.create	Adds auxiliary volumes to the target site
power-iaas.volume-onboarding.list	Lists all volume onboarding operations
power-iaas.volume-onboarding.read	Reads information about a volume onboarding operation
power-iaas.volume-snapshot.list	Lists all volume snapshots in a workspace
power-iaas.volume-snapshot.read	Reads the details of a volume snapshot
power-iaas.volume.clone	Creates a volume clone for specified volumes
power-iaas.volume.configure	Attaches, detaches, or updates a volume that is attached to a Power Virtual Server instance

power-iaas.volume.create	Creates a volume
power-iaas.volume.delete	Deletes a volume
power-iaas.volume.list	Lists all volumes
power-iaas.volume.operation	Triggers an action on a volume
power-iaas.volume.read	Reads a volume, flash copy mappings of a volume, and remote copy volume information
power-iaas.volume.start	Starts an operation on a volume
power-iaas.volume.update	Updates a volume
Data volumes events	

Storage capacity events

The following events are generated when you list or read storage capacity information for a workspace or pod:

Event	Description
power-iaas.storage-capacity.list	Lists storage capacity information for a workspace
power-iaas.storage-capacity.read	Reads storage capacity information
power-iaas.pod-capacity.list	Lists system and storage capacity for a client location pod
Storage capacity events	

Storage pool events

The following event is generated when you list storage pools for an IBM data center:

Event	Description
power-iaas.system-pools.list	Lists storage pools of an IBM data center
Storage pool events	

Tenant events

The following events are generated when you read or update tenant information and manage tenant SSH keys:

Event	Description
power-iaas.tenant-sshkey.create	Creates a tenant SSH key
power-iaas.tenant-sshkey.delete	Deletes a tenant SSH key
power-iaas.tenant-sshkey.list	Lists the SSH keys of a tenant
power-iaas.tenant-sshkey.read	Reads a tenant SSH key
power-iaas.tenant-sshkey.update	Updates a tenant SSH key
power-iaas.tenant.read	Reads a tenant

power-iaas.tenant.update	Updates a tenant
--------------------------	------------------

Tenant events

Cloud instance job events

The following events are generated when you create, delete, list, or read cloud instance jobs:

Event	Description
power-iaas.job.create	Creates a cloud instance job
power-iaas.job.delete	Deletes a cloud instance job
power-iaas.job.list	Lists the last five jobs that were run for the cloud instance
power-iaas.job.read	Reads the details of a job

Cloud instance job events

Network port events

The following events are generated when you create, update, or delete network ports in a workspace:

Event	Description
power-iaas.port.create	Creates a network port
power-iaas.port.delete	Deletes a network port
power-iaas.port.list	Lists all network ports
power-iaas.port.read	Reads a network port
power-iaas.port.update	Updates a network port

Network port events

SAP events

The following events are generated when you create or read SAP Power Virtual Server instances:

Event	Description
power-iaas.sap.create	Creates an SAP Power Virtual Server instance
power-iaas.sap.list	Lists all SAP Power Virtual Server instances
power-iaas.sap.read	Reads an SAP Power Virtual Server instance

SAP events

Cloud connections events

The following events are generated when you create, update, or delete cloud connections in a workspace:

Event	Description
power-iaas.cloud-connection.create	Creates a cloud connection
power-iaas.cloud-connection.delete	Deletes a cloud connection

power-iaas.cloud-connection.list	Lists all cloud connections
power-iaas.cloud-connection.read	Reads a cloud connection
power-iaas.cloud-connection.update	Updates a cloud connection

Cloud connections events

Placement groups events

The following events are generated when you create, update, or delete placement groups for Power Virtual Server instances:

Event	Description
power-iaas.placement-group.create	Creates a placement group
power-iaas.placement-group.delete	Deletes a placement group
power-iaas.placement-group.list	Lists all placement groups
power-iaas.placement-group.read	Reads a placement group
power-iaas.placement-groups.update	Updates a placement group

Placement groups events

Shared processor pool (SPP) placement groups events

The following events are generated when you create, update, or delete SPP placement groups:

Event	Description
power-iaas.spp-placement-group.create	Creates an SPP placement group
power-iaas.spp-placement-group.delete	Deletes an SPP placement group
power-iaas.spp-placement-group.list	Lists all SPP placement groups
power-iaas.spp-placement-group.read	Reads an SPP placement group
power-iaas.spp-placement-groups.delete	Deletes all SPP placement groups in a workspace
power-iaas.spp-placement-groups.update	Updates an SPP placement group

SPP placement groups events

Data center events

The following events are generated when you list or read information and capabilities for IBM Power Virtual Server in IBM data center or in Client location:

Event	Description
power-iaas.datacenter-private.list	Lists all data center information at a client location
power-iaas.datacenter-private.read	Reads the data center information at a client location
power-iaas.datacenter.list	Lists all IBM data center information
power-iaas.datacenter.read	Reads the IBM data center information

Data center events

Host events

The following events are generated when you create, read, update, delete, or list dedicated hosts and host groups in a workspace:

Event	Description
power-iaas.host-group.create	Creates a host group with one or more hosts
power-iaas.host-group.delete	Deletes a host group
power-iaas.host-group.list	Lists all host groups for your workspace
power-iaas.host-group.read	Reads the details of a host group
power-iaas.host-group.update	Updates a host group
power-iaas.host.delete	Releases a host from its host group
power-iaas.host.list	Lists all hosts that are available to a workspace
power-iaas.host.read	Reads the details of a host
power-iaas.host.update	Updates the display name of a host
power-iaas.available-hosts.list	Lists all hosts that can be reserved

Host events

Route events

The following events are generated when you create, update, or delete routes in a workspace:

Event	Description
power-iaas.route.create	Creates a network route
power-iaas.route.delete	Deletes a network route
power-iaas.route.list	Lists all network routes in a workspace
power-iaas.route.read	Reads information about a network route
power-iaas.route.update	Updates information for a network route

Route events

SPP events

The following events are generated when you create, read, update, delete, or list SPPs for a cloud instance:

Event	Description
power-iaas.shared-processor-pool.create	Creates an SPP
power-iaas.shared-processor-pool.delete	Deletes an SPP from a cloud instance
power-iaas.shared-processor-pool.list	Lists all SPPs for a cloud instance
power-iaas.shared-processor-pool.read	Reads the details of an SPP for a cloud instance

power-iaas.shared-processor-pool.update	Updates an SPP for a cloud instance
---	-------------------------------------

SPP events

Snapshot events

The following events are generated when you list, read, update, or delete workspace snapshots:

Event	Description
power-iaas.snapshot.list (deprecated)	Lists all snapshots in a workspace
power-iaas.snapshot.read (deprecated)	Reads the details of a snapshot
power-iaas.snapshot.delete	Deletes a snapshot
power-iaas.snapshot.update	Updates a snapshot

Snapshot events

Stock image events

The following events are generated when you list or read available stock images for a workspace:

Event	Description
power-iaas.stock-image.list	Lists all available stock images
power-iaas.stock-image.read	Reads the details of an available stock image

Stock image events

Virtual serial number events

The following events are generated when you list, read, update, or delete virtual serial numbers:

Event	Description
power-iaas.virtual-serial-number.delete	Unreserves a retained virtual serial number
power-iaas.virtual-serial-number.list	Lists all used and retained virtual serial numbers
power-iaas.virtual-serial-number.read	Reads the information for a virtual serial number
power-iaas.virtual-serial-number.update	Updates the description of a reserved virtual serial number

Virtual serial number events

Workspace events

The following events are generated when you enable, disable, list, or read workspaces that belong to a tenant:

Event	Description
power-iaas.workspace.disable	Disables a workspace
power-iaas.workspace.enable	Enables a workspace
power-iaas.workspace.list	Lists all workspace information and capabilities
power-iaas.workspace.read	Reads information of a workspace

power-iaas.workspace.unknown	Triggers an unknown action on a workspace
------------------------------	---

Workspace events



Note: An `unknown` event is generated when an unrecognized action is triggered on a workspace.

Disaster recovery events

The following event is generated when you read disaster recovery site details for a data center location:

Event	Description
power-iaas.disaster-recovery.read	Reads the disaster recovery site details for the current data center location

Disaster recovery events

Service broker events

The following event is generated when you list supported storage tiers for a Power Virtual Server instance:

Event	Description
power-iaas.service-broker.read	Lists all supported storage tiers for a Power Virtual Server instance

Service broker events

Software tiers (IBM i licensing) events

The following event is generated when you list supported software tiers for IBM i licensing:

Event	Description
power-iaas.software-tier.list	Lists all supported software tiers for IBM i licensing

Software tiers (IBM i licensing) events

Metadata and identity events

The following events are generated when you create or request identity tokens or read metadata for a virtual server instance:

Event	Description
power-iaas.metadata-identity-token.create	Creates an identity token
power-iaas.metadata-computeresource-token.request	Requests an IBM Cloud Identity and Access Management (IAM) access token
power-iaas.metadata-instance.read	Reads information about the virtual server instance (VSI) that issued the request

Metadata and identity events

Viewing events

Events are automatically forwarded to North America, Europe, Tokyo, or Sydney geographic locations. You can access the IBM Cloud Logs as follows:

- All North America and South America data centers from Dallas.
- All Europe data centers from Frankfurt.
- All Sydney data center from Sydney, and
- All Japan data center from Tokyo.

IBM Cloud Logs sample response format

The response format that is used in IBM Cloud Logs follows the Cloud Auditing Data Federation (CADF) standard. This standard ensures that auditing events are collected and routed consistently across cloud platforms.

The CADF standard provides a framework for security auditing in cloud environments. It defines an event model that captures the data that is needed to certify, manage, and audit the security of cloud applications and services.

The following code snippet shows the response format.

```
$ {
  "logSourceCRN": "crn:v1:bluemix:public:power-iaas:us-east:a/xxxxxxxxxxxxxxxxxxxx:yyyyyyyyyyyyyyyyyyyy::",
  "saveServiceCopy": true,
  "dataEvent": false,
  "outcome": "success",
  "eventTime": "2022-06-30T03:12:49.63+0000",
  "action": "power-iaas.tenant.read",
  "correlationId": "xxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "severity": "normal",
  "initiator": {
    "id": "IBMid-xxxxxxxx",
    "name": "xxxxm@us.ibm.com",
    "typeURI": "service/security/account/user",
    "authnId": "",
    "authnName": "",
    "host": {
      "agent": "PostmanRuntime/7.28.4",
      "address": "127.0.0.1",
      "addressType": "IPv4"
    },
    "credential": {
      "type": "user"
    }
  },
  "target": {
    "id": "crn:v1:bluemix:public:power-iaas:us-east:a/xxxxxxxxxxxxxxxxxxxx:yyyyyyyyyyyyyyyyyyyy::",
    "name": "testName",
    "typeURI": "power-iaas/tenant",
    "resourceGroupId": "crn:v1:bluemix:public:resource-controller::a/xxxxxxxxxxxxxxxxxxxx::resource-group:zzzzzzzzzzzzzzzzzzzz"
  },
  "reason": {
    "reasonCode": 200,
    "reasonType": "OK"
  },
  "requestData": null,
  "responseData": {
    "cloudInstances": [
      {
        "capabilities": [],
        "cloudInstanceId": "yyyyyyyyyyyyyyyyyyyy",
        "enabled": true,
        "href": "/pcloud/v1/cloud-instances/yyyyyyyyyyyyyyyyyyyy",
        "initialized": false,
        "name": "testName",
        "region": "us-east"
      }
    ],
    "creationDate": "2019-05-21T21:32:00.746Z",
    "enabled": true,
    "sshKeys": [],
    "tenantID": "xxxxxxxxxxxxxxxxxxxx"
  },
  "message": "Power Virtual Server: read tenant xxxxxxxxxxxxxxxxxxxx ",
  "observer": {
    "name": "IBM Cloud Logs"
  }
}
```

IBM Cloud Logs regions

You can create a cloud log instance and provision it in the same region where your data center is located. For more information, see [Provisioning an instance](#).

The Power Virtual Server workspaces that run in various regions or data centers send events to IBM Cloud Logs instances in their respective regions. You

must create and provision instances of IBM Cloud Logs in the regions where your workspaces are located to maintain access to Power Virtual Server log events. If you want to export IBM Cloud Logs data, see [Exporting data](#).

The following table lists the data centers and their corresponding regions where you can deploy an IBM Cloud Logs instance:

Data centers	Current IBM Cloud Logs region	New IBM Cloud Logs region
WDC04	us-south	us-east
WDC06	us-south	us-east
WDC07	us-south	us-east
MON01	us-south	ca-tor
TOR04	us-south	ca-tor
SAO01	us-south	br-sao
SAO04	us-south	br-sao
LON04	eu-de	eu-gb
LON06	eu-de	eu-gb
OSA21	jp-tok	jp-osa

Data centers and their corresponding IBM Cloud Logs instance regions

Understanding data portability for Power Virtual Server

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

On IBM Cloud, data portability refers to a set of tools and procedures. You can use data to transfer data from IBM® Power® Virtual Server to on-premises or on a different service provider that supports the Power architecture to implement a similar workload or to process data.

Power Virtual Server is an Infrastructure as a Service (IaaS) offering. As a Power Virtual Server user, you own the data, operating system images, and configuration parameters. The Power Virtual Server architecture follows Power deployments within infrastructure to facilitate data portability. The Power Virtual Server architecture is identical to a certified private cloud infrastructure. The infrastructure is designed such that the Power Virtual Server can maintain key enterprise software certification and support.

Responsibilities

Power Virtual Server provides interfaces and instructions to guide you to copy and store your content on your selected location. The content can include data that is stored on logical volumes, operating system images, and configuration parameters.

You are responsible for the following actions:

- Use of the exported data
- Use of the configuration to enable data portability to other infrastructures
- Create and execute the plan to set up the infrastructure on a different service provider that has capabilities similar to the Power Virtual Server services

For more information about the responsibility assignment matrix for Power Virtual Server, see [High-level architecture](#).

Migrating data

To migrate data from one infrastructure to another, you must create a plan and have a strategy to achieve the expected performance. Complete the steps that are defined in the following topics to migrate Power workloads to a Power Virtual Server:

- [Planning a workload migration to IBM® Power® Virtual Server](#)
- [Migration strategies for AIX](#)
- [Migration strategies for IBM i](#)
- [Additional migration strategies](#)
- [Using third-party tools for migration](#)
- [Back up and restore](#)

You can use Power Virtual Server to capture full or specific copies of the logical volumes or data sets. By using IBM FlashCopy® feature and Power Virtual Server APIs, you can create delta snapshots, volume clones, and restore your disks. For more information, see [Snapshots, cloning, and restoring](#).

Exported data formats

Power Virtual Server provides mechanisms to capture and export a virtual machine (VM). For more information about the available options to capture and export a VM, see [Capturing and exporting a virtual machine \(VM\)](#).

Data ownership

The data that is exported to your on-premises resources or on service providers resources is classified as customer content. Customer ownership and licensing rights are applicable to the exported content as stated in the [IBM Cloud Service Agreement](#).

IBM Power Virtual Server maintenance operations overview

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

IBM® Power® Virtual Server maintenance operations ensure the reliability, security, and performance of the infrastructure components in the data centers, including hosts, networks, and storage systems. IBM performs periodic maintenance to maintain optimal service availability and avoid disruptions.

Data center maintenance

IBM maintains its data centers by updating the network components, storage systems, and compute server infrastructure. Your workloads usually remain unaffected during the data center maintenance. Power Virtual Server maintenance includes hosts, dedicated hosts, storage, and network infrastructure resources. The maintenance ensures that the IBM Power technology stack remains secure and up-to-date with appropriate security fixes and feature updates.

Most updates do not affect the workloads that are running on the Power Virtual Server. If the maintenance is expected to affect your workloads, IBM notifies you in advance.

Host maintenance

Host maintenance activities, such as firmware updates, require a live partition migration of a virtual service instance (VSI) from the compute host to another host. In such cases, the VSIs continue to run without interruption. Therefore, IBM does not send a notification. For more information, see [Viewing notifications](#).

When you use the VSI pinning options, the VSI's availability during maintenance activities is affected. VSI pinning prevents Power Virtual Server from migrating the VSI to another host during the maintenance operation. As a result, the live migration of the VSI is not possible, and the VSI must be in the *Shutoff* state for the host maintenance operation to complete. Therefore, you must use the VSI pinning options cautiously. Any downtime that is caused by VSI pinning during the host maintenance is excluded from the service-level agreements. For more information, see [Service Level Agreements for IBM Cloud](#).

For more information about VSI pinning, see [What does VSI pinning do?](#).

Network maintenance

Power Virtual Server infrastructure includes redundant network paths that support nondisruptive network maintenance. Network maintenance operation targets only one network fabric at a time from the set of redundant network fabrics. When one of the network fabrics is under maintenance, the network automatically redirects traffic to the alternate fabric. Some network packet loss might occur during any traffic rerouting, including when traffic is redirected from one network fabric to another.



Important: If you use IBM Cloud Connections instead of Power Edge Router (PER), you must configure two Cloud Connections to reduce network disruptions during the maintenance activities. For more information about how to migrate to PER, see [Migrating to PER](#).

Storage maintenance

Power Virtual Server infrastructure provides redundant storage I/O paths that allow nondisruptive upgrades of the storage hardware. During the storage maintenance, some of the redundant paths become temporarily unavailable. This behavior aligns with the standard multipath I/O management practices that are supported by AIX, IBM i, and Linux operating systems. Refer to your operating system guidelines for information about the multipath I/O management during the maintenance events.



Note: Verify that all the disk paths are online after maintenance. If any disk paths remain offline, initiate an operating system device scan to restore the offline paths.

Command reference

Getting started with the IBM® Power® Virtual Server CLI plug-in

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

The IBM Power Virtual Server command-line interface (CLI) provides a set of commands that are grouped by namespace for you to interact with IBM Power Virtual Server in IBM data center and in Client location. For more information, see the following topics:

- [IBM Power Virtual Server CLI version 1.10.0 for IBM data center](#)
- [IBM Power Virtual Server CLI version 1.10.0 for Client location](#)

After you switch accounts, target the workspace that is associated with the new account before you run additional commands. Most CLI commands require a targeted workspace to run.

You can also use the application programming interfaces (APIs) to interact with the IBM Power Virtual Server. For more information, see [API references](#).

IBM® Power® Virtual Server CLI change log

In this change log, you can learn about the latest changes, improvements, and updates for the IBM® Power® Virtual Server CLI plug-in.

June 2026

CLI v1.10.0

The CLI plug-in version v1.10.0 is available for IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location.

New commands

The following CLI commands are added:

- [ibmcloud pi route](#): You can use this command to create, delete, get, and list routes for IBM Power Virtual Server Private Cloud in Client location.
- [ibmcloud pi vpmem-volume update](#): You can use this command to update the names of the virtual persistent memory (vPMEM) volumes.

New options

The following options are added to the existing CLI commands:

- `--advertise` is added to the [ibmcloud pi subnet create](#) and [ibmcloud pi subnet update](#) commands for IBM Power Virtual Server Private Cloud in Client location. You can use this option to determine whether to advertise the private networks.
- `--arp-broadcast` is added to the [ibmcloud pi subnet create](#) and [ibmcloud pi subnet update](#) commands for IBM Power Virtual Server Private Cloud in Client location. You can use this option to enable or disable the ARP broadcast.
- `--enable-dhcp` is added to the [ibmcloud pi subnet create](#) and [ibmcloud pi subnet update](#) commands for IBM Power Virtual Server Private Cloud in Client location. You can use this option to enable or disable subnet support for Dynamic Host Configuration Protocol (DHCP).
- `--allow-remote-restart` is added to the [ibmcloud pi instance create](#), [ibmcloud pi instance update](#), and [ibmcloud pi instance sap create](#) commands. You can use this option to enable for an instance.
- `--default-trusted-profile` is added to the [ibmcloud pi instance create](#), [ibmcloud pi instance update](#), and [ibmcloud pi instance sap create](#) commands. You can use this option to specify the trusted profile cloud resource name (CRN), ID, or name of the default identity and access management (IAM) trusted profile to use for a virtual server instance (VSI).
- `--default-trusted-profile-autolink` is added to the [ibmcloud pi instance create](#), [ibmcloud pi instance update](#), and [ibmcloud pi instance sap create](#) commands. You can use this option to enable the system to create a link to the specified trusted profile when you create or update the server.
- `--metadata-service` is added to the [ibmcloud pi instance create](#), [ibmcloud pi instance update](#), and [ibmcloud pi instance sap create](#) commands. You can use this option to specify whether the metadata service endpoint is available to the VSI.
- `--metadata-service-force` is added to the [ibmcloud pi instance update](#) command. You can use this option to disable the metadata service while the VSI is active.

Other enhancements

The following enhancements are included in this release:

- Added the `CRN` field to the output of the [ibmcloud pi workspace create](#) command.
- Added the `Deprecated` field to the output of the [ibmcloud pi instance sap profile](#) command.
- Updated the error message when `--net-type` is used in the [ibmcloud pi subnet create](#) command.
- Improved the readability of volume clone displays.
- Refined translations for improved clarity and consistency.

March 2026

CLI v1.9.0

The CLI plug-in version v1.9.0 is available for IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location.

New commands

The following CLI command is added to the Power Virtual Server Private Cloud workspaces:

- [ibmcloud pi instance vpmem-volume](#): You can use this command to attach, detach, view, and list vPMEM volumes for a Power Virtual Server instance.

New options

The following option is added to the existing CLI command:

- `--vpmem-volumes` is added to the [ibmcloud pi instance create](#) command for private workspaces. You can use this option to attach vPMEM volumes to a Power Virtual Server instance. You can attach up to 1 vPMEM volume to an SAP instance and up to 4 PMEM volumes to other instances.

Other enhancements

The following enhancements are included in this release:

- Added `Accelerated SAP Application Performance Standard (aSAPs)` field to the output of the [ibmcloud pi instance sap profile](#) and [ibmcloud pi instance sap list](#) commands.
- Added `External IP Address` field to the output of the [ibmcloud pi network-interface get](#), [ibmcloud pi network-interface create](#), and [ibmcloud pi network-interface update](#) commands.
- Enhanced the output of the [ibmcloud pi instance](#) commands to group fields more clearly and omit fields that are not returned in the command output.

CLI v1.8.1

The CLI plug-in version v1.8.1 is available. This version fixes an issue where the `--storage-pool-affinity` option in the [ibmcloud pi instance create](#) command did not apply correctly.

December 2025

CLI v1.8.0

The CLI plug-in version v1.8.0 is available. This version is available for the IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location.

New commands

The following CLI command is added for Power Virtual Server workspaces in the IBM data center.

- [ibmcloud pi instance vpmem-volume](#): You can use this command to attach, detach, view, and list virtual persistent memory (vPMEM) volumes for an instance.

New options

The following options are added to the existing CLI commands:

- `--vpmem-volumes` is added to the [ibmcloud pi instance create](#) and [ibmcloud pi instance sap create](#) commands for Power Virtual Server workspaces in the IBM data center. You can use this option to specify a list of vPMEM volumes that you can associate with an instance. The maximum allowed vPMEM volumes vary by the instance type:
 - SAP instances: 1
 - Other instances: 4
- `--parameters` is added to the [ibmcloud pi workspace create](#) command. You can use this option to configure additional key-value pairs in a workspace.

Updated commands

The following commands are updated:

- Updated the usage instructions for the [ibmcloud pi instance operation](#) command.
- Updated the [ibmcloud pi instance action](#) command by adding *dumpprestart* as a valid value.

Other enhancements

The following enhancements are included in this release:

- Resolved issues where some object responses were incomplete.
- Enhanced the output of the [ibmcloud pi image create](#) and [ibmcloud pi image get](#) commands to include *Shared* and *Type* properties, and to dynamically hide properties that are not available in the response.
- Updated command messages and descriptions to improve translation accuracy and readability across supported languages.

September 2025

CLI v1.7.0

CLI plug-in version 1.7.0 is available for the IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location. In this release, the CLI is updated to use the latest version of the internationalization package. In addition, messages and options are updated to use the new guidelines for writing the descriptions.

New commands

The following CLI commands are added to the IBM Power Virtual Server Private Cloud workspaces:

- [ibmcloud pi network-peer](#). You can use this command to create, delete, get, list, and update network peering connection. In addition, you can use this command to list peer network interfaces.
- [ibmcloud pi network-peer route-filter](#). You can use this command to create, delete, and get the details of the route filters in the network peering connection.

New options

The following options are added to the existing CLI commands:

The `--preferred-processor-compatibility-mode` option is added to the [ibmcloud pi instance create](#) and [ibmcloud pi instance sap create](#) commands. You can use this option to set the instance to your preferred processor compatibility mode. The processor must be compatible with the system in use.

Updated commands

The following commands are updated:

- Added `Effective Processor Compatibility Mode` and `Preferred Processor Compatibility Mode` to the instance output.
- Added `Creation Date` to the shared-processor-pool output.

- Added `State` to the image output.
- Added `CRN` to the volume snapshot output.
- Added the `sr3` filter based on the [ibmcloud pi instance sap list --prefix](#) option to filter on s1122 and e1150 profiles.
- Updated the [ibmcloud pi snapshot update](#) output.
- Updated the use of `Creation Date` and `Last Update` dates to be consistent across all outputs.

CLI v1.6.1

The CLI version v1.6.1 is available with updates related to translation.

July 2025

CLI version 1.6.0

The CLI plug-in version 1.6.0 supports Power11 systems. This version is available for the IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location.

New commands

The following CLI commands are added:

- [ibmcloud pi route](#): You can use this command to create, delete, get, list, report, and update the custom network routes within the Power Virtual Server workspaces in the IBM data center.
- [ibmcloud pi virtual-serial-number software-tiers](#): You can use this command to list all the supported software-tiers.
- [ibmcloud pi instance operation](#): You can use this command to set the server boot mode, the server operating mode, a job task, or an operation type to an IBM i virtual server instance. This command can be used for the Power Virtual Server Private Cloud.

New options

The following options are added to the existing CLI commands:

- `--family`: Added to the [ibmcloud pi instance sap list](#) command. You can use this option to filter the SAP profiles by the family type.
- `--prefix`: Added to the [ibmcloud pi instance sap list](#) command. You can use this option to filter the SAP profiles by their prefix.
- `--user-tags`: Added to the [ibmcloud pi placement-group create](#) and [ibmcloud pi shared-processor-pool placement-group create](#) commands. You can use this option to add user tags to the Power Virtual Server resources.
- `--description`: Added to the [ibmcloud pi ssh-key create](#) command. You can use this option to add or update the description of an SSH key.
- `--visibility` option is added to the [ibmcloud pi ssh-key create](#) and [ibmcloud pi ssh-key update](#) commands. You can use this option to set the visibility of an SSH key at `account` or `workspace` level.
- `--advertise`: Added to the [ibmcloud pi subnet create](#) and [ibmcloud pi subnet update](#) commands. You can use this option to determine whether or not to advertise the private networks.
- `--arp-broadcast`: Added to the [ibmcloud pi subnet create](#) and [ibmcloud pi subnet update](#) commands. You can use this option to enable or disable the ARP broadcast.
- `--software-tier`: Added to the [ibmcloud pi instance virtual-serial-number update](#) command. You can use this option to set a software tier on an IBM i instance.
- `--name` and `--key`: Added to the [ibmcloud pi ssh-key update](#) command. These options replace the `--new-name` and `--new-key` options.

Deprecated commands or options

You can no longer use the following commands or options as the Power Virtual Server VPN as a Service (VPNaaS) reached its end of life and is not available for use:

- Deprecated the [ibmcloud pi vpn](#), [ibmcloud pi ike-policy](#), and [ibmcloud pi ipsec-policy](#) command as the Power Virtual Server VPN as a Service (VPNaaS) has reached its end of life and is not available for use.
- `--access-config`, `--jumbo`, `--peer-id`, `--peer-type`, and `--source-ip` options from the [ibmcloud pi subnet create](#) command.
- Removed the DHCP option as a subnet type in the Power Virtual Server Private Cloud.

CLI version 1.5.2

The CLI plug-in version 1.5.2 for the IBM Power Virtual Server in IBM data center and IBM Power Virtual Server Private Cloud in Client location is available. In this version, a bug in the CLI plug-in metadata is fixed.

April 2025

New CLI version v1.5.1 is available. The Power Virtual Server CLI plug-in is available with the following features:

What's changed

- Security update with CLI plugin dependencies.

March 2025

New CLI version v1.5.0 is available. The Power Virtual Server CLI plug-in is available with the following features:

New commands

- Added [ibmcloud pi instance virtual-serial-number](#) family of commands (assign, get, unassign, and update) for IBM Power Virtual Server Private Cloud in Client location workspaces.
- Added [ibmcloud pi virtual-serial-number](#) family of commands (delete, get, list, and update) for IBM Power Virtual Server Private Cloud in Client location workspaces.

New options

- Added `--virtual-serial-number` option to [ibmcloud pi instance create](#) command for IBM Power Virtual Server Private Cloud in Client location workspaces. Use this option to add virtual serial number information to an IBM i instance.
- Added `--retainVSN` option to [ibmcloud pi instance delete](#) command for IBM Power Virtual Server Private Cloud in Client location workspaces. Use this option to confirm whether the virtual serial number must be retained in your account after it is removed from the instance.

What's changed

- In the IBM Power Virtual Server Private Cloud in Client location, if you enable the `sap-subscription-mode` option for the workspace, different options are displayed in the image list-catalog.
- Added a default system to the instance SAP profile and the instance SAP list.

February 2025

New CLI version v1.4.2 is available. The Power Virtual Server CLI plug-in is available with the following features:

New commands

- Added [ibmcloud pi network-address-group](#) group of commands (create, delete, get, list, member-add, member-remove, and update) for IBM Power Virtual Server in IBM data center workspaces.
- Added [ibmcloud pi network-security-group](#) group of commands (clone, create, delete, get, list, member-add, member-move, member-remove, rule-add, rule-remove, and update) for IBM Power Virtual Server in IBM data center workspaces.

New options

- Added `--network-security-group-id` option to [ibmcloud pi instance subnet attach](#) for IBM Power Virtual Server in IBM data center workspaces. Use this option to specify the network security group of which the network interface is a member.
- Added `--deployment-target` option to [ibmcloud pi instance sap create](#) command for IBM Power Virtual Server in IBM data center workspaces. Use this option to specify the deployment of the dedicated host.
- Added `--user-tags` option to the [ibmcloud pi volume onboarding create](#) command for IBM Power Virtual Server in IBM data center and [ibmcloud pi volume onboarding create](#) command for IBM Power Virtual Server Private Cloud in Client location. Use this option to add a list of user tags to be associated with the volumes when you onboard them.

What's changed

- Deprecated [ibmcloud pi cloud-connection](#) create command.

- Updated `--subnets` option for [ibmcloud pi instance create](#) to allow users to specify a network-security-group for IBM Power Virtual Server in IBM data center workspaces.
- Removed `--sys-type` default option for [ibmcloud pi instance create](#) and [ibmcloud pi instance sap create](#). This should not impact existing scripts.

January 2025

New CLI version `v1.4.1` is available. The Power Virtual Server CLI plug-in is available with the following features:

What's changed

- Security update with CLI plugin dependencies.
- Fixed the [ibmcloud pi workspace action](#) command to use local context.

December 2024

New CLI version `v1.4.0` is available. The Power Virtual Server CLI plug-in is available with the following features:

New commands

- Added `ibmcloud pi instance virtual-serial-number` family of commands (assign, unassigned, get, and update).
- Added `ibmcloud pi network-peer` family of commands (list) for Client location workspaces.
- Added `ibmcloud pi virtual-serial-number` family of commands (delete, get, list, and update).

New options

- Added `--private` option to [ibmcloud pi datacenter get and list commands](#). Use this option to retrieve additional details about a private datacenter that you own.
- Added `--virtual-serial-number` option to [ibmcloud pi instance create](#) command. Use this option to add virtual serial number information to an IBM i instance.
- Added `--retainVSN` option to [ibmcloud pi instance delete](#) command. Use this option to determine whether the virtual serial number is retained after it is removed from the instance.

What's changed

- In the [ibmcloud pi subnet create](#) command `--access-config` option is deprecated for Client location workspaces. Alternatively, you can use `--peer-id`, `--peer-type`, and `--source-id` options.
- The `che01` region is supported in the following commands:
 - [ibmcloud pi image export](#)
 - [ibmcloud pi image import](#)
 - [ibmcloud pi instance capture](#)
- The `per-migrate-status` option is removed from the [ibmcloud pi workspace action](#) command.
- Commands and options are updated with wording improvements.

September 2024

New CLI version `v1.3.0` is available. The Power Virtual Server CLI plug-in is available with the following features:

New commands

- Added [ibmcloud pi instance volume bulk-detach](#) command to detach multiple volumes from an instance at once.
- Added [ibmcloud pi instance snapshot update](#) command to update name and description of a snapshot.
- Added [ibmcloud pi network-interface](#) family of commands (create, delete, get, list, update).

- Added [ibmcloud pi volume bulk-delete](#) command to delete multiple volumes at once.
- Added [ibmcloud pi volume clone-async](#) *create* and *get* commands to asynchronously create clone tasks and query their status.
- Added [ibmcloud pi volume snapshot](#) *get* and *list* to retrieve information about snapshots in a workspace.
- Added [ibmcloud pi workspace action](#) command to perform an action in a workspace.

New options

- Added `--user-tags` option to [ibmcloud pi host reserve](#), [ibmcloud pi image create](#), [ibmcloud pi image import](#), [ibmcloud pi instance capture create](#), [ibmcloud pi instance create](#), [ibmcloud pi shared-processor-pool create](#), [ibmcloud pi instance snapshot create](#), [ibmcloud pi subnet create](#), [ibmcloud pi volume clone execute](#), [ibmcloud pi volume create](#), and [ibmcloud pi workspace create](#) commands. Use this option to add a comma separated list of user tags to be attached to the created resource.
- Added user tag option to `--hosts` option in [ibmcloud pi host-group create](#) command. Option is now also repeatable.
- Added `--boot-volume-replication-enabled` option to [ibmcloud pi instance create](#), and [ibmcloud pi instance sap create](#) command. Use this to enable storage replication on the boot volume.
- Added `--replication-sites` option to [ibmcloud pi instance create](#) and [ibmcloud pi instance sap create](#) command. Use this to indicate the replication site of the boot volume.
- Added `--replication-sites` option to [ibmcloud pi volume create](#). Use this to indicate the replication site of the volume.
- Added "maxVolumeSupport" valid value to `--storage-connection` option in [ibmcloud pi instance create](#) command.
- Added `--boot-volume` option to [ibmcloud pi instance volume attach](#) command. Use this to attach a boot volume to an instance.

What's changed

- [ibmcloud pi snapshot](#) command and subcommands have been deprecated. Please use [ibmcloud pi instance snapshot](#) command family instead.
- Clarified usage of [ibmcloud pi instance console list](#), [ibmcloud pi instance capture create](#) and [ibmcloud pi job list](#) commands.
- Minor wording improvements to several command and options.

June 2024

New CLI version `v1.2.0` available. The Power Virtual Server CLI plug-in is available with the following features:

New commands

- Added [ibmcloud pi available-hosts](#) command to list of hosts available for reservation.
- Added [ibmcloud pi host](#) family of commands (get, list, release, reserve, update).
- Added [ibmcloud pi host-group](#) family of commands (create, get, list, update).
- Added [ibmcloud pi instance snapshot](#) family of commands (create, delete, get, update, and restore).
- Enabled [ibmcloud pi shared-processor-pool](#) create, delete, get, list, update commands on IBM Power Virtual Server Private Cloud in Client location workspaces.
- Enabled [ibmcloud pi shared-processor-pool placement-group](#) create, delete, get, list, member-add, and member-remove commands on IBM Power Virtual Server Private Cloud in Client location workspaces.

New options

- Added `--import-details` option to [ibmcloud pi image import](#) command. Use this to import details for SAP image. You must include a license, product, and vendor. Valid license value: byol. Valid product value: Hana, Netweaver. Valid vendor value: SAP.
- Added `--storage-connection` option to [ibmcloud pi instance create](#) command. Use this to set the connection type. Valid value is `vSCSI` (more will be added in the future).
- Added `--storage-pool-affinity` option to [ibmcloud pi instance create](#) and [ibmcloud pi instance update](#) command. Use this to indicate whether all volumes attached to the server must reside in the same storage pool.
- Added `--deployment-target` option to [ibmcloud pi instance create](#) command. Use this to set the deployment of the dedicated host.
- Added `--host-id` option to [ibmcloud pi shared-processor-pool](#) create command. This option is only available for dedicated hosts.
- Added `--checksum` option to [ibmcloud pi image import](#), [ibmcloud pi image export](#), and [ibmcloud pi instance capture](#) create on IBM Power Virtual Server Private Cloud in Client location workspaces. This creates a checksum file after the operation is complete.
- Enabled `--shared-processor-pool`, `--IBMiCSS-license`, `--IBMiPHA-license`, and `--IBMiRDS-users` options on [ibmcloud pi instance create](#) command on IBM Power Virtual Server Private Cloud in Client location workspaces.
- Enabled `--IBMiCSS-license`, `--IBMiPHA-license`, and `--IBMiRDS-users` options on [ibmcloud pi instance update](#) command on IBM Power Virtual

Server Private Cloud in Client location workspaces.

What's changed

The [ibmcloud pi instance snapshot list](#) command can now list all snapshots of an instance or all snapshots on the workspace depending on arguments.

April 2024

New CLI version `v1.1.1` available. The Power Virtual Server CLI plug-in is available with the following features:

- Added virtual core support. You can now use `--virtual-cores` option in the [instance update](#) command.
- Added the "eu-es" region to the [image export](#), [image-import](#), and [instance-capture](#) commands.

March 2024

New CLI version `v1.1.0` available. The Power Virtual Server CLI plug-in is available with the following features:

- Added virtual core support. You can now use `--virtual-cores` option in the [ibmcloud pi instance create](#) command.
- Added mtu support. You can now use `--mtu` in the [ibmcloud pi subnet create](#) command. The jumbo option is deprecated.
- The updated [ibmcloud pi volume clone](#) commands now use the latest API.
- Reworks [ibmcloud pi volume clone cancel](#) and [ibmcloud pi volume clone create](#) commands.
- Added a new [ibmcloud pi volume clone delete](#) command.

January 2024

- New CLI version `0.7.1` available. Here are the changes for the new CLI version:

New options

- "list-catalog": `--sap` Include SAP images.
- "list-catalog": `--vtl` Include VTL images.

What's Changed

Removed `--IBMiDBQ-license` option from `instance-create` command.

Bug Fixes

Fixed issue in `instance-update` command when `--virtual-optical-device` option is not used.

December 2023

New CLI version `0.7.0` available. Here are the changes for the new CLI version:

New command

- [List all storage tiers for the targeted region](#): List all storage tiers for the targeted region.

New options

- A `--virtual-optical-device` option is added in [Update a server instance](#) command: Use this to attach a virtual optical device to this instance. Valid value is "attach".
- A `--mtu` option is added in [Create a private network](#) command: Use this is to define the Maximum Transmission Unit. The default value is 1450.
- A `--mtu` option is added in [Create a public network](#) command: Use this is to define the Maximum Transmission Unit. The default value is 1450.
- A `--target-tier` option is added in [Perform an action on a volume](#) command: Use this to change the storage tier of the volume (use [List all storage tiers for the targeted region](#) to see available storage tiers in the targeted region). `Tier5k` volumes cannot exceed 200 GB.

What's Changed

- New custom deployment type - `VMNoStorage` for [Create a server instance](#) command.

- Deprecate `--jumbo` option in [Create a public network](#) and [Create a private network](#) commands.
- New Power Edge Router (PER) details field when using the `workspace` command.

November 2023

New CLI version `0.6.0` available. Here are the new changes for the new CLI version:

- New commands - [Create a workspace](#) and [Delete a workspace](#) added.

New CLI version `0.5.0` available. Here are the new changes for the new CLI version:

- New `--user-data` option added in the [instance-create](#) command.
- New command [datacenter](#) and [datacenters](#) added.
- Deprecated `service-list` command in favor of new [workspace](#) and [workspaces](#) commands.

September 2023

- Added s1022 in `sys-type value` for [Create a server instance](#) and [Create a virtual tape library](#) commands.

2022

December 2022

- You can now get new error messages for undefined response codes for new service endpoint response codes.

September 2022

- You can now use shared processor pool that uses CLI. The following are new commands for shared processor pools:
 - [View details of a shared processor pool](#)
 - [Create a shared processor pool](#)
 - [Delete a shared processor pool](#)
 - [Update a shared processor pool](#)
 - [List all shared processor pools](#)
 - [View details of a shared processor pool placement group](#)
 - [Create a shared processor pool placement group](#)
 - [Delete a shared processor pool placement group](#)
 - [Add a shared processor pool to the placement group](#)
 - [Remove a shared processor pool from the placement group](#)
 - [List all shared processor pool placement groups](#)
- The command [Create a server instance](#) is updated to include the following new options for SPP and Epic:
 - *shared-processor-pool value*
 - *deployment-type value*
- You can now use global replication service by using CLI. The following commands are added new for global replication service:
 - [List disaster recovery locations for the current region or all regions](#)
 - [Perform an action on a volume](#)
 - [Get a list of flash copy mappings of a volume directly from primary storage host.](#)
 - [View details of a volume group](#)
 - [Create a volume group](#)
 - [Delete a volume group](#)
 - [Get all remote copy relationships for each volume in a volume group](#)
 - [Reset a volume group](#)
 - [List all volume groups](#)
 - [Start a volume group](#)
 - [Stop a volume group](#)

- [View storage details of a volume group](#)
- [Update a volume group](#)
- [Get the information of volume onboarding operation](#)
- [Create a volume onboarding operation](#)
- [List all volume onboarding operations](#)
- [Get the remote copy relationship information of a volume](#)
- The command [Create a volume](#) is updated to include a new option *replication-enabled*.
- The description of [Create a new SAP PVM Instance](#) is changed for HANA images.

July 2022

- You can now use [Transit Gateway](#) to interconnect your Power Virtual Server to the IBM Cloud classic and Virtual Private Cloud (VPC).

April 2022

- Provisioning VTL instances is temporarily disabled.

2021

December 2021

- You can now use [Storage pools](#) to set affinity policies by using CLI.
- You can now use [Create connection](#) to set a 10 Gbps speed for your Cloud connection by using CLI.
- You can now use [Placement Groups](#) to create placement groups and add VMs to set policies by using CLI.

October 2021

- You can now use [VPN](#) to create VPN connection by using CLI.
- You can now use [VPN IKE policies](#) to create an IKE policy for the VPN connection by using CLI.
- You can now use [VPN IKE policies](#) to create an IPsec policy for the VPN connection by using CLI.

September 2021

- You can now use [Import Image](#) to Import an image from IBM Cloud Object Storage by using CLI.
- You can now use [Jobs](#) to View details of a job by using CLI.
- You can now use [Console language](#) to update Language to Japanese.

May 2021

- You can now use [Cloud connection](#) to automate the way you connect your Power Systems Virtual Server instances to the IBM Cloud resources by using CLI.

March 2021

- You can now manage [snapshots](#) of a cloud instance by using the CLI.
- You can create and list [SAP profiles](#) by using CLI.
- You can [list of available system pools](#) within a particular data center by using CLI.
- You can [attach](#), [detach](#), or [list all the attached networks](#) to an instance.
- Added image-import progress (task) monitoring.

Installing the IBM Power Virtual Server CLI plug-in

IBM Power Virtual Server in [IBM data center](#)

IBM Power Virtual Server Private Cloud in [Client location](#)

To install, update, or view the IBM Power Virtual Server CLI plug-in, complete the following steps:

1. Install the [IBM Cloud® CLI](#).

2. Install or update the `power-iaas` plug-in.

To install:

```
$ ibmcloud plugin install power-iaas
```

To update:

```
$ ibmcloud plugin update
```

To view your installed plug-ins and versions:

```
$ ibmcloud plugin list
```

3. Log in to the [IBM Cloud](#).

The power-iaas command-line plug-in uses the region that `ibmcloud login` targets to determine the IBM Power Virtual Server endpoint. For example, to use the IBM Power Virtual Server endpoint `https://cloud.ibm.com` you must use the `ibmcloud login -a https://cloud.ibm.com` command to target the `us-east` region. If you have a federated account, use the following command:

```
$ ibmcloud login -a https://cloud.ibm.com -sso
```

4. Run one of the following commands to list all the services under your account:

- For IBM Power Virtual Server in IBM data center, run the `ibmcloud pi ws ls` command. The **Cloud Resource Name** (CRN) under **ID** contains your **Tenant ID** and **Cloud Instance ID**. The following example shows a typical CRN:

```
$ crn:v1:staging:public:power-iaas:us-east:a/abcdefghijklmnopqrstuvwxyzabcdef:121d5ee5-b87d-4a0e-86b8-aaff422135478::
```

- For IBM Power Virtual Server Private Cloud in Client location, run the `ibmcloud pi ws ls -p` command.

5. Target your service by entering `ibmcloud pi ws tg <CRN>` command:

- Following is an example of the output for IBM Power Virtual Server in IBM data center:

```
$ ibmcloud pi ws tg crn:v1:staging:public:power-iaas:us-east:a/abcdefghijklmnopqrstuvwxyzabcdef:121d5ee5-b87d-4a0e-86b8-aaff422135478::
```

- Following is an example of the output for IBM Power Virtual Server Private Cloud in Client location:

```
$ ibmcloud pi ws tg crn:v1:staging:public:power-iaas:satloc_dal_clp3foc20dd82387pe83:a/c7e6bd2517ad44eabbd61fcc25cf68d5:dc815dc0-e0ba-4576-b286-a3aeedaa9e60::
```



Note: The IBM Power Virtual Server CLI plug-in requires a valid IAM token authorization before each use. Use the `ibmcloud login` command to renew authorization if your token expires.

IBM Power Virtual Server CLI version 1.10.0 for IBM data center

IBM Power Virtual Server in `IBM data center`

The following list of commands are available with command-line interface (CLI) for IBM Power Virtual Server in IBM data center.

ibmcloud pi

Alias: `pi`

Description: Manage IBM Cloud Power Virtual Server service.

Usage: `pi`

Available Commands:

- `available-hosts` : List of hosts available for reservation.
- `cloud-connection` : IBM Cloud Power Virtual Server Cloud Connections.

- `datacenter` : IBM Cloud Power Virtual Server Datacenters.
- `disaster-recovery` : List disaster recovery locations for the current region or all regions.
- `host` : IBM Cloud Power Virtual Server Host.
- `host-group` : IBM Cloud Power Virtual Server Host Group.
- `ike-policy` : [DEPRECATED] IBM Cloud Power Virtual Server Internet Key Exchange policies.
- `image` : IBM Cloud Power Virtual Server Images.
- `instance` : IBM Cloud Power Virtual Server Instances.
- `ipsec-policy` : [DEPRECATED] IBM Cloud Power Virtual Server Internet Protocol Security policies.
- `job` : IBM Cloud Power Virtual Server Jobs.
- `network-address-group` : IBM Cloud Power Virtual Server Network Address Groups.
- `network-interface` : IBM Cloud Power Virtual Server Network Interfaces.
- `network-security-group` : IBM Cloud Power Virtual Server Network Security Groups.
- `placement-group` : IBM Cloud Power Virtual Server Placement Groups.
- `route` : IBM Cloud Power Virtual Server Network Routes.
- `shared-processor-pool` : IBM Cloud Power Virtual Server Shared Processor Pools.
- `snapshot` : [DEPRECATED] IBM Cloud Power Virtual Server Snapshots.
- `ssh-key` : IBM Cloud Power Virtual Server SSH-Keys.
- `storage-pools` : List all storage pools for the targeted region.
- `storage-tiers` : List all storage tiers for the targeted region.
- `subnet` : IBM Cloud Power Virtual Server Subnets.
- `virtual-serial-number` : IBM Cloud Power Virtual Server Virtual Serial Number.
- `volume` : IBM Cloud Power Virtual Server Volumes.
- `volume-group` : IBM Cloud Power Virtual Server Volume Groups.
- `vpn` : [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking.
- `workspace` : IBM Cloud Power Virtual Server Workspaces.

ibmcloud pi available-hosts

Alias: `available-hosts`, `ahost`

Description: List of hosts available for reservation.

Usage: `available-hosts`

ibmcloud pi cloud-connection

Alias: `cloud-connection`, `cc`

Description: IBM Cloud Power Virtual Server Cloud Connections.

Usage: `cloud-connection`

Available Commands:

- `create` : [DEPRECATED] Create a cloud connection.
- `delete` : Delete a cloud connection.
- `get` : View details of a cloud connection.
- `list` : List all cloud connections.
- `subnet` : IBM Cloud Power Virtual Server Virtual Cloud Connection Subnets.
- `update` : Update a cloud connection.
- `vpcs` : List all virtual private clouds.

ibmcloud pi cloud-connection create

Alias: `create`, `cr`

Description: [DEPRECATED] Create a cloud connection.

Usage:

```
$ create CONNECTION_NAME --speed SPEED
  ([--classic=True|False [--subnets "SUBNET_ID1..SUBNET_IDn" --gre-tunnel "CIDR DEST-IP"]] | [--subnets "SUBNET_ID1..SUBNET_IDn"])
  [--global-routing=True|False] [--metered=True|False] [--vpc=True|False --vpcIDs "VPC-ID"]
pi cloud-connection create CONNECTION_NAME --speed SPEED --transit-enabled=True|False
  [--subnets "SUBNET_ID1..SUBNET_IDn"] [--global-routing=True|False] [--metered=True|False]

CONNECTION_NAME: The unique name of the cloud connection.
```

Available Options :

```
$ -c, --classic      Enable "Classic" cloud connection endpoint.
-g, --global-routing  Global routing option.
-t, --gre-tunnel string Space separated "cidr" and "destinationIPAddress". Use with "--classic" option.
                        GRE tunnel cannot be configured with speeds above 5000.
-m, --metered        Metered cloud connection option.
-s, --speed int       Speed of the cloud connection. Speed is in megabits per second.
                        Valid values are: 50, 100, 200, 500, 1000, 2000, 5000, 10000.
-n, --subnets strings  Comma separated subnet identifiers.
-r, --transit-enabled  Enable transit gateway.
-v, --vpc            Enable "VPC" cloud connection endpoint.
-p, --vpcIDs strings   VPC ID (i.e. crn:v1:..) to add to cloud connection. Use with "--vpc" option.
```

Examples:

```
$  ibmcloud pi cloud-connection create test-cloud-connection --speed 100 --classic
  ibmcloud pi cloud-connection test-cloud-connection --speed 500 --classic --gre-tunel "192.0.0.1 192.0.0.1/32" --subnets test-subnet
```

ibmcloud pi cloud-connection delete

Alias: delete, del

Description: Delete a cloud connection.

Usage:

```
$ delete CONNECTION_ID

CONNECTION_ID: The unique identifier of the cloud connection.
```

ibmcloud pi cloud-connection get

Alias: get

Description: View details of a cloud connection.

Usage:

```
$ get CONNECTION_ID

CONNECTION_ID: The unique identifier of the cloud connection.
```

ibmcloud pi cloud-connection list

Alias: list, ls

Description: List all cloud connections.

Usage: list

ibmcloud pi cloud-connection subnet

Alias: subnet, snet

Description: IBM Cloud Power Virtual Server Virtual Cloud Connection Subnets.

Usage: subnet

Available Commands:

- attach : Attach a subnet to a cloud connection.
- detach : Detach a subnet from a cloud connection.

ibmcloud pi cloud-connection subnet attach

Alias: attach, att

Description: Attach a subnet to a cloud connection.

Usage:

```
$ attach CONNECTION_ID --subnet SUBNET_ID
```

CONNECTION_ID: The unique identifier of the cloud connection.

Available Options:

```
$ -s, --subnet string Subnet ID to attach to the cloud connection.
```

ibmcloud pi cloud-connection subnet detach

Alias: detach, det

Description: Detach a subnet from a cloud connection.

Usage:

```
$ detach CONNECTION_ID --subnet ID
```

CONNECTION_ID: The unique identifier of the cloud connection.

Available Options:

```
$ -s, --subnet string Subnet ID to detach from the cloud connection.
```

ibmcloud pi cloud-connection update

Alias: update, upd

Description: Update a cloud connection.

Usage:

```
$ update CONNECTION_ID [--classic=True|False [--gre-tunnel "CIDR DEST-IP"]] [--global-routing=True|False]
  [--metered=True|False] [--name NAME] [--speed SPEED] [--vpc=True|False [<--vpcIDs "VPC-ID">]]
```

CONNECTION_ID: The unique identifier of the cloud connection.

Available Options:

```
$ -c, --classic      Enable "Classic" cloud connection endpoint.
-g, --global-routing Global routing option.
-t, --gre-tunnel string Space separated "cidr" and "destinationIPAddress". Use with "--classic" option.
                        GRE tunnel cannot be configured with speeds above 5000.
-m, --metered       Metered cloud connection option.
-n, --name string    Name of the cloud connection.
-s, --speed int      New speed value for the cloud connection.
                        Valid values are: 50, 100, 200, 500, 1000, 2000, 5000.
                        Speeds currently at 10000 cannot be downgraded lower and speeds cannot be increased to 10000.
-v, --vpc            Enable "VPC" cloud connection endpoint.
```


-p, --vpclDs strings VPC ID (i.e. crn:v1:...) to add to cloud connection. Use with "--vpc" option.

Examples:

```
$ ibmcloud pi cloud-connection update test-cloud-connection --classic --global-routing --gre-tunnel "192.0.0.1 192.0.0.1/32" --metered --name new-test-cloud-connection --speed 1000 --vpc --vpclDs test-vpc
```

ibmcloud pi cloud-connection vpcs

Alias: vpcs

Description: List all virtual private clouds.

Usage: vpcs

ibmcloud pi datacenter

Alias: datacenter, dat

Description: IBM Cloud Power Virtual Server Datacenters.

Usage: datacenter

Available Commands:

- get : View details of a datacenter.
- list : List all datacenter details.

ibmcloud pi datacenter get

Alias: get

Description: View details of a datacenter.

Usage:

```
$ get DATACENTER
```

DATACENTER: The name of the datacenter.

ibmcloud pi datacenter list

Alias: list, ls

Description: List all datacenter details.

Usage: list [--long=True|False]

Available Options:

```
$ -l, --long Retrieve additional details for datacenters.
```

ibmcloud pi disaster-recovery

Alias: disaster-recovery, drl

Description: List disaster recovery locations for the current region or all regions.

Usage: disaster-recovery [--all-regions=True|False]

Available Options:

```
$ -a, --all-regions List disaster recovery locations for all regions.
```

ibmcloud pi host

Alias: `host`, `hs`

Description: IBM Cloud Power Virtual Server Host.

Usage: `host`

Available Commands:

- `get` : View details of a host.
- `list` : List all hosts.
- `release` : Release a host from a host group.
- `reserve` : Reserve a host.
- `update` : Update a host.

ibmcloud pi host get

Alias: `get`

Description: View details of a host.

Usage:

```
$ get HOST_ID
```

HOST_ID: The unique identifier of the host.

ibmcloud pi host list

Alias: `list`, `ls`

Description: List all hosts.

Usage: `list`

ibmcloud pi host release

Alias: `release`, `rel`

Description: Release a host from a host group.

Usage:

```
$ release HOST_ID
```

HOST_ID: The unique identifier of the host.

ibmcloud pi host reserve

Alias: `reserve`, `res`

Description: Reserve a host.

Usage: `reserve "DISPLAYNAME1 SYSTYPE1"["DISPLAYNAMEn SYSTYPEn"] --host-group HOST_GROUP_ID [--user-tags USER_TAG1[,USER_TAGn]]`

Available Options:

```
$ -g, --host-group string  Host group to add the host to.
-u, --user-tags strings    Comma separated list of user tags to be attached to the host.
```

Examples:

```
$ ibmcloud pi host reserve "testHost s922" --host-group 43064761-948f-469d-ac8e-b8e5f0d6056f --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi host update

Alias: `update`, `upd`

Description: Update a host.

Usage:

```
$ update HOST_ID --display-name NAME
```

HOST_ID: The unique identifier of the host.

Available Options:

```
$ -n, --display-name string  New name of the host.
```

ibmcloud pi host-group

Alias: `host-group`, `hg`

Description: IBM Cloud Power Virtual Server Host Group.

Usage: `host-group`

Available Commands:

- `create` : Create a host group.
- `get` : View details of a host group.
- `list` : List all host groups.
- `update` : Update a host group.

ibmcloud pi host-group create

Alias: `create`, `cr`

Description: Create a host group.

Usage:

```
$ create HOST_GROUP_NAME <--hosts "DISPLAYNAME SYSTYPE [USER_TAG1,[USER_TAGn]]"> [--secondaries "WORKSPACE_ID1 [HOST_GROUP_NAME1]"[, "WORKSPACE_IDn [HOST_GROUP_NAMEn]]
```

HOST_GROUP_NAME: The desired name of the host group.

Available Options:

```
$ --hosts stringArray  List of hosts to be included in the host group. Hosts must include a display name and system type.  
-s, --secondaries strings  List of secondaries to be included in the host group. Secondaries must include a workspace ID and optionally a name for the host group.
```

Examples:

```
$ ibmcloud pi host-group create test-host-group --hosts "/v1/hosts/421 s922 project:customer-poc,env:dev,dataresidency:germany" --hosts "/v1/hosts/422 s922 project:customer-poc,env:test,dataresidency:france"
```

ibmcloud pi host-group get

Alias: `get`

Description: View details of a host group.

Usage:

```
$ get HOST_GROUP_ID
```

HOST_GROUP_ID: The unique identifier of the host group.

ibmcloud pi host-group list

Alias: list, ls

Description: List all host groups.

Usage: list

ibmcloud pi host-group update

Alias: update, upd

Description: Update a host group.

Usage:

```
$ update HOST_GROUP_ID [--add-workspaces "WORKSPACE_ID1 [HOST_GROUP_NAME1]"[, "WORKSPACE_IDn [HOST_GROUP_NAMEn]] | [--remove-workspace WORKSPACE_ID])
```

HOST_GROUP_ID: The unique identifier of the host group.

Available Options:

```
$ -a, --add-workspaces strings  List of workspace IDs to share the host group with. Input must be a workspace ID and optionally a name for the host group separated by a space. Multiple workspace additions are separated by commas.
-r, --remove-workspace string  A workspace ID to stop sharing the host group with.
```

ibmcloud pi ike-policy

Alias: ike-policy, ike

Description: [DEPRECATED] IBM Cloud Power Virtual Server Internet Key Exchange policies.

Usage: ike-policy

Available Commands:

- `create`: [DEPRECATED] Create a VPN IKE policy.
 - `delete`: [DEPRECATED] Delete a VPN IKE policy.
 - `get`: [DEPRECATED] View details of a VPN IKE policy.
 - `list`: [DEPRECATED] List all VPN IKE policies.
 - `update`: [DEPRECATED] Update a VPN IKE policy.
-

ibmcloud pi ike-policy create

Alias: create, cr

Description: [DEPRECATED] Create a VPN IKE policy.

Usage:

```
$ create IKE_POLICY_NAME --authentication AUTHENTICATION --dh-group DH_GROUP --encryption ENCRYPTION --key-lifetime SECONDS --preshared-key KEY --version VERSION
```

IKE_POLICY_NAME: A unique name of the VPN IKE policy. The maximum name length is 47 characters.

Available Options:

```
$ -a, --authentication string  Authentication algorithm of the IKE policy. Valid values are: "sha-256", "sha-384", "sha1", "none".
-d, --dh-group int            DH group number of the IKE policy. Valid values are: "1", "2", "5", "14", "19", "20", "24".
-e, --encryption string       Encryption algorithm of the IKE policy.
                               Valid values are: "aes-256-cbc", "aes-192-cbc", "aes-128-cbc", "aes-256-gcm", "aes-128-gcm", "3des-cbc".
                               When using "aes-128-gcm" or "aes-256-gcm" authentication should be set to "none".
-k, --key-lifetime int         Key lifetime of the IKE policy in seconds. Valid range is 180 to 86400 seconds.
-p, --preshared-key string     Preshared key used in this VPN connection. The key length must be even.
```

-v, --version int	Version number of the IKE policy. Valid values are: "1", "2".
-------------------	---

Examples:

```
$ ibmcloud pi ike-policy create test-ike-policy --authentication sha-256 --encryption aes-256-cbc --dh-group 2 --key-lifetime 190 --preshared-key sample-key --version 2
```

ibmcloud pi ike-policy delete

Alias: delete, del

Description: [DEPRECATED] Delete a VPN IKE policy.

Usage:

```
$ delete IKE_POLICY_ID

IKE_POLICY_ID: The unique identifier of the VPN IKE policy.
```

ibmcloud pi ike-policy get

Alias: get

Description: [DEPRECATED] View details of a VPN IKE policy.

Usage:

```
$ get IKE_POLICY_ID

IKE_POLICY_ID: The unique identifier of the VPN IKE policy.
```

ibmcloud pi ike-policy list

Alias: list, ls

Description: [DEPRECATED] List all VPN IKE policies.

Usage: list

ibmcloud pi ike-policy update

Alias: update, upd

Description: [DEPRECATED] Update a VPN IKE policy.

Usage:

```
$ update IKE_POLICY_ID [--authentication AUTHENTICATION] [--dh-group DH_GROUP] [--encryption ENCRYPTION] [--key-lifetime SECONDS] [--name NAME] [--preshared-key KEY] [--version VERSION]

IKE_POLICY_ID: The unique identifier of the VPN IKE policy.
```

Available Options:

```
$ -a, --authentication string  Authentication algorithm of the IKE policy. Valid values are: "sha-256", "sha-384", "sha1", "none".
-d, --dh-group int            DH group number of the IKE policy. Valid values are: "1", "2", "5", "14", "19", "20", "24".
-e, --encryption string       Encryption algorithm of the IKE policy.
                               Valid values are: "aes-256-cbc", "aes-192-cbc", "aes-128-cbc", "aes-256-gcm", "aes-128-gcm", "3des-cbc".
                               When using "aes-128-gcm" or "aes-256-gcm" authentication should be set to "none".
-k, --key-lifetime int        Key lifetime of the IKE policy in seconds. Valid range is 180 to 86400 seconds.
-n, --name string             New unique name of the IKE Policy. The maximum name length is 47 characters.
-p, --preshared-key string    Preshared key used in this VPN connection. The key length must be even.
-v, --version int            Version number of the IKE policy. Valid values are: "1", "2".
```

ibmcloud pi image

Alias: image, img

Description: IBM Cloud Power Virtual Server Images.

Usage: image

Available Commands:

- create : Create a copy of an available stock image into this account; stock image names cannot be changed.
- delete : Delete an image.
- export : Export an image to IBM Cloud Object Storage.
- export-show : View details of the last image export job.
- get : View details of an image.
- import : Import an image from IBM Cloud Object Storage.
- import-show : View details of the last image import job.
- list : List all images in a workspace.
- list-catalog : List images available in the regional image catalog.

ibmcloud pi image create

Alias: create, cr

Description: Create a copy of an available stock image into this account; stock image names cannot be changed.

Usage:

```
$ create IMAGE_ID [--user-tags USER_TAG1[,USER_TAGn]]
```

IMAGE_ID: The unique identifier or name of the image.

Available Options:

```
$ -u, --user-tags strings  Comma separated list of user tags to be attached to the created image.
```

Examples:

```
$ ibmcloud pi image create 7100-05-05 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi image delete

Alias: delete, del

Description: Delete an image.

Usage:

```
$ delete IMAGE_ID
```

IMAGE_ID: The unique identifier or name of the image.

ibmcloud pi image export

Alias: export, ex

Description: Export an image to IBM Cloud Object Storage.

Usage:

```
$ export IMAGE_ID --access-key KEY --bucket BUCKET_NAME --region REGION_NAME --secret-key KEY
```

IMAGE_ID: The unique identifier or name of the image.

Available Options:


```
$ -a, --access-key string    Cloud Object Storage HMAC access key.
-b, --bucket string         Cloud Object Storage bucket name.
-r, --region string         Cloud Object Storage region au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
-s, --secret-key string     Cloud Object Storage HMAC secret key.
```

Examples:

```
$ ibmcloud pi image export 43064761-948f-469d-ac8e-b8e5f0d6056f --bucket "bucket/stock-images" --region us-east --access-key access-key-sample --secret-key secret-key-sample
```

ibmcloud pi image export-show

Alias: export-show, exs

Description: View details of the last image export job.

Usage:

```
$ export-show IMAGE_ID

IMAGE_ID: The unique identifier or name of the image.
```

ibmcloud pi image get

Alias: get

Description: View details of an image.

Usage:

```
$ get IMAGE_ID

IMAGE_ID: The unique identifier or name of the image.
```

ibmcloud pi image import

Alias: import, im

Description: Import an image from IBM Cloud Object Storage.

Usage:

```
$ import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --storage-pool POOL [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --affinity-policy affinity (--affinity-instance INSTANCE | --affinity-volume VOLUME) [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --affinity-policy anti-affinity (--anti-affinity-instances "INSTANCE1 [INSTANCEn]" | --anti-affinity-volumes "VOLUME1 [VOLUMEn]") [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --storage-pool POOL [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --affinity-policy affinity (--affinity-instance INSTANCE | --affinity-volume VOLUME) [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --affinity-policy anti-affinity (--anti-affinity-instances "INSTANCE1 [INSTANCEn]" | --anti-affinity-volumes "VOLUME1 [VOLUMEn]") [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME

IMAGE_NAME: The desired name of the image.
```

Available Options:

\$ -k, --access-key string	Cloud Object Storage HMAC access key.
-i, --affinity-instance string	PVM instance identifier or name to base volume affinity policy against. This is required if "--affinity-policy affinity" is specified and "--affinity-volume" is not provided.
-a, --affinity-policy string	Affinity policy for data volume being created. Valid values are: "affinity" and "anti-affinity". If "--storage-pool" is provided then this cannot be specified.
-v, --affinity-volume string	Volume identifier or name to base volume affinity policy against. This is required if "--affinity-policy affinity" is specified and "--affinity-instance" is not provided.
-j, --anti-affinity-instances strings	Comma separated list of instance identifiers or names to base volume anti-affinity policy against. This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-volumes" is not provided.
-w, --anti-affinity-volumes strings	Comma separated list of volume identifiers or names to base volume anti-affinity policy against. This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-instances" is not provided.
-b, --bucket string	Cloud Object Storage bucket name.
-u, --bucket-access string	Indicates the bucket access type. Valid values are: "public", "private". Private access requires access and secret keys. Default is "private".
-n, --image-file-name string	The image file name.
-d, --import-details strings	Import details for SAP image. Must include a license, product and vendor. Valid license values: byol. Valid product values: Hana, Netweaver. Valid vendor values: SAP.
-o, --os-type string	Operating system contained in the image. Valid values are: "aix", "ibmi", "rhel", "sles". Required when importing a raw image.
-r, --region string	Cloud Object Storage region au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
-s, --secret-key string	Cloud Object Storage HMAC secret key.
-p, --storage-pool string	Storage pool where the image will be imported to. Use "ibmcloud pi storage-pools" to see available storage pools. If "--storage-pool" is provided then "--affinity-policy" cannot be specified.
-t, --storage-tier string	Tier of the disk storage. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region. Default to tier3 if not provided.
--user-tags strings	Comma separated list of user tags to be attached to the imported image.

Examples:

```
$ ibmcloud pi image import import-image-name --bucket-access public --image-file-name imageFileName.ova --bucket test-bucket --region us-east
ibmcloud pi image import import-image-name --bucket-access private --storage-tier tier1 --image-file-name imageFileName.ova --bucket test-bucket --region us-east --
access-key access-key-sample --secret-key secret-key-sample
```

ibmcloud pi image import-show

Alias: import-show, ims

Description: View details of the last image import job.

Usage: import-show

ibmcloud pi image list

Alias: list, ls

Description: List all images in a workspace.

Usage: list

ibmcloud pi image list-catalog

Alias: list-catalog, lc

Description: List images available in the regional image catalog.

Usage: list-catalog

Examples:

```
$ ibmcloud pi image list-catalog --sap
```

ibmcloud pi instance

Alias: instance, ins

Description: IBM Cloud Power Virtual Server Instances.

Usage: instance

Available Commands:

- action : Perform an operation in a PVM server instance.
- capture : IBM Cloud Power Virtual Server Instance Capture.
- console : IBM Cloud Power Virtual Server Instance Console.
- create : Create a server instance.
- delete : Delete a server instance.
- get : View details of a server instance.
- list : List all server instances.
- operation : Perform an operation on an IBMi server instance.
- sap : IBM Cloud Power Virtual Server Instance SAP.
- snapshot : IBM Cloud Power Virtual Server Instance Snapshots.
- subnet : IBM Cloud Power Virtual Server Instance Subnets.
- update : Update a server instance.
- virtual-serial-number : IBM Cloud Power Virtual Server Instance Virtual Serial Number.
- volume : IBM Cloud Power Virtual Server Instance Volumes.
- vpmem-volume : IBM Cloud Power Virtual Server Instance vPMEM Volumes.

ibmcloud pi instance action

Alias: action, act

Description: Perform an operation in a PVM server instance.

Usage:

```
$ action INSTANCE_ID --operation OPERATION
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -o, --operation string  Operation to be done in a PVM server instance.
                        Valid values are: dumprestart, hard-reboot, immediate-shutdown, reset-state, soft-reboot, start, stop.
```

Examples:

```
$  ibmcloud pi instance action 43064761-948f-469d-ac8e-b8e5f0d6056f --operation immediate-shutdown
```

ibmcloud pi instance capture

Alias: capture, cap

Description: IBM Cloud Power Virtual Server Instance Capture.

Usage: capture

Available Commands:

- create : Create a capture of a server instance.
- show : View the job details of the last server instance capture.

ibmcloud pi instance capture create

Alias: create, cr

Description: Create a capture of a server instance.

Usage:

```
$ create INSTANCE_ID --destination (cloud-storage | both) --name NAME --access-key KEY --image-path PATH --region REGION --secret-key KEY [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEn]]
pi instance capture create INSTANCE_ID --destination image-catalog --name NAME [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEn]]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -a, --access-key string    Cloud Object Storage HMAC access key. Required if destination is cloud-storage.
-d, --destination string    Destination for the deployable image. Valid values are: "cloud-storage", "image-catalog", "both".
-i, --image-path string     Cloud Object Storage image path. Required if destination is cloud-storage. E.g. bucket-name[optional/folder].
-n, --name string           Name of the deployable image created for the captured instance.
-r, --region string         Cloud Object Storage regions au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
                             Required if destination is cloud-storage.
-s, --secret-key string     Cloud Object Storage HMAC secret key. Required if destination is cloud-storage.
-u, --user-tags strings     Comma separated list of user tags to be attached to the instance capture.
-v, --volumes strings       Comma separated list of volume identifiers or names to include in the captured instance.
```

Examples:

```
$ ibmcloud pi instance capture create 43064761-948f-469d-ac8e-b8e5f0d6056f --destination both --name deployable-image-name --access-key access-key-sample --secret-key secret-key-sample --region us-south --image-path "imageBucket/imageLocation"
```

ibmcloud pi instance capture show

Alias: show, sh

Description: View the job details of the last server instance capture.

Usage:

```
$ show INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance console

Alias: console, con

Description: IBM Cloud Power Virtual Server Instance Console.

Usage: console

Available Commands:

- get: Get the console of an instance.
- list: List the available console languages for an IBMi instance.
- update: Update the console language of an instance. This update may take some time to take affect.

ibmcloud pi instance console get

Alias: get

Description: Get the console of an instance.

Usage:

```
$ get INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance console list

Alias: list, ls

Description: List the available console languages for an IBMi instance.

Usage:

```
$ list INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance console update

Alias: update, upd

Description: Update the console language of an instance. This update may take some time to take affect.

Usage:

```
$ update INSTANCE_ID --code CODE

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -c, --code string Language code to set. Use "ibmcloud pi instance console list" to see available codes.
```

Examples:

```
$ ibmcloud pi instance console update 43064761-948f-469d-ac8e-b8e5f0d6056f --code e1399
```

ibmcloud pi instance create

Alias: create, cr

Description: Create a server instance.

Usage:

```
$ create INSTANCE_NAME --image IMAGE --subnets "SUBNET1 [IP1] [NSG]"[, "SUBNETn [IPn] [NSGn]"]
  [--allow-remote-restart=True|False] [--boot-volume-replication-enabled=True|False]
  [--default-trusted-profile (CRN | TRUSTED_PROFILE_ID | TRUSTED_PROFILE_NAME)
  [--default-trusted-profile-autolink=True|False]]
  [--deployment-target ("HOST_GROUP_ID,hostGroup" | "HOST_ID,host")] [--deployment-type DEPLOYMENT_TYPE]
  [--IBMiCSS-license=True|False] [--IBMiPHA-license=True|False] [--IBMiRDS-users NUMBER_USERS]
  [--key-name NAME] [--memory MEMORY] [--metadata-service=True|False] [--pin-policy POLICY] [--placement-group GROUP_ID]
  [--preferred-processor-compatibility-mode MODE] [--processor-type PROC_TYPE] [--processors PROCESSORS]
  [--replicant-affinity-policy AFFINITY_POLICY] [--replicant-scheme SCHEME] [--replicants NUMBER]
  [--replication-sites SITE1[,SITE n]] [--shared-processor-pool SHARED_PROCESSOR_POOL] [--storage-affinity STORAGE_AFFINITY_POLICY]
  [--storage-affinity-instance INSTANCE] [--storage-affinity-volume VOLUME]
  [--storage-anti-affinity-instances INSTANCE1[,INSTANCE n]] [--storage-anti-affinity-volumes VOLUME1[,VOLUME n]]
  [--storage-connection STORAGE_CONNECTION] [--storage-pool STORAGE_POOL] [--storage-pool-affinity=True|False]
  [--storage-tier STORAGE_TIER] [--sys-type TYPE] [--user-data USER_DATA] [--user-tags USER_TAG1[,USER_TAG n]]
  [--virtual-serial-number "(SERIAL | 'auto-assign')[,DESCRIPTION]" [--software-tier SOFTWARE_TIER]]
  [--virtual-cores ASSIGNED_CORES] [--volumes VOLUME1[,VOLUME n]] [--vpmem-volumes "NAME SIZE[, "NAME n SIZE n"]]]

INSTANCE_NAME: The name of the instance.
```

Available Options:

\$ --IBMiCSS-license	IBMi CSS software license associated with the instance.
--IBMiPHA-license	IBMi PHA IASP Management software license associated with the instance.
--IBMiRDS-users int	Number of IBMi RDS user software licenses associated with the instance. Default is 0.
--allow-remote-restart	Enables automated remote restart for the instance.
-b, --boot-volume-replication-enabled	Enables storage replication on the boot volume. Default is "false".
--default-trusted-profile string	Trusted profile CRN, ID or name of the default IAM trusted profile to use for this instance.

-a, --default-trusted-profile-autolink	If set to true, the system will create a link to the specified trusted profile during server creation.
--deployment-target string	The deployment of the dedicated host. Deployment must include either a host group id and "hostGroup", or a host id and "host".
-d, --deployment-type string	The custom deployment type. Valid values are: "EPIC" or "VMNoStorage". If '--deployment-type "VMNoStorage"' is set, then "--image" must be one of "AIX-EMPTY", "IBMI-EMPTY", "RHEL-EMPTY", or "SLES-EMPTY".
-i, --image string	Operating system image identifier or name.
-k, --key-name string	Name of SSH key.
-m, --memory float	Amount of memory in GiB to allocate to the instance. Default is 2 GiB.
--metadata-service	Indicates whether the metadata service endpoint will be available to the virtual server.
--pin-policy string	Pin policy. Valid values are: "none", "soft", "hard". Default is "none".
--placement-group string	The placement group ID of the group that the server will be added to.
-c, --preferred-processor-compatibility-mode string	The preferred processor compatibility mode. Valid values are: default, POWER7, POWER8, POWER9, POWER9_Base, POWER10, POWER11
-r, --processor-type string	Type of processors: "shared" or "dedicated" or "capped". Default is "dedicated".
-p, --processors float	Amount of processors to allocate to the instance. Default is 1 core.
--replicant-affinity-policy string	Affinity policy to use when multicreate is used. Valid values are: affinity, anti-affinity, none.
--replicant-scheme string	Naming scheme to use for duplicate VMs. Valid values are: prefix, suffix.
--replicants int	The number of replicant instances to be created with this request. Values greater than 1 will result in the creation of multiple instances.
--replication-sites strings	Indicates the replication sites of the boot volume. See "ibmcloud pi disaster-recovery" command to get a list of replication sites.
--shared-processor-pool string	The shared processor pool ID of the pool that the server will be in.
--software-tier string	IBMi licensing software tiers. Must be set with "--virtual-serial-number" option. Valid values are: P05, P10, P20, P30.
--storage-affinity string	Affinity policy for storage pool selection. Valid values are: "affinity" and "anti-affinity". If "--storage-pool" is provided, then this option cannot be specified.
--storage-affinity-instance string	PVM instance identifier or name to base storage affinity policy against. This is required if "--storage-affinity affinity" is specified and "--storage-affinity-volume" is not provided.
--storage-affinity-volume string	Volume identifier or name to base storage affinity policy against. This is required if "--storage-affinity affinity" is specified and "--storage-affinity-instance" is not provided.
--storage-anti-affinity-instances strings	Comma separated list of PVM instance identifiers or names to base storage affinity policy against. This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-volumes" is not provided.
--storage-anti-affinity-volumes strings	Comma separated list of volume identifiers or names to base storage affinity policy against. This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-instances" is not provided.
--storage-connection string	The storage connection type. Valid values are: maxVolumeSupport.
--storage-pool string	Storage pool for server deployment (use "ibmcloud pi storage-pools" to see available storage pools). Only valid when you deploy one of the IBM supplied stock images. Storage tier and pool for a custom image (an imported image or an image that is created from a PVMInstance capture) defaults to the storage tier and pool the image was created in.
--storage-pool-affinity	Indicates if all volumes attached to the server must reside in the same storage pool. If set to false, then volumes from any storage tier and pool can be attached to the PVM instance. This setting affects PVM instance snapshot, capture, and clone operations. For capture and clone operations, only data volumes that are of the same storage tier and in the same storage pool of the boot volume of the PVM instance can be included. For snapshot operations, all data volumes to be included in the snapshot must reside in the same storage tier and storage pool. Once set to false, cannot be set back to true unless all volumes attached reside in the same storage tier and pool.
-t, --storage-tier string	Storage tier for server deployment when deploying a stock or custom image (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region). Default to tier3 if not provided.
-n, --subnets strings	Comma separated list of subnet identifiers or names, an optional IP address, and network security group id that the network interface is a member of to associate with the instance. If no network group security group id is specified, default one is used. Example: --subnets "subnet-name1 192.168.0.1 nsg-id,subnet-name2 192.168.0.2,subnet-name3"
-s, --sys-type string	System type used to host the instance. Valid values are: s922, e980, s1022, e1080, s1122, e1150, e1180.
-u, --user-data string	The user data passed into the instance. Strings and file names are supported. File names must be prepended with "@".
--user-tags strings	Comma separated list of user tags to be attached to the instance.
--virtual-cores int	The number of virtual cores assigned.
--virtual-serial-number string	IBMi virtual serial number information added with the instance. Must include an existing virtual serial number or 'auto-assign' and optionally a description.
-v, --volumes strings	Comma separated list of volume identifiers or names to associate with the instance.
--vpmem-volumes strings	Comma separated list of vPMEM volume names and sizes to associate with the instance. The maximum number of vPMEM volumes for SAP instances is 1 otherwise it is 4.

Examples:

```
$  ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d
  ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f --storage-tier tier1 --processors 1 --processor-type dedicated --deployment-type EPIC --sys-type e980
  ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --storage-tier tier1 --replicants 2 --replicant-scheme prefix --replicant-affinity-policy affinity
  ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --boot-volume-replication-enabled --replication-sites dal10
```

ibmcloud pi instance delete

Alias: delete, del

Description: Delete a server instance.

Usage:

```
$ delete INSTANCE_ID [--delete-data-volumes=True|False] [--retainVSN=True|False]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -d, --delete-data-volumes  Indicates whether all data volumes attached to the instance must be deleted.
                             Shared data volumes will be deleted if no other instances are attached.
-r, --retainVSN              Determines if the virtual serial number will be retained after being removed from the instance. Default is "false".
```

ibmcloud pi instance get

Alias: get

Description: View details of a server instance.

Usage:

```
$ get INSTANCE_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance list

Alias: list, ls

Description: List all server instances.

Usage: list

ibmcloud pi instance operation

Alias: operation, op

Description: Perform an operation on an IBMi server instance.

Usage:

```
$ operation INSTANCE_ID (--operation-type "boot" --boot-mode MODE --boot-operating-mode MODE | --operation-type "job" --job-task TASK)
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -b, --boot-mode string      Name of the server boot mode. Valid values are: a, b, c, d.
-m, --boot-operating-mode string  Name of the server operating mode. Valid values are: manual, normal.
-j, --job-task string         Name of the job task to execute.
                             Valid values are: consoleservice, dston, dumprestart, iopdump, iopreset, remotestoff, remoteston, retrydump.
-o, --operation-type string    Name of the operation to execute. Valid values are: boot, job.
```

Examples:

```
$ ibmcloud pi instance operation 43064761-948f-469d-ac8e-b8e5f0d6056f --operation-type boot --boot-mode c --boot-operating-mode normal
```

ibmcloud pi instance sap

Alias: sap

Description: IBM Cloud Power Virtual Server Instance SAP.

Usage: sap

Available Commands:

- create : Create a new SAP PVM Instance. This command is for use with Linux for SAP (HANA) images.
- list : List all SAP profiles for the targeted region.
- profile : Get the information on a SAP profile.

ibmcloud pi instance sap create

Alias: create, cr

Description: Create a new SAP PVM Instance. This command is for use with Linux for SAP (HANA) images.

Usage:

```
$ create SAP_INSTANCE_NAME --image IMAGE --profile-id PROFILE_ID --subnets "SUBNET1 [IP1] [NSG]"[, "SUBNETn [IPn] [NSGn]"]
[--allow-remote-restart=True|False] [--boot-volume-replication-enabled=True|False]
[--default-trusted-profile (CRN | TRUSTED_PROFILE_ID | TRUSTED_PROFILE_NAME)
[--default-trusted-profile-autolink=True|False]]
[--deployment-target ("HOST_GROUP_ID,hostGroup" | "HOST_ID,host")] [--key-name KEY-NAME] [--metadata-service=True|False]
[--pin-policy POLICY] [--placement-group PLACEMENT_GROUP_ID] [--preferred-processor-compatibility-mode MODE]
[--replication-sites SITE1[,SITE n]] [--storage-affinity STORAGE_AFFINITY_POLICY]
[--storage-affinity-instance INSTANCE] [--storage-affinity-volume VOLUME] [--storage-anti-affinity-instances INSTANCE1[,INSTANCEn]]
[--storage-anti-affinity-volumes VOLUME1[,VOLUMEn]] [--storage-pool STORAGE_POOL] [--storage-tier STORAGE_TIER]
[--sys-type TYPE] [--user-data USER_DATA] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEn]]
[--vpmem-volumes "NAME SIZE[,NAME n SIZE n]]
```

SAP_INSTANCE_NAME: The name of the SAP instance.

Available Options:

\$	--allow-remote-restart	Enables automated remote restart for the instance.
-b,	--boot-volume-replication-enabled	Enables storage replication on the boot volume. Default is "false".
	--default-trusted-profile string	Trusted profile CRN, ID or name of the default IAM trusted profile to use for this instance.
-a,	--default-trusted-profile-autolink	If set to true, the system will create a link to the specified trusted profile during server creation.
	--deployment-target string	The deployment of the dedicated host. Deployment must include either a host group id and "hostGroup", or a host id and "host".
-i,	--image string	Operating system image identifier or name.
-k,	--key-name string	Name of SSH key.
	--metadata-service	Indicates whether the metadata service endpoint will be available to the virtual server.
	--pin-policy string	Pin policy. Valid values are: "none", "soft", "hard". Default is "none".
	--placement-group string	The placement group ID of the group that the server will be added to.
-c,	--preferred-processor-compatibility-mode string	The preferred processor compatibility mode. Valid values are: default, POWER7, POWER8, POWER9, POWER10, POWER11
-p,	--profile-id string	The unique identifier of the SAP profile.
	--replication-sites strings	Indicates the replication sites of the boot volume. See "ibmcloud pi disaster-recovery" command to get a list of replication sites.
	--storage-affinity string	Affinity policy for storage pool selection. Valid values are: "affinity" and "anti-affinity".
		If "--storage-pool" is provided, then this option cannot be specified.
	--storage-affinity-instance string	PVM instance identifier or name to base storage affinity policy against.
		This is required if "--storage-affinity affinity" is specified and "--storage-affinity-volume" is not provided.
	--storage-affinity-volume string	Volume identifier or name to base storage affinity policy against.
		This is required if "--storage-affinity affinity" is specified and "--storage-affinity-instance" is not provided.
	--storage-anti-affinity-instances strings	Comma separated list of PVM instance identifiers or names to base storage affinity policy against.
		This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-volumes" is not provided.
	--storage-anti-affinity-volumes strings	Comma separated list of volume identifiers or names to base storage affinity policy against.
		This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-instances" is not provided.
	--storage-pool string	Storage pool for SAP PVM instance deployment. Only valid when you deploy one of the IBM supplied stock images.
-t,	--storage-tier string	Storage tiers for SAP PVM instance deployment when deploying a stock or custom image
		Use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region.
		Default to tier3 if not provided.
-n,	--subnets strings	Comma separated list of subnet identifiers or names, an optional IP address, and network security group id that the network interface is a member of to associate with the instance.
		If no network group security group id is specified, default one is used.
-s,	--sys-type string	System type used to host the instance. Valid values are: e980, e1080.

-u, --user-data string	The user data passed into the instance. Strings and file names are supported. File names must be prepended with "@".
--user-tags strings	Comma separated list of user tags to be attached to the instance.
-v, --volumes strings	Comma separated list of volume identifiers or names to associate with the instance.
--vpmem-volumes strings	Comma separated list of vPMEM volume names and sizes to associate with the instance.
The maximum number of vPMEM volumes for SAP instances is 1.	

Examples:

```
$ ibmcloud pi instance sap create test-sap-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --profile-id test-profile-id
    ibmcloud pi instance sap create test-sap-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --profile-id test-profile-id --storage-tier tier1 --storage-affinity anti-affinity --storage-anti-affinity-volumes 6a8b77c3-db9d-29d5-be1c-a59f09dc6b55
    ibmcloud pi instance sap create test-sap-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --profile-id test-profile-id --storage-tier tier3 --boot-volume-replication-enabled --replication-sites dal10
```

ibmcloud pi instance sap list

Alias: list, ls

Description: List all SAP profiles for the targeted region.

Usage: list [--family FAMILY] [--prefix PREFIX]

Available Options:

```
$ -f, --family string  SAP profile family to filter by. Valid values are: balanced, compute, memory, sap-rise, sap-rise-app, small, ultra-memory.
-p, --prefix string   SAP profile prefix to filter by. Valid values are: bh1, bh2, ch1, ch2, mh1, mh2, umh, ush1, sh2, sr2, sr3.
```

ibmcloud pi instance sap profile

Alias: profile, prof

Description: Get the information on a SAP profile.

Usage:

```
$ profile SAP_PROFILE_ID

SAP_PROFILE_ID: The unique identifier or name of the SAP profile.
```

ibmcloud pi instance snapshot

Alias: snapshot, snap

Description: IBM Cloud Power Virtual Server Instance Snapshots.

Usage: snapshot

Available Commands:

- create : Create a snapshot for an instance.
- delete : Delete a snapshot from an instance.
- get : Get a snapshot for an instance
- list : List all snapshots for an instance or all snapshots in the workspace.
- restore : Restore a snapshot for an instance.
- update : Update a snapshot for an instance.

ibmcloud pi instance snapshot create

Alias: create, cr

Description: Create a snapshot for an instance.

Usage:

```
$ create INSTANCE_ID --name SNAPSHOT_NAME [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEen]]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

```
$ -d, --description string  Snapshot description.
-n, --name string          Name of the snapshot.
-u, --user-tags strings    Comma separated list of user tags to be attached to the snapshot.
-v, --volumes strings      Comma separated list of volume identifiers or names to include in the snapshot.
                           If you do not specify this option or if the volumes list is empty,
                           all the volumes that are attached to the instance are included in the snapshot.
```

Examples:

```
$ ibmcloud pi instance snapshot create 85716e61-948f-309d-de8b-c4e5f0d3126d --name test-snapshot-name
```

ibmcloud pi instance snapshot delete

Alias: delete, del

Description: Delete a snapshot from an instance.

Usage:

```
$ delete SNAPSHOT_ID
```

SNAPSHOT_ID: The unique identifier of the snapshot.

ibmcloud pi instance snapshot get

Alias: get

Description: Get a snapshot for an instance

Usage:

```
$ get SNAPSHOT_ID
```

SNAPSHOT_ID: The unique identifier of the snapshot.

ibmcloud pi instance snapshot list

Alias: list, ls

Description: List all snapshots for an instance or all snapshots in the workspace.

Usage:

```
$ list [INSTANCE_ID]
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance snapshot restore

Alias: restore, res

Description: Restore a snapshot for an instance.

Usage:

```
$ restore INSTANCE_ID --snapshot SNAPSHOT_ID [--force] [--restore VALUE]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -f, --force          By default the VM must be shutoff during a snapshot restore, force set to true will relax the VM shutoff pre-condition.
-r, --restore string   Action to take on a failed snapshot restore. Valid values for "--restore" are: retry, rollback.
-s, --snapshot string  The unique identifier of the snapshot.
```

Examples:

```
$ ibmcloud pi instance snapshot restore 85716e61-948f-309d-de8b-c4e5f0d3126d --snapshot 43064761-948f-469d-ac8e-b8e5f0d6056f --force --restore retry
```

ibmcloud pi instance snapshot update

Alias: `update`, `upd`

Description: Update a snapshot for an instance.

Usage:

```
$ update SNAPSHOT_ID [--description DESCRIPTION] [--name NAME]

SNAPSHOT_ID: The unique identifier of the snapshot.
```

Available Options:

```
$ -d, --description string  New snapshot description.
-n, --name string          New name of the snapshot.
```

ibmcloud pi instance subnet

Alias: `subnet`, `snet`

Description: IBM Cloud Power Virtual Server Instance Subnets.

Usage: `subnet`

Available Commands:

- `attach` : Attach a subnet to a server instance.
- `detach` : Detach a specific subnet or mac address from a server instance.
- `list` : List all the attached subnets.

ibmcloud pi instance subnet attach

Alias: `attach`, `att`

Description: Attach a subnet to a server instance.

Usage:

```
$ attach INSTANCE_ID --subnet SUBNET_ID [--ip-address IP_ADDRESS] [--network-security-group-id NETWORK_SECURITY_GROUP_ID]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -i, --ip-address string      The requested ip address of this subnet interface (192.168.1.0).
    --network-security-group-id string  The network security group that the network interface is a member of.
                                         If not specified, the default network security group is used.
-n, --subnet string           The subnet ID.
```

Examples:

```
$ ibmcloud pi instance subnet attach 85716e61-948f-309d-de8b-c4e5f0d3126d --subnet 43064761-948f-469d-ac8e-b8e5f0d6056f --ip-address 192.168.0.12
```


ibmcloud pi instance subnet detach

Alias: detach, det

Description: Detach a specific subnet or mac address from a server instance.

Usage:

```
$ detach INSTANCE_ID --subnet SUBNET_ID [--mac-address MAC_ADDRESS]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -m, --mac-address string  The mac address of the subnet interface to be removed. Default is all mac addresses.
-n, --subnet string        The subnet ID.
```

Examples:

```
$ ibmcloud pi instance subnet detach 85716e61-948f-309d-de8b-c4e5f0d3126d --subnet 43064761-948f-469d-ac8e-b8e5f0d6056f --mac-address eb:26:8f:ba:5f:62
```

ibmcloud pi instance subnet list

Alias: list, ls

Description: List all the attached subnets.

Usage:

```
$ list INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance update

Alias: update, upd

Description: Update a server instance.

Usage:

```
$ update INSTANCE_ID [--allow-remote-restart=True|False]
  [--default-trusted-profile (CRN | TRUSTED_PROFILE_ID | TRUSTED_PROFILE_NAME)
  [--default-trusted-profile-autolink=True|False]]
  [--IBMiCSS-license=True|False] [--IBMiPHA-license=True|False]
  [--IBMiRDS-users NUMBER_USERS] [--memory AMOUNT]
  [--metadata-service=True|False] [--metadata-service-force=True|False]]
  [--name NAME] [--pin-policy POLICY] [--preferred-processor-compatibility-mode MODE]
  [--processor-type TYPE] [--processors NUMBER] [--profile-id SAP_PROFILE_ID]
  [--storage-pool-affinity=True|False] [--virtual-cores ASSIGNED_CORES]
  [--virtual-optional-device ("attach" | "detach")]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

\$	--IBMiCSS-license	New IBMi CSS software license associated with the instance.
	--IBMiPHA-license	New IBMi PHA IASP Management software license associated with the instance.
	--IBMiRDS-users int	New number of IBMi RDS user software licenses associated with the instance.
	--allow-remote-restart	Enables automated remote restart for the instance.
	--default-trusted-profile string	Trusted profile CRN, ID or name of the default IAM trusted profile to use for this instance.
-a,	--default-trusted-profile-autolink	If set to true, the system will create a link to the specified trusted profile during server update.
-m,	--memory float	New amount of memory for the server instance.
	--metadata-service	Indicates whether the metadata service endpoint will be available to the virtual server.
	--metadata-service-force	If set to true, this allows the metadata service to be disabled while the VM is active, which may require manual interface cleanup by the OS administrator. This option is only supported for disabling the metadata service.

-n, --name string	New name of the server instance.
--pin-policy string	New pin policy for the server instance. Valid values are: "none", "soft", "hard".
-c, --preferred-processor-compatibility-mode string	The preferred processor compatibility mode. Valid values are: default, POWER7, POWER8, POWER9, POWER9_Base, POWER10, POWER11
-r, --processor-type string	New processor type for the server instance.
-p, --processors float	New amount of processors for the server instance.
--profile-id string	SAP profile ID.
-s, --storage-pool-affinity	Indicates if all volumes attached to the server must reside in the same storage pool. If set to false, then volumes from any storage tier and pool can be attached to the PVM instance. This setting affects PVM instance snapshot, capture, and clone operations. For capture and clone operations, only data volumes that are of the same storage tier and in the same storage pool of the boot volume of the PVM instance can be included. For snapshot operations, all data volumes to be included in the snapshot must reside in the same storage tier and storage pool. Once set to false, cannot be set back to true unless all volumes attached reside in the same storage tier and pool.
--virtual-cores int	New number of virtual cores assigned.
-v, --virtual-optical-device string	Attach or detach a virtual optical device to this instance. Valid values are: attach, detach.

Examples:

```
$ ibmcloud pi instance update 43064761-948f-469d-ac8e-b8e5f0d6056f --memory 2 --name new-instance-name --pin-policy hard --processors 2 --processor-type shared --storage-pool-affinity
```

ibmcloud pi instance virtual-serial-number

Alias: virtual-serial-number, vsn

Description: IBM Cloud Power Virtual Server Instance Virtual Serial Number.

Usage: virtual-serial-number

Available Commands:

- assign : Assign a virtual serial number to a virtual server instance.
- get : Get a virtual serial number assigned to a virtual server instance.
- unassign : Unassign a virtual serial number from a virtual server instance.
- update : Update a virtual serial number assigned to a virtual server instance.

ibmcloud pi instance virtual-serial-number assign

Alias: assign, asn

Description: Assign a virtual serial number to a virtual server instance.

Usage:

```
$ assign INSTANCE_ID [--description DESCRIPTION] [--serial SERIAL]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -d, --description string  Description for the virtual serial number.
-s, --serial string        Virtual serial number to attach. If no serial is specified or 'auto-assign' is set, a virtual serial number will be automatically generated.
```

Examples:

```
$ ibmcloud pi instance virtual-serial-number assign 43064761-948f-469d-ac8e-b8e5f0d6056f --description "new-virtual-serial-number-description"
```

ibmcloud pi instance virtual-serial-number get

Alias: get

Description: Get a virtual serial number assigned to a virtual server instance.

Usage:

```
$ get INSTANCE_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance virtual-serial-number unassign

Alias: `unassign, uasn`

Description: Unassign a virtual serial number from a virtual server instance.

Usage:

```
$ unassign INSTANCE_ID [--retainVSN=True|False]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -r, --retainVSN    Determines if virtual serial number will be retained after being unassigned from the instance. Default is "false".
```

Examples:

```
$ ibmcloud pi instance virtual-serial-number unassign 43064761-948f-469d-ac8e-b8e5f0d6056f --retainVSN
```

ibmcloud pi instance virtual-serial-number update

Alias: `update, upd`

Description: Update a virtual serial number assigned to a virtual server instance.

Usage:

```
$ update INSTANCE_ID (--description DESCRIPTION | --software-tier SOFTWARE_TIER)
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -d, --description string    New virtual serial number description.
    --software-tier string    IBMi licensing software tiers. Valid values are: P05, P10, P20, P30.
```

ibmcloud pi instance volume

Alias: `volume, vol`

Description: IBM Cloud Power Virtual Server Instance Volumes.

Usage: `volume`

Available Commands:

- `attach` : Attach one or more volumes to an instance.
- `bulk-detach` : Detach multiple volumes from an instance.
- `detach` : Detach a volume from an instance.
- `list` : List all the attached volumes.

ibmcloud pi instance volume attach

Alias: `attach, att`

Description: Attach one or more volumes to an instance.

Usage:

```
$ attach INSTANCE_ID --volumes VOLUME1[,VOLUMEn] [--boot-volume BOOT_VOLUME]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

```
$ -b, --boot-volume string  Boot volume to be attached to an instance.
-v, --volumes strings      Comma separated list of volume identifiers or names to associate with the instance.
```

Examples:

```
$ ibmcloud pi instance volume attach 85716e61-948f-309d-de8b-c4e5f0d3126d --boot-volume e5d394ef-3156-21a3-8b12-de15c6d5e276 --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f,32064761-298e-469a-bc6f-d8e5f0d6067a
```

ibmcloud pi instance volume bulk-detach

Alias: bulk-detach, bdet

Description: Detach multiple volumes from an instance.

Usage:

```
$ bulk-detach INSTANCE_ID (--detach-all=True|False | --volumes VOLUME1[,VOLUMEn]) [--detach-primary=True|False]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

```
$ -a, --detach-all      Indicates if all volumes, except primary boot volume, attached to the instance should be detached. Required if "--volumes" is not provided.
-p, --detach-primary    Indicates if primary boot volume attached to the instance should be detached.
-v, --volumes strings   List of volumes to be detached from an instance. Required if "--detach-all" is not provided.
```

Examples:

```
$ ibmcloud pi instance volume bulk-detach 85716e61-948f-309d-de8b-c4e5f0d3126d --detach-all --detach-primary
```

ibmcloud pi instance volume detach

Alias: detach, det

Description: Detach a volume from an instance.

Usage:

```
$ detach INSTANCE_ID --volume VOLUME_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

```
$ -v, --volume string  Volume to detach from the instance.
```

Examples:

```
$ ibmcloud pi instance volume detach 85716e61-948f-309d-de8b-c4e5f0d3126d --volume 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi instance volume list

Alias: list, ls

Description: List all the attached volumes.

Usage:

```
$ list INSTANCE_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance vpmem-volume

Alias: vpmem-volume, vpmem, vpmemvol

Description: IBM Cloud Power Virtual Server Instance vPMEM Volumes.

Usage: vpmem-volume

Available Commands:

- attach: Attach a vPMEM volume to an instance.
- detach: Detach a vPMEM volume from an instance.
- get: Get a vPMEM volume attached to an instance.
- list: List all vPMEM volumes attached to an instance.
- update: Update a server instance.

ibmcloud pi instance vpmem-volume attach

Alias: attach, att

Description: Attach a vPMEM volume to an instance.

Usage:

```
$ attach INSTANCE_ID --vpmem-volumes "NAME SIZE[,\"NAMEn SIZEn\"]" [--user-tags USER_TAG1[,USER_TAGn]]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -u, --user-tags strings      User tags to add to the created vPMEM volume(s).
  --vpmem-volumes strings     Comma separated list of vPMEM volume names and sizes to associate with the instance.
                               The maximum number of vPMEM volumes for SAP instances is 1 otherwise it is 4.
```

ibmcloud pi instance vpmem-volume detach

Alias: detach, det

Description: Detach a vPMEM volume from an instance.

Usage:

```
$ detach INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -v, --vpmem-volume string  vPMEM volume ID that is associated with the PVM instance.
```

ibmcloud pi instance vpmem-volume get

Alias: get

Description: Get a vPMEM volume attached to an instance.

Usage:

```
$ get INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

\$ -v, --vpmem-volume string vPMEM volume ID that is associated with the PVM instance.

ibmcloud pi instance vpmem-volume list

Alias: list, ls

Description: List all vPMEM volumes attached to an instance.

Usage:

\$ list INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance vpmem-volume update

Alias: update, upd

Description: Update a server instance.

Usage:

\$ update INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID --name NAME

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

\$ -n, --name string New name of the vPMEM volume.
-v, --vpmem-volume string vPMEM volume ID that is associated with the PVM instance.

ibmcloud pi ipsec-policy

Alias: ipsec-policy, ips

Description: [DEPRECATED] IBM Cloud Power Virtual Server Internet Protocol Security policies.

Usage: ipsec-policy

Available Commands :

- create : [DEPRECATED] Create a VPN IPSec policy.
- delete : [DEPRECATED] Delete a VPN IPSec policy.
- get : [DEPRECATED] View details of a VPN IPSec policy.
- list : [DEPRECATED] List all IPSec policies.
- update : [DEPRECATED] Update a VPN IPSec policy.

ibmcloud pi ipsec-policy create

Alias: create, cr

Description: [DEPRECATED] Create a VPN IPSec policy.

Usage:

\$ create IPSEC_POLICY_NAME --authentication AUTHENTICATION --dh-group DH_GROUP --encryption ENCRYPTION --key-lifetime SECONDS [--pfs=True|False]

IPSEC_POLICY_NAME: A unique name of the VPN IPSec policy. The maximum name length is 47 characters.

Available Options :


```
$ -a, --authentication string  Authentication encryption type of the IPSec policy. Valid values are: "hmac-sha-256-128", "hmac-sha1-96", "none".
-d, --dh-group int            DH group number of the IPSec policy. Valid values are: "1", "2", "5", "14", "19", "20", "24".
-e, --encryption string       Connection encryption policy of the IPSec policy.
                               Valid values are: "aes-256-cbc", "aes-192-cbc", "aes-128-cbc", "aes-256-gcm", "aes-192-gcm", "aes-128-gcm", "3des-cbc".
                               When using "aes-128-gcm", "aes-192-gcm" or "aes-256-gcm" authentication should be set to "none".
-k, --key-lifetime int        Key lifetime of the IPSec policy in seconds. Valid range is 180 to 86400 seconds.
-p, --pfs                     Enable perfect forward secrecy. Disabled if not specified.
```

Examples:

```
$ ibmcloud pi ipsec-policy create test-ipsec-policy --authentication hmac-sha-256-128 --encryption aes-256-cbc --dh-group 2 --key-lifetime 190 --pfs
```

ibmcloud pi ipsec-policy delete

Alias: delete, del

Description: [DEPRECATED] Delete a VPN IPSec policy.

Usage:

```
$ delete IPSEC_POLICY_ID

IPSEC_POLICY_ID: The unique identifier of the VPN IPSec policy.
```

ibmcloud pi ipsec-policy get

Alias: get

Description: [DEPRECATED] View details of a VPN IPSec policy.

Usage:

```
$ get IPSEC_POLICY_ID

IPSEC_POLICY_ID: The unique identifier of the VPN IPSec policy.
```

ibmcloud pi ipsec-policy list

Alias: list, ls

Description: [DEPRECATED] List all IPSec policies.

Usage: list

ibmcloud pi ipsec-policy update

Alias: update, upd

Description: [DEPRECATED] Update a VPN IPSec policy.

Usage:

```
$ update IPSEC_POLICY_ID [--authentication AUTHENTICATION] [--dh-group DH_GROUP] [--encryption ENCRYPTION] [--key-lifetime SECONDS] [--name NAME] [--pfs=True|False]]

IPSEC_POLICY_ID: The unique identifier of the VPN IPSec policy.
```

Available Options:

```
$ -a, --authentication string  Authentication encryption type of the IPSec policy. Valid values are: "hmac-sha-256-128", "hmac-sha1-96", "none".
-d, --dh-group int            DH group number of the IPSec policy. Valid values are: "1", "2", "5", "14", "19", "20", "24".
-e, --encryption string       Connection encryption policy of the IPSec policy.
                               Valid values are: "aes-256-cbc", "aes-192-cbc", "aes-128-cbc", "aes-256-gcm", "aes-192-gcm", "aes-128-gcm", "3des-cbc".
                               When using "aes-128-gcm", "aes-192-gcm" or "aes-256-gcm" authentication should be set to "none".
```


-k, --key-lifetime int	Key lifetime of the IPSec policy in seconds. Valid range is 180 to 86400 seconds.
-n, --name string	New unique name of the IPSec policy. The maximum name length is 47 characters.
-p, --pfs	Enable perfect forward secrecy. Disabled if not specified.

ibmcloud pi job

Alias: job

Description: IBM Cloud Power Virtual Server Jobs.

Usage: job

Available Commands:

- delete : Delete a job.
- get : View details of a job.
- list : List up to the last 5 jobs in a workspace.

ibmcloud pi job delete

Alias: delete, del

Description: Delete a job.

Usage:

```
$ delete JOB_ID

JOB_ID: The unique identifier of the job.
```

ibmcloud pi job get

Alias: get

Description: View details of a job.

Usage:

```
$ get JOB_ID

JOB_ID: The unique identifier of the job.
```

ibmcloud pi job list

Alias: list, ls

Description: List up to the last 5 jobs in a workspace.

Usage: list [--operation-action ACTION] [--operation-id ID] [--operation-target TARGET]

Available Options:

```
$ -a, --operation-action string  Operation action to filter returned jobs. Valid values are: epalmageImport, epaVMCapture, imageExport, imageImport, storage, vmCapture.
-i, --operation-id string       Operation ID to filter returned jobs.
-t, --operation-target string    Operation target to filter returned jobs. Valid values are: cloudConnection, image, pvmInstance.
```

ibmcloud pi network-address-group

Alias: network-address-group, nag

Description: IBM Cloud Power Virtual Server Network Address Groups.

Usage: network-address-group

Available Commands:

- `create` : Create a network address group.
- `delete` : Delete a network address group.
- `get` : View details of a network address group.
- `list` : List all network address groups.
- `member-add` : Add a member to a network address group.
- `member-remove` : Remove a member from a network address group.
- `update` : Update a network address group.

ibmcloud pi network-address-group create

Alias: `create, cr`

Description: Create a network address group.

Usage:

```
$ create NETWORK_ADDRESS_GROUP_NAME [--user-tags USER_TAG1[,USER_TAGn]]

NETWORK_ADDRESS_GROUP_NAME: The desired name of the network address group.
```

Available Options:

```
$ -u, --user-tags strings  Comma separated list of user tags to be attached to the network address group.
```

Examples:

```
$  ibmcloud pi network-address-group create test-network-address-group --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi network-address-group delete

Alias: `delete, del`

Description: Delete a network address group.

Usage:

```
$ delete NETWORK_ADDRESS_GROUP_ID

NETWORK_ADDRESS_GROUP_ID: The unique identifier of the network address group.
```

ibmcloud pi network-address-group get

Alias: `get`

Description: View details of a network address group.

Usage:

```
$ get NETWORK_ADDRESS_GROUP_ID

NETWORK_ADDRESS_GROUP_ID: The unique identifier of the network address group.
```

ibmcloud pi network-address-group list

Alias: `list, ls`

Description: List all network address groups.

Usage: `list`

ibmcloud pi network-address-group member-add

Alias: member-add, ma

Description: Add a member to a network address group.

Usage:

```
$ member-add NETWORK_ADDRESS_GROUP_ID --cidr-block CIDR

NETWORK_ADDRESS_GROUP_ID: The unique identifier of the network address group.
```

Available Options:

```
$ -c, --cidr-block string  The member CIDR to add. Valid values are in CIDR notation (192.168.1.0/22).
```

Examples:

```
$ ibmcloud pi network-address-group member-add 43064761-948f-469d-ac8e-b8e5f0d6056f --cidr-block "198.168.0.0/24"
```

ibmcloud pi network-address-group member-remove

Alias: member-remove, mr

Description: Remove a member from a network address group.

Usage:

```
$ member-remove NETWORK_ADDRESS_GROUP_ID --member-id NETWORK_ADDRESS_GROUP_MEMBER_ID

NETWORK_ADDRESS_GROUP_ID: The unique identifier of the network address group.
```

Available Options:

```
$ -m, --member-id string  The id of the network address group member to be removed.
```

Examples:

```
$ ibmcloud pi network-address-group member-remove 43064761-948f-469d-ac8e-b8e5f0d6056 --member-id 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi network-address-group update

Alias: update, upd

Description: Update a network address group.

Usage:

```
$ update NETWORK_ADDRESS_GROUP_ID --name NAME

NETWORK_ADDRESS_GROUP_ID: The unique identifier of the network address group.
```

Available Options:

```
$ -n, --name string  New name of the network address group.
```

ibmcloud pi network-interface

Alias: network-interface, ni

Description: IBM Cloud Power Virtual Server Network Interfaces.

Usage: network-interface

Available Commands:

- `create` : Create a network interface for a network.
- `delete` : Delete a network interface from a network.
- `get` : View details of a network interface for a network.
- `list` : List all network interfaces for a network (use "ibmcloud pi subnet list" to see available networks).
- `update` : Update a network interface for a network.

ibmcloud pi network-interface create

Alias: `create`, `cr`

Description: Create a network interface for a network.

Usage:

```
$ create NETWORK_INTERFACE_NAME --network-id NETWORK_ID [--ip-address IP_ADDRESS] [--user-tags USER_TAG1[,USER_TAGn]]
```

NETWORK_INTERFACE_NAME: The desired name of the network interface.

Available Options:

```
$ -i, --ip-address string  The requested ip address of this network interface (192.168.1.0).
-n, --network-id string   Network ID that is associated with the network interface.
-u, --user-tags strings   Comma separated list of user tags to be attached to the network interface.
```

Examples:

```
$  ibmcloud pi network-interface create test-network-interface --network-id 43064761-948f-469d-ac8e-b8e5f0d6056f --ip-address 195.168.0.43 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi network-interface delete

Alias: `delete`, `del`

Description: Delete a network interface from a network.

Usage:

```
$ delete NETWORK_INTERFACE_ID --network-id NETWORK_ID
```

NETWORK_INTERFACE_ID: The unique identifier of the network interface.

Available Options:

```
$ -n, --network-id string  Network ID that is associated with the network interface.
```

ibmcloud pi network-interface get

Alias: `get`

Description: View details of a network interface for a network.

Usage:

```
$ get NETWORK_INTERFACE_ID --network-id NETWORK_ID
```

NETWORK_INTERFACE_ID: The unique identifier of the network interface.

Available Options:

```
$ -n, --network-id string  Network ID that is associated with the network interface.
```

ibmcloud pi network-interface list

Alias: list, ls

Description: List all network interfaces for a network (use "ibmcloud pi subnet list" to see available networks).

Usage: list --network-id NETWORK_ID

Available Options:

```
$ -n, --network-id string    Network ID that is associated with the network interface.
```

ibmcloud pi network-interface update

Alias: update, upd

Description: Update a network interface for a network.

Usage:

```
$ update NETWORK_INTERFACE_ID --network-id NETWORK_ID [--instance-id INSTANCE_ID] [--name NAME]

NETWORK_INTERFACE_ID: The unique identifier of the network interface.
```

Available Options:

```
$ -i, --instance-id string    The ID of the instance that will be added to the network interface.
                               If this value is set to "", the network interface detaches from the instance.
    --name string             New name of network interface.
-n, --network-id string      Network ID that is associated with the network interface.
```

ibmcloud pi network-security-group

Alias: network-security-group, nsg

Description: IBM Cloud Power Virtual Server Network Security Groups.

Usage: network-security-group

Available Commands:

- action : Perform actions on network security groups.
- clone : IBM Cloud Power Virtual Server Network Security Group Clones.
- create : Create a network security group.
- delete : Delete a network security group.
- get : View details of a network security group.
- list : List all network security groups.
- member-add : Add a member to a network security group.
- member-move : Move a member to another network security group.
- member-remove : Remove a member from a network security group.
- rule-add : Add a rule to a network security group.
- rule-remove : Remove a rule from a network security group.
- update : Update a network security group.

ibmcloud pi network-security-group action

Alias: action, act

Description: Perform actions on network security groups.

Usage:

```
$ action NETWORK_SECURITY_GROUP_ACTION

NETWORK_SECURITY_GROUP_ACTION: The desired action to be performed on network security groups. Valid values are "enable" and "disable".
```


If "enable" is chosen, the currently targeted workspace will have network-security-group functionality added.
If "disable" is chosen, the currently targeted workspace will have network-security-group functionality removed.
Prior to choosing the "disable" action, all network-security-groups and network-address-groups must be to be removed.

Examples:

```
$ ibmcloud pi network-security-group action enable
```

ibmcloud pi network-security-group clone

Alias: clone, cl

Description: IBM Cloud Power Virtual Server Network Security Group Clones.

Usage: clone

Available Commands:

- create : Create a network security group using an existing network security group id.

ibmcloud pi network-security-group clone create

Alias: create, cr

Description: Create a network security group using an existing network security group id.

Usage:

```
$ create NETWORK_SECURITY_GROUP_ID --name CLONE_NAME
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

Available Options:

```
$ -n, --name string The name of the cloned network security group.
```

ibmcloud pi network-security-group create

Alias: create, cr

Description: Create a network security group.

Usage:

```
$ create NETWORK_SECURITY_GROUP_NAME [--user-tags USER_TAG1[,USER_TAGn]]
```

NETWORK_SECURITY_GROUP_NAME: The desired name of the network security group.

Available Options:

```
$ -u, --user-tags strings Comma separated list of user tags to be attached to the network security group.
```

Examples:

```
$ ibmcloud pi network-security-group create test-network-security-group --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi network-security-group delete

Alias: delete, del

Description: Delete a network security group.

Usage:

```
$ delete NETWORK_SECURITY_GROUP_ID
```


NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

ibmcloud pi network-security-group get

Alias: `get`

Description: View details of a network security group.

Usage:

```
$ get NETWORK_SECURITY_GROUP_ID
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

ibmcloud pi network-security-group list

Alias: `list, ls`

Description: List all network security groups.

Usage: `list`

ibmcloud pi network-security-group member-add

Alias: `member-add, ma`

Description: Add a member to a network security group.

Usage:

```
$ member-add NETWORK_SECURITY_GROUP_ID --member-type ("ipv4-address" | "network-interface") --target TARGET
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

Available Options:

```
$ -m, --member-type string  The type of member. Valid values are: ipv4-address, network-interface.
-t, --target string         The target member to add. This is an IP4 address if type is set to 'ipv4-address'. This is a network interface ID if type is set to 'network-interface'.
```

Examples:

```
$  ibmcloud pi network-security-group member-add 43064761-948f-469d-ac8e-b8e5f0d6056f --member-type "network-interface" --target 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi network-security-group member-move

Alias: `member-move, mm`

Description: Move a member to another network security group.

Usage:

```
$ member-move NETWORK_SECURITY_GROUP_ID --member-id NETWORK_SECURITY_GROUP_MEMBER_ID --network-security-group-id-destination
NETWORK_SECURITY_GROUP_ID_DESTINATION
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

Available Options:

```
$ -m, --member-id string      The id of the network security group member to be move.
-n, --network-security-group-id-destination string  The id of the network security group the member will move to.
```

ibmcloud pi network-security-group member-remove

Alias: member-remove, mr

Description: Remove a member from a network security group.

Usage:

```
$ member-remove NETWORK_SECURITY_GROUP_ID --member-id NETWORK_SECURITY_GROUP_MEMBER_ID

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.
```

Available Options:

```
$ -m, --member-id string  The id of the network security group member to be removed.
```

Examples:

```
$ ibmcloud pi network-security-group member-remove 43064761-948f-469d-ac8e-b8e5f0d6056 --member-id 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi network-security-group rule-add

Alias: rule-add, ra

Description: Add a rule to a network security group.

Usage:

```
$ rule-add NETWORK_SECURITY_GROUP_ID --action ("allow" | "deny")
  (--protocol "all" | --protocol "icmp" [--icmp-type TYPE] |
  --protocol "tcp" [--dest-port-min MIN] [--dest-port-max MAX] [--src-port-min MIN] [--src-port-max MAX] [--tcp-flags FLAG1[,FLAGn]] |
  --protocol "udp" [--dest-port-min MIN] [--dest-port-max MAX] [--src-port-min MIN] [--src-port-max MAX])
  (--remote-type "default-network-address-group" |
  --remote-type "network-address-group" --remote-group-id REMOTE_NETWORK_ADDRESS_GROUP_ID |
  --remote-type "network-security-group" --remote-group-id REMOTE_NETWORK_SECURITY_GROUP_ID)

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.
```

Available Options:

\$ -a, --action string	The action to take if the rule matches network traffic. Valid action values are: allow, deny.
--dest-port-max int	The end of the destination port range. If the value is not present then the default value of 65535 will be the maximum port number.
--dest-port-min int	The start of the destination port range. If the value is not present then the default value of 1 will be the minimum port number.
-i, --icmp-type string	The ICMP packet type affected by ICMP rules. Valid values are: all, echo, echo-reply, source-quench, time-exceeded, destination-unreach.
-p, --protocol string	The protocol of the network traffic. Valid values are: all, icmp, tcp, udp.
-g, --remote-group-id string	The unique identifier of the remote network address group or network security group.
-r, --remote-type string	Specifies the type of remote group that defines the origin of network traffic for rule matching. Remote groups represent MAC addresses, IP addresses, CIDRs, or external CIDRs. Valid values are: default-network-address-group, network-address-group, network-security-group. The "default-network-address-group" is a network address group that specifies all external network traffic.
--src-port-max int	The end of the source port range. If the value is not present then the default value of 65535 will be the maximum port number.
--src-port-min int	The start of the source port range. If the value is not present then the default value of 1 will be the minimum port number.
-t, --tcp-flags strings	Comma separated list of TCP flag options. Valid values are: syn, ack, fin, rst.

Examples:

```
$ ibmcloud pi network-security-group rule-add 43064761-948f-469d-ac8e-b8e5f0d6056f --action allow --protocol all --remote-type network-address-group --remote-group-id 85716e61-948f-309d-de8b-c4e5f0d3126d
  ibmcloud pi network-security-group rule-add 43064761-948f-469d-ac8e-b8e5f0d6056f --action allow --protocol tcp --dest-port-min 1200 --dest-port-max 37465 --src-port-min 1000 --src-port-max 20000 --tcp-flags "syn,ack,fin" --remote-type network-security-group --remote-group-id 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi network-security-group rule-remove

Alias: rule-remove, rr

Description: Remove a rule from a network security group.

Usage:

```
$ rule-remove NETWORK_SECURITY_GROUP_ID --rule-id RULE_ID
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

Available Options:

```
$ -r, --rule-id string  The id of the network security group rule to be removed.
```

Examples:

```
$ ibmcloud pi network-security-group rule-remove 43064761-948f-469d-ac8e-b8e5f0d6056 --rule-id 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi network-security-group update

Alias: `update`, `upd`

Description: Update a network security group.

Usage:

```
$ update NETWORK_SECURITY_GROUP_ID --name NAME
```

NETWORK_SECURITY_GROUP_ID: The unique identifier of the network security group.

Available Options:

```
$ -n, --name string  New name of the network security group.
```

ibmcloud pi placement-group

Alias: `placement-group`, `pg`

Description: IBM Cloud Power Virtual Server Placement Groups.

Usage: `placement-group`

Available Commands:

- `create` : Create a server placement group.
- `delete` : Delete a server placement group.
- `get` : View details of a server placement group.
- `list` : List all server placement groups.
- `server-add` : Add a server to a server placement group.
- `server-remove` : Remove a server from a server placement group.

ibmcloud pi placement-group create

Alias: `create`, `cr`

Description: Create a server placement group.

Usage:

```
$ create PLACEMENT_GROUP_NAME --policy POLICY [--user-tags USER_TAG1[,USER_TAGn]]
```

PLACEMENT_GROUP_NAME: A unique name of the placement group.

Available Options:

```
$ -p, --policy string  Affinity policy for placement group being created. Valid values are affinity and anti-affinity.  
-u, --user-tags strings  Comma separated list of user tags to be attached to the placement group.
```

Examples:

```
$ ibmcloud pi placement-group test-placement-group --policy anti-affinity
```

ibmcloud pi placement-group delete

Alias: delete, del

Description: Delete a server placement group.

Usage:

```
$ delete PLACEMENT_GROUP_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

ibmcloud pi placement-group get

Alias: get

Description: View details of a server placement group.

Usage:

```
$ get PLACEMENT_GROUP_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

ibmcloud pi placement-group list

Alias: list, ls

Description: List all server placement groups.

Usage: list

ibmcloud pi placement-group server-add

Alias: server-add, sa

Description: Add a server to a server placement group.

Usage:

```
$ server-add PLACEMENT_GROUP_ID --server INSTANCE_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options:

```
$ -s, --server string The server instance ID of the server to add to the placement group.
```

Examples:

```
$ ibmcloud pi placement-group server-add 43064761-948f-469d-ac8e-b8e5f0d6056f --server 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi placement-group server-remove

Alias: server-remove, sr

Description: Remove a server from a server placement group.

Usage:

```
$ server-remove PLACEMENT_GROUP_ID --server INSTANCE_ID
```

PLACEMENT_GROUP_ID: The unique identifier of the placement group.

Available Options:

\$ -s, --server string The server instance ID of the server to add to the placement group.

Examples:

\$ ibmcloud pi placement-group server-remove 43064761-948f-469d-ac8e-b8e5f0d6056f --server 85716e61-948f-309d-de8b-c4e5f0d3126d

ibmcloud pi route

Alias: route, rt

Description: IBM Cloud Power Virtual Server Network Routes.

Usage: route

Available Commands:

- create : Create a network custom route.
- delete : Delete a network custom route.
- get : View details of a network custom route.
- list : List all network custom routes.
- report : View details of network custom route report.
- update : Update a network custom route.

ibmcloud pi route create

Alias: create, cr

Description: Create a network custom route.

Usage:

\$ create ROUTE_NAME --destination DESTINATION --next-hop NEXT_HOP
[--action ACTION] [--advertise ("enable" | "disable")] [--destination-type DESTINATION_TYPE]
[--enabled=True|False] [--next-hop-type NEXT_HOP_TYPE] [--user-tags USER_TAG1[,USER_TAGn]]

ROUTE_NAME: A unique name of the network custom route.

Available Options:

\$ -a, --action string	The action of the route to take. Valid values are: deliver. Default is "deliver".
--advertise string	Enable the route to be advertised. Valid values are: enable, disable. Default is "enable".
-d, --destination string	The route destination.
--destination-type string	The route destination type. Valid values are: ipv4-address. Default is "ipv4-address".
-e, --enabled	Indicates if the route should be enabled in the fabric.
--next-hop string	The IP address to route the packet to.
--next-hop-type string	The next hop type. Valid values are: ipv4-address. Default is "ipv4-address".
-u, --user-tags strings	Comma separated list of user tags to be attached to the network security group.

Examples:

\$ ibmcloud pi route create test-route --advertise disable --destination 192.168.1.23 --next-hop 192.168.1.24

ibmcloud pi route delete

Alias: delete, del

Description: Delete a network custom route.

Usage:


```
$ delete ROUTE_ID
```

ROUTE_ID: The unique identifier of the network custom route.

ibmcloud pi route get

Alias: `get`

Description: View details of a network custom route.

Usage:

```
$ get ROUTE_ID
```

ROUTE_ID: The unique identifier of the network custom route.

ibmcloud pi route list

Alias: `list, ls`

Description: List all network custom routes.

Usage: `list`

ibmcloud pi route report

Alias: `report, rp`

Description: View details of network custom route report.

Usage: `report`

ibmcloud pi route update

Alias: `update, upd`

Description: Update a network custom route.

Usage:

```
$ update ROUTE_ID [--action ACTION] [--advertise ("enable" | "disable")]
  [--destination DESTINATION] [--destination-type DESTINATION_TYPE] [--enabled=True|False]
  [--name NAME] [--next-hop NEXT_HOP] [--next-hop-type NEXT_HOP_TYPE]
```

ROUTE_ID: The unique identifier of the network custom route.

Available Options:

\$ -a, --action string	The action of the route to take. Valid values are: deliver.
--advertise string	Enable the route to be advertised. Valid values are: enable, disable.
-d, --destination string	The route destination.
--destination-type string	The route destination type. Valid values are: ipv4-address.
-e, --enabled	Indicates if the route should be enabled in the fabric.
-n, --name string	New name of the route.
--next-hop string	The IP address to route the packet to.
--next-hop-type string	The next hop type. Valid values are: ipv4-address.

ibmcloud pi shared-processor-pool

Alias: `shared-processor-pool, spp`

Description: IBM Cloud Power Virtual Server Shared Processor Pools.

Usage: `shared-processor-pool`

Available Commands:

- `create` : Create a shared processor pool.
- `delete` : Delete a shared processor pool.
- `get` : View details of a shared processor pool.
- `list` : List all shared processor pools.
- `placement-group` : IBM Cloud Power Virtual Server Placement Groups.
- `update` : Update a shared processor pool.

ibmcloud pi shared-processor-pool create

Alias: `create`, `cr`

Description: Create a shared processor pool.

Usage:

```
$ create SHARED_PROCESSOR_POOL_NAME --host-group HOST_GROUP --reserved-cores NUMBER_OF_CORES [--host-id HOST_ID] [--placement-group-id PLACEMENT_GROUP_ID] [--user-tags USER_TAG1[,USER_TAGn]]
```

SHARED_PROCESSOR_POOL_NAME: A unique name of the shared processor pool. A minimum of 2 characters and a maximum of 12 are allowed. The only special character allowed is the underscore '_'.

Available Options:

```
$ -g, --host-group string      The host group where the host will be chosen. See "ibmcloud pi host-group list" command to get valid values.
-i, --host-id string          The identifier of a host in a host group. This option is only available for dedicated hosts.
-p, --placement-group-id string The identifier of the shared processor pool placement group the pool will be added to.
-r, --reserved-cores int      The integer amount of reserved processor cores for the shared processor pool.
-u, --user-tags strings       Comma separated list of user tags to be attached to the placement-group.
```

Examples:

```
$ ibmcloud pi shared-processor-pool create test-shared-processor-pool --host-group e980 --reserver-cores 2 --placement-group-id 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi shared-processor-pool delete

Alias: `delete`, `del`

Description: Delete a shared processor pool.

Usage:

```
$ delete SHARED_PROCESSOR_POOL_ID
```

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.

ibmcloud pi shared-processor-pool get

Alias: `get`

Description: View details of a shared processor pool.

Usage:

```
$ get SHARED_PROCESSOR_POOL_ID
```

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.

ibmcloud pi shared-processor-pool list

Alias: `list`, `ls`

Description: List all shared processor pools.

Usage: `list`

ibmcloud pi shared-processor-pool placement-group

Alias: placement-group, pg

Description: IBM Cloud Power Virtual Server Placement Groups.

Usage: placement-group

Available Commands:

- create : Create a shared processor pool placement group.
- delete : Delete a shared processor pool placement group.
- get : View details of a shared processor pool placement group.
- list : List all shared processor pool placement groups.
- member-add : Add a shared processor pool to a placement group.
- member-remove : Remove a shared processor pool from a placement group.

ibmcloud pi shared-processor-pool placement-group create

Alias: create, cr

Description: Create a shared processor pool placement group.

Usage:

```
$ create PLACEMENT_GROUP_NAME --policy POLICY [--user-tags USER_TAG1[,USER_TAGn]]
```

PLACEMENT_GROUP_NAME: A unique name of the placement group.

Available Options:

```
$ -p, --policy string      Affinity policy for placement group being created. Valid values are affinity and anti-affinity.
-u, --user-tags strings    Comma separated list of user tags to be attached to the shared processor pool placement group.
```

Examples:

```
$ ibmcloud pi shared-processor-pool placement-group create spp-pg-test --policy affinity
```

ibmcloud pi shared-processor-pool placement-group delete

Alias: delete, del

Description: Delete a shared processor pool placement group.

Usage:

```
$ delete PLACEMENT_GROUP_ID
```

PLACEMENT_GROUP_ID: The unique identifier of the placement group.

ibmcloud pi shared-processor-pool placement-group get

Alias: get

Description: View details of a shared processor pool placement group.

Usage:

```
$ get PLACEMENT_GROUP_ID
```

PLACEMENT_GROUP_ID: The unique identifier of the placement group.

ibmcloud pi shared-processor-pool placement-group list

Alias: list, ls

Description: List all shared processor pool placement groups.

Usage: list

ibmcloud pi shared-processor-pool placement-group member-add

Alias: member-add, ma

Description: Add a shared processor pool to a placement group.

Usage:

```
$ member-add PLACEMENT_GROUP_ID --shared-processor-pool POOL_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options:

```
$ -s, --shared-processor-pool string  The shared processor pool ID of the pool to add to the placement group.
```

Examples:

```
$  ibmcloud pi shared-processor-pool placement-group member-add 43064761-948f-469d-ac8e-b8e5f0d6056 --shared-processor-pool 32bec9b0-560a-4baa-8cc0-7c68e41e5843
```

ibmcloud pi shared-processor-pool placement-group member-remove

Alias: member-remove, mr

Description: Remove a shared processor pool from a placement group.

Usage:

```
$ member-remove PLACEMENT_GROUP_ID --shared-processor-pool POOL_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options:

```
$ -s, --shared-processor-pool string  The shared processor pool ID of the pool to remove from the placement group.
```

Examples:

```
$  ibmcloud pi shared-processor-pool placement-group member-remove 43064761-948f-469d-ac8e-b8e5f0d6056 --shared-processor-pool 32bec9b0-560a-4baa-8cc0-7c68e41e5843
```

ibmcloud pi shared-processor-pool update

Alias: update, upd

Description: Update a shared processor pool.

Usage:

```
$ update SHARED_PROCESSOR_POOL_ID [--name SHARED_PROCESSOR_POOL_NAME] [--reserved-cores NUMBER_OF_CORES]

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.
```

Available Options:

```
$ -n, --name string      New name of the shared processor pool. A minimum of 2 characters and a maximum of 12 are allowed. The only special character allowed is the
```

```
underscore '_'.
-r, --reserved-cores int  The integer amount of reserved processor cores for the shared processor pool.
```

ibmcloud pi snapshot

Alias: `snapshot`, `snap`

Description: [DEPRECATED] IBM Cloud Power Virtual Server Snapshots.

Usage: `snapshot`

Available Commands:

- `create`: [DEPRECATED] Create a snapshot.
- `delete`: [DEPRECATED] Delete a snapshot.
- `get`: [DEPRECATED] View details of a snapshot.
- `list`: [DEPRECATED] List all snapshots.
- `restore`: [DEPRECATED] Restore a snapshot.
- `update`: [DEPRECATED] Update a snapshot.

ibmcloud pi snapshot create

Alias: `create`, `cr`

Description: [DEPRECATED] Create a snapshot.

Usage:

```
$ create INSTANCE_ID --name SNAPSHOT_NAME [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEen]]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -d, --description string  Snapshot description.
-n, --name string          Name of the snapshot.
-u, --user-tags strings    Comma separated list of user tags to be attached to the snapshot.
-v, --volumes strings      Comma separated list of volume identifiers or names to include in the snapshot.
                           If you do not specify this option or if the volumes list is empty,
                           all the volumes that are attached to the instance are included in the snapshot.
```

Examples:

```
$  ibmcloud pi snapshot create 85716e61-948f-309d-de8b-c4e5f0d3126d --name test-snapshot-name
```

ibmcloud pi snapshot delete

Alias: `delete`, `del`

Description: [DEPRECATED] Delete a snapshot.

Usage:

```
$ delete SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi snapshot get

Alias: `get`

Description: [DEPRECATED] View details of a snapshot.

Usage:

```
$ get SNAPSHOT_ID
```

SNAPSHOT_ID: The unique identifier of the snapshot.

ibmcloud pi snapshot list

Alias: `list`, `ls`

Description: [DEPRECATED] List all snapshots.

Usage: `list`

ibmcloud pi snapshot restore

Alias: `restore`, `res`

Description: [DEPRECATED] Restore a snapshot.

Usage:

```
$ restore INSTANCE_ID --snapshot SNAPSHOT_ID [--force] [--restore VALUE]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -f, --force           By default the VM must be shutdown during a snapshot restore, force set to true will relax the VM shutdown pre-condition.
-r, --restore string    Action to take on a failed snapshot restore. Valid values are: "retry", "rollback".
-s, --snapshot string   The unique identifier of the snapshot.
```

Examples:

```
$  ibmcloud pi snapshot restore 85716e61-948f-309d-de8b-c4e5f0d3126d --snapshot 43064761-948f-469d-ac8e-b8e5f0d6056f --force --restore retry
```

ibmcloud pi snapshot update

Alias: `update`, `upd`

Description: [DEPRECATED] Update a snapshot.

Usage:

```
$ update SNAPSHOT_ID [--description DESCRIPTION] [--name NAME]
```

SNAPSHOT_ID: The unique identifier of the snapshot.

Available Options:

```
$ -d, --description string  New snapshot description.
-n, --name string           New name of the snapshot.
```

ibmcloud pi ssh-key

Alias: `ssh-key`, `ssh`

Description: IBM Cloud Power Virtual Server SSH-Keys.

Usage: `ssh-key`

Available Commands:

- `create`: Create an SSH-Key with an imported RSA public key.
- `delete`: Delete an SSH-Key.
- `get`: View details of an SSH-Key.

- `list` : List all SSH-Keys.
- `update` : Update an SSH-Key.

ibmcloud pi ssh-key create

Alias: `create, cr`

Description: Create an SSH-Key with an imported RSA public key.

Usage:

```
$ create KEY_NAME --key KEY [--description DESCRIPTION] [--visibility ("account" | "workspace")]
```

KEY_NAME: The name of the key.

Available Options :

```
$ -d, --description string  Description of the SSH-Key.
-k, --key string            SSH RSA key string within a double quotation marks. For example, "ssh-rsa AAA... "
-v, --visibility string     Visibility of the SSH-Key. Valid values are: account, workspace.
                             Value of "workspace" means SSH-Key is only accessible in a workspace.
                             Value of "account" means SSH-Key is accessible throughout an account.
                             Default is "workspace".
```

Examples:

```
$  ibmcloud pi ssh-key create test-ssh-key --key ssh-key-sample
```

ibmcloud pi ssh-key delete

Alias: `delete, del`

Description: Delete an SSH-Key.

Usage:

```
$ delete KEY_NAME
```

KEY_NAME: The name of the key.

ibmcloud pi ssh-key get

Alias: `get`

Description: View details of an SSH-Key.

Usage:

```
$ get KEY_NAME
```

KEY_NAME: The name of the key.

ibmcloud pi ssh-key list

Alias: `list, ls`

Description: List all SSH-Keys.

Usage: `list`

ibmcloud pi ssh-key update

Alias: `update, upd`

Description: Update an SSH-Key.

Usage:

```
$ update KEY_NAME [--description DESCRIPTION] [--key KEY] [--name NAME] [--visibility ("account" | "workspace")]

KEY_NAME: The name of the key.
```

Available Options:

```
$ -d, --description string  Description of the SSH-Key.
-k, --key string           SSH RSA key string.
-n, --name string          SSH-Key name.
-v, --visibility string     Visibility of the SSH-Key. Valid values are: account, workspace.
                           Value of "workspace" means SSH-Key is only accessible in a workspace.
                           Value of "account" means SSH-Key is accessible throughout an account.
```

ibmcloud pi storage-pools

Alias: storage-pools, spools

Description: List all storage pools for the targeted region.

Usage: storage-pools

ibmcloud pi storage-tiers

Alias: storage-tiers, stiers

Description: List all storage tiers for the targeted region.

Usage: storage-tiers

ibmcloud pi subnet

Alias: subnet, snet

Description: IBM Cloud Power Virtual Server Subnets.

Usage: subnet

Available Commands:

- create : Create a subnet.
- delete : Delete a subnet.
- get : View details of a subnet.
- list : List all subnets in a workspace.
- update : Update a subnet.

ibmcloud pi subnet create

Alias: create, cr

Description: Create a subnet.

Usage:

```
$ create SUBNET_NAME --cidr-block CIDR --net-type private [--advertise ("enable" | "disable")] [--arp-broadcast ("enable" | "disable")]
  [--dns-servers "DNS1,[DNSn]]" [--gateway GATEWAY] [--ip-range "startIP-endIP[,startIP-endIP]"
  [--mtu MTU] [--user-tags "USER_TAG1[,USER_TAGn]"]

SUBNET_NAME: The name of the subnet.
```

Available Options:

```
$ --advertise string  Enable the subnet to be advertised. Valid values are: enable, disable. Default is "enable" for private networks.
```

-b, --arp-broadcast string Enable ARP Broadcast. Valid values are: enable, disable. Default is "disable" for private networks.

-c, --cidr-block string Subnet in CIDR notation (192.168.1.0/22).

-d, --dns-servers strings Comma separated list of DNS Servers to use for this subnet.
161.26.0.10 and 161.26.0.11 by default if DNS server is not specified for private subnet types in PER enabled workspaces else
127.0.0.1 by default if DNS server is not specified and workspace is not PER enabled.
9.9.9.9 by default for public subnet types.

-g, --gateway string Gateway to use for this subnet.

-i, --ip-range string IP Addresses range(s) for this subnet, format: "startIP-endIP[,startIP-endIP]".

-m, --mtu int Maximum Transmission Unit. MTU be between 1450 and 9000.

-n, --net-type string Subnet type. Either "public" or "private".

-u, --user-tags strings Comma separated list of user tags to be attached to the subnet.

Examples:

```
$ ibmcloud pi subnet create test-private-subnet --net-type private --cidr-block "192.168.0.0/24" --ip-range "192.168.0.10-192.168.0.20" --dns-servers "8.8.8.8" --gateway "192.168.0.1"
ibmcloud pi subnet create test-public-subnet --net-type public --dns-servers "8.8.8.8" --mtu 2000
```

ibmcloud pi subnet delete

Alias: delete, del

Description: Delete a subnet.

Usage:

```
$ delete SUBNET_ID

SUBNET_ID: The unique identifier or name of the subnet.
```

ibmcloud pi subnet get

Alias: get

Description: View details of a subnet.

Usage:

```
$ get SUBNET_ID

SUBNET_ID: The unique identifier or name of the subnet.
```

ibmcloud pi subnet list

Alias: list, ls

Description: List all subnets in a workspace.

Usage: list

ibmcloud pi subnet update

Alias: update, upd

Description: Update a subnet.

Usage:

```
$ update SUBNET_ID [--advertise ("enable" | "disable")] [--arp-broadcast ("enable" | "disable")]
  [--dns-servers "DNS1,[DNSn]"] [--gateway GATEWAY] [--ip-range "startIP-endIP[,startIP-endIP]"]
  [--name SUBNET_NAME]

SUBNET_ID: The unique identifier or name of the subnet.
```

Available Options:

```
$ --advertise string      Enable the subnet to be advertised. Valid values are: enable, disable.
-b, --arp-broadcast string Enable ARP Broadcast. Valid values are: enable, disable.
-d, --dns-servers strings Comma separated list of DNS Servers to use for this subnet.
-g, --gateway string      Gateway to use for this subnet.
-i, --ip-range string      IP Addresses range(s) for this subnet, format: "startIP-endIP[,startIP-endIP]".
-n, --name string          New name of the subnet.
```

ibmcloud pi virtual-serial-number

Alias: virtual-serial-number, vsn

Description: IBM Cloud Power Virtual Server Virtual Serial Number.

Usage: virtual-serial-number

Available Commands:

- delete : Delete a virtual serial number.
- get : View details of a virtual serial number.
- list : List all virtual serial number assigned to an instance or all virtual serial numbers in the workspace.
- software-tiers : List all supported software tiers.
- update : Update a retained virtual serial number.

ibmcloud pi virtual-serial-number delete

Alias: delete, del

Description: Delete a virtual serial number.

Usage:

```
$ delete VIRTUAL_SERIAL_NUMBER

VIRTUAL_SERIAL_NUMBER: The virtual serial number.
```

ibmcloud pi virtual-serial-number get

Alias: get

Description: View details of a virtual serial number.

Usage:

```
$ get VIRTUAL_SERIAL_NUMBER

VIRTUAL_SERIAL_NUMBER: The virtual serial number.
```

ibmcloud pi virtual-serial-number list

Alias: list, ls

Description: List all virtual serial number assigned to an instance or all virtual serial numbers in the workspace.

Usage:

```
$ list ([PVM_INSTANCE_ID] | [--retainVSN=True|False])

PVM_INSTANCE_ID: The unique identifier or name of the PVM instance.
```

Available Options:

```
$ -r, --retainVSN  Lists only retained virtual serial numbers. Default is "false".
```

ibmcloud pi virtual-serial-number software-tiers

Alias: software-tiers, st

Description: List all supported software tiers.

Usage: software-tiers

ibmcloud pi virtual-serial-number update

Alias: update, upd

Description: Update a retained virtual serial number.

Usage:

```
$ update VIRTUAL_SERIAL_NUMBER [--description DESCRIPTION]
```

VIRTUAL_SERIAL_NUMBER: The virtual serial number.

Available Options:

```
$ -d, --description string  New virtual serial number description.
```

ibmcloud pi volume

Alias: volume, vol

Description: IBM Cloud Power Virtual Server Volumes.

Usage: volume

Available Commands:

- **action** : Perform an action on a volume.
 - **bulk-delete** : Delete multiple volumes.
 - **clone** : IBM Cloud Power Virtual Server Volume Clone Requests. This command can be used to create clone requests whose lifecycle must be managed. It cannot be used to clone single volumes.
 - **clone-async** : IBM Cloud Power Virtual Server Volume Clones. This command asynchronously creates clone tasks whose status can be queried. It can be used to clone one or more volumes.
 - **create** : Create a volume.
 - **delete** : Delete a volume.
 - **flash-copy-mapping** : Get a list of flash copy mappings of a volume directly from primary storage host.
 - **get** : View details of a volume.
 - **list** : List all storage volumes in a workspace.
 - **onboarding** : IBM Cloud Power Virtual Server Volume Onboarding.
 - **remote-copy-relationship** : Get the remote copy relationship information of a volume.
 - **snapshot** : IBM Cloud Power Virtual Server Volume Snapshot.
 - **update** : Update a volume.
-

ibmcloud pi volume action

Alias: action, act

Description: Perform an action on a volume.

Usage:

```
$ action VOLUME_ID [--replication-enabled=True|False] [--target-tier STORAGE_TIER]
```

VOLUME_ID: The unique identifier or name of the volume.

Available Options:

```
$ -r, --replication-enabled  Enables storage replication on the volume. Default is "false".
-t, --target-tier string    Change the storage tier of the volume (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region).
                             "Tier5k" volumes cannot exceed 200 GiB.
```

Examples:

```
$  ibmcloud pi volume action test-volume --replication-enabled
```

ibmcloud pi volume bulk-delete

Alias: `bulk-delete, bdel`

Description: Delete multiple volumes.

Usage: `bulk-delete --volumes VOLUME1[,VOLUMEn]`

Available Options:

```
$ -v, --volumes strings  List of volumes to be deleted.
```

ibmcloud pi volume clone

Alias: `clone, cl`

Description: IBM Cloud Power Virtual Server Volume Clone Requests. This command can be used to create clone requests whose lifecycle must be managed. It cannot be used to clone single volumes.

Usage: `clone`

Available Commands:

- `cancel` : Cancel a volume clone request.
- `create` : Create a volume clone request using the specified volumes.
- `delete` : Delete a volume clone request.
- `execute` : Execute a volume clone request using the specified options.
- `get` : View details of a volume clone request.
- `list` : List all volume clone requests in a workspace.
- `start` : Starts a volume clone request.

ibmcloud pi volume clone cancel

Alias: `cancel, ca`

Description: Cancel a volume clone request.

Usage:

```
$ cancel VOLUME_CLONE_REQUEST_ID [--force=True|False]

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Available Options:

```
$ -f, --force  Force the cancellation of a volume clone request. If false, cancel will only be allowed if the status is 'prepared', or 'available'. If true, cancel will be allowed when the status is NOT completed, cancelling, cancelled, or failed. Default false.
```

Examples:

```
$  ibmcloud pi volume clone cancel 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone create

Alias: `create, cr`

Description: Create a volume clone request using the specified volumes.

Usage:

```
$ create --name VOLUME_CLONE_REQUEST_NAME --volumes VOLUME1[,VOLUMEn]

VOLUME_CLONE_REQUEST_NAME: The name of a volume clone request.
```

Available Options:

```
$ -n, --name string      Name to assign to volume clones request.
-v, --volumes strings    Comma separated list of volume identifiers or names to be cloned.
                          A minimum of 2 Volume IDs are required and at least one of them must be in the in-use state.
```

Examples:

```
$  ibmcloud pi volume clone create --name test-volume-clone --volumes "43064761-948f-469d-ac8e-b8e5f0d6056f,85716e61-948f-309d-de8b-c4e5f0d3126d"
```

ibmcloud pi volume clone delete

Alias: `delete, del`

Description: Delete a volume clone request.

Usage:

```
$ delete VOLUME_CLONE_REQUEST_ID

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Examples:

```
$  ibmcloud pi volume clone delete 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone execute

Alias: `execute, ex`

Description: Execute a volume clone request using the specified options.

Usage:

```
$ execute VOLUME_CLONE_REQUEST_ID --name BASE_NAME [--replication-enabled=True|False] [--rollback-prepare=True|False] [--target-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Available Options:

```
$ -n, --name string      Base name of the new cloned volume(s).
  --replication-enabled  Enables replication for the cloned volume. By default, the replication property of the source volume will be used to determine the replication property of the cloned target volume.
  --rollback-prepare     Determines rollback behavior. If false, execute failure rolls back clone activity but leaves prepared snapshot. If true, Execute failure rolls back clone activity and removes the prepared snapshot. Default false.
-t, --target-tier string Target storage tier for the cloned volumes. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region.
                          Default to the storage tier of the original volume(s) if not specified.
-u, --user-tags strings   Comma separated list of user tags to be attached to the cloned volume.
```

Examples:

```
$  ibmcloud pi volume clone execute 43064761-948f-469d-ac8e-b8e5f0d6056f --name test-volume-clone --rollback-prepare --target-tier tier3 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```


ibmcloud pi volume clone get

Alias: `get`

Description: View details of a volume clone request.

Usage:

```
$ get VOLUME_CLONE_REQUEST_ID

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

ibmcloud pi volume clone list

Alias: `list`, `ls`

Description: List all volume clone requests in a workspace.

Usage: `list [--filter FILTER]`

Available Options:

```
$ -f, --filter string  Volume clone filter to limit list items.
                        cancel - includes status values (cancelling)
                        cancelled - includes status values (cancelled)
                        completed - includes status values (completed)
                        execute - includes status values (executing, available-rollback)
                        failed - includes status values (failed)
                        finalized - included status values (completed, failed, cancelled)
                        prepare - includes status values (preparing, prepared)
                        start - includes status values (starting, available)
```

ibmcloud pi volume clone start

Alias: `start`, `st`

Description: Starts a volume clone request.

Usage:

```
$ start VOLUME_CLONE_REQUEST_ID

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Examples:

```
$  ibmcloud pi volume clone start 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone-async

Alias: `clone-async`, `cla`

Description: IBM Cloud Power Virtual Server Volume Clones. This command asynchronously creates clone tasks whose status can be queried. It can be used to clone one or more volumes.

Usage: `clone-async`

Available Commands:

- `create` : Create a volume clone for specific volumes.
- `get` : Get the status of a clone request for the specified clone task ID.

ibmcloud pi volume clone-async create

Alias: `create`, `cr`

Description: Create a volume clone for specific volumes.

Usage:

```
$ create CLONE_NAME --volumes VOLUME1[,VOLUMEn] [--replication-enabled=True|False] [--target-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]

CLONE_NAME:The name of a clone.
```

Available Options:

```
$ -r, --replication-enabled  Enables replication for the cloned volume.
                             If no value is provided, it will default to the replication status of the source volume.
-t, --target-tier string     Target storage tier for the cloned volumes. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region.
                             Default to the storage tier of the original volume(s) if not specified.
-u, --user-tags strings      Comma separated list of user tags to be attached to the cloned volume.
-v, --volumes strings        Comma separated list of volume identifiers or names to be cloned.
```

Examples:

```
$ ibmcloud pi volume clone-async create test-volume --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f --target-tier tier1 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi volume clone-async get

Alias: get

Description: Get the status of a clone request for the specified clone task ID.

Usage:

```
$ get CLONE_TASK_ID

CLONE_TASK_ID:The unique identifier of a clone task.
```

ibmcloud pi volume create

Alias: create, cr

Description: Create a volume.

Usage:

```
$ create VOLUME_NAME --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEen]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
  pi volume create VOLUME_NAME --size SIZE --storage-pool POOL [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEen]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
  pi volume create VOLUME_NAME --affinity-policy affinity (--affinity-volume VOLUME | --affinity-instance INSTANCE) --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEen]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
  pi volume create VOLUME_NAME --affinity-policy anti-affinity (--anti-affinity-volumes "VOLUME1[,VOLUMEn]" | --anti-affinity-instances "INSTANCE1[,INSTANCEn]") --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEen]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]

VOLUME_NAME: The name of the volume.
```

Available Options:

```
$ -i, --affinity-instance string      PVM instance identifier or name to base volume affinity policy against.
                                     This is required if "--affinity-policy affinity" is specified and "--affinity-volume" is not provided.
-a, --affinity-policy string          Affinity policy for data volume being created. Valid values are "affinity" and "anti-affinity". If "--storage-pool" is provided then this cannot be specified.
-v, --affinity-volume string          Volume identifier or name to base volume affinity policy against.
                                     This is required if "--affinity-policy affinity" is specified and "--affinity-instance" is not provided.
-j, --anti-affinity-instances strings Comma separated list of instance identifiers or names to base volume anti-affinity policy against.
                                     This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-volumes" is not provided.
-w, --anti-affinity-volumes strings  Comma separated list of volume identifiers or names to base volume anti-affinity policy against.
                                     This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-instances" is not provided.
```

-c, --count int	Number of volumes to create. Default is 1.
-r, --replication-enabled	Enables storage replication on the volume. Default is "false".
--replication-sites strings	List of replication sites for volume replication. See "ibmcloud pi disaster-recovery" command to get a list of replication sites.
-e, --shareable	Whether the volumes can be attached to multiple VMs. Default is "false".
-s, --size int	Size of the volume in GiB. Size cannot exceed 200 GiB for storage tier "Tier5k".
-p, --storage-pool string	Volume pool where the volume will be created. If the "--storage-pool" option is set then the "--affinity-policy" option cannot be specified.
-t, --storage-tier string	Tier of the volume (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region). Default to tier3 if not provided.
-u, --user-tags strings	Comma separated list of user tags to be attached to the volume.

Examples:

```
$ ibmcloud pi volume create test-volume --size 1
ibmcloud pi volume create test-volume --size 1 --storage-tier tier1 --affinity-policy anti-affinity --anti-affinity-volumes anti-affinity-volume
ibmcloud pi volume create test-volume --size 1 --storage-tier tier1 --count 2
ibmcloud pi volume create test-volume --size 1 --replication-enabled --replication-sites dal10
```

ibmcloud pi volume delete

Alias: delete, del

Description: Delete a volume.

Usage:

```
$ delete VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume flash-copy-mapping

Alias: flash-copy-mapping, fcm

Description: Get a list of flash copy mappings of a volume directly from primary storage host.

Usage:

```
$ flash-copy-mapping VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume get

Alias: get

Description: View details of a volume.

Usage:

```
$ get VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume list

Alias: list, ls

Description: List all storage volumes in a workspace.

Usage: list [--auxiliary=True|False] [--long=True|False] [--replication-enabled=True|False]

Available Options:

```
$ -a, --auxiliary      Filter auxiliary volumes if set to "true" or non-auxiliary volumes if set to "false". Default is "true".
-l, --long            Retrieve all volume details.
-r, --replication-enabled  Filter replication-enabled volumes if set to "true" or non-replication-enabled volumes if set to "false". Default is "true".
```

ibmcloud pi volume onboarding

Alias: onboarding, on

Description: IBM Cloud Power Virtual Server Volume Onboarding.

Usage: onboarding

Available Commands:

- `create` : Create a volume onboarding operation.
 - `get` : Get the information of volume onboarding operation.
 - `list` : List all volume onboarding operations.
-

ibmcloud pi volume onboarding create

Alias: create, cr

Description: Create a volume onboarding operation.

Usage: create <--auxiliary-volumes "AUXVOLUMENAME1 [NAME1]"> --source-crn SOURCE_CRN [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]]

Available Options:

```
$ -a, --auxiliary-volumes strings  Comma separated list of identifiers of the volume(s) at storage host level. Repeat this option to add more auxiliary volumes.
-d, --description string          Volume onboarding description.
-s, --source-crn string           The CRN of the workspace in which the primary volume is located.
-u, --user-tags strings           Comma separated list of user tags to be attached to the volume onboarding.
```

Examples:

```
$ ibmcloud pi volume onboarding create --auxiliary-volumes "test-auxiliary-volume-name" --source-crn "crn:v1:staging:public:power-iaas:dal13:a/mockcrn::"
```

ibmcloud pi volume onboarding get

Alias: get

Description: Get the information of volume onboarding operation.

Usage:

```
$ get VOLUME_ONBOARDING_ID

VOLUME_ONBOARDING_ID: The unique identifier of the onboarding operation.
```

ibmcloud pi volume onboarding list

Alias: list, ls

Description: List all volume onboarding operations.

Usage: list

ibmcloud pi volume remote-copy-relationship

Alias: remote-copy-relationship, rcr

Description: Get the remote copy relationship information of a volume.

Usage:

```
$ remote-copy-relationship VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume snapshot

Alias: snapshot, snap

Description: IBM Cloud Power Virtual Server Volume Snapshot.

Usage: snapshot

Available Commands:

- get : Get a snapshot for a volume.
- list : List all snapshots in the workspace.

ibmcloud pi volume snapshot get

Alias: get

Description: Get a snapshot for a volume.

Usage:

```
$ get SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi volume snapshot list

Alias: list, ls

Description: List all snapshots in the workspace.

Usage: list

ibmcloud pi volume update

Alias: update, upd

Description: Update a volume.

Usage:

```
$ update VOLUME_ID [--bootable=True|False] [--name NAME] [--size SIZE] [--shareable=True|False]

VOLUME_ID: The unique identifier or name of the volume.
```

Available Options:

```
$ -b, --bootable      New value for whether the volume can be booted.
-n, --name string     New name of the volume.
-t, --shareable       New value for whether the volume can be attached to multiple VMs.
-s, --size int        New size of the volume. Size cannot exceed 200 GiB for storage tier "Tier5k".
```

ibmcloud pi volume-group

Alias: volume-group, vg

Description: IBM Cloud Power Virtual Server Volume Groups.

Usage: volume-group

Available Commands:

- action : Perform an action on a volume group.
- create : Create a volume group.
- delete : Delete a volume group.
- get : View details of a volume group.

- `list` : List all storage volume groups in a workspace.
- `remote-copy-relationships` : Get the remote copy relationship information of a volume group.
- `storage-details` : Get storage details for a volume group.
- `update` : Update a volume group.

ibmcloud pi volume-group action

Alias: `action`, `act`

Description: Perform an action on a volume group.

Usage:

```
$ action VOLUME_GROUP_ID --operation reset [--status STATUS]
pi volume-group action VOLUME_GROUP_ID --operation start [--source SOURCE]
pi volume-group action VOLUME_GROUP_ID --operation stop [--allow-read-access=True|False]
```

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

Available Options :

```
$ -a, --allow-read-access  Allow the auxiliary volume to be accessible after stopping the volume group. Default is "false".
-o, --operation string    Operation to be done in a volume group. Valid values are: reset, start, stop.
    --source string       The copying volume source. Valid values are: auxiliary, master.Default is "master".
    --status string       New status to be set for a volume group. Default is "available".
```

Examples:

```
$  ibmcloud pi volume-group action test-volume-group --operation reset
   ibmcloud pi volume-group action test-volume-group --operation start --source master
```

ibmcloud pi volume-group create

Alias: `create`, `cr`

Description: Create a volume group.

Usage:

```
$ create (--volume-group-name VOLUME_GROUP_NAME | --consistency-group-name CONSISTENCY_GROUP_NAME) --member-volume-ids "VOLUME1[,VOLUMEn]"
```

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

Available Options :

```
$ -c, --consistency-group-name string  Storage volume group name. This is required to onboard existing volume group on the target site for DR set up.
-m, --member-volume-ids strings       Comma separated member volume identifiers.
-v, --volume-group-name string        Storage volume group name. This is required for the creation of new volume group.
```

Examples:

```
$  ibmcloud pi volume-group create --volume-group-name test-volume-group-name --member-volume-ids "43064761-948f-469d-ac8e-b8e5f0d6056f,85716e61-948f-309d-de8b-c4e5f0d3126d"
```

ibmcloud pi volume-group delete

Alias: `delete`, `del`

Description: Delete a volume group.

Usage:

```
$ delete VOLUME_GROUP_ID
```


VOLUME_GROUP_ID: The unique identifier or name of the volume group.

ibmcloud pi volume-group get

Alias: `get`

Description: View details of a volume group.

Usage:

\$ get VOLUME_GROUP_ID [--long=True|False]

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

Available Options :

\$ -l, --long Retrieve additional details for the volume group.

ibmcloud pi volume-group list

Alias: `list`, `ls`

Description: List all storage volume groups in a workspace.

Usage: `list [--long=True|False]`

Available Options :

\$ -l, --long Retrieve additional details for the volume group.

ibmcloud pi volume-group remote-copy-relationships

Alias: `remote-copy-relationships`, `rcr`

Description: Get the remote copy relationship information of a volume group.

Usage:

\$ remote-copy-relationships VOLUME_GROUP_ID

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

ibmcloud pi volume-group storage-details

Alias: `storage-details`, `sd`

Description: Get storage details for a volume group.

Usage:

\$ storage-details VOLUME_GROUP_ID

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

ibmcloud pi volume-group update

Alias: `update`, `upd`

Description: Update a volume group.

Usage:

\$ update VOLUME_GROUP_ID [--add-member-volume-ids "VOLUME1[,VOLUMEn]"] [--remove-member-volume-ids "VOLUME1[,VOLUMEn]"]

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

Available Options:

\$ -a, --add-member-volume-ids strings Comma separated volume identifiers to add as members of the volume group.
-r, --remove-member-volume-ids strings Comma separated volume identifiers to remove as members of the volume group.

ibmcloud pi vpn

Alias: vpn

Description: [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking.

Usage: vpn

Available Commands:

- create : [DEPRECATED] Create a VPN connection.
- delete : [DEPRECATED] Delete a VPN connection.
- get : [DEPRECATED] View details of a VPN connection.
- list : [DEPRECATED] List all VPN connections.
- peer-subnet : [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking Peer-Subnets.
- subnet : [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking Subnets.
- update : [DEPRECATED] Update a VPN connection.

ibmcloud pi vpn create

Alias: create, cr

Description: [DEPRECATED] Create a VPN connection.

Usage:

\$ create VPN_CONNECTION_NAME --ike-policy-id IKE_POLICY_ID --ipsec-policy-id IPSEC_POLICY_ID --mode MODE --peer-gateway-address PEER_GATEWAY --peer-subnet-cidrs "CIDR1 [CIDRn]" --subnets "ID1 [IDn]"

VPN_CONNECTION_NAME: A unique name of the VPN connection.

Available Options:

\$ -k, --ike-policy-id string Unique ID of IKE policy selected for this VPN connection.
-p, --ipsec-policy-id string Unique ID of IPSec policy selected for this VPN connection.
-m, --mode string Mode to be used by the VPN connection and cannot be updated later. Valid values are 'route' and 'policy'.
-g, --peer-gateway-address string IP address of the peer gateway attached to this VPN connection.
-c, --peer-subnet-cidrs strings Comma separated list of peer subnet CIDRs.
-s, --subnets strings Comma separated list of subnet IDs attached to this VPN connection.

ibmcloud pi vpn delete

Alias: delete, del

Description: [DEPRECATED] Delete a VPN connection.

Usage:

\$ delete VPN_CONNECTION_ID

VPN_CONNECTION_ID: The unique identifier of the VPN connection.

ibmcloud pi vpn get

Alias: get

Description: [DEPRECATED] View details of a VPN connection.

Usage:

```
$ get VPN_CONNECTION_ID

VPN_CONNECTION_ID: The unique identifier of the VPN connection.
```

ibmcloud pi vpn list

Alias: list, ls

Description: [DEPRECATED] List all VPN connections.

Usage: list

ibmcloud pi vpn peer-subnet

Alias: peer-subnet, pnet

Description: [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking Peer-Subnets.

Usage: peer-subnet

Available Commands:

- attach : [DEPRECATED] Attach a peer subnet to a specific VPN connection.
- detach : [DEPRECATED] Detach a peer subnet from a specific VPN connection.
- list : [DEPRECATED] Get a list of peer subnets attached to a specific VPN connection.

ibmcloud pi vpn peer-subnet attach

Alias: attach, att

Description: [DEPRECATED] Attach a peer subnet to a specific VPN connection.

Usage:

```
$ attach VPN_CONNECTION_ID --peer-subnet-cidr CIDR

VPN_CONNECTION_ID: The unique identifier of the VPN connection.
```

Available Options:

```
$ -p, --peer-subnet-cidr string Peer subnet CIDR to attach to the VPN connection.
```

Examples:

```
$ ibmcloud pi vpn peer-subnet attach 43064761-948f-469d-ac8e-b8e5f0d6056f --peer-subnet-cidr 128.170.1.0/20
```

ibmcloud pi vpn peer-subnet detach

Alias: detach, det

Description: [DEPRECATED] Detach a peer subnet from a specific VPN connection.

Usage:

```
$ detach VPN_CONNECTION_ID --peer-subnet-cidr CIDR

VPN_CONNECTION_ID: The unique identifier of the VPN connection.
```

Available Options:

```
$ -p, --peer-subnet-cidr string Peer subnet CIDR to detach from this VPN connection.
```

Examples:

```
$ ibmcloud pi vpn peer-subnet detach 43064761-948f-469d-ac8e-b8e5f0d6056f --peer-subnet-cidr 128.170.1.0/20
```

ibmcloud pi vpn peer-subnet list

Alias: list, ls

Description: [DEPRECATED] Get a list of peer subnets attached to a specific VPN connection.

Usage: list VPN_CONNECTION_ID

ibmcloud pi vpn subnet

Alias: subnet, snet

Description: [DEPRECATED] IBM Cloud Power Virtual Server Virtual Private Networking Subnets.

Usage: subnet

Available Commands:

- attach : [DEPRECATED] Attach a subnet to a specific VPN connection.
- detach : [DEPRECATED] Detach a subnet to a specific VPN connection.
- list : [DEPRECATED] Get a list of subnets attached to a specific VPN connection.

ibmcloud pi vpn subnet attach

Alias: attach, att

Description: [DEPRECATED] Attach a subnet to a specific VPN connection.

Usage:

```
$ attach VPN_CONNECTION_ID --subnet ID

VPN_CONNECTION_ID: The unique identifier of the VPN connection.
```

Available Options:

```
$ -s, --subnet string Subnet ID to attach to the VPN connection.
```

Examples:

```
$ ibmcloud pi vpn subnet attach 43064761-948f-469d-ac8e-b8e5f0d6056f --subnet 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi vpn subnet detach

Alias: detach, det

Description: [DEPRECATED] Detach a subnet to a specific VPN connection.

Usage:

```
$ detach VPN_CONNECTION_ID --subnet ID

VPN_CONNECTION_ID: The unique identifier of the VPN connection.
```

Available Options:

```
$ -s, --subnet string Subnet ID to detach from the VPN connection.
```

Examples:

```
$ ibmcloud pi vpn subnet detach 43064761-948f-469d-ac8e-b8e5f0d6056f --subnet 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi vpn subnet list

Alias: list, ls

Description: [DEPRECATED] Get a list of subnets attached to a specific VPN connection.

Usage: list VPN_CONNECTION_ID

ibmcloud pi vpn update

Alias: update, upd

Description: [DEPRECATED] Update a VPN connection.

Usage:

```
$ update VPN_CONNECTION_ID [--name VPN_CONNECTION_NAME] [--peer-gateway-address PEER_GATEWAY] [--ike-policy-id IKE_POLICY_ID] [--ipsec-policy-id IPSEC_POLICY_ID]
```

VPN_CONNECTION_ID: The unique identifier of the VPN connection.

Available Options:

```
$ -k, --ike-policy-id string      New ID of IKE policy selected for this VPN connection.
-p, --ipsec-policy-id string     New ID of IPSec policy selected for this VPN connection.
-n, --name string               New unique name of this VPN connection.
-g, --peer-gateway-address string New IP address of the peer gateway attached to this VPN connection.
```

ibmcloud pi workspace

Alias: workspace, ws

Description: IBM Cloud Power Virtual Server Workspaces.

Usage: workspace

Available Commands:

- action**: Perform an action on a workspace.
- context**: Display the CRN of the currently targeted workspace.
- create**: Create a workspace.
- delete**: Delete a workspace.
- get**: View details of a workspace.
- list**: List all workspaces for this account.
- target**: Target a workspace.

ibmcloud pi workspace action

Alias: action, act

Description: Perform an action on a workspace.

Usage:

```
$ action WORKSPACE_ID --operation OPERATION
```

WORKSPACE_ID: The unique identifier of the workspace.

Available Options:

```
$ -o, --operation string  Operation to be done in a workspace. Valid values are: per-migrate-start, per-migrate-validate.
```


Examples:

```
$ ibmcloud pi workspace action 85716e61-948f-309d-de8b-c4e5f0d3126d --operation per-migrate-start
```

ibmcloud pi workspace context

Alias: context, con

Description: Display the CRN of the currently targeted workspace.

Usage: context

ibmcloud pi workspace create

Alias: create, cr

Description: Create a workspace.

Usage:

```
$ create WORKSPACE_NAME --datacenter DATACENTER --group RESOURCE_GROUP --plan PLAN [--parameters "KEY1 VALUE1"["KEYn VALUEn"]] [--user-tags USER_TAG1[,USER_TAGn]]

WORKSPACE_NAME: The desired name of the workspace.
```

Available Options:

```
$ -d, --datacenter string    The datacenter location where the instance should be hosted.
                             Use the "ibmcloud pi datacenter list" command to see values for public datacenters,
                             and the "ibmcloud sat location ls" command to see values for private datacenters.
                             When using the "ibmcloud sat location ls" to get a private datacenter,
                             please prefix "satloc_<REGION>_" to the displayed id.
-g, --group string          The ID of the resource group. Use the "ibmcloud resource groups" command to see possible values.
--parameters strings       List of key-value parameter pairs. Each pair must be a quoted string in the format "KEY VALUE".
                             Example: --parameters "env dev" "version 1.0"
-p, --plan string           Plan associated with the offering. Valid values are "public"/"off-prem" or "private"/"on-prem".
-u, --user-tags strings     Comma separated list of user tags to be attached to the workspace.
```

Examples:

```
$ ibmcloud pi workspace create test-workspace --datacenter dal12 --group 393843beabcdea9d803840a384093841 --plan public --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi workspace delete

Alias: delete, del

Description: Delete a workspace.

Usage:

```
$ delete (WORKSPACE_CRN | WORKSPACE_ID)

WORKSPACE_CRN: The cloud resource name that uniquely identifies IBM Cloud resources.

WORKSPACE_ID: The unique identifier of the workspace.
```

ibmcloud pi workspace get

Alias: get

Description: View details of a workspace.

Usage:


```
$ get WORKSPACE_ID
```

WORKSPACE_ID: The unique identifier of the workspace.

ibmcloud pi workspace list

Alias: list, ls

Description: List all workspaces for this account.

Usage: list ([--long=True|False | --private=True|False])

Available Options:

```
$ -l, --long    Retrieve all workspace details.
-p, --private  List private workspaces.
```

ibmcloud pi workspace target

Alias: target, tg

Description: Target a workspace.

Usage:

```
$ target WORKSPACE_CRN
```

WORKSPACE_CRN: The cloud resource name that uniquely identifies IBM Cloud resources.

IBM Power Virtual Server CLI version 1.10.0 for Client location

IBM Power Virtual Server Private Cloud in Client location

The following list of commands are available with command-line interface (CLI) for IBM Power Virtual Server in Client location.

ibmcloud pi

Alias: pi

Description: Manage IBM Cloud Power Virtual Server service.

Usage: pi

Available Commands:

- datacenter : IBM Cloud Power Virtual Server Datacenters.
- disaster-recovery : List disaster recovery locations for the current region or all regions.
- image : IBM Cloud Power Virtual Server Images.
- instance : IBM Cloud Power Virtual Server Instances.
- job : IBM Cloud Power Virtual Server Jobs.
- network-peer : IBM Cloud Power Virtual Server Network Peers.
- placement-group : IBM Cloud Power Virtual Server Placement Groups.
- route : IBM Cloud Power Virtual Server Network Routes.
- shared-processor-pool : IBM Cloud Power Virtual Server Shared Processor Pools.
- snapshot : [DEPRECATED] IBM Cloud Power Virtual Server Snapshots.
- ssh-key : IBM Cloud Power Virtual Server SSH-Keys.
- storage-pools : List all storage pools for the targeted region.
- storage-tiers : List all storage tiers for the targeted region.
- subnet : IBM Cloud Power Virtual Server Subnets.
- virtual-serial-number : IBM Cloud Power Virtual Server Virtual Serial Number.

- `volume` : IBM Cloud Power Virtual Server Volumes.
- `volume-group` : IBM Cloud Power Virtual Server Volume Groups.
- `workspace` : IBM Cloud Power Virtual Server Workspaces.

ibmcloud pi datacenter

Alias: `datacenter`, `dat`

Description: IBM Cloud Power Virtual Server Datacenters.

Usage: `datacenter`

Available Commands:

- `get` : View details of a datacenter.
- `list` : List all datacenter details.

ibmcloud pi datacenter get

Alias: `get`

Description: View details of a datacenter.

Usage:

```
$ get DATACENTER [--private=True|False]

DATACENTER: The name of the datacenter.
```

Available Options:

```
$ -p, --private  Retrieve additional details about a private datacenter that you own.
                  Required if getting information from a private/on-prem datacenter.
```

ibmcloud pi datacenter list

Alias: `list`, `ls`

Description: List all datacenter details.

Usage: `list [--long=True|False] [--private=True|False]`

Available Options:

```
$ -l, --long      Retrieve additional details for datacenters.
-p, --private    Retrieve additional details for private datacenters that you own.
```

ibmcloud pi disaster-recovery

Alias: `disaster-recovery`, `drl`

Description: List disaster recovery locations for the current region or all regions.

Usage: `disaster-recovery [--all-regions=True|False]`

Available Options:

```
$ -a, --all-regions  List disaster recovery locations for all regions.
```

ibmcloud pi image

Alias: `image`, `img`

Description: IBM Cloud Power Virtual Server Images.

Usage: `image`

Available Commands:

- `create` : Create a copy of an available stock image into this account; stock image names cannot be changed.
- `delete` : Delete an image.
- `export` : Export an image to IBM Cloud Object Storage.
- `export-show` : View details of the last image export job.
- `get` : View details of an image.
- `import` : Import an image from IBM Cloud Object Storage.
- `import-show` : View details of the last image import job.
- `list` : List all images in a workspace.
- `list-catalog` : List images available in the regional image catalog.

`ibmcloud pi image create`

Alias: `create`, `cr`

Description: Create a copy of an available stock image into this account; stock image names cannot be changed.

Usage:

```
$ create IMAGE_ID [--user-tags USER_TAG1[,USER_TAGn]]
```

IMAGE_ID: The unique identifier or name of the image.

Available Options:

```
$ -u, --user-tags strings  Comma separated list of user tags to be attached to the created image.
```

Examples:

```
$ ibmcloud pi image create 7100-05-05 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

`ibmcloud pi image delete`

Alias: `delete`, `del`

Description: Delete an image.

Usage:

```
$ delete IMAGE_ID
```

IMAGE_ID: The unique identifier or name of the image.

`ibmcloud pi image export`

Alias: `export`, `ex`

Description: Export an image to IBM Cloud Object Storage.

Usage:

```
$ export IMAGE_ID --access-key KEY --bucket BUCKET_NAME --region REGION_NAME --secret-key KEY [--checksum=True|False]
```

IMAGE_ID: The unique identifier or name of the image.

Available Options:

```
$ -a, --access-key string  Cloud Object Storage HMAC access key.
-b, --bucket string       Cloud Object Storage bucket name.
-c, --checksum            Creates a checksum file. Default is "false".
```

```
-r, --region string      Cloud Object Storage region au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
-s, --secret-key string  Cloud Object Storage HMAC secret key.
```

Examples:

```
$ ibmcloud pi image export 43064761-948f-469d-ac8e-b8e5f0d6056f --bucket "bucket/stock-images" --region us-east --access-key access-key-sample --secret-key secret-key-sample
```

ibmcloud pi image export-show

Alias: export-show, exs

Description: View details of the last image export job.

Usage:

```
$ export-show IMAGE_ID

IMAGE_ID: The unique identifier or name of the image.
```

ibmcloud pi image get

Alias: get

Description: View details of an image.

Usage:

```
$ get IMAGE_ID

IMAGE_ID: The unique identifier or name of the image.
```

ibmcloud pi image import

Alias: import, im

Description: Import an image from IBM Cloud Object Storage.

Usage:

```
$ import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --storage-pool POOL [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --affinity-policy affinity (--affinity-instance INSTANCE | --affinity-volume VOLUME) [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME [--bucket-access private] [--storage-tier STORAGE_TIER] --affinity-policy anti-affinity (--anti-affinity-instances "INSTANCE1 [INSTANCEn]" | --anti-affinity-volumes "VOLUME1 [VOLUMEn]") [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --access-key KEY --secret-key KEY --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --storage-pool POOL [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --affinity-policy affinity (--affinity-instance INSTANCE | --affinity-volume VOLUME) [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME
pi image import IMAGE_NAME --bucket-access public [--storage-tier STORAGE_TIER] --affinity-policy anti-affinity (--anti-affinity-instances "INSTANCE1 [INSTANCEn]" | --anti-affinity-volumes "VOLUME1 [VOLUMEn]") [--os-type OSTYPE] [--import-details "LICENSE,PRODUCT,VENDOR"] [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] --image-file-name IMAGE_FILE_NAME --bucket BUCKET_NAME --region REGION_NAME

IMAGE_NAME: The desired name of the image.
```

Available Options:

\$ -k, --access-key string	Cloud Object Storage HMAC access key.
-i, --affinity-instance string	PVM instance identifier or name to base volume affinity policy against. This is required if "--affinity-policy affinity" is specified and "--affinity-volume" is not provided.
-a, --affinity-policy string	Affinity policy for data volume being created. Valid values are: "affinity" and "anti-affinity". If "--storage-pool" is provided then this cannot be specified.
-v, --affinity-volume string	Volume identifier or name to base volume affinity policy against. This is required if "--affinity-policy affinity" is specified and "--affinity-instance" is not provided.
-j, --anti-affinity-instances strings	Comma separated list of instance identifiers or names to base volume anti-affinity policy against. This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-volumes" is not provided.
-w, --anti-affinity-volumes strings	Comma separated list of volume identifiers or names to base volume anti-affinity policy against. This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-instances" is not provided.
-b, --bucket string	Cloud Object Storage bucket name.
-u, --bucket-access string	Indicates the bucket access type. Valid values are: "public", "private". Private access requires access and secret keys. Default is "private".
-c, --checksum	Checks the checksum file from the COS bucket against the one computed on the downloaded image.
-n, --image-file-name string	The image file name.
-d, --import-details strings	Import details for SAP image. Must include a license, product and vendor. Valid license values: byol. Valid product values: Hana, Netweaver. Valid vendor values: SAP.
-o, --os-type string	Operating system contained in the image. Valid values are: "aix", "ibmi", "rhel", "sles". Required when importing a raw image.
-r, --region string	Cloud Object Storage region au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
-s, --secret-key string	Cloud Object Storage HMAC secret key.
-p, --storage-pool string	Storage pool where the image will be imported to. Use "ibmcloud pi storage-pools" to see available storage pools. If "--storage-pool" is provided then "--affinity-policy" cannot be specified.
-t, --storage-tier string	Tier of the disk storage. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region. Default to tier3 if not provided.
--user-tags strings	Comma separated list of user tags to be attached to the imported image.

Examples:

```
$ ibmcloud pi image import import-image-name --bucket-access public --image-file-name imageFileName.ova --bucket test-bucket --region us-east
ibmcloud pi image import import-image-name --bucket-access private --storage-tier tier1 --image-file-name imageFileName.ova --bucket test-bucket --region us-east --
access-key access-key-sample --secret-key secret-key-sample
```

ibmcloud pi image import-show

Alias: import-show, ims

Description: View details of the last image import job.

Usage: import-show

ibmcloud pi image list

Alias: list, ls

Description: List all images in a workspace.

Usage: list

ibmcloud pi image list-catalog

Alias: list-catalog, lc

Description: List images available in the regional image catalog.

Usage: list-catalog [--sap=True|False]

Available Options:

```
$ -s, --sap Include SAP images.
```

Examples:


```
$ ibmcloud pi image list-catalog --sap
```

ibmcloud pi instance

Alias: instance, ins

Description: IBM Cloud Power Virtual Server Instances.

Usage: instance

Available Commands:

- action : Perform an operation in a PVM server instance.
- capture : IBM Cloud Power Virtual Server Instance Capture.
- console : IBM Cloud Power Virtual Server Instance Console.
- create : Create a server instance.
- delete : Delete a server instance.
- get : View details of a server instance.
- list : List all server instances.
- operation : Perform an operation on an IBMi server instance.
- snapshot : IBM Cloud Power Virtual Server Instance Snapshots.
- subnet : IBM Cloud Power Virtual Server Instance Subnets.
- update : Update a server instance.
- virtual-serial-number : IBM Cloud Power Virtual Server Instance Virtual Serial Number.
- volume : IBM Cloud Power Virtual Server Instance Volumes.
- vpmem-volume : IBM Cloud Power Virtual Server Instance vPMEM Volumes.

ibmcloud pi instance action

Alias: action, act

Description: Perform an operation in a PVM server instance.

Usage:

```
$ action INSTANCE_ID --operation OPERATION

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -o, --operation string  Operation to be done in a PVM server instance.
                        Valid values are: dumprestart, hard-reboot, immediate-shutdown, reset-state, soft-reboot, start, stop.
```

Examples:

```
$ ibmcloud pi instance action 43064761-948f-469d-ac8e-b8e5f0d6056f --operation immediate-shutdown
```

ibmcloud pi instance capture

Alias: capture, cap

Description: IBM Cloud Power Virtual Server Instance Capture.

Usage: capture

Available Commands:

- create : Create a capture of a server instance.
- show : View the job details of the last server instance capture.

ibmcloud pi instance capture create

Alias: create, cr

Description: Create a capture of a server instance.

Usage:

```
$ create INSTANCE_ID --destination (cloud-storage | both) --name NAME --access-key KEY --image-path PATH --region REGION --secret-key KEY [--checksum=True|False] [-
-user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEen]]
pi instance capture create INSTANCE_ID --destination image-catalog --name NAME [--checksum=True|False] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes
VOLUME1[,VOLUMEen]]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -a, --access-key string    Cloud Object Storage HMAC access key. Required if destination is cloud-storage.
-c, --checksum              Creates a checksum file. Default is "false".
-d, --destination string    Destination for the deployable image. Valid values are: "cloud-storage", "image-catalog", "both".
-i, --image-path string     Cloud Object Storage image path. Required if destination is cloud-storage. E.g. bucket-name[/optional/folder].
-n, --name string           Name of the deployable image created for the captured instance.
-r, --region string         Cloud Object Storage regions au-syd, br-sao, ca-tor, che01, eu-de, eu-es, eu-gb, jp-osa, jp-tok, us-east, us-south.
                             Required if destination is cloud-storage.
-s, --secret-key string     Cloud Object Storage HMAC secret key. Required if destination is cloud-storage.
-u, --user-tags strings     Comma separated list of user tags to be attached to the instance capture.
-v, --volumes strings       Comma separated list of volume identifiers or names to include in the captured instance.
```

Examples:

```
$ ibmcloud pi instance capture create 43064761-948f-469d-ac8e-b8e5f0d6056f --destination both --name deployable-image-name --access-key access-key-sample --secret-
key secret-key-sample --region us-south --image-path "imageBucket/imageLocation"
```

ibmcloud pi instance capture show

Alias: show, sh

Description: View the job details of the last server instance capture.

Usage:

```
$ show INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance console

Alias: console, con

Description: IBM Cloud Power Virtual Server Instance Console.

Usage: console

Available Commands :

- get : Get the console of an instance.

ibmcloud pi instance console get

Alias: get

Description: Get the console of an instance.

Usage:

```
$ get INSTANCE_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance create

Alias: create, cr

Description: Create a server instance.

Usage:

```
$ create INSTANCE_NAME --image IMAGE --subnets "SUBNET1 [IP1]"["SUBNETn [IPn]"
  [--boot-volume-replication-enabled=True|False]
  [--default-trusted-profile (CRN | TRUSTED_PROFILE_ID | TRUSTED_PROFILE_NAME)
  [--default-trusted-profile-autolink=True|False]]
  [--IBMiCSS-license=True|False] [--IBMiPHA-license=True|False] [--IBMiRDS-users NUMBER_USERS]
  [--key-name NAME] [--memory MEMORY] [--metadata-service=True|False] [--pin-policy POLICY] [--placement-group GROUP_ID]
  [--preferred-processor-compatibility-mode MODE] [--processor-type PROC_TYPE] [--processors PROCESSORS]
  [--replicant-affinity-policy AFFINITY_POLICY] [--replicant-scheme SCHEME] [--replicants NUMBER]
  [--replication-sites SITE1[,SITEn]] [--shared-processor-pool SHARED_PROCESSOR_POOL]
  [--storage-affinity STORAGE_AFFINITY_POLICY] [--storage-affinity-instance INSTANCE]
  [--storage-affinity-volume VOLUME] [--storage-anti-affinity-instances INSTANCE1[,INSTANCEn]]
  [--storage-anti-affinity-volumes VOLUME1[,VOLUMEn]] [--storage-connection STORAGE_CONNECTION]
  [--storage-pool STORAGE_POOL] [--storage-pool-affinity=True|False] [--storage-tier STORAGE_TIER]
  [--sys-type TYPE] [--user-data USER_DATA] [--user-tags USER_TAG1[,USER_TAGn]]
  [--virtual-serial-number "(SERIAL | 'auto-assign')[,DESCRIPTION]" [--software-tier SOFTWARE_TIER]]
  [--virtual-cores ASSIGNED_CORES] [--volumes VOLUME1[,VOLUMEn]] [--vpmem-volumes "NAME SIZE[,NAMEn SIZEn"]]
```

INSTANCE_NAME: The name of the instance.

Available Options:

\$	--IBMiCSS-license	IBMi CSS software license associated with the instance.
	--IBMiPHA-license	IBMi PHA IASP Management software license associated with the instance.
	--IBMiRDS-users int	Number of IBMi RDS user software licenses associated with the instance. Default is 0.
-b,	--boot-volume-replication-enabled	Enables storage replication on the boot volume. Default is "false".
	--default-trusted-profile string	Trusted profile CRN, ID or name of the default IAM trusted profile to use for this instance.
-a,	--default-trusted-profile-autolink	If set to true, the system will create a link to the specified trusted profile during server creation.
-i,	--image string	Operating system image identifier or name.
-k,	--key-name string	Name of SSH key.
-m,	--memory float	Amount of memory in GiB to allocate to the instance. Default is 2 GiB.
	--metadata-service	Indicates whether the metadata service endpoint will be available to the virtual server.
	--pin-policy string	Pin policy. Valid values are: "none", "soft", "hard". Default is "none".
	--placement-group string	The placement group ID of the group that the server will be added to.
-c,	--preferred-processor-compatibility-mode string	The preferred processor compatibility mode. Valid values are: default, POWER7, POWER8, POWER9, POWER9_Base, POWER10, POWER11
-r,	--processor-type string	Type of processors: "shared" or "dedicated" or "capped". Default is "dedicated".
-p,	--processors float	Amount of processors to allocate to the instance. Default is 1 core.
	--replicant-affinity-policy string	Affinity policy to use when multicreate is used. Valid values are: affinity, anti-affinity, none.
	--replicant-scheme string	Naming scheme to use for duplicate VMs. Valid values are: prefix, suffix.
	--replicants int	The number of replicant instances to be created with this request.
		Values greater than 1 will result in the creation of multiple instances.
	--replication-sites strings	Indicates the replication sites of the boot volume. See "ibmcloud pi disaster-recovery" command to get a list of replication sites.
	--shared-processor-pool string	The shared processor pool ID of the pool that the server will be in.
	--software-tier string	IBMi licensing software tiers. Must be set with "--virtual-serial-number" option. Valid values are: P05, P10, P20, P30.
	--storage-affinity string	Affinity policy for storage pool selection. Valid values are: "affinity" and "anti-affinity".
		If "--storage-pool" is provided, then this option cannot be specified.
	--storage-affinity-instance string	PVM instance identifier or name to base storage affinity policy against.
		This is required if "--storage-affinity affinity" is specified and "--storage-affinity-volume" is not provided.
	--storage-affinity-volume string	Volume identifier or name to base storage affinity policy against.
		This is required if "--storage-affinity affinity" is specified and "--storage-affinity-instance" is not provided.
	--storage-anti-affinity-instances strings	Comma separated list of PVM instance identifiers or names to base storage affinity policy against.
		This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-volumes" is not provided.
	--storage-anti-affinity-volumes strings	Comma separated list of volume identifiers or names to base storage affinity policy against.
		This is required if "--storage-affinity anti-affinity" is specified and "--storage-anti-affinity-instances" is not provided.
	--storage-connection string	The storage connection type. Valid values are: maxVolumeSupport.
	--storage-pool string	Storage pool for server deployment (use "ibmcloud pi storage-pools" to see available storage pools).
		Only valid when you deploy one of the IBM supplied stock images.
		Storage tier and pool for a custom image (an imported image or an image that is created from a PVMInstance capture) defaults to the storage tier and pool the image was created in.

--storage-pool-affinity	Indicates if all volumes attached to the server must reside in the same storage pool. If set to false, then volumes from any storage tier and pool can be attached to the PVM instance. This setting affects PVM instance snapshot, capture, and clone operations. For capture and clone operations, only data volumes that are of the same storage tier and in the same storage pool of the boot volume of the PVM instance can be included.
-t, --storage-tier string	For snapshot operations, all data volumes to be included in the snapshot must reside in the same storage tier and storage pool. Once set to false, cannot be set back to true unless all volumes attached reside in the same storage tier and pool. Storage tier for server deployment when deploying a stock or custom image (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region). Default to tier3 if not provided.
-n, --subnets strings	Comma separated list of subnet identifiers or names and optional IP address to associate with the instance. Example: --subnets "subnet-name2 192.168.0.2,subnet-name3"
-s, --sys-type string	System type used to host the instance. Valid values are: s1022, e1050, e1080, s1122, e1150, e1180.
-u, --user-data string	The user data passed into the instance. Strings and file names are supported. File names must be prepended with "@".
--user-tags strings	Comma separated list of user tags to be attached to the instance.
--virtual-cores int	The number of virtual cores assigned.
--virtual-serial-number string	IBMi virtual serial number information added with the instance. Must include an existing virtual serial number or 'auto-assign' and optionally a description.
-v, --volumes strings	Comma separated list of volume identifiers or names to associate with the instance.
--vpmem-volumes strings	Comma separated list of vPMEM volume names and sizes to associate with the instance. The maximum number of vPMEM volumes for SAP instances is 1 otherwise it is 4.

Examples:

```
$ ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d
ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f --storage-tier tier1 --processors 1 --processor-type dedicated --deployment-type EPIC --sys-type e980
ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --storage-tier tier1 --replicants 2 --replicant-scheme prefix --replicant-affinity-policy affinity
ibmcloud pi instance create test-instance --image 43064761-948f-469d-ac8e-b8e5f0d6056f --subnets 85716e61-948f-309d-de8b-c4e5f0d3126d --boot-volume-replication-enabled --replication-sites dal10
```

ibmcloud pi instance delete

Alias: delete, del

Description: Delete a server instance.

Usage:

```
$ delete INSTANCE_ID [--delete-data-volumes=True|False] [--retainVSN=True|False]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -d, --delete-data-volumes Indicates whether all data volumes attached to the instance must be deleted.
Shared data volumes will be deleted if no other instances are attached.
-r, --retainVSN Determines if the virtual serial number will be retained after being removed from the instance. Default is "false".
```

ibmcloud pi instance get

Alias: get

Description: View details of a server instance.

Usage:

```
$ get INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance list

Alias: list, ls

Description: List all server instances.

Usage: `list`

ibmcloud pi instance operation

Alias: `operation, op`

Description: Perform an operation on an IBMi server instance.

Usage:

```
$ operation INSTANCE_ID (--operation-type "boot" --boot-mode MODE --boot-operating-mode MODE | --operation-type "job" --job-task TASK)
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -b, --boot-mode string      Name of the server boot mode. Valid values are: a, b, c, d.
-m, --boot-operating-mode string  Name of the server operating mode. Valid values are: manual, normal.
-j, --job-task string         Name of the job task to execute.
                               Valid values are: consoleservice, dston, dumprestart, iopdump, iopreset, remotedstoff, remotedston, retrydump.
-o, --operation-type string    Name of the operation to execute. Valid values are: boot, job.
```

Examples:

```
$ ibmcloud pi instance operation 43064761-948f-469d-ac8e-b8e5f0d6056f --operation-type boot --boot-mode c --boot-operating-mode normal
```

ibmcloud pi instance snapshot

Alias: `snapshot, snap`

Description: IBM Cloud Power Virtual Server Instance Snapshots.

Usage: `snapshot`

Available Commands:

- `create` : Create a snapshot for an instance.
- `delete` : Delete a snapshot from an instance.
- `get` : Get a snapshot for an instance
- `list` : List all snapshots for an instance or all snapshots in the workspace.
- `restore` : Restore a snapshot for an instance.
- `update` : Update a snapshot for an instance.

ibmcloud pi instance snapshot create

Alias: `create, cr`

Description: Create a snapshot for an instance.

Usage:

```
$ create INSTANCE_ID --name SNAPSHOT_NAME [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEn]]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -d, --description string  Snapshot description.
-n, --name string           Name of the snapshot.
-u, --user-tags strings     Comma separated list of user tags to be attached to the snapshot.
-v, --volumes strings       Comma separated list of volume identifiers or names to include in the snapshot.
                               If you do not specify this option or if the volumes list is empty,
                               all the volumes that are attached to the instance are included in the snapshot.
```

Examples:

```
$ ibmcloud pi instance snapshot create 85716e61-948f-309d-de8b-c4e5f0d3126d --name test-snapshot-name
```

ibmcloud pi instance snapshot delete

Alias: delete, del

Description: Delete a snapshot from an instance.

Usage:

```
$ delete SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi instance snapshot get

Alias: get

Description: Get a snapshot for an instance

Usage:

```
$ get SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi instance snapshot list

Alias: list, ls

Description: List all snapshots for an instance or all snapshots in the workspace.

Usage:

```
$ list [INSTANCE_ID]

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance snapshot restore

Alias: restore, res

Description: Restore a snapshot for an instance.

Usage:

```
$ restore INSTANCE_ID --snapshot SNAPSHOT_ID [--force] [--restore VALUE]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -f, --force      By default the VM must be shutoff during a snapshot restore, force set to true will relax the VM shutoff pre-condition.
-r, --restore string Action to take on a failed snapshot restore. Valid values for "--restore" are: retry, rollback.
-s, --snapshot string The unique identifier of the snapshot.
```

Examples:

```
$ ibmcloud pi instance snapshot restore 85716e61-948f-309d-de8b-c4e5f0d3126d --snapshot 43064761-948f-469d-ac8e-b8e5f0d6056f --force --restore retry
```


ibmcloud pi instance snapshot update

Alias: update, upd

Description: Update a snapshot for an instance.

Usage:

```
$ update SNAPSHOT_ID [--description DESCRIPTION] [--name NAME]
```

SNAPSHOT_ID: The unique identifier of the snapshot.

Available Options:

```
$ -d, --description string  New snapshot description.
-n, --name string          New name of the snapshot.
```

ibmcloud pi instance subnet

Alias: subnet, snet

Description: IBM Cloud Power Virtual Server Instance Subnets.

Usage: subnet

Available Commands:

- `attach`: Attach a subnet to a server instance.
- `detach`: Detach a specific subnet or mac address from a server instance.
- `list`: List all the attached subnets.

ibmcloud pi instance subnet attach

Alias: attach, att

Description: Attach a subnet to a server instance.

Usage:

```
$ attach INSTANCE_ID --subnet SUBNET_ID [--ip-address IP_ADDRESS]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -i, --ip-address string  The requested ip address of this subnet interface (192.168.1.0).
-n, --subnet string       The subnet ID.
```

Examples:

```
$ ibmcloud pi instance subnet attach 85716e61-948f-309d-de8b-c4e5f0d3126d --subnet 43064761-948f-469d-ac8e-b8e5f0d6056f --ip-address 192.168.0.12
```

ibmcloud pi instance subnet detach

Alias: detach, det

Description: Detach a specific subnet or mac address from a server instance.

Usage:

```
$ detach INSTANCE_ID --subnet SUBNET_ID [--mac-address MAC_ADDRESS]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:


```
$ -m, --mac-address string  The mac address of the subnet interface to be removed. Default is all mac addresses.
-n, --subnet string        The subnet ID.
```

Examples:

```
$  ibmcloud pi instance subnet detach 85716e61-948f-309d-de8b-c4e5f0d3126d --subnet 43064761-948f-469d-ac8e-b8e5f0d6056f --mac-address eb:26:8f:ba:5f:62
```

ibmcloud pi instance subnet list

Alias: `list`, `ls`

Description: List all the attached subnets.

Usage:

```
$ list INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance update

Alias: `update`, `upd`

Description: Update a server instance.

Usage:

```
$ update INSTANCE_ID
  [--default-trusted-profile (CRN | TRUSTED_PROFILE_ID | TRUSTED_PROFILE_NAME)
  [--default-trusted-profile-autolink=True|False]]
  [--IBMiCSS-license=True|False] [--IBMiPHA-license=True|False]
  [--IBMiRDS-users NUMBER_USERS] [--memory AMOUNT]
  [--metadata-service=True|False [--metadata-service-force=True|False]]
  [--name NAME] [--pin-policy POLICY] [--preferred-processor-compatibility-mode MODE]
  [--processor-type TYPE] [--processors NUMBER] [--storage-pool-affinity=True|False]
  [--virtual-cores ASSIGNED_CORES] [--virtual-optional-device ("attach" | "detach")]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

\$	--IBMiCSS-license	New IBMi CSS software license associated with the instance.
	--IBMiPHA-license	New IBMi PHA IASP Management software license associated with the instance.
	--IBMiRDS-users int	New number of IBMi RDS user software licenses associated with the instance.
	--default-trusted-profile string	Trusted profile CRN, ID or name of the default IAM trusted profile to use for this instance.
-a,	--default-trusted-profile-autolink	If set to true, the system will create a link to the specified trusted profile during server update.
-m,	--memory float	New amount of memory for the server instance.
	--metadata-service	Indicates whether the metadata service endpoint will be available to the virtual server.
	--metadata-service-force	If set to true, this allows the metadata service to be disabled while the VM is active, which may require manual interface cleanup by the OS administrator.
		This option is only supported for disabling the metadata service.
-n,	--name string	New name of the server instance.
	--pin-policy string	New pin policy for the server instance. Valid values are: "none", "soft", "hard".
-c,	--preferred-processor-compatibility-mode string	The preferred processor compatibility mode. Valid values are: default, POWER7, POWER8, POWER9, POWER9_Base, POWER10, POWER11
-r,	--processor-type string	New processor type for the server instance.
-p,	--processors float	New amount of processors for the server instance.
-s,	--storage-pool-affinity	Indicates if all volumes attached to the server must reside in the same storage pool.
		If set to false, then volumes from any storage tier and pool can be attached to the PVM instance.
		This setting affects PVM instance snapshot, capture, and clone operations. For capture and clone operations, only data volumes that are of the same storage tier and in the same storage pool of the boot volume of the PVM instance can be included.
		For snapshot operations, all data volumes to be included in the snapshot must reside in the same storage tier and storage pool.
		Once set to false, cannot be set back to true unless all volumes attached reside in the same storage tier and pool.
	--virtual-cores int	New number of virtual cores assigned.
-v,	--virtual-optical-device string	Attach or detach a virtual optical device to this instance. Valid values are: attach, detach.

Examples:

```
$ ibmcloud pi instance update 43064761-948f-469d-ac8e-b8e5f0d6056f --memory 2 --name new-instance-name --pin-policy hard --processors 2 --processor-type shared --storage-pool-affinity
```

ibmcloud pi instance virtual-serial-number

Alias: virtual-serial-number, vsn

Description: IBM Cloud Power Virtual Server Instance Virtual Serial Number.

Usage: virtual-serial-number

Available Commands:

- assign : Assign a virtual serial number to a virtual server instance.
- get : Get a virtual serial number assigned to a virtual server instance.
- unassign : Unassign a virtual serial number from a virtual server instance.
- update : Update a virtual serial number assigned to a virtual server instance.

ibmcloud pi instance virtual-serial-number assign

Alias: assign, asn

Description: Assign a virtual serial number to a virtual server instance.

Usage:

```
$ assign INSTANCE_ID [--description DESCRIPTION] [--serial SERIAL]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options:

```
$ -d, --description string  Description for the virtual serial number.
-s, --serial string        Virtual serial number to attach. If no serial is specified or 'auto-assign' is set, a virtual serial number will be automatically generated.
```

Examples:

```
$ ibmcloud pi instance virtual-serial-number assign 43064761-948f-469d-ac8e-b8e5f0d6056f --description "new-virtual-serial-number-description"
```

ibmcloud pi instance virtual-serial-number get

Alias: get

Description: Get a virtual serial number assigned to a virtual server instance.

Usage:

```
$ get INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance virtual-serial-number unassign

Alias: unassign, uasn

Description: Unassign a virtual serial number from a virtual server instance.

Usage:

```
$ unassign INSTANCE_ID [--retainVSN=True|False]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -r, --retainVSN    Determines if virtual serial number will be retained after being unassigned from the instance. Default is "false".
```

Examples:

```
$  ibmcloud pi instance virtual-serial-number unassign 43064761-948f-469d-ac8e-b8e5f0d6056f --retainVSN
```

ibmcloud pi instance virtual-serial-number update

Alias: `update`, `upd`

Description: Update a virtual serial number assigned to a virtual server instance.

Usage:

```
$ update INSTANCE_ID (--description DESCRIPTION | --software-tier SOFTWARE_TIER)

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -d, --description string    New virtual serial number description.
    --software-tier string    IBMi licensing software tiers. Valid values are: P05, P10, P20, P30.
```

ibmcloud pi instance volume

Alias: `volume`, `vol`

Description: IBM Cloud Power Virtual Server Instance Volumes.

Usage: `volume`

Available Commands :

- `attach` : Attach one or more volumes to an instance.
- `bulk-detach` : Detach multiple volumes from an instance.
- `detach` : Detach a volume from an instance.
- `list` : List all the attached volumes.

ibmcloud pi instance volume attach

Alias: `attach`, `att`

Description: Attach one or more volumes to an instance.

Usage:

```
$ attach INSTANCE_ID --volumes VOLUME1[,VOLUMEn] [--boot-volume BOOT_VOLUME]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -b, --boot-volume string    Boot volume to be attached to an instance.
-v, --volumes strings        Comma separated list of volume identifiers or names to associate with the instance.
```

Examples:

```
$  ibmcloud pi instance volume attach 85716e61-948f-309d-de8b-c4e5f0d3126d --boot-volume e5d394ef-3156-21a3-8b12-de15c6d5e276 --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f,32064761-298e-469a-bc6f-d8e5f0d6067a
```

ibmcloud pi instance volume bulk-detach

Alias: bulk-detach, bdet

Description: Detach multiple volumes from an instance.

Usage:

```
$ bulk-detach INSTANCE_ID [--detach-all=True|False | --volumes VOLUME1[,VOLUMEn]] [--detach-primary=True|False]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -a, --detach-all      Indicates if all volumes, except primary boot volume, attached to the instance should be detached. Required if "--volumes" is not provided.
-p, --detach-primary    Indicates if primary boot volume attached to the instance should be detached.
-v, --volumes strings   List of volumes to be detached from an instance. Required if "--detach-all" is not provided.
```

Examples:

```
$  ibmcloud pi instance volume bulk-detach 85716e61-948f-309d-de8b-c4e5f0d3126d --detach-all --detach-primary
```

ibmcloud pi instance volume detach

Alias: detach, det

Description: Detach a volume from an instance.

Usage:

```
$ detach INSTANCE_ID --volume VOLUME_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -v, --volume string   Volume to detach from the instance.
```

Examples:

```
$  ibmcloud pi instance volume detach 85716e61-948f-309d-de8b-c4e5f0d3126d --volume 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi instance volume list

Alias: list, ls

Description: List all the attached volumes.

Usage:

```
$ list INSTANCE_ID

INSTANCE_ID: The unique identifier or name of the instance.
```

ibmcloud pi instance vpmem-volume

Alias: vpmem-volume, vpmem, vpmemvol

Description: IBM Cloud Power Virtual Server Instance vPMEM Volumes.

Usage: vpmem-volume

Available Commands :

- attach : Attach a vPMEM volume to an instance.

- detach : Detach a vPMEM volume from an instance.
- get : Get a vPMEM volume attached to an instance.
- list : List all vPMEM volumes attached to an instance.
- update : Update a server instance.

ibmcloud pi instance vpmem-volume attach

Alias: attach, att

Description: Attach a vPMEM volume to an instance.

Usage:

```
$ attach INSTANCE_ID --vpmem-volumes "NAME SIZE[,\"NAMEn SIZEn\"]" [--user-tags USER_TAG1[,USER_TAGn]]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -u, --user-tags strings      User tags to add to the created vPMEM volume(s).
    --vpmem-volumes strings    Comma separated list of vPMEM volume names and sizes to associate with the instance.
                                The maximum number of vPMEM volumes for SAP instances is 1 otherwise it is 4.
```

ibmcloud pi instance vpmem-volume detach

Alias: detach, det

Description: Detach a vPMEM volume from an instance.

Usage:

```
$ detach INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -v, --vpmem-volume string    vPMEM volume ID that is associated with the PVM instance.
```

ibmcloud pi instance vpmem-volume get

Alias: get

Description: Get a vPMEM volume attached to an instance.

Usage:

```
$ get INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -v, --vpmem-volume string    vPMEM volume ID that is associated with the PVM instance.
```

ibmcloud pi instance vpmem-volume list

Alias: list, ls

Description: List all vPMEM volumes attached to an instance.

Usage:


```
$ list INSTANCE_ID
```

INSTANCE_ID: The unique identifier or name of the instance.

ibmcloud pi instance vpmem-volume update

Alias: `update`, `upd`

Description: Update a server instance.

Usage:

```
$ update INSTANCE_ID --vpmem-volume VPMEM_VOLUME_ID --name NAME
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options:

```
$ -n, --name string      New name of the vPMEM volume.
-v, --vpmem-volume string vPMEM volume ID that is associated with the PVM instance.
```

ibmcloud pi job

Alias: `job`

Description: IBM Cloud Power Virtual Server Jobs.

Usage: `job`

Available Commands:

- `delete`: Delete a job.
- `get`: View details of a job.
- `list`: List up to the last 5 jobs in a workspace.

ibmcloud pi job delete

Alias: `delete`, `del`

Description: Delete a job.

Usage:

```
$ delete JOB_ID
```

JOB_ID: The unique identifier of the job.

ibmcloud pi job get

Alias: `get`

Description: View details of a job.

Usage:

```
$ get JOB_ID
```

JOB_ID: The unique identifier of the job.

ibmcloud pi job list

Alias: `list`, `ls`

Description: List up to the last 5 jobs in a workspace.

Usage: `list [--operation-action ACTION] [--operation-id ID] [--operation-target TARGET]`

Available Options:

```
$ -a, --operation-action string  Operation action to filter returned jobs. Valid values are: epalmageImport, epaVMCapture, imageExport, imageImport, storage, vmCapture.
-i, --operation-id string       Operation ID to filter returned jobs.
-t, --operation-target string   Operation target to filter returned jobs. Valid values are: cloudConnection, image, pvmlInstance.
```

ibmcloud pi network-peer

Alias: `network-peer, np`

Description: IBM Cloud Power Virtual Server Network Peers.

Usage: `network-peer`

Available Commands:

- `create` : Create a network peer.
- `delete` : Delete a network peer.
- `get` : View details of a network peer.
- `interfaces` : List all network peer interfaces.
- `list` : List all network peers.
- `route-filter` : IBM Cloud Power Virtual Server Route Filters.
- `update` : Update a network peer.

ibmcloud pi network-peer create

Alias: `create, cr`

Description: Create a network peer.

Usage:

```
$ create NETWORK_PEER_NAME --customer-asn ASN --customer-cidr CIDR
  --ibm-asn ASN --ibm-cidr CIDR --peer-interface-id PEER_INTERFACE_ID --type TYPE --vlan VLAN
  [--default-export-route-filter ("allow" | "deny")] [--default-import-route-filter ("allow" | "deny")]

NETWORK_PEER_NAME: The desired name of the network peer.
```

Available Options:

```
$  --customer-asn int           The ASN number at customer network side.
  --customer-cidr string       The IP address used for configuring customer network interface with network subnet mask.
                                "--customer-cidr" and "--ibm-cidr" must have matching network and subnet mask values.
-e, --default-export-route-filter string  Default action for export route filter. Valid values are: allow, deny. Default is "allow".
-i, --default-import-route-filter string  Default action for import route filter. Valid values are: allow, deny. Default is "allow".
  --ibm-asn int               The ASN number at IBM PowerVS side.
  --ibm-cidr string           The IP address used for configuring IBM network interface with network subnet mask.
                                "--customer-cidr" and "--ibm-cidr" must have matching network and subnet mask values.
-p, --peer-interface-id string  Peer interface ID.
-t, --type string              Type of the peer network. Valid values are: dcnetwork_bgp.
-v, --vlan int                 The VLAN configured at the customer network.
```

Examples:

```
$  ibmcloud pi network-peer create test-network-peer --customer-asn 64512 --customer-cidr 192.168.91.1/30 --ibm-asn 64513 --ibm-cidr 192.168.91.2/30 --peer-interface-id 85716e61-948f-309d-de8b-c4e5f0d3126d --type dcnetwork_bgp --vlan 10
```

ibmcloud pi network-peer delete

Alias: `delete, del`

Description: Delete a network peer.

Usage:

```
$ delete NETWORK_PEER_ID
```

NETWORK_PEER_ID: The unique identifier of the network peer.

ibmcloud pi network-peer get

Alias: `get`

Description: View details of a network peer.

Usage:

```
$ get NETWORK_PEER_ID
```

NETWORK_PEER_ID: The unique identifier of the network peer.

ibmcloud pi network-peer interfaces

Alias: `interfaces, int`

Description: List all network peer interfaces.

Usage: `interfaces`

ibmcloud pi network-peer list

Alias: `list, ls`

Description: List all network peers.

Usage: `list`

ibmcloud pi network-peer route-filter

Alias: `route-filter, rf`

Description: IBM Cloud Power Virtual Server Route Filters.

Usage: `route-filter`

Available Commands:

- `create`: Create a network peer route-filter.
- `delete`: Delete a network peer route-filter.
- `get`: View details of a network peer route-filter.

ibmcloud pi network-peer route-filter create

Alias: `create, cr`

Description: Create a network peer route-filter.

Usage:

```
$ create NETWORK_PEER_ID --action ("allow" | "deny") --direction ("export" | "import")
--index INDEX --prefix PREFIX [--ge PREFIX-SET-MIN] [--le PREFIX-SET-MAX]
```

NETWORK_PEER_ID: The unique identifier of the network peer.

Available Options:

```
$ -a, --action string      Action of the filter. Valid values are: allow, deny.
-d, --direction string    Direction of the filter. Valid values are: export, import.
-g, --ge int              The minimum matching length of the prefix-set. Values range from 1 to 32 inclusive,
```

	and must be less than or equal to the value of the "--le" option.
-i, --index int	Priority or order of the filter.
-l, --le int	The maximum matching length of the prefix-set. Values range from 1 to 32 inclusive, and must be greater than or equal to the value of the "--ge" option.
-p, --prefix string	IP prefix representing an address and mask length of the prefix-set.

ibmcloud pi network-peer route-filter delete

Alias: delete, del

Description: Delete a network peer route-filter.

Usage:

```
$ delete ROUTE_FILTER_ID --network-peer-id NETWORK_PEER_ID
```

ROUTE_FILTER_ID: The unique identifier of the route filter.

Available Options:

```
$ -n, --network-peer-id string  Network peer ID that is associated with the route filter.
```

ibmcloud pi network-peer route-filter get

Alias: get

Description: View details of a network peer route-filter.

Usage:

```
$ get ROUTE_FILTER_ID --network-peer-id NETWORK_PEER_ID
```

ROUTE_FILTER_ID: The unique identifier of the route filter.

Available Options:

```
$ -n, --network-peer-id string  Network peer ID that is associated with the route filter.
```

ibmcloud pi network-peer update

Alias: update, upd

Description: Update a network peer.

Usage:

```
$ update NETWORK_PEER_ID [--customer-asn ASN] [--customer-cidr CIDR]
  [--default-export-route-filter ("allow" | "deny")]
  [--default-import-route-filter ("allow" | "deny")]
  [--ibm-asn ASN] [--ibm-cidr CIDR] [--name NAME] [--type TYPE] [--vlan VLAN]
```

NETWORK_PEER_ID: The unique identifier of the network peer.

Available Options:

\$	--customer-asn int	The ASN number at customer network side.
	--customer-cidr string	The IP address used for configuring customer network interface with network subnet mask.
		"--customer-cidr" and "--ibm-cidr" must have matching network and subnet mask values.
-e, --default-export-route-filter string		Default action for export route filter. Valid values are: allow, deny.
-i, --default-import-route-filter string		Default action for import route filter. Valid values are: allow, deny.
	--ibm-asn int	The ASN number at IBM PowerVS side.
	--ibm-cidr string	The IP address used for configuring IBM network interface with network subnet mask.
		"--customer-cidr" and "--ibm-cidr" must have matching network and subnet mask values.
-n, --name string		New name of the network peer.
-t, --type string		Type of the peer network. Valid values are: dcnetwork_bgp.
-v, --vlan int		The VLAN configured at the customer network.

ibmcloud pi placement-group

Alias: placement-group, pg

Description: IBM Cloud Power Virtual Server Placement Groups.

Usage: placement-group

Available Commands:

- `create` : Create a server placement group.
 - `delete` : Delete a server placement group.
 - `get` : View details of a server placement group.
 - `list` : List all server placement groups.
 - `server-add` : Add a server to a server placement group.
 - `server-remove` : Remove a server from a server placement group.
-

ibmcloud pi placement-group create

Alias: create, cr

Description: Create a server placement group.

Usage:

```
$ create PLACEMENT_GROUP_NAME --policy POLICY [--user-tags USER_TAG1[,USER_TAGn]]
```

PLACEMENT_GROUP_NAME: A unique name of the placement group.

Available Options:

```
$ -p, --policy string      Affinity policy for placement group being created. Valid values are affinity and anti-affinity.
-u, --user-tags strings    Comma separated list of user tags to be attached to the placement group.
```

Examples:

```
$ ibmcloud pi placement-group test-placement-group --policy anti-affinity
```

ibmcloud pi placement-group delete

Alias: delete, del

Description: Delete a server placement group.

Usage:

```
$ delete PLACEMENT_GROUP_ID
```

PLACEMENT_GROUP_ID: The unique identifier of the placement group.

ibmcloud pi placement-group get

Alias: get

Description: View details of a server placement group.

Usage:

```
$ get PLACEMENT_GROUP_ID
```

PLACEMENT_GROUP_ID: The unique identifier of the placement group.

ibmcloud pi placement-group list

Alias: `list, ls`

Description: List all server placement groups.

Usage: `list`

ibmcloud pi placement-group server-add

Alias: `server-add, sa`

Description: Add a server to a server placement group.

Usage:

```
$ server-add PLACEMENT_GROUP_ID --server INSTANCE_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options:

```
$ -s, --server string  The server instance ID of the server to add to the placement group.
```

Examples:

```
$  ibmcloud pi placement-group server-add 43064761-948f-469d-ac8e-b8e5f0d6056f --server 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi placement-group server-remove

Alias: `server-remove, sr`

Description: Remove a server from a server placement group.

Usage:

```
$ server-remove PLACEMENT_GROUP_ID --server INSTANCE_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options:

```
$ -s, --server string  The server instance ID of the server to add to the placement group.
```

Examples:

```
$  ibmcloud pi placement-group server-remove 43064761-948f-469d-ac8e-b8e5f0d6056f --server 85716e61-948f-309d-de8b-c4e5f0d3126d
```

ibmcloud pi route

Alias: `route, rt`

Description: IBM Cloud Power Virtual Server Network Routes.

Usage: `route`

Available Commands:

- `create` : Create a network custom route.
 - `delete` : Delete a network custom route.
 - `get` : View details of a network custom route.
 - `list` : List all network custom routes.
 - `report` : View details of network custom route report.
 - `update` : Update a network custom route.
-

ibmcloud pi route create

Alias: create, cr

Description: Create a network custom route.

Usage:

```
$ create ROUTE_NAME --destination DESTINATION --next-hop NEXT_HOP
  [--action ACTION] [--advertise ("enable" | "disable")] [--destination-type DESTINATION_TYPE]
  [--enabled=True|False] [--next-hop-type NEXT_HOP_TYPE] [--user-tags USER_TAG1[,USER_TAGn]]

ROUTE_NAME: A unique name of the network custom route.
```

Available Options :

```
$ -a, --action string      The action of the route to take. Valid values are: deliver. Default is "deliver".
  --advertise string      Enable the route to be advertised. Valid values are: enable, disable. Default is "enable".
-d, --destination string   The route destination.
  --destination-type string The route destination type. Valid values are: ipv4-address. Default is "ipv4-address".
-e, --enabled             Indicates if the route should be enabled in the fabric.
  --next-hop string        The IP address to route the packet to.
  --next-hop-type string   The next hop type. Valid values are: ipv4-address. Default is "ipv4-address".
-u, --user-tags strings    Comma separated list of user tags to be attached to the network security group.
```

Examples:

```
$ ibmcloud pi route create test-route --advertise disable --destination 192.168.1.23 --next-hop 192.168.1.24
```

ibmcloud pi route delete

Alias: delete, del

Description: Delete a network custom route.

Usage:

```
$ delete ROUTE_ID

ROUTE_ID: The unique identifier of the network custom route.
```

ibmcloud pi route get

Alias: get

Description: View details of a network custom route.

Usage:

```
$ get ROUTE_ID

ROUTE_ID: The unique identifier of the network custom route.
```

ibmcloud pi route list

Alias: list, ls

Description: List all network custom routes.

Usage: list

ibmcloud pi route report

Alias: report, rp

Description: View details of network custom route report.

Usage: report

ibmcloud pi route update

Alias: update, upd

Description: Update a network custom route.

Usage:

```
$ update ROUTE_ID [--action ACTION] [--advertise ("enable" | "disable")]
  [--destination DESTINATION] [--destination-type DESTINATION_TYPE] [--enabled=True|False]
  [--name NAME] [--next-hop NEXT_HOP] [--next-hop-type NEXT_HOP_TYPE]

ROUTE_ID: The unique identifier of the network custom route.
```

Available Options:

```
$ -a, --action string      The action of the route to take. Valid values are: deliver.
  --advertise string      Enable the route to be advertised. Valid values are: enable, disable.
-d, --destination string   The route destination.
  --destination-type string The route destination type. Valid values are: ipv4-address.
-e, --enabled             Indicates if the route should be enabled in the fabric.
-n, --name string         New name of the route.
  --next-hop string       The IP address to route the packet to.
  --next-hop-type string   The next hop type. Valid values are: ipv4-address.
```

ibmcloud pi shared-processor-pool

Alias: shared-processor-pool, spp

Description: IBM Cloud Power Virtual Server Shared Processor Pools.

Usage: shared-processor-pool

Available Commands:

- create : Create a shared processor pool.
- delete : Delete a shared processor pool.
- get : View details of a shared processor pool.
- list : List all shared processor pools.
- placement-group : IBM Cloud Power Virtual Server Placement Groups.
- update : Update a shared processor pool.

ibmcloud pi shared-processor-pool create

Alias: create, cr

Description: Create a shared processor pool.

Usage:

```
$ create SHARED_PROCESSOR_POOL_NAME --host-group HOST_GROUP --reserved-cores NUMBER_OF_CORES [--placement-group-id PLACEMENT_GROUP_ID] [--
user-tags USER_TAG1[,USER_TAGn]]

SHARED_PROCESSOR_POOL_NAME: A unique name of the shared processor pool. A minimum of 2 characters and a maximum of 12 are allowed. The only special
character allowed is the underscore '_'.
```

Available Options:

```
$ -g, --host-group string    The host group where the host will be chosen.
-p, --placement-group-id string The identifier of the shared processor pool placement group the pool will be added to.
-r, --reserved-cores int    The integer amount of reserved processor cores for the shared processor pool.
-u, --user-tags strings      Comma separated list of user tags to be attached to the placement-group.
```

Examples:

```
$ ibmcloud pi shared-processor-pool create test-shared-processor-pool --host-group e980 --reserver-cores 2 --placement-group-id 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi shared-processor-pool delete

Alias: delete, del

Description: Delete a shared processor pool.

Usage:

```
$ delete SHARED_PROCESSOR_POOL_ID

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.
```

ibmcloud pi shared-processor-pool get

Alias: get

Description: View details of a shared processor pool.

Usage:

```
$ get SHARED_PROCESSOR_POOL_ID

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.
```

ibmcloud pi shared-processor-pool list

Alias: list, ls

Description: List all shared processor pools.

Usage: list

ibmcloud pi shared-processor-pool placement-group

Alias: placement-group, pg

Description: IBM Cloud Power Virtual Server Placement Groups.

Usage: placement-group

Available Commands:

- create : Create a shared processor pool placement group.
- delete : Delete a shared processor pool placement group.
- get : View details of a shared processor pool placement group.
- list : List all shared processor pool placement groups.
- member-add : Add a shared processor pool to a placement group.
- member-remove : Remove a shared processor pool from a placement group.

ibmcloud pi shared-processor-pool placement-group create

Alias: create, cr

Description: Create a shared processor pool placement group.

Usage:

```
$ create PLACEMENT_GROUP_NAME --policy POLICY [--user-tags USER_TAG1[,USER_TAGn]]

PLACEMENT_GROUP_NAME: A unique name of the placement group.
```

Available Options :

```
$ -p, --policy string      Affinity policy for placement group being created. Valid values are affinity and anti-affinity.
-u, --user-tags strings   Comma separated list of user tags to be attached to the shared processor pool placement group.
```

Examples:

```
$ ibmcloud pi shared-processor-pool placement-group create spp-pg-test --policy affinity
```

ibmcloud pi shared-processor-pool placement-group delete

Alias: delete, del

Description: Delete a shared processor pool placement group.

Usage:

```
$ delete PLACEMENT_GROUP_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

ibmcloud pi shared-processor-pool placement-group get

Alias: get

Description: View details of a shared processor pool placement group.

Usage:

```
$ get PLACEMENT_GROUP_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

ibmcloud pi shared-processor-pool placement-group list

Alias: list, ls

Description: List all shared processor pool placement groups.

Usage: list

ibmcloud pi shared-processor-pool placement-group member-add

Alias: member-add, ma

Description: Add a shared processor pool to a placement group.

Usage:

```
$ member-add PLACEMENT_GROUP_ID --shared-processor-pool POOL_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options :

```
$ -s, --shared-processor-pool string  The shared processor pool ID of the pool to add to the placement group.
```

Examples:

```
$ ibmcloud pi shared-processor-pool placement-group member-add 43064761-948f-469d-ac8e-b8e5f0d6056 --shared-processor-pool 32bec9b0-560a-4baa-8cc0-7c68e41e5843
```

ibmcloud pi shared-processor-pool placement-group member-remove

Alias: member-remove, mr

Description: Remove a shared processor pool from a placement group.

Usage:

```
$ member-remove PLACEMENT_GROUP_ID --shared-processor-pool POOL_ID

PLACEMENT_GROUP_ID: The unique identifier of the placement group.
```

Available Options :

```
$ -s, --shared-processor-pool string  The shared processor pool ID of the pool to remove from the placement group.
```

Examples:

```
$  ibmcloud pi shared-processor-pool placement-group member-remove 43064761-948f-469d-ac8e-b8e5f0d6056 --shared-processor-pool 32bec9b0-560a-4baa-8cc0-7c68e41e5843
```

ibmcloud pi shared-processor-pool update

Alias: update, upd

Description: Update a shared processor pool.

Usage:

```
$ update SHARED_PROCESSOR_POOL_ID [--name SHARED_PROCESSOR_POOL_NAME] [--reserved-cores NUMBER_OF_CORES]

SHARED_PROCESSOR_POOL_ID: The unique identifier or name of the shared processor pool.
```

Available Options :

```
$ -n, --name string      New name of the shared processor pool. A minimum of 2 characters and a maximum of 12 are allowed. The only special character allowed is the underscore '_'.
-r, --reserved-cores int  The integer amount of reserved processor cores for the shared processor pool.
```

ibmcloud pi snapshot

Alias: snapshot, snap

Description: [DEPRECATED] IBM Cloud Power Virtual Server Snapshots.

Usage: snapshot

Available Commands :

- create : [DEPRECATED] Create a snapshot.
- delete : [DEPRECATED] Delete a snapshot.
- get : [DEPRECATED] View details of a snapshot.
- list : [DEPRECATED] List all snapshots.
- restore : [DEPRECATED] Restore a snapshot.
- update : [DEPRECATED] Update a snapshot.

ibmcloud pi snapshot create

Alias: create, cr

Description: [DEPRECATED] Create a snapshot.

Usage:

```
$ create INSTANCE_ID --name SNAPSHOT_NAME [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]] [--volumes VOLUME1[,VOLUMEn]]
```

INSTANCE_ID: The unique identifier or name of the instance.

Available Options :

```
$ -d, --description string  Snapshot description.
-n, --name string          Name of the snapshot.
-u, --user-tags strings    Comma separated list of user tags to be attached to the snapshot.
-v, --volumes strings     Comma separated list of volume identifiers or names to include in the snapshot.
                           If you do not specify this option or if the volumes list is empty,
                           all the volumes that are attached to the instance are included in the snapshot.
```

Examples:

```
$ ibmcloud pi snapshot create 85716e61-948f-309d-de8b-c4e5f0d3126d --name test-snapshot-name
```

ibmcloud pi snapshot delete

Alias: delete, del

Description: [DEPRECATED] Delete a snapshot.

Usage:

```
$ delete SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi snapshot get

Alias: get

Description: [DEPRECATED] View details of a snapshot.

Usage:

```
$ get SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi snapshot list

Alias: list, ls

Description: [DEPRECATED] List all snapshots.

Usage: list

ibmcloud pi snapshot restore

Alias: restore, res

Description: [DEPRECATED] Restore a snapshot.

Usage:

```
$ restore INSTANCE_ID --snapshot SNAPSHOT_ID [--force] [--restore VALUE]

INSTANCE_ID: The unique identifier or name of the instance.
```

Available Options :

```
$ -f, --force          By default the VM must be shutdown during a snapshot restore, force set to true will relax the VM shutdown pre-condition.
-r, --restore string   Action to take on a failed snapshot restore. Valid values are: "retry", "rollback".
-s, --snapshot string  The unique identifier of the snapshot.
```


Examples:

```
$ ibmcloud pi snapshot restore 85716e61-948f-309d-de8b-c4e5f0d3126d --snapshot 43064761-948f-469d-ac8e-b8e5f0d6056f --force --restore retry
```

ibmcloud pi snapshot update

Alias: update, upd

Description: [DEPRECATED] Update a snapshot.

Usage:

```
$ update SNAPSHOT_ID [--description DESCRIPTION] [--name NAME]

SNAPSHOT_ID: The unique identifier of the snapshot.
```

Available Options :

```
$ -d, --description string  New snapshot description.
-n, --name string          New name of the snapshot.
```

ibmcloud pi ssh-key

Alias: ssh-key, ssh

Description: IBM Cloud Power Virtual Server SSH-Keys.

Usage: ssh-key

Available Commands :

- create : Create an SSH-Key with an imported RSA public key.
- delete : Delete an SSH-Key.
- get : View details of an SSH-Key.
- list : List all SSH-Keys.
- update : Update an SSH-Key.

ibmcloud pi ssh-key create

Alias: create, cr

Description: Create an SSH-Key with an imported RSA public key.

Usage:

```
$ create KEY_NAME --key KEY [--description DESCRIPTION] [--visibility ("account" | "workspace")]

KEY_NAME: The name of the key.
```

Available Options :

```
$ -d, --description string  Description of the SSH-Key.
-k, --key string           SSH RSA key string within a double quotation marks. For example, "ssh-rsa AAA... "
-v, --visibility string     Visibility of the SSH-Key. Valid values are: account, workspace.
                           Value of "workspace" means SSH-Key is only accessible in a workspace.
                           Value of "account" means SSH-Key is accessible throughout an account.
                           Default is "workspace".
```

Examples:

```
$ ibmcloud pi ssh-key create test-ssh-key --key ssh-key-sample
```

ibmcloud pi ssh-key delete

Alias: delete, del

Description: Delete an SSH-Key.

Usage:

```
$ delete KEY_NAME

KEY_NAME: The name of the key.
```

ibmcloud pi ssh-key get

Alias: get

Description: View details of an SSH-Key.

Usage:

```
$ get KEY_NAME

KEY_NAME: The name of the key.
```

ibmcloud pi ssh-key list

Alias: list, ls

Description: List all SSH-Keys.

Usage: list

ibmcloud pi ssh-key update

Alias: update, upd

Description: Update an SSH-Key.

Usage:

```
$ update KEY_NAME [--description DESCRIPTION] [--key KEY] [--name NAME] [--visibility ("account" | "workspace")]

KEY_NAME: The name of the key.
```

Available Options:

```
$ -d, --description string  Description of the SSH-Key.
-k, --key string            SSH RSA key string.
-n, --name string           SSH-Key name.
-v, --visibility string      Visibility of the SSH-Key. Valid values are: account, workspace.
                             Value of "workspace" means SSH-Key is only accessible in a workspace.
                             Value of "account" means SSH-Key is accessible throughout an account.
```

ibmcloud pi storage-pools

Alias: storage-pools, spools

Description: List all storage pools for the targeted region.

Usage: storage-pools

ibmcloud pi storage-tiers

Alias: storage-tiers, stiers

Description: List all storage tiers for the targeted region.

Usage: `storage-tiers`

ibmcloud pi subnet

Alias: `subnet, snet`

Description: IBM Cloud Power Virtual Server Subnets.

Usage: `subnet`

Available Commands:

- `create` : Create a subnet.
- `delete` : Delete a subnet.
- `get` : View details of a subnet.
- `list` : List all subnets in a workspace.
- `update` : Update a subnet.

ibmcloud pi subnet create

Alias: `create, cr`

Description: Create a subnet.

Usage:

```
$ create SUBNET_NAME --cidr-block CIDR --net-type private [--advertise ("enable" | "disable")] [--arp-broadcast ("enable" | "disable")]
  [--dns-servers "DNS1,[DNSn]]" [--enable-dhcp=True|False] [--gateway GATEWAY] [--ip-range "startIP-endIP[,startIP-endIP]"
  [--mtu MTU] [--user-tags "USER_TAG1[,USER_TAGn]"]

SUBNET_NAME: The name of the subnet.
```

Available Options:

```
$ --advertise string    Enable the subnet to be advertised. Valid values are: enable, disable. Default is "enable" for private networks.
-b, --arp-broadcast string  Enable ARP Broadcast. Valid values are: enable, disable. Default is "disable" for private networks.
-c, --cidr-block string    Subnet in CIDR notation (192.168.1.0/22).
-d, --dns-servers strings  Comma separated list of DNS Servers to use for this subnet.
                           161.26.0.10 and 161.26.0.11 by default if DNS server is not specified for
                           private subnet types in PER enabled workspaces else
                           127.0.0.1 by default if DNS server is not specified and workspace is not PER enabled.
                           9.9.9.9 by default for public subnet types.
--enable-dhcp            Indicates if the network will support DHCP or not. Default is "true" for private networks.
-g, --gateway string      Gateway to use for this subnet.
-i, --ip-range string     IP Addresses range(s) for this subnet, format: "startIP-endIP[,startIP-endIP]".
-m, --mtu int             Maximum Transmission Unit. MTU be between 1450 and 9000. Default is 1450.
-n, --net-type string     Subnet type.
-u, --user-tags strings   Comma separated list of user tags to be attached to the subnet.
```

Examples:

```
$ ibmcloud pi subnet create test-private-subnet --net-type private --cidr-block "192.168.0.0/24" --ip-range "192.168.0.10-192.168.0.20" --dns-servers "8.8.8.8" --gateway
"192.168.0.1"
ibmcloud pi subnet create test-public-subnet --net-type public --dns-servers "8.8.8.8" --mtu 2000
```

ibmcloud pi subnet delete

Alias: `delete, del`

Description: Delete a subnet.

Usage:

```
$ delete SUBNET_ID

SUBNET_ID: The unique identifier or name of the subnet.
```

ibmcloud pi subnet get

Alias: `get`

Description: View details of a subnet.

Usage:

```
$ get SUBNET_ID
```

SUBNET_ID: The unique identifier or name of the subnet.

ibmcloud pi subnet list

Alias: `list`, `ls`

Description: List all subnets in a workspace.

Usage: `list`

ibmcloud pi subnet update

Alias: `update`, `upd`

Description: Update a subnet.

Usage:

```
$ update SUBNET_ID [--advertise ("enable" | "disable")] [--arp-broadcast ("enable" | "disable")]
  [--dns-servers "DNS1,[DNSn]"] [--enable-dhcp=True|False] [--gateway GATEWAY]
  [--ip-range "startIP-endIP[,startIP-endIP]"] [--name SUBNET_NAME]
```

SUBNET_ID: The unique identifier or name of the subnet.

Available Options:

```
$  --advertise string    Enable the subnet to be advertised. Valid values are: enable, disable.
-b, --arp-broadcast string  Enable ARP Broadcast. Valid values are: enable, disable.
-d, --dns-servers strings  Comma separated list of DNS Servers to use for this subnet.
  --enable-dhcp            Indicates if the network will support DHCP or not.
-g, --gateway string      Gateway to use for this subnet.
-i, --ip-range string      IP Addresses range(s) for this subnet, format: "startIP-endIP[,startIP-endIP]".
-n, --name string          New name of the subnet.
```

ibmcloud pi virtual-serial-number

Alias: `virtual-serial-number`, `vsn`

Description: IBM Cloud Power Virtual Server Virtual Serial Number.

Usage: `virtual-serial-number`

Available Commands:

- `delete` : Delete a virtual serial number.
- `get` : View details of a virtual serial number.
- `list` : List all virtual serial number assigned to an instance or all virtual serial numbers in the workspace.
- `software-tiers` : List all supported software tiers.
- `update` : Update a retained virtual serial number.

ibmcloud pi virtual-serial-number delete

Alias: `delete`, `del`

Description: Delete a virtual serial number.

Usage:

```
$ delete VIRTUAL_SERIAL_NUMBER

VIRTUAL_SERIAL_NUMBER: The virtual serial number.
```

ibmcloud pi virtual-serial-number get

Alias: get

Description: View details of a virtual serial number.

Usage:

```
$ get VIRTUAL_SERIAL_NUMBER

VIRTUAL_SERIAL_NUMBER: The virtual serial number.
```

ibmcloud pi virtual-serial-number list

Alias: list, ls

Description: List all virtual serial number assigned to an instance or all virtual serial numbers in the workspace.

Usage:

```
$ list ([PVM_INSTANCE_ID] | [--retainVSN=True|False])

PVM_INSTANCE_ID: The unique identifier or name of the PVM instance.
```

Available Options:

```
$ -r, --retainVSN  Lists only retained virtual serial numbers. Default is "false".
```

ibmcloud pi virtual-serial-number software-tiers

Alias: software-tiers, st

Description: List all supported software tiers.

Usage: software-tiers

ibmcloud pi virtual-serial-number update

Alias: update, upd

Description: Update a retained virtual serial number.

Usage:

```
$ update VIRTUAL_SERIAL_NUMBER [--description DESCRIPTION]

VIRTUAL_SERIAL_NUMBER: The virtual serial number.
```

Available Options:

```
$ -d, --description string  New virtual serial number description.
```

ibmcloud pi volume

Alias: volume, vol

Description: IBM Cloud Power Virtual Server Volumes.

Usage: `volume`

Available Commands:

- `action` : Perform an action on a volume.
- `bulk-delete` : Delete multiple volumes.
- `clone` : IBM Cloud Power Virtual Server Volume Clone Requests. This command can be used to create clone requests whose lifecycle must be managed. It cannot be used to clone single volumes.
- `clone-async` : IBM Cloud Power Virtual Server Volume Clones. This command asynchronously creates clone tasks whose status can be queried. It can be used to clone one or more volumes.
- `create` : Create a volume.
- `delete` : Delete a volume.
- `flash-copy-mapping` : Get a list of flash copy mappings of a volume directly from primary storage host.
- `get` : View details of a volume.
- `list` : List all storage volumes in a workspace.
- `onboarding` : IBM Cloud Power Virtual Server Volume Onboarding.
- `remote-copy-relationship` : Get the remote copy relationship information of a volume.
- `snapshot` : IBM Cloud Power Virtual Server Volume Snapshot.
- `update` : Update a volume.

ibmcloud pi volume action

Alias: `action, act`

Description: Perform an action on a volume.

Usage:

```
$ action VOLUME_ID [--replication-enabled=True|False] [--target-tier STORAGE_TIER]
```

VOLUME_ID: The unique identifier or name of the volume.

Available Options:

```
$ -r, --replication-enabled  Enables storage replication on the volume. Default is "false".
-t, --target-tier string    Change the storage tier of the volume (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region).
                           "Tier5k" volumes cannot exceed 200 GiB.
```

Examples:

```
$  ibmcloud pi volume action test-volume --replication-enabled
```

ibmcloud pi volume bulk-delete

Alias: `bulk-delete, bdel`

Description: Delete multiple volumes.

Usage: `bulk-delete --volumes VOLUME1[,VOLUMEn]`

Available Options:

```
$ -v, --volumes strings  List of volumes to be deleted.
```

ibmcloud pi volume clone

Alias: `clone, cl`

Description: IBM Cloud Power Virtual Server Volume Clone Requests. This command can be used to create clone requests whose lifecycle must be managed. It cannot be used to clone single volumes.

Usage: clone

Available Commands:

- cancel : Cancel a volume clone request.
- create : Create a volume clone request using the specified volumes.
- delete : Delete a volume clone request.
- execute : Execute a volume clone request using the specified options.
- get : View details of a volume clone request.
- list : List all volume clone requests in a workspace.
- start : Starts a volume clone request.

ibmcloud pi volume clone cancel

Alias: cancel, ca

Description: Cancel a volume clone request.

Usage:

```
$ cancel VOLUME_CLONE_REQUEST_ID [--force=True|False]

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Available Options:

```
$ -f, --force Force the cancellation of a volume clone request. If false, cancel will only be allowed if the status is 'prepared', or 'available'. If true, cancel will be allowed when the status is NOT completed, cancelling, cancelled, or failed. Default false.
```

Examples:

```
$ ibmcloud pi volume clone cancel 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone create

Alias: create, cr

Description: Create a volume clone request using the specified volumes.

Usage:

```
$ create --name VOLUME_CLONE_REQUEST_NAME --volumes VOLUME1[,VOLUMEn]

VOLUME_CLONE_REQUEST_NAME: The name of a volume clone request.
```

Available Options:

```
$ -n, --name string Name to assign to volume clones request.
-v, --volumes strings Comma separated list of volume identifiers or names to be cloned.
                        A minimum of 2 Volume IDs are required and at least one of them must be in the in-use state.
```

Examples:

```
$ ibmcloud pi volume clone create --name test-volume-clone --volumes "43064761-948f-469d-ac8e-b8e5f0d6056f,85716e61-948f-309d-de8b-c4e5f0d3126d"
```

ibmcloud pi volume clone delete

Alias: delete, del

Description: Delete a volume clone request.

Usage:


```
$ delete VOLUME_CLONE_REQUEST_ID
```

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.

Examples:

```
$ ibmcloud pi volume clone delete 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone execute

Alias: `execute`, `ex`

Description: Execute a volume clone request using the specified options.

Usage:

```
$ execute VOLUME_CLONE_REQUEST_ID --name BASE_NAME [--replication-enabled=True|False] [--rollback-prepare=True|False] [--target-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
```

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.

Available Options:

```
$ -n, --name string      Base name of the new cloned volume(s).
    --replication-enabled Enables replication for the cloned volume. By default, the replication property of the source volume will be used to determine the replication property of the cloned target volume.
    --rollback-prepare   Determines rollback behavior. If false, execute failure rolls back clone activity but leaves prepared snapshot. If true, Execute failure rolls back clone activity and removes the prepared snapshot. Default false.
-t, --target-tier string Target storage tier for the cloned volumes. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region.
                        Default to the storage tier of the original volume(s) if not specified.
-u, --user-tags strings  Comma separated list of user tags to be attached to the cloned volume.
```

Examples:

```
$ ibmcloud pi volume clone execute 43064761-948f-469d-ac8e-b8e5f0d6056f --name test-volume-clone --rollback-prepare --target-tier tier3 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi volume clone get

Alias: `get`

Description: View details of a volume clone request.

Usage:

```
$ get VOLUME_CLONE_REQUEST_ID
```

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.

ibmcloud pi volume clone list

Alias: `list`, `ls`

Description: List all volume clone requests in a workspace.

Usage: `list` `[--filter FILTER]`

Available Options:

```
$ -f, --filter string  Volume clone filter to limit list items.
                        cancel - includes status values (cancelling)
                        cancelled - includes status values (cancelled)
                        completed - includes status values (completed)
                        execute - includes status values (executing, available-rollback)
                        failed - includes status values (failed)
```

finalized - included status values (completed, failed, cancelled)
prepare - includes status values (preparing, prepared)
start - includes status values (starting, available)

ibmcloud pi volume clone start

Alias: start, st

Description: Starts a volume clone request.

Usage:

```
$ start VOLUME_CLONE_REQUEST_ID

VOLUME_CLONE_REQUEST_ID: The unique identifier of a volume clone request.
```

Examples:

```
$ ibmcloud pi volume clone start 43064761-948f-469d-ac8e-b8e5f0d6056f
```

ibmcloud pi volume clone-async

Alias: clone-async, cla

Description: IBM Cloud Power Virtual Server Volume Clones. This command asynchronously creates clone tasks whose status can be queried. It can be used to clone one or more volumes.

Usage: clone-async

Available Commands:

- create : Create a volume clone for specific volumes.
- get : Get the status of a clone request for the specified clone task ID.

ibmcloud pi volume clone-async create

Alias: create, cr

Description: Create a volume clone for specific volumes.

Usage:

```
$ create CLONE_NAME --volumes VOLUME1[,VOLUMEn] [--replication-enabled=True|False] [--target-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]

CLONE_NAME:The name of a clone.
```

Available Options:

```
$ -r, --replication-enabled Enables replication for the cloned volume.
                             If no value is provided, it will default to the replication status of the source volume.
-t, --target-tier string    Target storage tier for the cloned volumes. Use "ibmcloud pi storage-tiers" to see available tiers in the targeted region.
                             Default to the storage tier of the original volume(s) if not specified.
-u, --user-tags strings     Comma separated list of user tags to be attached to the cloned volume.
-v, --volumes strings       Comma separated list of volume identifiers or names to be cloned.
```

Examples:

```
$ ibmcloud pi volume clone-async create test-volume --volumes 43064761-948f-469d-ac8e-b8e5f0d6056f --target-tier tier1 --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi volume clone-async get

Alias: get

Description: Get the status of a clone request for the specified clone task ID.

Usage:

```
$ get CLONE_TASK_ID

CLONE_TASK_ID:The unique identifier of a clone task.
```

ibmcloud pi volume create

Alias: `create`, `cr`

Description: Create a volume.

Usage:

```
$ create VOLUME_NAME --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEn]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
pi volume create VOLUME_NAME --size SIZE --storage-pool POOL [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEn]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
pi volume create VOLUME_NAME --affinity-policy affinity (--affinity-volume VOLUME | --affinity-instance INSTANCE) --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEn]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]
pi volume create VOLUME_NAME --affinity-policy anti-affinity (--anti-affinity-volumes "VOLUME1[,VOLUMEn]" | --anti-affinity-instances "INSTANCE1[,INSTANCEn]" ) --size SIZE [--count COUNT] [--replication-enabled=True|False] [--replication-sites SITE1[,SITEn]] [--shareable=True|False] [--storage-tier STORAGE_TIER] [--user-tags USER_TAG1[,USER_TAGn]]

VOLUME_NAME: The name of the volume.
```

Available Options:

```
$ -i, --affinity-instance string      PVM instance identifier or name to base volume affinity policy against.
                                     This is required if "--affinity-policy affinity" is specified and "--affinity-volume" is not provided.
-a, --affinity-policy string          Affinity policy for data volume being created. Valid values are "affinity" and "anti-affinity". If "--storage-pool" is provided then this cannot be specified.
-v, --affinity-volume string          Volume identifier or name to base volume affinity policy against.
                                     This is required if "--affinity-policy affinity" is specified and "--affinity-instance" is not provided.
-j, --anti-affinity-instances strings Comma separated list of instance identifiers or names to base volume anti-affinity policy against.
                                     This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-volumes" is not provided.
-w, --anti-affinity-volumes strings   Comma separated list of volume identifiers or names to base volume anti-affinity policy against.
                                     This is required if "--affinity-policy anti-affinity" is specified and "--anti-affinity-instances" is not provided.
-c, --count int                      Number of volumes to create. Default is 1.
-r, --replication-enabled             Enables storage replication on the volume. Default is "false".
    --replication-sites strings        List of replication sites for volume replication. See "ibmcloud pi disaster-recovery" command to get a list of replication sites.
-e, --shareable                      Whether the volumes can be attached to multiple VMs. Default is "false".
-s, --size int                       Size of the volume in GiB. Size cannot exceed 200 GiB for storage tier "Tier5k".
-p, --storage-pool string             Volume pool where the volume will be created. If the "--storage-pool" option is set then the "--affinity-policy" option cannot be specified.
-t, --storage-tier string             Tier of the volume (use "ibmcloud pi storage-tiers" to see available storage tiers in the targeted region). Default to tier3 if not provided.
-u, --user-tags strings               Comma separated list of user tags to be attached to the volume.
```

Examples:

```
$  ibmcloud pi volume create test-volume --size 1
   ibmcloud pi volume create test-volume --size 1 --storage-tier tier1 --affinity-policy anti-affinity --anti-affinity-volumes anti-affinity-volume
   ibmcloud pi volume create test-volume --size 1 --storage-tier tier1 --count 2
   ibmcloud pi volume create test-volume --size 1 --replication-enabled --replication-sites dal10
```

ibmcloud pi volume delete

Alias: `delete`, `del`

Description: Delete a volume.

Usage:

```
$ delete VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume flash-copy-mapping

Alias: flash-copy-mapping, fcm

Description: Get a list of flash copy mappings of a volume directly from primary storage host.

Usage:

```
$ flash-copy-mapping VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume get

Alias: get

Description: View details of a volume.

Usage:

```
$ get VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume list

Alias: list, ls

Description: List all storage volumes in a workspace.

Usage: list [--auxiliary=True|False] [--long=True|False] [--replication-enabled=True|False]

Available Options:

```
$ -a, --auxiliary      Filter auxiliary volumes if set to "true" or non-auxiliary volumes if set to "false". Default is "true".
-l, --long            Retrieve all volume details.
-r, --replication-enabled  Filter replication-enabled volumes if set to "true" or non-replication-enabled volumes if set to "false". Default is "true".
```

ibmcloud pi volume onboarding

Alias: onboarding, on

Description: IBM Cloud Power Virtual Server Volume Onboarding.

Usage: onboarding

Available Commands:

- create : Create a volume onboarding operation.
- get : Get the information of volume onboarding operation.
- list : List all volume onboarding operations.

ibmcloud pi volume onboarding create

Alias: create, cr

Description: Create a volume onboarding operation.

Usage: create <--auxiliary-volumes "AUXVOLUMENAME1 [NAME1]"> --source-crn SOURCE_CRN [--description DESCRIPTION] [--user-tags USER_TAG1[,USER_TAGn]]

Available Options:

```
$ -a, --auxiliary-volumes strings  Comma separated list of identifiers of the volume(s) at storage host level. Repeat this option to add more auxiliary volumes.
-d, --description string          Volume onboarding description.
-s, --source-crn string           The CRN of the workspace in which the primary volume is located.
-u, --user-tags strings           Comma separated list of user tags to be attached to the volume onboarding.
```

Examples:

```
$ ibmcloud pi volume onboarding create --auxiliary-volumes "test-auxiliary-volume-name" --source-crn "crn:v1:staging:public:power-iaas:dal13:a/mockcrn::"
```

ibmcloud pi volume onboarding get

Alias: `get`

Description: Get the information of volume onboarding operation.

Usage:

```
$ get VOLUME_ONBOARDING_ID

VOLUME_ONBOARDING_ID: The unique identifier of the onboarding operation.
```

ibmcloud pi volume onboarding list

Alias: `list, ls`

Description: List all volume onboarding operations.

Usage: `list`

ibmcloud pi volume remote-copy-relationship

Alias: `remote-copy-relationship, rcr`

Description: Get the remote copy relationship information of a volume.

Usage:

```
$ remote-copy-relationship VOLUME_ID

VOLUME_ID: The unique identifier or name of the volume.
```

ibmcloud pi volume snapshot

Alias: `snapshot, snap`

Description: IBM Cloud Power Virtual Server Volume Snapshot.

Usage: `snapshot`

Available Commands:

- `get` : Get a snapshot for a volume.
- `list` : List all snapshots in the workspace.

ibmcloud pi volume snapshot get

Alias: `get`

Description: Get a snapshot for a volume.

Usage:

```
$ get SNAPSHOT_ID

SNAPSHOT_ID: The unique identifier of the snapshot.
```

ibmcloud pi volume snapshot list

Alias: `list, ls`

Description: List all snapshots in the workspace.

Usage: `list`

ibmcloud pi volume update

Alias: `update`, `upd`

Description: Update a volume.

Usage:

```
$ update VOLUME_ID [--bootable=True|False] [--name NAME] [--size SIZE] [--shareable=True|False]
```

VOLUME_ID: The unique identifier or name of the volume.

Available Options:

```
$ -b, --bootable      New value for whether the volume can be booted.
-n, --name string     New name of the volume.
-t, --shareable      New value for whether the volume can be attached to multiple VMs.
-s, --size int        New size of the volume. Size cannot exceed 200 GiB for storage tier "Tier5k".
```

ibmcloud pi volume-group

Alias: `volume-group`, `vg`

Description: IBM Cloud Power Virtual Server Volume Groups.

Usage: `volume-group`

Available Commands:

- `action` : Perform an action on a volume group.
- `create` : Create a volume group.
- `delete` : Delete a volume group.
- `get` : View details of a volume group.
- `list` : List all storage volume groups in a workspace.
- `remote-copy-relationships` : Get the remote copy relationship information of a volume group.
- `storage-details` : Get storage details for a volume group.
- `update` : Update a volume group.

ibmcloud pi volume-group action

Alias: `action`, `act`

Description: Perform an action on a volume group.

Usage:

```
$ action VOLUME_GROUP_ID --operation reset [--status STATUS]
pi volume-group action VOLUME_GROUP_ID --operation start [--source SOURCE]
pi volume-group action VOLUME_GROUP_ID --operation stop [--allow-read-access=True|False]
```

VOLUME_GROUP_ID: The unique identifier or name of the volume group.

Available Options:

```
$ -a, --allow-read-access  Allow the auxiliary volume to be accessible after stopping the volume group. Default is "false".
-o, --operation string     Operation to be done in a volume group. Valid values are: reset, start, stop.
    --source string        The copying volume source. Valid values are: auxiliary, master.Default is "master".
    --status string        New status to be set for a volume group. Default is "available".
```

Examples:

```
$ ibmcloud pi volume-group action test-volume-group --operation reset
ibmcloud pi volume-group action test-volume-group --operation start --source master
```

ibmcloud pi volume-group create

Alias: `create, cr`

Description: Create a volume group.

Usage:

```
$ create (--volume-group-name VOLUME_GROUP_NAME | --consistency-group-name CONSISTENCY_GROUP_NAME) --member-volume-ids "VOLUME1[,VOLUMEn]"

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

Available Options:

```
$ -c, --consistency-group-name string  Storage volume group name. This is required to onboard existing volume group on the target site for DR set up.
-m, --member-volume-ids strings      Comma separated member volume identifiers.
-v, --volume-group-name string        Storage volume group name. This is required for the creation of new volume group.
```

Examples:

```
$ ibmcloud pi volume-group create --volume-group-name test-volume-group-name --member-volume-ids "43064761-948f-469d-ac8e-b8e5f0d6056f,85716e61-948f-309d-de8b-c4e5f0d3126d"
```

ibmcloud pi volume-group delete

Alias: `delete, del`

Description: Delete a volume group.

Usage:

```
$ delete VOLUME_GROUP_ID

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

ibmcloud pi volume-group get

Alias: `get`

Description: View details of a volume group.

Usage:

```
$ get VOLUME_GROUP_ID [--long=True|False]

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

Available Options:

```
$ -l, --long  Retrieve additional details for the volume group.
```

ibmcloud pi volume-group list

Alias: `list, ls`

Description: List all storage volume groups in a workspace.

Usage: `list [--long=True|False]`

Available Options:

```
$ -l, --long Retrieve additional details for the volume group.
```

ibmcloud pi volume-group remote-copy-relationships

Alias: remote-copy-relationships, rcr

Description: Get the remote copy relationship information of a volume group.

Usage:

```
$ remote-copy-relationships VOLUME_GROUP_ID

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

ibmcloud pi volume-group storage-details

Alias: storage-details, sd

Description: Get storage details for a volume group.

Usage:

```
$ storage-details VOLUME_GROUP_ID

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

ibmcloud pi volume-group update

Alias: update, upd

Description: Update a volume group.

Usage:

```
$ update VOLUME_GROUP_ID [--add-member-volume-ids "VOLUME1[,VOLUMEn]"] [--remove-member-volume-ids "VOLUME1[,VOLUMEn]"]

VOLUME_GROUP_ID: The unique identifier or name of the volume group.
```

Available Options:

```
$ -a, --add-member-volume-ids strings Comma separated volume identifiers to add as members of the volume group.
-r, --remove-member-volume-ids strings Comma separated volume identifiers to remove as members of the volume group.
```

ibmcloud pi workspace

Alias: workspace, ws

Description: IBM Cloud Power Virtual Server Workspaces.

Usage: workspace

Available Commands:

- context : Display the CRN of the currently targeted workspace.
- create : Create a workspace.
- delete : Delete a workspace.
- list : List all workspaces for this account.
- target : Target a workspace.

ibmcloud pi workspace context

Alias: context, con

Description: Display the CRN of the currently targeted workspace.

Usage: context

ibmcloud pi workspace create

Alias: create, cr

Description: Create a workspace.

Usage:

```
$ create WORKSPACE_NAME --datacenter DATACENTER --group RESOURCE_GROUP --plan PLAN [--parameters "KEY1 VALUE1"["KEYn VALUEn"]] [--user-tags USER_TAG1[,USER_TAGn]]
```

WORKSPACE_NAME: The desired name of the workspace.

Available Options:

```
$ -d, --datacenter string    The datacenter location where the instance should be hosted.
                             Use the "ibmcloud pi datacenter list" command to see values for public datacenters,
                             and the "ibmcloud sat location ls" command to see values for private datacenters.
                             When using the "ibmcloud sat location ls" to get a private datacenter,
                             please prefix "satloc_<REGION>_" to the displayed id.
-g, --group string          The ID of the resource group. Use the "ibmcloud resource groups" command to see possible values.
--parameters strings       List of key-value parameter pairs. Each pair must be a quoted string in the format "KEY VALUE".
                             Example: --parameters "env dev" "version 1.0"
-p, --plan string           Plan associated with the offering. Valid values are "public"/"off-prem" or "private"/"on-prem".
-u, --user-tags strings     Comma separated list of user tags to be attached to the workspace.
```

Examples:

```
$ ibmcloud pi workspace create test-workspace --datacenter dal12 --group 393843beabcdea9d803840a384093841 --plan public --user-tags "project:customer-poc,env:dev,dataresidency:germany"
```

ibmcloud pi workspace delete

Alias: delete, del

Description: Delete a workspace.

Usage:

```
$ delete (WORKSPACE_CRN | WORKSPACE_ID)
```

WORKSPACE_CRN: The cloud resource name that uniquely identifies IBM Cloud resources.

WORKSPACE_ID: The unique identifier of the workspace.

ibmcloud pi workspace list

Alias: list, ls

Description: List all workspaces for this account.

Usage: list ([--long=True|False | --private=True|False])

Available Options:

```
$ -l, --long    Retrieve all workspace details.
-p, --private  List private workspaces.
```

ibmcloud pi workspace target

Alias: target, tg

Description: Target a workspace.

Usage:

```
$ target WORKSPACE_CRN
```

WORKSPACE_CRN: The cloud resource name that uniquely identifies IBM Cloud resources.

FAQs for Power Virtual Server

IBM Power Virtual Server in **IBM data center**

IBM Power Virtual Server Private Cloud in **Client location**

What is IBM® Power® Virtual Server Private Cloud?

IBM Power Virtual Server Private Cloud is an as-a-service offering that includes a prescriptive set of physical infrastructure (compute, network, and storage). The infrastructure is deployed in your own data center. IBM site reliability engineers (SREs) fully maintain and operate your Client location infrastructure and manage it through the IBM Cloud. Also, you can adjust your workloads by using pay-as-you-use billing. For more information, see [What is IBM Power Virtual Server](#).

What is the difference between the IBM data center and Client location offerings of IBM Power Virtual Server?

The primary difference between the two is where the physical infrastructure resides. The Client location infrastructure resides in your data center, while Power Virtual Server infrastructure resides in the IBM data centers.

Which Power servers are supported?

IBM data center IBM Power S922, IBM Power E980, IBM Power S1022, IBM Power S1122.

Client location IBM Power S1122, IBM Power E1150, IBM Power E1180.

For complete specifications, see [Hardware specifications for Power Virtual Server](#) and [Hardware and software specifications for IBM Power Virtual Server Private Cloud](#).

What versions of AIX, IBM i, and Linux® are supported?

For details about supported AIX, IBM i, and Linux operating systems on Power Virtual Server, see [Operating systems supported in IBM Power Virtual Server](#).

Where can I find cost estimates for Power Virtual Server infrastructure?

To generate an estimated price, use the [Power Virtual Server Estimate cost](#) tool. For more information, see [Generating an estimate](#). For other pricing-related questions for IBM Power Virtual Server Private Cloud, see [Pricing FAQs](#).

Can IBM Power Virtual Server Private Cloud in Client location pods be expanded with additional compute nodes?

Yes, you can add up to the maximum number of compute nodes for a specific configuration capacity. For example, you can start with 5 nodes and then add 3 more nodes.

Can IBM Power Virtual Server Private Cloud in Client location pods be expanded with additional storage?

Yes, you can expand the pod with additional storage capacity. But you cannot add more storage controllers.

Are the IBM Power Virtual Server Private Cloud in Client location pods equipped with spare compute nodes for maintenance?

In each pod, one spare node is available that is exclusively usable for IBM operational purposes, such as to perform system maintenance. The system type of the spare node matches the largest client-usable node. For example, if you have a pod with 4X S1022 and 1X E1080 client-usable hosts, then the spare node is E1080.

Where can I find the logs for the pod software or operator access logs?

As a security officer, auditor, or manager, you can use the IBM Cloud Logs service to manage general purpose application logs, platform logs, or structured audit events. IBM Cloud Logs can be used with logs from both IBM Cloud services and customer applications. For more information, see [Getting started with IBM Cloud Logs](#)

Will the pod disconnect from the IBM Cloud if there is an unplanned network outage?

If an unplanned network outage occurs for the management network that is connecting the IBM Cloud instance to the pod infrastructure, the VMs continue to run within the pod.

See Table 1 for the implications of a pod that is running in a disconnected mode due to an unplanned network outage. Also, the primary and secondary management connections (Direct Link or site-to-site VPN) to the IBM Cloud are lost.

Capability	Behavior in disconnected mode
Client workload and data	Client workload is operational and you can access all the data.
GUI or API for read operations	GUI is operational and uses the most recent cached data.
GUI or API for write operations, such as creating virtual machine (VM) or volumes	You cannot perform write operations to the Power Virtual Sever resources until the connectivity to the control plane network is restored.
Command-line interface (CLI)	You can perform read operations. However, you cannot perform write operations until the connectivity to the control plane network is restored.
Billing and metering	Metering uses the most recent cached data. If the infrastructure is disconnected, you cannot perform write operations until the connectivity to the control plane network is restored.
IBM remote support	IBM remote support team cannot remotely connect to the IBM Power Virtual Server Private Cloud infrastructure until the connectivity to the control plane network is restored.
Dynamic Host Configuration Protocol (DHCP) services for client data networks	DHCP services are provided by the resident network in the IBM Power Virtual Server Private Cloud infrastructure and do not require a connection to the IBM Cloud.

Impact of unplanned network outage

Can I use my own AIX, IBM i, or Linux image?

Yes. This feature is known as **bring your own image**. For more information, see [Deploying a custom image within IBM Power Virtual Server](#).

What versions of stock images are available?

For each major version (example: Technology Level) of the operating system (OS) that is enabled through the offering, Power Virtual Server provides a single stock image. Power Virtual Server typically provides stock images for the last three major versions of the supported OS. Any update to the OS stock image is planned only when the image level is certified for Power Virtual Server environment.

When are stock images removed from the catalog?

Any unsupported and older stock images are periodically removed from the offering. You are notified three weeks before the images are removed.

What happens to virtual machines deployed by using stock images that are removed?

If the stock images that are used to deploy the virtual machines are removed, the virtual machines can continue to operate without any issue. You are recommended to update the operating system by following the vendor’s guidelines specific to your operating system.

What formats can I use to upload a custom image?

Currently, you can import a custom image in the following formats: *.ova*, *.ova.gz*, *.tar*, *.tar.gz* and *.tgz*.

What storage types are available in the storage area network (SAN)?

Each volume has a storage tier, which defines how many I/O operations per second (IOPS) can be started against that volume. These tiers can scale

according to the capacity of the volume.

The following tiers are supported:

- Tier 0 (25 IOPS/GB)
- Tier 1 (10 IOPS per GB)
- Tier 3 (3 IOPS per GB)
- Fixed 5000 IOPS

If you find the storage tiers are over or under-provisioned, you can change the storage tier of an existing volume. For more information, see [Storage tiers](#).

How do I extend my AIX rootvg?

By default, the system deploys 20 GBs for the AIX `rootvg`. You can extend the AIX `rootvg` by using the [extendvg](#) command to add a physical volume.

What's the difference between shared capped and shared uncapped processor performance? How are they compared with dedicated processor performance?

When you deploy a VM, you can choose between **Dedicated**, **Shared capped**, or **Shared uncapped** cores. The following list provides a simplified breakdown of their differences:

- **Shared uncapped**: shared among other clients
- **Shared capped**: shared, but resources do not expand beyond the requested capacity (used mostly for licensing)
- **Dedicated**: resources are allocated for a specific client (used for specific third-party considerations)

The core-to-virtual core ratio is 1:1. For shared processors, fractional cores round up to the nearest whole number. For example, 1.25 cores equal 2 virtual cores. For more information, see [How does shared processor performance compare to dedicated processors](#), [Pricing for IBM Power Virtual Server Private Cloud](#), and [Pricing for Power Virtual Server](#).

Dedicated processors

The hypervisor makes a 1:1 binding between the processor of the partition and a physical processor core. After a VM is activated, the 1:1 binding is static. In the VM, the operating system (OS) logical thread runs on the physical processor core that is bound with the processor. With a dedicated processor partition, you must resize the number of cores to meet the **peak** demand of the partition. For example, on a typical workday, the CPU consumption is around four cores. But, because of the **peak** demand, the processor requires around eight cores. So, configure the partition with eight cores to handle the **peak** demand and avoid any queuing delays in dispatching the applications.

Dedicated processors

Shared processors

Shared processors have two sharing modes: capped or uncapped. For a capped partition, the amount of CPU time is capped to the value specified for the entitlement. For example, a capped partition with processing units set to 0.5, can use up to 30 seconds of CPU time every minute. For an uncapped partition, the number of virtual processors defines the upper-limit of CPU consumption and not the value that is specified for processing units. For example, if the number of virtual processors are set to 3, the partition can use up to 180 seconds of CPU time every minute (three virtual processors each running at 100% utilization are equivalent of three physical cores worth of CPU time). The server must have unused capacity available for a partition to use more than its configured processing units.

Shared processors

How does my current environment compare to what's available through the Power Virtual Server?

If you like to compare your current environment's performance to what's available through the Power Virtual Server offering, see the [IBM Power Performance Report](#).

How do I migrate my VSI from one data center to another (WDC04 to DAL13)?

To migrate your VSI from one data center to another, you must capture and export your VSI to Cloud Object Storage. After you successfully capture and export your VSI, copy it to the Cloud Object Storage in the destination region, then do an import followed by a deployment.

What does VSI pinning do?

You can pin a VSI to the host on which the VSI is running by selecting the **Soft** or **Hard** pinning policy. If you do not set a pinning policy or if you select **Soft**, Power Virtual Server automatically migrates or remotely restarts the VSI during maintenance windows or if a host fails.

When you select **Soft** pinning, Power Virtual Server automatically migrates the VSI to the original host after the host returns to its operating state.

When you select **Hard** pinning, Power Virtual Server restricts VSI movement during remote restart, automated remote restart, dynamic resource optimization, and live partition migration. Power Virtual Server shuts off hard-pinned VSIs when the host is in a **Shutoff**, **Standby**, or **Error** state due to maintenance or host failure. When you use **Hard** pinning, consider protecting your workload by using a separate *High availability* (HA) and *Disaster recovery* (DR) solution.



Important: When you change the pin policy of a VSI from **Hard** to **Soft** or disable the pin policy, consider enabling **Automated remote restart** on the VSI. The automated remote restart setting restarts the VSI on another available host if an IBM Power host fails.

What does it mean to set an affinity or anti-affinity rule?

You can apply affinity and anti-affinity policies to both VMs and volumes.

VM affinity and anti-affinity policy allow you to spread a group of VMs across different hosts or keep them on a specific host.

Volume affinity and anti-affinity policy allow you to control the placement of a new volume based on an existing PVM instance (VM) or volume. When you set an affinity policy for a new storage volume, the volume is created within the same storage provider as an existing PVM instance or volume. With an anti-affinity policy, the new volume is created in a different storage provider other than the storage provider the existing PVM instance or volume is located in.

The use of volume affinity policy (affinity or anti-affinity) requires the availability of multiple storage providers. You might experience the following errors when you use a volume affinity policy:

- If an additional storage provider is not available to fulfill the requested policy, you might receive an error. The error indicates the inability to locate a storage provider to create a volume by using the requested volume affinity policy.
- If additional storage providers exist but the storage providers do not have sufficient space to fulfill the requested policy, you might receive an error. The error indicates the inability to locate a storage provider with enough free capacity to satisfy the requested volume size.

How to set a PVM instance to allow attaching mixed storage?

You can now attach storage volumes to a PVM instance from different storage tiers and pools, other than the storage pool the PVM instance's root (boot) volume is deployed in. To attach storage volumes to a PVM, modify the PVM instance and set the *storagePoolAffinity* property of the new PVM instance to false. By default, the *storagePoolAffinity* property of the PVM instance is set to true when the PVM instance is deployed and can be changed only by using the modified PVM instance API. Attaching mixed storage to a PVM instance has implications on the PVM instance capture, clone, and snapshot features. For more information about modifying a PVM instance API, see [Modify PVM Instance](#).

Does IBM provide maintenance for the AIX, IBM i, or Linux operating systems?

No. It is the customer's responsibility to maintain, update, and manage the AIX, IBM i, or Linux operating system.

How does licensing work for the AIX, IBM i, or Linux operating systems?

The license for the AIX and IBM i operating systems is part of the overall cost for the workspace. You cannot use an existing license that you already purchased. Refer to the AIX section to learn how to [create an AIX VM](#).

You can use the movable IBM i (IBM i MOL) to move your existing on premises entitlements to Power Virtual Server. Contact support to know more about the IBM i MOL, see [Getting help and support](#).

Power Virtual Server supports multiple levels of RHEL and SLES. You can either use IBM provided stock Linux images with IBM Full Linux Subscription or bring your own custom Linux image with vendor-provided subscription.

For more information about supported versions of OS, see [Operating systems supported in IBM Power Virtual Server](#).

How does third-party licensing work?

Clients are responsible for third-party licensing.

What are the hardware specifications?

For more information, see [Hardware specifications for Power Virtual Server](#) and [Hardware and software specifications for IBM Power Virtual Server](#)

Do Power Virtual Server run in a multi-tenant environment?

IBM data center

The Power Virtual Server runs in a multi-tenant environment. If you have signed up for a dedicated host, you can get single-tenant capabilities.

Are there bare-metal options?

No, the bare-metal options are not available. The Power Virtual Server offering focuses on virtual instances.

Can you tell me more about the snapshotting, cloning, and restoring capabilities?

Power Virtual Server provides the capability to capture full and point-in-time copies of entire logical volumes or data sets. Using IBM's *FlashCopy* feature, the [Power Virtual Server API](#) lets you create delta snapshots, volume clones, and restore your disks. To learn more, see [Snapshotting, cloning, and restoring](#).

What are the key differences between a snapshot and a clone?

The key differences are as follows:

Context	Snapshot	Clone
Definition	A snapshot is a thin-provisioned group of volumes that cannot be attached to a host or accessed or manipulated.	A clone is created from a snapshot and results in independent volumes which surface in the GUI and can be attached to hosts.
Primary function	Revert or restore the source disks to a desired state	Create a complete volume clone
Ease of creation	Easy and quick process	Three-step process and takes a long time
Pricing	Charged 30% of the regular storage rate	target volume storage plus the GRS costs

Differences between a snapshot and clone

See [Snapshots, cloning, and restoring](#) for more detailed information.

Are there any initial snapshot requirements in terms of storage?

None. The storage is allocated on demand.

Does the snapshot and volume clone support any safeguard policy?

None. Power Virtual Server does not (currently) provide any options to safeguarded copy (such as cyber protection).

Can you tell me more about the backup process by using the PowerHA Toolkit for IBM i?

The PowerHA Toolkit for IBM i provides the 5250 user interfaces and automation to use them for backups. In a nutshell, it does the following tasks:

- Automates the memory flush
- Create the volumes-clone
- Attach the clones to a host
- Start the host
- Kick-off the backups
- Move the BRMS data back to the production VM and then shut down the backup VM
- Remove the cloned volumes

Using the PowerHA toolkit, you can create an intermediate snap-shot and a volumes-clone before the process enters the long-running volume detach or attach phase. You can pause the process immediately before the volumes are attached.

How do you set up private networks between Intel® Virtual Servers (x86) and Power Virtual Server?

IBM data center

See the tutorial on [IBM Power Virtual Server integration with x86-based workloads](#).

How do you set up customer site access to a private network by using VPN?

IBM data center

For a complete tutorial about site-to-site Virtual Private Network (VPN) connectivity from a private cloud environment to Power Virtual Server, see [IBM Power Virtual Server Virtual Private Network Connectivity](#). For more information on VPN, see [Managing VPN connections](#).

What firewall options are there around VPN connectivity?

IBM data center

You must set your own firewall in your IBM Cloud account.

How do you connect a server instance between two data centers (DAL13 to WDC04)?

You can use IBM Cloud Connect to connect two data centers. IBM Cloud Connect is a software-defined network interconnect service that brings secure connectivity to client locations around the world.



Important: IBM Cloud Connect is only available to IBM clients within the US.

How is network bandwidth billed?

IBM data center

IBM Cloud Classic environment: Inbound bandwidth is unlimited and not charged. Outbound bandwidth is charged per GB tier with bandwidth offered as an allotment for each month. As an example, for your compute instances, 250 GB is included with each monthly virtual server and 20 TB is included with each monthly bare metal server. Extra bandwidth can also be purchased per package. For more information, see [Bandwidth packages](#).

IBM Power Virtual Server environment: Inbound bandwidth is unlimited and not charged. Bandwidth is not charged when you use a public network. If you are using a private network with DirectLink Connect, you are charged **IBM Cloud Classic environment** rates.

What monitoring services are available?

IBM data center

IBM provides performance monitoring and status monitoring services for Power Virtual Server only if the Power Virtual Server workspace is registered as an observability instance on the IBM Cloud Monitoring system. The IBM Cloud Monitoring system is a cloud-native and container-intelligence management system. You can get the visibility to the performance and health of your applications, services, and platforms through the IBM Cloud Monitoring system. A Power Virtual Server workspace is registered as an observability instance on the IBM Cloud Monitoring system when the **Monitoring** option is set to for the workspace. By default, the **Monitoring** option is set to on when you create an IBM Power Virtual Server workspace. You can set the **Monitoring** option to on or off for an existing workspace on the Workspace details page. For more information, see [Monitoring a workspace](#).

For more information about the regions that support IBM Cloud Monitoring, see [Regions for IBM Cloud Monitoring](#).

What performance and capacity planning services do you provide for IBM i?

IBM uses the same tools that are on a private cloud system.

IBM i and solution certification

You can find self-certification and listing information on the [IBM Global Solutions Directory](#).

How to delete a workspace?

To delete your workspace and its resources from IBM Power Virtual Server, complete the following steps:

1. Log in to [IBM Cloud catalog](#) with your credentials. IBM Cloud Dashboard screen is displayed.
2. Type **Power Virtual Server** in the search field of the IBM Cloud Dashboard to view the Catalog Results.

3. Select **Power Virtual Server** from the Catalog Results. Power Virtual Server window is displayed.
4. Click **Workspaces** on the left navigation pane of the **Power Virtual Server** window. A list of workspaces that are available in your account is displayed.
5. Click the overflow menu of the workspace that must be deleted to view the delete option.
6. Click **Delete** from the overflow menu. **Delete workspace** pop-up window with a confirmation message for deleting is displayed.
7. Type the workspace name in the text field provided to confirm the deletion.
8. Click **Delete**.



Note: You must first delete the resources of the workspace to increase the success of the workspace deletion.

How do I delete a single virtual server instance?

Deleting a virtual server instance is a manual process. To delete all VSIs, delete the workspace or delete a subset of the virtual server instance.

- Delete a single virtual server instance from the Virtual server instances page. Click the overflow menu (icon with 3 vertical dots) on the far right of each virtual server instance entry on the table. From the pull-down menu, click **Delete** to open the deleted confirmation modal. Click **Delete instance** to initiate the deletion request. This action cannot be undone.
- Delete a single virtual server instance from the details page. On the Virtual server instances page, click the virtual server instance name present on the table, and go to the virtual server instance details page. Find and click the trash icon on the upper right of the screen. Confirm the request by clicking **Delete instance**. This action cannot be undone.

How does the processor compatibility mode work in a VSI?

You can set the preferred processor compatibility mode for a Virtual server instance (VSI) during its creation by using the GUI, CLI, API, or Terraform. For VSIs that are already deployed, you can change the preferred processor compatibility mode by modifying the VSI settings.

On the **Overview** tab of the VSI details page in the Power Virtual Server user interface, you can view the preferred and effective processor compatibility modes. The following processor compatibility modes are displayed:

- **Preferred:** The processor mode in which you want the VSI to operate. By default, Power Virtual Server sets the preferred processor compatibility mode to the highest mode that is supported by the targeted host type for the VSI.
- **Effective:** The processor mode that is currently in use for the VSI. The physical host on which the VSI runs determines the effective processor compatibility mode.



Important: You cannot dynamically change the effective processor compatibility mode of a VSI. To change the effective processor compatibility mode, you must first change the preferred processor compatibility mode of the VSI, shut down the VSI, and then start the VSI. When you start the VSI, the hypervisor attempts to set the effective processor compatibility mode to match the preferred mode that you have specified for the VSI.



Note: The effective processor compatibility mode for the VSI might not match the preferred mode that you have selected. If the operating system installed in the VSI does not support the preferred processor compatibility mode, the hypervisor can set the effective mode to a lower mode than the preferred mode. However, the hypervisor cannot set the effective mode to a higher mode than the preferred mode.

When the VSI is deployed and activated, the hypervisor on the host server checks the preferred processor compatibility mode and determines whether the operating system that is running in the VSI supports that mode. If the operating system supports the preferred processor compatibility mode, the hypervisor assigns the preferred processor compatibility mode to the VSI. If the operating system does not support the preferred processor compatibility mode, the hypervisor assigns the highest processor compatibility mode to the VSI that is supported by the operating system. After the VSI is activated, you can check the effective processor compatibility mode on the VSI details page.

For example, if a VSI runs on a POWER11 processor-based host and you set **POWER11** as the preferred processor compatibility mode, but the operating system installed in the VSI supports only **POWER10**, the hypervisor sets **POWER10** as the effective processor compatibility mode. This mode is the highest mode that the operating system and the host firmware support and is lower than the preferred mode of **POWER11**.

If you set the preferred processor compatibility mode to **Default**, the hypervisor does not consider **POWER9** as a valid effective processor compatibility mode. In the default mode, the effective modes can be **POWER9_Base**, **POWER10**, or **POWER11**, but it cannot be **POWER9**.

Use the **Default** mode as the preferred processor compatibility mode with caution. If the VSI for which the preferred mode is set to **Default** migrates to a different host during maintenance operations (or for other reasons) and you subsequently shut down and start the VSI again, the hypervisor might assign a different mode than the one the VSI was previously running in. This reassignment of the processor compatibility mode can prevent the VSI from migrating back to its original host.

For more information about how to change the preferred processor compatibility mode for a VSI, see [Changing the preferred processor compatibility mode](#).

How do I open a support ticket for the Power Virtual Server workspace?

To open a support ticket, see [Getting help and support](#).

What are the supported databases that I can run for SAP on Power Virtual Server?

On an AIX VM, the following databases are supported:

- IBM Db2 for LUW (Linux, UNIX, and Windows) version 10.5, or later
- Oracle Database version 12.1.0.2, or later
- SAP Adaptive Server Enterprise version 16.0 SP03, or later

On a Linux VM, the following database is supported:

- SAP HANA Platform 2.0 SPS 04, or later

You can find an up-to-date list at [SAP Apps on IBM Power Virtual Server](#).

How can I get the WebSphere Application Server that is included with the Web Enablement for i packages?

If you have an IBM i VM instance with the licensed program bundle in the Power Virtual Server offering, you can download the WebSphere Application Server at no extra cost. This is available in the Web Enablement for i software at the Entitled System Support (ESS) website by completing the following steps:

1. Go to the [ESS website](#).
2. Sign in. If this is the first time you are using ESS, refer to the **Help** section on the left menu. Download and read the **ESS_Registration_IBM_Customers_Guidelines** PDF.
3. Go to **My Entitled Software > IBM i evaluation and NLV download**.
4. Find the required software that you can download, install, and use. For example:

Web Enablement for i (5722-WE2) - WebSphere Express V8.5.5 **Web Enablement for i (5733-WE3)** - WebSphere V9

How do I run Red Hat OpenShift Container Platform (OCP) on Power Virtual Server?

You can find a complete tutorial at the IBM Developer site: [Deploying Red Hat OpenShift Container Platform 4.x on IBM Power Virtual Server](#).

What is the network latency over Direct Link?

IBM data center Network latency over Direct link is less than 1 millisecond in every location. To know more about network latency, see [Understanding latency](#).

What must be the network latency between the data center and the corresponding IBM Cloud region?

Client location The network latency between your data center and the corresponding IBM Cloud region must maintain a network round-trip time (RTT) of less than or equal to 200 milliseconds. For more information, see [Network latency](#).

Are we notified about any planned maintenance activities?

For planned maintenance and disruptive changes, the Power Virtual Server operations team sends you notifications at least 7 days in advance. Watch the notifications space in the IBM Cloud dashboard for these alerts. You can receive a copy of these notifications directly in your inbox if your email is subscribed for notifications.

How do I convert existing volumes to replication-enabled volumes?

You can retype the volume to toggle the `replicationEnable` flag of the volume by using [Perform an action on a Volume](#) request. This is possible only when the volume pool of existing volumes supports replication.

How can I check whether volume is already replication enabled?

You need to check the `replicationEnabled` attribute of the volume. A volume is `replicationEnabled` when it is true.

How can I check whether a volume is a primary or an auxiliary volume?

Volume is an auxiliary when `isAuxiliary` field of volume is true. When `replicationEnabled` is true and `isAuxiliary` is false then the volume is a primary volume.

Can I update the storage tiers for the Global Replication Services (GRS) enabled volumes?

You cannot update the storage tiers for the GRS enabled volumes. To change the storage tier type, complete the following steps:

1. Remove the volume from the volume group and disable GRS by completing the steps provided in the [Disabling GRS](#) topic.
2. Update the volume to the required storage tier type.
3. Enable the replication on the volume by setting the `replicationEnabled` flag as `True`.
4. Add the replication-enabled volume back to the volume group.

How can I check the serial number of my virtual server instance?

The serial number is available after you deploy your virtual server instance and you can choose to display the serial number system value.

What should I do if I do not see the latest information in the UI?

Consider the following if you do not see an update in the User Interface(UI):

1. Power Virtual Server uses a new caching mechanism for some resources to ensure that UI refresh operations complete in a timely manner.
2. In some scenarios out-dated information might be shown while the cache is updated, for approximately four minutes.
3. You can refresh the page to trigger an update to the cached data, eventually leading to the updated information's display.
4. When the DC has a heavy amount of traffic, and the cache is not refreshed within the last four minutes, Power Virtual Server UI might display an error message. A subsequent page refresh retrieves and displays the updated information.

Why can't I see the storage pool and tier of my boot images?

IBM improved the performance of copying a stock image into customers' accounts. As a result of this new feature, the newly copied stock image acts like an image reference, where volumes are not accessible to the user. The improved process now offers:

1. Faster copy of stock image to your private project.
2. The stock image cannot be exported. One can do VM capture and export on a deployed VM that uses the stock image.
3. Storage pool and tier of a stock image shows "Empty" (API) or "Any" (UI) as VM can be deployed to any tier or pool by using the stock image.

Can I select a specific resource group when I create a cloud connection?

IBM data center No. When you create a cloud connection by using Power Virtual Server, the cloud connection is always created in the default resource group even if you choose a specific resource group.

What is the Maximum Transmission Unit (MTU) capacity that is supported in Power Virtual Server networks?

The Power Virtual Server supports an MTU capacity of 1476 bytes for the public network interfaces and for the private network interfaces that are attached to a Power Virtual Server VPN.

Can I automate the Maximum Transmission Unit (MTU) configuration?

Yes, you can automate the network configurations such as the Maximum Transmission Unit (MTU).

To automate the MTU configuration, you need to customize your cloud-init network configuration. For more information, see the [Cloud-init docs on network configuration](#).

Both AIX and IBM i support the configurations for custom cloud-init at the time of Power Virtual Server instance (VM) deployment.

You can customize the cloud-init configurations only through the Power Virtual Server API. The `userData` request parameter specifies the custom cloud-init. For more information, see [Create a new Power VM Instance](#).

Client location The automation of MTU is not supported. The admin must update the MTU value on the virtual machine manually.

Can I add a user interface to an existing virtual machine?

Yes, you can add a user interface to an existing virtual machine by performing Operation System administration steps to configure the required adapter settings.

How can I search for Power Virtual Server resources using the assigned user tags?

To view or search for resources that are provisioned in Power Virtual Server by using the assigned `user tags`, see [Searching for Resources](#). Note that the `user tags` are not included in the response for GET API and CLI requests.

How can I share images across IBM Cloud accounts?

In advanced image management scenarios, Power Virtual Server supports sharing images between IBM Cloud accounts. To enable this feature, open an IBM Cloud support ticket and state your business need. For more information about opening a support ticket, see [Getting Help and Support](#).

Troubleshooting

Troubleshooting AIX-related issues

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

Learn how to troubleshoot Power® Virtual Server AIX-related issues.

What can I do if my AIX virtual machine (VM) does not initially boot?

Symptoms: The AIX boot disk that you are using to provision an AIX VM is not successfully booting. As a result, the console displays a blank screen without standard debugging options.

Causes: The AIX boot disk might be corrupted.

Resolve: If the AIX VM does not boot, you must provision an extra AIX VM and use it as a Network Installation Management (NIM) server. Without a NIM server, you cannot debug the boot issue and must reimage your disk.

1. Determine the hostname and IP address of the system by using the following two commands:
 - `hostname` and
 - `ifconfig -a`
2. Using the information from the previous step, add an entry for your hostname and IP address into `/etc/hosts`. For example, `echo "192.168.0.15 aix-7100-05-04" >> /etc/hosts`.
3. Run the following command:

```
$ nim_master_setup -a device=/usr/sys/inst.images -a mk_resource=no
```

4. Once completed, the NIM master file set is installed and the basic resource objects created. The administrator is now able to add more NIM clients and define resources.

For more information, see [Setting up NIM to boot into maintenance mode](#). If you are unfamiliar with this process, create a [new support case](#).

How can I improve the performance of my AIX VM Tier 1 NVMe-based Flash Storage disk?

Symptoms: Your AIX VM with a Tier 1 (NVMe-based Flash Storage) disk is running the following NVMe specifications.

Causes: The AIX Tier 1 (NVMe-based Flash Storage) disk's default settings are hindering optimal performance.

Resolve: To improve disk performance, increase its **queue_depth** to *64* or *128* and change the algorithm to **shortest_queue** with a **no_reserve policy**. These modifications can improve performance by up to 50%.

To look at your AIX disk's current settings, enter the following commands:

- `lsattr -El hdiskX -a algorithm`
- `lsattr -El hdiskX -a reserve_policy`
- `lsattr -El hdiskX -a queue_depth`

 **Tip:** In the output of these `lsattr` commands, you will see either **True** or **True+** at the very end. **True** indicates that the attribute is user-changeable. **True+** indicates that the attribute is user-changeable, and can be modified while the disk is open by using the `-U` flag.

You can set your disk's **queue_depth** to *64* and change the algorithm to **shortest_queue** with a **no_reserve policy** by using the following command:

```
$ chdev -l hdiskX -a algorithm=shortest_queue -a reserve_policy=no_reserve -a queue_depth=64
```

If the AIX disk is open, an error message appears that states that the disk is busy. To prevent this error from occurring, add a `-U` flag to the `chdev` command and the values are updated without disruption (assuming the Object Data Manager (ODM) for the disk has that feature that is enabled).

Troubleshooting snapshot, clone and restore issues

Learn how to troubleshoot Power® Virtual Server snapshot, clone and restore issues.

How do I resolve the issue where both volumes in my AIX operating system have the same PVID?

Symptoms: When you clone and attach a volume to the same PVM as the PVM containing the original volume, the PVIDs of both volumes are found to be the same on your AIX operating system.

Causes: There's a problem with the underlying technology that is supporting the clone feature.

Resolve: To resolve this issue, enter the following commands:

1. `chdev -a pv=clear -l hdisk2`
2. `recreatevg -y copyvg hdisk2`
3. `mount /fs/<original-mount-point>`

In this example, `hdisk2` refers to the cloned volume name and `copyvg` is the name to be given to the newly created volume group. The `recreatevg` command creates a new file system with the same name as the original but in the `/fs` path. For more information, see "Section B" under [Possible Cause and Solution 2: The Disks Are Copies of Each Other](#).

Configuring and troubleshooting metadata service connectivity

Learn how to configure the metadata service network interface on your virtual server instance (VSI) and troubleshoot connectivity issues.

You must perform the following procedures while you are logged in to the operating system (OS) on your VSI. You can log in by using SSH or by accessing the OS through the Power Virtual Server console.

The metadata service uses a link-local network with IP address 169.254.169.253 on your VSI and connects to the metadata service endpoint at 169.254.169.254. This network is system-created and must not be modified unless required to correct connectivity issues or as a cleanup activity, as described in this topic.

Power Virtual Server uses `cloud-init` package to configure connectivity from a VSI to the metadata service. The `cloud-init` package is required for each OS. If you use a custom image and your OS does not have the IP address 169.254.169.253 configured after you enable metadata service access for your VSI, ensure that your custom image includes the `cloud-init` package.

Reconfiguring the metadata service interface on IBM i after an OS disk overwrite

Learn how to restore the metadata service network interface on an IBM i VSI after you overwrite the OS disk with a backup image.

Reconfiguring the metadata service interface after an OS disk overwrite

When you overwrite the OS disk of an IBM i VSI, the network configuration, including the metadata service interface, might be lost. Before you overwrite the disk, record the resource name and hardware address of the metadata service interface. After the overwrite, reconfigure the metadata service interface by using the same hardware address.

Before overwriting your image

To record the resource name and hardware address, complete the following steps:

1. From a 5250 session, run the following command to identify the metadata service interface:

```
$ NETSTAT *IFC
```

2. Look for the interface with IP address `169.254.169.253`. Record the metadata service interface name (for example, `ETH01`, `ETH02`).
3. Run the following command to display the line description details. Record the hardware address from the line description details.

```
$ DSPLIND LIND(ETH01)
```

Alternatively, you can use the following command:

```
$ WRKHDWRSC *CMN
```

Then select option 5 (Display details) for the Ethernet adapter.

- Record the following information:
 - Resource name (for example, `CMN05`)
 - Hardware address (MAC address, for example, `FA163EA1B2C3`)
- Save this information to use it after the image overwrite.

After overwriting your image

To reconfigure the metadata service interface by using the information that you recorded, complete the following steps:

- From a 5250 session, run the following command to display the hardware resources:

```
$ WRKHDWRSC *CMN
```

- Select option 5 (Display details) for each Ethernet adapter to identify which resource has the hardware address that you recorded.
- Run the following command to create a line description for the metadata service interface:

```
$ CRTLINETH LIND(METADATA) RSRcname(CMN05) LINESPEED(1G) DUPLEX(*FULL)
```

Replace `CMN05` with the resource name that you recorded. Set the `LINESPEED` value to match your the speed of your Ethernet adapter.

- Run the following command to activate the line description:

```
$ VRYCFG CFGOBJ(METADATA) CFGTYPE(*LIN) STATUS(*ON)
```

- Run the following command to add the TCP/IP interface:

```
$ ADDTCPIFC INTNETADR('169.254.169.253') LIND(METADATA) SUBNETMASK('255.255.255.252') AUTOSTART(*YES)
```

Verifying the metadata service interface configuration

To verify that the metadata service interface is configured correctly and active, complete the following steps:

- Run the following command to verify the metadata interface configuration:

```
$ NETSTAT *IFC
```

Look for the metadata service interface with IP address `169.254.169.253`. The status of the metadata service interface must be `Active`.

- Run the following command to test connectivity to the metadata service interface:

```
$ PING RMTSYS('169.254.169.254')
```

Press F3 to end the ping after confirming connectivity.



Tip: The metadata service network uses the link-local IP range `169.254.169.254/30`. The metadata service interface IP address is `169.254.169.253`. No gateway is required for this network.

Reconfiguring the metadata service interface on Linux after an OS disk overwrite

When you overwrite the OS disk of a Linux VSI with a previously created disk image, the network configuration, including the metadata service interface, might be removed. You can take preventative measures before you capture the image or use the MAC address to restore the configuration afterward.

Preventing network configuration loss

You can take one of the following preventative measures to avoid losing connectivity to the metadata service endpoint:

- Remove the network persistence rules before you perform the capture
- Note the MAC address of the interface to restore the configuration

Removing network persistence rules before capture

To prevent network configuration loss by removing network persistence rules before you capture the image, complete the following steps:

1. Run the following command to back up the file that contains network persistence rules, including the MAC address:

```
$ cp /etc/udev/rules.d/70-persistent-net.rules /home/admin/.
```



Note: The `admin` user might not exist in your environment. Adjust the path as needed for your environment.

2. Delete the contents of the `/etc/udev/rules.d/70-persistent-net.rules` file. Ensure that file permissions are retained.
3. Run the following command to back up the file that generates network persistence rules:

```
$ cp /lib/udev/rules.d/85-persistent-net-generator.rules /home/admin/.
```

4. Delete the contents of the `/lib/udev/rules.d/85-persistent-net-generator.rules` file. Ensure that file permissions are retained.

Before overwriting your image

If you prefer to keep the network persistence rules in place, record the MAC address of the interface that communicates with the metadata service endpoint. After you overwrite the boot disk of your VSI with the saved image, you can reconfigure the interface by using the MAC address.

To record the MAC address before you overwrite your image, complete the following steps:

1. Run the following command to identify the metadata service interface:

```
$ ip addr show | grep -B 2 "169.254.169.253"
```

The output shows the metadata service interface details, including the MAC address and IP address.

```
2: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
    link/ether fa:16:3e:a1:b2:c3 brd ff:ff:ff:ff:ff:ff
    inet 169.254.169.253/30 brd 169.254.169.255 scope link eth1
```

2. Record the MAC address from the output. In this example: `fa:16:3e:a1:b2:c3`.

Alternatively, run the following command to retrieve only the MAC address:

```
$ ip link show eth1 | grep link/ether | awk '{print $2}'
```

Replace `eth1` with the interface name from the output in step 1.

3. Save this MAC address to use it after the image overwrite.

After overwriting your image

If you removed the network persistence rules before the image capture, no further configuration is required. If you kept the network persistence rules in place and recorded the MAC address, complete the following steps to reconfigure the metadata service interface:

1. Run the following command to identify the interface that has the MAC address that you recorded:

```
$ ip link show | grep -B 1 "fa:16:3e:a1:b2:c3"
```

Replace `fa:16:3e:a1:b2:c3` with the MAC address that you recorded in step 2 of the [Before overwriting your image](#) section.

The output shows the metadata service interface name (for example, `eth1`, `ens4`).

2. Configure the interface by using the following procedures for your Linux distribution.

For RHEL or CentOS 7 and earlier

To configure the metadata service interface on RHEL or CentOS 7 and earlier, complete the following steps:

1. Create or edit the `/etc/sysconfig/network-scripts/ifcfg-eth1` file with the following content:

```
DEVICE=eth1
BOOTPROTO=none
ONBOOT=yes
```

```
IPADDR=169.254.169.253
NETMASK=255.255.255.252
```

Replace `eth1` with your metadata service interface name.

2. Run the following command to restart the network service:

```
$ systemctl restart network
```

For RHEL or CentOS 8 and later (using NetworkManager)

To configure the metadata service interface on RHEL or CentOS 8 and later with NetworkManager, run the following commands:

```
$ nmcli connection add type ethernet con-name trusted-profile ifname eth1 ip4 169.254.169.253/30
nmcli connection up trusted-profile
```

Replace `eth1` with your metadata service interface name.

For SLES (using wicked)

To configure the metadata service interface on SLES by using wicked, complete the following steps:

1. Edit the `/etc/sysconfig/network/ifcfg-eth1` file with the following content:

```
BOOTPROTO='static'
IPADDR='169.254.169.253/30'
STARTMODE='auto'
```

Replace `eth1` with your metadata service interface name.

2. Run the following command to restart the network:

```
$ wicked ifup eth1
```

Replace `eth1` with your metadata service interface name.

Verifying the metadata service interface configuration

To verify that the metadata service interface is configured correctly and active, complete the following steps:

1. Run the following command to verify the interface configuration:

```
$ ip addr show eth1
```

Replace `eth1` with your metadata service interface name.

The output must include the following IP address configuration:

```
inet 169.254.169.253/30 brd 169.254.169.255 scope link eth1
```

2. Verify that the metadata service interface is configured to start automatically on boot:
 - For RHEL or CentOS 7 and earlier: Verify that the `ONBOOT` parameter is set to `yes` in the interface configuration file.
 - For NetworkManager-based systems: NetworkManager configures the connection to start automatically on boot.
3. Run the following command to verify that the VSI can reach the metadata service and request an identity token:

```
$ curl -X PUT "https://api.metadata.power-iaas.cloud.ibm.com/identity/v1/token" -H "Metadata-Flavor: ibm" -H "Accept: application/json"
```

Steps to perform within AIX after disabling access to the metadata service endpoint

When you disable access to the metadata service on an AIX VSI, the network interfaces are not automatically removed. Network interfaces that are left in a "Defined" state require cleanup.

To clean up the metadata service network interfaces, complete the following steps:

1. Run the following command to list all network adapters:

```
$ lsdev -Cc adapter
```

2. From the output, identify the adapter that is in the "Defined" state.
3. Run the following command to display the adapter attributes:

```
$ lsattr -El en<X>
```

Replace `<X>` with the adapter number that you identified in step 2.

4. Verify that the IP address shown in the output is `169.254.169.253`.
5. Run the following commands to remove the logical and physical device entries:

```
$ rmdev -dl en<X>
rmdev -dl ent<X>
```

Replace `<X>` with the adapter number from step 2.

Cleaning up network interfaces after disabling the metadata service on Linux

If you use the `force_disable` flag during a VSI edit, you must clean up the network interface that the metadata service used. The tool that configures the metadata service interface varies based on the Linux distribution. The following example uses NetworkManager and the `nmcli` command to remove the connection.

To clean up the metadata service network interface, complete the following steps:

1. Run the following command to identify the interface with the IP address `169.254.169.253`:

```
$ ip a
```

2. Run the following command to list all connections:

```
$ nmcli con show
```

The output displays connection information in columns:

NAME	UUID	TYPE	DEVICE
trusted-profile	1875ae2a-0235-48de-b451-777154597ea9	ethernet	eth1
lo	fc819d11-25f8-456a-a026-a926639ba021	loopback	lo

Locate the connection that uses the interface name from step 1, and record the UUID (the second column).

Alternatively, you can filter the output by using the interface name:

```
$ nmcli con show | grep <interface_name>
```

Replace `<interface_name>` with the interface name from step 1. Record the UUID from the second column of the output.

3. Run the following command to delete the connection from your OS configuration:

```
$ nmcli con delete <UUID>
```

Replace `<UUID>` with the UUID that you recorded from step 2.

 **Important:** Ensure that you do not delete the connection for your primary network interface.

Troubleshooting metadata service connectivity on AIX

If your AIX VSI cannot reach the metadata service endpoint, complete the following steps to validate and restore connectivity

Validating and restoring network connectivity

To validate and restore network connectivity to the metadata service endpoint, complete the following steps:

1. Run the following command to verify whether the IP address `169.254.169.253` is configured on any interface:

```
$ ifconfig -a
```

2. If the IP address is not present in the output, identify an interface that does not have an IP address assigned.
3. Run the following command to configure the IP address and bring the interface up:

```
$ chdev -l en2 -a netaddr=169.254.169.253 -a netmask=255.255.0.0 -a state=up
```

Replace `en2` with the interface name that you identified in step 2.

4. Run the following command to ensure that the interface is active:

```
$ ifconfig en2 up
```

Replace `en2` with your interface name.

5. Run the following command to validate connectivity to the metadata service endpoint:

```
$ curl -X PUT "https://api.metadata.power-iaas.cloud.ibm.com/identity/v1/token" -H "Metadata-Flavor: ibm" -H "Accept: application/json"
```

This command retrieves an instance identity token from the metadata service.

If you still cannot communicate with the metadata service endpoint, contact [IBM Support](#).

Troubleshooting metadata service connectivity on Linux

If your Linux VSI cannot reach the metadata service endpoint, complete the following steps to validate and restore connectivity.

Validating the network configuration

To validate the metadata service network configuration, complete the following steps:

1. Run the following command to verify whether the IP address `169.254.169.253` is configured on any interface:

```
$ ip a
```

2. If the IP address is not present in the output, identify an interface that does not have an IP address assigned. Then, configure the IP address on that interface by following the steps in [After overwriting your image](#).

Troubleshooting when the interface is configured but unreachable

If the metadata service interface is configured but you cannot reach the metadata service endpoint at `169.254.169.254`, complete the following steps:

1. Run the following command to verify that the route exists:

```
$ ip route show dev eth1
```

Replace `eth1` with your metadata service interface name.

The output should show a route for the `169.254.169.252/30` network.

2. Run the following commands to check firewall rules:

For `firewalld`:

```
$ firewall-cmd --list-all
```

For `iptables`:

```
$ iptables -L -n
```

If necessary, add a rule to allow traffic to IP address `169.254.169.254`.

3. Run the following command to verify that no conflicting routes exist:

```
$ ip route show | grep 169.254
```



Important: If you see conflicting routes, carefully review the routes before making any changes. Your custom image might include routes that were intentionally configured for your environment. Removing routes without understanding their purpose might disrupt other network connectivity. Ensure that removing a route does not break any other intentional network configurations.

Troubleshooting metadata service connectivity on IBM i

If your IBM i VSI cannot reach the metadata service endpoint, use these procedures to diagnose and resolve the connectivity issue.

Resolving interface activation issues

If the interface configuration is in place but the interface does not activate, complete the following steps:

1. Run the following command to check the line description status:

```
$ WRKLIND LIND(METADATA)
```

The status should show **VARIED ON**.

2. If the line is varied off, run the following command to vary it on:

```
$ VRYCFG CFGOBJ(METADATA) CFGTYPE(*LIN) STATUS(*ON)
```

3. Run the following command to check for error messages in the job log:

```
$ DSPJOBLOG
```

Look for messages that are related to the line description or TCP/IP interface.

4. Run the following command to verify that the resource name is correct and not in use by another line description:

```
$ WRKHDWRSC *CMN
```

Ensure that the resource is not allocated to another line description.

5. Run the following command to check if the interface is set to autostart:

```
$ NETSTAT *IFC
```

Look at the Auto-Start column. If it shows **NO**, run the following command to update the interface:

```
$ CHGTCPIFC INTNETADR('169.254.169.253') AUTOSTART(*YES)
```

Validating network connectivity to the metadata service endpoint

If the metadata service interface is configured but you cannot reach the metadata service endpoint at **169.254.169.254**, complete the following steps:

1. Run the following command to verify that the route exists:

```
$ NETSTAT *RTE
```

The output should show a route for the **169.254.169.252/30** network associated with your metadata interface.



Important: If you see conflicting routes or interface configurations, carefully review the routes before making any changes. Your custom image might include configurations that were intentionally set for your environment. Removing or modifying configurations without understanding their purpose might disrupt other network connectivity. Ensure that any changes do not break other intentional network configurations.

2. Run the following command to check if the interface is active:


```
$ NETSTAT *IFC
```

The interface with IP address `169.254.169.253` should show status `Active` .

3. Run the following command to verify that the line description is varied on:

```
$ WRKLIND LIND(METADATA)
```

The line description status should show `VARIED ON` .

4. Run the following command to access the TCP/IP configuration menu:

```
$ CFGTCP
```

Select option 1 (Work with TCP/IP interfaces) and verify that the metadata service interface configuration is correct.

5. If the previous steps do not restore connectivity, configure the metadata service IP address on another `CMN*` device and test connectivity again. The RMC interface that is used by Power Virtual Server might not have been configured on your OS, or another `CMN*` device might be available but unconfigured.

If you are still unable to communicate with the metadata service endpoint, contact [IBM Support](#).

Getting Help and Support

IBM Power Virtual Server in IBM data center

IBM Power Virtual Server Private Cloud in Client location

You can choose a Basic, Advanced, or Premium support plan to customize your IBM Cloud® support experience for your business needs. The level of support that you select determines the severity that you can assign to support cases and your level of access to the tools available in the Support Center. For more information about different support plans, see [Basic, Advanced, and Premium support plans](#).



Note: Users with a Lite or Trial account can create support cases but are limited to non-technical issues.

Before you open a support ticket against Power® Virtual Server, review the [FAQ](#) and [Getting support](#).

Opening PowerVS infrastructure cases

To open a support ticket for IBM Power Virtual Server in IBM data center or in Client location, complete the following steps:

1. Log in to the [IBM Cloud](#) with your IBM Cloud account credentials.
2. In the menu bar, click **Help** (?) icon. Select the **Support center** option.
3. On the **Support Center** page, click **Create a case**. The **Create a case** page opens.
4. From the **Topic** menu, select **Power Virtual Server** option.
5. From the **Subtopic** menu, select the topic that is related to your issue. For example, Power VS Compute Related or Power VS On Premises Related.
6. Provide a short description of the issue in the **Subject** field and a detailed description of the issue in the **Description** field.
7. (Optional) Attach any helpful files in the **Attachments** field.
8. (Optional) From the **Watchlist** menu, select the users in your account who are interested to get the updates about the issue.
9. (Optional) From the **Add resources** menu, select your impacted PowerVS Private Cloud resource.
10. (Optional) From the **Data center settings** menu, select your data center.
11. (Optional) To get the email notifications, select **Email me updates about this case** checkbox.
12. Review your support ticket summary, and click **Submit case**. You will receive an email verification. Follow the instructions on the email for further communication on the issue.

Opening Operating System (AIX®, Linux®, or IBM i) related cases IBM data center

If you are facing any AIX, Linux, or IBM i operating system-specific issue, you must directly engage with the operating system support by using the portal as follows:

1. Log in to the [IBM Support](#) portal by using your IBMid.
2. Click **Open a case**.
3. On the **Product** field, enter **AIX on PowerVS** or **IBM i on PowerVS**.
4. Complete all the other required information, and click **Submit Case**.

Opening Operating System (AIX, Linux, or IBM i) related cases Client location

If you are facing any AIX, Linux, or IBM i operating system-specific issue, you must directly engage with the operating system support by using the portal as follows:

1. Log in to the [IBM Support](#) portal by using your IBMid.
2. Click **Open a case**.
3. On the **Product** field, enter AIX, Red Hat Enterprise Linux Server, or IBM i on Cloud. For AIX, enter any 7-digit serial number in the **Machine Serial Number** field. For Red Hat Enterprise Linux Server or IBM I on Cloud, no serial number is required.
4. In the **Account** field, provide the account or customer number that is listed on your Client location invoice to ensure that the case is addressed quickly.
5. Complete all the other required information, and click **Submit Case**.

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